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Yann Bouchery
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Sustainable Supply Chains

A Research-Based Textbook
on Operations and Strategy

Second Edition



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Christopher S. Tang, University of California, Los Angeles, CA, USA

Supply Chain Management (SCM), long an integral part of Operations Management, focuses on all elements of creating a product or service, and delivering that product or service, at the optimal cost and within an optimal timeframe. It spans the movement and storage of raw materials, work-in-process inventory, and finished goods from point of origin to point of consumption. To facilitate physical flows in a time-efficient and cost-effective manner, the scope of SCM includes technology-enabled information flows and financial flows.

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Editors

Sustainable Supply Chains

A Research-Based Textbook on Operations
and Strategy

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Preface to the Second Edition: About This Book

Objectives and Intended Audience

Who is this book for? In this preface, we provide a brief introduction to the objectives and intended audience of this book, outline the guiding principles we followed during its assembly, discuss the changes made since the first edition, and describe its structure. In the Introduction (Chap. 1, by Bouchery et al. 2024), we offer some more broader perspectives on the current and future state of sustainability in supply chains.

Sustainability is increasingly seen as a supply chain issue, not something that a single firm can deal with effectively. We provide below some examples of relevant work related to sustainable supply chain management and sustainable operations. Each of these reviews and books has their place, but none of them were produced with more technical graduate-level courses on sustainable supply chains in mind. That is our objective, to provide a textbook that can be used in MSc or relatively technical MBA courses (or advanced undergraduate courses) in this area. We have observed an increase in the number of such courses being offered, but without a corresponding increase in teaching materials. This book aims to draw on the latest research to support courses on sustainable supply chains for graduate programs such as Business, Management, Industrial Engineering, and Industrial Ecology.

Several authors have produced review articles on sustainable supply chains, as well as special issues of journals, including Linton et al. (2007), Srivastava (2007), Seuring and Müller (2008), Carter and Easton (2011), Brandebourg et al. (2014), Reefke and Sundaram (2017), Barbosa-Póvoa et al. (2018). Reviews on sustainable operations, often including some discussion of supply chains, include Angell and Klassen (1999), Kleindorfer et al. (2005) and Corbett and Klassen (2006), Drake and Spinler (2013), Jaehn (2016), and Atasu et al. (2020). Several books with similar titles have also appeared, each with their own focus. Some are aimed primarily at practitioners, such as *Greening the Supply Chain*, edited by Sarkis (2006); *Sustainable Supply Chains: Models, Methods, and Public Policy Implications*, edited by Boone et al. (2012); *Sustainable Supply Chain Management:*

Practical Ideas for Moving Towards Best Practice, edited by Cetinkaya et al. (2011); *Green Supply Chains: An Action Manifesto* by Emmett and Sood (2010); *Green Supply Chain Management: Product Life Cycle Approach* by Wang and Gupta (2011) and *Sustainable Operations and Supply Chain Management* by Belvedere and Grando (2017). The books on *Green Logistics: Improving the Environmental Sustainability of Logistics*, edited by McKinnon et al. (2015), and *Sustainable Logistics and Supply Chain Management: Principles and Practices for Sustainable Operations and Management*, authored by Grant et al. (2017) focus more in depth on logistics rather than on supply chains in the broader sense of the word. The *Handbook on the Sustainable Supply Chain*, edited by Sarkis (2019), mainly target scholars and advanced business and management students.

Rather than write a mediocre book ourselves, we turned to leading experts to provide overviews of their respective fields. We asked them to describe the current state of affairs and to identify future needs and trends, rather than presenting new results not yet published elsewhere. Nevertheless, we anticipate that the book should also be of interest for researchers in the broader sustainable supply chain space, whether from the operations management and industrial engineering side or more from the industrial ecology and life cycle assessment side, as a vehicle to learn from another community and to identify new research opportunities.

We worked with Springer, and with the series editor Professor Chris Tang, to keep the price of the book as low as possible. Although the standard purchase price was still higher than we liked, the book was available for free to institutions that subscribed to Springer's ebook package, and it was available at a low price (currently USD 39.99) in MyCopy Softcover version. The biggest boost came from Springer's decision to include the first edition in their collection of over 500 "key textbooks" that they made available online for free in response to the covid-19 pandemic in Spring 2020. We were delighted with the book's reception: chapters have been accessed 2.65 million times (as of February 15, 2024). That was one of the reasons we decided a second edition might be worthwhile.

Philosophy Behind This Book

In assembling this book, we followed a few key principles (discussed at more length in the Introduction). First, we use the term "sustainable" in a loose sense, meaning that one considers the triple bottom line in making decisions, i.e., taking into account economic, environmental, and social aspects. Second, we emphasize that sustainability is multi-dimensional and that measurement is key, which is why the book starts with several chapters on measurement. Third, we recognize the need to address a wide range of aspects of "sustainability" in the book, but do not necessarily aim to cover them all in every chapter. Fourth, and most importantly, we decided to let the experts who contributed these chapters speak: it is more important for readers to be exposed to a wide view of experts than just to hear our opinions.

As a result, some topics may be covered in multiple chapters, while other topics are not covered at all.

The Second Edition

In preparing the second edition, we invited almost all authors from the first edition to update their chapters. The vast majority agreed, and while some revisions were relatively light updates, others were quite substantive. Sadly, a few authors from the first edition are no longer with us. In the first edition of this book, Jacqueline Bloemhof (1969–2020) contributed to a chapter on sustainable food supply chain design together with Mehmet Soysal; Brian Fynes (1959–2020) contributed to a chapter on risk and uncertainty management for sustainable supply chains together with Kirstin Scholten; Arjan Hoekstra (1967–2019) contributed to a chapter on water footprinting. For this second edition, we have reproduced Arjan Hoekstra's chapter. We have invited Renzo Akkerman and Frans Cruijssen to present a deep dive into food loss and waste. We dedicate this chapter to Jacqueline's memory in the confidence that she would have appreciated this focus on food loss and waste. Finally, Kirstin Scholten invited Mark Stevenson to contribute to the update of the chapter about risk and uncertainty management for sustainable supply chains. We also identified several topics that have grown so much in importance since the first edition that they now merit their own chapter: demand management for sustainable supply chains, collaboration in supply chains, the circular economy, and reusable packaging.

Structure of This Book

The structure of the book is straightforward. Part I starts with several chapters on measurement and reporting. Separate chapters provide introductions to life cycle assessment, carbon footprinting, and water footprinting. Part II covers core operational aspects of sustainable supply chains, with separate chapters devoted to green logistics, green inventory management, green network design and facility location, operational implications of environmental regulation, food loss and waste in supply chains, and demand management for sustainable supply chain operations. Part III revolves around issues related to business models and strategy in sustainable supply chains, with chapters on supply chain collaboration for sustainability, green technology choice for deep decarbonization, business implications of sustainability practices, moving from a product-based to a service-based economy, a strategic overview of closed-loop supply chains, a guiding framework for circular supply chains, reusable packaging management, sustainable non-renewable materials management, and managing risk and uncertainty for sustainable supply chains. Part IV focuses more on the social dimension of sustainability, with chapters on

how to improve social and environmental performance in global supply chains, perspectives on social responsibility in supply chains, and how to manage cross-sector partnerships with NGOs in sustainable supply chains. Inevitably, some chapters could have been arranged differently, and they can be read in any sequence. All chapters are also available from the publisher separately.

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Acknowledgments

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Between the four of us, we have too many parents, spouses, partners, children, colleagues, friends, and others to whom we are deeply indebted for their support (voluntary or involuntary, knowing or otherwise) to name them all individually. We wish we could promise that with the completion of (the second edition of) this book we will have more time for all of you, but we cannot rule out that something else will come along to preoccupy us.

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Chapter 1

Sustainable Supply Chains: Introduction



Yann Bouchery , Charles J. Corbett , Jan C. Fransoo , and Tarkan Tan 

1.1 Sustainable Supply Chains: What's New?

Are supply chains becoming more or less sustainable over time? The answer is not obvious. In the 1960s, we became aware of the dangers inherent in various chemicals including those used in pesticides, specifically DDT. Some would point to Rachel Carson's 1962 book *Silent Spring* as a defining moment in the history of the environmental movement, contributing to the creation of the environmental movement and the Environmental Protection Agency in the USA. In the 1970s, we started to learn that seemingly innocent aerosols in spray cans were eating a hole in the ozone layer, removing a crucial protection against being toasted and eventually burned by the sun. Various regulatory and voluntary actions followed, and depletion of the ozone layer seems to have slowed down recently. In the 1980s, we found out that acid rain was killing our lakes and eating our cities. Various regulatory responses have since helped to reduce emissions of chemicals leading to acidification. During the 1990s, we heard horror stories of how workers in

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sweatshops in China, Vietnam, and elsewhere were treated. Firms such as Nike initially responded that they did not own the factories, and hence it was not their problem, but by the 2000s, at least some firms (including Nike) had completely changed their tone. Throughout these decades, it transpired increasingly clearly that anthropogenic emissions were likely contributing to rising levels of greenhouse gases in the atmosphere, with a variety of effects on sea level and climate. A plethora of regulatory and voluntary responses is underway to mitigate climate change, with decidedly mixed results so far.

But how much of all this is new? Insecticides have been regulated in the USA as far back as 1910, though what is now the UK already passed a rule in about 1236 AD that “forbade the addition of anything to the food supply which was ‘not wholesome’” (Aspelin 2003, Chap. 2, pp. 10–11). The ozone layer was discovered in 1913, and its properties and thickness were already increasingly measured during the subsequent decades (e.g., Solomon 1999). The link between acid rain and atmospheric pollution was established in 1852, with the term “acid rain” appearing in 1872; damaging effects of pollution had been observed even earlier (see, for example, Cowling 1982). Early anti-sweatshop movements gave rise to the Factory Act in the UK in 1833 (Nardinelli 1980) and more recently to the Fair Labor Standards Act in the USA in 1938 (Samuel 2000).

Climate change is no different. Arrhenius (1896) already reported on the connection between atmospheric carbon dioxide concentrations and global temperature, predicting that a doubling of the CO₂ concentration would lead to a temperature increase of 4.9 to 6.1 °C (depending on the latitude). This estimate is remarkably close to more recent ones by the IPCC, who report a range of predicted temperature rises between 1.5 and 4.5 °C (Houghton et al. 1995). Arrhenius was keenly aware of the geological importance of his calculations, writing (p. 267) “I should certainly not have undertaken these tedious calculations if an extraordinary interest had not been connected with them.” His primary motivation however was explaining the Ice Age, rather than predicting future global warming.

These timelines have a clear implication for any book on “sustainable supply chains”: scientists of all kinds are undoubtedly already making observations and constructing theories of environmental and social aspects that we are blissfully unaware of today but that will come to haunt us in the decades to come. In other words, there can be no such thing as a “sustainable supply chain,” as we do not and cannot know today what will be critical elements of “sustainability” tomorrow.

Even if we limit ourselves to the environmental and social factors that we are already aware of, a truly “sustainable” supply chain is difficult to envision. This is certainly true when one adopts a strong sustainability perspective, which does not allow manufactured capital to substitute for natural capital. Almost any supply chain, at some level, converts natural capital into manufactured capital, rendering them inherently unsustainable under that perspective. In the weak sustainability view, it is acceptable to deplete natural resources, as long as there is a corresponding increase in manufactured capital. Deforestation is acceptable as long as technological progress delivers substitutes for wood. Increasing water pollution is acceptable as long as decontamination technology advances in parallel. In this perspective, a

supply chain could be sustainable if its “net” impact on people and planet is neutral, still a tall order. As Pagell and Wu (2024) argue in Chap. 13, sustainability in supply chains should be thought of in terms of trajectories, rather than absolute states.

Our quick historical sketches above underline that such trajectories can never have a well-defined endpoint. Instead, a more sustainable supply chain is one that is better at identifying current and future environmental and social impacts and finding ways to mitigate those. That is the, admittedly very loose, definition we use in this book for “sustainable.” We take no position on weak vs. strong sustainability, on whether one should be more concerned about climate change, water availability, or social impacts. We do take the position that for a firm to thrive, it is increasingly imperative that it be aware of economic, environmental, and social dimensions of the entire supply chain it belongs to and that it proactively monitor and manage those.

The arguments so far might point toward the need for a firm to engage with its economic, environmental, and social impacts. But why would it need to do so for the entire supply chain within which it operates? As firms become progressively more tightly coupled in global supply chains, rather than being large vertically integrated monoliths, risks and opportunities associated with activities upstream or downstream will increasingly impinge upon their own well-being. Referring to “risks and opportunities associated with activities upstream or downstream” clearly signifies an equally broad and loose definition of “supply chains”: we will not attempt to draw a clear boundary of what we do and do not consider part of a supply chain, but, in this book, we do include not just suppliers and customers but also consumers, society, employees, communities, NGO partners, and others who in one way or another participate in the “activities upstream or downstream.” For many firms, some of the main risks and opportunities do lie outside their own boundaries, hence necessitating this broader supply chain perspective.

To summarize, this book should not be called “Sustainable Supply Chains,” but rather something like “Gradual but Never-Ending Paths Toward Less Unsustainable Global Value Networks.”

1.2 Sustainable Supply Chains: What’s Really New?

Having argued that in some sense “sustainability” is not new, one might still ask: What *is* new? Why did we assemble the first edition of this book in 2017 and why a second edition at this particular point in time?

First, the scientific understanding of many aspects of sustainability is advancing faster than ever before. That was true in 2017 and is still true today. In the past, many environmental and social issues were largely treated independently from one another, but we are increasingly understanding that climate change interacts heavily with health and poverty. The links between energy, food, and water are sinking in. Climate change threatens biodiversity which in turn endangers continuity of supply of various crops, putting already poor constituencies at further risk, etc.

Second, the political landscape is evolving in parallel. While there is a wide variation in regulation and enforcement across countries and even across cities or regions within countries, the global trend is undeniably toward more and more comprehensive environmental and social regulation. In particular, the “Paris Agreement” has initiated many regulatory and subsidy initiatives, in particular in the European Union and to a lesser extent also in the USA and China.

Third, the landscape within which business operates is undergoing similar evolution. Firms no longer treat environmental concerns as single-issue one-off distractions but are increasingly taking a comprehensive approach to managing their full spectrum of environmental and social (as well as economic) impacts. That is triggered in part by the evolution in scientific understanding and political agendas but presumably most of all by the maturing realization that, done judiciously, a proactive approach to sustainability can go hand in hand with increased profitability (more on which below). This is the result of market forces, the changing regulatory environment, employee engagement, investor action, and more. The term “ESG,” for environment, social, and governance, was already in use in 2017 and has become a major force in financial markets, encouraging firms and investors to consider firms’ performance along all three criteria. The concept of ESG is already becoming the target of substantial skepticism of the decisions made under its label and the data used to inform those decisions, but the underlying desire to better understand firms’ performance more holistically is unmistakable.

Finally, and perhaps most importantly in terms of this book, we have noticed a recent surge in the number of courses on “sustainable supply chain management” or related topics but a relative lack of corresponding teaching materials. While sustainability has been widely studied at the level of company strategy and extensive pedagogical materials exist at that more strategic level, there is still a relative lack of materials on sustainability with a supply chain management perspective.

Taken together, we hope that these observations jointly indicate that the time is ripe for a book of this nature. At the same time, these observations point to some principles that we adopted in assembling this book, outlined in the next section.

1.3 Principles Underlying This Book

Having established that neither “sustainability” nor “supply chains” are necessarily new in themselves, we made a number of deliberate choices in designing this book, each of which we discuss in more details below:

1. There are economic reasons for firms to adopt a proactive approach to sustainability (but not every sustainability initiative is necessarily profitable).
2. Merely measuring environmental and social impacts is often already enough to bring about improvements in performance.
3. Supply chain operations are key to boosting sustainability performance.
4. Sustainability is not a state, but (at best) a direction.

5. Sustainability involves engaging with a range of stakeholders, not limited to the traditional supply chain partners.

1.3.1 Economic Drivers of Sustainability

There is clearly a strong ethical dimension to sustainability. However, in this book, we emphasize the economic side. Many studies seek to determine whether firms that are more sustainable are also more profitable. That question, as stated, is probably unanswerable, as “more sustainable” can mean too many different things and even for any specific dimension of sustainability the answer will depend on too many conditions, such as geography, industry, market conditions, timing, and more. There are some meta-analyses of this literature that generally do point toward a positive link between sustainability and profitability; see, for instance, Orlitzky et al. (2003) for a meta-analysis of the link between corporate social performance and financial outcomes and Horváthová (2010) for the link between financial and environmental performances. The lingering ambiguity about the nature of the link between sustainability and financial performances is highlighted by the analysis in Berchicci and King (2020), who show how sensitive results in this field can be to modeling assumptions. Moreover, the fact that the weight of the evidence tilts toward a positive link does not mean that every sustainability initiative will be profitable by every metric.

This discussion might suggest that becoming more sustainable is optional. Sometimes, it is not: firms are facing increasing pressure to become more sustainable and to take their supply chains with them. That pressure comes from customers, both from individual consumers (especially in developed countries) and from organizations in the B2B context. Regulations continue to target larger segments of the supply chain and on more dimensions. Governments are moving more and more to a cradle-to-grave (or even cradle-to-cradle) perspective in designing regulation, which makes the supply chain the inevitable unit of analysis. In addition, nongovernmental organizations (NGOs) find various ways to encourage or pressure firms to adopt more sustainable behavior, whether related to labor practices, eliminating various ingredients in food, reducing emissions, or otherwise. Add to this pressure from investors, banks, insurance companies, employees, and more, and it is clear that sustainability is often an imperative more than a luxury.

Sustainability is also strongly connected to risk management. Blass et al. (2024, Chap. 18) discuss the numerous raw materials that can be sources of concern and highlight how these nonrenewable materials can be managed in a more sustainable way to mitigate risk. Besides, in Chap. 19, Scholten and Stevenson (2024) discuss the relationship between risk, resilience, and sustainability.

1.3.2 *What You Measure Is What You Get*

There are good reasons to believe that the mere act of measuring something triggers ways to improve it. The fact that Walmart, of all firms, was able to find ways to reduce both GHG emissions and costs by installing auxiliary power units in their trucks once they started taking a close look at their carbon footprint is a prime example of this. That is one reason why this book contains several chapters on measurement. Moreover, measurement is not at all straightforward and inherently requires a supply chain perspective. A firm seeking to reduce its “own” emissions faces two big challenges. First, it is practically impossible to unambiguously measure or even closely estimate a firm’s responsibility for the emissions associated with a final product or service. A typical product goes through numerous manufacturing and transportation activities, often jointly operated by a number of companies in a supply chain. Second, the direct emissions that a firm might abate typically constitute a small part of the supply chain emissions. Matthews et al. (2008) report that across all industries, companies’ direct emissions average only 14% of their supply chain emissions *prior to* use and disposal. Therefore, sustainability measurement should take a supply chain perspective, rather than limiting itself to the firm’s own emissions and impacts.

In Chap. 2, Guinée and Heijungs (2024) provide an introduction to life cycle assessment, which can be thought of as the overarching framework for measurement of environmental (and perhaps in the future also social) impacts. Of all the environmental impacts, those related to climate change currently receive the most attention. Therefore, Chap. 3, by Boukherroub et al. (2024), focuses on measurement of carbon footprints specifically. Water is increasingly recognized as another key resource that is under threat, due to overuse, climate change, and pollution. Although the term “water footprint” insinuates a parallel with “carbon footprint,” the measurement issues are in fact quite different, among others due to water not being time- and space-invariant the way that GHG emissions are. Hoekstra (2024) delves deeper into this in Chap. 4.

This series of chapters highlights that measurement, although highly valuable, is often not trivial. Challenges include the multiple dimensions of impacts that need to be consolidated, the appropriate acknowledgment of how impacts differ across space and time, the challenge of setting appropriate boundaries, how to deal with variation and uncertainty, and more. Many of these questions are the topic of ongoing research and unresolved. Nevertheless, we believe that the most frequent sustainability-related tasks which students will find themselves confronted with revolve around measurement.

1.3.3 Supply Chain Operations Are Key to Sustainability

One could argue that there are two broad domains within which one can strive for more sustainable outcomes. One is consumer behavior: this can be getting consumers to launder their clothes at lower temperatures and less frequently or to consume less goods and services in the first place. While customer behavior is largely outside the scope of this book, Chap. 10, by Agatz and Fleischmann (2024), discusses how demand management can influence the sustainability of operations. The other domain is the supply chain leading up to the consumer. Here, we contend genuine improvements in sustainability outcomes ultimately require changes in the supply chain's operations, broadly defined. Operation forms the core of every organization, whether in manufacturing, service, or nonprofit industries. In the past few decades, firms have experienced increasing globalization and a shifting focus to competition among networks of firms. Due to these trends, the field of supply chain management has become an increasingly central domain within operations. This can refer to greener logistics (Chap. 5 by Blanco and Sheffi 2024), inventory management (Chap. 6 by Marklund and Berling 2024), or network design and facility location (Chap. 7 by Velázquez Martínez and Fransoo 2024). Operations can involve adopting greener technologies (Ovchinnikov and Mackinnon 2024, Chap. 12). Besides, the specific challenges and opportunities vary by industry, as Akkerman and Cruijssen (2024, Chap. 9) discuss in the context of the food supply chains.

Although inevitably these topics are discussed in separate chapters, one of the challenges in practice is that they are all interrelated. For example, inventory management practices have a significant impact on logistics. A product or component design that serves the same functionality with less material use will not just help the manufacturer due to a cut down in the material costs, but it will also decrease all transport- and storage-related requirements downstream in the supply chain and hence also the related costs and environmental burden (such as emissions due to transport) in the supply chain. This calls for supply chain collaboration as discussed in Chap. 11, by Tan et al. (2024). Similarly, developing technologies that increase fuel efficiency of vehicles or use renewable energy sources benefits also supply chain actors through decreased fuel usage.

Sometimes, these operational changes result from regulation. Huang and Atasu (2024, Chap. 8) point out that even well-intended regulations can have adverse consequences, as the devil is often in the detail; an immediate consequence is that regulators also need a deeper understanding of the operations of the firms they are trying to regulate and that firms need a more nuanced understanding of the exact effect certain regulations will have on their supply chains. In other instances, operational changes are driven by changes in business model. This can include shifting the emphasis from products to services, as outlined in Bellos et al. (2024, Chap. 14), developing closed-loop supply chains (Abbey et al. 2024, Chap. 15), or developing reusable packaging solutions (Transchel et al. 2024, Chap. 17). Saman

et al. (2024, Chap. 16) propose a guiding framework for circular supply chain implementation.

1.3.4 Sustainability as a Direction, Not a State

As noted above, a supply chain is unlikely to ever be “sustainable” (even by the weak definition). It is nevertheless a worthwhile goal to constantly strive for, as some firms are aggressively doing. The number of firms setting net-zero targets for greenhouse gas emissions has skyrocketed since the first edition of this book with more than 2000 companies and financial institutions that have committed to science-based targets for carbon reduction (Boukherroub et al. 2024, Chap. 3), sometimes including their entire supply chain in those goals. Other examples of objectives related to water, labor conditions, human slavery, biodiversity, and other dimensions of sustainability abound. In order to determine whether a firm or supply chain is making progress toward that goal, one has to have the right measures in place, hence our focus on measurement at the beginning of this book.

Recognizing that sustainability is a path, not a state; it is helpful to analyze the stages that firms go through on that path. Pagell and Wu (2024, Chap. 13) outline four main trajectories that firms can follow and point out that choices firms make today determine the options available to them in the future. This notion of path dependency, while well-known in itself, is often neglected in the literature on sustainable supply chains.

1.3.5 Sustainability Involves Wider Range of Stakeholders

An inescapable consequence of the rising expectations firms face related to sustainability is the expanding set of stakeholders they have to interact with and the more complex ways in which they have to interact with each of them. Keeping shareholders happy has long been a prime objective, but that now requires far more extensive reporting than ever before. This trend is further encouraged by an emerging set of NGOs who are, depending on one’s perspective, either representing investors by asking these wide-ranging questions or using investors to exert pressure on firms to expand their reporting. Customers, employees, and regulators are expecting more from firms. At the same time, firms are expected to be aware of environmental and social issues faced by communities around the world that are (positively or negatively) affected by their supply chains, whether directly or through local regulators and NGOs. Gillai et al. (2024, Chap. 20) explain the importance of sensing and responding to social and environmental problems in global supply chains. Sodhi and Tang (2024, Chap. 21) ask what social responsibility firms bear and argue that firms can and should take the utilities of a wide range of stakeholders into account when making decisions. Sometimes,

firms choose to partner proactively with NGOs to address environmental or social problems in their supply chains; Balaisyte et al. (2024, Chap. 22) discuss factors that contribute to the success of such partnerships.

1.4 The Future of Sustainable Supply Chains

We started this chapter by arguing that, in many ways, the themes currently being discussed under the heading “sustainability” are not new. Therefore, one easy prediction about what the future holds for sustainable supply chains is “more of the same.” At the same time, it is certain that the future will bring new challenges and opportunities, some of which may already be visible today even if not yet recognized as major issues. Here, we first elaborate on what “more of the same” will look like and then speculate about what else may be around the corner.

1.4.1 *More of the Same*

We are convinced that “sustainability” will continue to become a larger part of the conversation between supply chain partners worldwide. These conversations will be increasingly wide-ranging, moving from focusing on a single issue (such as carbon, child labor, or fire safety) to a more systematic multidimensional view. The emphasis may shift over time from one set of issues to another, depending on (unpredictable) world events, but we would be surprised if the overall emphasis on sustainability did not increase.

As a result of this and of the constant expansion of resources dedicated to measuring and reporting sustainability performance, this will become an increasingly quantitative and standardized affair, ever more tightly integrated within firms’ decision processes. This will enable firms to make more analytical, proactive, and long-term decisions related to sustainability, rather than (often) having to respond to “the issue of the day.” Accounting as an organized profession emerged in the nineteenth century, and hence financial accounting has a head start of at least 100 years over sustainability accounting; one could use the advances that have occurred in financial accounting since the 1900s as a guideline for predicting what might happen with sustainability accounting and reporting over the coming 100 years. Despite those advances in financial accounting, the profession still faces challenges that are relevant for operations and supply chains.

One is that the data that are generated for financial accounting purposes are often unsuitable for operational decision-making but nevertheless widely used for precisely that as there is no other easy substitute available. One can anticipate the same happening with sustainability-related information: data on social impacts that is collected for the firm’s annual sustainability report may not be suitable to support day-to-day decisions on how to operate a supply chain, but unfortunately it is likely

that the errors that currently exist in how financial accounting information is used will be replicated in sustainability accounting (see also Blass and Corbett (2018)).

A second challenge is that financial accounting data are still subject to various degrees of manipulation, ranging from mild and innocent to outright fraud. As sustainability reporting gains importance, the stakes become higher and hence the likelihood of biases and actual fraud increase. The fact that sustainability-related information may be even harder to audit than financial information further exacerbates this. The recent attention to, and skepticism of, ESG data (Berg et al. 2022; Kotsantonis and Serafeim 2019) is consistent with this prediction.

1.4.2 What's Around the Corner

Climate change may be the most-discussed environmental (and social) issue facing the world today. However, some might argue that availability of fresh water may overtake climate change as the dominant topic, either because society tires of climate change or because it genuinely becomes a more imminent threat. Even if, as seems very unlikely today, renewable energy grows fast enough to prevent dangerous climate change, will we discover negative unintended consequences of photovoltaic panels, solar farms, and wind turbines? Some are already known, and as adoption becomes more widespread, others will inevitably crop up.

As our historical analogies at the beginning of this chapter point out, there are undoubtedly chemicals being used today, or even new materials being introduced today, that, at some point in the future, will turn out to have been mistakes. The continued opposition to genetically modified organisms (GMOs), especially in Europe, suggest that many fear they will fall into this category. Nanomaterials have great promise, but their properties are far from fully understood. The problems of plastics have become much more widely recognized, which is why we invited a new chapter on reusable packaging by Transchel et al. (2024, Chap. 17).

Improving working and living conditions for employees and local communities in faraway countries upstream in a global supply chain seems very noble (and should be encouraged). Is there a risk that firms, in doing so, contribute to creating a population that is more educated and less despondent, more prone to travel or migrate around the globe, and therefore less accepting of local governments, democratic or otherwise? Will firms that find ways to reduce poverty end up contributing to increasing inequality and hence fomenting social problems that do not currently exist?

None of these questions have easy answers. The fact that there may be unintended negative consequences of well-intended actions today does not mean that we should not take those well-intended actions, quite the contrary. But it does mean that we should constantly be vigilant, looking for what new issues may emerge, recognizing that more sustainable supply chains can be a force for good but that they will never be truly sustainable. We hope that this second edition will continue to contribute to educating future supply chain managers and business leaders about many of

the issues they will face today and tomorrow and equip them with the tools and frameworks that will help them on this journey.

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Part I
Measuring Environmental Impacts in
Supply Chains

Chapter 2

Introduction to Life Cycle Assessment



Jeroen Guinée  and Reinout Heijungs 

2.1 LCA: What Does It Address and How Has It Developed?

Today's society is highly consumption-based. Through advertisements and marketing campaigns, we are stimulated on a daily basis to consume products, services, and, if possible, more and more. Products and services are the key selling items of our economic system. At the same time, we are facing huge sustainability challenges with respect to, for example, climate change, land-use change, water shortages, toxic pollution, and resource scarcity. Products and services are key concepts in addressing these sustainability challenges. Refraining from any consumption is not an option, so we have to strive for a more sustainable production and consumption pattern. Environmental policy today focuses at the transition to such sustainable production and consumption patterns. This is taking place in various ways and at various levels.

For this purpose, we often will have to compare the sustainability of products against each other. We then first need a few definitions: what is sustainability, what are comparable products, and what exactly are products?

This chapter is partly based on Heijungs, R., and Guinée, J.B. (2012). An Overview of the Life Cycle Assessment Method – Past, Present, and Future. In: Curran, M.A. (Ed.), *Life Cycle Assessment Handbook; A Guide for Environmentally Sustainable Products*, pp. 15–42. Beverly: Scrivener Publishing.

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- Sustainability has at least three dimensions: the economic, the environmental, and the social dimensions; LCA focuses on the environmental dimensions only.
- Products fulfill functions or services. We buy a car, motor, or bike to travel from A to B; we eat and drink to satisfy our nutritional needs, etc. We generally have different alternative products to fulfill these functions.
- We are not so much looking into products as such but rather at a system of economic or industrial processes needed for the functioning of that product. Here, the term “product system” (or even better “function system”) enters the arena. A product system refers to the entire life cycle of a product, from extraction of natural resources to final waste management of the disposed product, from “cradle to grave.” We don’t just look at driving a car, but we include the total system of industrial/economic processes needed for constructing all components of the car, the maintenance of the car, the use of the car including the total life cycle of the fuels needed to drive the car, the roads needed, etc. until the final disposal of the car at the end of its life which may include recycling.

Knowledge of the environmental impacts of such product systems is indispensable if we are aiming for improving the environmental performance of these systems. We preferably need numbers for all relevant environmental impacts of product systems, from the cradle to the grave (whole life cycle), in order to find best solutions for their improvement without shifting impacts to other fields or to other phases of the life cycle (trade-offs). One of the assessment methods widely used for this is environmental life cycle assessment, abbreviated LCA.

Do we really need a potentially complex method as LCA? Isn’t driving an electric vehicle simply better than driving a gasoline-based vehicle? If it was as simple as that, we wouldn’t need LCA, but unfortunately the reality is much more complex if we take a system approach mapping the whole life cycle and all potentially relevant environmental impacts. Electric vehicles need a lot of batteries to store electricity, and these batteries need a lot of scarce resources that may potentially leak to the environment as hazardous substances, but depending on the source of electricity, they may perform better in the use phase compared to gasoline-based cars. Only by analyzing the full life cycle of these two functionally equivalent “car driving systems” we can determine which one is environmentally better performing. But even then, it’s often not possible to simply conclude that one option is better than the other as it may still depend on consumer’s behavior or on the source of electricity (e.g., renewables or fossil fuels) as in the case of the electric car (Hawkins et al. 2012). The same ratio can be applied to comparing plastic and carton disposable mugs with porcelain mugs for drinking coffee, disposable diapers versus cotton diapers, biofuel versus fossil fuel-based electricity systems, etc.

2.1.1 LCA in a Nutshell

LCA offers a method for quantitatively compiling and evaluating the inputs, the outputs, and the potential environmental impacts of a product system throughout its life cycle (ISO 1996), and its results may be used to support decision-making in this area. LCA refers to a method, but it also refers to the result of this method. In this chapter, we will mainly focus on the method that is used to obtain an “LCA result.”

LCA has made a long way, and it is still changing. But since a decade or so, there is a broadly accepted set of principles that can claim to be the present-day LCA framework based on a series of standards and technical reports issued by the International Organization for Standardization (ISO), the 14,040 series (ISO 1996, 1998, 2000a, b, 2000c, 2002, 2003, 2006a, b, 2012a, b, c, 2017, 2020a, b). This series consists of the documents listed in Table 2.1.

The standards are organized into the different phases of an LCA study. These are:

- Goal and scope definition
- Inventory analysis
- Life cycle impact assessment
- Life cycle interpretation

Table 2.1 ISO documents on LCA

Number	Type	Title	Year
14,040	International standard	Principles and framework	1996, 2006, 2020 ^a
14,041	International standard	Goal and scope definition and inventory analysis	1998 ^b
14,042	International standard	Life cycle impact assessment	2000a ^b
14,043	International standard	Life cycle interpretations	2000b ^b
14044	International standard	Requirements and guidelines	2006 ^c , 2017 ^a , 2020 ^a
14047	Technical report	Examples of application of ISO 14042	2003
14047	Technical report	Examples of ISO 14044 impact assessment application	2012a
14048	Technical specification	Data documentation format	2002
14049	Technical report	Examples of application of ISO 14041	2000c
14049	Technical report	Examples of ISO 14044 goal and scope definition and inventory analysis application	2012b
14067	Technical report	Carbon footprint of products, requirements and guidelines	2012c

^aAmendments to the current versions

^bUpdated in 2006 and merged into 14044

^cReplaces 14,041, 14,042, and 14043

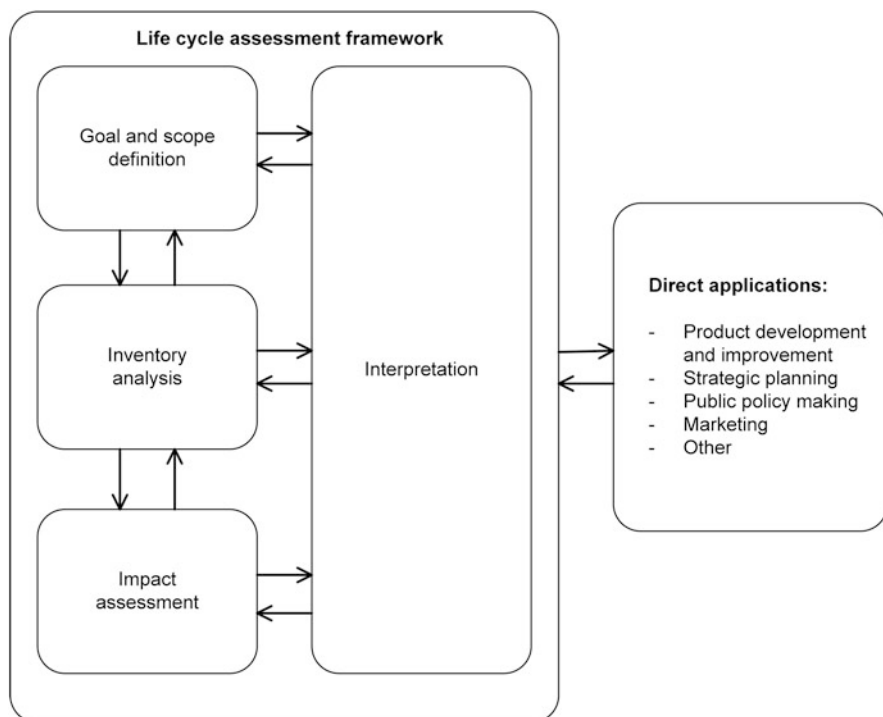


Fig. 2.1 The general methodological framework for LCA (ISO 1996)

The relations between these phases have been illustrated in a figure, and this figure has become a sort of logo of LCA (Fig. 2.1).

2.1.2 History of LCA

As mentioned, LCA has made a long way. This section discusses the historical development of LCA in terms of its past and present. The text is largely based on Guinée et al. (2011).

2.1.2.1 The Past of LCA (1970–2000)

Firstly, we will briefly discuss and evaluate LCA as developed and applied in the past while distinguishing two periods: (1) 1970–1990 and (2) 1990–2000.

1970–1990: Decades of Conception The first studies that are now recognized as (partial) LCAs date from the late 1960s and early 1970s, a period in which

environmental issues like resource and energy efficiency, pollution control, and solid waste became issues of broad public concern (Assies 1992). One of the first (unfortunately unpublished) studies quantifying the resource requirements, emission loadings, and waste flows of different beverage containers was conducted by Midwest Research Institute (MRI) for The Coca-Cola Company in 1969. Together with several follow-ups, this marked the beginning of the development of LCA as we know it today. After a period of diminishing public interest in LCA and a number of unpublished studies, there has been rapidly growing interest in the subject from the early 1980s on. In 1984, the Swiss published a report (BUS 1984) that presented a comprehensive list of the data needed for LCA studies, thus catalyzing a broader application of LCA.

The period 1970–1990 comprised the decades of conception of LCA with widely diverging approaches, terminologies, and results. There was a clear lack of international scientific discussion and exchange platforms for LCA. During the 1970s and the 1980s, LCAs were performed using different methods and without a common theoretical framework. LCA was repeatedly applied by firms to substantiate market claims. The obtained results differed greatly, even when the objects of the study were the same, which prevented LCA from becoming a more generally accepted and applied analytical tool (Guinée et al. 1993).

1990–2000: Decade of Standardization The 1990s saw a remarkable growth of scientific and coordination activities worldwide. Also, the first scientific journal papers started to appear in the *Journal of Cleaner Production*, in *Resources, Conservation and Recycling*, in the *International Journal of Life Cycle Assessment*, in *Environmental Science & Technology*, in the *Journal of Industrial Ecology*, and in other journals.

The Society of Environmental Toxicology and Chemistry (SETAC) started playing a leading and coordinating role in bringing LCA practitioners, users, and scientists together to collaborate on the continuous improvement and harmonization of LCA framework, terminology, and methodology. Next to SETAC, the International Organization for Standardization (ISO) has been involved in LCA since 1994. Whereas SETAC working groups focused at development and harmonization of methods, ISO adopted the formal task of standardization of methods and procedures.

The period of 1990–2000 can therefore be characterized as a period of *convergence* through SETAC's coordination and ISO's standardization activities, providing a standardized framework and terminology and platform for debate and harmonization of LCA methods. In other words, the 1990s was a decade of standardization. During this period, LCA also became part of policy documents and legislation. The main focus was on packaging legislation.

Although this decade is mainly one of convergence, it is also the stage of scientific scrutiny, research into the foundations of LCA, and exploring the connections with existing disciplines. For instance, we observe sprouting ideas on consequential LCA and related allocation methods (Weidema 2000; Ekvall 2000) marking the transition to the present decade of LCA.

2.1.2.2 2000–Present: The Era of Elaboration

The first decade of the twenty-first century has shown an ever-increasing attention to LCA. In 2002, the United Nations Environment Programme (UNEP) and the Society of Environmental Toxicology and Chemistry (SETAC) launched an International Life Cycle Partnership, known as the Life Cycle Initiative (<http://www.lifecycleinitiative.org/>). In 2005, the European Platform on Life Cycle Assessment (<http://eplca.jrc.ec.europa.eu/>) was established to promote the availability, exchange, and use of quality-assured life cycle data, methods, and studies for reliable decision support in (EU) public policy and in business. In the USA, the US Environmental Protection Agency started promoting the use of LCA, and also US environmental policy got increasingly life cycle-based all over the world (e.g., <http://www.gpo.gov/fdsys/pkg/BILLS-110hr6enr/pdf/BILLS-110hr6enr.pdf>).

In this same period, several life cycle-based carbon footprint standards have been, or are being, established (<https://pre-sustainability.com/articles/product-carbon-footprint-standards-which-standard-to-choose/>). Although the footprint concept is broader than LCA (it, for instance, includes organization footprints and national footprints as well), a rich literature has spawned around the product environmental footprint (PEF; https://ec.europa.eu/environment/eussd/smgp/dev_methods.htm), which basically is an LCA, and in addition carbon footprints (CF), which are LCAs focusing on greenhouse gases only. CF is now often associated with consumer products, and in an increasing number of cases, we find the CF score on the packaging or the label. The introduction of carbon footprint standards for supporting policies and (bioenergy) performance-based regulation raised some severe problems, which have often not yet been solved adequately (Matthews et al. 2008). Life cycle-based carbon footprint results need to be robust and “lawsuit-proof.” This implies that the freedom of methodological choices for the handling of, for example, biogenic carbon balances and allocation should be reduced to an absolute minimum and that uncertainties are dealt with appropriately, which is unfortunately not yet a common practice. Another issue is the limited scope of carbon footprints toward climate change impacts neglecting the possible trade-off to other impacts. Finally, we reiterate that the term footprint not necessarily refers to a full-fledged LCA; see also Hoekstra’s (2024) Chap. 4 in this volume.

The second decade of the twenty-first century has shown an increasing focus on the so-called ex ante and prospective LCA modes. Both these modes of LCA have in common that they “aim to assess the environmental life-cycle performance of a future system on the short, mid, or long term” (Guinée et al. 2018). Van der Giesen et al. (2020) define ex ante LCA as “performing an environmental life cycle assessment of a new technology before it is commercially implemented in order to guide R&D decisions to make this new technology environmentally competitive as compared to the incumbent technology mix.” Spielmann et al. (2005) and Walser et al. (2011) adopt the term prospective LCA to indicate that their studies estimate future life cycle environmental impacts using scenarios. The difference between these future modes of LCAs and traditional LCA is not so much in the general framework of LCA but rather in the specific implementation of some of its steps.

We further observe the adoption of life cycle thinking in other disciplines, leading to cross fertilizations and more complex models—just to mention a few examples, agent-based models (Davis et al. 2009; Baustert and Benetto 2017), product design (Basbagill et al. 2017; Mahmood et al. 2018), multicriteria decision theory (Myllyviita et al. 2012; Zanghelini et al. 2018), and probability and statistics (Mendoza Beltran et al. 2018; Igos et al. 2019) but also supply chain management (Guillén-Gosálbez and Grossmann 2010; Pozo et al. 2012; Liu et al. 2014; Khoo et al. 2019).

The period 2000–present can be characterized as the decades of elaboration. While the demand on LCA increases, the period is characterized by a *divergence* in methods again. As ISO never aimed to standardize LCA methods in detail and as there is no common agreement on how to interpret some of the ISO requirements, diverging approaches have been developed with respect to system boundaries and allocation methods, dynamic LCA, spatially differentiated LCA, environmental input-output-based LCA (EIO-LCA), and hybrid LCA, which combines the strengths of classical LCA and environmentally extended input-output analysis (EE-IOA). On top of this, various life cycle costing (LCC; cf. Hunkeler et al. 2008) and social life cycle assessment (SLCA; cf. Benoît and Mazijn 2009) approaches, as well as the integration into eco-efficiency (Kicherer et al. 2007; Vercalsteren et al. 2010) and life cycle sustainability analysis (Kölsch et al. 2008; Finkbeiner et al. 2010), have been proposed and/or developed.

2.2 Overview of LCA

Typically, LCA starts by defining goal and scope, then proceeds to the inventory analysis, then optionally continues to impact assessment, and ends with the interpretation. However, as indicated in Fig. 2.1, an LCA study is a highly iterative process, so that the LCA practitioner may need to go back to goal and scope after the preliminary inventory work, to move back from impact assessment to inventory analysis, to have a look at the interpretation in an early stage, etc.

Below, we will discuss the main idea and content of the four phases distinguished in Fig. 2.1 in separate subsections. All quotations are taken from the ISO documents, unless otherwise indicated.

2.2.1 Goal and Scope Definition

There is no explicit ISO definition of the first phase of LCA. However, it obviously centers around formulating the question and stating the context of answering this question. In the goal and scope definition, no data is collected and no results are calculated. Rather, it is a place where the plan of the LCA study is defined as clearly and unambiguously as possible. Likewise, in an LCA report, it should help

the reader to quickly find out the precise question addressed and main principles chosen.

The goal of the LCA should deal with the following topics:

- The intended application
- The reasons for carrying out the study
- The intended audience
- Whether the results are to be used in comparative assertions disclosed to the public

The choices made here have an influence on the rest of the LCA procedure. For instance, depending on the intended audience, a critical review may be needed, and it may be important that an external expert takes this task.

In the scope definition, a number of major choices are made—first of all, the product system or systems to be studied and the function the system delivers (or in case of a comparative LCA, the functions the systems deliver). For instance, one might be interested in the product system incandescent light bulb versus the LED bulb, with the function of lighting a room.

An important aspect of the scope definition is the functional unit. It is obviously pointless to compare an incandescent bulb with an LED bulb: the life spans and performances differ considerably, and the function is not having a light bulb, but having light of a certain quality. The functional unit expresses the function of the products and thereby offers a way to equalize differences in performance. A functional unit for analyzing lighting systems could thus better be phrased in terms of the function, for instance, “lighting a standard room of 15 square meter with 1000 lumen for 1 h.” As LCA employs mathematically a linear calculation rule, the results will scale by choosing a numerically different functional unit (say, “lighting a standard room of 20 square meter with 800 lumen for 3 h”), but the alternatives considered will scale up or down consistently, so this will not affect the conclusions. Due to this and some temporal constraints of LCA (Guinée et al. 2017, 2022), LCA cannot tell if a product is “sustainable” or “environmentally friendly”; LCA can only indicate if product X is “more sustainable” or “more environmentally friendly” than product Y or that the use phase is the “least sustainable” or “least environmentally friendly” part of the life cycle for product Z.

The scope definition further sets the main outline on a number of subjects that are discussed and further refined in more details in the later phases. These include, among others:

- System boundaries
- Impact categories
- Treatment of uncertainty

The ISO standard and some other texts at places suggest that these topics are implemented in detail in the scope definition. This is wrong: the goal and scope definition is not concerned with collecting data or calculating results, so no concrete details on such topics can be specified at this phase.

2.2.2 Inventory Analysis

ISO defines inventory analysis (LCI) as the “phase of life cycle assessment involving the compilation and quantification of inputs and outputs for a product throughout its life cycle.” It will be clear that quantification is an important aspect here and numbers, in terms of data and calculations, are of central concern in the inventory analysis.

The LCI is built up on the basis of the unit process. A unit process is the “smallest element considered in the life cycle inventory analysis for which input and output data are quantified.” Examples of unit process are coal mining, steel production, refining of oil, production of furniture, use of a TV, recycling of wastepaper, and transport by lorry. Each of these processes is described in quantitative terms as having inputs and outputs. As a matter of fact, a unit process is in LCA considered as a black box that converts a bundle of inputs into a bundle of outputs. Inputs come in several types: products (including components, materials, and services), waste for treatment, and natural resources (including fossils, ores, biotic resources, and land). Outputs come in several types as well: again products (including components, materials, and services), waste for treatment, and residuals to the environment (including pollutants to air, water, and soil, waste heat, and noise); see Fig. 2.2.

Unit processes form the building blocks of an LCA. This is because products are not harmful for the environment as such, except for the processes involved in products. Producing, using, and disposing products create the burden to the environment. Therefore, these processes assume a central position in LCA.

The essential feature of LCA in which it distinguishes itself from the analysis of an industrial or agricultural process is that it connects different unit processes into a system. A flow diagram is a graphical representation of the system of connected unit processes. Figure 2.3 shows a fragment of such a flow diagram.

As we can see, some unit processes are connected with one another in simple upstream-downstream connections, e.g., TV production is upstream connected to semiconductor production. But there are also more complicated connections, e.g., electricity linking to different parts of the system and recycling feeding back to production. Flow diagrams are in fact huge webs of interconnected unit processes.

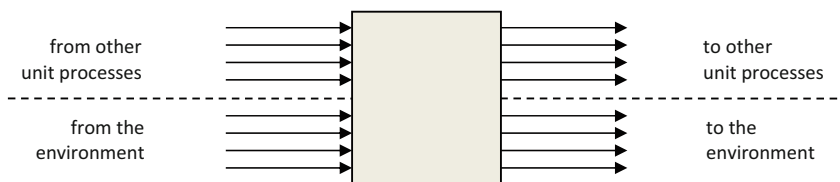


Fig. 2.2 General template of a unit process. The process (grey rectangle) is considered as a black box, having inputs (left-hand side) and outputs (right-hand side) from and to other unit processes (top lines) and from and to the environment (bottom lines)

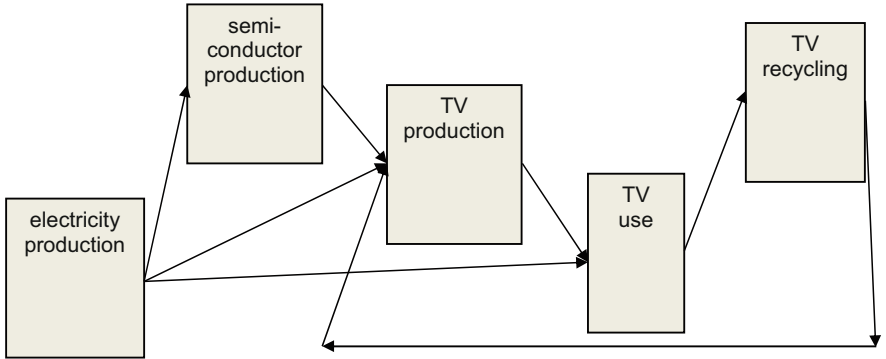


Fig. 2.3 Fragment of a simplified flow diagram for an LCA on television sets. Because the purpose is to show how unit processes are connected, only the flows from and to other unit processes are displayed, and flows from and to the environment are hidden. All transport, packaging, etc. has been left out as well

Table 2.2 Example of a unit process specification

Type of flow	Name	Amount	Unit
Inputs from other unit processes	Aluminum oxide	2	Kg
Inputs from other unit processes	Electricity	20	kWh
Outputs to other unit processes	Aluminum	1	Kg
Outputs to the environment	CO ₂	0.2	Kg

In the present era of digital databases, LCA studies can easily comprise several thousands of unit processes.

LCA is primarily a quantitative model. In the LCI, all unit processes included have to be quantified. This means that we have to specify the sizes of the inflows and outflows, per unit process. As an example, let us take the unit process of aluminum production.

An aluminum plant may specify their technology in terms of inputs and outputs by stating input requirements (e.g., 2-kg aluminum oxide and 20-kWh electricity per kg produced aluminum) and emissions (e.g., 200-g CO₂ per kg produced aluminum). We must translate this into our template for unit processes; see Table 2.2.

For each of the unit processes included, quantitative data should be collected. Moreover, in order to be able to process the data and perform the calculations automatically, a clear and unambiguous representation is needed. This implies, among others, harmonization of nomenclature (e.g., not using “carbon dioxide” for one unit process and “CO₂” for another one) and harmonization of units (e.g., not mixing up kg and lbs).

In Table 2.2, the unit process data is given per unit of output, here per kg of aluminum. In an LCA, we must next find out how much we need. For instance, the product may need 3 kg of aluminum, not 1 kg. The basic assumption of the LCA

model is that technologies are linear. This means that we can scale the data of a unit process by a simple multiplication. In the example, 3 kg of aluminum would require 6 kg of aluminum oxide and 60 kWh of electricity, while it would release 600 g of CO₂. The assumption of linear technology is an important restriction of LCA; yet it is an important step in making the calculation and data collection feasible.

In scaling the unit processes, the weblike nature of the system quickly creates complications, as everything depends upon everything. The calculation of the scaling factors, and with that of the emissions to and extraction from the environment, is greatly simplified by considering the problem as a system of linear equations: one unknown (the scaling factor) for every unit process and one equation (a balance) for every flow. Thus, solutions may be obtained by matrix algebra. The details of this are not discussed here; see Heijungs and Suh (2002) for a detailed exposition.

The approach mentioned above may fail in a number of cases. We mention two complications:

- For some products, upstream production processes or downstream disposal process may be difficult to quantify.
- For some unit processes, the balance equations become impossible due to the fact these processes produce not just one product but several products.

The first issue can be solved by a procedure known as cutoff and the second one by allocation.

Cutoff is a solution to the problem that the system is theoretically infinitely large. To produce a TV, we need machines, and these machines are produced by machines, and these machines in turn need machines, etc. But of course, we have an intuitive idea that some very distant upstream processes will be quite unimportant. This means that we will cut off certain inputs: although we know that something is needed and we sometimes even know how much is needed, we do not go into the trouble of specifying how these inputs are produced. It turns out to be difficult to specify reliable criteria when cutoff is allowed or to estimate how large the error is when a cutoff is made. Criteria on the basis of negligible contribution to mass or cost (e.g., smaller than 1%) often work pretty well but occasionally have been shown to yield large errors. Alternatively, estimates of missing parts by means of similar processes (e.g., estimating production of a freezer by production of a fridge) or by economic input-output tables may be helpful. Another approach is to do a difference analysis: in comparing a CRT and an LCD TV, we may leave out the broadcasting processes.

The second problem has given rise to one of the biggest controversies in LCA theory. The problem can be stated simply: if a transportation process needs gasoline, the upstream unit process is a refinery that produced not only gasoline but also diesel, kerosene, heavy oils, and some more. The direct impacts (from pollutants like CO₂) but also the flows to and from other processes that may lead to impacts (e.g., from oil drilling) may be argued not be attributable to gasoline only but in need to be distributed over gasoline, diesel, and all other coproducts. This is hardly contested, but the debate focuses on how to do this. To make it more concrete, how much of the CO₂ from a refinery is allocated to the gasoline? Different schools have

provided different arguments, and none of these have been completely compelling so far. Some solutions lead to strange results, other solutions may be very difficult to carry out (e.g., for lack of data or appropriate software), and still others are rejected by a large number of experts. To complicate the issue, the problem does occur not only in unit processes that produce several coproducts but also in unit processes that treat more than one type of waste or that recycle a waste into a good. It is even not agreed if the multi-output case, the multi-input case, and the recycling case must be treated in the same way or not nor how to determine which process is multifunctional particularly regarding recycling cases (Guinée et al. 2021).

Within ISO, a preference order for solving the multifunctionality problem has been designed. It distinguishes several solutions (dividing the unit process into two or more sub-processes, expanding the system to include the additional functions, partitioning on the basis of a physical parameter, partitioning on the basis of an economic parameter), separated by clauses like “wherever possible” and “where ... cannot be established.” This stepwise procedure is a clear comprise, and in practice, it leaves so much freedom that LCA studies that are according to the ISO standard can give conflicting results. One peculiarity deserves to be mentioned: besides the ISO-based “expanding the system to include the additional functions,” we often see a method that is best described as “subtracting the avoided impacts from additional functions” but that is more commonly known as the substitution method or the avoided burden method. For instance, when a waste treatment activity co-produces electricity, the emissions from the regular way of producing the same amount of electricity are subtracted. This method has similarities with that of system expansion, but of course it is not identical. Many LCA studies employing the substitution method claim to be ISO-compliant, even though strictly speaking ISO 14044 does not mention this method, let alone recommend it. That does not necessarily mean that these studies are incorrect, of course. Compliance with ISO is not a sufficient quality guarantee, but also not a necessary one.

After appropriate cutoff and allocation steps, the final inventory results can be calculated. Typically, this is a table with the quantified inputs from and outputs to the environment, for each of the alternative systems considered, expressed in relation to the functional unit. With the present-day software and databases, this inventory table may be 1000 lines long or more. It contains not only the familiar pollutants and resources, such as CO₂, NO_x, and crude oil, but also more exotic items, such as 1-pentanol, cyprodinil, and dolomite. Typically, these so-called elementary flows are aggregated over the entire system, so that the CO₂ number is the life cycle emission of CO₂.

2.2.3 Impact Assessment

Life cycle impact assessment (LCIA), or impact assessment in short, is the “phase of life cycle assessment aimed at understanding and evaluating the magnitude and

significance of the potential environmental impacts for a product system throughout the life cycle of the product.” Its motivation comes from two observations:

- The final result of the inventory analysis, the inventory table, is too long (e.g., 1000 different items) to handle.
- The inventory table contains many items that require expert knowledge (such as 2-methyl-2-butene) to understand in terms of importance.

Impact assessment, and in particular the characterization step, solves both issues: it “involves the conversion of LCI results to common units and the aggregation of the converted results within the same impact category.”

While the unit process is the central element of the inventory analysis, the central element in impact assessment is the impact category. ISO defines it as a “class representing environmental issues of concern to which life cycle inventory analysis results may be assigned.” Perhaps more helpful are some examples: climate change, toxicity, and depletion of fossil energy carriers.

As climate change (often used interchangeably with global warming) is a well-known issue, we will illustrate the main ideas of impact assessment with this case. The inventory table contains a number of greenhouse gases: CO₂, CH₄, N₂O, etc. These are known to contribute all to the phenomenon of climate change. Climate change involves a long sequence of causal mechanisms: emissions of greenhouse gases lead to changes in the composition of the atmosphere, which lead to a change in the radiation balance, which in turn leads to a change in the temperature distribution, which leads to changes in climate, which leads to changes in ecosystems and human activities, etc. The further we proceed in this causal chain, the more uncertain and speculative our knowledge becomes. While quite some scientific evidence is available with respect to the composition of the atmosphere, the impacts on biodiversity are debated. Many of these later impacts are even conditional on our future activities, including future emission scenarios and mitigating actions. To be able to quantitatively model the emissions of different greenhouse gases into an impact indicator for climate change, we must do several things.

First, we must choose a certain point in the causal mechanism. This can be at the front end (change in radiation balance), at the back end (change of biodiversity), or somewhere in between (change in temperature). In LCA, two main schools have emerged:

- Those that focus on the front end, the so-called midpoint approach.
- Those that focus on the back end, the so-called end-point approach.

The midpoint approach has the advantage that it includes fewer debatable assumptions and less-established facts; the end-point approach has the advantage that it provides more intuitive metrics (like loss of life years instead of kg CO₂ equivalent). Regardless of the choice between midpoint and endpoint, the indicator chosen is referred to as the impact category indicator or category indicator in short.

Second, a way must be found to convert the emission data into the chosen impact indicator. Scientists in chemistry, meteorology, ecology, etc. have developed model

fragments to estimate the atmospheric lifetimes of greenhouse gases, their effect on the radiation balance and the formation of clouds, the effects of temperature on the distribution of species, etc. These fragments have been combined by workgroups from the UN-based Intergovernmental Panel on Climate Change (IPCC) into quantitative models of the impacts of greenhouse gas emissions. Part of this is the global warming potentials (GWPs), which are quantitative measures of the strength of different greenhouse gases. Many midpoint LCIA methods apply GWPs for climate change. We will illustrate their usage below. For now, it suffices to mention that GWPs provide one example of a set of characterization factors and that the IPCC model from which they are derived is an example of a characterization model. Note, by the way, that IPCC has not developed this model as a characterization model for LCIA but that the LCA community has adopted this model as such and its derived GWPs as characterization factors. Also note that the characterization model itself is not used by LCA practitioners; only the characterization factors that have been derived from it as a one-time exercise are used. Characterization factors are often tabulated in LCA guidebooks and are implemented in many LCA software packages, while the characterization models often require supercomputers and expert knowledge.

In fact, there is one element before one can select a category indicator and a characterization model with associated characterization factors. It is the selection of impact categories to be addressed. Some LCA studies concentrate on just one impact category. For instance, the carbon footprint (of a product, not of a company or country) is a form of LCA that addresses just climate change at the midpoint level through GWPs. Some chapters in this book go into more depth on a single impact category: see Chap. 3 by Boukherroub et al. (2024) for more on carbon footprinting and Chap. 4 by Hoekstra (2024) for more on water footprinting. At the other extreme, some LCA studies incorporate 15 or more impact categories. For consistency reasons, the choice of impact categories is often made on the basis of a recommended impact assessment guidebook or its implementation in software. Thus, in practice, one often sees LCA studies reporting the use of “IMPACT2002+,” “TRACI,” “CML-IA,” “ReCiPe,” “ILCD,” etc. All these methods comprise a recommended set of impact categories with a category indicator and set of characterization factors. ISO does not specify any choice in these matters. Table 2.3 gives an overview of some often-used impact categories and category indicators. We see that the column with end-point indicators contains many times the same term (e.g., “loss of life years”). This suggests that impact categories can be aggregated into fewer end-point indicators than midpoint indicators.

As a concrete example of how characterization works, let us study a fragment of a hypothetical inventory table, containing the following information: emission of CO₂ 100 kg, emission of CH₄ 1 kg, and emission of SO₂ 1 kg. Characterizing greenhouse gases with GWPs requires a table with GWPs. In such a table, one can find that the GWP of CO₂ is 1 (by definition) and that the GWP of CH₄ is 28 (kg CO₂ equivalent/kg CH₄). SO₂ has no GWP; it is assumed not to contribute to climate change. Characterization now proceeds in the case of climate change by

Table 2.3 Overview of widely used impact categories with examples of category indicators at midpoint and end-point levels

Impact category	Midpoint category indicator	End-point category indicator
Climate change	Infrared radiative forcing	Loss of life years, fraction of disappeared species
Ozone layer depletion	Change in tropospheric ozone concentration	Loss of life years
Acidification	H ⁺ concentration	Fraction of disappeared species
Eutrophication	Biomass potential	Fraction of disappeared species
Human toxicity (sometimes split into carcinogens, noncarcinogens, respiratory effects, etc.)	Time-integrated exposure, corrected for hazard	Loss of life years
Eco-toxicity (sometimes split into aquatic toxicity, terrestrial toxicity, marine toxicity, etc.)	Time-integrated exposure, corrected for hazard	Fraction of disappeared species
Depletion of energy carriers	Primary energy requirement	Decreased availability
Depletion of material resources	Amount of material used, corrected for availability and/or importance	Decreased availability
Land-use impacts	Amount of land occupied or transformed	Fraction of disappeared species
Water-use impacts	Amount of water used or displaced	Decreased availability

calculating

$$1 \times 100 + 28 \times 10 = 380 \text{ kg CO}_2 - \text{equivalent.}$$

For the more general case, this can be written as

$$GW = \sum_s GWP_s \times m_s$$

where GW is the global warming score, s the substance (the different greenhouse gases), GWP_s the GWP of substance s , and m_s the emitted amount of substance s in kg. This may be further generalized as

$$I_c = \sum_s CF_{c,s} \times m_s$$

where c codes for the impact category, I represents the indicator result for category c , and $CF_{c,s}$ is the characterization that links substance s to impact category

c. This formula is the operational formula for characterization. With a table of characterization factors specified, it makes clear

- That LCIA builds on the results of LCI (as is clear from the term m_s)
- That characterization converts the results of LCI into a common metric (as is clear from the multiplication by CF)
- That characterization aggregates the converted LCI results (as is clear from the summation symbol)

The result from characterization is a list of numbers, for instance, a score for climate change, a score for toxicity, etc. ISO refers to such numbers as “category indicator results,” but most LCA practitioners prefer names like “score,” sometimes expanded with the name of the impact (like in “toxicity score”). The complete list is known by names like “LCIA profile,” “characterization table,” etc.

An optional next step is normalization referring to calculating “the magnitude of the category indicator results relative to some reference information.” It is an optional step for ISO, and indeed, many LCIA studies stop at the characterization. The reference information is in most cases that total impact in a certain region in a certain time period, e.g., in the country of decision in 1 year. Normalization is done “to understand better the relative magnitude for each indicator result.” Without normalization, the indicator results are in quite different units, e.g., kg CO₂ equivalent for climate change and MJ primary energy for fossil energy depletion. To put these results in perspective, the normalization expresses them as a share of the total impact size in the region. Arbitrary differences due to a choice of units disappear, and it becomes clear to which impact category a product contributes relatively much. The units of the normalized indicator results are equal; nevertheless, such numbers cannot meaningfully be added because the severity of the different impact categories has not yet been accounted for. This can be done in the weighting step; see below. Normalization fulfills several functions: it provides insight into the meaning of the impact indicator results, it helps to check for errors, and it prepares for a possible weighting step. However, due to the fact that many data for normalization are missing, current normalization results are often biased (Norris 2001; Heijungs et al. 2007).

Weighting is a final step of the impact assessment phase. Weighting, like characterization, converts and aggregates, but while characterization does so for the LCI results, weighting starts with the characterization (or normalization) results. Typically, weighting factors are applied, either to the characterization indicator results or to their normalized version. The weighting factors themselves are supposed to reflect value judgments, such as social and political priorities. Weighting typically produces one final number, by means of

$$W = \sum_c WF_c \times I_c$$

where I_c again symbolizes the impact score (or normalized impact score) for impact category c , WF_c the weighting factor for this impact category, and W the weighted

result. Well-known examples of such weighted results are the eco-indicator and the ELU (environmental load unit).

2.2.4 Interpretation

ISO defines the interpretation as the “phase of life cycle assessment in which the findings of either the inventory analysis or the impact assessment, or both, are evaluated in relation to the defined goal and scope in order to reach conclusions and recommendations.” Several elements are mentioned by ISO:

- Identification of significant issues
- An evaluation that considers completeness, sensitivity, and consistency checks
- Conclusions, limitations, and recommendations
- Appropriateness of the definitions of the system functions, the functional unit, and the system boundary
- Limitations identified by the data quality assessment and the sensitivity analysis

The text of ISO on interpretation is very concise, and no details are given on procedures and techniques to be employed. The same applies to most guidebooks on LCA. They mention carrying out an uncertainty analysis but give no clear guidance on how this should be done.

In another context, we have introduced the distinction between procedural and numerical approaches (Heijungs and Kleijn 2001):

- Procedural approaches include all types of analyses that deal with the data and results in relation to other sources of information, like expert judgments, reports on similar products, intuition, reputation of data suppliers, and so on.
- Numerical approaches include those approaches that somehow deal with the data that is used during the calculations, without reference to those other sources of information, but as algorithms that use and process the data in different ways, so as to produce different types of “smart” data reduction that provide an indication of reliability, key issues, discernibility, robustness, and so on.

This distinction helps to understand some important roles of interpretation. On the one hand, it is about comparing the data and results with previous findings and putting the results in the context of decision-making and limitations. On the other hand, it is devoted to a systematic analysis with the help of statistical and other decision-analytic techniques. The latter type may be incorporated in software, and indeed, an increasing number of software packages contain options for running Monte Carlo analysis, doing sensitivity analysis, carrying out statistical significance tests, etc. For instance, in the CMLCA software, we have implemented, among others,

- Contribution analysis
- Comparative analysis

- Uncertainty analysis
- Perturbation analysis
- Key issue analysis
- Discernibility analysis

The iterative nature of the ISO framework (Fig. 2.1) shows up in this context. Whenever the uncertainties are too high, we may go back to collect better data. Whenever sensitivity analysis shows that some decisions are crucial, we may go back and do a more refined analysis. In this way, the interpretation helps to prepare for a balanced decision but also helps to improve the LCA.

The development of methods in this area is booming (see Henriksson et al. 2013, and 2015), but current practice is quite meager, unfortunately. We still see many LCA studies without uncertainty or sensitivity analysis, even though methods and software increasingly facilitate this. There is of course a psychological argument that a contractor pays for finding out something, not for increasing the doubt. And as many LCA practitioners spend several months on collecting data, it is never a nice thing to waste this effort in a last-minute uncertainty analysis. But decision-making obviously means also taking into account the limits of knowledge. Moreover, as discussed before, a proper analysis of uncertainties and sensitivities helps to prioritize the steps earlier on in the framework: collecting data, setting boundaries, and making choices.

2.2.5 *LCA in Practice*

In the text above, the emphasis has been on the generally accepted practice. This is a mix of the ISO standards and a not precisely defined set of guidebooks (e.g., Wenzel et al. 1997; Guinée et al. 2002; Baumann and Tillman 2004; ILCD 2010; Curran 2012; Klöpffer and Grahl 2014; Jolliet et al. 2015; Hauschild et al. 2017). All these texts interpret, add, refine, or modify the ISO standards. As has been indicated at a few places, the practice in LCA is sometimes different from what the ISO standards prescribe. There are differences in terminology (e.g., one seldom sees the term intermediate products), in method (cf. the frequent use of the substitution method), in quality control (judged by frequent absence of uncertainty analyses), etc.

There are also de facto additional standards, dictated by the use of software (for overviews, see Bach et al. 2019; Olanunju and Olanrewaju 2020; and Karunaratne and Dharmarathna 2022) and databases (for an overview, see Ciroth and Burhan 2021). Many software packages for LCA have built-in options for impact assessment and uncertainty analysis, but nearly always in a restricted form, allowing some variants and prohibiting other variants. LCI databases are often constructed with predefined allocation methods and cutoff rules, so the user cannot choose otherwise and cannot carry out sensitivity analyses.

2.3 Examples of LCA Applications

LCA has been applied to a wide range of products and services. Until the late 1990s bibliographies of LCA case studies performed were kept up to date (Grotz and Rubik 1997) on a continuous basis. Using Internet search machines results in a list of hundreds of LCA case studies documented in scientific papers or reports. Even more studies have been made for company-internal purposes, without publication in the scientific literature or on the web. LCA has been applied to simple products as shopping bags and packaging to more complex products such as mobile phones, PCs, cars, and buildings. Studies may involve both an environmental comparison between existing products and the development of new products (ecodesign). LCA has also been applied to services such as LCAs on hazardous waste site cleanup options, on waste management strategies, and on different modes of freight transport (road, rail, water). As in the case of product LCAs, it is the function provided which is the core object of these service LCAs, but in this case, the function is cleaning up a hazardous waste site, waste management, or freight transport.

The results of these case studies were often in line with general expectations, but there were also numerous counterintuitive results. We randomly provide a selection of examples of the latter below.

Fargione et al. (2008) stirred the biofuel debate by introducing the concept of “biofuel carbon debt.” The increasing demand for biofuels was initially increasing the production of biofuels from food crops such as corn, sugarcane, soybeans, and palms. As a result, land in undisturbed ecosystems, especially in the Americas and Southeast Asia, was being converted to biofuel production as well as to crop production (indirect land-use change) when existing agricultural land was diverted to biofuel production (direct land-use change). This land clearing releases huge amounts of CO₂ as a result of burning or microbial decomposition of organic carbon stored in plant biomass and soils over a long time. Fargione et al. (2008) called the amount of CO₂ released during the first 50 years of this process the “carbon debt” of land conversion. Over time, biofuels can afterward repay this carbon debt if their production and combustion have less net GHG emissions compared to the life cycle emissions of the fossil fuels they displace. Their conclusion was that “converting rainforests, peatlands, savannas, or grasslands to produce food crop-based biofuels in Brazil, Southeast Asia, and the United States creates a ‘biofuel carbon debt’ by releasing 17 to 420 times more CO₂ than the annual greenhouse gas (GHG) reductions that these biofuels would provide by displacing fossil fuels. In contrast, biofuels made from waste biomass or from biomass grown on degraded and abandoned agricultural lands planted with perennials incur little or no carbon debt and can offer immediate and sustained GHG advantages.” As policies throughout the world were increasingly promoting biofuels, this publication significantly influenced reconsidering these policies.

Bovea et al. (2010) compared the environmental life cycle performance of different alternatives for the management of municipal solid. They analyzed and evaluated 24 different waste management scenarios. They concluded that “fuel consumed

during the collection, transport and waste sorting stages makes a contribution to the impact in all the categories that were analysed, since there is not any avoided environmental burden attributable to those processes.” Moreover, they conclude that “recycling allows the pollution burden to be avoided for all impact categories, since it avoids the consumption of virgin material according to the substitution rate of 1:1; the contribution made by landfilling depends on whether it is carried out with or without energy recovery [...]” This study shows the dependency of results of methodological choices such as the choice of allocation approach, which is particularly a problem in waste management studies (see also Ekvall et al. 2007).

Hawkins et al. (2012) developed a very comprehensive and transparent LCA study comparing the life cycle environmental performance of conventional and electric vehicles. They found that “EVs powered by the present European electricity mix offer a 10% to 24% decrease in global warming results relative to conventional diesel or gasoline vehicles assuming lifetimes of 150,000 km. However, EVs exhibit the potential for significant increases in human toxicity, freshwater ecotoxicity, freshwater eutrophication, and metal depletion impacts, largely emanating from the vehicle supply chain. Results are sensitive to assumptions regarding electricity source, use phase energy consumption, vehicle lifetime, and battery replacement schedules.” For EVs, production impacts thus become more important, while for conventional cars the use phase is by far the most important phase. The environmental performance of the EV can be improved by extending the lifetime of the EV, by reducing the impacts of the EV production supply chain, and by wider adoption of cleaner electricity sources.

Gregory et al. (2013) evaluated and compared the environmental impact (focusing on climate change) of five hand-drying systems: hands-under dryers, high-speed hands-under dryers, high-speed hands-in dryers, cotton roll towels, and paper towels. They also developed a method for incorporating uncertainty in the comparison of these hand-drying systems to understand the statistical robustness of the difference between the environmental impacts of the five hand-drying systems. They conclude “with a high degree of confidence that the high-speed dryers have a lower impact than paper towels and cotton roll towels.”

Van der Giesen et al. (2014) published an LCA study focusing on global warming impacts on a subject that receives increasing interest as part of the discussion on the transition toward renewable energy sources: using CO₂ as a resource to produce sustainable liquid hydrocarbon fuels. Producing these fuels by only using solar energy labels them as solar fuels. Today, new technologies for energy production are often claimed to be “sustainable” before they have been developed properly and/or subjected any kind of sustainability assessment. Therefore, Van der Giesen et al. (2014) performed a quantitative LCA “to investigate some of the claims made in this discussion.” They concluded “that producing liquid hydrocarbon fuels starting from CO₂ by using existing technologies requires much more energy than existing fuels. An improvement in life cycle CO₂ emissions is only found when solar energy and atmospheric CO₂ are used. Producing fuels from CO₂ is a very long-term niche at best, not the panacea suggested in the recent public discourse.”

2.4 Developments in and Challenges to LCA

LCA is an active research area in terms of methodology as well as in practical aspects. Methodology developers are working hard on further improving different parts of the LCA method while also putting efforts in keeping the methods practically feasible. We here briefly discuss a subjective selection of ongoing developments and challenges that LCA is facing:

1. Development of consequential LCA
2. Broadening LCA to life cycle sustainability assessment (LCSA)
3. Dealing with uncertainty in LCA
4. Streamlining LCA and LCA for ecodesign
5. Standardization of LCA methods beyond ISO
6. Ex ante and prospective LCA
7. Bridging LCA and supply chain analysis

2.4.1 *Development of Consequential LCA*

Since the beginning of the present century, consequential LCA has significantly grown in terms of the number of case study applications (Zamagni et al. 2012; Guinée and Heijungs 2011). Consequential LCA is a modeling approach that aims to describe the consequences of a decision and often models various scenarios to examine possible consequences. Consequential LCAs include unit processes in the product system to the extent that they are expected to change as a consequence of a change in the demand for the product (Weidema et al. 2009). Attributional—as opposed to consequential—LCA aims to describe the environmentally relevant physical flows to and from a life cycle and its subsystems as they are, were, or are expected to be. According to Schmidt (2010) and Weidema (2003), the core differences between CLCA and attributional LCA (ALCA) are that (1) consequential LCA includes the suppliers actually affected by a change in demand instead of averages as in ALCA and (2) coproduct allocation is avoided by system expansion instead of applying allocation factors. This change is modeled not over time but as a comparison of the situation with and without a specific demand; various product-related future scenarios are actually modeled. Future scenarios imply forecasting of future technologies, affluence, and consumer behavior and thus include large uncertainties, which increase as the term of the forecast becomes longer and the scope of the change becomes larger. Such uncertainties are inherent in modeling the future and thus also apply to an attributional LCA of a future system (Guinée and Heijungs 2011). However, due to the higher ambition of modeling how the world will look like, consequential LCA studies may have much bigger challenges in terms of data demand and models.

The modeling principles of attributional LCA (ALCA) and CLCA are the same, what distinguishes the two modes of LCA is the choice of the processes to be included in the system (Zamagni et al. 2012). Although the differences seem small, the application of the two modes to one case study (which can actually not be done, because they basically address different questions) may give significantly different results, not only between CLCA and ALCA but also between different (scenario) assumptions within CLCA (see Schmidt 2010). Bearing in mind that ALCA and CLCA can result in potentially significant differences, Plevin et al. (2014a) recently argued that “using ALCA to estimate climate-change mitigation benefits misleads policy-makers” basically claiming that CLCA is conceptually superior to ALCA for supporting robust decision-making. Plevin et al. received a lot of criticism (Anex and Lifset 2014; Brandão et al. 2014; Dale and Kim 2014; Hertwich 2014; Plevin et al. 2014b; Suh and Yang 2014; Plevin et al. 2014c), but the debate has not been cleared yet, if ever (Baitz 2017; Prox and Curran 2017).

2.4.2 *Broadening LCA to Life Cycle Sustainability Assessment (LCSA)*

As disclosed in Sec. 2.1.2, environmental life cycle assessment (LCA) has developed fast over the last three decades. LCA developed from merely energy analysis to a comprehensive environmental burden analysis in the 1970s, full-fledged life cycle impact assessment and life cycle costing (LCC) models were introduced in the 1980s and 1990s, and—among many other things—social LCA (S-LCA) and particularly consequential LCA gained ground in the first decade of the twenty-first century. With these latter developments, LCA broadened itself from a merely environmental LCA into a more comprehensive life cycle sustainability analysis (LCSA). Based on definitions by Klöpffer and Renner (2007); see also Klöpffer (2008) and Guinée et al. (2011), we can distinguish three dimensions along which LCSA expanded or may expand when compared to environmental LCA:

1. *Broadening* the scope of current LCA from mainly environmental *impacts* only to covering all three dimensions of sustainability (people, planet, and prosperity):

$$\text{LCSA} = \text{LCA} + \text{LCC} + \text{S-LCA}$$
2. *Broadening* the scope from predominantly product-related *questions* (product-level) to questions related to sector (sector-level) or even economy-wide levels (economy level)
3. *Deepening* current LCA *methods* to also include other than just technological relations, e.g., physical relations (including limitations in available resources and land), economic and behavioral relations (including rebound effects), etc.

LCSA is a developing area of research and needs further research, particularly in designing practical methods. Nevertheless, several articles on LCSA have already been published in international journals, professional sections have been established

such as the ISIE-LCSA section (<https://is4ie.org/sections/lifecycle>), and LCSA has become a subject area in, for example, the *International Journal of Life Cycle Assessment* (<http://www.springer.com/environment/journal/11367>), and first efforts for harmonizing practical applications have been undertaken (Valdivia et al. 2021).

2.4.3 Dealing with Uncertainty in LCA

A huge challenge for environmental LCA, and even huger for LCSA and CLCA, is how to deal with the many uncertainties related to LCA and LCA results. This issue was—and still is—largely underexposed by methodology developers, practitioners, databases, and software.

LCA is an integral method, bringing together data on many processes, data and models on impacts, assumptions on functional unit, system boundaries, allocation, etc. All of these data and (impact) models already bring many uncertainties with them, and in LCA, we lump these uncertainties. LCA practitioners are usually already happy when they get the main part of the data they need. Getting a grip on the uncertainties related to these data and to the models they apply is generally a bridge too far, let alone to also quantify the influence on the results of different methodological choices and different assumptions.

The importance of addressing uncertainty was recognized long ago (Fava et al. 1994), and it is also stressed by the ISO standards. During the past 25 years, many proposals have been developed (Heijungs 1996; Weidema and Wesnæs 1996; Chevalier and Le Téno 1996; Kennedy et al. 1996), and demonstration implementations were published as well (Copius Peereboom et al. 1999; Maurice et al. 2000). Yet, in a review in 2022 (Lo Piano and Benini 2022), it is concluded that “uncertainty is only appraised in a few studies on life cycle assessment.” There are several reasons that the practice is lagging behind:

- The limited availability of uncertainty information in LCA databases (theecoinvent database has primarily estimates of this, not observed data)
- The virtual absence of uncertainty information for LCIA data
- The limited options of most LCA software systems to process uncertainty (most packages offer only Monte Carlo analysis for inventory data)
- The lack of guidance by the ISO standards and general guidebooks (for instance, ISO 14044 states that “an analysis of results for sensitivity and uncertainty shall be conducted” but gives no information how this should be done)
- An unfamiliarity of LCA practitioners about what is a good practice
- The absence of guidance on how to interpret results from an uncertainty analysis (Mendoza et al. 2018)

The situation for sensitivity analysis is equally troublesome. For recent reviews on uncertainty and sensitivity in LCA, we refer to Wei et al. (2015), Groen et al. (2017), Igos et al. (2019), Bamber et al. (2020), Michiels and Geeraerd (2020), and Lo Piano and Benini (2022).

Main challenges now are to harmonize methods developed, to consistently elaborate and apply them over all sources of uncertainties throughout the whole LCA method, and to collect the relevant input uncertainty data. Particularly the latter requires substantial efforts and can only be achieved by reporting LCA data comprehensively, transparently, and in a publicly accessible way.

2.4.4 Simplifying and Streamlining LCA

Improving the LCA method is one thing, but keeping it practically feasible for users is another. Options for simplifying and streamlining LCA have been reviewed by Zamagni et al. (2008) and Pesonen and Horn (2013). They distinguished three different types of simplified approaches: qualitative (e.g., qualitative matrices, checklists, expert panels), semiquantitative (e.g., partly quantified matrices), and quantitative (simplified LCA). For all three categories, practical methods have been developed. There is unfortunately no “one-size-fits-all” solution for streamlining and simplification, and streamlined methods generally don’t produce conclusions similar to those reached through full LCAs (Hunt et al. 1998). Zamagni et al. (2008) conclude that “the choice of the most suitable simplified method, or combination of simplified methods, depends on the type of results users are looking for.” For a study, “aiming at supporting a choice between several alternatives and a study aimed at identifying critical aspects and suggesting mitigation strategies different methods may be suitable. In the first case, it is much more important to have quantitative data than in the second case. In fact, in the first case the lack of a quantitative dimension would hinder the comparison and make it difficult to differentiate between products. On the other hand, problems could arise when in quantitative LCA, aspects that are difficult to quantify are handled qualitatively, because this qualitative information is often overlooked” (Hoschorn and Finnveden 2003; Zamagni et al. 2008). For this first type of question, the challenge is to simplify LCA models based on detailed LCA experiences for specific product (categories). The EU fifth framework project OMNIITOX developed models calculating characterization factors for assessing the potential toxic impacts of chemicals within the framework of LCA. The project developed two interrelated models in order to be able to provide LCA impact assessment characterization factors for toxic releases for an as broad a range of chemicals as possible: (1) a base model representing a state-of-the-art multimedia model and (2) a simple model derived from the base model using statistical tools. Most simplified LCA models up till now have been developed in a rather “isolated” way, independent from detailed LCAs. The experiences in OMNIITOX (Birkved and Heijungs 2011) and reported by Huijbregts et al. (2006) seem useful to copy to simplified LCA models based on detailed LCAs for specific products. However, in some cases, simplification may be much simpler than this. In the 1990s, LCAs were performed on light bulbs, and the results were clearly dominated by the electricity needed for using the bulb. That steered all new bulb designs later on.

Simplification and streamlining are particularly relevant for ecodesigners. Ecodesign and LCA have always had a natural connection as they both strive to improve the environmental performance of product systems. Although the natural connection between the two seems obvious, there is also a natural tension between them. Whereas product designers in early stages often only have limited ideas—and thus data—on the eventual material and chemical composition of the product, an LCA study needs this information in order to produce useful supporting environmental information for the designer at stake. Designers' rules of thumb should be based on LCAs. In that perspective, we point to studies that try to use full-scale LCA in order to derive relationships with design parameters (Caduff et al. 2012; Padey et al. 2013). However, rules of thumb always have the danger of being wrong. In particular, they may turn out to be self-denying in the long run. For TV sets with a cathode ray tube, energy use was the most successful predictor of overall life cycle impact. With the development of flat screens, the value of this rule of thumb has decreased dramatically. Now, the presence of the rare metals may be a better predictor. This and an increasing focus on innovation for environmental sustainability have resulted in a new field of LCA: *ex ante* and prospective LCA.

2.4.5 *Ex Ante and Prospective LCA*

In the twentieth century, LCA was mostly applied to existing product systems. Due to the increasing attention to sustainability, particularly in the beginning of the twenty-first century, we are now in the middle of a transition toward more sustainable societies. As part of this transition, new technology systems are proposed of which the environmental sustainability still needs to be proven. These emerging technologies often only function at lab or pilot scale, and process data are also only available at these scales, and not at observed full-market scales. The application of LCA to novel technologies implies a change from *ex post* to *ex ante* assessments (Villares et al. 2017; Cucurachi et al. 2018). The goal of *ex ante* LCA is also different. Its main aim is to provide recommendations to the developers of the novel technology during the development process while comparing and checking its environmental life cycle performance to comparable relevant technology systems. *Ex ante* LCAs must deal with considerable challenges compared to conventional LCAs for which data is readily available or can be measured or accounted for based on full-scale existing technologies. For novel technologies, system specifications and process data are very uncertain and largely unknown. System specification and data from early phases of technology development originate from lab experiments, prototyping, or pilot plants. System specifications and data may rapidly change in a relatively short time due to new ideas or further development of the novel technology. Data for laboratory-scale (lab-scale) pilot plants are generally far from representative for any commercial-scale application and thus need to be upscaled through explorative scenarios to represent the technological performance in the future. In the second decade of the twenty-first century, an increasing number

of studies have addressed these challenges of ex ante LCA (Walser et al. 2011; Hetherington et al. 2013; Piccinno et al. 2016; Arvidsson et al. 2017; Villares et al. 2017; Cucurachi et al. 2018; van der Giesen et al. 2020; Thonemann et al. 2020). In the same period also, prospective LCA raised increasing attention (Spielmann et al. 2005; Mendoza et al. 2018; Cox et al. 2018; Hertwich et al. 2014; Adrianto et al. 2021). Prospective LCA studies future technological systems and their environmental implications developing future datasets for upcoming technologies using scenarios. For example, Hertwich et al. (2014) assess the environmental impacts of long-term, wide-scale implementation of electricity generation from renewable sources and of carbon dioxide capture and storage for fossil power generation applying LCA for the climate-change-mitigation and business-as-usual scenarios of the International Energy Agency up to 2050, while Mendoza et al. (2018) and Cox et al. (2018) developed and applied an approach systematically integrating scenario information of the Integrated Model to Assess the Global Environment (IMAGE) (Stehfest et al. 2014) to the ecoinvent database (Wernet et al. 2016).

2.4.6 Standardization of LCA Methods Beyond ISO

As disclosed in Sec. 2.1.2., the past first decade of the twenty-first century showed basically divergence in LCA methods again. On top of this, LCA started increasingly supporting policies and (bioenergy) performance-based regulations. For that, LCA results needed to be robust and reliable. Both robustness and reliability can be challenged. For example, some of these performance-based regulations in the field of bioenergy adopted attributional LCA, others consequential LCA, and others a hybrid of these approaches, likely resulting in different ratings and even altering preference orders (cf. Schmidt 2010) for transportation fuels that were the subject of these regulations. In addition, most regulations lack reporting requirements on the uncertainty of the LCA results and subjective choices by the LCA practitioner, which can result in LCA results seeming more certain and scientifically objective than they really are. It became clear that ISO standards were not sufficient to tackle these problems and thus product carbon footprint standards were developed all around the world (e.g., BSI 2011) and for EU policy applications the International Reference Life Cycle Data System (ILCD) Handbook was developed (ILCD 2010).

Standardizing and harmonizing methods for supporting policies and performance-based regulations are of utmost importance, but it will remain a challenge to properly balance this need with the equally justified need of science to progress and improve itself. It may thus make sense to distinguish between LCAs for science and LCAs for policy. “Whereas the former aims for completeness, the latter aims for robustness. The use of LCA in the policy context will benefit largely from the acceptance of this difference and by drawing up a guideline that is based on the aim of robustness” (Wardenaar et al. 2012) with an update every now and then to include progresses in science, the balance could be established. Eventually,

also here, there will be no “one-size-fits-all” solution, and we will have to accept that different questions require different approaches.

A complication that is getting more and more pervasive is that software and databases de facto dictate the methodology. For instance, if scientists prove that one method for uncertainty calculations is superior to another but the available software continues to offer only the inferior method, practice will not change. Similarly, scientists may develop noise impact methods, but if unit process databases do not offer information on sound emissions, such developments will remain within the academic domain only.

2.4.7 Bridging LCA and Supply Chain Analysis

LCA considers a product system, which to a large extent coincides with the traditional notions of a value chain and a supply chain. In supply chain analysis, many concepts have been developed and found a place in the daily work of operations managers. Subsequent chapters in this book describe such practices from the point of view of sustainable supply chains. But the relationship between LCA and supply chain analysis (SCA) deserves a special mention, also given that LCA is often descriptive while SCA often focuses on optimization and decisions. Blass and Corbett (2017) offer an interesting view on the bridge between the two. Progress is currently made in this field (e.g., Ramadhan et al. 2014; Gao and You 2017; Mota et al. 2018; Negri et al. 2021). We expect life cycle optimization (LCO) to be an active field of research and a fruitful tool toward more sustainable supply chains.

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Chapter 3

Carbon Footprinting in Supply Chains: Measurement, Reporting, and Disclosure



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3.1 Introduction

Climate change, a key issue in sustainability, has already begun. We now know it is responsible for worldwide losses of ice, ocean heat content increases, rising sea levels, and deep ocean acidification (IPCC 2021a). Lines of evidence further show that climate change is influenced by human activities through the release of greenhouse gases (GHGs). “Carbon” is often used as a shorthand for GHGs, as carbon dioxide (CO₂) is the main GHG released by human activities. Consequently, measuring GHG emissions is often referred to as “carbon footprinting,” the term we use in the remainder of this chapter. A carbon footprint may pertain to an

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organization, a value chain, or a product (Carbon Trust 2014). The organizational carbon footprint accounts for emissions from all activities across an organization (e.g., building energy use, industrial processes, company vehicles). The value chain (or supply chain) carbon footprint also includes emissions outside the organization's own operations (i.e., emissions from both suppliers and consumers, including product use and end-of-life emissions). Finally, the product carbon footprint includes emissions over the whole life cycle of a given unit of product or service, from the extraction of raw materials and manufacturing to its use and final reuse, recycling, or disposal. This chapter focuses on the organization and value chain footprints; product-level carbon footprinting is more closely related to life cycle assessment (LCA), which is covered in Chap. 2 by Guinée and Heijungs (2024).

This chapter is organized into four sections: Section 3.1 provides a brief scientific background on climate change, which is not intended to be exhaustive but only to the extent necessary to explain the methodology behind carbon footprinting. It also introduces the main motivations for carbon footprinting. Section 3.2 explains how carbon footprints can be measured and describes several carbon accounting methodologies. Section 3.3 focuses on supply chain carbon footprinting and provides an example from the telecommunications industry. Finally, Section 3.4 discusses some challenges related to supply chain carbon footprinting in practice.

3.1.1 The Science of Climate Change

According to the Intergovernmental Panel on Climate Change (IPCC 2007),¹ “climate change” refers to any change in climate over time due to natural variability or as a result of human activity. The scientific community has collected substantial evidence that the Earth's climate is changing (IPCC 2013) as a result of the increased concentration of GHGs in the atmosphere, which is due in part to human activity. The main GHGs are water vapor, CO₂, methane, nitrous oxide, hydrofluorocarbons, perfluorocarbons, and sulfur hexafluoride, some of which are naturally present in the atmosphere. These GHGs are responsible for the greenhouse effect, a natural phenomenon responsible for warming the atmosphere and allowing life on Earth. In recent times, GHG emissions have increased notably due to industrialization and changes in agriculture and land use. Carbon dioxide, for example, is emitted by the combustion of fossil fuels such as coal, oil, and gas; methane mainly comes from agriculture, livestock, and landfills; and nitrous oxide is found in large quantities in nitrogen fertilizer and chemical processes. These human-made GHGs, known as “anthropogenic GHGs,” intensify the greenhouse effect, and their emissions have increased continually from 2010 to 2019 (IPCC 2022a).

¹ IPCC is the leading international body for the assessment of climate change (<http://www.ipcc.ch/>).

Table 3.1 Atmospheric lifetime and global warming potentials of some GHGs

Gas	Atmospheric lifetime	GWP-100	GWP-20
CH ₄ (methane)	11.8	27.9	81.2
HFC-134a (hydrofluorocarbon-134a)	14	1530	4140
CFC-11 (trichlorofluoromethane)	52	5560	7430
N ₂ O (nitrous oxide)	109	273	273

Source: Extracted from IPCC (2021a, Table 7.SM.7)

To measure the climate impact of GHG emissions, the LCA community (see Chap. 2 by Guinée and Heijungs (2024) for more details) has developed an impact category called the “global warming potential” (GWP), which has become one of the most prominent metrics to measure the climate impact of GHG emissions (IPCC 2021a). It refers to “an index measuring the radiative forcing following an emission of a unit mass of a given substance, accumulated over a chosen time horizon, relative to that of the reference substance, carbon dioxide (CO₂). The GWP thus represents the combined effect of the differing times these substances remain in the atmosphere and their effectiveness in causing radiative forcing” (IPCC 2021b). Put simply, GWP is the heat trapped in the atmosphere by a given amount of GHG over a given period, relative to that trapped by an equivalent amount of CO₂ during the same period. Table 3.1 shows the GWP of some GHGs over 100-year and 20-year periods, referred to as GWP-100 and GWP-20, respectively. For example, the GWP-100 of methane (CH₄) is about 28 (27.9), meaning that 1 metric ton (hereinafter referred to as simply “ton”) of methane in the atmosphere over 100 years traps the same amount of heat as 28 tons of CO₂ over 100 years. Carbon dioxide serves as a reference for evaluating GWP because it is the most important anthropogenic GHG in quantity and in total impact (based on GWP-100 calculations). Consequently, GHG emissions are often referred to as “carbon emissions.”

Table 3.1 shows that the GWP and the atmospheric lifetime vary widely among GHGs. This variation means that the time period chosen to calculate the GWP may lead to significant differences. For instance, the atmospheric lifetime of methane is about 12 years, much lower than that of CO₂; consequently, the GWP-20 of methane is much higher than the GWP-100. The opposite effect occurs when the atmospheric lifetime of a GHG is higher than the lifetime of CO₂. According to IPCC (2021b), the choice of a metric (including its time horizon) should be consistent with the objectives for which the metric is used. In practice, however, many organizations use the GWP-100 by convention. In the United Nations Framework Convention on Climate Change (UNFCCC), GWP-100 is required in reports of GHG emissions (UNFCCC 2022a). For more details on the effect of time horizons on LCA, see Dyckhoff and Kasah (2014).

Using the GWP enables us to aggregate GHG emissions into a single metric, commonly expressed in either CO₂ equivalent (CO₂e) or carbon equivalent. Note that these two metrics should not be confused: 3 million tons of carbon equivalent is equal to 11 million tons of CO₂e. The conversion between carbon equivalent and

Table 3.2 Changes in GWP estimates for methane and nitrous oxide for the three latest IPCC assessment reports

GHGs	20-year horizon			100-year horizon		
	AR4 (2007)	AR5 (2014)	AR6 (2021)	AR4 (2007)	AR5 (2014)	AR6 (2021)
CH ₄ (fossil sources)			82.5			29.8
CH ₄ (biogenic sources)	72	84	80.8	25	28	27.2
N ₂ O	289	264	273	298	265	273

Source: AR4, AR5, and AR6—technical summary

CO₂e is related to the ratio of the atomic mass of a CO₂ molecule to the atomic mass of a carbon atom, that is, 44/12 (Environmental Protection Agency [EPA] 2005).

Carbon emissions expressed in CO₂e (or carbon equivalent) are often considered an unambiguous measure of the effect of GHGs on global warming. However, this measure is based on various assumptions and imperfect models. Any recommendation based on carbon equivalent calculations should acknowledge that the results are subject to some scientific uncertainty; according to the IPCC (2022b), “all GHG emission metrics have limitations and uncertainties, given that they simplify the complexity of the physical climate system and its response to past and future GHG emissions.” The GWPs get revised periodically as the models used in the calculations evolve (Carbon Trust 2014), as shown in Table 3.2, which presents three different estimates of GWP for CO₂, methane, and nitrous oxide over 100 and 20 years for 2007, 2014, and 2021. IPCC (2021a) consider methane from fossil fuel sources to have slightly higher emission metric values than those from biogenic sources (e.g., emissions from vegetation and soils), which was not the case in previous assessment reports.

These varying measurements have immediate implications for carbon footprinting and reporting, in that the carbon footprints for different companies and at different points in time may be based on different GWPs. An analogy in financial accounting is the effect of currency exchange rates: Financial statements are published in a single currency, using whatever collection of exchange rates is appropriate at that time, but changes in reported financial metrics may result from changes in exchange rates rather than in actual performance. Although GWPs may not be as volatile as currency exchange rates, Table 3.2 shows that, for example, the GWP-20 for methane changed by about 13% between 2007 and 2021 IPCC reports. Despite these shortcomings, using the GWP to aggregate different GHGs into a single metric expressed in CO₂e is the most common approach to carbon footprinting.

3.1.2 Motivations for Carbon Footprinting and Reporting

Carbon footprinting has become more widely used (e.g., Minx et al. (2009)) than other environmental footprints, such as the ecological footprint, land footprint, or water footprint (see Chap. 4 by Hoekstra (2024) for more on water footprinting). The main reasons for this can be linked to legislation around carbon emissions; public awareness of climate change risks, especially among youth; and investors' expectations of carbon emission reporting. Some companies ask suppliers and subcontractors to provide data on their emissions. For example, DHL requires all its carriers to enter data on vehicles used, distance travelled, fuel efficiency, and so on, not only to calculate total carbon emissions but also to screen the carriers for environmental performance (WRI and WBCSD 2004). Reducing carbon emissions can also lead to lower costs. A survey conducted by the Consumer Electronics Association indicated that companies measuring their carbon footprint were able to reduce their electricity consumption by 5%–25% (Vasan et al. 2014).

Regarding regulations on carbon emissions, many governments require carbon emitters to report their emissions annually. Other countries have established carbon and energy taxes (e.g., the CRC Energy Efficiency Scheme in the United Kingdom) that link financial penalties with carbon emissions. Most countries have signed the Kyoto Protocol,² which entered into force in 2005; in December 2015, 196 parties adopted the Paris Agreement on climate change at COP21, which entered into force in November 2016. The goal was to limit global warming to below 2 (preferably 1.5) degrees Celsius. To this end, countries aim to reach global peak of GHG emissions as soon as possible to achieve a climate-neutral world by 2050 (UNFCCC 2022b). As a result, many governments are taking further steps to reduce carbon emissions through regional or national policies, including the introduction of emission trading programs (e.g., European Union Emission Trading Scheme [EU ETS]). Under such a trading system, permits (allowances) are required for a given company to be allowed to emit GHGs (i.e., a cap), and the number of available allowances in the market (regional, national, or international) is limited. To achieve emission reductions, this number should decrease gradually over time (Nissen et al. 2021). European companies covered by the EU ETS, such as Lafarge and Rockwool International, report their carbon emissions on a mandatory basis (Carbon Disclosure Project [CDP], 2014³). Between 2009 and 2013, the number of available allowances exceeded the demand. The surplus of emission allowances accumulated in this period led to lower allowance prices. Consequently, a number

² The Kyoto Protocol committed its parties by setting internationally binding carbon emission reduction targets: 5% against 1990 levels during the first commitment period (2008–2012) and at least 18% below 1990 levels during the second commitment period (2013–2020).

³ The CDP is a nonprofit international organization that runs the global disclosure system for businesses, cities, and governments to measure and manage their environmental impacts (<https://www.cdp.net/>). Adding the cities, states, and regions that also disclosed data to CDP in 2021, the total number of disclosing entities was over 14,000 (CDP 2022).

of allowances initially planned to be allocated through auctions between 2014 and 2016 were not allocated, which resulted in a considerable decrease of the total number of allowances available. The COVID-19 pandemic had a major impact on emission trading in 2020 (e.g., overall emissions were down 63.5% in the aviation sector compared with 2019), and due to the first lockdown, the price of allowances fell to 14.60 euros/ton in March 2020. However, from that time, the price started increasing until the end of 2021 to 80 euros/ton and has deviated around that level all through 2022.

In 2021, more than 13,000 companies reported on their environmental impact through the CDP, which is 37% higher than in 2020 and 135% higher than in 2015. More than 1500 of these companies are Chinese (CDP 2022). According to CDP, its disclosure process shows that environmental leadership is consistent with business success: “the 300+ companies that scored most highly by CDP in 2020—the so-called ‘A-List’—are worth a combined \$15 trillion in market cap and outperform the stock market by 5.3% on average” (CDP 2022).

Another incentive for carbon footprinting has arisen from pressure exerted by the public, which has become increasingly aware of the risks of climate change. In September 2019, 3 days prior to a UN Climate Action Summit, millions of people around the world, including a notable proportion of youths, went on strike (known as the September 2019 climate strikes), demanding that governments act more urgently to end the use of fossil fuels and cut carbon emissions. Several reports demonstrate that climate change and global warming are currently considered among the risks of highest concerns worldwide. For example, in a global survey, DHL (2010) states that 60% of all respondents identified climate change as being among the top three most serious current world problems. The ninth Global Risks Assessment Report released by the World Economic Forum (WEF 2014) stated “failure of climate change mitigation and adaptation” and “greater incidence of extreme weather events (e.g. floods, storms, fires)” were ranked fifth and sixth, respectively, among the top ten global risks. In the latest WEF (2022) survey, respondents indicated that societal and environmental risks are the most concerning for the next 5 years and the health of the planet is the dominant concern for the next 10 years. Environmental risks are perceived to be the five most critical long-term threats. “Climate action failure,” “extreme weather,” and “biodiversity loss” are ranked the top three most severe risks (WEF 2022). In 2014, the European Commission’s (EC 2014) second survey on climate change revealed that climate change is perceived as the third most serious issue worldwide, after “poverty, hunger and lack of drinking water” and “the economic situation.” Its latest survey shows that climate change is the most serious problem facing the world according to European citizens, with 93% of the respondents considering it a serious problem and 78% considering it a very serious problem (EC 2022). Companies need to account for this public pressure to maintain their license to operate. One option followed by many of them is to make public commitments about carbon footprint reduction, such as becoming carbon-neutral.

The Science Based Targets initiative lists more than 2000 companies and financial institutions that have committed to science-based targets for carbon reduction.⁴

Investors also require that the long-term risks related to environmental externalities are managed to protect their long-term investments. For example, the CDP Investor Initiatives, backed in 2015 by more than 822 institutional investors representing over US\$95 trillion in assets,⁵ provide investors with a global source of annual information to support long-term objective analysis, including evidence and insight into companies' carbon footprint and strategies for managing climate change. The data published annually by the CDP are used in global investment research, products, indices, and ratings, including Bloomberg, STOXX, S&P Global Trucost, FTSE/Russell, MSCI ESG, ISS ESG, and Goldman Sachs (CDP 2022). The CDP's Carbon Action Initiative (backed by 190 investors) asks companies in heavy-emitting industries to reduce carbon emissions every year by setting emission targets and making reductions while still generating returns on investment (CDP 2016). According to the CDP, market demand for corporate environmental transparency is now higher than ever: in 2020, 590 investors (US\$110 trillion in assets) and more than 200 large purchasers (US\$4 trillion in buying power) supported the CDP's request for disclosure (CDP 2022).

In summary, the first step toward managing supply chain carbon emissions (and disclosing them) is to measure the carbon footprint, because "if you can't measure it, you can't manage it" (Kaplan and Norton 1992). The next section is devoted to questions of how.

3.2 How Can a Carbon Footprint Be Measured?

This section consists of two parts: The first presents the GHG Protocol, the most widely used framework to account for carbon emissions. Then, we discuss the main methods applied in practice for carbon footprinting.

3.2.1 *The GHG Protocol*

The GHG Protocol⁶ (www.ghgprotocol.org/) is the main framework for carbon emissions accounting worldwide. Government and business leaders use it to understand, quantify, and manage their carbon footprint. It also serves as the foundation for nearly every GHG standard and program in the world, as well as hundreds of GHG inventories prepared by individual companies (WRI and

⁴ See <https://sciencebasedtargets.org/companies-taking-action#table>.

⁵ See <https://www.cdp.net/en-US/WhatWeDo/Pages/investors.aspx>.

⁶ Where appropriate, we quote directly from the GHG Protocol throughout this chapter.

WBCSD 2004). For example, it serves as the basis for the ISO 14064-1 standard (EPA 2014) and ISO 14067.⁷ The GHG Protocol is also the foundation for the Global Logistics Emissions Council framework, the main standard for measuring carbon footprints in transportation.⁸ The GHG Protocol provides guidance to help companies, governments, and cities prepare their carbon inventory while reducing compilation costs with standardized approaches. The GHG Protocol recommends setting organizational boundaries and operational boundaries.

- *Organizational boundaries*

Two distinct approaches can be used: the *equity share* and the *control* approaches. Following equity share, a company accounts for carbon emissions from operations according to its share of equity in the operation (i.e., economic interest). Typically, the share of economic risks and rewards in an operation is aligned with the company's percentage ownership of that operation, and equity share will be the same as the ownership percentage. With the control approach, the company accounts for 100% of the carbon emissions from operations over which it has control. Control can be defined in either financial terms (ability to direct the financial and operating policies) or operational ones (full authority to introduce and implement operating policies at the operation).

- *Operational boundaries*

The GHG Protocol distinguishes three “scopes” for operational boundaries (see Fig. 3.1):

- *Scope 1: direct carbon emissions* – Direct carbon emissions occur from sources owned or controlled by the company (e.g., combustion in owned or controlled boilers, furnaces, vehicles; chemical production in owned or controlled process equipment).
- *Scope 2: purchased energy indirect carbon emissions* – It accounts for carbon emissions from the generation of purchased or acquired electricity, steam, heating, or cooling consumed by the reporting company. Scope 2 emissions physically occur at a facility outside the organizational boundary of the reporting company.
- *Scope 3: other indirect carbon emissions* – This scope allows for the treatment of all other indirect emissions. These emissions occur as a consequence of the company's activities but are from sources owned or controlled by other entities in the supply chain. They include both upstream (e.g., purchased products, purchased product transportation) and downstream (e.g., use or disposal of sold products) emissions. Note that Scope 3 emissions are often excluded from regulatory requirements as well as voluntary commitments from companies; however, this is likely to evolve soon. Indeed, the International Sustainability

⁷ See <https://www.iso.org/standard/71206.html>.

⁸ See <https://ghgprotocol.org/blog/glec-framework-universal-method-logistics-emissions-accounting>.

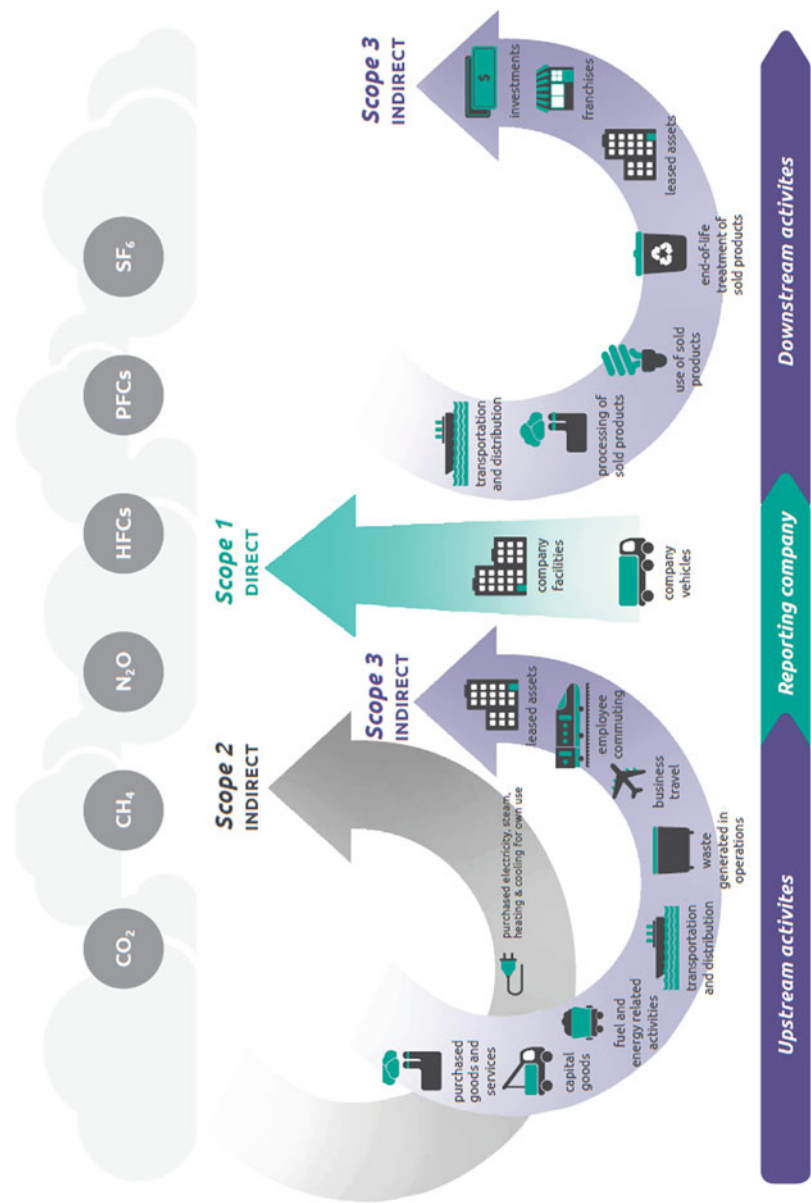


Fig. 3.1 Overview of scopes and emissions across a value chain. Source: WRI and WBCSD (2011a), p. 5, Fig. 3.1)

Standards Board (ISSB) announced recently that businesses will be required to include Scope 3 emissions in their disclosures.⁹ There are similar initiatives led by the European Union¹⁰ and by the United States.¹¹ Section 3.3 shows that Scope 3 emissions are often pivotal and that companies need to measure them to curb overall emissions.

These three scopes are mutually exclusive for the reporting company; for example, Scope 3 emissions do not include any emissions already accounted for as Scope 1 or 2. When boundaries are defined, the carbon footprint can be measured. In Sect. 3.3.2, we present an example from a telecommunications company to illustrate how the GHG Protocol could be used in practice. The next subsections provide an overview of the main methods used to measure carbon emissions.

3.2.2 Methodologies for Measuring the Carbon Footprint

Researchers and practitioners have proposed various methods for measuring carbon emissions. We classify them by the level of extrapolation involved, from the most direct measurement methodology to the one relying most heavily on extrapolation:

- Direct measurement
- Energy-based calculations
- Activity-based calculations
- Economic input-output life cycle assessment (EIO-LCA)

Direct Measurement

The direct measurement methodology is usually applied to production sites. The measurement of emissions is achieved by continuously measuring the pollutants emitted into the atmosphere in exhaust gases from combustion or industrial processes, often through the use of continuous emission monitoring systems (EPA 2008). These systems are mainly installed in production sites subject to environmental regulations and process monitoring applications, such as the US EPA 40 CFR Part 75 and 40 CFR Part 60. Apart from these sites, direct measurement of carbon emissions by monitoring concentration and flow rate is not common in supply chains. Direct measurement is generally only applicable for a share of Scope 1 emissions. Another option for direct measurement of carbon emissions is using

⁹ See <https://www.ipe.com/news/issb-to-require-businesses-to-report-scope-3-greenhouse-gas-emissions/10062925.article>.

¹⁰ See <https://www.europarl.europa.eu/news/en/press-room/20221107IPR49611/sustainable-economy-parliament-adopts-new-reporting-rules-for-multinationals>.

¹¹ See <https://www.cdp.net/en/articles/media/in-bold-new-move-biden-administration-makes-cdps-model-the-law> and <https://www.allenoverly.com/en-gb/global/news-and-insights/publications/us-securities-and-exchange-commission-proposes-extensive-climate-related-disclosure-regime-covering-all-sec-registrants>.

satellite monitoring, which is usually applied at the regional scale but can also be used for large sources of carbon emissions (for a review, see Pan et al. (2021)).

Energy-Based Calculations

Energy-based calculations determine carbon emissions based on mass balance or theoretical combustion specific to a facility or a process. Several levels of analysis may be conducted, depending on the information available. Take fuel as an example: the amount of fuel combusted is generally not directly monitored, and extrapolation based on the amount of fuel consumed is commonplace. In addition, the carbon content of a given combusted fuel is often estimated based on average values. Energy-based calculations may also be applied for indirect emissions from electricity consumption, as many providers release the average level of carbon emissions per unit of electricity produced. These calculations require information that is generally available for a share of Scope 1 and 2 emissions and are most applicable to process-related emissions, such as those from cement, aluminum, and waste processing (Department for Environment, Food and Rural Affairs [DEFRA] 2009) or transportation-related emissions for logistics providers.

Consider the example of energy-based carbon footprint calculation for a large warehouse in France: the warehouse is about 43,000m² and serves as storage for grocery products for a French retailer (with a mix of ambient temperature and refrigeration). We collected 6-month energy consumption data, and we extrapolated for an entire year. The warehouse consumed 6,060,584kWh of electricity, 127,410kWh of natural gas, and 99,174 L of diesel. The carbon footprint for French energy consumption is 0.0571kgCO₂e/kWh for electricity, 0.205kgCO₂e/kWh for natural gas, and 3.25kgCO₂e/L for diesel,¹² which leads to an annual carbon footprint of 695 tons of CO₂e—that is, the yearly average carbon footprint of about 158 French inhabitants. In addition, about 592,000 pallets were shipped from the warehouse during the same year, which leads to a carbon footprint from warehousing of about 1.17kgCO₂e/pallet.

Activity-Based Calculations

The most common approach used to calculate carbon emissions, the activity-based calculation method, aims at deriving the carbon emissions from activity information by using conversion factors that relate carbon emissions to a proxy measure of activity at a given source (often referred to as “emissions factors”). Available methods differ in their level of aggregation, some of which require more detailed information than others (Velazquez-Martinez et al. 2014). As an example, consider a reporting company that uses truck transportation for inbound deliveries. Assume that transportation is outsourced to a third-party logistics provider and therefore related carbon emissions are included in Scope 3. Data on fuel consumption (required for energy-based calculation) are generally not shared by third-party logistics providers, because fuel consumption is a key element in the cost structure

¹² Bilan Carbone ADEME (French equivalent of the US EPA): [https://data.ademe.fr/datasets/base-carbone\(r\)](https://data.ademe.fr/datasets/base-carbone(r)).

of truck transportation and carriers are usually not willing to share information about this cost structure with their customers. The reporting company thus must apply activity-based calculation by converting the weight transported over a given distance and using a given type of truck to estimate average carbon emissions based on emission factors. Several initiatives provide such emission factors for main supply chain activities. Chapter 5 by Blanco and Sheffi (2024) provides more details in the context of logistics. As an example, the emission factor for heavy-duty truck transportation in France is $175\text{gCO}_2\text{e/t.km}^{12}$, which gives the average carbon footprint for transporting 1 ton over 1 km by taking into account average load factors and average empty running.¹³ This calculation implies that transporting one pallet of 750 kg over 300 km (average distance for big-box store distribution in France) leads to a carbon footprint of $39.38\text{kgCO}_2\text{e/pallet}$, about 34 times more than carbon emissions from warehousing, as calculated previously.

EIO-LCA

The EIO-LCA allows companies to convert the expenditures in their industry sector into an average amount of carbon emissions; \$X spent in the transport industry sector leads on average to Y tons of CO_2e (for a detailed exposition, see Hendrickson et al. (2010); for an application of EIO-LCA to estimate the carbon footprint of all US economic sectors, see Huang et al. (2009)). This method is a relatively common way to measure Scope 3 emissions due to its ease of use. However, it exclusively accounts for upstream emissions. Besides, it is rather meant for estimation than facilitating the identification of improvement opportunities.

Methodologies that rely less on extrapolation provide a more accurate estimation of the true carbon footprint of operations. For example, Matthews et al. (2008) state that firm-level data such as electricity and energy use produce more accurate footprint results than EIO-LCA. However, the data required to apply a methodology that relies less on extrapolation may be more difficult to gather or simply not available. O'Rourke (2014) mentions that accessing data from full supply chains can be expensive, time-consuming, and even impossible. An inherent trade-off exists between the scope of measurement chosen (and consequently the portion of the total footprint analyzed) and the accuracy of the estimation (we return to the issue of accuracy in Sect. 3.4.2.). Defining the right scope for carbon emission measurement is of crucial importance, because this decision has strong implications for the type of measurement methods that can be implemented. One of the key questions here is assessing the importance of Scope 3 emissions for the reporting company. If Scope 3 emissions represent a small share of total supply chain emissions for the company, focusing on Scopes 1 and 2 emissions and applying a more direct methodology may make more sense. However, Scope 3 emissions can be of great importance when they represent a large share of a company's carbon footprint, as is true for most firms. We discuss the importance of Scope 3 emissions in more depth next.

¹³ Note that there exist more detailed methods for calculating the carbon footprint of transportation; please refer to Chap. 5 by Blanco and Sheffi (2024).

3.3 Supply Chain Carbon Footprinting

This section is divided into three parts: First, we identify the importance of Scope 3 emissions in supply chain carbon footprinting. Second, we illustrate the process of carbon footprinting in a supply chain using a case study in the telecommunications industry. Third, we conclude with a discussion of some issues related to carbon footprint allocation among different supply chain partners.

3.3.1 Importance of Scope 3 Emissions

To understand the environmental impact related to a company's activities, focusing on direct emissions and risks is not enough; rather, the entire value chain must be considered to evaluate its performance and improve it (CDP 2021). Matthews et al. (2008) estimate that on average, 74% of an industry sector's carbon footprint can be attributed to upstream Scope 3 emissions (without accounting for downstream Scope 3 emissions). Ozawa-Meida et al. (2013) measure the carbon emissions of a university and show that 79% fall under Scope 3. Dragomir (2012) notes that BP estimates its Scope 3 emissions to be 15 times higher than those from its operations. Huang et al. (2009) focus on upstream emissions of US economic sectors and provide a sector-specific repartition of emissions from Scope 1 to Scope 3; the impact of upstream Scope 3 emissions emerges as substantial for most US industry sectors, and a large share of US companies' upstream Scope 3 emissions can be attributed to their top ten suppliers. This result may help a company understand which upstream Scope 3 category contributes most to its total carbon footprint, such that it can focus its measurement efforts on relevant suppliers. Recently, CDP (2021) estimated that Scope 3 emissions are on average 11.4 times higher than Scope 1 and 2 emissions (more than double previous estimates).

Many industrial sectors are increasingly recognizing that not considering Scope 3 emissions in carbon footprinting may lead to serious gaps in results at the corporate and industrial levels (Radonjić and Tompa 2018). To the CDP (2021), "real climate leadership cannot be achieved without tackling Scope 3 impacts." In practice, companies often focus on measuring the portion of their Scope 3 emissions related to employee commuting and business travel. According to CDP (2013), 72% of the global 500 companies reporting to CDP reported emissions from business travel, even if these emissions accounted for only 0.2% of total reported Scope 3 emissions. In turn, CDP (2013) concludes that "instead of measuring carbon-intensive activities in their supply chain, companies often focus on relatively insignificant opportunities for carbon reduction."

Huang et al. (2009) also show that the portion of Scope 3 emissions varies greatly from one industry sector to another, which explains why providing generic guidance on which emissions to include in the inventory is difficult. The authors recommend that protocol organizations develop sector-specific Scope 3 guidelines.

Even companies that voluntarily disclose Scope 3 emissions are under no obligation to be comprehensive. To estimate how (in)complete current Scope 3 emissions reports are, Blanco et al. (2016) compare CDP disclosures by US firms with the predicted breakdown of emissions in Huang et al. (2009). They estimate that US firms that disclosed any Scope 3 emissions in 2013 only reported 22% of their full upstream supply chain emissions to CDP. In addition, Blanco et al. (2016) mention that Scope 3 reporting varies significantly from one industry to another. In summary, Scope 3 reporting remains underdeveloped, even if companies are progressively improving. The process of measuring and managing Scope 1 and 2 emissions is well established; this is not the case for Scope 3 emissions (Shrimali 2022). In the next section, we provide a case study to illustrate how to perform supply chain carbon footprinting. It also reflects the difficulty (and importance) of measuring Scope 3 emissions.

3.3.2 An Example from a Telecommunications Company

To better understand how carbon footprinting works in practice, we briefly summarize Radonjič and Tompa's (2018) case study and describe some calculations they performed. The authors note that the information and communication technology sector is developing quickly and, therefore, its energy consumption is increasing, leading to an increase of GHG emissions. Large companies in this sector, such as British Telecommunications and Deutsche Telekom AG, have undertaken initiatives related to GHG emission reduction. The authors focus on Telekom Slovenije d.d., the largest telecommunications company in Slovenia and one of the largest in southeastern Europe. The authors followed the GHG Protocol guidelines to calculate its carbon footprint in four steps: definition of organizational and operational boundaries; identification of GHG emission sources for Scopes 1, 2, and 3; data collection on energy consumption and GHG emissions; and selection of emission factors and GWP calculation. We explain the application of these steps to the case study company in more details in the following subsections.

Definition of the Organizational/Operational Boundaries and Identification of GHG/Energy Sources

To define the organizational boundaries, Radonjič and Tompa (2018) used the control approach (one of the two approaches allowed by the GHG Protocol). In this case, the company is accountable for 100% of GHG emissions from operations over which it has control. Two approaches could be used: operational or financial control. The authors chose the former and used the GHG Protocol guidelines as well as the supporting service sector guide. According to the characteristics of the company, they included the following: company-owned buildings; car fleet; employee activities; telecommunications network and equipment; heating, ventilation, and air conditioning systems; rented buildings/car fleet; and leased premises of building, equipment, and cars (note that the authors excluded subsidiaries,

Table 3.3 Categorization of activities and processes within each scope and GHG/energy source types: Telekom Slovenije company

Scopes	Activities and processes	Energy source types
Scope1	Use of vehicles for transportation	Fuel (gasoline and diesel)
	Use of diesel aggregates for electricity production	Fuel (diesel)
	Air conditioning (owned spaces)	HFC gases (HFC-134a)
	Heating (owned spaces)	Fuel oil, natural gas, liquefied petroleum gas
Scope 2	Purchased electricity consumption	Electricity
	District heating	Heat energy
Scope 3	Use of employee-owned car for business travel	Fuel (gasoline and diesel)
	Air travel of employees	Fuel (kerosene)
	Employee commuting	Fuel (gasoline and diesel)
	Electricity consumption of premises rented to others	Electricity
	Heating of premises rented to business partners	Fuel oil
	Heating of premises rented to business partners	Natural gas, liquefied petroleum gas, heat energy (district heating), fuel (gasoline, diesel)
	Renting cars to business partners	

Source: Radonjič and Tompa (2018)

associates, and joint ventures in the country of origin [Slovenia] and abroad). Within operational boundaries, the authors categorized GHG emissions as direct and indirect emissions, and they identified the three scopes. Table 3.3 presents activities and processes included in each scope, as well as associated GHG/energy sources used. Regarding which activities to include in Scope 3, the authors chose according to the availability of high-quality data and time and human resources available (i.e., they focused on the company’s own in-house accounting statistics). For example, some maintenance service is outsourced, but the subsidiary company had not yet established a carbon footprinting method at a level that would allow for properly including their GHG emissions as part of the telecommunications company’s Scope 3 emissions; therefore, the authors excluded GHG emissions related to maintenance from Scope 3. For the same reason, the authors did not include telework emissions in Scope 3.

Data Collection on Energy Consumption/GHG Emissions and GWP Calculation

Radonjič and Tompa (2018) selected 2012 as the reference year for GHG emission calculations, such that GHG emissions calculated over subsequent years can be compared with those calculated for 2012. They followed the GHG Protocol, which recommends choosing the reference year on the basis of the earliest relevant point in time for which accurate, reliable, and comprehensive data are available. The authors also mention that 2012 was the year the company started providing both fixed and mobile telecommunications services, which is considered an important criterion for future GHG emission reduction comparisons. For data collection, the authors

considered all six GHGs covered by the GHG Protocol, which led to data obtained for the period 2012–2014 (3 years):

- *Scope 1*

Radonjič and Tompa (2018) obtained data related to Scope 1 from the company's fuel purchase and fuel use records (transportation and heating). They calculated fuel consumption data related to car transportation by considering total distance travelled, the type and total amount of fuel used, and the average fuel efficiency (obtained from car manufacturer's technical specifications). They collected data related to fuel consumption for diesel aggregates and to HFC-134a gas emissions from the company's maintenance records. The authors considered the electricity required air conditioning equipment functioning as a part of total purchased electricity (i.e., Scope 2).

- *Scope 2*

The firm purchases electricity to be used for lighting, building, automation, heating, information technology infrastructure, and workstations. For office and technical facilities, the authors obtained data from direct meter readings and from monthly electric consumption bills, and they monitored daily electricity consumption online for about 50 locations. They also collected these data for leased and rented assets (Scope 3). More than half of the company's facilities are connected to district heating networks; therefore, the authors obtained related natural gas consumption data from the gas supplier's website and the company's heat consumption records.

- *Scope 3*

To measure employee-owned car use for business travel, Radonjič and Tompa (2018) used travel orders generated by the company together with receipts issued for travel expenses (for estimating distances travelled by the employees). They used the company's statistics on the share of cars equipped with diesel versus gasoline engines to estimate fuel consumption, and they used the same approach for cars rented to business partners. For employee commuting, the authors gathered data on the distances employees travelled between home and work from the company's documentation. The firm records all commuting travels in a database of the company's information system (presence at worksites in different locations), and the authors estimated the travelled distances according to the commuting costs paid by the company to employees (total cost, retrieved from travel orders, divided by the country's official financial value of 1 km travelled by a vehicle). The company does not track the transportation modes used by its employees; thus, the authors used governmental statistics on passenger transportation to and from work (by car, bus, and train), including the share of cars equipped with diesel versus gasoline engines in the country. They gathered data on commercial air travel based on the distances travelled by each employee (expressed in passenger-kilometers), using the company's documentation on business travels, as well as data provided by travel agencies (distance travelled per flight). Calculation parameters provided by

Table 3.4 Emission factors
(averages for the period
2012–2014)

Energy source	Unit	Emission factor (CO ₂ e)
Fuel oil	t/TJ	74.4390
Liquefied petroleum gas	t/TJ	63.1968
Natural gas	t/TJ	55.4211
Gasoline	t/TJ	72.0300
Gas oil	t/TJ	74.3013
Diesel fuel	t/TJ	81.8930
District heating	Kg/kWh	0.3134
Electrical energy	Kg/kWh	0.5720

Source: Calculations based on Radonjič and Tompa (2018) and AR5 GWP-100 indexes (IPCC's AR5 values, Table 3.2)

the GHG Protocol determined GHG emissions for all mobile combustion modes. The estimation of energy consumption in leased assets was based on the square footage expressed as a percentage of the total square footage of the premises in every building owned or rented by the company. In Table 3.4, we present emission factors used for the case study company.

Radonjič and Tompa (2018) collected the data used to calculate the emission factors from extant literature, governmental and research institutions, and companies in the energy sector (note that these data are valid for the Slovenian conditions [energy and power systems]). They used Eq. 3.1 (GHG Protocol) to calculate the carbon footprint for each activity/process:

$$CF = \sum_i AD_i * EF_i, \quad (3.1)$$

where CF is the carbon footprint, AD_i is the activity data from source i (based on a unit of measurement), and EF_i is the emission factor for source i (kg CO₂e/unit of measurement).

Emission factors for activities within Scope 3, transport of employees (to and from work and air travel), were based on GHG Protocol tables summarizing emission factor generic values (version 2.5). To determine fuel quantities, expressed in tons or m³ (gasoline, diesel fuel, fuel oil, liquefied petroleum gas, and natural gas), the authors calculated the amount of energy per ton using net calorific values obtained from the Slovenian Environment Agency. Table 3.5 presents the carbon footprints of the various activities and processes (Scopes 1, 2, and 3) of the telecommunications company during the period 2012–2014.

The total cumulative carbon footprint of the company is 176,690.31 tCO₂e over 2012–2014. Activities within Scope 1 contribute to 6.04% of the total emissions, and activities within Scopes 2 and 3 contribute 78.04% and 15.92%, respectively. Indirect emissions contribute the most to the GHG emissions of the studied company (almost 94%). In this case study, Scope 3 emissions are low compared with Scope 2 (and total) emissions. One reason for this result is the exclusion of a large number

Table 3.5 Carbon footprint by activity/process obtained for Telekom Slovenije (3-year cumulative period: 2012–2014)

Activity/process	Carbon footprint (tCO ₂ e)
Electricity consumption	131,906.31
Electricity consumption of leased assets	15,584.18
Employees commuting	10,748.90
Business travel (use of company-owned cars)	8222.87
Heating	7748.58
Heating of leased assets	993.47
Business travel (air)	640.39
Air conditioning	473.85
Use of diesel aggregates	211.67
Business travel (employee-owned cars)	159.65
Renting cars to other companies	0.43

Source: Radonjič and Tompa (2018) and AR5 GWP-100 indexes (IPCC's AR5 values, Table 3.2)

of activities and processes from the analysis: the company collected only 3 of the 15 activities proposed by GHG Protocol within its Scope 3 emission category. In contrast, three other European telecommunications companies that reported on their carbon footprint (following GHG Protocol) in 2014, 2015, and 2016 included 10, 11, and 12 activities, respectively; for example, Deutsche Telekom reported 64.94% of its GHG emissions under Scope 3, and BT and Swisscom reported, respectively, 93.73% and 93.00% of their GHG emissions under Scope 3. Although these companies have different characteristics (e.g., size, geographical location, product, and service portfolio), even within the same industrial sector, we can observe a substantial gap between the results obtained for the case study company and the three other companies regarding Scope 3 emissions. For these three companies, the top two activities that contributed most to Scope 3 emissions were “purchased goods and services” and “use of sold products,” representing 78.8%, 65.8%, and 78.9% of the total Scope 3 emissions of BT, Deutsche Telekom, and Swisscom, respectively. These two activities were excluded in the case study.

While the Telekom Slovenije case study demonstrates how carbon footprinting can be applied in practice, it also shows how a lack of reliable and accurate data can impede the accurate measurement of Scope 3 emissions and how omitting key activities can lead to inaccurate conclusions. The main point of carbon footprinting is to identify action points for improvement. By comparing the Telekom Slovenije case to other three companies in the same sector, it appears that significant sources of Scope 3 emissions, such as purchased goods and services and the use of sold products, were excluded, while relatively insignificant sources, such as rented cars, were included (note that this specific item emitted 0.43 tCO₂e, equivalent to about 163 liters of diesel or about the amount needed to fill three fuel tanks). This aligns with the findings of the CDP (2013) report, which stated that companies tend to concentrate on relatively minor opportunities for carbon reduction. However, it is important to note that ignoring Scope 3 emissions entirely due to data limitations

would be detrimental to making informed decisions about reducing supply chain GHG emissions.

3.3.3 *Emission Allocation in the Supply Chain*

Allocation is a thorny issue in LCA in general (see also Chap. 2 by Guinée and Heijungs (2024)) and, thus, for carbon footprinting. We discuss two key settings here. First, it is often necessary to allocate shared emissions from an activity among its various outputs. Truck emissions should be allocated to all the products transported on that truck, and emissions caused by heating, cooling, and lighting in a plant should be allocated to the range of products and customers that it serves. Second, supply chain emissions are often the result of joint processes by multiple parties, and these emissions should be allocated among supply chain members.

Definition of the Organizational/Operational Boundaries and Identification of GHG/Energy Sources Allocating Shared Emissions to Separate Units¹⁴

The WRI and WBCSD (2011a, Chap. 8) provide guidance on allocating shared emissions to separate units. Allocation is necessary when a single activity produces multiple outputs and emissions are only quantified for the entire activity. When allocation is required, companies should select the allocation approach that (1) best reflects the causal relationship between the production of the outputs and the resulting emissions, (2) results in the most accurate and credible emissions estimates, (3) best supports effective decision-making and GHG reduction activities, and (4) otherwise adheres to the principles of relevance, accuracy, completeness, consistency, and transparency (WRI and WBCSD 2011a). The general method that WRI and WBCSD propose to allocate emissions from a facility is to multiply total facility emissions by the reporting company's purchases as a fraction of total production (excluding waste)—for example, using mass as the allocation factor (Eq. 3.2):

$$\text{Allocated facility emissions} = \frac{\text{Mass of Products Purchased}}{\text{Total Mass of Products Produced} \times \text{Total Emissions}} \quad (3.2)$$

Using a physical relationship between the multiple inputs/outputs and the quantity of emissions generated is preferable, according to allocation factors such as mass, volume, energy, chemical, number of units, and so on (e.g., protein content of food coproducts, floor space occupied by products). The other option is to use

¹⁴ This section draws on “The GHG Protocol Corporate Value Chain (Scope 3) Accounting and Reporting Standard” (WRI and WBCSD 2011a), which we cite at various points throughout this section.

economic factors (by value) or other relationships; these options are less preferable, because physical factors likely best reflect the causal relationship between the production of the outputs and the resulting emissions. Furthermore, it is preferable to let the reporting company select the allocation method to ensure consistency, even if doing so might have some practical limitations in terms of data disclosure. Clearly, different allocation methods are prone to yielding significantly different results (for an in-depth discussion on the implications of the method chosen to allocate emissions among coproducts, see Sunar and Plambeck (2016)).

Allocating Emissions from Joint Processes

Many GHG emissions are the result of joint processes by multiple parties in a supply chain (Caro et al. 2013). A typical product goes through numerous manufacturing and transportation stages operated by multiple companies in a supply chain. Consequently, further reductions in emissions—beyond those of a firm's own operations—can be achieved by joint effort of multiple parties in a supply chain through collaboration, coordination, or information sharing. These joint efforts may bring to light additional cost-saving opportunities (we refer to Chap. 11 by Tan et al. for some examples of supply chain collaboration for sustainability). The CDP (2015) supply chain report notes that companies that engage with one or more of their suppliers, consumers, or other partners are more than twice as likely to see a financial return on their emission reduction investments, as well as almost twice as likely to reduce emissions, compared with those that do not engage with their value chain.

Nevertheless, when multiple firms jointly affect total emissions, they face a critical and nontrivial challenge in measuring their share of the responsibility for emissions (or that of the emission reductions). In LCA and carbon footprinting literature, various guidelines exist for how to allocate shared emissions. A common attribute in those guidelines (including the GHG Protocol covered in this section) is that the sum of the allocated emissions for each output of a system should equal 100% of emissions from the system. Because LCA is aimed at making product and process design decisions based on an accurate inventory of environmental impacts, it is natural that LCA literature seeks to avoid over-allocation, or “double counting,” of impacts (e.g., Lenzen (2008)). Similarly, in trading schemes, avoiding double counting of GHG emissions is crucial due to the financial consequences involved.

Therefore, LCA and operations management literature have begun investigating how to reconcile allocating responsibility for impacts while avoiding double counting. For example, Lenzen et al. (2007), building on Gallego and Lenzen's (2005) work, propose a scheme by which producers and consumers share responsibility for emissions in such a way that adding total emissions across all producers and consumers yields an accurate account of economy-wide emissions. Leenders et al.'s (2017) study of how to allocate carbon emissions to a specific shipment in transportation routing proposes four allocation mechanisms. Gopalakrishnan et al. (2021b) use cooperative game theory to allocate the upstream emissions attributable to a project among the various supply chain members involved. They show that the nucleolus allocation avoids over-allocation while exhibiting a certain consistency

property that is especially important in a regulatory context. Gopalakrishnan et al. (2021a) also apply cooperative game theory to derive an emission responsibility-allocation scheme for distributing the total carbon emissions among the firms in the supply chain. This scheme, which corresponds to the Shapley value of the underlying cooperative game, has several desirable properties. These contributions all propose allocation rules that avoid over-allocation.

In contrast, Caro et al. (2013) argue that whenever emissions (or reductions in emissions) result from multiple parties, double counting is necessary to induce the optimal level of abatement effort among the supply chain members. Accordingly, even if the true social cost of carbon emissions is internalized (e.g., by applying a carbon tax to all supply chain members), the abatement efforts implemented would be less than the optimal level if double counting is avoided. Even in the absence of an optimal allocation rule (which would require double counting), firms with an interest in overall supply chain efficiency should include, at minimum, the full cost of all GHG emissions that they can influence when they decide where to focus their efforts. The fact that double counting is unlikely to be implemented on a large scale in practice should not preclude firms from identifying where their efforts may have the greatest effect, and consequently, managers should not neglect opportunities that exist elsewhere in the supply chain (Blanco et al. 2017). For example, if the greatest return on firm 1's effort is on emissions currently allocated to firm 2, then firm 1 could explore mechanisms to share the costs and benefits of reducing emissions with firm 2. Without at least allowing for double counting in a pro forma fashion, many valuable opportunities for joint improvement will go unexploited.

3.4 Discussion

This section presents several issues related to supply chain carbon footprinting. We first introduce challenges related to gathering information necessary to calculate a carbon footprint and then consider the issue of accuracy. Finally, we highlight the need to extend the horizon of sustainable supply chains beyond carbon emissions.

3.4.1 *How Can We Get Information in Practice?*

The amount of data required to calculate the carbon footprint of a company is substantial, and these data are often difficult to gather, especially when performing a value chain carbon footprint. For example, when Fujitsu carried out a carbon footprint analysis on its desktop PC and servers to show customers its products' superiority through reduced carbon emissions, it faced some challenges to gather data in the use and disposal phases, and it had to make numerous assumptions to be able to conduct a cradle-to-grave LCA (Vasan et al. 2014). The Telekom Slovenije case study presented in Subsection 3.3.2 provides another clear example. Even when

the data exist, most companies are not organized to collect these data systematically or automatically, though some enterprise resource planning (ERP) systems include information on carbon footprint and other sustainability metrics. In this regard, SAP, IBM, SAS, and other software vendors have built tools to extract energy data from supply chain procurement systems (O'Rourke 2014). Accordingly, DEFRA (2009) recommends that companies include carbon emission reporting in their existing reporting tools and processes.

However, currently, most companies must manually combine data from disparate sources to compile carbon footprints, often in spreadsheets. Furthermore, data collection and compilation are typically not standardized, so the process must be repeated every year—often a costly and time-consuming task subject to errors and/or approximations. Moreover, the multiplicity of competing reporting standards and organizations challenges reporting companies, due to their diverse content, requirements, and formats, despite efforts of the main reporting initiatives to align the information required and facilitate translation across standards (Bateman et al. 2017). The manual, time-consuming compilation of data from separate sources also causes a transparency issue. Keeping all calculations transparent is difficult, and introducing mistakes is easy. In addition, reporting companies often have loose contacts with their tier-two and more upstream suppliers. Therefore, when a reporting company requests information about a supplier's practices, upstream suppliers are often required to survey their own suppliers and so on, leading to what is referred to as the "survey waterfall" (Bateman et al. 2017), in which one information chain member's lack of expertise could lead to the wide propagation of incorrect information.

To address these challenges, third-party organizations can support reporting companies in two ways: First, they can facilitate the data collection and reporting process. EcoVadis proposes to manage the entire information-gathering process to obtain all the data a company needs to assess its supply chain impacts (Bateman et al. 2017). Second, the reporting company can rely on third-party audits. Many companies worldwide offer carbon auditing services and have developed carbon footprint certification programs to enable companies to report a verified carbon footprint. For example, DHL states that the reporting methodologies and calculation tools used in its express and air, ocean, and road freight divisions have been verified by the Swiss-based Société Générale de Surveillance since 2011 (DHL 2015). However, even third-party audits may miss errors, because they focus on ensuring the use of correct conversion factors rather than whether the input data are correct and complete. The question of how to collect and organize verifiable information efficiently, without having to redo it every year, is of crucial importance to companies and deserves particular attention (please refer to the Philips case in Chap. 11 by Tan et al. (2024) for a more detailed discussion). That said, these challenges should not deter companies from measurement and disclosure, considering the various internal and external benefits associated with measurement and reporting (Blanco et al. 2017).

Some emerging digital technologies could improve companies' capabilities for information traceability and sharing (Montecchi et al. 2021), and these technologies

can also be applied to carbon footprinting. Harbich et al. (2021) highlight that the Internet of Things (IoT), blockchain, and artificial intelligence (AI) are particularly noteworthy for this purpose: IoT can improve real-time tracking and monitoring of primary data; blockchain can improve trust, information sharing, and traceability of data; and AI can be used to elicit missing data and for decision support. While these technologies are promising, companies should not underestimate their implementation and integration costs or the related technical challenges. Besides, these technologies will also lead to additional emissions associated with their energy requirement.

3.4.2 How Accurate Is Accurate Enough?

Carbon footprinting is inherently associated with a certain level of uncertainty. More particularly, because Scope 3 covers activities that are not under the reporting company's ownership or control, companies are likely to face additional challenges that contribute to uncertainty in Scope 3 accounting (WRI and WBCSD 2011a). Lee (2011) reports that, regarding Hyundai Motor Company's process of carbon footprinting, one senior manager stated "we had some difficulties in terms of scopes and measurement of carbon footprint. In our case, Scope 1 and 2 carbon footprint measurements are completed with over 95% confidence. But Scope 3 CO₂ measurement is still limited with regard to the supply chain network. We should explore further feasible methods and practices to track CO₂ emissions from the supply chain." The Telekom Slovenije case study presented in Sect. 3.3.2 presents some notable limitations that can arise in practice in the telecommunications sector when dealing with Scope 3 activities and related GHG emissions, especially at the start of carbon footprinting process implementation. The availability of reliable and high-quality data, as well as a lack of time and human resources, represent main obstacles (Radonjič and Tompa 2018).

Uncertainties related to carbon inventories can be categorized into scientific uncertainty and estimation uncertainty (WRI and WBCSD 2004). Scientific uncertainty arises when the science of the actual emission is not completely understood. Many direct and indirect factors associated with GWP values involve significant scientific uncertainty (IPCC 2000). The evolution of the GWP-100 of methane from 25 in the fourth IPCC report to 29.8 in the sixth IPCC report (see Table 3.2) is an example. Analyzing and quantifying scientific uncertainty are challenging and best addressed by the scientific community rather than companies (IPCC 2000).

Estimation uncertainty occurs any time carbon emissions are quantified, such that all carbon footprint measurements are associated with some degree of estimation uncertainty. Estimation uncertainty can be further classified into two types: model uncertainty and parameter uncertainty (IPCC 2000). Model uncertainty occurs when the emissions are not directly measured (i.e., using energy-based calculations, activity-based calculations, or EIO-LCA). In this case, a model translates a given input into a certain amount of carbon emissions. How this translation is handled

is subject to uncertainty, referred to as “model uncertainty.” Estimating emissions from truck transportation using an energy-based calculation methodology would lead to uncertainty in the precise amount of carbon emissions released by the motor, because this calculation depends on the quality of the combustion and thus the vehicle’s operating conditions. Parameter uncertainty is associated with quantifying the parameters used as inputs for estimation models (IPCC 2000) (e.g., amount of fuel a truck consumes).

Given that uncertainty is an intrinsic part of any carbon footprint assessment, an immediate question is what level of uncertainty is acceptable. On the one hand, high-quality information has greater value and more uses, and even if a company does not anticipate future regulatory mechanisms, internal and external stakeholders may demand high-quality inventory information (WRI and WBCSD 2004). On the other hand, in the context of carbon emission regulations, low-quality information may have little or no value and may even incur penalties. Defining the level of accuracy depends on the carbon footprinting analysis objective, as well as on the capability of the company. However, this decision may greatly influence a company’s estimated carbon footprint. This decision is referred to as the application of a cut-off threshold in LCA literature.

After the desired level of accuracy has been determined, the next question to address is how to report uncertainties in carbon footprints. Given that only parameter uncertainty is within the feasible scope of most companies, uncertainty estimates for carbon inventories will necessarily be imperfect (IPCC 2000). Parameter uncertainties can be evaluated through statistical analysis, measurement equipment precision determinations, and expert judgment (IPCC 2000). Statistical analysis may be difficult, due to a lack of complete and robust sample data. Most of the time, a single data point is available (e.g., liters of fuel for truck transportation). If the data are obtained from an instrument, precision and/or calibration information should be included. However, IPCC (2000) states that expert judgment is often the only possibility for companies. Experts can either be the source of the necessary data or help identify and explain uncertainties. The problem with expert judgment is that it is difficult to obtain in a comparable and consistent manner across parameters, source categories, or companies (WRI and WBCSD 2004).

The GHG Protocol Corporate Standard has developed a supplementary guidance document on uncertainty assessments, along with an uncertainty calculation tool (WRI and WBCSD 2011b). This document describes how to use the calculation tool to aggregate uncertainties. It also outlines in depth the types of uncertainties, limitations of quantitative uncertainty assessments, and appropriate methods for interpreting uncertainty estimates (for additional guidance and information on assessing uncertainty, see EPA (1999) and IPCC (2000)).

3.4.3 *How Can We Extend the Horizons Beyond Carbon?*

In Sect. 3.1.2, we highlighted some of the reasons the focus on carbon emissions is currently strong; however, sustainability considerations cannot be limited to carbon emissions. For example, water scarcity, its quality, and the regulations affecting it are growing business problems.¹⁵ Other environmental dimensions of sustainability and social impacts should not be overlooked in the quest for carbon emission reductions. We refer to Chap. 4 by Hoekstra (2024) for more on water footprinting, to Chap. 18 by Blass et al. (2024) for more on managing nonrenewable materials, and to Chapters 20 and 21 by Gillai et al. (2024) and Sodhi and Tang (2024) for more on socially responsible supply chains.

The strong current focus on carbon emissions may be an opportunity for other environmental indicators to be developed and adopted, in the sense that platforms and accumulated experience related to carbon footprinting can be beneficial. For example, once acquainted with environmental reporting through carbon emissions, a company may be more prone and able to develop an overall sustainability assessment. This trend is also reflected by the CDP's commitment to use the experience and reputation obtained from carbon footprinting to develop new initiatives related to water use and forest management.

However, an important observation here is that the capability developed through carbon footprinting might not directly transpose to other sustainability aspects. Companies, nongovernmental organizations, and governments need to acknowledge that these aspects could have different characteristics than carbon emissions. For example, location and timing play a major role in water footprinting but not in carbon footprinting. Extending the capabilities developed for carbon footprinting to other dimensions of sustainability presents an exciting opportunity—but one that should be approached thoughtfully.

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¹⁵ See <https://www.unicef.org/reports/critical-business-actions-achieving-water-secure-world>.

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Chapter 4

Water Footprint Assessment in Supply Chains



Arjen Y. Hoekstra

4.1 Introduction

The World Economic Forum has listed water scarcity as one of the three global systemic risks of highest concern, an assessment based on a broad global survey on risk perception among representatives from business, academia, civil society, governments, and international organizations (WEF 2014). Freshwater scarcity manifests itself in the form of declining groundwater tables, reduced river flows, shrinking lakes, and heavily polluted waters but also in increasing costs of supply and treatment, intermittent supplies, and conflicts over water (Hoekstra 2014a). Future water scarcity will grow as a result of various drivers: population and economic growth, increased demands for animal products and biofuels, and climate change (Ercin and Hoekstra 2014). The private sector is becoming aware of the problem of freshwater scarcity but is facing the challenge of formulating effective responses. Even companies operating in water-abundant regions can be vulnerable to water scarcity, because the supply chains of most companies stretch across the globe. An estimated 22% of global water consumption and pollution relates to the production of export commodities (Hoekstra and Mekonnen 2012). Countries such as the USA, Brazil, Argentina, Australia, India, and China are big virtual water exporters, which means that they intensively use domestic water resources for producing export commodities. In contrast, countries in Europe, North Africa, and the Middle East as well as Mexico and Japan are dominated by virtual water import, which means that they rely on import goods produced with water resources elsewhere. The water use behind those imported goods is often not sustainable, because many of the export regions overexploit their resources.

Increasingly, companies start exploring their water footprint, thereby looking at both their operations and supply chain. Key questions that industry leaders pose

themselves are as follows: where is my water footprint located, what risks does water scarcity impose to my business, how sustainable is the water footprint in the catchments where my operations and supply chain processes are located, where and how can water use efficiency be increased, and what is good water stewardship? The demand for new sorts of data emerges, types of data that were usually not collected. The focus shifts from relatively simple questions—whether the company has got sufficient water abstraction permits and whether wastewater disposal standards are met—to the more pressing question how the company actually contributes to the overexploitation and pollution of water resources, not only through its own facilities but also through its supply chain. Having permits and meeting standards do not imply sustainability. Most experience with collecting the new sorts of data required and with addressing questions about good water stewardship is within the food and beverage sector, which depends most clearly on water. In other industries, the connection with water is not always clear, because indirect and mostly through the supply chain. The aim of this chapter is to introduce the water footprint concept, review experiences with water footprint assessment, and reflect on future challenges.

In the next section, I will start with discussing and comparing three methods to trace resource use and pollution over supply chains: environmental footprint assessment, life cycle assessment, and environmentally extended input output analysis. Next, I will discuss what new perspective the water footprint concept brings to the table, compared to the traditional way of looking at water use. In the third section, I will reflect on direct and indirect water footprints of the different sectors of the economy, with examples for two specific sectors: food and beverage and transport. In the last section, I will discuss future challenges, such as the issue of data gathering and reporting, the demand for water stewardship and greater product transparency, and the need to establish water footprint benchmarks.

4.2 Footprints, Life Cycle Assessment, and Input-Output Modeling

4.2.1 Methods to Trace Natural Resource Use and Pollution over Supply Chains

Three different methods have been developed to analyze direct and indirect natural resource use and emissions in relation to products or economic sectors: environmental footprint assessment (EFA), life cycle assessment (LCA), and environmentally extended input output analysis (EE-IOA). All three methods have been applied also in the field of water, to trace direct and indirect water use and pollution over supply chains. Each of the three methods has its specific goal, approach, and focus, but there are commonalities across the methods as well. They all focus on understanding natural resource use and emissions along supply or value chains. EFA

focuses on macro-questions on resource use sustainability, efficiency, equitability, and security. LCA concentrates on the comparative analysis of environmental impacts of products. EE-IOA focuses on understanding how natural resource use and environmental impacts can be traced throughout the economy.

The field of EFA comprises methods to quantify and map land, water, material, carbon, and other environmental footprints and assesses the sustainability of these footprints as well as the efficiency, equitability, and security of resource use (Hoekstra and Wiedmann 2014). Water footprint assessment (WFA) can be regarded as a specific branch of this field and refers to the full range of activities to quantify and locate the water footprint of a process, product, producer, or consumer or quantify in space and time the water footprint in a specified geographic area; assess the environmental sustainability, economic efficiency, and social equitability of water footprints; and formulate a response strategy (Hoekstra et al. 2011). Broadly speaking, the goal of assessing water footprints is to analyze how human activities or specific products relate to issues of water scarcity and pollution and to see how consumption, production, trade, and specific products can become more sustainable from a water perspective.

LCA is a method for estimating and assessing the environmental impacts attributable to the life cycle of a product, such as climate change; stratospheric ozone depletion; tropospheric ozone (smog) creation; eutrophication; acidification; toxicological stress on human health and ecosystems; the depletion of resources, water use, land use, and noise; and others (Rebitzer et al. 2004). The assessment includes all stages of the life cycle of a product, from cradle to grave (from material extraction to returning of wastes to nature). An LCA study includes four phases: setting goal and scope, inventory accounting, impact assessment, and interpretation. Water use and pollution can be considered as specific impact categories within LCA (Kounina et al. 2013). LCA focuses on *comparing* the environmental impacts of alternative processes, materials, products, or designs. (See Chap. 2 by Guinée and Heijungs (2024) for more on LCA.)

Environmentally extended input-output analysis (EE-IOA) is a method for studying the relation between different sectors of the economy and indirect natural resource use and environmental impacts. It combines the classical monetary input-output formalism with satellite accounts containing data on resource use and emissions into the environment. Over the past decade, we have seen quite a number of applications of EE-IOA to analyze “embodied” water flows through the economy (Daniels et al. 2011). Applications have been carried out, for example, for Australia (Lenzen and Foran 2001), Spain (Duarte et al. 2002; Cazarro et al. 2013), the UK (Yu et al. 2010; Feng et al. 2011a), China (Zhao et al. 2009; Zhang and Anadon 2014), and the City of Beijing (Zhang et al. 2011). Input-output models basically show monetary flows between sectors within the economy; environmentally extended input-output models usually express water use in terms of liter per dollar (or other currency). Most environmentally extended input-output models also have some form of accounting of product flows in physical units, but due to the aggregation of specific economic activities into sectors, it remains difficult to reach the same high level of detail as achieved in a process-based WFA or LCA.

Both WFA and LCA enable an analysis of water use in all processes of the value chain and attribution of the water use along value chains to specific products. A promising path in this respect is the method of the so-called *hybrid* environmentally extended input-output modeling, in which physical flows are integrated into the model (Ewing et al. 2012; Steen-Olsen et al. 2012).

Process-based WFA and LCA are generally constrained by the fact that parts of the value chain have to be left out from the analysis for practical reasons. This problem does not occur in input-output modeling. Therefore, there is a development to enhance process-based WFA and LCA with the advantage of input-output modeling. In the case of LCA, this results in the so-called hybrid LCA approach (Finnveden et al. 2009). In hybrid LCA, the environmental impacts of flows that were not included in the process-based LCA are estimated with an environmentally extended input-output model. This hybrid approach is also called environmental input-output-based LCA (EIO-LCA). In the case of WFA, a similar development can be expected (Feng et al. 2011b).

The difference between EFA and LCA is the focus on sustainability of production and consumption at macro-level of the former and the focus on comparing environmental impact at process and product level of the latter (Hoekstra 2015b). This is explained in Box 4.1 for the example of cutting trees. Typical questions in EFA studies relate to how different processes and products contribute to the overall footprints at larger scales, how different consumption patterns influence the overall footprint, whether footprints at the larger scales remain within their maximum sustainable levels, how footprints can be reduced by better technology, whether different people have equitable shares in the total footprint of humanity, and what externalization of footprints may imply for resource security (Hoekstra and Wiedmann 2014). LCA is designed to compare the environmental impact of one product over its full value chain with the overall impact of another product or to compare the differences in environmental impact between different product designs and alternative production processes.

Box 4.1 The Sustainability of Cutting Trees: The Fundamental Difference Between LCA and EFA

Is it sustainable to cut a tree? Although a relevant question, it is impossible to answer this question in isolated form. On the one hand, it is hard to argue that cutting just one tree is not sustainable. After a tree has been cut, a new one will grow, so that it is sustainable. On the other hand, if one takes this insight on the sustainability of cutting one tree to conclude that one can cut all forests, one cannot maintain that this is sustainable. The reason why answering a simple question like this tree-cutting question causes a fundamental problem is that sustainability is a concept that cannot be applied at the level of single activities but only at the level of a system as a whole. Still, there is a strong wish among

(continued)

people to measure the sustainability of single activities, because individuals undertake single activities and consume goods and services that relate to series of single activities to produce them. The methods of LCA and EFA deal with this problem in fundamentally different ways. In LCA, the approach is to leave the larger question on sustainability and look at *comparative* contributions of different activities to natural resource appropriation, emissions, and potential impacts at the larger scale. In other words, LCA addresses the question how cutting one tree compares to cutting two trees, a question that is not hard to answer. In EFA, the approach is to estimate humanity's total natural resource appropriation and emissions and compare that to the Earth's carrying or assimilation capacity. Both methods struggle in a similar way with how to compare apples and pears, for example, how to compare cutting trees with polluting water. The approach in LCA is to *weigh* different types of primary resource use or emissions according to their potential final impact on human health and ecosystem health. The approach in EFA is to compare the different types of resource use and pollution to their respective maximum sustainable levels. The great similarity between LCA and EFA is that resource use and emissions are analyzed per process (activity) and per product (by analyzing the processes along supply chains). The difference comes when LCA starts weighing different types of resource use and emissions based on their potential impact and comparing alternative processes or products according to their overall potential environmental impact. In contrast, EFA adds the resource use and emissions of different activities in order to get a complete picture, analyze the sustainability of the whole, and study the relative contribution of different processes, products, and consumers to the total. In many applications, though, the difference between LCA and EFA is not so clear. By comparing the footprints of two different processes or products, EFA also allows for comparative analysis. However, the comparative analysis is partial in this case, because different footprints are not weighted and added to get a measure of "overall potential environmental impact." On the other hand, one can also extend an LCA from comparing products to comparing consumption patterns, which is at the larger scale typically for EFA. The fundamental difference between LCA and EFA in the way they treat the tree-cutting question, however, remains.

At the level of basic data, EFA and LCA require similar data. The data collection and analysis required in the accounting stage of a product-focused WFA (as opposed to a geographic- or consumption-focused WFA) are very much similar to what is needed in the inventory stage of a water-focused LCA (Boulay et al. 2013).

EFA, LCA, and EE-IOA are not static analytical methods, but still young fields under development. We can observe a development in the past few years in which

a fruitful exchange between the three fields leads to the adoption of approaches from one field into the other. In EFA studies, we have seen the adoption of life cycle accounting procedures from LCA and the exploration of using input-output models to calculate national and sector footprints, in addition to the already existing bottom-up and top-down trade-balance approaches. In LCA, we recently observe, fed by experiences in EFA, an interest to develop methods to carry out an LCA for a whole organization instead of for a product and to carry out LCAs for consumer lifestyles or for national consumption as a whole (Hellweg and Milà i Canals 2014). Additionally, based on experiences in EE-IOA, the LCA community is exploring hybrid LCA methods as already mentioned above. The EE-IOA practice improves in the direction of hybrid methods that include physical accounting and have greater granularity in the analysis, fed by the practices in the EFA and LCA fields. This mutual enrichment and to some extent convergence of approaches does not imply that the three methods will grow into one. They may develop into a more consistent framework of coherent methods, but the fact that different sorts of questions will remain implies that different approaches will continue to be necessary.

All three methods—EFA, LCA, and EE-IOA—have a focus on environmental issues, leaving out social issues (like labor conditions, human rights). Principally, though, there is nothing that necessarily restricts the methods to environmental issues. Broadly speaking, one can trace all sorts of process characteristics along supply chains. The oldest forms of accounting along supply chains are the accounting of monetary added value and the accounting of material flows and energy use along supply chains. Material flow analysis (MFA) or substance flow analysis (SFA) aims at the quantification of stocks and flows of materials or substances in a well-defined system, drawing mass balances for each sub-system and the system as a whole. Energy flow analysis aims at quantifying the energy content of flows within an economy. The innovation of EFA, LCA, and EE-IOA lies in the attribution of resource use, emissions, or impacts along supply chains to products and final consumption. In this context, one speaks about the embodied, embedded, indirect, or virtual land, water and energy in a product or consumption pattern, or indirect emissions, etc. When doing so, the method of EE-IOA is linked to traditional economic accounting, which is a strong point of this method. The methods of EFA and LCA are rather linked to physical accounting, which is their strength. In all three fields, we observe efforts to enhance the methods and broaden the scope, with an increasing number of hybrid approaches.

4.2.2 The Water Footprint Concept

The water footprint (WF) is a measure of freshwater appropriation underlying a certain product or consumption pattern. Three components are distinguished: the blue, green, and gray WF (Hoekstra et al. 2011). The blue WF measures the volume of water abstracted from the ground or surface water system minus the volume of water returned to the system. It thus refers to the sum of the water

flow that evaporates during the process of production, the water incorporated into a product, and the water released in another catchment. The blue WF differs from the conventional way of measuring freshwater use by looking at net rather than gross water withdrawal. This is done because it makes more sense to look at net water withdrawal if one is interested in the effect of water use on water scarcity within a catchment. Return flows can be reused within the catchment, unlike the water flow that evaporates or is captured within a product. The green WF refers to the volume of rainwater consumed in a production process. This is particularly relevant in agriculture and forestry, where it refers to the total rainwater evapotranspiration (from fields and plantations) plus the water incorporated into the harvested crop or wood. The gray WF is an indicator of freshwater pollution and defined as the volume of freshwater required to assimilate a load of pollutants based on natural background concentrations and existing ambient water quality standards. The advantage of expressing water pollution in terms of the water volume required for assimilating the pollutants, rather than in terms of concentrations of contaminants, is that this brings water pollution into the same unit as consumptive use. In this way, the use of water as a drain and the use of water as a resource, two competing uses, become comparable. The WF refers thus to both consumptive water use (of rainwater, the green WF, and of surface and groundwater, the blue WF) and degenerative or degradative water use (the gray WF).

As a measure of freshwater use, the WF differs from the classical measure of “water withdrawal” in several ways. The term “water withdrawal”—also called “water abstraction” or often simply “water use”—refers to the extraction of water from the groundwater or a surface water body like a river, lake, or artificial storage reservoir. It thus refers to what we call *blue* water use. The WF is not restricted to measuring blue water use but also measures the use of green water resources (the green WF) and the volume of pollution (the gray WF). Another difference between the WF and the classical way of measuring water use was mentioned already above: the classical measure of “water use” always refers to gross blue water abstraction, while the blue WF refers to net blue water abstraction. Another difference between the classical way of measuring water use and the WF is that the latter concept can be used to measure water use over supply chains. When we talk about the WF of a product, we refer to the water consumption and pollution in all stages of the supply chain of the product. When we speak about the WF of a producer or a consumer, we refer to the full WF of all the products produced or consumed.

The WF thus offers a wider perspective on how a product, producer, or consumer relates to the use of freshwater systems. It is a volumetric measure of water consumption and pollution. WF accounts give spatiotemporally explicit information on how water is appropriated for various human purposes. The local environmental impact of a certain amount of water consumption and pollution depends on the vulnerability of the local water system and the number of water consumers and polluters that make use of the same system. The WF within a catchment needs to be compared to the maximum sustainable WF in the catchment in order to understand the sustainability of water use. The WF of a specific process or product needs to be compared to a WF benchmark based on best available technology and

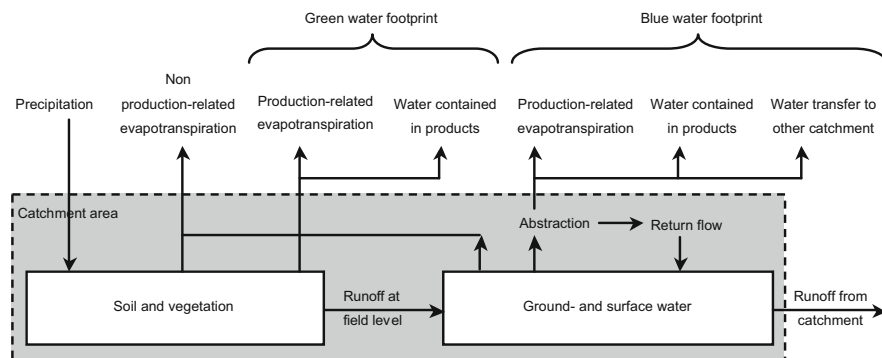


Fig. 4.1 Definition of the green and blue water footprint in relation to the water balance of a catchment area. Source: Hoekstra et al. (2011)

practice in order to understand the efficiency of water use. The WF per capita for a community can be compared to the WF of other communities to understand the degree of equitable sharing of limited water resources. WF accounts can thus feed the discussion about the sustainability, efficiency, and equitability of water use and allocation (Hoekstra 2013, 2014b).

The definition of the green and blue WF can best be understood by considering the water balance of a river basin (Fig. 4.1). The total annual water availability in a catchment area is given by the annual volume of precipitation, which will leave the basin partly through evapotranspiration and partly through runoff to the sea. Both the evaporative flow and the runoff can be appropriated by humans. The green WF refers to the human use of the evaporative flow from the land surface, mostly for growing crops or production forest. The blue WF refers to the consumptive use of the runoff flow, i.e., the net abstraction of runoff from the catchment. The term “water consumption” can be confusing, because many people—particularly those not aware of the big difference between gross and net water abstraction—use the term for gross water abstraction. Specialists, though, define water consumption as net blue water abstraction (gross abstraction minus return flow). Evaporation is generally considered as a loss to the catchment. Even though evaporated water will always return in the form of precipitation at global scale, this will not alleviate the water scarcity in the catchment in the period that the river is emptied due to net water abstractions. Moisture recycling at smaller spatial scales is generally only modest.

The definition of the gray WF is clarified in Fig. 4.2. The basis for the calculation is the anthropogenic load of a substance into a freshwater body (groundwater, river, lake), that is, the additional load caused by a human activity (e.g., a production process). We should acknowledge that the effluent from an industry might contain certain amounts of chemicals that were already in the water abstracted. Therefore, we should look at the *additional* load to a freshwater body as a result of a certain activity. Furthermore, we should look at the load of a substance that really enters the river, lake, or groundwater, which means that if an effluent is treated before

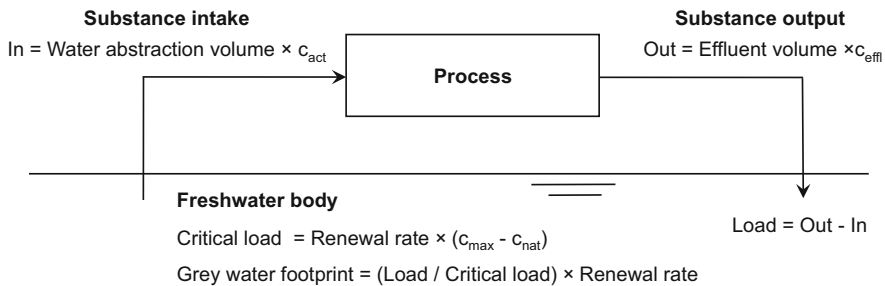


Fig. 4.2 Definition of the gray water footprint based on the load of a chemical into a freshwater body. The symbols c_{act} , c_{nat} , and c_{max} refer to the actual, natural, and maximum allowable concentration of the chemical in the freshwater body; c_{eff} refers to the concentration of the chemical in the effluent

disposal, we have to consider the load of chemicals in the effluent that remains *after* treatment. The critical load in a freshwater body is defined as the difference between maximum acceptable and natural concentrations of a chemical for the receiving water body times the renewal rate of the freshwater body. Note that as for the maximum allowable concentration, we have to take the ambient water quality standard for the receiving freshwater body, not the effluent standard (Franke et al. 2013). In a river, the renewal rate is equal to runoff; in a groundwater reservoir, the renewal rate is equal to groundwater recharge, which (over the longer term) is the same as groundwater runoff. In a lake, the renewal rate equals the flow through the lake. The gray WF is calculated as the pollutant load to a freshwater body divided by the critical load, times the renewal rate of that freshwater body. Defined in this way, it means that when the gray WF onto a freshwater body becomes as big as the renewal rate of this freshwater body, the assimilation capacity has been fully used. When the size of the gray WF in a catchment exceeds the size of runoff from this catchment, pollution is bigger than the assimilation capacity, resulting in a violation of the maximum acceptable concentration. When an effluent contains different sorts of pollutants, as is usually the case, the gray WF is determined by the pollutant that is most critical, which is the one that gives the largest pollutant-specific gray WF. Thermal pollution can be dealt with in a similar way as pollutants, whereby the load consists of heat and the assimilation capacity depends on the accepted temperature increase of the receiving water body (Hoekstra et al. 2011).

4.3 Direct and Indirect Water Footprints of Different Sectors of the Economy

4.3.1 *The Importance of Water Use in the Primary Sector*

Usually, economic activities are categorized into three different sectors. The primary sector of the economy, the sector that extracts or harvests products from the earth, has the largest water footprint on Earth. This sector includes activities like agriculture, forestry, fishing, aquaculture, mining, and quarrying. The green WF of humanity is nearly entirely concentrated within the primary sector. It has been estimated that about 92% of the blue WF of humanity is just in agriculture alone (Table 4.1).

The secondary sector covers the manufacturing of goods in the economy, including the processing of materials produced by the primary sector. It also includes construction and the public utility industries of electricity, gas, and water. Sometimes, the public utility industries are also mentioned under the tertiary (service) sector, because they do not only produce something (electricity, gas, purified water) but also supply it to customers (as a service). Water utilities could even partly fall under the primary sector, because part of the activity is the abstraction of water from the environment (rivers, lakes, groundwater). The work of water utilities comprises water collection, purification, distribution and supply, wastewater collection (sewerage), wastewater treatment, materials recovery, and wastewater disposal. It's rather common to categorize the whole water utility sector under the secondary sector. The tertiary industry is the service industry and covers services to both businesses and final consumers. This sector includes activities like retail and wholesale sales, transportation and distribution, entertainment, restaurants, clerical services, media, tourism, insurance, banking, healthcare, defense, and law. Even though sometimes categorized into another, quaternary sector, one can also list here activities related to government, culture, libraries, scientific research, education, and information technology. The secondary and tertiary sectors have much smaller WFs than the primary sector.

The contribution of agriculture to water scarcity is underestimated by conventional water use statistics, which show gross blue water abstractions. In agriculture, most of the gross water use will evaporate from storage reservoirs, from irrigations canals, or finally from the field. The water abstracted for irrigation in agriculture is thus largely unavailable for reuse within the basin. In industrial water use, the ratio of net to gross abstraction is estimated at less than 5%. In municipal water use, this ratio varies from 5 to 15% in urban areas and from 10 to 50% in rural areas (FAO 2014). Water that returns to the catchment after use can be reused. Presenting gross or net water abstractions thus makes a huge difference for industries and households and less in agriculture.

Even though the primary sector is the largest water user, governmental programs to create public awareness of water scarcity often focus on public campaigns calling for water saving at home. This is little effective at large given the fact that the major

Table 4.1 Global water footprint within different water-using categories, Period: 1996–2005

Economic sector	Water use category	Global water footprint (10 ⁹ m ³ /year)				%	Remark
		Green	Blue	Gray	Total		
Primary sector	Crop farming	5771	899	733	7404	81.5	
	Pasture	913	–	–	913	10.0	
	Animal farming	–	46	–	46	0.5	Water for drinking and cleaning
	Agriculture total	6684	945	733	8363	92.0	
	Aquaculture	?	?	?	?	?	No global data
	Forestry	?	?	?	?	?	No global data
	Mining, quarrying	?	?	?	?	?	No global data
Secondary sector	Industry (self-supply)	–	38	363	400	4.4	Water use in manufacturing, electricity supply, and construction
	Municipal water supply	–	42	282	324	3.6	Water supply to consumers and (small) users in primary, secondary, and tertiary sector
Tertiary sector	Self-supply	?	?	?	?	?	No global data
Consumers	Self-supply	?	?	?	?	?	No global data
	Total	6684	1025	1378	9087	100	

Data sources: Mekonnen and Hoekstra (2011) for crop farming, Mekonnen and Hoekstra (2012) for pasture and animal farming, and Hoekstra and Mekonnen (2012) for industry and municipal water supply. Note that the blue WF figure for crop farming relates to evapotranspiration of irrigation water at field level; it excludes losses from storage reservoirs and irrigation canals. The blue WF figure for “industry” presented here includes water use in mining, which is part of the primary sector. The figure excludes water lost from reservoirs for hydroelectric generation. All gray WF figures are conservative estimates. Forestry is not included as a water use sector because of a lack of data

share of water use in most places relates to agriculture and in second instance to industry. Water scarcity is thus generally caused mostly by excessive water use in agriculture. Installing water-saving showerheads and dual flush toilets in households will have hardly any impact in mitigating water scarcity at all, but still this is what most water-saving campaigns advocate for. It would be more useful to make people aware of the water use and pollution underlying the food items and other products they buy and to advocate for product labels that show the sustainability of the WF of a product.

4.3.2 Food and Beverage Products

The food and beverage sector is the manufacturing sector with the largest WF, maybe not the largest operational WF but definitely the largest supply chain WF. Chapter 9 by Akkerman and Cruijsen (2024) focuses on food loss and waste in supply chains. The reason is that the food and beverage sector is the largest client of the agricultural sector, which is responsible for the largest share in global water consumption (Table 4.1). Interesting WF studies carried out in the beverage sector are, for example, the studies carried out by SABMiller (SABMiller and WWF-UK 2009; SABMiller et al. 2010), Coca-Cola (TCCC and TNC 2010; Coca-Cola Europe 2011), and the Beverage Industry Environmental Roundtable (BIER 2011). Some nice examples of WF studies in the food sector come from Unilever (Jefferies et al. 2012), Dole (Sikirica 2011), Mars (Ridoutt et al. 2009), and Barilla (Ruini et al. 2013).

Traditionally, the beverage industry focuses on the so-called water-use ratio (WUR), which is defined as the total water use divided by the total production at a bottling facility, expressed in terms of liter of water used per liter of beverage produced. Water use stands here for gross blue water abstraction, not net blue water abstraction (blue WF). In a global benchmarking study for the period 2009–2011, BIER (2012) reports a WUR of 1.2 to 2.2 L/L (with an average of 1.5) for bottled water, a WUR of 1.5 to 4.0 (average 2.1) for carbonated soft drinks, a WUR of 3.2 to 6.6 (average 4.3) for beer breweries, a WUR of 8 to 126 (average 36) for distilleries, and a WUR of 2.0 to 18.5 (average 4.4) for wineries. The WUR is of limited value, because the operational WF of bottling factories is very small when compared to the full WF of a beverage, as shown by Ercin et al. (2011) for a carbonated soft drink. They show that the WF of a half-liter bottle of soft drink resembling cola can range between 150 and 300 L, of which 99.7 to 99.8% refers to the supply chain.

4.3.3 Transport

Transport is always considered as an important sector in carbon footprint assessment, since transport can significantly contribute to the overall carbon footprint of a final product, measured over its full supply chain. In the case of the WF of a final

product, the contribution of transport will generally be relatively small, because not much freshwater is being consumed or polluted during transport. It is worth considering the indirect WF of transport related to materials (trucks, trains, boats, airplanes) and energy used, but particularly materials will generally contribute very little, due to the fact that the WF of transport vehicles can be distributed over all goods transported over the lifetime of a vehicle. The WF of energy may be more relevant, but even that can be small compared to other components of the WF of goods, particularly in the case of agricultural goods. The key determinant in the WF of transport is probably the energy source (Gerbens-Leenes et al. 2009a; King and Webber 2008). The WF of bioenergy in terms of cubic meter per GJ is generally two to three orders of magnitude larger than that for energy from fossil fuels or wind or solar power. However, in all energy categories, WFs per unit of energy can widely vary, depending on the precise source and production technology. The technique of hydraulic fracturing (fracking) to mine natural gas or petroleum reserves, for example, has a larger blue and gray WF than when mining reserves that are easier accessible using more conventional techniques. In the case of bioenergy, it matters greatly whether one speaks about biodiesel from oil crops, bioethanol from sugar or starch crops (Gerbens-Leenes et al. 2009b; Dominguez-Faus et al. 2009), biofuel from cellulosic fractions of crops or waste materials (Chiu and Wu 2012), and biofuel from algae (Gerbens-Leenes et al. 2014) or about bioelectricity. In the latter case, it makes a large difference what is burnt: biomass grown for the purpose of organic waste. As an illustration of the large differences between different bioenergy forms, Table 4.2 shows the WF of different modes of passenger and freight transport when based on first-generation biofuel produced in the European Union. Governmental policies to replace substantial percentages of fossil fuels by biofuels will lead to a rapid growth of the WF of the transport sector (Gerbens-Leenes et al. 2012).

4.4 Challenges

4.4.1 *The Need to Take a Supply Chain Perspective*

The supply chain WF of most companies is many times greater than their operational footprint, but most companies restrict their efforts to reducing the latter, leaving the supply chain WF out of scope. Studies carried out by companies like Coca-Cola, PepsiCo, SABMiller, and Heineken have shown that the supply chain WF for beverage companies can easily be over 99% of their total WF. Nevertheless, all these companies apply a “key performance indicator” for water that refers to the water use in their own operations only. Common reduction targets in the beverage industry, such as going from 2 to 1.5 liters of water use in the bottling plant per liter of beverage, have little effect on the larger scale given that the supply chain WF of most beverages is of the order of 100 liters of water per liter of beverage or even

Table 4.2 The water footprint of different modes of passenger and freight transport when based on first-generation biofuel produced in the European Union

Transport mode	Energy source	Green+blue water footprint of passenger transport (L per passenger km)	Green+blue water footprint of freight transport (L per 1000 kg of freight per km)
Airplane	Biodiesel from rapeseed	142–403	576–1023
	Bioethanol from sugar beet	42–89	169–471
Car (large)	Biodiesel from rapeseed	214–291	–
	Bioethanol from sugar beet	138–289	–
Car (small)	Biodiesel from rapeseed	65–89	–
	Bioethanol from sugar beet	24–50	–
Bus / lorry	Biodiesel from rapeseed	67–126	142–330
	Bioethanol from sugar beet	20–58	–
Train	Biodiesel from rapeseed	15–40	15–40
Ship (inland)	Biodiesel from rapeseed	–	36–68
Ship (sea, bulk)	Biodiesel from rapeseed	–	8–11
Electric train	Bioelectricity from maize	3–12	2–7
Electric car	Bioelectricity from maize	4–7	–
Walking	Sugar from sugar beet	3–6	–
Bike	Sugar from sugar beet	1–2	–

Source of data: Gerbens-Leenes and Hoekstra (2011). The total water footprint of transport based on first-generation biofuel mainly relates to the water volumes consumed in growing the crop

more (Hoekstra 2013). Water sustainability investments by beverage companies are geared to better perform in their own operations, which means that investments are made that aim to reduce that 1% of their total WF. It is difficult to imagine that these investments will be most cost-effective if really sustainability is the goal. Incorporating sustainability principles into a company's business model would include the adoption of mechanisms to secure sustainable water use in the supply chain. For companies, moving toward a sustainable supply chain is a much bigger challenge than greening their own operations, because the WF of the supply chain can be influenced only indirectly. Mechanisms to reduce the supply chain WF are, for example, the application of water criteria for sustainable procurement and reaching agreements with suppliers about footprint reduction over time.

4.4.2 The Need to Incorporate Temporal and Spatial Variability in Water Footprint Assessment

When formulating WF reduction targets for processes in their operations or supply chain, companies should look not only at the numbers but also at the geographic locations where their WF is sited. Priority is to be given to WF reduction in catch-

Table 4.3 Example of how overexploitation in a water-stressed river basin (A) can be solved by increasing water productivity in a water-abundant basin (B). Source: Hoekstra (2013)

Parameter	Unit	Current situation		Possible solution	
		Basin A	Basin B	Basin A	Basin B
Max. sustainable water footprint	Water units/unit of time	50	250	50	250
Water footprint	Water units/unit of time	100	200	50	200
Production	Product units/unit of time	100	100	50	200
Water footprint per product unit	Water units/product unit	1	2	1	1
Water productivity	Product units/water unit	1	0.5	1	1

ments in which the overall footprint exceeds the carrying capacity or assimilation capacity of the catchment. It has been argued that reduction in water-abundant catchments does not even make sense (Pfister and Hellweg 2009), but this is based on a misunderstanding. Since the WF ($\text{m}^3/\text{product unit}$) is simply a reverse of water productivity (product units per m^3), it is difficult to see why one would not set targets regarding the reduction of the WF of a product, which is the same as setting targets regarding the increase of water productivity. The relevance of increased water productivities worldwide, also in water-abundant places, can be illustrated with the following example (Hoekstra 2013). Suppose the hypothetical case of two river basins, with the same surface (Table 4.3). Basin A is relatively dry and has, on an annual basis, 50 water units available, the maximum sustainable WF. The maximum level, however, is exceeded by a factor of two. Farmers in the basin consume 100 water units per year to produce 100 crop units. Basin B has more water available, 250 water units per year. Water is more abundant than in the first basin, but water is used less efficiently. Farmers in the basin consume 200 water units per year, to produce 100 crop units, the same amount as in the first basin, but using two times more water per crop unit. A geographic analysis shows that in basin B, the WF (200) remains below the maximum level (250), so this is sustainable. In basin A, however, the WF (100) by far exceeds the maximum sustainable level (50), so this is clearly unsustainable. The question is now the following: should we categorize the crops originating from basin A as unsustainable and the crops from basin B as sustainable? From a geographic perspective, the answer is affirmative. In basin A, the WF of crop production needs to be reduced, which seems to be the crux. However, when we take a product perspective, we observe that the WF per crop unit in basin B is two times larger than in basin A. If the farmers in basin B would use their water more productively and reach the same water productivity as in basin A, they would produce twice as many crops without increasing the total WF in the basin. It may well be that farmers in basin A cannot easily further increase their water productivity, so that—if the aim is to keep global production at the same level—the only solution is to bring down the WF in basin A to a sustainable level by cutting production by half while enlarging production in basin B by increasing the water productivity. If basin B manages to achieve the same water productivity level

as in basin A, the two basins together could even increase global production while halving the total WF in basin A and keeping it at the same level in basin B.

4.4.3 Measuring and Reporting

It is difficult to get water use statistics organized along the same structure of economic sector classifications. Many countries and regions have their own classification of economic activities, distinguishing main sectors, subsectors, etc. One of the international standard classifications is the Industrial Classification of All Economic Activities of the United Nations (UN 2008). Conventional water use statistics mostly show gross blue water withdrawals and distinguish three main categories: agricultural, industrial, and municipal water use (FAO 2014). Also, WF statistics distinguish between agricultural, industrial, and municipal sectors. These three sectors cannot be mapped one to one onto the primary, secondary, and tertiary sector. “Agricultural water use” obviously is about water use in the primary sector, while “industrial water use” is about water use in the secondary sector. However, water use in mining—part of the primary sector—will generally be categorized under “industrial water use” as well. Industrial water use refers to self-supplied industries not connected to the public distribution network. It includes water for the cooling of thermoelectric plants, but it does not include hydropower (which is often left out of the water use accounts altogether). Municipal water use—often alternatively called domestic water use or public water supply—refers to the water use by water utilities and distributed through the public water distribution network. Water utilities provide water directly to consumers but also to water users in the primary, secondary, and tertiary sector.

The mismatch between the three main categories in water use statistics and the different sectors as usually distinguished in the economy can be quite confusing. The “water supply sector” as distinguished in economic classifications refers to water utilities delivering municipal water to households and others connected to the public water supply system. Unfortunately, the category of municipal water use lumps water use for a great variety of water users: final consumers (households) and users in all economic sectors. Specifications by type of user are not always available. Additionally confusing is that even though the “water supply sector” serves all sorts of users, the sector refers to only a minor fraction of total water use. Most of the water use in agriculture, the largest water user, is not part of the “water supply sector.” Furthermore, water self-supply by industries does not fall within this sector and neither do self-supply in the tertiary sector and self-supply by final consumers. Given that only an estimated 3.6% of the total WF of humanity relates to what we call the “water supply sector” (Hoekstra and Mekonnen 2012), the sector receives disproportionate attention in public debates about water use and scarcity, diverting the necessary attention on water use in agriculture and industry.

For companies, much confusion exists as to what needs to be measured and reported. Traditionally, companies have focused on monitoring gross water abstrac-

tions and compliance with legal standards. However, net water abstractions are more relevant than gross abstractions, and meeting wastewater quality standards is not enough to discard the contribution to water pollution made by a company. Regarding terminology and calculation standards, the Water Footprint Network—a global network of universities, nongovernmental organizations, companies, investors, and international organizations—developed the global water footprint standard (Hoekstra et al. 2011). The International Organization for Standardization is developing a reporting standard based on life cycle assessment (ISO 2014). Both standards emphasize the need to incorporate the temporal and spatial variability in WFs and the need to consider the WF in the context of local water scarcity and water productivity. In practice, companies face a huge challenge in tracing their supply chain. Apparel companies, for example, have generally little idea about where their cotton is grown or processed, yet both cotton growing and processing are notorious water consumers and polluters. It is difficult to see quick progress in the field of supply chain reporting if governments don't force companies to do it.

4.4.4 Water Stewardship and Transparency

There is an increasing call for good water stewardship and transparency in the private sector, driven by increased public awareness, demands from investors, and perceived water risks by the sector itself. Water stewardship is a comprehensive concept that includes the evaluation of the sustainability of water use across the entire value chain, the formulation of water consumption and pollution reduction targets for both the company's operations and supply chain, the implementation of a plan to achieve these targets, and the proper reporting on all of this (Hoekstra 2014a). In priority catchments, it requires the pursuit of collective action and community engagement (Sarni 2011). Large priority river basins are, for example, the Colorado and San Antonio basins in North America, the Lake Chad, Limpopo and Orange basins in Africa, the basins of the Jordan, Tigris, Euphrates, Indus, Ganges, Krishna, Cauvery, Tarim and Yellow rivers, the Yongding River basins in Asia, and the Murray-Darling Basin in Australia (Hoekstra et al. 2012).

The increasing interest in how companies relate to unsustainable water use calls for greater transparency on water consumption and pollution. Openness is required at different levels: the company, product, and facility level. Driven by environmental organizations and the investment community, businesses are increasingly urged to disclose relevant data at company level on how they relate to water risks (Deloitte 2013). Simultaneously, there is an increasing demand for product transparency through labeling or certification. Despite the plethora of existing product labels related to environmental sustainability, none of these includes criteria on sustainable water use. Finally, there is a movement to develop principles and certification schemes for sustainable site or facility management, such as the initiatives of the European Water Partnership and the Alliance for Water Stewardship. But despite progress in awareness, still hardly any companies in the world report on water

consumption and pollution in their supply chain or reveal information about the sustainability of the WF of their products.

Another concern regarding good water stewardship is the extent to which a company pays for the full cost of its water use. Water use is subsidized in many countries, either through direct governmental investments in water supply infrastructure or indirectly by agricultural subsidies, promotion of crops for bioenergy, or fossil-energy subsidies to pump water. Besides, water scarcity and pollution remain unpriced (Hoekstra 2013). In order to give the right price signal, users should pay for their pollution and consumptive water use, with a differentiated price in time and space based on water vulnerability and scarcity.

4.5 The Need to Establish WF Benchmarks

WFs per unit of product strongly vary across different production locations and production systems. Therefore, we need to establish WF benchmarks for water-intensive products such as food and beverages, cotton, flowers, and biofuels. The benchmark for a product will depend on the maximum reasonable water consumption in each step of the product's supply chain, based on best available technology and practice. Benchmarks for the various water-using processes along the supply chain of a product can be taken together to formulate a WF benchmark for the final product. An end-product point of view is particularly relevant for the companies, retailers, and consumers who are not directly involved in the water-using processes in the early steps of the supply chains of the products they are manufacturing, selling, or consuming but still interested in the water performance of the product over the chain as a whole. WF benchmarks will offer a reference for companies to work toward and a reference for governments in allocating WF permits to users. Besides, manufacturers, retailers, and final consumers in the lower end of the supply chain get an instrument to compare the actual WF of a product to a certain reference level. Business associations within the different sectors of economy can develop their own regional or global WF benchmarks, though governments can take initiatives in this area as well, including the development of regulations or legislation. The latter will be most relevant to completely ban worst practices.

4.6 Water Footprint Reduction Goals and Possible Trade-Offs

Companies should strive toward zero WF in industrial operations, which can be achieved through nullifying evaporation losses, full water recycling, and recapturing chemicals and heat from used water flows. The problem is not the fact that water is being used, but that it is not fully returned to the environment or not returned

clean. The WF measures exactly that: the consumptive water use and the volume of water polluted. As the last steps toward zero WF may require more energy, the challenge will be to find a balance between reducing the water and the carbon footprint. Furthermore, companies should set reduction targets regarding the WF of their supply chain, particularly in areas of great water scarcity and in cases of low water productivity. In agriculture and mining, achieving a zero WF will generally be impossible, but in many cases the water consumption and pollution per unit of production can be reduced easily and substantially (Brauman et al. 2013).

4.7 Conclusion

Spatial patterns of water depletion and contamination are closely tied to the structure of the global economy. As currently organized, the economic system lacks incentives that promote producers and consumers to move toward wise use of our limited freshwater resources. In order to achieve sustainable, efficient, and equitable water use worldwide, we need greater product transparency, international cooperation, water footprint ceilings per river basin, water footprint benchmarks for water-intensive commodities, water pricing schemes that reflect local water scarcity, and some agreement about equitable sharing of the limited available global water resources among different communities and nations.

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Operational Aspects of Sustainable Supply Chains

Part II

Chapter 5

Green Logistics



Edgar E. Blanco and Yossi Sheffi

5.1 Introduction

Logistics encompasses the business processes that plan, control, and implement the flow of goods and related information between points of origin and points of consumption to meet customer demand. It does so by managing transportation, warehousing, and inventory decisions across the company and, whenever possible, across its supply chain.

Traditionally, logistics decisions have been driven by minimizing cost, maximizing profitability, and/or achieving customer service targets. As companies have added sustainability goals to their business objectives, there has been an increased interest in mitigating the social and environmental impact of their products and operations. This new focus has also impacted the field of logistics: transportation providers are expected to reduce greenhouse gas emissions from their vehicles, warehouse managers have focused on waste and energy reduction strategies, and products are redesigned to increase recyclability and reuse, which require different inventory planning needs.

Green logistics refers to the systematic measurement, analysis, and, ultimately, mitigation of the environmental impact of logistics activities. This effort to mitigate environmental externalities in logistics activities includes reducing of consumption of nonrenewable energy sources, air emissions (e.g., particulate matter), greenhouse gas emissions, and waste. Some of these efforts may be technological, such as replacing vehicle fleets from diesel to hybrid or replacing cardboard boxes with

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returnable totes. Other strategies involve better ways to plan and execute the movement of goods, such as increasing the utilization of trucks while maintaining inventory levels under control or using modes of transportation that have lower greenhouse gas emissions. Finally, some green logistics initiatives may be in support of larger business environmental goals, such as increasing reverse logistics activities to recover and reuse more of the products delivered to customers.

The International Energy Agency (IEA 2022a) estimates that passenger and freight transport as a whole accounted for 37% of worldwide energy-related carbon dioxide (CO₂) emission in 2021. As the world recovered from the pandemic, the transport of CO₂ emissions rebounded by 8%. They are expected to return to their pre-pandemic levels and settle back to their all-sector record annual growth of 1.7%. Under current policies and technology trends, these emissions are expected to grow 16% by 2030 and between 30% and 100% by 2050 compared to 2021 levels. Following Paris Agreement, states and economic unions set policies such as the EU banning sales of petrol and diesel vehicles by 2035. The impacts of these policies are still to be measured, but the IEA states that strong regulations, fiscal incentives, and considerable investment are still needed to achieve the Net Zero Emissions by 2050 Scenario.

Within the freight transportation sector, trucking represents 75% of all CO₂ emissions. Even though electric vehicles have shown promising improvements over the past years, they still represent less than 1% of the fleet (IEA 2022b). Figure 5.1 provides an overall estimate of CO₂ emissions from freight transport and logistics activities. Transport is responsible for about 90% of these emissions. The main focus of this chapter on green logistics consequently relates to freight transportation.

This chapter is organized as follows: Section 5.2 describes the main environmental impacts of logistic operations, namely, greenhouse gases, pollution, noise, vibration, and packaging waste. Details on how these impacts could be estimated and their relative importance are discussed. Section 5.3 focuses on the importance

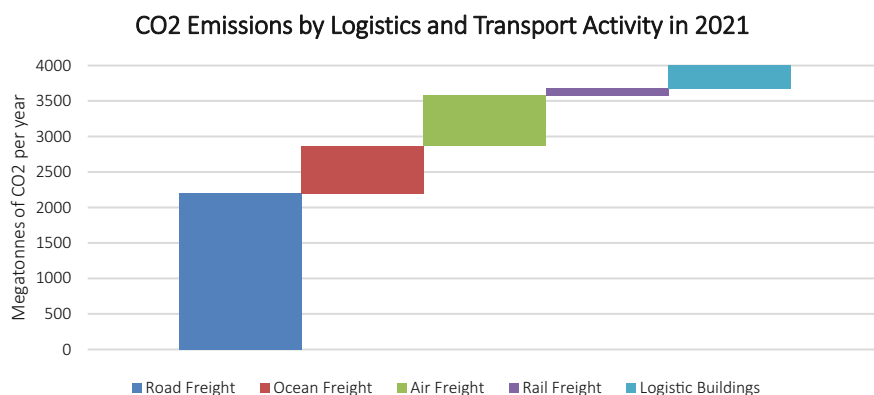


Fig. 5.1 GHG emissions of logistics and transport activity. *Data source:* IEA 2022a and author projections baselining from WEF (2009)

and subtleties of measuring green logistics. Section 5.4 introduces the various green logistics strategies available to mitigate these impacts. These strategies are presented within a framework of decision-making, including a discussion on how to leverage two important modeling approaches to achieve green logistics: network design and vehicle routing. Using real-life success stories, Sect. 5.5 discusses how organizations collaborate to implement green logistics in practice. We conclude with a discussion on other strategies and relevant aspects of a sound green logistics strategy.

5.2 The Environmental Impact of Logistics

As goods flow from origins to destinations through the logistics network, they are moved in conveyances (e.g., planes, trucks, ships, motorcycles) powered by fossil fuels (e.g., diesel, petrol). During the engine combustion process, visible and invisible gasses are emitted through exhaust pipes that impact the local, regional, and global atmospheric composition, ranging from local air, water, or soil pollution to global climate change. Energy used during storage and handling of goods also impacts the atmosphere, albeit not always directly but indirectly through nonrenewable energy use. The transport conveyances also generate noise and vibration as they navigate roads, highways, and waterways, thereby affecting human and wildlife quality of life. Finally, additional packaging and materials is used to preserve the integrity of products before they reach customers. Inadequate disposal or excess waste of this additional protective packaging is another potential environmental impact of logistics.

In this section, we will describe the four main environmental impacts of transportation operations in logistics: greenhouse gas (GHG) emissions that affect global climate, pollution of air quality and water ways, noise and vibration that affect human health, and packaging waste that increases pressure on landfills.

5.2.1 GHG Emissions

Greenhouse gases trap heat, making the planet warmer. The Intergovernmental Panel on Climate Change (IPCC) identifies transportation activities as producing three direct greenhouse gases: carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O). The combustion of transport fuels produces relatively little carbon in non-CO₂ gases. The impact of each of these greenhouse gases is expressed in terms of carbon dioxide equivalents (CO₂e), a process commonly referred to as carbon footprinting. For more on carbon footprinting, see Chap. 3 by Boukherroub et al. (2024). The GLEC Framework (Smart Freight Centre 2019) is also a good reference for emissions accounting across supply chains and geographies.

As mentioned earlier, more than 90% of the GHG emissions in logistics are due to freight transportation. For each mode of transportation in a logistics network (road, railways, waterborne navigation, aviation), the IPCC recommends a fuel-based approach to measuring emissions due to the fairly consistent estimates of the amount of greenhouse gases produced by combustion of each type of fuel. This approach is also known as energy-based calculation, and it is the most reliable for CO₂, the primary greenhouse gas from transportation, representing an estimated 97% of GHG emissions from road (IPCC 2019) and 97% from marine transportation (IMO 2014).

If total fuel consumption is known, CO₂ emissions can be computed as described in Eq. 5.1 by multiplying the total fuel used by the conveyance multiplied by the emission factor for that fuel.

$$\text{Emission} = \sum_a [\text{Fuel}_a \times \text{EF}_a]$$

Where :

Emission = Emissions of CO₂

Fuel_a = fuel sold

EF_a = emission factor equal to carbon content of the fuel multiplied by 44/12

a = type of fuel

(5.1)

CH₄ and N₂O are best estimated using distance traveled plus emissions produced during cold start of the conveyance. It necessitates a more detailed breakdown of the data, requiring distance traveled and emission factors by fuel type, vehicle type, emission control technology, and operating conditions such as road types. This is shown in Eq. 5.2.

$$\text{Emission} = \sum_{a,b,c,d} [\text{Distance}_{a,b,c,d} \times \text{EF}_{a,b,c,d}] + \sum_{a,b,c,d} C_{a,b,c,d} \quad (5.2)$$

where:

Emission = emission of CH₄ or N₂O

EF_{a,b,c,d} = emission factor

Distance_{a,b,c,d} = distance travelled during thermally stabilized engine operation phase for a given mobile source activity

C_{a,b,c,d} = emissions during warm-up phase

a = fuel type

b = vehicle type

c = emission control technology

d = operationg conditions

Table 5.1 illustrates the resulting emissions for 1000 gal of fuel, using emission factors from various sources. Note that the resulting calculations are affected by the type of GHG gas included, as well as the scope of emissions included in the

Table 5.1 GHG emissions calculated for 1000 gal of fuel

Fuel	GHGs	Source	Emissions included	Results	Units
Diesel	CO ₂	UK Defra	Pump to wheel	9998	Kg CO ₂
Diesel	CO ₂	US EPA	Pump to wheel	10,150	Kg CO ₂
Diesel	CO ₂ , CH ₄ , N ₂ O	GHG Protocol (EPA)	Pump to wheel	10,159	Kg CO ₂ e
Biodiesel	CO ₂	GHG Protocol (EPA)	Pump to wheel	0	Kg CO ₂
Biodiesel	CO ₂ (biomass)	GHG Protocol (EPA)	Pump to wheel	9460	Kg CO ₂
Diesel	CO ₂ , CH ₄ , N ₂ O	REET	Well to wheel	12,933	Kg CO ₂ e
Biodiesel	CO ₂ , CH ₄ , N ₂ O	REET	Well to wheel	2964	Kg CO ₂ e

Source: Craig et al. (2013)

calculation, from the pump to wheel or well to wheel/life cycle (see Sect. 5.3.2 for a discussion on emission scopes).

5.2.1.1 Activity-Based Calculations

Equations 5.1 and 5.2 from the IPCC guidelines assume total fuel consumption numbers are readily available or easy to estimate. Although this may be the case at the national or regional level (e.g., total oil imports, total petrol sales) or to conveyance owners that keep track of fuel purchases, this data is often not accessible to third-party logistics providers, manufacturers, or retailers who make logistics decisions. Moreover, logistics decisions are not made at the conveyance level (e.g., truck, vessel, locomotive) but rather at the shipment level (e.g., box, carton, pallet) or at another planning metric (e.g., kilogram, cubic feet, or ton).

Activity-based methods work by estimating the fuel consumed during transportation based on vehicle characteristics or combining fuel consumption data with activity data to calculate average efficiency numbers. Like fuel-based methods, these methods will be sensitive to the choice of fuel emission factors.

5.2.1.1.1 Distance-Based

The simplest approach to estimating emissions from activity data is to use the distance traveled multiplied by the average fuel consumption of the vehicle or conveyance. Together, these produce an estimate of the fuel consumed, which can then be used to estimate GHG emissions by choosing an appropriate factor, as discussed in the fuel-based methods. A number of different approaches are used in practice to estimate vehicle distance emission factors, generally varying in the level of precision they provide.

The GHG Protocol provides default emission factors per mile for a number of vehicle types, using both US and UK numbers. The emission factors for US vehicles are based on assumed average vehicle efficiency for a variety of vehicle types (heavy duty, light duty, passenger cars, motorbikes, etc.) to determine fuel consumption and

the standard factors for CO₂, CH₄, and N₂O from the US Environmental Protection Agency (EPA) discussed in the fuel-based section. Numbers in the UK are based on surveys of fuel consumption in vehicle fleets. The fuel consumption data is combined with the UK Department for Environment, Food and Rural Affairs (Defra) standard CO₂ factor to produce an emission factor considering only CO₂ on a per-kilometer basis.

Other sources have focused more on a single-mode type to provide more precise levels of emission factors. The EPA's SmartWay program (see Sect. 5.5.4) collects data from a number of different carriers. It employs a fuel-based methodology to calculate emissions from the carriers and combines this with activity data supplied by the carriers to calculate distance-based emission factors at the individual carrier level. This is then used to create a hierarchy of emission factors, from which a user can select emission factors for a mode (truck, rail, multimodal, logistics), a category within the mode (such as package, truckload/dry van, refrigerated, and others within the truck category), and finally a specific carrier within that category. A single company may have a number of different emission factors, one for each category of business for which it reported data.

The Network for Transport and Environment (NTM) program does not collect specific data from carriers but rather uses the ARTEMIS simulation tool to calculate fuel consumption for a number of different scenarios (NTM 2010). These scenarios account for different sizes of vehicles, percent loaded, road type, and driving conditions. By using these scenarios and an associated fuel-based emission factor, a range of emission factors can be calculated. In each case, the emissions are calculated using a straightforward multiplication of the distance and the vehicle-specific emission factor.

Table 5.2 shows a summary of the results of using a number of different types of factors to calculate the emissions from a 1000-mile trip

Despite little variation between emission factors for diesel fuel, the emissions estimated for a specific trip using activity-based distance methods can vary considerably. This is true even for vehicles in the same class, as the NTM factors shown for a truck + semitrailer range from 1.6 to 2.3 depending on the load factor and road type. The EPA SmartWay factors illustrate the correlation of emissions by specific carrier and type of freight.

This demonstrates important points about the precision of the emission factors used. Estimations of fuel consumed can vary considerably, and therefore even if consistent fuel-based factors are used, the results obtained from activity-based data are sensitive to the assumptions regarding vehicle operating conditions (e.g., terrain, amount of cargo, driver proficiency).

5.2.1.1.2 Weight-Distance-Based

Despite the ease of using vehicle distance factors and the availability of a wide range of emission factors, it is still inadequate for logistics analysis when using shared modes of transportation or when only the bare minimum of information is known

Table 5.2 Estimated emissions for a 1000-mile distance for various modes

Source	Emission factor	Value	Units	GHGs	Total	Units
GHG Protocol	Heavy goods vehicle-articulated-diesel-year 1960-present (US EPA)	1.722	Kg CO ₂ e/mile	CO ₂ , CH ₄ , N ₂ O	1722	Kg CO ₂ e
GHG Protocol	HGV-articulated-engine size unknown (UK Defra)	1.560	Kg CO ₂ /mile	CO ₂	1560	Kg CO ₂
GHG Protocol	HGV-rigid-engine size 7.5–17 tons-50% weight laden (UK Defra)	1.235	Kg CO ₂ /mile	CO ₂	1235	Kg CO ₂
SmartWay	Flatbed, carrier A ^a	1.700	Kg CO ₂ /mile	CO ₂	1700	Kg CO ₂
SmartWay	TL/dry van, carrier A ^b	1.750	Kg CO ₂ /mile	CO ₂	1750	Kg CO ₂
SmartWay	TL/dry van, carrier B ^b	1.550	Kg CO ₂ /mile	CO ₂	1550	Kg CO ₂
NTM	Small lorry/truck, motorway, 100% loaded	0.583	Kg CO ₂ /mile	CO ₂	583	Kg CO ₂
NTM	Lorry/truck + semitrailer, motorway, 100% loaded	2.296	Kg CO ₂ /mile	CO ₂	2296	Kg CO ₂
NTM	Lorry/truck + semitrailer, urban roads, 0% loaded	1.569	Kg CO ₂ /mile	CO ₂	1569	Kg CO ₂

Source: Craig et al. (2013)

^aSpecific carrier names and factors are available for download

^b Assumes default Defra factor for diesel fuel

about the shipment. In the first case, the emissions of the vehicle as a whole are not of concern, rather the share of emissions related to a specific amount of goods shipped. In the second case, the logistics decision-maker may not know the specific vehicle, the loading factors, or the exact route used.

In these situations, weight-distance methods are generally used, though in some cases a volume-distance method may be more appropriate. Emission factors for weight-distance methods are generally expressed in terms of ton-miles of goods moved (or perhaps TEU-miles for ocean containers where volume may be more important than weight). These methods provide a quick and easy method of calculating emissions, relying only on the weight of the goods shipped, the distance, and a general knowledge of the mode of transport used. They are also useful in comparing between modes, when efficiency is measured not just in the amount of emissions produced but the total amount of goods moved.

The GHG Protocol provides emission factors in terms of ton-miles for a variety of transportation modes, using factors derived from both the EPA and Defra. These factors introduce another layer of assumptions beyond those of fuel-based and vehicle-distance-based methods, as now the factors must include assumptions regarding the total amount of goods on the vehicle. This can lead to a wide range of emission factors, depending on the assumptions used. This is illustrated in Table 5.3, where emission factors for different modes and types of transportation are compared for a shipment consisting of 10,000 short ton-miles (equivalent to a 10-t shipment being moved 1000 miles).

Table 5.3 also shows the wide variation not just between modes, in which ocean shipping may be as much as 200 times more efficient than air transport but also between sources. The EPA's numbers, for instance, are based on high-level data and do not distinguish between types of transport within a mode. Thus, there is no distinction between heavy-duty trucks or light-duty vehicles within road transport or between large container ships and small tankers in watercraft. This is in contrast to the Defra numbers that are generated at a greater level of precision and show the range of values that can exist between different types of transport.

It is important to note that some organizations, such as NTM or EcoTransIT, provide calculators with the ability to adjust some of the assumptions behind weight-distance-based emission factors, such as vehicle loading factors, to provide more realistic estimates. This fine-tuning requires more data, which is not always available to logistics decision-makers. The EPA SmartWay program also collects similar data by carrier, providing another layer of detail to emission factors. Finally, some logistics operators and transportation companies, such as DHL, FedEx, or Maersk, use their internal proprietary systems and their transportation network information (e.g., total weight or cube moved on a particular transportation lane during a year) and combine it with fuel consumption records of the vehicles or vessels, to provide lane-specific distance-weight emission factors and, in some cases, shipper-specific factors for dedicated customers. These factors are often updated annually, although there is a trend toward more frequent reporting. Even these organizations rely on emission factors like the ones listed in Table 5.3, because

Table 5.3 CO₂ emissions for a 10,000 short ton-mile shipment across multiple modes

Source	Emission factor	Value	Units	GHGs	Total (kg CO ₂)
GHG Protocol	Air-long haul (US EPA)	1.527	Kg CO ₂ /ton-mile	CO ₂	15,270
GHG Protocol	Air-long haul (UK Defra)	0.346	Kg CO ₂ /ton-mile	CO ₂	3460
GHG Protocol	Air-domestic (US EPA)	1.527	Kg CO ₂ /ton-mile	CO ₂	15,270
GHG Protocol	Air-domestic (UK Defra)	1.105	Kg CO ₂ /ton-mile	CO ₂	11,050
GHG Protocol	Watercraft-shipping-large container vessel (20,000 tons deadweight) (US EPA)	0.048	Kg CO ₂ /ton-mile	CO ₂	480
GHG Protocol	Watercraft-shipping-large container vessel (20,000 tons deadweight) (UK Defra)	0.007	Kg CO ₂ /ton-mile	CO ₂	70
GHG Protocol	Watercraft-shipping-small tanker (844 tons deadweight) (US EPA)	0.048	Kg CO ₂ /ton-mile	CO ₂	480
GHG Protocol	Watercraft-shipping-small tanker (844 tons deadweight) (UK Defra)	0.019	Kg CO ₂ /ton-mile	CO ₂	190
GHG Protocol	Road vehicle-HGV-articulated-engine size >33 tons (US EPA)	0.297	Kg CO ₂ /ton-mile	CO ₂	2970
GHG Protocol	Road vehicle-HGV-articulated-engine size >33 tons (UK Defra)	0.049	Kg CO ₂ /ton-mile	CO ₂	490
GHG Protocol	Road vehicle-light goods vehicle-petrol-engine size 1.305–1.74 tons (US EPA)	0.297	Kg CO ₂ /ton-mile	CO ₂	2970
GHG Protocol	Road vehicle-light goods vehicle-petrol-engine size 1.305–1.74 tons (UK Defra)	0.462	Kg CO ₂ /ton-mile	CO ₂	4620
GHG Protocol	Rail (US EPA)	0.025	Kg CO ₂ /ton-mile	CO ₂	250
GHG Protocol	Rail (UK Defra)	0.016	Kg CO ₂ /ton-mile	CO ₂	160

Source: Craig et al. (2013)

their shipments may move between trucks, ships, and airplanes for which fuel records are not available or that belong to third parties with less detailed information.

5.2.1.2 Mode-Specific Adjustments

All of the GHG calculation approaches and methods discussed above are applicable to all modes of transportation (road, rail, waterborne, and air). Most salient differences, besides variations in engine technology and type of fuels, are related to the GHG gases included, the quality of data, the allocation of emissions to freight due to capacity sharing (e.g., same vehicle moving people and boxes), and strategies to overcome data limitations.

5.2.1.2.1 Rail

The most important variation in rail has to do with the variability of the number of railcars (empty and full) being pulled by a single locomotive. In theory, the amount of fuel consumed during any journey or leg can be tracked and allocated to the cargo being hauled on that specific leg, thereby creating movement-specific factors. In practice, however, rail operators plan their movements and balance on a network perspective. Thus, it only makes sense to look at emissions from a series of interconnected networks or services and to measure the total amount of fuel and cargo moved through that rail network, as opposed to individual legs. These calculations are often done annually but could also be done on a monthly or quarterly basis, aligned with rail operator planning cycles.

5.2.1.2.2 Waterborne Navigation

Some waterborne navigation providers, such as barges, operate in a similar fashion as long-haul vehicles, albeit through a smaller transportation network: rivers and canals. For these providers, trip-based estimations are sensible: measuring the total fuel consumed between origin and destination and allocating the emissions to the amount of cargo loaded. Because adjustments for empty journeys need to be added and assigned to various trips, emission factors that span more than one journey over a time horizon are often needed.

For large ocean-going cargo ships, the situation is very different. Unlike river waterways, ocean-shipping companies plan their transportation networks by trade lanes between continents and subcontinents, stringing together multiple trade lanes to provide regular service to various ports. Thus, similar to rail transportation, even though it is mathematically possible, it doesn't make sense to compute GHG emissions by looking at port-to-port distances without full understanding of the overall trade lane dynamics. In addition, given the size of the vessels and the relatively few number of carriers (compared to road transport), the industry has been

developing joint efforts to calculate emission factors by trade lane in a homogeneous way. The Clean Cargo Working Group has been collecting data in collaboration with major shipping companies to provide a homogeneous calculation approach of emissions by trade lane and ship assignment, if possible.

An additional feature of some waterborne navigation has to do with the relevant unit of measures used by shipping companies to determine the amount of freight being transported: the container. After the invention and wide adoption of the container, waterborne navigation was transformed to leverage the economic and efficiency advantages of the container. As a consequence, all planning and pricing decisions are made in TEU (twenty-foot equivalent units), a volumetric unit of measure equivalent to the total cube of a standard 20 feet container. Although cargo owners may know the total weight inside a particular container, shipping companies often only know (and care) about the TEU. Thus, emission factors computed by waterborne transportation are often originally computed in kilograms of CO₂-e per TEU-kilometer and then converted into ton-kilometers by using a pre-agreed conversion factor.

5.2.1.2.3 Aviation

Sources of emissions for aviation are all civil commercial airplanes, including general aviation such as agricultural aircraft, private jets, and helicopters. The fuel-based methodology again uses only fuel consumption data and average emission factors to estimate emissions and is suitable for aircraft using aviation gasoline or when operational data for jet-fueled vehicles are not available. A fuel-based approach can also be estimated by calculating emissions separately for the cruise phase of a flight and the landing/takeoff (LTO) phase. This requires knowing the number of LTOs and separating the fuel consumed during this phase from the cruise phase, but it allows for using emission factors that capture differences in emissions, specially CH₄ and N₂O, during these phases.

Distance-based methods can be based on origin-destination (OD) data or full-flight trajectory information. The OD approach accounts for different flight distances, which changes the relative impact of the LTO phase compared to the cruise phase. The full-flight trajectory model uses aircraft—and engine-specific performance information over the entire flight, requiring engine performance modeling.

An additional complexity in aviation has to do with the allocation of emissions between people and freight, because they share the same airplane when cargo is loaded onto commercial flights. The most accurate way to allocate emissions is to use the ratio of weight used by passengers (and their bags) vs. the weight of freight, because weight is what determines the amount of fuel consumed during the flight. However, airlines do not plan routes using these criteria; instead, they evaluate the economics of each flight and the revenue from the various services they offer. Thus, emissions could be allocated based on the revenue of passenger vs. freight as a proxy to the planning approach. Even this approach has further complications because first class, business, and economy fares are sold at different rates on every

flight as part of yield management strategy, varying the revenue profile of each flight and “underutilizing” the maximum weight potential of a flight. A third approach would be to allocate a fixed share of emissions to freight on a flight, recognizing that airlines often plan and balance their network using revenue targets. The EN 16258 (see Sect. 5.3.3) has favored this latter approach, recommending a factor between 70% and 80% of emissions to be allocated to freight on commercial flight, regardless of actual load capacity or revenue. This number, although arbitrary, is a result of a consultation process with experts from academia and industry trying to balance accuracy of emissions with practical business matters of data collection and consistency in reporting.

5.2.1.3 Carbon Footprint Calculations in Transportation: A Primer

As mentioned at the beginning of this section, the most accurate method for calculation carbon footprints of transportation emissions is to use fuel records of the conveyance (see Eq. 5.1).

However, since most transportation activities often involve multiple organizations (e.g., shipper and carrier) and may further involve other intermediaries such as freight forwarders or 3PLs, weight-distance activity-based calculations are more commonly used in practice (see Sect. 5.2.1.1.2). The carbon footprint calculation using this approach is often estimated as a function of the shipment weight (w) or volume (v), the distance (d), and a mode-specific emission factor (EF). The most basic relationship is multiplicative as follows:

$$e(d, w) = d \times w \times EF \text{ or } e(d, v) = d \times v \times EF.$$

The shipment weight is the gross weight of the product being transported (including all primary and secondary packaging). This information is often well known by the shipper. The distance is the total over-the-road, over-the-air, over-the-track, or over-the-waterway distance traveled by the shipment. This number may not always be accurate or easily available for the shipper, but it is often known by the carrier or asset owner or can be approximated by using over-the-air distances multiplied by an adjustment factor. Such approximations should be avoided (if possible) since they add another level of uncertainty to the calculation.

The final component, the emission factor EF , is the most critical element of this computation. As discussed earlier in this section, there are multiple sources that publish values that are commonly used by practitioners (see Table 5.2). Ideally, these emission factors should be gathered directly from fuel consumption records from the carrier or vessel operator (see Sect. 5.5.4) but are most commonly a result of carrier surveys, econometric models, or engine fuel consumption modeling. Fuel consumption models allow for more accurate functional forms of $e(d, w)$.

The modeling and data collection undertaken by NTM is one of the most comprehensive and detailed methodologies available to estimate emission factors of transportation. Based on the excellent summary by Hoen et al. (2014a) of

NTM recommended calculations, the following sections summarize functional forms of carbon footprint calculations for different modes of transportation. For all calculations, the resulting calculation emissions are expressed in kilograms of CO₂. Distance d is expressed in kilometers, weight w in kilograms, and volume v in m^3 .

5.2.1.3.1 Air Transport

Emissions of air transportation can be estimated as follows:

$$e_{air}(d, w) = (E_1 + \epsilon_1 \cdot d) * \frac{w}{(W_1 \cdot \lambda_1)}$$

where

W_1 is the aircraft maximum payload in kg.

λ_1 is the aircraft payload utilization % for the specific trip.

E_1 are the emissions (kg of CO₂) corresponding to takeoff and landing. These emissions are a function of the actual aircraft payload $W_1 \cdot \lambda_1$.

ϵ_1 is kilograms of CO₂ per kilometer (kg of CO₂/km).

For a Boeing 757-200SF, the maximum payload is W_1 is 29,029 kg. When fully loaded (e.g., $\lambda_1 = 100\%$), $E_1 = 4,531.182$ and $\epsilon_1 = 15.363$. For a payload of $\lambda_1 = 75\%$ of maximum capacity, $E_1 = 4041.709$ and $\epsilon_1 = 15.351$. Thus, the share of emissions associated with moving a 500-kg shipment for 1000 km will be 342.66 kg CO₂ in a fully loaded Boeing 757-200SF and 445.36 kg CO₂ if the aircraft will be loaded at 75% payload capacity.

As mentioned earlier, aircraft payload capacity between passenger and cargo needs to be adequately accounted for. Also, air shipments are often priced volumetrically. In that case, $w = v \cdot \rho$, where ρ is the shipment density. This density will vary by product and may also be adjusted for pricing purposes.

5.2.1.3.2 Road Transport

Emissions of road transportation can be estimated as follows:

$$e_{road}(d, w) = \epsilon_2 \cdot d \cdot \frac{w}{(W_2 \cdot \lambda_2)}$$

where

W_2 is the vehicle maximum payload in kg.

λ_2 is the vehicle payload utilization % for the specific trip.

E_2 are the fixed emissions (kg of CO₂) of moving an empty tractor and trailer.

ϵ_2 is kilograms of CO₂ per kilometer (kg of CO₂/km).

Commonly used transport vehicles (tractor + trailer) have a maximum payload W_2 of 26,000 kg. To estimate ϵ_2 , NTM recommends taking into account fuel efficiency, type of road, load factor, and terrain slope. NTM estimates that diesel consumption for an unloaded freight vehicle with a payload capacity of 26,000 kg is 0.226, 0.230, and 0.288 liters per km for highway, rural, and city environments, respectively. At the other extreme, a fully loaded vehicle consumes 0.360, 0.396, and 0.504 liters per km in highway, rural, and city roads. Other load factors may be linearly interpolated between these numbers. Thus, for a load factor of 70%, fuel consumption will be 0.3198, 0.3462, and 0.4392 liters per km, for highway, rural, and city roads, respectively. Assuming diesel emissions are 2.621 kg CO₂ per liter, ϵ_2 for a $\lambda_2 = 70\%$ loaded vehicle will be given by $2.621 \cdot 0.3198 = 0.8382$ kg CO₂ per km. An additional adjustment to fuel consumption may be applied to take into account the steepness of the terrain. For instance, Hoen et al. (2014a) estimate a European-wide adjustment of an additional 5% to account to terrain. Thus, $\epsilon_2 = 0.8382 \cdot 1.05 = 5.0.8801$.

Therefore, the share of emissions associated with moving a 500-kg shipment for 1000 km will be 11.85 kg CO₂ in a 70% utilized 26,000 kg diesel-powered vehicle. Note that for road transport, the allocation of empty miles has a noticeable impact on final shipment emission calculations.

5.2.1.3.3 Rail Transport

Since only a handful of national rail operators provide rail transport service by country, it is often most reliable to obtain the emission factor directly from rail transport companies. These factors take into account overall locomotive efficiency, including electric- and diesel-powered technology, as well as required boxcar repositioning throughout the network. Thus, emissions can be simply calculated as follows:

$$e_{rail}(d, w) = \epsilon_3 \cdot d \cdot w$$

where

ϵ_3 is kilograms of CO₂ per kilometer (kg of CO₂/kg-km).

The EPA estimates this number to be $1.713 \cdot 10^{-5}$ kg of CO₂ per kg-km for the US rail network. Hoen et al. (2014a) present a detailed derivation of rail emission factor for Europe, which takes into account a combination of electric and diesel power distance and hilly terrain. The resulting value was $2.223 \cdot 10^{-5}$ kg of CO₂ per kg-km.

5.2.1.3.4 Water Transport

Emissions of inland water transportation can be estimated as follows:

$$e_{water}(d, w) = FC_4 \cdot FE_4 \cdot d \cdot \frac{w}{(W_4 \cdot \lambda_4)}$$

where

W_4 is the total vessel capacity in kg, TEU, or meters (for RORO vessels).

λ_4 is the vessel payload utilization % for the specific trip.

FC_4 is the vehicle fuel consumption in liters per km.

FE_4 are the fuel emissions in kg of CO₂ per liter.

The cargo capacity of an inland cargo vessel is 3,840,000 kg. Inland cargo vessels often have low utilization levels, close to 50%. Vessel fuel consumption is in the order of 0.007 tons of diesel per km and diesel emissions of approximately 3178 kg of CO₂ per ton of diesel. Using these parameters, the share of emissions associated with moving a 500-kg shipment for 1000 km via inland waterways will be 5.79 kg CO₂.

In the case of ocean shipping, it is often advised to use carrier-specific emission factors, such as the ones published by the Clean Cargo Working Group (CCWG), although similar calculations as the ones used above can be estimated by adjusting the fuel type used (often bunker fuel oil).

5.2.2 Pollution

Pollution is the introduction of a substance—solid, liquid, or gas—into a system that can have adverse consequences on humans or the natural ecosystem. In the case of logistics and transportation, the most important environmental impacts are due to air and water pollution generated during the operation of trucks, airplanes, locomotives, and vessels. Unlike GHG emissions that have global effects, pollution impacts tend to be local to cities, ports, trade lanes, or freight corridors, although pollutants can also travel long distances and have global effects.

5.2.2.1 Air Pollution

The use of internal combustion engines in trucks, airplanes, ships, and locomotive engines that move freight is the major source of air pollution. There are six common air pollutants, also known as “criteria pollutants” in combustion: particle pollution (often referred to as particulate matter), ground-level ozone, carbon monoxide, sulfur oxides, nitrogen oxides, and lead (EPA 2019):

- Ground-level ozone is not emitted directly into the air, but it is created by chemical reactions between oxides of nitrogen (NO_x) and volatile organic compounds (VOC) in the presence of sunlight.
- Particulate matter (PM) is a mixture of extremely small particles and liquid droplets. PM is made up of a number of components, including acids (such as nitrates and sulfates), organic chemicals, metals, and soil or dust particles. The size of particles is directly linked to their potential for causing health problems. The EPA regulations focus on particles that are $10\text{ }\mu\text{m}$ (PM10) in diameter or smaller because those generally pass through the throat and nose and enter the lungs.
- Carbon monoxide (CO) is a colorless, odorless gas emitted from combustion processes. Nationally, and particularly in urban areas, the majority of CO emissions to ambient air come from transportation activities (both passenger and freight). CO can cause harmful health effects by reducing oxygen delivery to the body's organs (like the heart and brain) and tissues. At extremely high levels, CO can cause death.
- Nitrogen oxides (NO_x) are a family of seven compounds, of which NO_2 is the most prevalent form. About 50% of all NO_x come from mobile sources including automobiles, trucks, and vessels (EPA 2019). They are generated during the combustion process of engines as a function of the ratio of fuel and oxygen. In addition to contributing to the formation of ground-level ozone and fine particle pollution, NO_x are linked with a number of adverse effects on the respiratory system. The amount of NO_x can be controlled through several means: engine design, regulating the oxygen and fuel mix, maintaining optimal temperature levels within the engine, changing fuel types, or adding a catalytic converter. All of these actions have an impact on fuel economy, sometimes positive or negative, and they are very dependent on individual engine configurations (EPA 2019).
- Sulfur oxides (SO_x) are highly reactive gases of which sulfur dioxide (SO_2) is the most prevalent form in transportation. SO_x is linked with a number of adverse effects on the respiratory system. The largest sources of SO_x emissions come from fossil fuel combustion at power plants (73%) and other industrial facilities (20%). Smaller sources of SO_x emissions include industrial processes such as extracting metal from ore and burning of high-sulfur-containing fuels by locomotives, large ships, and non-road equipment. Although maritime transportation represents a small share of global SO_x emissions, the emissions tend to accumulate in higher concentrations near ports and then travel to neighboring population centers. SO_x emissions are increasingly being regulated across Europe and are part of the focus areas of the maritime industry as a whole (IMO 2021).
- Lead is a naturally occurring element that can be harmful to humans when ingested or inhaled. Lead poisoning is particularly detrimental to the neurological development of children. The major sources of lead emissions have historically been from fuels in on-road motor vehicles (such as cars and trucks) and industrial sources. As a result of US and EU regulatory efforts to remove lead from on-road motor vehicle gasoline, emissions of lead from the transportation sector dramatically declined by 95% between 1980 and 1999, and levels of lead in

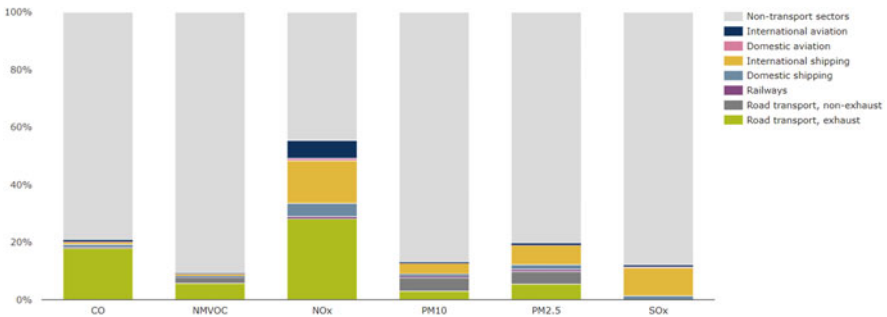


Fig. 5.2 Contribution of the transport sector to total air pollution. Source: European Environment Agency (2020a, b)

Table 5.4 Emission rates (grams per mile) for heavy-duty diesel trucks

Speed (mph)	NO _x	Sox	PM ₁₀
35	14.76	0.576	1.527
40	15.16	0.576	1.527
45	16.12	0.576	1.527
50	17.77	0.576	1.527
55	20.29	0.576	1.527

Source: Forkenbrock (1999)

the air decreased by 94% between 1980 and 1999. The major sources of lead emissions to the air today are ore and metal processing and piston-engine aircraft operating on leaded aviation gasoline.

Reliable global figures are not available, but the European Environment Agency (see Fig. 5.2) estimates that international shipping represents 10% of SO_x emissions, 15% of NO_x emissions, 7% of PM_{2.5}, and 4% of PM₁₀. Road transport (both passenger and freight) represents 29% of NO_x emissions, 10% of PM_{2.5}, and 7% PM₁₀. Aviation mainly contributes to NO_x emissions (7%), while railways have negligible emissions compared to other sources (EEA 2020a, b).

The levels of air pollution generated by engines are heavily dependent on the vehicle engine technology as well as operational conditions such as speed, road geometry, wind speeds, and altitude. Table 5.2 compares emission rates (grams per mile) of a diesel heavy-duty vehicle. PM and SO_x emissions don't vary by speed, but NO_x emissions increase significantly at higher speeds (Table 5.4).

It is important to highlight that noticeable progress has been made in reducing all sources of air pollutants in the transport sector since 1990 (see Fig. 5.3).

An important consideration related to air pollution is the trade-off between GHG emissions, fuel efficiency, and cost. A comprehensive field study commissioned by Defra in the UK illustrates this complexity. Figure 5.4 shows the overall progress achieved through legislation: a steady reduction of the ratio of NO_x vs. CO₂ emissions. Although diesel fuels have achieved significant reductions in absolute

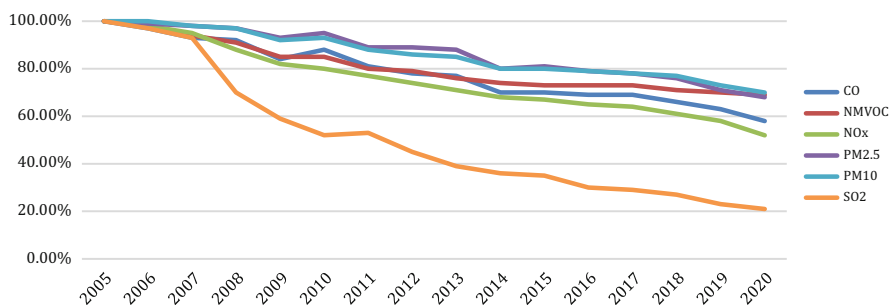


Fig. 5.3 2005–2020 trends in EU emissions of air pollutants as percentages of 2005 levels. Source: European Environment Agency (2020a, b)

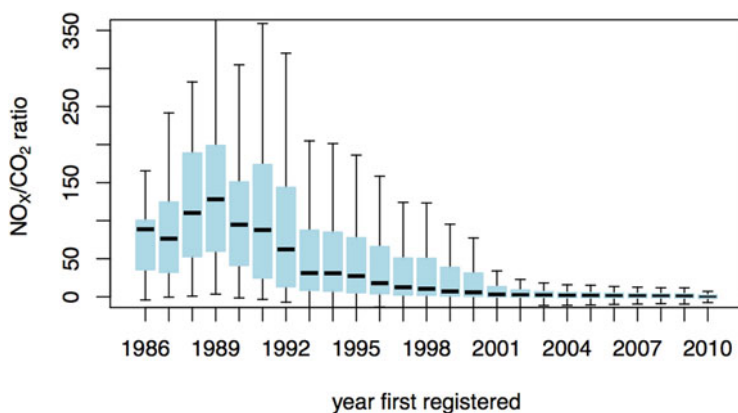


Fig. 5.4 Box plot of the volume ratio of NO_x/CO₂ for petrol cars. Source: Carslaw et al. (2011)

CO₂ emissions and are considered a “greener” alternative, they have not improved at the same rate of NO_x reductions compared to petrol (Fig. 5.5).

5.2.2.2 Water Pollution

Water pollution occurs during waterborne transportation due to four main causes:

- Release of oil and chemicals through accidental spills and operational discharges.
- Release of biocides from toxic chemicals used in antifouling paints. No estimates are available for the impact of these chemicals.
- Dumping of waste such as garbage and sewage.
- Transfer of invasive aquatic species due to ballast water.

Table 5.5 summarizes the total estimated water pollution (in tons per year) from seabed activities. Ships and small craft account for more than 40% of seabed pollution, while coastal facilities, including port activity, add another 9.2% of

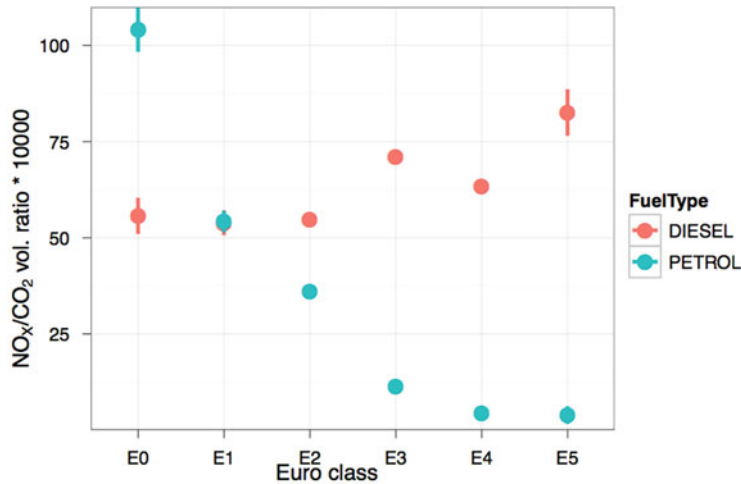


Fig. 5.5 Volume ratio of estimated NO_x/CO₂ for petrol and diesel cars by Euro classification. The error bars show the 95% confidence intervals in the mean. Source: Carslaw et al. (2011)

Table 5.5 Distribution of pollution from seabed activities, 2011

Seabed activity	Tons/year	%
Ships	457,000	36.7%
Offshore explorationand production	20,000	1.6%
Coastal facilities	115,000	9.2%
Small craft activity	53,000	4.3%
Natural seeps	600,000	48.2%
Unidentified sources	200	0.0%
Total	1,245,200	100%

Source: IMO (2012)

pollution. Any efforts taken to reduce the amount of this pollution, often through better technology and environmentally aware operations, have a direct impact on these figures.

Given the economic and political importance of waterborne transportation—90% of global trade is transported via international shipping (IMO 2015)—multi-stakeholder action is the main strategy to address water pollution. The MARPOL Convention signed in 1973 remains the most important international treaty instrument covering the prevention of pollution by ships. It sets out regulations dealing with pollution from ships by oil, by noxious liquid substances carried in bulk, by harmful substances carried by sea in packaged form, by sewage, by garbage, and with the prevention of air pollution from ships.

Table 5.6 Noise levels of common activities

Activity	Noise level
Whisper	30 dB
Normal conversation/laughter	50–65 dB
Vacuum cleaner at 10 feet	70 dB
Washing machine/dishwasher	78 dB
Midtown Manhattan traffic noise	70–85 dB
Motorcycle	88 dB
Train	100 dB
Jackhammer/power saw	85–90 dB
Thunderclap	110 dB
Stereo/BoomBox	110–120 dB
Nearby jet takeoff	130 dB

Source: NYCDEP (2008)

5.2.3 Noise and Vibration

The goal of establishing acceptable noise levels is to avoid hearing loss in people over their lifespan, as well as to allow for a comfortable environment to work and rest. In 1974, the EPA determined that a 24-h exposure level of 70 decibels (dB) be the threshold that will prevent any measurable hearing loss over a lifetime. Levels of 55 dB outdoors and 45 dB indoors were considered as acceptable levels for normal activity. These levels are not peak levels but 8–24-h averages.

Table 5.6 presents some reference activities and their noise levels. Noise levels related to freight transportation activities (traffic 70 dB, trains 100 dB, and airplanes 130 dB) are above the recommended 70-dB threshold level. This limits the amount of time that freight activities should be allowed near heavily population centers. Localities often require stricter noise levels after 6 pm and before 6 am in residential areas to further mitigate disruption to audible quality of life.

5.2.4 Packaging Waste

Packaging is used to sell, inform, contain, protect, preserve, and transport products (Soroka 1999). After product use, all packaging joins the waste stream.

- There are three main types of packaging (Saphire 1994):
- *Manufacturer-provided packaging.* This is the primary packaging that protects and preserves the product. In some cases, this packaging also informs and helps to sell the product contained within. As a customer decides which product to purchase, the packaging has the ability to draw the customer in with its design, image, and attractiveness, regardless of the quality and necessity of a product (Paine 2002).

Table 5.7 Generation and recovery of containers and packaging

Container and packaging material	Weight generated (thousand tons)	Weight recovered (thousand tons)	Recovery of generation (%)
Steel	2210	1630	73.7%
Aluminum	1920	0,670	34.9
Glass	9790	3060	31.2%
Paper and cardboard	41,90	33,89	80.8%
Plastics	14,530	1980	13.6%
Wood	11,530	3100	26.9
Other materials	340	Negligible	Negligible
Total	82,220	44,330	53.9%

Source: US EPA (2018)

- *Transport packaging or secondary packaging.* This type of packaging is used for the sole purpose of moving product around. Most commonly, it is used for bulk handling of product, usually in pallet sizes to facilitate the easy transfer from warehouse to truck or container for shipment across land, air, or sea. Its main function is to protect the contents within from damage from the elements or rough handling.
- *Parcel packaging or tertiary packaging.* This is used mainly to group primary packages together. It is most frequently used in the retail delivery industry to aggregate customer orders into one box, to facilitate easy delivery through the fulfillment system.

Logistics and transportation activities have direct influence on the design, use, and disposal of secondary and tertiary packaging. Any unnecessary levels or inadequately disposed packaging is an additional source of waste.

Although detailed statistics are not available, product containers and packaging represent 29% of the 250 million tons of waste generated in 2010 (EPA 2015). Approximately 49% of this waste is recovered (see Table 5.7), which leaves a 51% opportunity to either reduce packaging use or make sure that it reaches the right recycling facilities.

5.3 Measuring Green Logistics Impacts

Logistics decisions are metrics-driven, which means that green logistics models and initiatives require having the right measurements of the various environmental impacts.

Pollution, noise, vibration, and waste measurements are technical in nature and can be estimated through specialized equipment. For example, the EPA or the European Union standards (see Table 5.1) are enforced by subjecting technologies to lab and road tests under standard conditions. Driving, weather, terrain, congestion,

and operational conditions can dramatically change the actual environmental impact of freight operations. In a detailed study of drayage trucks in the port of Genoa, Zamboni et al. (2015) were able to observe differences of more than 50% in total fuel consumption and NO_x and SO_x emissions under a variety of speeds and stop patterns. Environmental analysis of logistics operations needs to be very aware of the various assumptions underlying commonly used emission factors. This is particularly important when estimating GHG emissions.

5.3.1 GHG Emission Measurement in Logistics

As discussed in Sect. 5.2.1, there are important variations in emission factors across sources (see Table 5.3). For instance, depending on the assumed (or observed) utilization factor of the conveyance and the granularity of data available (e.g., surveys or fuel records), the average fuel consumption per ton of cargo transported will vary significantly. The table below includes some of the emission factors included in the GHG Protocol. Note that the EPA reference numbers are the same for trucks of various engine sizes, while the Defra numbers vary dramatically between large (over 33 t) and smaller trucks (1.3–1.7 t). This difference is due to the level of detail of data collected by these two agencies at the moment of publication of the GHG Protocol: the EPA was using sector-level aggregated data, while Defra had access to vehicle-level surveys. Moreover, these numbers are regularly challenged and updated as more and better data become available across the logistics sector (Table 5.8).

Another complexity, especially relevant to GHG measurements, is what is and what is not included in the emission factors and calculations.

Table 5.8 Selected emission factors from the GHG Protocol

Emission factor	Value	Units
Road vehicle-HGV-articulated-engine size >33 tons (US EPA)	0.297	Kg CO ₂ /ton-mile
Road vehicle-HGV-articulated-engine size >33 tons (UK Defra)	0.049	Kg CO ₂ /ton-mile
Road vehicle-light goods vehicle-petrol-engine size 1.305–1.74 tons (US EPA)	0.297	Kg CO ₂ /ton-mile
Road vehicle-light goods vehicle-petrol-engine size 1.305–1.74 tons (UK Defra)	0.462	Kg CO ₂ /ton-mile

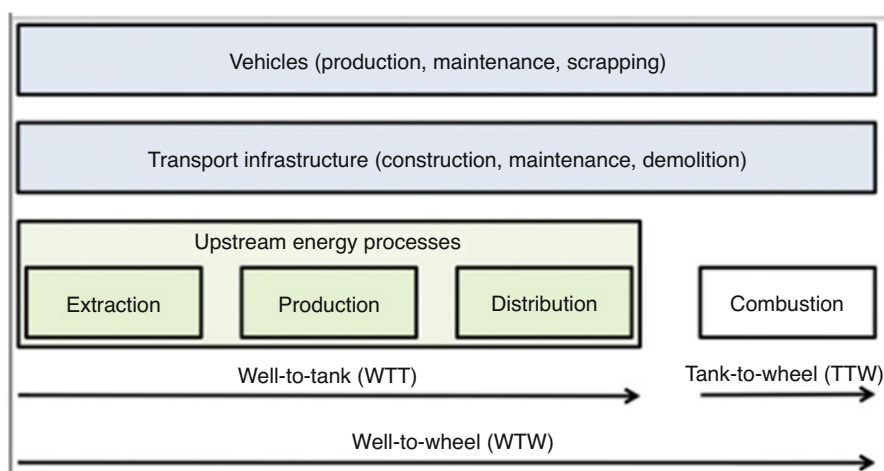


Fig. 5.6 Life cycle phases of transport. *Source:* VTT as reported by VTT (2012)

5.3.2 GHG Standards and Scopes¹

There are three types of standards that cover GHG impact estimations. If the intent of the measurement is an absolute quantity for a whole company, this is known as *corporate carbon footprinting*. The most widely known and adopted standard is the GHG Protocol followed by the ISO 14064. If the intent is to measure the GHG impact of an individual product, often from cradle to grave, it is known as *product carbon footprinting*. The PAS-2050, the ISO 14040, and the GHG Protocol Life Cycle Accounting and Reporting Standards are well-known references. Finally and most recently, the EN 16258 (VTT 2012) standard for quantifying greenhouse gas emissions from freight focuses specifically on carbon footprinting in the transportation sector.

All of these standards use emission factors and variations of the equations presented in Sect. 5.2.1. One of the main differences, besides the goal itself, is the scope of the GHG calculations. Figure 5.6 shows all the life cycle sources of GHG emissions of transportation services, from cradle to grave. These emissions are generated at different moments in time and points in space and allocated throughout the delivery of the service.

The first layer of emissions is all the GHG emissions generated during the manufacturing and maintenance of the trucks, airplanes, or boats. These emissions include the energy used in the assembly plants, the extraction of raw materials,

¹ The GHG Protocol Corporate Standard also includes a scope definition. Although conceptually related, it does not correspond to the scope definition when applied to logistics activities. See Chap. 3 by Boukherroub et al. (2024) for an in-depth discussion of carbon footprinting, including more on the GHG Protocol Scope definitions.

and the operation of maintenance operations. Next are all the emissions related to the construction and maintenance of the transportation infrastructure, including roadways, ports, airports, and intermodal terminals. Some of this infrastructure is shared with other services (e.g., roads are shared with bicycles and personal vehicles, ports are shared with government and military operations) and needs to be properly allocated.

In order to be able to do a complete calculation of the impact of freight transportation, some share of the vehicle and infrastructure emissions should be added to each logistic operation. This is a fairly complex and uncertain calculation that requires very large amounts of data and assumptions. With the exception of life cycle analysis or product carbon footprint calculations, most of these emissions are not included in GHG calculations in the logistics sector.

This leaves emissions from “well to wheel” (WTW): these are all the emissions related to the extraction, production, and distribution of fuels used during transportation operations, up to a point when the fuel is placed in the vehicle—“well to tank” (WTT)—plus all emissions generated during the combustion of this fuel during transport operations, “tank to wheel” (TTW). The EN 16258 explicitly requires that all transport GHG calculations include WTW emissions. Other standards, including the GHG corporate protocol, only required TTW emissions, although WTW were encouraged. Similarly, the EPA SmartWay program initially only required TTW emissions and is now expanding to WTW emissions. Table 5.9 shows the emission factor differences between TTW and WTW by fuel.

5.3.3 *GHG Allocation*

The other source of complexity is the allocation of the emissions from a corporate or individual route level to an individual shipment. The GHG Protocol and the EPA program favor “simple” activity-based approaches: First, compute all the emissions of a company or service using, for instance, fuel records. Then calculate the total distance (including empty miles) and weight moved during the transportation activities. Once WTW or TTW emission factors are applied to fuel consumption data, they can be divided by the total ton-mile of the logistics or transportation provider to estimate an emission factor per ton-mile. This approach ignores the fact that individual routes or services may have different efficiency levels, or that specific vehicles may have different performance metrics, in favor of a simple and consistent calculation that can be easily adopted by many organizations. The EPA SmartWay program does allow for tracking emissions by the type of fleet (e.g., flatbed trucks vs. drayage trucks), but it still recommends an aggregate approach for computing emission factors.

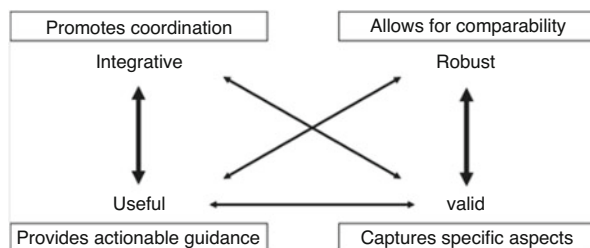
The EN 16258, on the other hand, favors a detailed approach. It recommends using the product of the weight of the consignment and the actual distance traveled—i.e., the transport capacity measured in ton-kilometers—as the allocation parameter. It aims at providing as accurate as possible assignment of emission

Table 5.9 GHG emissions for various types of fuels (kg of CO₂ equivalents per L) in accordance with EN 16258

	TTW	WTW
	kg CO₂e/L	kg CO₂e/L
Petrol	2.42	2.88
Ethanol	0.00	1.24
Petrol E5 (5 vol.% Ethanol)	2.30	2.80
Petrol E10 (10 vol.% Ethanol)	2.18	2.72
Diesel	2.67	3.24
Biodiesel	0.00	1.92
Diesel D5 (5 vol.-% biofuel)	2.54	3.17
Diesel D7 (7 vol.-% biofuel)	2.48	3.15
Compressed natural gas	2.68*	3.07*
Liquefied petroleum gas	1.70	1.90
Jet kerosene	2.54	3.10
Heavy fuel oil (HFO)	3.05	3.31
Marine diesel oil (MDO)	2.92	3.53
Marine gas oil (MGO)	2.88	3.49
*Per Kg		

factors to individual packages, taking into account logistics network configurations such as the relative location of the warehouse with respect to customers in a delivery route. Variations of this calculation are allowed in the standard, based on the quality of the information and type of service. An in-depth discussion of the EN 16258 standard is beyond this chapter, but this requires consistently collecting data across every transportation route, including shipment information, sequence, and distance. This is a gargantuan task, especially since logistics operations often include multiple providers with various levels of sophistication. Nevertheless, for large organizations like UPS, DHL, FedEx, or Maersk, the EN 16258 does provide a framework to develop information systems that can provide very accurate shipment-level allocation of GHG emissions.

Fig. 5.7 Trade-offs between criteria. *Source:* Caplice and Sheffi (1994)



5.3.4 GHG Metric Trade-Offs

As discussed on this section, GHG metrics will vary due to assumptions, scope, or data availability. Standards can guide organizations' choices, but designing metrics that excel in all dimensions is not practically possible. Instead, firms must choose metrics that trade off between certain criteria. Two of the primary trade-offs are between integrative and useful metrics and between robust and valid metrics (Caplice and Sheffi 1994) (Fig. 5.7).

Integrative metrics promote coordination across functions, while useful metrics are easily understood and provide managers with direct guidance. Providing managers with actionable guidance requires a level of specificity that makes promoting coordination across functions difficult. In this sense, measuring the carbon footprint of transportation is a useful metric, because it provides guidance on one specific aspect but not across functions. As such, it must be incorporated as one metric in an entire performance measurement system that covers both environmental and non-environmental aspects across the functions of the supply chain.

The other trade-off is between a robust metric that allows for comparability and a valid metric that captures specific aspects. A valid metric, such as the ones favored by the EN16258, provides help with making a specific decision but is less suitable to external uses where it might be compared with similar metrics for other organizations, unlike the GHG Protocol or EPA approaches.

5.4 Green Logistics Strategies

As discussed in the previous sections, logistics activities, though integral to the economic and social development, negatively impact the environment on multiple dimensions. The goal of *green logistics* is to mitigate the environmental impact of logistics-related activities.

As governments and companies have increased their focus on green logistics, numerous "best practices" and frameworks have been proposed (Craig et al. 2013). However, as the impacts outlined in the previous section show, there are five

main logistics variables that, when combined, drive the environmental impact of logistics:

- *Distance*. How far are products being moved? Where are they loaded/unloaded?
- *Mode*. Which mode of transportation is being used?
- *Equipment*. What kind of equipment is being used for the logistics operation? What kind of fuel and how much fuel does it consume?
- *Load*. How much product is being loaded into the conveyance? How efficiently is it loaded?
- *Operation*. How skillful is the driver in operating the vehicle? How optimal is the logistics plan?

Each of these variables is a lever that can be used toward designing greener logistics systems: distance reduction, modal shift, cleaner equipment, better load planning, and operational excellence.

All business decisions, including logistics and transportation, are made at the strategic, tactical, or operational level (Stank and Goldsby 2000). Strategic decisions are revisited every 3–5 years, tactical decisions are often done with a 6-month to 2-year horizon in mind, and operational-level decisions are made on a daily and weekly basis. Thus, decisions at strategic, tactical, and operational level are opportunities to mitigate the environmental impact of logistics by flexing one or more of the five green logistics levers.

Table 5.10 shows a non-exhaustive list of green logistics strategies for each of the five levers across the three decision levels. These strategies require a combination of business and analytical approaches in logistics. Some of these strategies may appear simplistic or obvious, but they matter in practice. For instance, everyone recognizes that idling vehicles consumes fuel unnecessarily. According to the EPA SmartWay program, long-duration idling of truck and locomotive engines consumes more than 1 billion gallons of diesel fuel per year; emits 11 million tons of carbon dioxide, 200,000 t of oxides of nitrogen, and 5000 t of particulate matter; and elevates noise levels (EPA 2015). But only when there is a managerial commitment to reduce these impacts are the required actions taken, even when they are economically sound. Walmart, for examples, outfitted its 7000-vehicle truck fleet with auxiliary power units (APUs). Walmart estimated an 18-month payback period through fuel and engine wear savings, in addition to all associated environmental benefits.

The next two sections will expand on two well-studied modeling approaches—network design and vehicle routing—that are commonly used in logistics decision-making and that have direct applicability on all levers at the strategic, tactical, and operational level. Business-centric dimensions will be further explored in the case studies of Sect. 5.4, focusing on real-life implementation challenges.

Table 5.10 Green logistics strategies

	Strategic	Tactical	Operational
Reduce distances	• Include environmental impacts in network design • Local sourcing	• Flexible territories/service contracts that allow for increased density of pickup/delivery networks • Identify cross-industry partnerships to reduce empty miles	• Advanced vehicle routing that includes congestion, fuel consumption modeling, and flexible time windows
Modal shift	• Evaluate network design incorporating facilities alongside intermodal terminals • Design networks to support flexible inventory and service levels to allow various network speeds	• Collaborate with customers/suppliers to adjust order quantities, inventory levels, lead times and service levels to allow multiple modes in lanes • Develop multimodal third-party logistics providers	• Define a clear hierarchy of preferred modes by lane • Develop multimodal experience by operating lanes across multiple modes
Cleaner equipment	• Joint investment in cleaner technologies, including early trials to foster equipment innovation	• Incentivize capital investments to regularly upgrade/replace aging equipment • Pilot new technologies to obtain “real” operational environmental performance	• Track equipment performance (fuel consumption, emissions, noise) • Develop environmentally aware preventive maintenance plans
Load planning	• Redesign product packaging to improve conveyance utilization	• Add environmental metrics to logistics planning reports • Review “green scenarios” in load planning	• Track and report the environmental impact of every move • Optimize conveyance loading using analytical approaches (OR)
Operational excellence	• Develop an environmentally aware logistics culture	• Benchmark environmental operational performance • Recognize top environmental performers regularly	• Develop operational environmental dashboards (e.g., fuel consumption, idling) • Establish targets and incentives

5.4.1 Network Design

Logistics network design is a strategic decision that has direct impact on two of the most important levers for green logistics: distance reduction and mode shift. It includes decisions related to the location of manufacturing plants, assembly facilities and multiple tiers of warehouses, as well as deciding how products flow through the network from suppliers to customers.

As elegantly summarized by Magnanti and Wong (1984), the basic ingredients of all network design models are a set of nodes (N) and a set of directed arcs (A) that are available to design the network. There are two types of decisions in network design models: (a) discrete choice decisions relating to selecting which nodes and which arcs should be included in the final network and (b) decisions about the flow of one or multiple commodities from supply to demand nodes along the selected network. To find the optimal solution, mathematical models trade off a variety of fixed and variable costs, as well as minimum and maximum flows, through each arc and node.

Network design models are widely used and studied in logistics planning decisions, ranging from global flows (see Goetschalckx et al. (2002)) to locating single facilities (see Melo et al. (2009)). These models often include cost and service trade-offs.

Adding environmental considerations can be achieved by augmenting a traditional network flow model with an environmental cost or an environmental constraint.

For instance, when deciding to open or close a facility in a network, there is often a fixed cost plus a variable cost driven by productivity, capacity, or labor choices. Because operating facilities may also generate environmental impacts that may vary by these same choices (e.g., pollution, GHG emissions), an “environmental cost” may be added to each candidate facility and then added as part of the objective function to minimize total emissions or as constraint to limit the total environmental impacts (see Chap. 7 by Velázquez and Fransoo (2024)).

Another common variation to network flow models to take into account green impacts is related to the flows through the network. Since every arc in the network represents the movement of one or more commodities via a mode of transportation, the model could explicitly estimate the environmental impact of this move using mode-specific emission factors. For example, Table 5.3 includes the amount of CO₂ emitted per ton-mile for various modes. By multiplying these factors to the corresponding arc-flow variables, the model can estimate the total CO₂ emissions of a particular transportation network configuration. These total transportation emissions could also be added to an objective function or part of a constraint to limit total emission costs. Because different transportation modes have different speeds, they will also have inventory impacts. To fully capture the network environmental impact, factors that measure the increase in emissions due to extra holding inventory may also be needed (e.g., extra energy required to hold extra inventory, extra waste generated due to increase obsolescence).

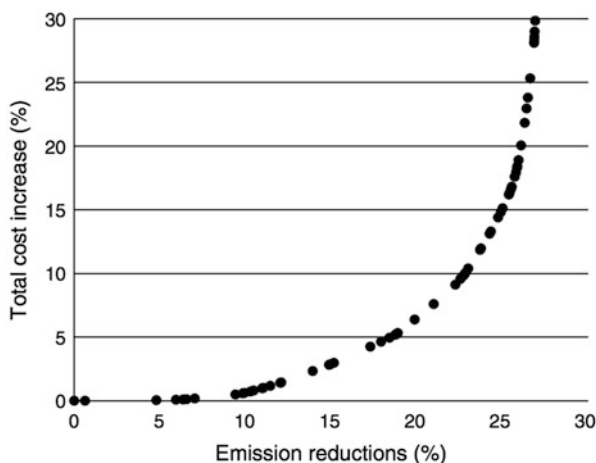


Fig. 5.8 Cost vs. CO₂ emission trade-off curve. *Source:* Hoen et al. (2014b)

Hoen et al. (2014b) conducted a comprehensive modeling and analysis of the economic and environmental impact of selecting various transportation modes to fulfill customer orders for Cargill. They modeled the impact on revenue, inventory, and costs and traded it off with total CO₂ emissions. Figure 5.8 shows their results as a trade-off curve between reduction of emissions and cost increase.

Although the trade-off curve will vary depending on the specific network configuration, economics, and mode choices available, it often has a similar shape: it is possible to achieve noticeable environmental reductions without adding significant extra cost to the logistics network. In the case study analyzed by Hoen et al., 10% CO₂ reductions can be achieved by adding less than 1% in cost. Achieving higher CO₂ reductions (e.g., 25%), however, will require a significant cost increase (e.g., 15%).

In addition to the cost vs. environment trade-off analysis, analyzing various scenarios can yield important insights to the network topology. Figure 5.9 shows the different network configurations for an apparel manufacturer in the USA under different scenarios. In this case, the optimal cost scenario will open four warehouses, compared to five or six facilities if the company would like to reduce 10% and 25% of total CO₂ emissions in the network, respectively. More importantly, by looking at which facilities are selected under each scenario, logistics planners may elect to evaluate hybrid solutions that balance economic and environmental objectives.

5.4.2 Vehicle Routing

At the other end of the spectrum of network design is the daily execution of logistics activities. This includes warehouse operations (e.g., receiving, unloading, loading,

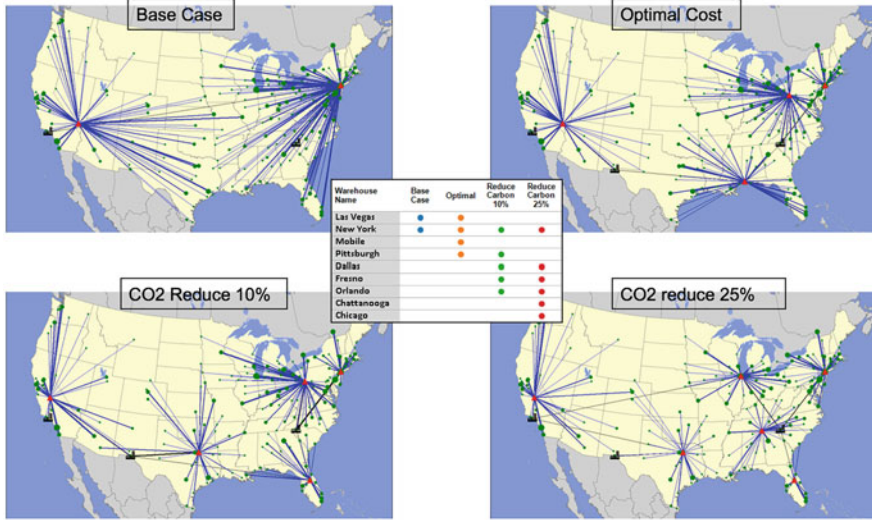


Fig. 5.9 Network variations under different CO₂ reductions. *Source:* Blanco and Simchi-Levi (2008)

storing, picking, and packing) and distribution operations (e.g., load building, product delivery, and product collection).

The design of the network and customer requirements limit the mode choices available for distribution operations, but there are still several levers that could be influenced directly, such as distance, equipment, and load building. Vehicle routing and scheduling is often the most environmentally intensive activity (from an energy and GHG perspective) because it is during the physical delivery of products that fuel is consumed and that the majority of transport emissions are generated (see Fig. 5.1).

There is a large body of literature devoted to solving vehicle routing problems (VRP) (see Laporte (2009) for a comprehensive review). These models almost always focus on minimizing distance, time, or cost of the planned routes. Although highly correlated, solving for minimum distance does not always translate into minimizing environmental impacts (Bektaş and Laporte 2011).

A similar approach to add environmental measurements to network design problem can be used to modify vehicle routing models: explicitly calculate the factors affecting fuel consumption and pollution such as equipment characteristics, customer time windows, product loaded during each leg of the route, speed of travel, slope of the road network, and congestion. These variations of VRP models are known as “pollution-routing problems” (PRP) (Bektaş and Laporte 2011; Koç et al. 2014).

PRP problems explicitly capture the environmental impacts of vehicle routing operations. Assigning a vehicle to a route is one example. Besides differences in

Table 5.11 EU emission standards for heavy-duty diesel engines (steady-state testing)

Stage	Date	CO	HC	NOx	PM
		g/kWh			
Euro I	1992, ≤ 85 kW	4.5	1.1	8.0	0.612
	1992, > 85 kW	4.5	1.1	8.0	0.36
Euro II	1996.10	4.0	1.1	7.0	0.25
	1998.10	4.0	1.1	7.0	0.15
Euro III	1999.10 <i>EEV only</i>	1.5	0.25	2.0	0.02
	2000.10	2.1	0.66	5.0	0.10a
Euro IV	2005.10	1.5	0.46	3.5	0.02
Euro V	2008.10	1.5	0.46	2.0	0.02
Euro VI	2013.01	1.5	0.13	0.40	0.01

Source: <https://www.dieselnet.com> compiled from the *Official Journal of the European Union* (1992, 1996, 1999, 2001, 2005, 1009, 2019)

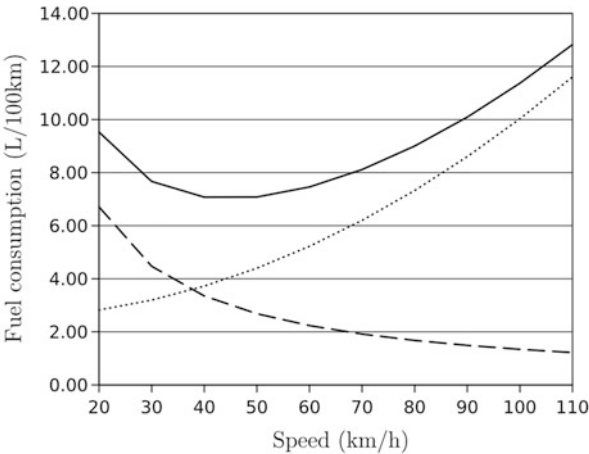


Fig. 5.10 Schematic fuel consumption rates by speed of travel. Source: Bektaş and Laporte (2011)

fuel consumption (that are part of the variable costs of operating a vehicle), different types of vehicle technologies have different environmental impacts (see Table 5.11).

Speed of travel is another dimension of vehicle operations that is relevant in PRP problems. Figure 5.10 shows fuel consumption by speed for a lightweight vehicle. We can notice the U-shape of the curve leading to an optimal speed at 40 km/h. As vehicles are assigned to various road segments in a network with varying speeds, the total emissions generated may vary (e.g., highways or local roads). For instance, Staples was able to achieve an increase in fuel efficiency from 8.5 to 10.4 miles per gallon by limiting driver speeds to 60 miles per hour without impacting delivery performance (Staples 2012). Also, for vehicle-routing problems with time windows, tracking environmental impact at various speeds is more relevant due to the interaction with customer service expectations.

Although fuel efficiency equations by vehicle type and speed are already complex, they can also be augmented to add payload impacts: the amount of fuel consumed while traveling at a certain speed increases with the amount of cargo hauled (Barth and Boriboonsomsin 2009). Thus, it is theoretically possible to reduce environmental impacts on vehicle routes by giving priority to deliver large payloads early in the route, due to the potential to continue the route with a lighter load. Furthermore, by modeling road slopes or stop lights, one can further explore route variations that can further reduce fuel consumption and emissions. UPS delivery routes, for example, are designed to minimize left turns, which require vehicles to wait at an intersection for traffic to clear before making the turn. In 2020, UPS had lowered CO₂ emissions by 14% compared to 2010. Their pledge is to reach a 50% reduction by 2035 (UPS 2020).

5.5 Case Studies: Implementing Green Logistics Strategies

The strategies included in Table 5.10 are known to reduce emissions and, in some cases, also drive cost savings. Nevertheless, companies often have trouble implementing them.

One of the challenges is focus: these initiatives are not always management priority. Another key obstacle is that they require internal and external collaboration. The logistics function interfaces with multiple business activities and supports a wide range of decisions. Client-supplier relationships, outsourcing arrangements, cost sharing, and coordination between trading partners are a few examples.

The following four case studies illustrate how, through collaboration, green logistics initiatives were put into practice.² The case studies focus on some of the unique business considerations and details required to make green logistics successful in practice. The first case study discusses how Boise was able to shift from truck to rail shipments and increase its load utilization by working with its customers on service and inventory impacts and through internal improvements of package design. The second case study also discusses modal shift, but this time achieved by two competitors sharing “empty miles” of a rail backhaul leg. The third case study is an example of a package redesign in Caterpillar’s inbound network that reduces shipment weight and thus consumes less transportation fuel. The section concludes by describing the EPA SmartWay program, a multi-stakeholder voluntary collaboration that has incentivized companies to prioritize green logistics efforts by providing the right incentives and transparency to the process.

² The first three case studies in this section closely follow “Delivering on the Promise of Green Logistics,” by Edgar Blanco and Ken Cotrill (Blanco & Cotrill, 2013).

5.5.1 Customer Collaboration: OfficeMax and Boise, Inc.

Boise, Inc. manufactures a wide range of packaging and paper products, with reported earnings of \$2.56 billion in 2012.³ The company operates mainly in the USA and has long-term relationships with many customers, including retailer OfficeMax. Boise supplies the majority of OfficeMax's paper products due to a long tradition of business and commercial relationships.

Truck transportation offers Boise the flexibility and speed it needs to meet customer delivery promises. However, moving products by truck also accounts for the largest percentage of the CO₂ emissions associated with logistics operations. Rail transport is more cost-effective and emits much less carbon per equivalent weight and distance, but it's also slower and less flexible than trucking.

As part of its cost and environmental efforts, Boise developed two initiatives. In the first one, called Carload Direct, Boise needed to coordinate with its long-time partner OfficeMax. Prior to the initiative, shipments from Boise were routinely moved by truck to the retailer's facilities, even when the destinations were accessible by rail. OfficeMax became a test case for switching to direct-by-rail shipments. The two companies' long-standing relationship helped Boise negotiate the change, but the Boise team was still tasked to make sure the transition did not disrupt OfficeMax's ordering process.

The benefits of the change, carbon-wise, were significant: a traditional truckload shipment comprises about 20 t of paper product, whereas a railcar carries around 70 t. The main drawback to the process is that a rail solution required customer orders to be warehoused until there was enough freight to fill a boxcar. This meant that the time from order placement to delivery was potentially extended, while running the risk that inventory needed by stores would be stuck in a warehouse. By coordinating the ordering process, the Boise team was able to determine when enough OfficeMax orders were available to "pool" into a Carload Direct shipment or when a regular truck shipment was the best option.

The solution may seem straightforward, but it required extensive cooperation between the transportation departments at Boise and OfficeMax. For example, ordering processes had to be adjusted and coordinated to take account of the change in product flows. Not every SKU was suitable for a Carload Direct shipment, so the teams had to select products that could be part of the pooling system. And the changes had to be made without altering the inventory positions or service levels of either enterprise, especially OfficeMax. The shift from using a mix of truck and rail to primarily rail between the major Boise and OfficeMax distribution centers eliminated more than 2600 t of CO₂, a 70% reduction compared to truckload shipments, or the equivalent of saving over 264,000 gal of fuel consumed by road

³ The Boise organization described in this case study refers to Boise Cascade LLC, the paper and forest products. It should not be confused with Boise Cascade Corporation that is now OfficeMax Incorporated.

vehicles. Both companies benefitted from this greener supply chain as the reduced transportation costs were shared between them.

For the second initiative, Boise looked at its packaging and loading processes. Paper products are often ordered in large quantities and shipped in full pallets. Boise field operations noticed that full pallets did not always maximize the space in the rail boxcars. Boise redesigned its pallets and loading procedures to accommodate a half-pallet, which allowed Boise to reorganize pallet stacking and maximize shipping capacities for its loads. Once the operational configuration was solved, Boise needed to work with its customers to allow orders that included a half-pallet size.

Boise created new SKUs and modified ordering and receiving systems to allow for the new half-pallet product configuration. The company realized that this was a win-win situation; the half-pallet solution was a perfect fit for companies that shipped seasonal and low-demand products. Using just 930 railcars in 2011, Boise reduced the company's CO₂ emissions by 190 t or 6.8% of its annual rail shipment emissions without any extra cost. This is equal to the CO₂ emissions from 21,600 gal of fuel or the annual GHG emissions from 38 passenger vehicles. In addition to improving the environmental performance of Boise and participating customers, the smaller, the half-pallet unit gave Boise and its customers more order flexibility.

5.5.2 Competitor Collaboration: Ocean Spray and Tropicana

Identifying opportunities for using intermodal can be challenging. The respective transportation services must mesh seamlessly in order to maintain service standards and avoid delays that could result in the costly buildup of inventory. Ocean Spray captured such an opportunity in an unconventional way: it collaborated with a competitor.

Ocean Spray is an agricultural cooperative owned by more than 700 cranberry growers in Massachusetts, Wisconsin, New Jersey, Oregon, Washington, Canada, and Chile, as well as 35 Florida grapefruit growers. The organization posted fiscal 2012 gross sales of \$2.2 billion and net proceeds of \$338 million.

One of Ocean Sprays' most significant logistics expenses—and a major contributor to its carbon footprint—was the need to ship product from its Bordentown, New Jersey, distribution center over 1000 miles to another distribution center in Lakeland, Florida. By coincidence, both of these distribution centers were located a short distance (60–65 miles) from rail yards used by a competitor, Tropicana, which shipped orange juice north from Florida in special refrigerated boxcars.

Tropicana also had a key difficulty when it came to logistics. Although it could send product direct from Florida up to the CSX rail terminal in New Jersey, once offloaded, the refrigerated rail boxcars were often traveling empty back to Florida.

Tropicana's third-party logistics (3PL) provider, Wheels Clipper, saw an opportunity to help its client while earning the trust of a new customer. The company approached Ocean Spray and proposed that they operate an intermodal lane from

New Jersey to Florida that would put Tropicana's empty orange juice boxcars to use on Ocean Spray's behalf. The goal was to save Ocean Spray money (and reduce emissions) by allowing them to switch from trucking their cranberry juice to sending it more efficiently by rail while also saving Tropicana the cost and emissions associated with empty cars returning south.

The plan provided environmental benefits and lowered costs for both companies. But there were potential downsides. To make the idea work, Ocean Spray would have to switch logistics providers and work with a new 3PL, transporting via a mode with which the company was unfamiliar. Even more challenging, the arrangement required Ocean Spray to collaborate with one of the company's strongest competitors in the beverage business, Tropicana.

Using the intermodal solution would save Ocean Spray over 40% in transportation costs compared to the previous trucking method, and it would also reduce its GHG emissions by a 65%. Meanwhile, Tropicana would eliminate most of the costs and GHG emissions of the boxcars' return trip from its ledger—a significant reduction in each metric, at the price of some (admittedly complex) process changes.

Ocean Spray accepted that the plan did make sense, and the company reconfigured its load planning processes to make shipments compatible with the rail mode. Tropicana and Wheels Clipper also needed to change their operating procedures to communicate railcar availability to Ocean Spray. To guarantee the on-time delivery of all shipments, all the participants worked to improve supply chain visibility. For example, they improved the flow of information between the parties on the status of shipments and potential delays. As the intermediary, Wheels Clipper helped to ensure that this information was delivered to the relevant parties in a timely fashion.

During several pilot runs, the companies were able to coordinate load pickup and delivery within required time windows and to confirm that product was being handled properly. The trials also revealed potential problems. For example, the new intermodal solution added 1–2 days to transit times, and Ocean Spray had to adjust its logistics operations to avoid having a negative impact on customer service. Yet in the end, the fact that there was a clear ROI for both shippers gave incentives to the participants that helped them to move the project forward and make it successful.

5.5.3 Supplier Collaboration: Caterpillar and Part Suppliers

Caterpillar Global Mining Division makes specialized trucks for the mining industry. Over the past 30 years, Caterpillar has produced nearly three times as many trucks as its closest competitor. The company assembled its 50,000th rigid frame construction and mining truck in 2009.

Caterpillar worked on an initiative focused on the North American inbound part logistics for Caterpillar's manufacturing facility in Decatur, Illinois. The goal was to reduce the weight of the thousands of returnable packing containers that Caterpillar used to transport parts from suppliers. Replacing the heavy metal packing containers

with lightweight plastic units cuts the weight of shipments, which translates into less fuel consumption and reduced carbon emissions.

However, the effort required Caterpillar to coordinate with multiple suppliers in different countries. It was a daunting challenge, particularly when long-established practices have to be changed. Some of these containers had been in circulation for more than 50 years.

Prior to the carbon footprint study, Caterpillar's internal analysis had determined that the fuel savings from lightweighting inbound containers would be significant. Yet Caterpillar had been slow to adopt the plastic containers because of several internal organizational and budgetary constraints.

The first step was to carry out a detailed analysis of the inbound flow of containers. In partnership with MIT, Caterpillar analyzed 16 weeks of transportation data that included detailed information about North American suppliers, number of parts shipped, and delivery dates for three of the flagship mining trucks. The data included more than 15,000 truck deliveries of 1400 different item numbers from more than 200 suppliers that ship directly to the Decatur manufacturing plant. The Caterpillar and MIT teams determined that 9.5% of shipments were eligible for lightweight packaging.

By reducing the shipping containers from 130 to 200 lb. each to 20–40 lb. each, Caterpillar could reduce CO₂ emissions across the Caterpillar North American network by 130 t. Caterpillar could now present a strong environmental case, in addition to a 2-year ROI viable financial benefit to all internal and external stakeholders affected by the packaging changes.

5.5.4 The EPA SmartWay Program

The SmartWay Transport Partnership is an innovative voluntary collaborative program between the EPA and the freight industry designed to improve energy efficiency and to lower GHG emissions and air pollution. Started in February 2004, the partnership aims to create strong market-based incentives for companies shipping products, and for the truck and rail companies delivering those products, to improve the environmental performance of their freight operations.

5.5.4.1 History⁴

In 2003, staff at the EPA's Office of Transportation and Air Quality (OTAQ) developed an initial concept to build an innovative voluntary freight transportation program focused on energy savings and emission reductions. They worked for over

⁴ Based on interviews and research conducted by K. C. Tan and E. E. Blanco from the MIT Center for Transportation & Logistics in 2009

a year in collaboration with a group of public and private stakeholders to develop the foundation of the EPA SmartWay Transport Partnership. Members of the group included the American Trucking Associations, Business for Social Responsibility, Canon U.S.A., Coca Cola Enterprises, CSX, FedEx, H-E-B Grocery, Home Depot, IKEA, Interface, Nike, Norm Thompson Outfitters, Schneider National, Swift Transportation, UPS, and YRC Worldwide.

Together, these stakeholders and the EPA designed a freight transportation program that addressed the goals and needs of both the freight industry and the EPA. The freight industry was interested in an improved public image, recognition for its efforts, and fuel savings to help companies in an extremely competitive industry. The EPA was interested in reducing emissions from diesel engines and improving energy security in the freight industry. The companies provided critical operational and technical insight into freight management and supply chain logistics. Their input helped the EPA to identify appropriate fuel-saving technologies for heavy trucks and to develop a fuel and emission-tracking tool that carriers and shippers could use to track their performance.

Because program enrollment was going to be on a voluntary basis, it was important to make it attractive for companies to participate and to motivate them toward a collaborative solution in addressing energy and environmental issues in the freight sector. After much planning, the EPA formally launched the program with the full support of the trucking industry at the American Trucking Associations' annual conference on February 9, 2004, with 15 initial charter partners.

5.5.4.2 Program Structure

The SmartWay Transport Partnership is tailored to progressive corporations and organizations involved in shipping goods. Companies that provide and hire freight delivery services become SmartWay Transport Partners by committing to improve the environmental performance of their freight delivery operations. SmartWay Transport Carriers commit to integrate innovative cost-saving strategies into their fleet operations. SmartWay Transport Shippers commit to ship the majority of their goods with SmartWay Transport Carriers. Companies that meet SmartWay Transport Partnership requirements benefit from reduced operating costs and enhanced recognition and visibility: partners that demonstrate superior performance earn the right to display the SmartWay Transport logo.

The SmartWay program also provides technical assistance to help partners benchmark and evaluate the effectiveness of a broad range of technology, equipment controls, and fuel-saving logistics management strategies. The SmartWay program has also established connections with financial institutions to provide flexible, reduced interest loans to improve access to these fuel-saving technologies and pollution controls.

5.5.4.3 Impact

Since 2004, SmartWay partners have eliminated over 152 million metric tons of CO₂, saved over 357 million barrels of oil and \$47.6 billion in fuel costs, and reduced 2.7 million tons of NO_x and 112,000 t of PM (EPA 2021).

The success of SmartWay in the USA has led to its adoption in Canada, where Natural Resources Canada now administers the program for Canadian firms, using the same tools methods and metrics for assessing freight efficiency. In addition, the SmartWay public-private partnership model is now being replicated in other countries in Latin America, Asia, and the European Union. SmartWay is also being used as a template to inform the development of freight sustainability programs under the United Nations Climate Environment Programs and Clean Air Coalition (EPA 2021).

5.6 Beyond Green Logistics

This chapter has presented various levers available to improve the environmental impact of logistics operations. Most of the approaches discussed are familiar to logisticians: optimize distance, maximize loads, assign the right equipment and resource, and avoid waste. These are themes that have been used for many years to reduce cost and improve service. The same systematic approach can be used to make logistics better for the environment.

Underlying all the success stories and approaches presented throughout the chapter—from network design, vehicle routing, or multi-stakeholder collaboration—is the need for a solid measurement foundation of GHG, pollution, noise, and waste impacts. Unlike financial measurements that are common and well understood, environmental metrics are uncertain and continuously evolving. Green logistics is about not only finding the optimal solution to an environmental metric but also making sure that traditional strategies are reviewed with environmental lenses. New trade-offs may be uncovered, as well as new sources of value and cost reduction.

This chapter uses a narrow scope—just logistics activities—for all analyses of impacts and strategies for improving. In some transport sustainability cases, a broader scope of analysis, such as an end-to-end life cycle analysis, will reveal trade-offs between the greenest choices in logistics and greenest choices in sourcing, manufacturing, returns, product design, and so forth. For example, differences in manufacturing efficiencies or the energy portfolios of different regions may mean that higher impacts in transportation permit much lower environmental impacts overall. Or a slower mode of transport might reduce transportation emissions but increase the amount of waste due to spoilage, obsolescence, or excessive inventory.

Thus, the analyses and strategies in this chapter are an element of a broader process of analyzing and improving the sustainability of the business or supply chain at a more holistic level. As a consequence, there are more opportunities

related to green logistics. Reverse logistics and closed-loop supply chains (see Chap. 15 by Abbey et al. (2024)) adds a new dimension to managing waste by focusing on how to recover products delivered to customers and extract new sources of value. Technological improvements that completely avoid emissions, such as solar energy, could completely avoid the need to reconfigure logistics networks. Advances in nanomaterials and biodegradable materials may make packaging waste a nonissue. And in some cases, life cycle thinking may challenge logistics operations to increase their impact to allow for a holistic reduction of the environmental impact of products: it may be sometimes better to import a product from the other side of the world if energy sources are cleaner there than to manufacture it locally.

A new generation of green logistics choices is quickly unfolding, and the concepts discussed in this chapter provide a foundation for the journey ahead.

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Chapter 6

Green Inventory Management



Johan Marklund and Peter Berling

6.1 Introduction

Managing inventories and thereby controlling the material flows through a facility, or network of facilities, are matters of key importance in achieving efficient supply chains. Core issues in traditional inventory management are: *how much to order, when, and where to satisfy the customers' service needs at minimum cost?* In this context, reasons for keeping inventory are economies of scale in batch ordering/production and uncertainties in demand and supply. However, inventory management may also involve strategic decisions regarding design of distribution and production networks, facility location, procurement, supplier selection, choice of product range, assortments, etc. Many of these strategic issues are treated thoroughly in other chapters of this book, and they will not be further discussed in this chapter. Instead, we will focus on the core operational issues and define inventory management in line with the operations research and management science (OR/MS) literature on inventory control and inventory modeling. It should be noted that closed loop systems encompassing reverse logistics flows, which is part of this literature, are treated separately in Chap. 15 by Abbey et al. (2024).

In green (or sustainable) inventory management, the pure economic focus (traditionally measured in terms of costs) is complemented with an environmental

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perspective (typically measured in terms of emissions). This means that potential trade-offs need to be identified and resolved. Thus, green inventory management in the context of this chapter is concerned with the overarching question of how to efficiently manage inventories (and thereby material flows) with respect to both costs and emissions. An important insight is that inventory decisions are interlinked with production and transportation decisions that directly influence emissions. A total cost and emission perspective is therefore important in green inventory management to avoid suboptimization and achieve cost-efficient emission reductions. Crudely speaking, emissions are often divided into three categories: greenhouse gases (GHG), other harmful gases, and particles. Among these, GHG emissions, for convenience often measured in CO₂ equivalents, are getting most of the attention in media and society at large, because of its impact on global warming. This is also reflected in the OR/MS literature on green inventory management and in the contents of this chapter. (Chapter 3 by Boukherroub et al. (2024) focuses entirely on carbon footprinting.) However, it is important not to forget that environmental consequences are not limited to GHG emissions and may include, for example, noise; accidents; waste product accumulation in landfills and oceans; chemicals that pollute our air, ground, and water supply; deforestation; exploitation of natural resources that damage our ecological systems, etc. (Chapter 2 by Guinée and Heijungs (2024) introduces LCA, and in doing so, a full spectrum of environmental impact categories are considered.)

As indicated above, inventory decisions of when and how much to order are intricately connected to the issue of how ordered items are produced, handled, and transported. For example, larger lot (or order) sizes in manufacturing, especially in energy-intensive process industries, may have a significant impact on energy use and emissions. However, it also leads to more finished goods and work in process inventory. The importance of transportation from an emission point of view is apparent. For example, by choosing a greener transport mode such as sea or rail, instead of road or air, the transport emissions, and generally also the transportation costs, can be reduced. Similarly, by employing shipment consolidation strategies and shipping full truckloads and containers less frequently, the transportation capacity is better utilized. This means fewer vehicles are needed to ship the same volume suggesting less transportation emissions and typically lower transportation costs for operating the system. The problem is that these strategies, which appear attractive when transportation is considered in isolation, can have negative consequences for the system as a whole. For example, greener transportation modes are generally slower and often associated with larger fixed costs, suggesting longer transportation lead times and larger order sizes/shipment quantities. This means more inventory and thereby increasing costs and emissions for holding inventory. Similarly, shipment consolidation suggests larger, less frequent, and/or less flexible shipments and thereby more inventories. It is also noteworthy that these strategies are at odds with general principles and perceptions of just in time (JIT) and lean, where frequent shipments of small batches are emphasized.

To further analyze how inventory management decisions may influence emissions and costs and vice versa, it is important to recognize that emissions are

associated with energy consumption. Thus, decisions that affect the energy consumption of the supply chain will influence its emissions. Bearing this in mind, we may divide the emissions associated with operating an inventory system into three categories: *emissions associated with ordering* (i.e., *producing and transporting*) *items*, *emissions associated with holding items in stock*, and *emissions associated with not satisfying customer demand on time*. Note that this partition mirrors the cost components usually considered in inventory management: *costs for ordering and procuring items*, *costs for holding items in stock*, and *shortage costs for not satisfying customer demands on time*.

In the remainder of this introduction, we will take a closer look at these three categories of costs and emissions and use them as a basis for identifying key questions and challenges for green inventory management research. Basic inventory modeling structures, assumptions, and issues are also discussed. This is followed by an overview of the existing green inventory management literature in Sections 6.2 and 6.3, which illustrates what issues and challenges have been addressed in the literature so far and how. Section 6.2 considers deterministic demand models and Sect. 6.3 stochastic demand models. We also distinguish between single-echelon and multi-echelon inventory systems. Section 6.4 concludes with remarks about findings, practical implications, and what remains to be done.

6.1.1 Costs and Emissions Associated with Ordering Items

The costs and emissions associated with ordering items from an internal or external supplier depend on the activities the supplier needs to undertake to satisfy the order and the energy they consume. These activities vary if the supplier is a manufacturing plant, a local or distant warehouse, an external wholesaler, etc. They may also be affected by when and how much are ordered from where. However, in general terms, we may divide them into costs and emissions for producing, handling, and transporting the ordered items.

If we start with the costs for ordering items, they should include all relevant handling and transportation costs and in case of a production order also the costs for releasing and producing the order. Thus, these costs may depend on many different things and can be quite difficult to determine on a detailed level. However, both in practice and in the literature, it is common to divide them into fixed (or semifixed) ordering (or setup) costs and variable costs. The fixed setup costs are considered independent of the order size, at least for defined ranges of order sizes. Examples of such costs include the cost associated with the setup time of configuring or cleaning a machine or production line before a new order can be produced, the administrative costs of placing a purchase order, or the truck driver and fuel costs associated with driving the truck or train without any load. Conversely, the variable ordering costs depend on how many units are ordered. Most often a constant per unit cost is presumed, meaning that the variable cost for the order is proportional to the number of units ordered, at least over specified ranges for the order size. For example, the

variable transportation cost per unit may be constant as long as the same type of truck is used. But if the order quantity becomes so large that a different truck is needed, the costs per unit may change. Quantity discounts and transportation tariffs may also lead to similar stepwise changes in the variable and fixed ordering costs.

Turning to the emissions associated with ordering items, these are also commonly divided into fixed and variable components. For example, the emissions for shipping an order on a certain vehicle can be divided in a fixed part for moving the empty vehicle and a variable part which depends on the load that is put on the vehicle. The total transport emissions are related to the energy and fuel consumption of moving the vehicle and its load. Moving is here interpreted in a wide sense to include emissions for temperature-regulated transports, for example, refrigerated or frozen products, where emissions are generated while the items are on the vehicle whether it moves or not. In manufacturing settings, examples of fixed setup-related emissions independent of the order size may include the preheating of dies in a press shop or cleaning the equipment in process industries, say a brewery or dairy, in connection with product changeovers. Variable emissions that depend on the size of the production order are associated with the energy consumption of running the machinery to produce the entire batch, for example, in the aforementioned press shop, brewery or dairy.

6.1.2 Costs and Emissions Associated with Holding Items in Stock

The fact that there are costs of holding items in stock is undisputed. How these costs should be determined in practice is often more of an issue. The norm is to assume that they are proportional to the number of items in stock. However, as some of the cost components usually are fixed or tend to increase stepwise with the number of units in stock, the appropriateness of assuming linear holding costs may sometimes be questioned. The holding cost per unit and time unit can be divided into two main components: cost of capital and out-of-pocket holding costs. The former is generally determined as a holding cost rate multiplied by the monetary value of the item. It should reflect the opportunity cost of tying up capital in inventory instead of in other investment opportunities. Conversely, the out-of-pocket holding cost rate should reflect the costs of operating the inventory facilities (including energy costs), handling the items, and losing items because they perish, are stolen, are destroyed in accidents, are misplaced, etc.

The emissions associated with holding items in stock are clearly connected to the energy consumption of operating the inventory facilities and handling the items, but also to items that are discarded and wasted. An important example of the latter is food (typically with high CO₂ content) that perishes and is thrown away. For instance, the Metabolic Institute reports that over 30% of the food produced for human consumption is lost or wasted globally (Gladek et al. [2017](#)).

Furthermore, in high-income countries, one of the main reasons for the waste is the lack of coordination between demand and supply. For more in-depth discussions of challenges and issues in sustainable food supply chains, we refer to Chap. 9 (Akkerman and Cruijssen, 2024). Another less obvious emission source can be leakage of pollutants. For example, Walmart Global Responsibility Report 2018 states that leakage of highly potent GHG refrigerants account for 13% of their total GHG footprint, which was almost twice as much as the GHG footprint from the fuel to their trucks. This emphasizes the fact that the emissions of operating inventory facilities and stock handling are case-dependent and very much affected by the type of products considered. If a temperature-controlled environment is needed, as is the case for frozen or refrigerated products, the energy consumption and thereby emissions of holding items in stock are much higher and more important to consider than if items are stored in a non-heated warehouse or outdoors. Another example, in contrast to refrigerated products, is that many chemical companies, such as Eastman, at least for limited periods, store products in molten form, which requires a lot of energy.

Generally speaking, more inventories suggest increased costs and emissions, although exactly how and to what extent is often more difficult to say without a careful analysis. Assuming a proportional relationship, which is often done, is convenient but may not always be appropriate. Relating inventory holding costs and emissions to the strategies for reducing production and/or transport emissions and costs above (e.g., use of greener transport modes and shipment consolidation) illustrates important trade-offs. Reducing transport emissions and costs may lead to more inventory suggesting increased emissions and costs for holding items in stock. Resolving trade-offs like this to minimize *total costs and emissions* for the system is a key challenge in green (or sustainable) inventory management.

6.1.3 Costs and Emissions Associated with Not Satisfying Customer Demand on Time

The purpose of the inventory system is to provide adequate service to its customers. Not satisfying customer demands on time have a negative impact on the customers as they are not receiving the service they expect. This negative impact can be translated into shortage costs for the inventory system. The shortage cost may be a fixed cost per shortage, or it may be dependent on the time a customer has to wait for an item (it is then often referred to as a backorder cost). The shortage cost can reflect concrete monetary reimbursements or price discounts in case of delayed deliveries. It should also reflect loss of goodwill in terms of foregone future sales and revenues. Some shortage cost components may be relatively easy to determine, such as downtime costs in a machine or in a production facility. Other shortage cost components, such as loss of goodwill and future revenues, are more difficult to ascertain. These difficulties explain why service (level) constraints are more

popular in practice than shortage costs. On the other hand, shortage costs are the preferred choice in the research literature. A reason for this is that measuring all performance aspects in monetary terms enables a total cost perspective, making it less ambiguous to compare the performance of different policies where the exact same service cannot be achieved.

The negative consequences of not satisfying customer demands on time may also involve increased emissions. For example, consider a dairy where a machine breakdown halts the production process. If the necessary spare part is not available within hours, the semifinished milk products in the process must be discarded, and the facility needs to be cleaned before it can be restarted. Clearly, the emissions associated with throwing away a production batch of dairy products and steam cleaning the process facility are significant. Another example of emissions associated with not satisfying customer demand on time is the transport emissions of a customer making an extra trip, for example, to IKEA, because the product was stocked out when she initially went there to buy it. In situations like this, when poor service generates emissions at a firm's customer (Scope 3 emissions), there is a problem of how to allocate the emissions between the two parties. The problem is especially difficult if the customer, as in the IKEA example, has a choice to make the extra trip or not. Issues concerning allocations of emissions are thoroughly discussed in Chap. 3 (Boukherroub et al. 2024). A common situation, not complicated by emission allocation issues between the firm and its customers, is when the firm use emergency shipments to deal with inventory shortages. The extra transportation emissions for these emergency shipments may be interpreted as emissions for not satisfying customer demand on time through the regular inventory system. The emergency shipments are often associated with higher emissions than regular deliveries due to lower load efficiency in vehicles and use of faster, more energy-intensive, transportation modes.

Similar to shortage costs, the emissions associated with not satisfying customer demand on time can be more or less difficult to determine: In the dairy example above and in terms of emergency shipments, it is relatively straightforward although not without practical challenges. However, in the IKEA example, it is more difficult as the second trip to IKEA in some cases would have been made anyway because the customer was going there to buy other products. Also, the length of the trip and thereby the emissions are different across customers, making it difficult to estimate. Although the emissions and costs associated with not satisfying customer demand on time are sometimes challenging to determine, they can be very important to consider in green inventory management. Returning to the dairy example above, which stems from a real case, the spare parts provider initially had the ambition to eliminate air freight and switch to road and sea freight to reduce transport emissions. However, after further analysis, they concluded that these reductions were minute in comparison to the increased risk of not delivering spare parts fast enough to avoid production losses.

6.1.4 Challenges in Green Inventory Management

From the discussion above, we can conclude that key challenges in green inventory management are to understand how a decision impacts the costs and emissions for the inventory system and how to leverage this understanding to make the “best” decisions. The overarching goal is to minimize both costs and emissions, but decisions that minimize the system’s costs rarely minimize its emissions and vice versa. Thus, a key question is how these potentially conflicting objectives may be reconciled or, in other words, to define what the “best” decision means in different contexts. The main approaches used are to (i) translate emissions into environmental costs and minimize the total costs, (ii) minimize the costs while restricting the total emissions to not exceed a given maximum value or cap, (iii) minimize the costs under an emission cap that can be traded on a market, and (iv) use multiple criteria decision-making (MCDM) techniques to more elaborately combine the emission and cost perspectives. The latter may involve identifying efficient frontiers, Pareto optimal solutions, construction of utility functions, etc. The former three reflects a hierarchical view where the main goal is cost minimization. From a company perspective, emission restrictions and environmental costs can reflect regulations and taxes imposed by society, or it may be based on voluntary commitments. Another form of the latter is investing in different types of carbon offsets (e.g., planting of trees).

An aspect worth noting is that depending on how the supply chain is organized, the types of emissions discussed above may (according to the GHG Protocol) belong to Scope 1 (direct emissions, e.g., from company-owned vehicles), Scope 2 (indirect emissions from electricity, heating, cooling, etc. purchased by the company), or Scope 3 (other indirect emissions, e.g., from customers and outsourced services, including transportation and warehousing). As outsourcing is very common in logistics (e.g., companies today rarely have their own private transportation fleet), Scope 3 emissions are significant. The challenges of measuring and allocating (Scope 3) emissions in supply chains are thoroughly discussed in Boukherroub et al. (2024) and will not be further discussed in this chapter.

Considering the relative size of the emissions discussed above, emissions for producing and transporting items are in general much larger than those for holding items in stock. For instance, the World Economic Forum (Doherty and Hoyle 2009) estimates that about 90% of the emissions for logistics and transport activities, measured in CO₂ equivalents, are due to freight transportation and the remaining 10% from operating logistics buildings. Emissions associated with production, or not satisfying the customer on time, are not discussed in this report, which focuses on emission sources in logistics and transportation from a macro perspective. Similarly, Ries et al. (2017) estimate that in the USA, 80% of the emissions for logistics and transport activities are caused by freight transportation and 20% are caused by operating logistics buildings. Furthermore, the OECD (2017) estimates that without additional measures, CO₂ emissions from global freight could increase with 160% over the next 30 years. Looking back over the last decades, according to the IEA (2022), the transport emissions have grown with an average annual rate

of 1.7% from 1990 to 2021 which is faster than any other end-use sector. If one excludes the last 2 years, which was greatly affected by the COVID pandemic, the growth rate has been closer to 2% per year. Thus, from a climate perspective, it is important to revert this trend and reduce transport emissions. An obvious step in this direction is development of cleaner vehicle technologies. However, the World Economic Forum (Doherty and Hoyle 2009) also emphasizes several other important strategies including despedding the supply chain, optimizing transport networks, and switching of transport modes to greener alternatives. Important enablers for despedding the supply chain, without significant cost increases due to prolonged lead times, are improved collaboration and information sharing between suppliers, customers, and other parties in the supply chain. With respect to optimized networks, the World Economic Forum (Doherty and Hoyle 2009) estimates that 24% of the goods vehicle kilometers in the EU are associated with empty vehicles and, when carrying a load, vehicles are on average filled to 57% of their maximum gross weight. The OECD (2015, 2017) also emphasize the importance of achieving higher load factors and fewer empty transports, for instance, by sharing of trucks and warehouses and better routing decisions. It is estimated that such improved planning and control measures could reduce truck CO₂ emissions by up to one third. Altogether, this indicates a large potential for increasing the capacity utilization of vehicles, for example, through better shipment consolidation and multimodal transport solutions. From an inventory management perspective, all these strategies for greener transports (except perhaps cleaner vehicle technologies) suggest longer lead times, and thereby a need for more inventory, or a decline in customer service. Thus, an important question in green inventory management is how to enable reduction in transport emissions without compromising customer service or increasing total costs and emissions. Here, green (or sustainable) management of multi-echelon (or multistage) inventory systems, with inventories upstream and downstream in the supply chain that jointly can compensate for lead time changes, offers interesting opportunities.

From a modeling perspective, the added complexity and key challenges with green inventory management, compared to traditional cost-based inventory models, are to quantify also the emissions (or environmental consequences) and to find optimal decisions with respect to both costs and emissions. An important first step is then to be able to evaluate the expected emissions and costs associated with a decision. We note that in cases where the emissions accumulate in the same way as the costs, the expected costs and the expected emissions may be evaluated using the same method. Thus, if a traditional inventory management model already exists for evaluating the expected costs, the same model can be used also for determining the expected emissions. The remaining challenge is then to find the best joint decisions with regard to costs and emissions. Of course, when the emission and cost components cannot be modeled analogously, new evaluation methods are needed. Thus, an important question for green (or sustainable) inventory management modeling is to what extent it may be built on existing results. The fact that it requires careful balancing of costs against emissions emphasizes a need for precise models and evaluation methods in both dimensions. This means that traditional

approximations and model assumptions should be carefully analyzed before they are used.

6.2 Green Inventory Management Under Deterministic Demand

This section provides an overview of green inventory management models based on the assumption that future demand is known with certainty. Such an assumption of deterministic demand can be appropriate in some situations. However, most often, future demand is uncertain. Still, deterministic models are often used with good results as heuristics for determining order quantities (or lot sizes) also when demand is stochastic. Disregarding demand uncertainties simplifies the analysis. It is therefore a natural starting point for gaining some first insights of the cost and emission trade-offs that exist in green inventory management. An important aspect of this is the impact of different mechanisms for reducing emissions used by policy makers. Most of the literature on sustainable inventory control with deterministic demand is based on the well-known single-echelon economic order quantity (EOQ) model for lot sizing published already in 1913 by Ford W. Harris (Harris 1913). In the following, we will first present some of these results and then comment on some more elaborate models that extend and confirm the main insights from the basic EOQ model. Finally, we will also briefly discuss some deterministic models for green multi-echelon inventory control.

6.2.1 The EOQ Model with Environmental Considerations

The classic EOQ model considers a single inventory location and a single product with a constant and continuous demand, D , per time unit. The lead time is zero, and no backorders or lost sales are allowed so each time the inventory level goes to zero a new order is placed. Stock keeping in this simple setting is motivated by economies of scale in batch ordering/production as reflected by the fixed cost per order. The aim is to minimize the average cost per time unit of ordering and keeping this product in stock. Extensions to discrete demand, positive lead times, planned backorders, and volume-based prices per unit are straightforward (see, for example, Axsäter (2015)) but will not be further discussed in this chapter.

The EOQ model assumes that each time an order is placed, the firm incurs a fixed cost, A , per order and a variable cost, c , per unit purchased. The cost for an order of Q units is thus $A + cQ$, and such an order will be placed every Q/D time unit. There is a holding cost, h , per unit and time unit for inventory kept in stock. The average amount of inventory on hand if one orders Q units will be $Q/2$. The total cost per time unit, $Z(Q)$, can be divided into two parts – the per unit purchase cost, cD , and

the inventory-related cost, $Z'(Q)$:

$$Z(Q) = \frac{AD}{Q} + h\frac{Q}{2} + cD = Z'(Q) + cD$$

The cost-minimizing solution, which does not involve any environmental considerations, is of the well-known square root form:

$$\min_Q Z(Q) \Rightarrow Q_Z^* = \sqrt{\frac{2AD}{h}} \Rightarrow Z'(Q_Z^*) = \sqrt{2AhD}.$$

Emissions can be incorporated into this model using an emission structure that mirrors the cost structure. To do so, we introduce the emission parameters $\hat{A}$, $\hat{c}$ and $\hat{h}$ denoting the emission per placed order, per unit ordered, and per unit kept in stock for one unit of time. The average emissions per time unit, $E(Q)$, can then be expressed as

$$E(Q) = \frac{\hat{A}D}{Q} + \hat{h}\frac{Q}{2} + \hat{c}D = E'(Q) + \hat{c}D.$$

The associated emission minimizing solution, $\min_Q E(Q)$, is analogous to the cost-minimizing solution:

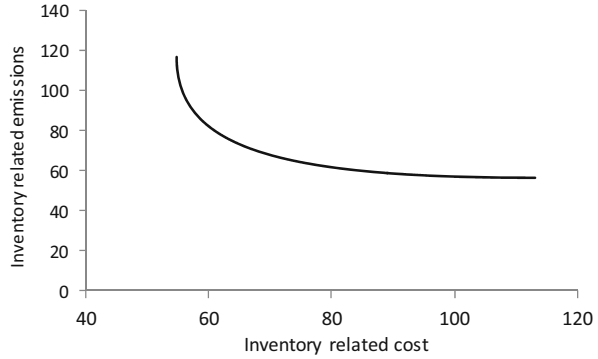
$$Q_E^* = \sqrt{\frac{2\hat{A}D}{\hat{h}}} \Rightarrow E'(Q_E^*) = \sqrt{2\hat{A}\hat{h}D}.$$

It follows that a reduction in emissions by adjusting the order quantity from the cost-minimizing solution can only be attained if the relationship

$$\alpha = \frac{\hat{A}/\hat{h}}{A/h} \neq 1$$

Note that if $\alpha = 1$, the two solutions, Q_Z^* and Q_E^* , will coincide. In all other cases, i.e., $\alpha \neq 1$, a reduction in emissions implies a deviation from the cost-minimizing order quantity, Q_Z^* , and thus an increase in cost. The best ordering decision from a joint emission and cost perspective is the one that gives the lowest cost for a given amount of emission or vice versa, i.e., a solution that belongs to the efficient cost-emission frontier as depicted in Fig. 6.1. The properties of this frontier for the classical EOQ problem are analyzed and discussed in Bouchery et al. (2012). Key results, illustrated in Fig. 6.1, are that the efficient frontier is convex and that substantial emission reductions can be attained with only a modest increase in cost as long as the cost is close to optimal. The opposite holds when the inventory-related cost is far from optimal, that is, a large increase in cost is needed to attain a small emission reduction.

Fig. 6.1 Efficient cost emission frontier for a single-echelon EOQ system (adapted from Bouchery et al. 2012)



The behavior can be explained by the well-established fact that for the EOQ model, the total cost is fairly insensitive to changes in the order quantity around its optimum Q_Z^* . Conversely, the total cost is highly sensitive to such changes when the order quantity is far from the optimum. This relationship between cost and emissions is formalized by Chen et al. (2013), which show that the relative reduction of the inventory-related emissions

$$\delta_{E'}(\delta_{Z'}, \alpha) = \frac{E'(Q_Z^*) - E'(Q(\delta_{Z'}, \alpha))}{E'(Q_Z^*)}$$

as a function of the relative increase in inventory-related costs

$$\delta_{Z'} = \frac{Z'(Q(\delta_{Z'}, \alpha)) - Z'(Q_Z^*)}{Z'(Q_Z^*)}$$

is

$$\delta_{E'}(\delta_{Z'}, \alpha) = \begin{cases} -\frac{(\delta_{Z'} - \sqrt{2\delta_{Z'} + \delta_{Z'}^2})(1 - \alpha + \delta_{Z'} - \sqrt{2\delta_{Z'} + \delta_{Z'}^2})}{(1 + \alpha)(1 + \delta_{Z'} - \sqrt{2\delta_{Z'} + \delta_{Z'}^2})}, & \text{if } \alpha < 1, \\ -\frac{(\delta_{Z'} + \sqrt{2\delta_{Z'} + \delta_{Z'}^2})(1 - \alpha + \delta_{Z'} + \sqrt{2\delta_{Z'} + \delta_{Z'}^2})}{(1 + \alpha)(1 + \delta_{Z'} + \sqrt{2\delta_{Z'} + \delta_{Z'}^2})}, & \text{if } \alpha > 1. \end{cases}$$

They also show that the maximum relative decrease in emissions compared to the cost-minimizing solution is

$$\delta_{E'} = (E'(Q_Z^*) - E'(Q_E^*)) / E'(Q_Z^*) = (1 - \sqrt{\alpha})^2 / (1 + \alpha)$$

which is attained with a relative increase in cost of

$$\delta_{Z'} = (Z'(Q_E^*) - Z(Q_Z^*)) / Z(Q_Z^*) = (1 - \sqrt{\alpha})^2 / 2\sqrt{\alpha}.$$

As mentioned earlier, multiple criteria decision-making (MCDM) techniques based on analyzing the efficient frontier, rendering expressions as the ones above, is one of several approaches used for combining emission and cost considerations in green inventory management. Other approaches, i.e., to translate the emissions into an environmental cost (e.g., in terms of an imposed carbon tax), to use an emissions constraint with a fixed cap, or to operate under cap-and-trade or cap-and-offset systems, have all been studied in the literature on green EOQ problems, for example, in Chen et al. (2013) and Hua et al. (2011). Some of these findings are summarized below.

If a tax of t is imposed for each emission unit, the average inventory-related cost per time unit is

$$Z'_t(Q) = Z'(Q) + tE'(Q) = \frac{(A + t\hat{A})D}{Q} + (h + t\hat{h})\frac{Q}{2},$$

and the associated optimal (cost minimizing) order quantity is

$$Q_t^*(t) = \sqrt{\frac{2(A + t\hat{A})D}{h + t\hat{h}}} \Rightarrow Z'_t(Q_t^*(t)) = \sqrt{2(A + t\hat{A})(h + t\hat{h})}D.$$

The order quantity $Q_t^*(t)$ is equal to the traditional cost-minimizing order quantity, Q_Z^* , if $t = 0$ (i.e., the tax rate is zero). Moreover, it approaches the emission-minimizing order quantity, Q_E^* , as $t \rightarrow \infty$ (from above when $\alpha < 1$ and from below when $\alpha > 1$). With $u = \hat{A}/A$ and $v = \hat{h}/h$, the relative shift in inventory-related costs and emissions as a function of the tax rate is

$$\frac{Z'_t(Q_Z^*) - Z'_t(Q_t^*(t))}{Z'_t(Q_Z^*)} = 1 - \frac{\sqrt{(1 + ut)(1 + vt)}}{1 + (u + v)t/2}$$

$$\frac{E'(Q_Z^*) - E'(Q_t^*(t))}{E'(Q_Z^*)} = 1 - \left(\frac{u}{u + v} \sqrt{\frac{1 + vt}{1 + ut}} + \frac{v}{u + v} \sqrt{\frac{1 + ut}{1 + vt}} \right).$$

Both these relative changes are increasing in the tax rate with the limiting value $(1 - \sqrt{\alpha})^2 / (1 + \alpha)$ as the tax rate goes to infinity (see Chen et al. (2013) for details). The relative cost increase compared to a system without any tax can be substantial and are equal to

$$\delta_{Z',t} = \frac{Z'_t(Q_t^*(t)) - Z'(Q_Z^*)}{Z'(Q_Z^*)} = \sqrt{\frac{(A + t\hat{A})(h + t\hat{h})}{Ah}} - 1.$$

However, most of this increase in cost is due to tax payments. The increase of the sum of holding and ordering costs is still limited to $(1 - \sqrt{\alpha})^2 / 2\sqrt{\alpha}$ in accordance with the findings for the efficient frontier.

If a cap, C , is imposed on the average emissions per time unit, we have

$$E'(Q) = \frac{\hat{A}D}{Q} + \hat{h} \frac{Q}{2} \leq C' = C - \hat{c}D.$$

This restricts the possible order quantities to

$$Q_L(C') \leq Q \leq Q_U(C')$$

where

$$Q_L(C') = \frac{C' - \sqrt{C'^2 - 2\hat{A}\hat{h}D}}{\hat{h}} \text{ and } Q_U(C') = \frac{C' + \sqrt{C'^2 - 2\hat{A}\hat{h}D}}{\hat{h}}$$

As $Z(Q)$ is convex in Q , the optimal (cost-minimizing) order quantity under a carbon constraint is

$$Q_C^*(C') = \begin{cases} Q_L(C') & \text{if } Q_Z^* < Q_L(C') \\ Q_Z^* & \text{if } Q_L(C') \leq Q_Z^* \leq Q_U(C') \\ Q_U(C') & \text{if } Q_Z^* > Q_U(C') \end{cases}$$

The inventory policy and the inventory-related emissions, $E'(Q_C^*(C')) = \min(E'(Q_Z^*), C')$, are independent of the cap as long as the constraint is not violated by the traditional cost-minimizing policy, Q_Z^* .

If a cap-and-trade system is used for regulating emissions, it is possible to purchase emission rights to relax the constraint or to sell unused emission rights if the firm's emissions are below the cap. If the buying and selling prices of the emission rights, p , are the same and potential revenue from trading is viewed as a negative cost, the average inventory-related cost under cap and trade is

$$Z'_{C\&T}(Q) = Z'(Q) + p(E'(Q) - C') = \frac{(A + p\hat{A})D}{Q} + (h + p\hat{h}) \frac{Q}{2} - pC'.$$

Comparing this cost expression with the one where an emission tax is applied, $Z'_t(Q)$, we can see that they are indeed very similar. The difference is that the price for emissions, p , has replaced the tax rate, t , and that an adjustment, pC' , is made as one only pays for the emission units in excess of the cap and receives revenues for emission units below the cap that are sold. Consequently, the optimal solution for a cap-and-trade system is also similar to an emission tax system:

$$\begin{aligned} Q_{C\&T}^*(p) &= \sqrt{\frac{2(A + p\hat{A})D}{h + p\hat{h}}} \Rightarrow Z'_{C\&T}(Q_{C\&T}^*(p)) \\ &= \sqrt{2(A + p\hat{A})(h + p\hat{h})D} - pC'. \end{aligned}$$

In fact, the EOQ model with an emission tax can be viewed as a special case of the same model with a cap-and-trade mechanism, with the cap set to zero and price of emission rights, p , set to the tax rate t . A major difference between a cap-and-trade system and tax system (also when $p = t$) is that the former implies less of a cost increase for the firms (except when $C' = 0$) as one gets a reimbursement equal to pC' . For some firms, the revenue from selling emission rights might even exceed their increase in inventory-related cost. This means they are better off under a cap-and-trade system than without any regulations.

Interestingly, from the expression for $Q_{C\&T}^*(p)$, we can see that there is no direct connection between the cap, C , and the optimal inventory policy. There should be an indirect connection though, as reduced caps imply an increased demand for emission rights. This should increase the price p and push $Q_{C\&T}^*(p)$ toward the emission-minimizing order quantity, Q_E^* . Some companies will benefit from a general reduction of the cap, whereas others will lose money.

A cap-and-offset system is similar to a cap-and-trade system, but it does not allow companies to sell emission rights. The optimal (cost-minimizing) policy in such a system is given by

$$Q_{C\&O}^*(p, C') = \begin{cases} Q_{C\&T}^*(p) & \text{if } Q_Z^* \leq Q_{C\&T}^*(p) \leq Q_L(C') \\ Q_L(C') & \text{if } Q_Z^* \leq Q_L(C') \leq Q_{C\&T}^*(p) \\ Q_Z^* & \text{if } Q_L(C') \leq Q_Z^* \leq Q_U(C') \\ Q_U(C') & \text{if } Q_{C\&T}^*(p) \leq Q_U(C') \leq Q_Z^* \\ Q_{C\&T}^*(p) & \text{if } Q_U(C') \leq Q_{C\&T}^*(p) \leq Q_Z^* \end{cases}.$$

This means that the cost-minimizing order quantity, Q_Z^* , is chosen if possible. Otherwise, the optimal order quantity is determined by the cap, via $Q_L(C')$ and $Q_U(C')$, or by the optimal cap-and-trade solution, $Q_{C\&T}^*(p)$. Note that $Q_{C\&T}^*(p)$ implies that emission rights are sold if $Q_L(C') < Q_{C\&T}^*(p) < Q_U(C')$. Hence, $Q_{C\&T}^*(p)$ is not a viable solution in a cap-and-offset system in this interval. The cap-and-offset system can be viewed as a special case of a cap-and-trade system with buying price of emission rights, p , and selling price of zero. Any cap and trade system with a difference in buying and selling price of emission rights will exhibit similar structure. Albeit with seven possible order quantities, as emission rights will be sold if the optimal order quantity when doing so falls between $Q_L(C')$ and $Q_{C\&T}^*(p)$, or $Q_U(C')$ and $Q_{C\&T}^*(p)$.

6.2.2 Extensions of the EOQ Model

Extensions of the single-echelon EOQ model with environmental considerations found in the literature include alternative cost structures, emission structures, and demand structures.

Several authors, including Battini et al. (2014), Bozorgi et al. (2014), and Konur and Schaefer (2014), consider nonlinear (stepwise increasing) ordering costs and emissions to capture the limited capacity of trucks and similar transportation resources. Problems with nonlinear ordering cost are not new, but one can argue that they are even more interesting under environmental considerations as much of the transport emissions are due to the movement of the actual vehicle. Konur (2014) elaborates further on this problem by integrating the design of the transportation fleet into the inventory management decisions.

Toptal et al. (2014) consider the EOQ model where there exists an alternative emission-reducing investment. In a cap-and-trade system, the optimal order quantity is still $Q_{C\&T}^*(p)$. The reason is that an emission reduction leads to the same savings or revenues no matter how it is achieved. In contrast, in a system with a fixed cap, the optimal ordering quantity and the alternative emission-reducing investment will be interdependent. To achieve an emission reduction, an adjustment of the order quantity will always be made. The alternative emission-reducing investment is used as a complement if the reduction required to reach the cap is substantial.

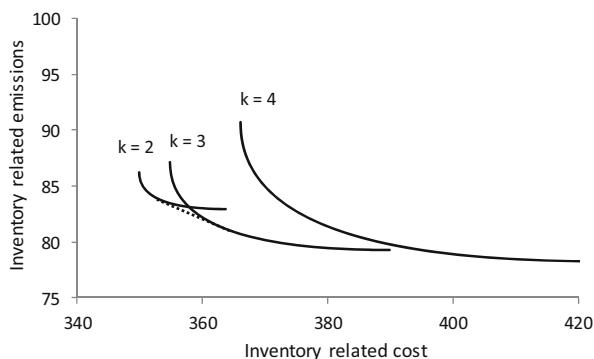
Inventory problems with deterministic nonstationary demand are considered, for example, in Benjaafar et al. (2013), Absi et al. (2013), and Retel Helmrich et al. (2015). Benjaafar et al. (2013) make a number of observations based on a series of numerical experiments that are in line with what can be deduced directly from the EOQ model discussed above. In addition, they point to the fact that if the emission cap is not based on the long run average but is imposed on a given period, then a tighter cap might paradoxically lead to higher total emissions (something that applies to the problem with stationary demand as well). Other alternative demand structures include price- and/or emission-dependent demand rates (e.g., Hovelaque and Bironneau (2015) and Taleizadeh et al. (2020)).

6.2.3 *Deterministic Multi-echelon Models with Environmental Considerations*

The literature on green inventory management in multi-echelon settings with deterministic demand is more limited. Bouchery et al. (2012) extend the analysis of the efficient cost emission frontier to a two-stage serial system. For a single-criteria problem, the optimal policy is a stationary-nested policy, i.e., the higher echelon always orders an integer multiple k of the lower echelon's order quantity, that fulfills the zero-inventory ordering property. This means orders are only placed so that the echelon inventory level just before delivery is equal to 0. For a multicriteria problem, the optimal policy is still nested but not necessarily stationary. This is illustrated in Fig. 6.2 that shows the frontier for the best combined policy as well as the efficient frontier of a number of stationary-nested policies.

The analysis in Bouchery et al. (2012) focuses on stationary policies arguing that such policies are easier to implement in practice. In the non-convex segments of the frontier of stationary policies, a better solution can be obtained using a linear

Fig. 6.2 Efficient cost emission frontier in a serial system (adapted from Bouchery et al. 2012). The solid lines depict the efficient frontier for stationary-nested policies with $k = 2, 3$, and 4, respectively. The efficient frontier for all policies is a combination of these and the dotted line which is a nonstationary policy



combination of two nested policies, represented by the dotted straight line in Fig. 6.2. This renders the true cost/emission frontier. Thus, in contrast to a single-echelon system, the optimal policy in a serial system is not stationary even under constant demand.

Bouchery et al. (2017) extends the previous model by considering nonlinear (stepwise increasing) ordering costs and emissions and limited shipping capacity, denoted $Q_{max,1}$ for shipments to the downstream echelon and $Q_{max,2}$ for shipments to the upstream echelon. They show that under decentralized control the optimal policy for the upper echelon might not fulfill the zero-inventory ordering property. More interestingly from an environmental perspective, they show that coordination from a cost perspective might increase the order quantity at the downstream facility and reduce it at the upstream facility. This implies higher utilization of the transport capacity and reduced number of transports between the two echelons. However, the number of transports to the upstream facility will increase due to the lower load factors ($Q_2/Q_{max,2}$) that come with the reduced order quantity at this stage. Whether cost-driven centralized solution will increase or decrease the total emissions will thus depend on which transportation leg has the largest emissions. A centralized solution with a more balanced aim showed great potential of reducing both the costs and the emissions.

Benjaafar et al. (2013) investigates the cost benefit of collaborating in a supply chain to meet emission caps. They show numerically that the relative benefits can be substantial, particularly for midrange emission caps. When the cap is high, it has little influence on the supply chain. If the cap is low, the cost for the supply chain to adhere to it will be very high, and the relative savings of collaborating to meet the cap will be low, even if the savings might be high in absolute numbers. They also show that, as always, it is important that the benefits of collaboration are not distributed in an unfair manner. Some parties might then prefer not to collaborate as this will increase their costs, even if the total cost for the supply chain decreases. Wahab et al. (2011) investigate supply chain collaboration in an EOQ setting considering imperfect items and environmental impact. The latter is captured as a tax on emissions, and they assume that there are no emissions linked to holding inventory. Their numerical examples show a reduction of the order quantity when

environmental considerations are taken. With more emissions linked to holding inventory, the results could, of course, be reversed.

6.3 Green Inventory Management Under Stochastic Demand

In this section, we extend the scope to green inventory management models that deal with demand uncertainty. In these models, the demand is represented by stochastic (or random) variables with known probability distributions. A stochastic variable can have many different outcomes, and the probability (or density, if it is a continuous variable) for each outcome is specified in a probability distribution. In reality, demand is almost always uncertain, and arguably one of the main reasons for stock keeping in practice is to buffer against this uncertainty and maintaining acceptable customer service. The importance of stochastic demand models in inventory management is therefore undisputed. In the following, we will consider a range of green inventory models available in the literature to illustrate and explain how the different challenges and trade-offs discussed in the beginning of this chapter are considered. We first focus on single-echelon models and then extend the scope to multi-echelon models involving networks of inventory facilities at given locations. (Green facility location is a separate topic examined in Chap. 7 by Velásquez Martínez and Fransoo (2024).) For reasons of space, some models are treated more briefly than others. This does not mean that they are less important or interesting. Mathematical details of more complex models are omitted.

6.3.1 Single-Echelon Models for Green Inventory Management

A single-echelon inventory model considers a single stock point in isolation. One of the simplest and most widely used single-echelon inventory models for stochastic demand is the newsvendor model. It considers a single product in a single period (or selling season) and may be appropriate for seasonal goods, fashion clothing, and high-tech products with short life cycles. Our exposé begins by considering green inventory management in newsvendor settings. The scope is then extended to more general inventory management situations with multiple periods and replenishment opportunities.

6.3.1.1 Newsvendor Models with Environmental Considerations

The classical newsvendor model assumes that the demand during the selling period is stochastic and at the end of the period there are overage (holding) costs associated with every unsold unit and underage (shortage) costs associated with every unsatisfied demand. The focus is on the decision of how many units to order

for the coming period, to minimize the expected overage (holding) and underage (shortage) costs. An alternative objective is to maximize the expected profit for the considered period by also taking the purchase price and selling price per unit into consideration.

To illustrate how the newsvendor framework can be used for green inventory management, we consider a dual-sourcing model analyzed in Rosič and Jammerneegg (2013) and Arıkan and Jammerneegg (2014). Together, these two papers consider costs and emissions associated with ordering, producing, and transporting items and with holding items in stock. They also consider different types of emission regulations and emission constraints for a profit-maximizing firm. More precisely, Rosič and Jammerneegg (2013) investigate the economic and environmental impact of dual sourcing for three scenarios of emission regulations: (i) no regulations, (ii) an emission (or carbon) tax, and (iii) a cap-and-trade system with fixed price emission allowances. In their model, a profit-maximizing firm has the option to source from a low-cost offshore supplier before a given selling season starts or to use a more expensive, fast, onshore supplier, when demand has been realized at the end of the selling season. Transporting goods from the offshore supplier is associated with transport emissions, while transport emissions from the onshore supplier are negligible. During the selling season, there is a stochastic demand for X units, and the firm sells the product at unit price p . Unsold products (in inventory) at the end of the season have a per unit salvage value z , which includes the inventory holding costs. The firm procures q units from the offshore supplier at the unit price c before the season starts and the rest from the onshore supplier at unit price $c + d$ at the end of the season when the demand is known. It is assumed that $p > c > z$ and $p > c + d$ so that the onshore supplier is an option to consider. Thus, if a demand of x units is realized, $\max(x - q, 0) = (x - q)^+$ units are purchased from the onshore supplier. This corresponds to the underage or lost sales in the classical newsvendor model. Moreover, the underage cost is d , the overage cost is $c - z$, and the firm's profit function, given q and x , is

$$P(q, x) = \begin{cases} (p - c)x - (c - z)(q - x) & \text{if } x \leq q \\ (p - c)x - d(x - q) & \text{if } x > q \end{cases}$$

The expected profit, given the offshore order quantity q , is obtained by taking the expectation with respect to the demand X :

$$P(q) = E[(p - c)X - (c - z)(q - X)^+ - d(X - q)^+].$$

Letting $F(x)$ denote the cumulative distribution function (cdf) of the continuous demand X and $F^{-1}(\cdot)$ its inverse, it follows from the classical newsvendor analysis that the optimal offshore order quantity is

$$q^* = F^{-1}\left(\frac{d}{d + c - z}\right).$$

Introducing a transportation emission tax, t , per unit shipped from the offshore supplier increases its per unit costs to $c + t$. This decreases the optimal amount procured offshore, q^t , and thereby the transport emissions, but it also decreases the expected profit:

$$P(q) = E[(p - c - t)X - (c + t - z)(q - X)^+ - (d - t)(X - q)^+].$$

$$q^t = F^{-1}\left(\frac{d - t}{d + c - z}\right)$$

As an alternative to the linear emission tax, the authors also consider an emission trading system, where the company, free of charge, receives emission allowances corresponding to an emission cap of L products transported from the offshore supplier. The firm can buy more emission allowances at the per unit price b and sell allowances it does not use at the per unit price s ($b \geq s$). This means that the transport emission costs are $b(q - L)^+$, the revenue of selling unused emission allowances is $s(L - q)^+$, and the expected profit for emission cap L and offshore order quantity q is

$$P_L(q) = \begin{cases} P_s(q) & \text{if } q \leq L \\ P_b(q) & \text{if } q > L \end{cases}, \text{ where } \begin{cases} P_s(q) = P(q) + s(L - q) \\ P_b(q) = P(q) - b(q - L) \end{cases}.$$

It is straightforward to show that the optimal offshore order quantity for an emission cap of L , q^L , is obtained as a two-sided control limit policy:

$$q^L = \begin{cases} q^b & \text{if } L < q^b \\ L & \text{if } q^b \leq L \leq q^s \\ q^s & \text{if } L > q^s \end{cases}.$$

Here, q^b and q^s maximize $P_b(q)$ and $P_s(q)$, respectively, and are easily obtained as

$$q^b = F^{-1}\left(\frac{d - b}{d + c - z}\right) \text{ and } q^s = F^{-1}\left(\frac{d - s}{d + c - z}\right)$$

Note that $b \geq s$ implies that $q^b \leq q^s$ and that $s = b = 0$ brings us back to the basic dual-sourcing model without emission regulations. Moreover, $L = 0$ and $b = t$ render the emission tax model.

From the above expressions, one can conclude that $q^L < q^*$, meaning that the offshore order quantity and thereby the emissions are reduced through the emission trading system. One can also see that the smallest offshore order quantity under the emission trading system is q^b . Thus, setting the cap, L , lower than this will not reduce the transport emissions; further, it will only decrease the firm's profit.

In a numerical study based on normally distributed demand, the authors illustrate that dual sourcing as described in this stylized model can reduce emissions and improve economic performance in comparison to single sourcing from an offshore supplier. By implementing environmental regulations in terms of a linear transport emission tax or a cap-and-trade system, the transport emissions can be further reduced. Both an emission tax and a cap-and-trade system are in the context of this model very effective tools to reduce emissions, but the emission tax tends to affect the firm's profit more severely. Under the cap-and-trade system, profits can even increase, as the firm may generate extra revenues by selling emission allowances it has received for free but do not need. These results indicate that from the firm's perspective a cap-and-trade system is preferred over an emission tax.

Arikan and Jammernegg (2014) consider the same dual-sourcing model but when a product carbon footprint (PCF) constraint, set voluntarily or by regulators, is used to limit the emissions. Hence, the management objective is to maximize the profit, $P(q)$, subject to the PCF constraint. The carbon footprint may be used to differentiate the firm's products, for example, by displaying it on an eco-label. The emissions considered in the model are production, transportation, and warehousing activities. Emissions caused by usage and consumption by the customers are excluded. The emissions related to holding items in stock (overage), including disposal of products, shipping them to secondary markets, etc., are modeled explicitly by the use of e^L , the per unit emissions for surplus inventory at the end of the period. Similarly, the target emission per unit is denoted e and the per unit emissions from the first (offshore) and second supply options are denoted e_1 and e_2 , respectively. Let $L(q)$ denote the expected overage given the offshore order quantity q and $Q(q)$ the expected underage:

$$L(q) = E[(q - X)^+] = \int_0^q (q - x) f(x) dx,$$

$$Q(q) = E[(X - q)^+] = \int_q^\infty (x - q) f(x) dx$$

The expected total emissions, $E(q)$, for the offshore order quantity q , are determined as

$$E(q) = e_1 q + e_2 Q(q) + e^L L(q).$$

Note that the model now considers emissions both from the first supply option (the offshore supplier), from the second supply option (which may be an onshore supplier), and from items in stock at the end of the selling period. The latter includes disposal of the unsold products.

The inventory management problem is to maximize the expected profit $P(q)$, defined as above, subject to the PCF constraint, $E(q) \leq B(q)$, where $B(q) = e(q + Q(q))$. Note that this PCF constraint concerns the expected emissions

and that the upper bound $B(q)$ is linear in the total order quantity ($q + Q(q)$) which corresponds to the total demand. (An alternative, for example, used in Song and Leng (2012) is to set a constant upper bound on the emissions independent of q , say, B . This is then analogous to a capacity constraint.) Under reasonable parameter values, the optimal offshore order quantity under the PCF constraint, q_{CF} is then

$$q_{CF} = \begin{cases} q_{l1} & \text{if } q^* < q_{l1} \\ q_{l2} & \text{if } q^* > q_{l2} \\ q^* & \text{otherwise.} \end{cases}$$

The lower and upper control limits, $q_{l1} < q_{l2}$, define the feasible region and are obtained as the intersection points of $E(q)$ and $B(q)$. Note however that the lower limit q_{l1} might be nonexistent, for example, if the condition $e_1 \leq e \leq e_2$ is not fulfilled. Recall that $q^* = F^{-1}(dl(d + c - z))$ is the unconstrained optimum to max $P(q)$ defined above.

Based on this analysis, Arıkan and Jammernegg (2014) consider two situations for the second supply option: (i) an environmentally friendly but expensive onshore supplier (similar to Rosi  and Jammernegg (2013) above) and (ii) a fast emergency transport option from the offshore supplier. The onshore supplier is characterized by lower per unit emissions than the offshore alternative (motivated by less transport emissions), i.e., $e_2 < e_1$. Conversely, fast emergency shipments from the offshore supplier suggest higher per unit emissions (and costs) than its regular shipments (e.g., air vs. sea) with $e_2 > e_1$.

For the onshore option with $q_{CF} = \min \{q^*, q_{l2}\}$, one can conclude that q_{CF} decreases if the per unit emission e_1 , e_2 , or e_L increases or if e (the per unit emission target) decreases. Thus, if the per unit emissions increase or if a tougher emission target is introduced, the firm buys less from the offshore supplier and more from the environmentally friendly onshore supplier. Recall that the model presumes that all demand is satisfied.

For the fast emergency transport option from the offshore supplier, the behavior of q_{CF} with respect to e_1 , e_2 , e_L , and e is the same as long as e_2 is smaller than e (i.e., the per unit emissions for emergency shipments are below the per unit target). In case $e_2 > e$, the behavior depends on the critical ratio $dl(d + c - z)$, and q_{CF} may either decrease or increase.

From a numerical study, the authors make a number of observations about trade-offs between the sourcing options, costs, and emissions. For example, within large ranges of the emission and cost parameters, the alternative with an onshore supplier offers significant reductions in emissions with only small reductions in expected profits. Moreover, if the onshore option offers lower per unit costs than the emergency shipment option, it is better both from environmental and economic perspectives. On the other hand, if the per unit cost for the onshore option is much higher than for the emergency shipment option, onshoring may lead to both higher costs and emissions. The latter is due to the model requirement of satisfying all demand. To avoid using the expensive onshore option (which must be used if there

are shortages), q_{CF} is increased significantly, which leads to more emissions for producing and transporting ($e_1 q_{CF}$) but also for handling and disposing of more units that are left over in inventory ($e_L L(q_{CF})$).

Finally, let us briefly summarize some of the other works based on the newsvendor framework in the green inventory management literature. Song and Leng (2012) consider the classical newsvendor model under three types of carbon emission regulations: a mandatory carbon emission constraint, a linear carbon emission tax, and a cap-and-trade system. Choi and Chiu (2012) consider a newsvendor model for a risk-averse fashion retailer where any items in stock at the end of the selling period is obsolete and wasted. They establish mean-variance (MR) and mean-downside risk (MDR) models and compare their levels of sustainability to the classical risk-neutral retailer, which maximizes the expected profit. The considered measures of sustainability are expected number of unsold items in stock, expected-sales-to-expected-coverage ratio, rate of return on investment, and the probability of achieving a predetermined profit target. They find that for these measures, the risk-averse MR retailer is more green than a risk-neutral retailer. Choi (2013) consider a dual-supply model in a newsvendor framework for a fashion retailer employing a quick response system. In contrast to Rosič and Jammerneegg (2013) above, both supply options (i.e., orders both to the offshore and onshore suppliers) occur before the selling season starts. Between the time of the first ordering opportunity (with the offshore supplier), O , and the time of second-order opportunity (with the local supplier), L , it is assumed that the fashion retailer can improve its demand forecast. The analysis shows how a properly designed per unit tax can be used to enhance the environmental sustainability by sourcing more (or solely) from the local (onshore) supplier with less emissions. However, the emissions are not modeled explicitly.

6.3.1.2 Multi-period Models with Environmental Considerations

The literature on inventory management of stochastic multi-period single-echelon inventory systems is vast. However, there are relatively few papers on green inventory management approaches that explicitly consider emissions or environmental concerns. Issues emphasized in this literature are impact of emission reduction approaches, both regulatory and voluntary, particularly with respect to modal choice and shipment consolidation. The focus on shipment-related decisions is logical as transportation is one of the major sources of emissions in logistics.

Hoen et al. (2014) consider a single-item, single-echelon, infinite horizon, periodic review inventory model, where replenishments are made in the beginning of each period according to an order-up-to-S (or base-stock) policy. Demand is assumed to be continuous and stochastic, and all unsatisfied demand is backordered. In this modeling context, the authors study how carbon emission regulations can affect a firm's choice of transportation mode for inbound deliveries. More precisely, the firm can choose between transport mode options with different constant lead times (L_i , $i = 1, 2, \dots, I$), costs, and emission characteristics. The firm can only choose one option involving a single transport mode (i.e., terminal-to-terminal

transport is assumed). The considered cost components for transport mode i are a backorder cost per unit and time unit (p_i), a holding costs per unit and time unit (h_i), and purchase and transportation costs per unit (c_i). The holding and transportation costs depend on the emissions for transport mode i via an emission cost, c^e , per unit of emissions. This emission cost can reflect an emission regulation, for example, in terms of a carbon tax. The emissions associated with shipping one unit on transport mode i , e_i , depend on the transportation distance, constant and variable emission factors, and load factors. The expected costs $C_i(S_i | c^e)$ for transport mode i , with base-stock level, S_i , and emission cost, c^e , can be expressed as

$$C_i(S_i | c^e) = h_i(c^e) [S_i - \mu(L_i + 1)] + (p + h_i(c^e)) G_i(S_i) + c_i(c^e) \mu$$

where

$h_i(c^e)$ = holding cost per unit and time unit for transport mode i and emission cost c^e .

$c_i(c^e)$ = transportation cost per unit for mode i and emission cost c^e .

$f_i(x)$ = density function of the demand for $L_i + 1$ periods.

$G_i(y) = \text{loss function} = \int_y^\infty (x - y) f_i(x) dx$.

μ = expected demand per period.

The optimal base-stock level, S_i , that minimizes $C_i(S_i | c^e)$ coincides with the solution to the traditional newsvendor problem (see, for example, Axsäter (2015)). The transport mode that minimizes the expected costs for a given emission cost, c^e , is easily obtained by comparing the cost-minimizing solutions for each transport mode:

$$\min_{i \in I} \left\{ \min_{S_i} C_i(S_i | c^e) \right\}$$

The authors investigate how the choice of transportation mode depends on the emission cost by deriving intervals for c^e where different modes are preferred. These intervals indicate how high a carbon tax needs to be in order for a certain transport mode to be selected. From a small numerical study, based on realistic values for cost parameters and emissions, they conclude that the emission cost typically needs to increase drastically for a greener transport mode to be preferred. This suggests that emission taxes are blunt weapons for achieving emission reductions through modal shifts. On the other hand, introducing emission constraints that force a change may lead to significantly higher costs. However, the authors are careful to note that in practice many other issues outside their model may influence the choice of transportation mode.

Tang et al. (2018) investigate how some of the most common approaches for reducing carbon emissions (carbon tax, cap and trade, constraints in the form of emission reduction targets, and carbon offsets) may be analyzed in the context of a classical single-echelon, continuous review (R, Q) system with complete backordering and normally distributed lead time demand. The use of a continuous review (R, Q) policy for inventory control means that an order of Q units are placed

as soon as the inventory position, i.e., the stock on hand plus outstanding orders minus backorders, reaches the reorder point R . Moreover, complete backordering means that all demand that cannot be satisfied directly from stock on hand is back-ordered and satisfied according to a first-come first-served (FCFS) sequence as units become available. The (R, Q) policy is commonly used in practice and in the literature, and it is closely related to principle of the deterministic EOQ model of ordering batches of Q units. The authors analyze the total costs and emissions in this stochastic inventory system, for the considered emission reduction approaches. The analysis assumes that the emission structure mirrors the cost structure with a fixed component associated with every batch Q and a variable part proportional to the number of items in stock.

Using the notation for the EOQ model in Sect. 6.2.1, the expected costs per time unit for such a policy (see, for example, Axsäter (2015)) can be expressed as

$$Z(R, Q) = \frac{AD}{Q} + h \cdot E[IL^+] + b \cdot E[IL^-] + cD = Z'(R, Q) + cD$$

where D represent the expected demand rate, $E[IL^+]$ is the average amount of inventory on hand, b is the backorder cost per unit and time unit, and $E[IL^-]$ is the average amount of backorders. It is well known that $Z(R, Q)$ is jointly convex in R and Q and that the optimal solution can be found through an efficient iterative procedure (see, for example, Axsäter (2015)).

The expected emissions can, in a similar fashion, be expressed as

$$E(R, Q) = \frac{\hat{A}D}{Q} + \hat{h} \cdot E[IL^+] + \hat{b} \cdot E[IL^-] + \hat{c}D = E'(R, Q) + \hat{c}D$$

Tang et al. (2018) focus on transport emissions associated with ordering and emissions for holding items in stock and do not include emissions linked to backorders (associated with not satisfying customer demand on time), which means that $\hat{b} = 0$ in their models. As a consequence, the reorder point, R , is restricted to nonnegative values (i.e., $R \geq 0$). Without this restriction, the emission-minimizing solution for $\hat{b} = 0$ would be an infinitely small negative R and an infinitely large Q .

The optimal inventory policy under an emission tax, t , can then be attained by minimizing the tax-adjusted total costs, $Z'_t(R, Q)$, with respect to R and Q , where

$$\begin{aligned} Z'_t(R, Q) &= Z'(R, Q) + tE'(R, Q) = \frac{(A+t\hat{A})D}{Q} \\ &\quad + (h + t\hat{h}) \cdot E[IL^+] + (b + t\hat{b}) \cdot E[IL^-], \end{aligned}$$

Because $Z'_t(R, Q)$ has the same structure as $Z(R, Q)$, the iterative procedure for minimizing the latter (see Axsäter (2015)) can be used also for finding the optimal solution that minimizes the former. The same analysis applies for a cap-and-trade scheme. Tang et al. (2018) also suggest new algorithms for finding the

optimal inventory policy when (voluntary) emission reduction targets are set as hard constraints and when this is combined with carbon offset opportunities.

A numerical study is used for comparing the performance of the considered emission reduction approaches. Based on this, the authors conclude, among other things, that moderate emission reduction targets can be reached without significant cost increases and carbon offset opportunities improve the situation further. Achieving the same emission reductions through a carbon tax or cap-and-trade mechanism leads to much higher costs for the company, although, among the two, the cap-and-trade mechanism has the potential to achieve the same reductions at lower costs. From a policy perspective, the authors also conclude that achieving substantial emission reductions through a carbon tax or cap-and-trade mechanism would require extremely high taxes or prices for tradable emission rights. Thus, similar to Hoen et al. (2014), they conclude that emission taxes (and cap-and-trade schemes) are relatively ineffective approaches to achieve substantial emission reductions.

Turning to issues of shipment consolidation and joint deliveries, Larsen and Turkensteen (2014) analyze a VMI (vendor-managed inventory) system by jointly modeling shipment consolidation, routing, and retailer inventory decisions using a JRP (joint replenishment problem) approach. A vendor with ample stock is supplying a product to a large number of geographically dispersed retailers. The objective is to minimize the total inventory and transportation costs. The retailers are assumed to be identical and uniformly distributed over a two-dimensional (circular) area or a one-dimensional line, for example, describing a transportation corridor. The retailers, experiencing Poisson-distributed demand, are grouped into zones, where each zone is replenished by joint deliveries on a single vehicle. Each retailer incurs holding and backorder costs and uses an order-up-to-S policy to replenish its inventory.

The replenishment orders from zone j accumulate at the vendor, and corresponding stock units are reserved and placed in a transportation bin. When the level V_j is reached (i.e., the bin is full), a dispatch is made (at the end of the considered time period). Hence, a quantity-based shipment release policy is used. The holding costs for units in the transportation bin are attributed to zone j . The transportation costs include a fixed cost per dispatch, a cost per distance unit traveled, and a fixed cost per retailer visited. As demand is stochastic, a shipment may contain replenishments to only a subset of the retailers in zone j . Because retailers are identical, it is easy to determine the probabilities for m retailers in zone j being replenished by a shipment. The expected length of a delivery route can then be estimated using continuous approximations. The detailed specification of the transportation costs enables an analysis of how the inventory decisions affect the vehicle utilization and transportation distances. Turkensteen and Larsen (2013) build on these results, and in conjunction with an engine emission model, they provide a method for evaluating the expected transport-related CO₂ emissions for the VMI system at hand. A numerical study illustrates that as the fixed dispatch costs increase, the CO₂

emissions decrease. This is a result of using shorter less frequent delivery routes (smaller zones) with increased load factor and comes at the price of more inventory.

Shipment consolidation may be done in many different ways. However, most policies studied in the literature can be categorized as either quantity-based policies (the policy above in Turkensteen and Larsen (2013) is one example), time-based policies, or hybrid policies (time- and quantity-based). There is a large body of literature on inventory management of single-echelon systems, analyzing different types of shipment consolidation (or release) policies belonging to these categories (e.g., Çetinkaya and Lee (2000), Axsäter (2001), Çetinkaya and Bookbinder (2003), Chen et al. (2005), Çetinkaya et al. (2006), Çetinkaya et al. (2008), Mutlu and Çetinkaya (2010), Mutlu et al. (2010), Kaya et al. (2012), and Satır et al. (2018)). Typically, these papers study the inventory management of a vendor, which dispatches consolidated shipments to a number of retailers with the objective to minimize holding and transportation costs (the inventory management decisions at the retailers are not included in the models). Emissions or environmental effects are in general not explicitly evaluated in this literature. An exception is Merrick and Bookbinder (2010) that use simulation to investigate how three commonly used shipment release policies can affect transport emissions. However, arguably, many of the cost models in this existing literature could be extended to include evaluations of expected emissions, particularly if the transport emissions can be modeled analogous to the transportation costs, i.e., by a fixed component per dispatch and a variable component proportional to the number of units on the shipment.

Another stream of literature of interest from a green inventory management perspective is the one focusing on inventory management of perishable products. Extensive reviews are provided in Nahmias (2011) and Karaesmen et al. (2011). A perishable item is characterized by having a limited life span during which it has a constant value (or utility). When that life span ends, the item loses its value and needs to be disposed of. This may fit many different types of products, but from an environmental perspective, food and pharmaceuticals are important examples where the disposal of outdated and wasted products may have a big financial and environmental impact, not only in terms of GHG emissions. The existing models for inventory management of perishable items (excluding the trivial special case of the newsvendor model) focus on cost minimization or profit maximization and do not, to our knowledge, explicitly consider emissions or environmental consequences. However, arguably, it is easy to incorporate evaluation of these sustainability issues as the analysis generally renders the expected number of perished items associated with certain inventory policy decisions. The challenge to reconcile the economic and environmental objectives and determine the best decision from a sustainable inventory management perspective still remains though. The perishable item literature is because of its complexity focused on inventory management of single-echelon systems. However, considering multi-echelon systems, as in Olsson (2010), is an interesting venue for future research in green inventory management.

6.3.2 *Stochastic Multi-echelon Models with Environmental Considerations*

In multi-echelon inventory management, the scope is extended from managing a single stock-point to jointly managing a system of connected inventory facilities, typically, upstream and downstream in a supply chain. The simplest multi-echelon structure is a serial system where each inventory location has exactly one predecessor and one successor. In distribution of finished goods, the divergent structure, where each inventory location has one predecessor but many successors, is very common. Conversely, assembly systems, where each inventory location can have many predecessors but only one successor, is commonly found manufacturing. Managing a network of connected inventory locations clearly increases the complexity but also the degrees of freedom and potential for green supply chain solutions. The substantial literature on stochastic multi-echelon inventory management still contains relatively few models that explicitly incorporate emissions or environmental consequences, but the field is growing.

Berling and Martínez-de-Albéniz (2015) use a continuous serial system (i.e., a serial system with infinite number of stages) to investigate the economic and environmental benefits of adjusting the speed at which a unit is transported through the system, based on the current inventory situation. This can be interpreted as the value of slow steaming or despeeding of the supply chain when the situation allows it. Applying the model to a road transport case study illustrates that substantial savings in both cost and emissions can be attained by adjusting the speed to the inventory situation rather than using a fixed speed policy (savings of 7% and 20%, respectively, were recorded).

Stenius et al. (2018) present a method for green inventory management of a distribution system with a central warehouse supplying goods to a number of retailers using a time-based shipment consolidation policy with periodic shipment intervals. In accordance with a VMI agreement, the central warehouse, which uses a continuous review (R, Q) policy to control its inventory, has access to inventory and point-of-sale information for the entire system in real time through an integrated IT system. The free flow of demand information means that there are no incentives for batch ordering at the retailers, which therefore behave according to continuous review base-stock policies. However, fixed costs for handling and shipping, together with ambitions to reduce transport emissions, create strong incentives for shipment consolidation and joint deliveries to groups of retailers. The presented model extends the work in Marklund (2011) by enabling exact evaluation of load-dependent nonlinear shipment costs and emissions. A key result is the derivation of the probability distribution for the number of units on each shipment. Based on these results, a setting emphasizing the model's usefulness from a sustainability perspective is further analyzed. It is motivated by industry applications and involves capacitated dual-transport options from the central warehouse to the retailers. Transportation capacity is reserved in advance on an intermodal shuttle train (or alternatively a fleet of modern low-emission trucks). If this capacity is

insufficient when a shipment is to be dispatched, truck transports available on demand (typically a more emission-intensive option) are used as a complement. The analysis, assuming Poisson demand, shows how to jointly optimize the reorder levels (at the warehouse and all retailers), the shipment intervals (to all retailer groups), and the capacity reservation quantities (for all shipments) to minimize the total expected costs. Emissions are considered through a constraint on the total expected emissions or alternatively by introducing emissions costs. The analysis is applicable to both single- and multi-item systems. A numerical example, based on realistic cost and emission data, illustrates how the model can be used for evaluating the cost impact of managerial decisions to reduce emissions. It shows that large reductions of the expected emissions can be achieved without severe cost increases. However, minimizing the emissions may increase the costs significantly. For instance, reducing the CO₂ emissions by 24% compared to the cost optimal solution only increases the expected costs by 3.5%. To achieve this, the shipment intervals (time between shipments) are increased, and more capacities are reserved on the train, resulting in an additional 10% of the total shipment volume to be moved from truck to train. The inventory is also increased to compensate for the longer lead times. On the other hand, minimizing the expected emissions leads to a maximum reduction of 34.5% compared to the cost optimal solution, while this causes the expected costs to increase by 17.5%.

Malmberg and Marklund (2023) analyze a continuous review, one-warehouse-multiple-retailer (OWMR) inventory distribution system, with quantity-based shipment consolidation to groups of nonidentical retailers facing Poisson demand. This means that shipments from the warehouse to a group of retailers may be restricted to full truckloads or containers to maximize the utilization of the transportation capacity and thereby reduce the transport emissions. Similarly, to the model in Stenius et al. (2018), the warehouse has free access to inventory and point-of-sale information for the entire system through an integrated IT system. Thus, there are no incentives for batch ordering at the retailers, which therefore use base-stock policies for controlling their inventories. The warehouse, on the other hand, orders from an outside supplier and manages its inventory using an (R, Q) policy. The authors present a method for exact analysis of the inventory-level distributions at all stock points, which allows them to evaluate the system's total expected costs and transport emissions for a given set of reorder levels, order and shipment quantities. They also show how to jointly optimize the shipment quantities and inventory reorder levels in the system to minimize the expected costs when there are strict emission constraints to adhere to. Alternatively, emissions may be included in the model as environmental costs. A numerical study illustrates that the method offers opportunities for achieving large reductions in transport emissions by increasing the shipment quantities beyond the unconstrained cost optimal solutions with only small increases in total costs. The explanation is that the total cost is insensitive to the exact choice of shipment quantities when the reorder levels at the warehouse and the retailers are optimized accordingly. Conversely, if the reorder levels are not carefully optimized for the shipment quantities used, the cost impact of reducing transport emissions through larger consolidated shipments can be significant. The

results emphasize the importance of jointly considering shipment and inventory decisions for obtaining economically and environmentally sustainable distribution systems. Examples of other papers that study joint optimization of inventory and shipment consolidation in multi-echelon inventory systems, but without explicitly incorporating transport emissions, include Marklund (2011), Howard and Marklund (2011), Stenius et al. (2016), and Johansson et al. (2020). All these papers focus on time-based shipment consolidation policies.

So far, there are few models that incorporate emissions associated with not satisfying customer demand on time; still in some cases, these emissions may be quite significant and most relevant to consider. One area where the impact of insufficient service often is quite severe is in the distribution of spare parts. The parts are often quite small suggesting quite limited emissions associated with transportation, inventory holding, and warehousing. However, not delivering them promptly may have serious consequences on costs and emissions. The former is accentuated by an increasing use of service contracts with high penalty costs and liability clauses in case of insufficient service fulfillment. A related issue is the use of emergency shipments to reduce the impact of shortages and associated downtime costs commonly seen in practice.

Motivated by collaboration with a spare part service provider for the dairy industry, Johansson and Olsson (2017) consider a model where failure to provide a spare part within a relatively short time limit after a machine breakdown leads to large environmental and economic consequences. The reason is that the entire production batch of milk or yogurt must be discarded and the facility must be cleaned before production may be resumed. The analysis deals with a one-warehouse- N -retailer inventory system where all stock points use continuous review base-stock policies to manage their inventories, which is often reasonable for spare parts with low demand. The authors derive exact closed form expressions of the inventory-level distributions and the customer waiting time distributions for all stock points. Based on this, they provide methods for evaluating and minimizing the expected holding and backorder costs by optimizing the base-stock levels at all facilities. The methods allow for nonlinear backorder costs that are function of the waiting times or time window service-level constraints. They also enable evaluation of the expected emissions associated with not satisfying the demand within a given time window. A numerical study illustrates that the emissions associated with the product waste of milk caused by not satisfying spare part demand on time can be significant even for small probabilities of exceeding the acceptable waiting time.

Johansson and Olsson (2018) extend the work of Johansson and Olsson (2017) by introducing the possibility for the retailers to request emergency shipments from an external supplier if a spare part demand cannot be satisfied within the predetermined time window. However, it is not certain that the emergency supplier has enough capacity at the moment when the order was requested. This means that it is not always possible to orchestrate an emergency delivery within the given time window. The uncertainty is modeled by a fixed probability that determines the fraction of requested shipments that cannot be carried out in time. Using an age-based inventory policy, the authors derive the stationary distribution of the age of

the oldest units at all locations, given the emergency replenishment policy. Based on this, approximate expressions for the total costs and emissions are provided together with an optimization procedure for determining the base-stock levels in the system. A numerical study shows that the obtained base-stock levels are near optimal. It also illustrates that careful use of more emission-intensive emergency shipments to complement regular replenishments may decrease the total expected CO₂ emissions. The transport emissions increase when the emergency shipments are introduced, but the expected emissions of the product waste for not satisfying the spare part demand on time decrease more. This emphasizes the importance of taking a holistic view on CO₂ emissions in green inventory management and logistics and not focus solely on transport emissions.

As indicated above, when shortages occur, it is often rational to use emergency shipments, even though they are generally more expensive and/or emission-intensive than regular shipments (at least if they are supplied from the same location as regular orders). The increased emissions may be caused by the use of faster and more emission-intensive modes of transportation (e.g., air instead of road or sea) or by reduced load efficiencies in vehicles used for emergency deliveries. As illustrated by Johansson and Olsson (2018), careful use of emergency shipments can lead to lower total costs and emissions. However, the excessive use of emergency shipments to compensate for inefficiencies in the inventory management and regular replenishment system typically leads to higher costs and emissions. Thus, avoiding *unnecessary* use of emergency shipments is an important aspect of green inventory management. A first step in this direction is to assure that the inventory control system provides the intended service. Berling and Marklund (2014b) use simulation to study this issue at a Scandinavian spare parts provider for agricultural machinery. The company's distribution system consists of a central warehouse (located in the south of Sweden), and a large number of retailers/dealers spread out across the Scandinavian countries. The objective is to investigate the potential of reducing total costs and transport emissions. This is done by simulating the system when applying the multi-echelon inventory control methods presented in Berling and Marklund (2013 and 2014a) instead of the methods currently used by the company. The simulation study, encompassing a stratified sample of 100+ representative items, shows that by using the reorder points obtained from the multi-echelon method, the transport-related CO₂ emissions were reduced by 57%, the inventory holding costs decreased by 18%, and the demand-weighted average fill rate across retailers and items increased by 34%. These results illustrate that avoiding unnecessary shortages, by implementing multi-echelon inventory control methods with better precision, can be a first low-hanging fruit in achieving more sustainable inventory management in practice.

Another aspect, discussed above, is that many existing models for evaluation and minimization of expected costs have a potential for incorporating evaluation of emissions or environmental aspects to some extent. We make no attempt to review the multi-echelon literature from this perspective, but we would like to highlight two areas with apparent sustainability potential: the use of advance demand information

and systems that allow for lateral transshipments or the use of emergency/quick response stocks.

Use of advance demand information essentially means that customers place (firm) orders before they need the products. The time between the order placement and demand realization represents a time buffer that offers possibilities to reduce inventories and/or use greener but slower transportation options. The use of different shipping alternatives by Amazon is one example motivating the work in Chen (2001). Other examples from the multi-echelon inventory literature on advance demand (or advance order) information include Özer (2003), Gallego and Özer (2003), and Marklund (2006). To our knowledge, the multi-echelon literature does not yet contain any sustainable inventory management models that explicitly consider advance demand (or advance order) information in conjunction with emissions or environmental consequences.

Inventory systems with lateral transshipments (sometimes referred to as complete pooling models) are characterized by multiple retailers sharing their inventory by agreeing to transship stock between them when needed. For a literature overview, we refer to Paterson et al. (2011). From an emission point of view, lateral transshipment systems offer interesting opportunities to combine short-distance transshipments by environmentally friendly electrical or biogas-powered vehicles, with long distance consolidated shipments by diesel truck, train, or ship to a geographical region. A related system structure seen in practice is one with partial pooling, where designated support warehouses with quick response stocks offer local warehouses fast emergency deliveries to avoid shortages (e.g., Kranenburg and van Houtum (2009), Axsäter et al. (2013), and Howard et al. (2015)). In principle, they offer the same possibilities to combine consolidated deliveries to a region with fast short-distance emergency shipments. The managerial complexity to coordinate decisions is reduced in these systems at the expense of reduced potential for inventory pooling. To our knowledge, no complete or partial pooling models that explicitly consider emissions are yet available in the multi-echelon literature.

6.4 Concluding Remarks About Findings, Practical Implications, and What Remains to Be Done

A recurring finding throughout this chapter is that models and methods for green inventory management can offer opportunities to substantially reduce emissions (from a cost-minimizing solution) with relatively small increases in total costs. However, attaining solutions that minimize emissions may be costly.

Several case studies have shown through simulation of real data that companies can improve the cost and service performance of the inventory management system by using better inventory control methods. More recent studies illustrate that this finding carry over to the environmental performance as well. A common problem in practice is that the inventory management system does not deliver the intended

service. This leads to additional shortages, and if these shortages leads to, for example, emergency shipments or goods being wasted, the additional emissions can be significant. Examples from work cited above show reductions of expected costs of 18% along with emission reductions of up to 57% linked solely to improved inventory control and better fulfillment of target service levels, not involving changes to order quantities, shipment frequencies, mode of transport, shipment consolidation policies, etc. This illustrates a promising improvement potential of inventory management practices to simultaneously reduce both costs and emissions.

With respect to the studied regulatory systems that policy makers may invoke to reduce emissions, the conclusion from several models is that a cap-and-trade system is more attractive from a company perspective (less costly) than carbon taxes or systems with fixed emission caps. Moreover, indications are that complete modal shifts to greener transportation alternatives are difficult to attain by the use of carbon taxes alone, unless they are extremely high. At the same time, other results suggest that multimodal transport solutions or dual-supply options may offer attractive compromise solutions.

A noteworthy finding is that most of the existing literature on green (or sustainable) inventory management is based on rather simple and restrictive inventory models, particularly the EOQ model and the newsvendor model. More general inventory models that incorporate emissions and other environmental consequences are needed to offer adequate tools for practical use. The literature is growing but much remains to be done. Existing inventory models with pure cost or profit focus can be a good starting point, if the emissions structure and cost structure coincide.

Finally, we note that the practical use of any green inventory management method requires access to accurate emission (and cost) data at appropriate levels of detail. To avoid suboptimization and achieve reductions in total costs and emissions for a supply chain, inventory management decisions should be considered in conjunction with transportation and production planning decisions and strategic decisions such as network configuration, modal shifts, etc. To facilitate this, standardization of emission measurements and calculations across different industries, companies, and activities that cause emissions (production, warehousing, transportation, handling, etc.) are important. Otherwise, evaluating trade-offs and total effects become ambiguous. Refinement and more widespread use of common tools for environmental impact calculations, such as NTM^{CALC} (n.d.) (by the organization Network for Transport Measures), EcoTransIT World (n.d.), Normative (n.d.), CarbonCare (n.d.), and others, are promising developments in this area.

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Chapter 7

Green Network Design and Facility Location



Josué C. Velázquez-Martínez and Jan C. Fransoo

7.1 Introduction

Transportation emissions comprise a large share of the world's overall emissions, and freight transport is responsible for a relatively large share of these emissions. Transportation emissions can be reduced by making different choices in logistics, such as changing the mode of transport or changing the routing or loading of the vehicles in the network (see Chap. 5 by Blanco and Sheffi (2024) for more on green logistics). These logistics choices are influenced significantly by the inventory policies that have been deployed in a company (see Chap. 6 by Marklund and Berling (2024) for more on green inventory management). For instance, allowing for a more carbon-friendly slow mode of transportation would typically require increasing or repositioning the inventory in the supply network.

Apart from logistics choices and inventory policies, transportation performance in terms of costs and emissions is strongly determined by the design of the network. In distribution networks, this refers in particular to the location of distribution centers or other transport hubs such as factories or cross-docks. In this chapter, we address the issue of locating such a transport hub.

The logistics problem that determines the configuration of a company's delivery of goods is the facility location problem. The facility location problem is to locate a set of facilities (e.g., factories, cross-docks, distribution centers) in a physical space,

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such that all the demands of the customers are assigned to at least one facility and the total transport cost is minimized. While the literature on facility location is well established and large in size, in this chapter, we focus on a variant of this problem that specifically includes the transport carbon emissions in the formulation. We refer to location problems that aim at minimizing transportation CO₂ emissions as green facility location problems.

By limiting the scope to emissions from *mobile* sources (i.e., transport), we do not consider emissions from *stationary* sources that could be influenced by the location decision. Without being exhaustive, these may include:

- *Emissions at the distribution center.* These relate to the energy usage of the distribution center. In most cases, this would be electricity for light and/or automation and for refrigeration. Potentially, economies of scale could exist that would be related to the design of the network. Industry data suggest that in most distribution networks the emissions at the distribution center are less than 10% of the total logistics-related emissions.
- *Availability of local energy sources.* In particular, for energy-intensive operations, the availability of renewable local energy may significantly impact the supply chain emissions. For instance, locating an aluminum plant in an area where geothermal electricity is available could reduce a supply chain's overall carbon emissions while still increasing its transport emissions.

Excluding the emissions from stationary sources from the models discussed in this chapter implies that effectively we are limiting ourselves to distribution networks and the location choice of distribution centers and cross-docks. However, in our discussion, we use the more general term “facility.”

Usually, companies designing their distribution channels select the locations of warehouses and distribution centers with the objective to serve the demand of the customers while minimizing distance (or transport costs). In this chapter, we review some models that include the transportation CO₂ emissions in the uncapacitated facility location. We then discuss the solutions we may obtain when the number of facilities to be located is fixed by using the p-Median problem. We present discussions and managerial implications for the green facility location. We are interested in learning whether location decisions obtained by cost minimization are different from those obtained by the green facility location model.

7.1.1 Facility Location and Carbon Emissions

Typically, facility location decisions are made by considering the associated costs that include transportation (from the facilities to the customers) and the operation of the facility (production and storage). As discussed above, we may split the main sources of CO₂ emissions associated with the location of facilities in a similar way: emissions from mobile sources (transportation) and emissions from stationary sources (production, storage, and handling). Having more facilities reduces the CO₂ emissions from mobile sources due to the fact that the distance from the facility

to the customer destinations decreases. This increases emissions from stationary sources due to their larger number. Therefore, the challenge in green facility location is to define the proper number and position of the facilities that will serve a set of customers while minimizing the overall CO₂ emissions.

Many studies show that transportation and production may substantially contribute to CO₂ emissions. For example, the three main contributing sectors to emissions in the developed world are electricity production, energy-intensive manufacturing, and transportation (European Environmental Agency 2018). While production of electricity and energy-intensive manufacturing are considered within Scopes 1 and 2 of the GHG inventory, transportation by service providers is considered within Scope 3 emissions (see Chap. 3 by Boukherroub et al. (2024) for further details). Scope 3 emissions often represent the largest source of GHG emissions accounting for more than 90% of a company's carbon footprint (World Business Council for Sustainable Development 2021). In addition, when the facility location problem consists of locating distribution centers instead of manufacturing plants, typically, the CO₂ emissions from mobile sources are much higher than those of the stationary sources, as the latter then only include the emissions at the distribution center. Storage and handling emissions are substantially smaller than transportation emissions, by a factor of 10 for some products (Cholette and Venkat 2009). Therefore, in this chapter, we focus on studying the location of distribution centers with a main emphasis on transportation carbon emissions.

Many practices exist in industry to reduce carbon emissions by implementing more efficient and sustainable practices into their logistics operations (e.g., Heineken Sustainability Report 2013; Groupe Danone 2014; MIT-EDF 2013). However, only a few have considered the location of distribution centers as a relevant alternative to reduce transport CO₂ emissions. For example, Unilever increased the number of regional hubs and located these hubs closer to the customers (Unilever Press Release 2013).

The location of facilities is critical to the efficient and effective operation of a supply chain; poorly placed plants can result in excessive costs and low service level no matter how well tactical decisions (e.g., vehicle routing, inventory management) are optimized (Daskin et al. 2005). In this chapter, we demonstrate that facility location choice may significantly impact mobile CO₂ emissions in the supply chain. Note that the main drivers of transportation carbon emissions are distance, vehicle load (Greenhouse Gas Protocol Standard 2015; Smart Freight Centre 2016), and the number of trips required to deliver a demand to each customer (Network for Transport Measures 2022). Changing the number and location of the facilities in the distribution network impacts all of these drivers.

7.1.2 *Trade-Off Between Cost and Carbon Emissions in the Facility Location*

Transportation costs (TC) in facility location problems typically take demand (w) and distance (d) into account. These costs are usually modeled as an objective function using the demand-weighted total distance ($TC = \alpha wd$) and assuming an α constant cost per distance per unit (Revelle et al. 2008). Notice that this formulation finds optimal solutions where the facilities are closer to regions with high demand. However, for minimizing transportation CO₂ emissions, good solutions may require a different analysis.

Transportation CO₂ emissions are affected by a variety of conditions related to the type of vehicle (e.g., engine power, torque, fuel-type, aerodynamic drag coefficient) and the characteristics of the delivery operation (e.g., road, slope, vehicle speed, load) (Akçelik and Besley 2003). Due to the lack of detailed information about the delivery operation (specific slopes, speed, aerodynamics, etc.) during the decision-making process, companies typically use more aggregate activity-based methods to estimate CO₂ emissions (see Chap. 3 by Boukherroub et al. (2024) for more background on carbon footprinting). Some of the most common activity-based methods are the GHG Protocol, the Global Logistics Emissions Council (GLEC) Framework (Greene and Lewis 2016), and the methodology developed by the Network for Transport Measures (NTM).

We notice that a facility location model based on the GHG Protocol or GLEC methodologies would provide optimal locations that are identical to cost minimization model solutions, i.e., optimal locations would tend to be closer to regions with high demand. This is because these estimation models multiply distance and cargo times a carbon intensity factor that assumes vehicle utilization, that is, it does not consider explicitly the number of trips required to deliver the specific cargo. This is not always the case, though, when a method like the NTM methodology is employed.

The NTM methodology requires more detailed parameters: fuel consumption efficiencies, distance traveled, and weight per shipment (NTM Road 2010). The fuel consumption efficiencies are a function of the type of truck, the load factor, and the type of road. NTM uses the European Assessment and Reliability of Transport Emission Models and Inventory Systems' database which developed a detailed emission model for all transport modes to provide consistent emission estimates at the national, international, and regional level (TRL 2010). The NTM estimation model is

$$E = l \left[d \left(f^e + \left(f^f - f^e \right) \frac{w}{W} \right) \right],$$

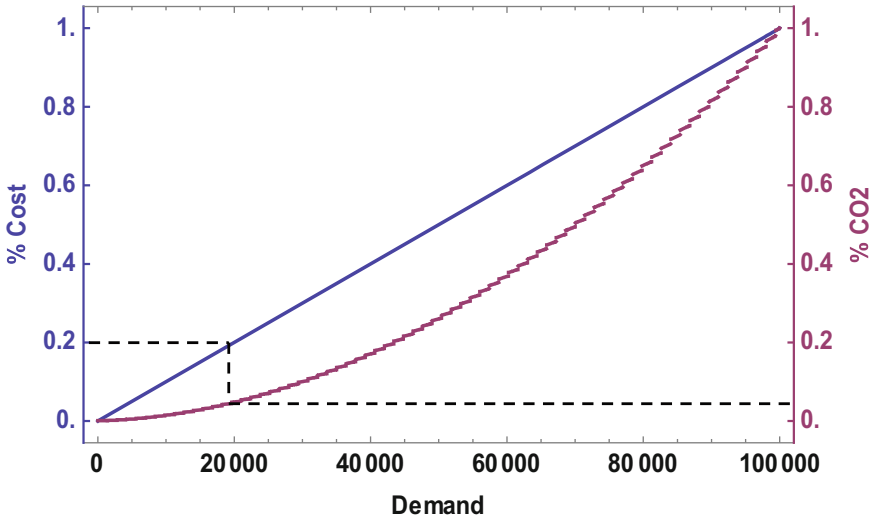


Fig. 7.1 Transport costs and CO₂ emissions over different demand levels

where

- E total emissions in grams of CO₂
- l constant emission factor (2621 grams of CO₂/liter)
- f^e fuel efficiency consumption of the empty vehicle (liters/km)
- f^f fuel efficiency consumption of the fully loaded vehicle (liters/km)
- W truck capacity

Comparing transport cost and CO₂ emissions, notice that the effect of distance is linear in both expressions. Demand and truck capacity, however, affect transportation costs differently than they affect CO₂ emissions. Figure 7.1 shows the comparison of transport costs and CO₂ emissions for different demand levels. For this example, we use a 14-ton truck for urban road type, and we set 100 demand units equivalent to 1 ton.

Note that the growth in demand does not translate into a linear increase in CO₂ emissions, as it is in cost. For example, a demand of 20,000 units increases the cost up to 20%, while the increase in CO₂ is approximate of 5%. The chart also shows that an increase in demand has an impact on CO₂ emissions, mainly when this growth implies more trips.

Because of these differences in the transport cost and CO₂ emission structures, intuitively, we may conclude that facility location models with either one or the other objective function may have different optimal solutions. While cost minimization models find optimal locations closer to high-demand nodes, CO₂ minimization models may also consider optimal locations closer to demand nodes where a larger number of trips are required to serve the customer's demand. This characteristic of CO₂ minimization models may be observed in both the high-demand nodes and for

restricted truck accessibility constraints in the nodes. Therefore, in facility location problems, solutions obtained by minimizing transportation costs are not necessarily equivalent to solutions obtained by minimizing transportation CO₂ emissions.

7.2 Green Facility Location Models

In this section, we present some general facility location models that are commonly studied in the logistics literature, including both continuous and discrete models. We later discuss some extensions of these models that study CO₂ emissions in location decisions.

7.2.1 Traditional Facility Location Models

The facility location problem has a very long history. It was first introduced by Weber (1909), and a large number of extensions and applications can be found in the literature. For a basic explanation of the facility location problems, we refer to Daskin (2008) and Daskin et al. (2005), and for recent reviews, we refer to Melo et al. (2009) and ReVelle et al. (2008). Typically, facility location problems are classified based on their solution space as *continuous* if the candidate locations can be located anywhere within the area or *discrete* if the candidate facilities are restricted to a finite set of locations (Daskin 2008). In addition, when continuous models assume that demands are distributed continuously across a service region, this approach is known as *analytical* location models.

The continuous and analytical approaches provide a general overview of the optimal locations and are commonly used for researchers to provide guidelines or insights (Geofrion 1976). A variety of applications can be found in the literature related to extensions of location models, such as the hub location problem (Saberi and Mahmassani 2013), freight transport network (Campbell 2013), and hub-and-spoke network design (Carlsson and Jia 2013). For analytical models, solution methods are derived by using mathematical analysis, while for continuous location models that are not analytically solvable, iterative numerical procedures ensure their convergence to optimal solutions, for example, the Weiszfeld (1936) algorithm for the Weber problem.

For practical applications, discrete formulations are more realistic to provide feasible and optimal locations but are more difficult to solve. For this type of model, candidate locations are pre-screened based on complementary information such as supplier's proximity, labor proximity, local regulations, and available physical space, among others. The basic model that locates the optimal facility among a set of candidate locations in a discrete space is known as the p-Median problem. The p-Median problem is defined as follows (ReVelle and Swain 1970):

Let I be a set of demand nodes and J be a set of candidate locations.

Parameters:

h_i demand at node $i \in I$

d_{ij} distance between candidate facility site $j \in J$ and customer location $i \in I$

Decision variables:

X_j 1 if we locate at site $j \in J$, 0 otherwise

Y_{ij} fraction of demand at customer location $i \in I$ that is served by facility at site $j \in J$

The p-Median problem is then formulated as follows (P1):

$$\text{Min} \rightarrow \sum_{j \in J} \sum_{i \in I} h_i d_{ij} Y_{ij}$$

Subject to

$$\sum_{j \in J} Y_{ij} = 1 \quad \forall i \in I \quad (1)$$

$$\sum_{j \in J} X_j = p \quad (2)$$

$$Y_{ij} - X_j \leq 0 \quad \forall i \in I \quad \forall j \in J \quad (3)$$

$$X_j \in \{0, 1\} \quad \forall j \in J \quad (4)$$

$$Y_{ij} \geq 0 \quad \forall i \in I, \forall j \in J \quad (5)$$

The objective function minimizes the demand-weighted total distance. Constraint (1) states that each demand node is covered. Constraint (2) establishes that p facilities are located. Constraint (3) states that the facility is opened when a demand node is assigned. Constraints (4) are the integrality constraints and (5) are the nonnegative constraints. When applied to a general network, the p-Median problem can be difficult to solve. However, since the single-sourcing condition holds in this formulation (i.e., Y_{ij} will naturally take values of zero or one), the property limits the potential facility locations to the network nodes, and therefore it reduces the number of possible location configurations to $n!/(n-p)!p!$, where n is the number of nodes (Owen and Daskin 1998). However, a total enumeration of all possible solutions may be computationally prohibited. Kariv and Hakimi (1979) showed that the p-Median problem is NP-hard.

The p-Median problem has been the basis of multiple extensions such as the fixed charge facility location problem, both uncapacitated and capacitated, and in other

problems such as multi-item and multi-echelon (Geoffrion and Graves 1974) (Pirkul and Jayaraman 1996). It also has multiple real-world applications such as plant location-allocation (Daskin and Dean 2005), network design (Kalpakis et al. 2001; Ruffolo et al. 2007; Stephens et al. 1994), sensor deployment (Greco et al. 2010), and data mining (Christou 2011). Other applications are presented in ReVelle et al. (2008). The p-Median problem has also attracted much research attention in combinatorial optimization, and many solution methods have been proposed to solve the problem. For instance, variable neighborhood search (Hansen and Mladenovi 1997), genetic algorithm (Hosage and Goodchild 1986), tabu search (Rolland et al. 1997), scatter search (García-López et al. 2003), ant colony optimization (Kochetov et al. 2005), and simulated annealing (Murray and Church 1996). Pullan (2008) finally presents a population-based hybrid search that was tested again multiple instances from literature, and the results show that the algorithm finds the optimal solutions for many problems, and for others, it was capable of finding improvements on the best-known solutions from the literature.

A natural extension of the p-Median problem is to relax the number of facilities to be opened p and include a fixed location cost f_j . This problem is called the fixed charge facility location problem (P2) (Balinski 1965):

$$\text{Min} \rightarrow \sum_{j \in J} \sum_{i \in I} f_j X_j + \alpha \sum_{j \in J} \sum_{i \in I} h_i d_{ij} Y_{ij}$$

Subject to (1)–(5)

When we also include a constraint (6) $\sum_{i \in I} h_i Y_{ij} - b_j X_j \leq 0, \forall j \in J$, that limits the assigned demand at facility $j \in J$ to a maximum of b_j , the resulting model (P3) is known as the capacitated facility location problem. Similar to the p-Median problem, the fixed charge facility location is also NP-hard. Previous approaches used to solve the p-Median problem may also be applicable in this case. Other solution heuristics methods are tabu search (Glover 1989; Glover and Laguna 1997) and the dual ascendant algorithm (Erlenkotter 1978), among others.

7.2.2 Carbon Emissions in Facility Location Models

We now discuss some models that include the estimation of CO₂ emissions in the facility location problem. As mentioned in Sect. 1, transportation CO₂ emissions in facility location models should be considered carefully, specifically because cost and CO₂ emission structures do not typically share the same structures. However, some studies demonstrate that even when taking into account the same structure, solutions obtained by minimizing transport costs are not always equivalent to solutions obtained by minimizing CO₂ emissions in more complex logistics networks (e.g., intermodal).

7.2.2.1 Analytical and Continuous Models

We start by discussing the study of Bouchery and Fransoo (2015) on intermodal hinterland network design. The authors present an analytical model that aims at finding the optimal location of one facility (in their example, an inland container terminal) with respect to cost, carbon emissions, and modal shift objectives. The demand is assumed to be uniform over a rectangular region representing the hinterland of the port under consideration. The density of the demand is equal to ρ containers per square kilometer, and the origin of the flows (the port) is located at the coordinates (0, 0).

The model assumes that transport cost and carbon emissions have the same structure and considers two transport mode options: direct shipment (shipment via truck directly from the origin to the customer) and intermodal transportation (shipment via rail to an intermodal terminal and subsequently from the terminal via truck to the customer). The cost and CO₂ emissions of serving a demand region i of size A_i by using direct shipment are expressed as follows: $Z_{0,i}^{DS} = \delta_{0,i} \rho A_i Z_1$ and $E_{0,i}^{DS} = \delta_{0,i} \rho A_i E_1$, respectively, where

- $\delta_{0,i}$ distance from the port to the gravity center of demand zone i (km)
- Z_1 truck transportation cost per container-kilometer
- E_1 carbon emissions from truck transportation (kg of CO₂ per container-km)

The cost and CO₂ emissions when using intermodal transportation are expressed as follows: $Z_{0,j}^{IT} = \delta_{0,T} (ZF_2 + \rho A_i Z_2) + \delta_{T,i} \rho A_i Z_1$ and $E_{0,i}^{IT} = \delta_{0,T} (EF_2 + \rho A_i E_2) + \delta_{T,i} \rho A_i E_1$, where

- $\delta_{0,T}$ distance from the port to the inland terminal (km)
- $\delta_{T,i}$ distance from the terminal to the gravity center of demand zone i (km)
- ZF_2 fixed train transportation cost per km
- Z_2 linear train transportation cost per container-km
- EF_2 fixed emissions associated with train transportation (kg of CO₂ per km)
- E_2 linear train transportation emissions (kg of CO₂ per container-km)

The authors identify optimal solutions based on European data. Their results show that the terminal is located closer to the port when optimizing cost and is located further away from the port when optimizing carbon emissions. This result shows that even when cost and CO₂ emissions have the same structure, there are significant differences in the optimal solutions for both formulations. This effect is clearly explained by the differences in the fixed train parameters, which are also consistent with the fact that train transportation under high utilization is more efficient from an emission perspective than road transportation, but it is more expensive in terms of cost. For more details, we refer to the full study (Bouchery and Fransoo 2015).

Although some other articles on continuous green facility location models can be found in the literature, the area is still very scarce. Buyuksaatci and Esnaf (2014) present a carbon emission-based facility location problem that considers

the minimization of CO₂ emissions by using the gravitational center method. The study uses a formulation based on the GHG protocol, but it does not discuss any managerial insight or implication derived from the proposed formulation.

7.2.2.2 Discrete Models

We now discuss the studies on green facility location models with discrete formulations. Diabat and Simchi-Levi (2010) present a two-level multi-commodity facility location problem with a carbon constraint. Their problem is to decide the optimal location of plants and distribution centers and the assignment, in such a way that the total costs are minimized and the carbon emissions do not exceed a specific carbon cap. The model assumes carbon emissions from distributions by using a distance emission factor (tons of CO₂ per km) and thus neglecting the impact of the load on CO₂ emissions (see Chap. 5 by Blanco and Sheffi (2024)). Despite this rather coarse assumption, the general conclusion seems in line with intuition: if carbon emission allowance decreases, supply chain cost increases.

Elhedhli and Merrick (2012) study a supply chain network design problem that takes CO₂ emissions into account. The objective of the study is to simultaneously minimize logistics costs and the environmental costs of CO₂ emissions by strategically locating warehouses within the distribution network. This model considers the GHG Protocol estimation of CO₂ emissions and uses a scaling parameter to convert the CO₂ into cost. This approach allows the inclusion of the cost of carbon emissions into the supply chain network design. The experimental results show that the addition of carbon costs drives solutions with more distribution centers being opened to decrease CO₂ emissions in transportation.

Although the study provides interesting managerial insights, the model uses the most aggregate approaches to estimate CO₂ emissions in transportation (i.e., the GHG Protocol with EPA emission factors). Velázquez-Martínez et al. (2014a) address the effects of using different aggregation levels to measure transport carbon emissions, and they show that errors associated with aggregation could be substantial and systematic. This suggests that increasing the level of detail in the facility location problem is necessary.

Cost may not necessarily be the only driver to reduce CO₂ emissions in transportation. For example, companies may be subject to a cap-and-trade system or may use carbon emission reductions as a driver for brand management, product differentiation, or employee motivation (CDP 2011a, b). This suggests that a practical formulation of green facility location models should potentially take simultaneously cost and CO₂ objectives into account.

A possible alternative to consider both objectives (cost and CO₂ emissions) is to model the green facility location problem using a multiobjective setting. Most real-world problems naturally involve multiple objectives (minimizing cost, maximizing service level, minimizing CO₂ emissions, etc.) A multiobjective approach allows to define a set of efficient solutions (or a Pareto frontier) which is defined as the set of solutions such that there is no other solution that dominates them, i.e., each

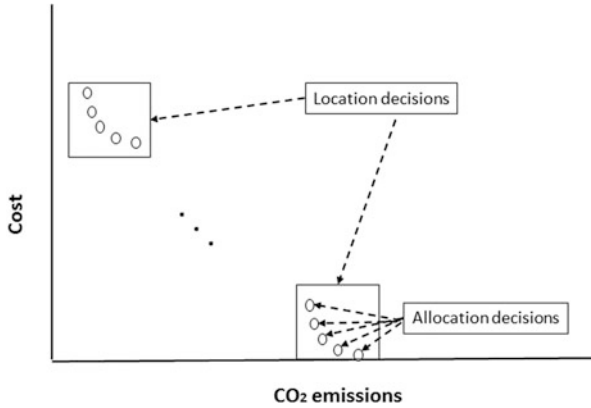


Fig. 7.2 Trade-off solutions for customer location-allocation decisions. Adapted from Harris et al. (2014)

solution of the set is strictly better than the rest of the solutions in at least one objective and is not worse than the rest of the solutions in all objectives (Coello 2009). These efficient solutions are often preferred to single solutions because they can be practical when considering real-life problems since the final solution of the decision-maker is always a trade-off (Konak et al. 2006).

In line with this stream of research, Harris et al. (2014) present a formulation of the fixed charge facility location model ($P3$) with two objective functions: costs and CO_2 emissions. Their facility location model considers individual depots with capacities b_j , where each customer is served directly by a single depot, thus forcing the “single-sourcing condition” to be held in the model. Therefore, it is possible to build a solution algorithm that first determines which facilities to open and then allocates the customers to the open facilities. The study proposes an expression to estimate transportation CO_2 emissions based on the GHG Protocol, i.e., transportation CO_2 emissions are linearly dependent on the distance traveled and demand.

The study discusses a multiobjective optimization solution method for the cost and CO_2 facility location model, in which a decision-maker can explore trade-off solutions for customer allocation based on the preselected facility location. Figure 7.2 (Harris et al. 2014) shows the different solutions of the location decision and, for each decision, the potential allocation assignment.

The article focuses on the solution methods and provides a framework to analyze trade-offs between cost and CO_2 emissions for location models.

Because we notice that all previous studies conclude that the increase in the number of open facilities implies a reduction in transportation CO_2 emissions (and typically more facilities also imply higher costs), we may argue that a practical approach to analyzing the trade-off between cost and CO_2 emissions in facility locations is to simplify the formulation by not including the fixed emission per open facility. Therefore, we are interested in studying the effect of transportation cost

versus transportation CO₂ emissions with a fixed amount of facilities previously defined (i.e., p-Median problem).

Velázquez-Martínez et al. (2014b) study the trade-off between cost and CO₂ emissions by using a multiobjective approach for the facility location problem. The model corresponds to the p-Median problem with cost and CO₂ objective functions. The general assumptions of the p-Median problem apply to this model; that is, deterministic demand and the candidate locations are known in advance. In addition, the model also assumes that the company may manage multiple trucks with different capacities, and the trucks are assigned according to demand node constraints (or company policy). These assumptions allow the model to include the possibility that certain customers are reachable only by certain types of trucks, with distinct cost structures.

To formulate the carbon emission objective function, the authors include the NTM methodology in the objective function (Velázquez-Martínez et al. 2014b). Note that $\left\lceil \frac{h_i}{W_i} \right\rceil$ represents the number of trips that are required to serve customer $i \in I$ and, thus, affects the total distance traveled:

$$l \sum_{j \in J} \sum_{i \in I} d_{ij} \left[f_i^e \left\lceil \frac{h_i}{W_i} \right\rceil + (f_i^f - f_i^e) \frac{h_i}{W_i} \right] Y_{ij}$$

This formulation enables us to understand in more details the trade-off between distance (d_{ij}) and utilization (h_i/W_i) while deciding the location-allocation decisions. For example, when serving customers with a homogeneous fleet (i.e., $W_i = W$ for all $i \in I$), the location solutions are the same as those that are obtained by *PI*, i.e., facilities are located closer to customers with the highest demand, thus minimizing transport cost is equivalent to minimizing CO₂ emissions. However, when serving customers with a nonhomogeneous fleet (e.g., caused by truck constraints due to regulations or transport infrastructure), facilities may be located closer to customers served by small trucks. This may be explained due to the fact that multiple trips are required and, thus, more distance is traveled to serve these customers.

7.3 Practical Implications of the Green Location Models

Transportation is one of the main contributing factors to global carbon emissions, and thus, when dealing with facility location models in a distribution context, transportation emissions may be substantially higher than the emissions due to production or storage. In addition, because facility location models define the configuration of deliveries, green location models become an important alternative to reduce CO₂ emissions in logistics. Because transportation usually is included in Scope 3 of the GHG inventory and usually represents the highest source of emissions in a supply chain, companies may start focusing more on increasing the

number of distribution centers while increasing the reachability to customers. We have seen this phenomenon at the biggest retailers, like Amazon, as a result of their fast-shipping business model and increased customer service speed (Dastin 2020).

While cost minimization solutions tend to locate facilities closer to high-demand customers, CO₂ emission minimization solutions tend to locate facilities closer to customers that have truck accessibility constraints. This is explained because truck constraints drive the number of trips required to serve customers, and this factor is larger than the increase in demand and/or utilization. This may be particularly important for companies managing nonhomogeneous vehicle fleets or for policymakers in large dense areas where demand is high (based on the high density of inhabitants and small stores), but heavy-duty vehicles are not allowed. New regulations may be needed to balance the accessibility of big trucks in certain periods to increase logistics efficiency and also reduce the number of small vehicles in those regions.

For some logistics problems, even when aggregate approaches are used to estimate transportation CO₂ emissions and thus this formulation shares the same structure with transportation cost, the location solutions may be substantially different. For companies that are interested in increasing modal shift or using more intermodal transport, these strategies may result in an increase in CO₂ emissions. Particularly when different modes are used like in intermodal networks, the difference in parameters for transportation cost and CO₂ emissions can lead to a completely different set of solutions for both objective functions.

A multiobjective setting for the green facility location models may provide decision-makers with a framework to analyze the trade-off between cost and CO₂ emissions. This approach may bring a new tool for companies to define better strategies to reduce CO₂ emissions. Because decision-makers likely seek alternatives that reduce emissions while keeping costs low, multiobjective modeling provides a set of trade-off solutions that were previously unknown in single-objective modeling. This may imply that new solutions may appear with a good offset between cost and CO₂ emissions. For example, locations with small increases in cost may imply high reductions of CO₂ emissions.

7.4 Directions for Future Work

The area of green facility location is still small in research. Because transportation costs and CO₂ emissions do not have the same structure, a specific formulation for the CO₂ emission minimization model for facility locations should be considered. Unfortunately, few studies consider the detailed expression to estimate transportation CO₂ emissions in location models, and most of them use GHG Protocol and GLEC framework; thus, the complete effect has not been studied and understood.

In addition, only a few companies have implemented strategies using facility locations to reduce their environmental impact. Thus, more applications of the models in practical cases are needed so more understanding of the models and trade-

offs can be achieved and validated in practice. In addition, a few articles from prior literature include in their formulations the emissions generated by the facilities and usually only the production of electricity. The models are mainly focused on the emissions caused by transportation, specifically for last-mile delivery. However, no research has been conducted to analyze the impact of the transportation of raw materials in facility locations, and thus, more model formulations are needed to address this gap.

In this chapter, we have limited our discussion to the impact of carbon emissions from mobile sources. However, considering the different sources of energy for the facilities (wind, fuels, etc.), and including them in the future green facility location models to understand the impact of energy sources on planned locations, is a fruitful research avenue, specifically in connection with the electrification strategy of the transport sector. Facilities may play a key role to support electrified transportation and serve as onsite storage for electric batteries or other types of sources, as well as energy hubs to allow for plug load, and charging stations.

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Chapter 8

Operational Implications of Environmental Regulation



Ximin (Natalie) Huang and Atalay Atasu

8.1 Introduction and Motivation

The fast expansion of industrial production and consumption activities has created heavy environmental burden and raised calls from the public, environmental groups, and non-governmental organizations (NGOs) for regulatory measures to promote sustainable practices in supply chains. As a result, many different forms of environmental regulations have been considered or enacted in recent years for various industries across the globe. Examples include the Restriction of Hazardous Substances (RoHS) (2002/95/EC) (Europa-Environment 2002), the End-of-Life Vehicle (ELV) (2000/53/EC) (Europa-Environment 2000), the Waste Electrical and Electronic Equipment (WEEE) (2003/108/EC) (Europa-Environment 2003) Directives of the European Union, the Regional Greenhouse Gas Initiative in the USA (RGGI 2009), and the regulation for recovery and disposal of electronic waste (e-waste) in China (State Council of China 2009).

Environmental regulation will inevitably influence different aspects of firm operations. Nevertheless, implementation choices for environmental regulation are often made based on certain implicit assumptions that do not take into consideration of firms' possible operational decisions induced by such regulation or other market dynamics on the ground. This has been demonstrated by existing research in operations management. In particular, the recent research has challenged numerous implicit assumptions made by high-impact regulations such as the WEEE Directive,

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identified the associated unintended consequences, and provided recommendations regarding the implications of different regulatory implementation choices on the environment, supply chains, and economy in general.

In this chapter, we review a number of research papers to highlight challenges associated with environmental regulation implementations, particularly in the context of take-back regulation, given its popularity in the last two decades. These papers cover a broad spectrum of possible issues that may arise in the transposition of regulatory principles into working systems. Some of them challenge the assumptions implicitly embedded in the forms or specifications of regulation, some point out possible loopholes of regulation, and others identify how differences across the target industries or nature of products will induce different supply chain responses to regulation.

Our discussions on these papers collectively show that devil is in the details: We highlight the importance of an operational perspective, which is crucial for regulators to set proper boundaries and scope for environmental regulation, and formulate appropriate implementation models. We further suggest that managers should have a clear understanding of the operational implications of different regulation implementations in order to achieve environmental and economic sustainability of their supply chains. In what follows, we first provide a basic background on general environmental regulation, followed by discussion on take-back regulation as our unit of analysis. We then dive into operational challenges associated with such regulation and conclude with insights for regulators and supply chains.

8.2 Background on Environmental Regulation

In the environmental policy literature, environmental regulations can be differentiated in multiple dimensions (see Richards 1999, for a detailed review): Some regulations are product-related (e.g., the WEEE and the RoHS Directives of the EU) and others are facility-related (e.g., the US Clean Water Act (US EPA 1972)). Some regulations impose restrictions (e.g., the RoHS Directive) and others require only disclosures (e.g., the US Conflict Minerals Rule (US SEC 2012)). Some regulations are international (e.g., Basel Convention (BAN 1992) and the US Responsible Recycling Act (US Congress 2011)), some are national (e.g., the WEEE Directive), and some are local (e.g., the 26 state-level e-waste recycling regulations in the USA (US EPA 2022)).

Among those classifications, the ones that focus on regulation implementation differences are relevant from a supply chain perspective. In this context, Fullerton (2001) is a useful reference that classifies environmental regulation in two basic dimensions: (i) the form of control imposed by regulation, and (ii) the type of incentives utilized by regulation. In the first classification dimension, one distinguishes between emission restrictions (sometimes called performance standards) and technology restrictions (sometimes referred to as design standards). Examples of the former include the greenhouse gas emission restrictions with cap-and-

trade systems that have been enacted in different parts of the world including the USA and EU (Islegen et al. 2015). Examples of the latter include the RoHS Directive and requirements to install flue-gas desulphurization in electric plants in the USA (Fullerton 2001). In the second classification dimension, one distinguishes between the types of incentives, i.e., taxes and subsidies. Examples of the former include the carbon taxes such as those used in Australia (Islegen et al. 2015), and examples of the latter include renewable energy credits (Ata et al. 2012). The unit of our analysis, e-waste take-back regulation, typically operates under a mix of these regulatory instruments. For example, the WEEE Directive uses performance standards including collection and recycling targets imposed on producers, and the take-back regulation in California charges unit disposal taxes to consumers (US EPA 2022). We provide a more detailed discussion on the e-waste take-back regulation below.

The momentum for the rapid development of take-back regulation in recent years is generated by the urgent need to manage the growing volume of e-waste. The average lifespan of a computer has shrunk from 6 years in 1997 to just 2 years in 2005 (Arwidsson and Allard 2010), and more than 151 million mobile phones are trashed in the USA every year (US EPA 2009). In 2019, 53.6 million tons of e-waste were generated worldwide, with a concerning annual rate of increase estimated at 3% to potentially reach 74.3 million tons by 2030. Less than 20% of the e-waste were recycled, and the rest were sent to landfills or incinerators (Forti et al. 2020). Landfilling and incineration, however, can release hazardous substances such as dioxins, lead, and mercury into the air, soil, or water, posing serious threats to the environment and public health. To deal with this problem, take-back regulation organizes proper treatment of e-waste through collection and recycling.

The majority of take-back regulation implementations utilize the Extended Producer Responsibility (EPR) principle, which directs the financial or physical responsibility of proper end-of-life product treatment to producers (see Walls 2006, for an extensive overview) via performance standards or disposal taxes as explained above. The rationale behind EPR is to have producers internalize the negative environmental impacts created by their products, so that they find incentives to redesign their products to mitigate such externalities. As such, take-back regulation based on EPR aims to not only divert end-of-life products from landfills but also incentivize producers to incorporate environmentally superior attributes in their products (OECD 2001). This potential of EPR appears to explain its popularity across the globe. The WEEE Directive of the EU, which probably has the largest scope in the world, is EPR-based. Moreover, despite the lack of federal-level regulation in the USA, 25 out of the 26 states (except for California) that have passed state-level e-waste bills stipulate producer responsibility (US EPA 2022).

It is important to note that the successful implementation of environmental regulation is a challenge (Drake and Just 2016). Be it with respect to take-back, emissions control, or any other environmental objectives, setting up the ground rules for regulatory compliance is a critical step and should not rely on convenient assumptions. Recent research has demonstrated that efficient implementations of cap-and-trade or tax-based emissions regulations need to carefully consider

the incentives they create for carbon leakage (Islegen et al. 2015; Sunar 2016). Likewise, it has been shown that environmental taxes can induce producers to invest in inferior environmental technologies (Krass et al. 2013), and disclosure mandates can drive producers toward not learning their actual environmental impacts (Kalkanci et al. 2016). Similarly, in the context of take-back regulation, the landfill diversion and product redesign potentials of EPR heavily rely on certain premises, which may make sense from a basic economic point of view. These fundamental premises include that (i) recycling end-of-life products (particularly electronics) must be costly, and hazardous waste will leak into the environment if left unregulated (see Sect. 8.3), (ii) take-back regulation will create incentives to recycle only end-of-life products that can be considered as waste (see Sect. 8.4), (iii) redesign incentives generated by take-back regulation will only focus on making products more recyclable (see Sect. 8.5), (iv) such regulation is appropriate for most product categories including both durables and consumables (see Sect. 8.6), (v) interaction among producers in the market is not a concern under the regulation (see Sect. 8.7), (vi) implementation details related to the existing reverse supply chain infrastructures (networks) will not matter (see Sect. 8.8), (vii) effectiveness and environmental benefits of the regulation depend on the redesign incentives for products only (see Sect. 8.9), and (viii) take-back and recycling take place in the country where the regulation is enacted (see Sect. 8.10). While these assumptions may appear reasonable in theory, they (or the invalidity of any of those) can be quite harmful to the efficacy of implementations on the ground and have important supply chain management and environmental implications, as we demonstrate in the following sections.

8.3 Trash or Treasure?

A fundamental assumption underlying the EPR-based take-back regulation is that environmentally sound treatment of end-of-life products is always costly. This also constitutes the reason why regulation is needed in the first place: to ensure that producers take the responsibility for proper collection and recycling. The thinking that logically follows is that increasing the stringency of regulatory requirements should strengthen incentives for landfill diversion and sustainable design of products, which appears to underlie the 2012 Recast of the WEEE Directive (2012/19/EU) to increase the collection target to 45% by 2016 and then to 65% by 2019. However, doubt has been cast on the assumption of costly recycling in recent years, especially with the roaring prices and scarcity of certain raw materials. For example, the amount of gold that can be recovered from printed circuit boards can be more than thirty times greater than that from a gold ore of equal weight (Kasper and Veit 2018); in other words, recycling some of the e-waste can be more profitable than metal ore mining. Consequently, it appears that producers may be motivated to recycle some electronics voluntarily even in the absence of take-back regulation. At the same time, it is important to note that this opportunity presents

itself to others as well, i.e., third-party recyclers may also be attracted into the product recovery business, introducing competition in the market for end-of-life products with recoverable value. These observations contradict the costly recycling assumption, and may imply challenges on the effectiveness of take-back regulation, which is examined by Esenduran et al. (2019).

In their paper, Esenduran et al. (2019) consider a stylized model of a recycling market in which producers and third-party recyclers may compete for recycling valuable waste. They focus on a single product category. After reaching the end of life, products remain in the hands of a set of waste-holders that may be geographically dispersed. The recoverable value in each waste-holder's product may vary depending on its proximity to the producers and recyclers. As a result, producers and recyclers may extract different values from end-of-life products, and they compete on acquiring the products from waste-holders by, respectively, offering recovery prices. Given the recovery prices and their respective value heterogeneity, waste-holders decide on whether and to whom to return their end-of-life products to maximize their utilities. In this context, the authors assume take-back regulation in the form of a collection and recycling target imposed only on the producers. They further assume that producers can enhance the product end-of-life value by increasing the recyclability at a cost.

The authors show the following: Even when waste has value, the existence of regulation may be necessary if the producers exhibit a strong position in terms of their access to the recoverable waste. The reason is that although the value of waste translates into some incentives for the producers to collect and recycle, its effect is only limited to the "low-hanging fruits" (i.e., items that are easier or cheaper to access). Consequently, the recycling level chosen by the producers remains below the social optimum. In this case, enacting regulation brings multiple benefits: It increases landfill diversion, encourages more recyclable product design, and also guarantees improvement of social welfare. However, in the presence of competition in the recycling market, regulation can lead to unintended consequences. First, increasing the regulation stringency can reduce landfill diversion. This is because the higher collection and recycling target imposed only on the producers drives the producers to act more aggressively in acquiring the products from waste-holders. This may lead to very high recovery prices offered to waste-holders, causing third-party recyclers to hold back from the recycling market. Effectively, this suggests that the overall collection and recycling volume may decrease because the portion contributed from the third-party recyclers may go down. Furthermore, this can happen to an extent that the reduction in the third-party recycled volume actually offsets the increase in the producer recycling induced by take-back regulation. Second, a higher collection and recycling target imposed on producers may discourage improvement on product design. This is because the spill-over of design benefits can be accentuated by competition, which makes design improvement a less appealing option for the producers. As such, the results suggest that more stringent regulation may not always be ideal, particularly when waste is valuable. Instead, reducing the stringency of collection and recycling targets imposed on producers, while regulating the third-party recycling market in terms of the standards of recycling

may be a more effective solution from both the economic and environmental perspectives.

The take-away from Esenduran et al. (2019) is that even the simplest and possibly the most reasonable assumption that underlies the need for environmental regulation may be challenged by market dynamics. When this happens, the consequence may appear straightforward: For the case of valuable e-waste, one may expect that the more value in waste, the more producers (or the responsible supply chain divisions) will recycle. Nevertheless, how the market competition on the ground functions may completely change the outcomes. As demonstrated by Esenduran et al. (2019), competing for geographically dispersed heterogeneous waste in the presence of design improvement options may necessitate the need for reverse supply chain managers to think differently about how they cope with the take-back regulation. In particular, they will play a key role in the compliance efforts, as they are the parties that can observe the waste competition related issues and inform firms about such challenges in advance. In sum, the operational considerations (i.e., devil in the detail) that matter in this case are the geographical dispersion of waste (see Dhanorkar et al. 2015 and Sunar 2016 for similar observations in the waste exchange and emissions regulation contexts) and whether and how producers and their reverse supply chains can effectively access the waste under competition with third parties in the recycling market.

8.4 End-of-Life or Used Product Recycling?

Another basic assumption of take-back regulation is that in order to fulfill the EPR obligation, producers acquire only end-of-life products from waste-holders. Intuitively sound as it is, this assumption may not directly extend to durable goods. In particular, durable goods producers can also acquire used (midlife) items from the secondary market for product recovery, despite the remaining lifespans these items have.

Producers may have incentives to pull used products off the secondary markets and use them as inputs for recycling due to two reasons. First, a secondary market offers used products as cheaper (although maybe inferior) alternatives to the new ones and hence potentially attracts consumers away from buying new products, i.e., it cannibalizes the new product sales. As such, reducing cannibalization constitutes a motivation for producers to interfere with, or even suppresses the secondary market. Second, acquiring used products also helps lower the need and cost of collecting end-of-life items to meet the collection target. This is not only a viable but also feasible option for producers: They can obtain used products through offering trade-ins or buyback programs.

However, from the environmental and regulatory standpoint, retiring used products before they reach the end-of-life can be unfavorable, because this practice shortens the average useful life of products and increases the total waste volume. This leads to two critical questions: How will the producers' optimal strategies to

interfere with the secondary market change when faced with take-back regulation? Consequently, how will such interference influence the effectiveness of regulation? Alev et al. (2020) shed light on these questions.

Alev et al. (2020) develop a stylized economic model to study the decisions of a monopolistic durable goods producer under EPR-based take-back regulation with (i) a collection target, which can be met by collecting both end-of-life or used products, (ii) requirements on collection infrastructure, which will be reflected in the unit collection cost of end-of-life products, and (iii) recycling standards, which will be reflected in the unit cost of recycling every product. Their analysis shows that the decision of the producer on interfering with the secondary market is indeed influenced by the regulation. This is evidenced by the fact that retiring used products can be purely induced by the need to fulfill the collection target and will not exist in the absence of regulation. When such interference occurs, it can lead to a lower reuse level and a higher new production volume, both commonly associated with unfavorable environmental consequences.

In this case, the effectiveness of take-back regulation is weakened by the existence of two types of collection that a durable goods producer can strategically utilize for compliance: In addition to collecting end-of-life products, the producer can also acquire used products from consumers in the secondary market for recycling. Since the alternative collection type is unique to durable goods, regulators and producers of durable goods should pay particular attention to the environmental and supply chain implications of take-back regulation.

From the regulator perspective, the results in Alev et al. (2020) suggest the following: First, recycling standards remain effective. Higher recycling standards help ensure the producer collects and recycles end-of-life products. As a result, premature retirement of used products decreases. Second, collection infrastructure requirements bring in similar benefits as the recycling standards, but only up to a certain stringency, beyond which their effect negates and leads to increased producer collection from the secondary market. Finally, the effect of the collection target is not always positive. Specifically, when the collection target increases within a low value range, it may incentivize the producer to collect from the secondary market and hence adversely results in increase in new production and decrease in reuse levels. These results suggest that the take-back regulation for durable goods, while achieving the objective of increasing the level of collection for recycling, may compromise other environmentally sound objectives such as reuse and consumption reduction. This outcome appears undesirable from the perspective of the well-established Waste Management Hierarchy, which prioritizes reduce over reuse, and reuse over recycling.

From the supply chain perspective, these observations once again highlight the importance of the reverse supply chain manager's role in compliance. In particular, the reverse supply chain manager who observes the costs of compliance and different product recovery options can help achieve better coordinated strategies for the firm. For instance, the reverse supply chain manager can bridge the connections between compliance and marketing departments to manage the acquisition of used products through trade-in programs and direct such flows toward fulfilling the take-

back obligation, which appears to be the essence of Apple's recent Recycling and Reuse program (Apple 2022).

8.5 Durability or Recyclability?

The durability embedded in products raises complexities in regulation in regard to not only how it influences the collection decisions for producers (by providing an additional collection channel as discussed above) but also how it influences the design decisions. Recall that incentivizing environmentally sound product designs is one of the basic goals of EPR take-back regulation. When regulation imposes the obligation of recycling, one expects that it should induce producers to make recycling cheaper by improving recyclability (e.g., incorporating easier disassembly features) in the product design. This assumption is also popular among regulators as reflected in regulatory statements. One example is the E-waste Recycling Law in Washington State, which spells out that "The legislature further finds that the system must encourage the design of electronic products that are less toxic and more recyclable" (New Section, Sec 1, SB4628). However, for durable products, enhancing recyclability to reduce the unit cost of recycling is not the only option for producers to alleviate the economic burden associated with the recycling obligation. Producers can also improve the product durability to lower the recycling volume. The underlying rationale is that making products more durable allows the producers to set a higher price for new products and sell less, leading to a smaller amount of end-of-life products and hence a lower total cost of recycling. As such, when recyclability and durability are both available design options for producers, it may be of interest to review the design implications of regulation. In particular, how do the producer's design choices and the regulation efficiency change in the presence of these two design options, especially when they are interdependent? This problem is explored by Huang et al. (2019).

Huang et al. (2019) consider a monopolist producer selling a single durable product and facing take-back regulation that imposes a collection target and a recycling target for end-of-life products. The producer, in response to the regulation, can choose the product recyclability and durability in the design stage to deal with the cost associated with the obligation. The authors find that when the product recyclability and durability are independent or complementary attributes, increasing the recycling target helps induce the desired outcome with improvements on both product attributes. However, when recyclability and durability are substitutes, the design trade-off between them can result in non-monotonic effects of the recycling target, which is in sharp contrast to common expectations: As the recycling target goes up, it first leads to recyclability improvements (while the durability may decrease) and then switches to durability improvements (while the recyclability may decrease). Meanwhile, in contrast to the recycling target, further increasing the stringency of a high collection target will instead lead to increase in the recyclability, while durability may be compromised due to the design trade-off. Note

that increasing recyclability in product design encourages recycling by bringing down the unit recycling cost, while increasing durability promotes waste reduction by lowering the production volume. Therefore, when recyclability compromises durability, the design trade-off suggests that recycling comes at the expense of waste reduction.

These conclusions send an important message to regulators: Design alternatives for producers to deal with the costs associated with the EPR obligation are critical operational factors that influence the outcomes of take-back regulation. As demonstrated by Huang et al. (2019), the existence of durability as an alternative design option to recyclability may not necessarily expand the effective scope of regulation (i.e., enhance both product attributes). On the contrary, the possible trade-off between design choices may diminish or even negate the effectiveness of regulation (see Plambeck and Wang 2009 and Krass et al. 2013 for similar observations in the context of new product introduction frequency and technology choices, respectively).

From the supply chain perspective, these results once again suggest that effective compliance with take-back regulation will require coordination between the reverse supply chain manager, the compliance department, and the design/engineering department. Consider photovoltaic panel (PVP) producers for instance, who face a clear trade-off between designing products for durability and recyclability (see Huang et al. 2019 for details). The 2012 Recast of the WEEE Directive has added PVPs into the regulated product categories. Even though compliance with such regulation may not be a concern at present for the producers because the lifespans of PVPs are estimated to be 25 years, it will be critical for the producers to prepare for future liabilities imposed by the take-back regulation. In this case, it is important for PVP producers to plan for a reverse supply chain infrastructure that will be able to accommodate different design profiles and help reduce the economic burden associated with PVP recycling.

8.6 Durable or Consumable Products?

While our discussion so far has been centered around durable products, EPR-based take-back regulation has also extended to consumable products in recent years, and examples include the Pharmaceutical Stewardship Act (US Congress 2017) in the USA and the regulation on product recovery and reclamation in Quebec after its 2017 amendment (Québec Government 2022).

In the context of take-back regulation for consumables, leveraging experiences from durable product markets may appear viable. However, producers and regulators must be aware of the implementation differences stemming from the consumable and perishable nature of this product category and adjust accordingly. In particular, in contrast to durable products, the volume of end-of-life waste does not equal the volume of production for consumables. Moreover, recovery options for consumables are limited because reuse and recycle are usually not applicable.

Therefore, regulation for consumables needs to focus more on reducing the volume of unused products that are not disposed of in an environmentally responsible way. In doing so, there exist two prevalent implementation approaches. The first is Source Reduction (SR), which aims to lower production by charging a fee based on production volume. The second is End-of-pipe Control (EC), which mandates the producer to organize or finance proper waste treatment. Given the product nature of consumables, further investigation is needed in order to determine how one can choose between the SR and EC approaches to effectively improve the environmental and economic impacts of regulation. Alev et al. (2022) pioneer in this area and their findings depart considerably from the existing results for durable products.

The products considered in Alev et al. (2022) are consumable pharmaceuticals. The decision makers in the pharmaceutical chain include a social planner, a producer, a doctor and a patient, whose interactions are captured by a sequential game model. The doctor acts as an intermediary between the producer and the consumer (the patient). Therefore, the sales volume of pharmaceuticals is not directly determined by the consumer behavior, but by the doctor's prescription instead. The prescription amount that exceeds the actual consumption by the patient (over-prescription) constitutes the main source of pharmaceutical waste. The doctor makes the prescription decision based on not only the cost of treatment and impacts of the prescription on patient's health but also the doctor's own utility from the prescription. The producer also influences the sales volume by setting the market price, as well as the promotion efforts on both the doctor and the patient. The social planner chooses between the two regulatory implementation approaches from below and sets the respective policy parameter.

- (i) EC: It requires the producer to collect a certain fraction of unused pharmaceuticals, and the producer also bears the associated collection cost. This implementation resembles a typical producer-operated collection and recycling system.
- (ii) SR: It imposes a fee on every unit of sales. After obtaining the fee payments, the regulator manages the collection (e.g., chooses the collection level) and the costs incurred. SR resembles a typical state-operated system.

The analysis finds that for pharmaceuticals that are consumable in nature and have a different demand structure with the doctor acting as an intermediary between production and actual consumption, the comparison between the two implementation approaches is very different from that for durable products. The paper shows that a producer-operated system (under EC) can be preferred by a social planner when the following conditions are met: (i) the cost of collection is high, (ii) the product use-phase impact (i.e., the health-improvement effect for the patient) is less significant, and (iii) the product impacts on the society (e.g., medicine abuses) and the environment (e.g., chemical contamination in the ecology) are high. Notably, (iii) is in direct contrast to the conclusion for durables, which suggests that under similar condition the state-operated system should work more efficiently (Atasu et al. 2013).

This study, although solely focuses on pharmaceuticals, delivers an insightful lesson that may be generalized for other consumables. The product nature is the operational detail that matters. While take-back regulation is devoted to dealing with products that are at the end of their useful life, the definition of useful life can vary substantially depending on the product nature. For a durable product, the end-of-life item has lost its functionality and will not be demanded by consumers. For a consumable, on the other hand, reaching the end of its useful life means it no longer provides the original designated usage, but it may still be in a functional state and may be consumed by others. This difference can significantly affect the efficiency and implications of different regulation implementation choices as demonstrated by Alev et al. (2022).

Therefore, the product nature must be carefully accounted for by regulators and producers in order to avoid implementation choices that would compromise operational or regulatory efficiency. It is especially important that the reverse supply chain manager understands these implementation issues. In particular, Alev et al. (2022) show that which implementation model will be preferred by a regulator depends on the unused pharmaceutical collection infrastructure accessible to the producer. For instance, while leveraging existing pharmacies or municipal collection systems may appear a low-cost option for the producer, in those circumstances the regulator may in fact favor an SR approach with higher taxes, eventually leading to a lower profit for the producer. As such, the reverse supply chain design for the compliance with take-back regulation needs not be the most cost-efficient one in the context of consumables.

Our discussion so far has demonstrated that the proper evaluation and development of policy instruments must build on correct and thorough understanding of the detailed operational characteristics of the target industries, including the commodity market dynamics, the alternative strategic responses available to producers, and the nature of products. Beyond that, the devil in the details also matters in the implementation stage, in which regulators and producers translate the regulation into a working system. We illustrate this phenomenon in the next sections.

8.7 Collective or Individual Producer Responsibility?

Although EPR-based take-back regulation sets a unified principle of assigning producer responsibility, it leaves discretion for regulators to choose the operationalization of product collection and recycling. Typically, two models of implementation can be observed: Individual, and Collective Producer Responsibility (abbreviated as IPR and CPR, respectively, see Plambeck and Wang 2009 for a similar discussion). Under IPR, each producer deals with only its own products. A somewhat similar model is adopted by the Personal Computer Recycling Law in Japan, where products are separated by brands after collection and then become the responsibility of the corresponding producers. Under CPR, all participating producers jointly conduct

end-of-life product treatment and share the total costs. Examples of this model can be observed in the WEEE Directive implementations in several EU member states.

Both the IPR and CPR models have certain pros and cons, and neither of them is dominantly better than the other. On the one hand, CPR is commonly considered superior in cost efficiency because it realizes economies of scale by aggregating the collection and recycling of a large volume of products and also saves on the costs associated with brand identification and separation. However, CPR is criticized for significantly undermining the eco-design incentives: Under CPR where costs are shared, the benefits (such as recycling cost reduction) from the individual design efforts spill over to other producers or even competitors, and hence producers may be less motivated to improve the product design for recyclability. The IPR model, on the other hand, overcomes this weakness of CPR and creates stronger design incentives, because it enables each producer to retain the returns from the investment on product design improvements with the independent processing cost. Therefore, the choice between IPR and CPR is largely driven by the trade-off between design incentives and recycling cost efficiency. At the same time, the impacts of these two factors also depend on the dynamics in the market. As such, it is a useful exercise to compare the two models, as in Atasu and Subramanian (2012) and Esenduran and Kemahlioğlu-Ziya (2015).

Atasu and Subramanian (2012) build a single-period model with a duopoly sales market in which high-end and low-end products are sold, respectively, by two producers. Consumers make purchasing decisions based on their heterogeneous product valuations for the differentiated products. The regulation mandates producer responsibility by imposing a binding recovery target for end-of-life products on both producers. The regulation can adopt either the IPR or the CPR model. Under IPR, each producer only pays the cost associated with its own products. Under CPR, the two producers share the total processing cost. The cost sharing can be set endogenously (i.e., according to the return shares, which also reflect the market shares in this context) or exogenously (which will be predetermined and fixed). In response to the regulation, each producer can invest in improving the product recyclability to reduce the cost of end-of-life recovery treatment of its products.

By comparing the outcomes under IPR and CPR, the paper shows that incentives to design more recyclable products exist under both models, but the incentives are always weaker under CPR than IPR. This is because CPR allows for one of the producers to free-ride on the other's design efforts. Although the existence of free-riding is less surprising, the detail as to who free-rides under CPR can be counter-intuitive. When the cost sharing is endogenous, the bigger contributor (whose processing cost has a higher weight in determining the final shared unit cost) may suffer from being free-ridden just as expected. However, when the cost sharing is determined exogenously, the bigger contributor can become the free-rider. This reversal of the outcome is caused by the competition dynamics in the primary market. Under IPR, in contrast, the market competition does not have direct influences on the design incentives. Moreover, the free-riding under CPR also has feedback effects on the primary market because the producer who benefits from free-riding can lower the price more aggressively and hence provides a higher surplus to

its consumers, while its competitor only offers a lower consumer surplus. Note that these observations are well aligned with perspectives from different fields such as industrial ecology (Lifset and Lindhqvist 2008) and practitioners (Atasu et al. 2010).

This analysis draws our attention to another operational aspect that matters in implementing take-back regulation: The scope of responsibility, shared under CPR versus independent under IPR, may have different influences on product design. Furthermore, the strategic dynamics including market competition and integration of the compliance-related considerations into the design and pricing decisions for producers are key drivers of take-back regulation efficiency.

Interestingly, while also looking at the comparison between IPR and CPR, Esenduran and Kemahlioğlu-Ziya (2015) take a very different perspective: They analyze how the coalition structure under the collective model matters. The composition of a coalition is a nontrivial factor because it can vary dramatically in reality. For example, the European Recycling Platform (ERP) that is set up in response to the WEEE Directive mainly consists of large (in terms of market/return share) firms including Braun, Electrolux, HP, and Sony. In contrast, the collective systems set up by local government authorities such as the one in Washington State covers a much wider range of both big and small firms. The authors argue that the coalition structure in a collective system may have important implications on the efficiency of the take-back regulation implementation.

Esenduran and Kemahlioğlu-Ziya (2015) look at multiple firms with different market shares in a collective system. All firms are subject to the same regulation that mandates a collection target. In response, each firm can choose the recyclability of its products. A higher recyclability helps reduce the unit cost of recycling, but it also incurs the incremental cost because it makes production more expensive. To stay focused on the take-back context, the paper focuses on compliance-related costs that include collection and recycling costs, as well as the cost of increased recyclability. Compliance schemes can be of two types. Under IPR, each firm sets the actual collection rate to minimize its own overall cost. Under CPR, the central entity of the collective system determines the actual collection rate to minimize the total cost of collection and recycling incurred to the entire system, which will then be split among all producers by market shares. Note that the option of over-compliance for producers, i.e., the actual collection being higher than the regulation target, is also considered. Economies of scale are reflected in the marginal recycling cost that is decreasing in the total recycling volume.

Esenduran and Kemahlioğlu-Ziya (2015) show that under CPR, the equilibrium recyclability level is lower for smaller firms, which is somewhat consistent with the free-riding result in Atasu and Subramanian (2012). In addition, the paper also reveals the following unexpected conclusions: One result shows that the equilibrium recyclability level can be higher under CPR than under IPR, which is in direct contrast to Atasu and Subramanian (2012). The recyclability being higher under CPR tends to happen when the collective system is formed by big firms. The underlying rationale is that under CPR, free-riding issues are less severe for the collaboration among big firms. Moreover, producers benefit from higher economies of scale. Therefore, the equilibrium collection rate is higher, which then drives

up the total return of the recyclability investment (reduced processing cost for more units) and hence encourages better designs. Another result points out that although CPR appears to facilitate economies of scale and hence should always lead to a higher collection rate (i.e., higher degree of over-compliance), it is in fact possible to have a lower collection rate under CPR than IPR, especially when the collective system consists of small firms. The reason is that besides the economies of scale, better designs can also induce more collection because of the resulting lower processing cost. Yet when a firm is in a collective system with other small firms, it has lower design incentives than when it acts alone, as it bears the risk of being free-ridden and cannot retain the full benefit of design improvements. The negative effects of diminished design incentives can even dominate the positive effects of the scale economies in processing and therefore result in a lower overall collection rate under CPR.

Another insightful finding is that a firm may incur a higher cost when it joins a collective system with other relatively smaller firms compared to when it collects and recycles individually. In that case, if the firm pulls out of the collective system and if the firm is big enough, then the compliance cost of the collective system goes up. Note that these observations have important implications for supply chain managers. In particular, these results suggest that the reverse supply chain manager needs to consider take-back compliance collaboration with other firms very strategically. Depending on the type of compliance partners, a reverse supply chain manager may increase or decrease the reach of its collection programs and inform the design/engineering department regarding the extent of take-back compliance cost savings that can be achieved by improved designs.

The take-away from this study is that the coalition structure in a collective system is another operational detail that makes a difference in implementation. Specifically, compliance behaviors of a producer in collection or product design can vary significantly depending on its relative size compared to the other participants in the coalition.

In sum, Atasu and Subramanian (2012) and Esenduran and Kemahlioğlu-Ziya (2015) collectively provide a comprehensive understanding of the collective and individual models for EPR implementation. Their results reveal certain weaknesses associated with each model and motivate the need for research toward more powerful regulatory tools for efficient implementation. One natural direction for such pursuit is to explore the possibility of integrating the most prominent advantages of the two models, namely, generating superior design incentives and achieving cost efficiency.

As such, we start with the collective model as a base to seek potential improvements. First and foremost, note that the consolidation of end-of-life products, while facilitating economies of scale, does not exclude the potential of a collective model to properly internalize the associated processing costs for incentivizing eco-design. Rather, the root of the problem resides in the cost allocation mechanism, which is usually based on return share or market share. Such proportional cost allocations gain popularity among both scholars and regulators mainly due to their simplicity and transparency. However, they are not ideal solutions because they give rise to

a major weakness of the collective model, which involves fairness and stability concerns: Producers who feel overcharged by a collective system may break away and operate in an individual system instead. This jeopardizes the stability of the collective system, which is a precondition to achieve the scale advantage. This is indeed observed in practice. One example is that in Washington State, where a centrally run standard plan is in place with a state-wide collective system, some producers file requests for switching to self-operated independent plans with the belief that their stand-alone cost will be lower. This is because the proportional cost allocation model employed by the collection system in Washington State falls short of accounting for recycling cost heterogeneity across producers. Note also that responsibility allocation problems are not unique to take-back regulation. See Caro et al. (2013) and Sunar (2016) for similar discussions in the emissions regulation context.

As such, a first step toward building a more robust collective system starts with a search for a cost allocation mechanism that accounts for product-specific metrics to help achieve fairness in cost allocation and guarantee stability. Meanwhile, note that the operations of collection and recycling in a collective system are likely to be conducted in a network, where the end-of-life items from numerous producers are processed by the aggregation of different recycling capacities. Therefore, the cost allocation model to guarantee fairness and stability should also properly incorporate network effects. Gui et al. (2016) present a fruitful attempt in this direction.

8.8 The Individual System or the Network?

Gui et al. (2016) study a set of producers that are faced with recycling standards imposed by regulation. A collection and recycling network (CRN), which is effectively a collaborative reverse supply chain, collectively handles all the end-of-life products from producers. The network comprises three sets of nodes that represent the collection points, the consolidators, and the processors, respectively. All items go through one node from each of these sets in sequence. The edges in the network stand for costly processes of collection, consolidation, processing, and the transportation in between. Product heterogeneity in the model is reflected by different processing costs of products from different producers. The collective system is stable when a grand coalition is formed and all producers choose to voluntarily stay in the system. The system stability requires a fair cost allocation, which is defined as one that ensures each producer to be better off by remaining in the grand coalition. If a producer can achieve lower cost outside the collective system, it will leave the grand-coalition to operate individually or form a sub-coalition with some of the other producers.

The processing capacities in the network consist of capacities that are privately owned (or contracted) by producers and capacities that are contracted by the central authority who runs the grand coalition. While the former type of capacities can be used by any (grand or sub-)coalition the owner producer belongs to, the latter can

only be used by the grand coalition at no additional cost. The total cost incurred by the collective system is calculated based on the socially optimal routing (i.e., the cost minimizing solution to the network flow problem) and will be shared by all members in the system.

The paper develops a fair cost allocation model based on the cost-corrected return share with capacity adjustment (hereafter referred to as the dual-based cost allocation), which makes two adjustments to the prevalent return share-based cost allocation. The first is that in calculating the return share, each producer's return volume at every collection point is weighted by the marginal cost of processing its products. This adjustment reflects the product heterogeneity in the network. The second adjustment is to grant a reward to each producer for the valuable capacities it contributes to the network, at a properly set unit reward price. The study further shows that this cost allocation can be widely applicable and retain fairness even when economies of scale enter the model or when the central authority-contracted capacities are open at a fee to producers outside the grand-coalition. These findings are also validated using the actual EPR implementation data from Washington State.

This research once again highlights the importance of the operational angle, this time in the form of network effects under the collective model. Since the key operational characteristic of a network is the pooling of available capacities for better utilization, the critical detail that matters here is how resource sharing can be reflected in the cost allocation. Notably, the dual-based cost allocation, in addition to maintaining system stability, also exhibits potential to make up for another deficiency of the prevalent return share-based allocation. To be specific, the dual-based allocation can help enhance design incentives in collective take-back regulation implementations, because it rewards the ease of recycling of products by accounting for the product heterogeneity. Gui et al. (2018) explore this further and show that not only the product design choices but also the process technology matters in a network setting.

In Gui et al. (2018), end-of-life product treatment is conducted collectively in a network with a similar setting to that in Gui et al. (2016). One crucial aspect of the network configuration highlighted in this paper is the differentiation among processors in not only capacity but also process cost efficiency (i.e., process technology). When there are sufficient low-cost capabilities, cost synergy in the collective network is high because the resource sharing in collection and recycling has a higher potential for cost savings. In the network, product design improvements and characteristics of the processors are inter-related, because the benefits of design are mostly realized through influencing the processing cost. Specifically, product design is a substitute (complement) for process technology when the enhancement of recyclability leads to higher (lower) cost reduction at the less efficient processors.

The paper focuses on a collective system under a stable cost allocation, such as the dual-based cost allocation that ensures the grand-coalition stability as discussed above. It shows that a collective implementation, in addition to having a cost advantage, can also stimulate design incentives that are comparable or even superior to those under an individual implementation. This environmentally and economically desirable outcome occurs when the cost synergy is low (high) and

the product design and the process technology are substitutes (complements). When these conditions are not met, the tension between achieving cost efficiency and creating design incentives is irreconcilable under a collective system, and hence it may not be both coalitionally stable (which facilitates economies of scale) and strong in motivating product recyclable design at the same time.

This paper points out the importance of another operational detail: Understanding the level of process technology in a collective system network and its relation with the product design is key in constructing effective regulation. It also suggests that the reverse supply chain manager needs to carefully account for the existing collection and recycling infrastructures in its strategic compliance with the regulation: The compliance system in which the firm collaborates with other producers and the central authority, as well as the technologies it possesses, can significantly influence the value added from the collection and recycling capacity investments or product design improvements.

8.9 Product Design or Recycling Process Improvements?

While improvements in eco-design of products have been a central focus of EPR regulation to increase the effectiveness of recycling, product design should not be assumed as the only lever. The recycling infrastructure and process technology development also play a critical role in recycling: The potential environmental benefits of better product designs need to be materialized through the availability of recycling infrastructures with sufficient capacity and efficiency, as well as process technologies with appropriate equipment and capability, for conducting the recycling activities on the ground. Meanwhile, the choice of EPR implementation models may also influence the investment decisions on recycling infrastructures. A natural question that arises is how do the individual and collective systems compare when we also account for their impacts on promoting the recycling infrastructure development and how does the comparison differ from that derived from the perspective of improving product designs (e.g., as discussed above in Atasu and Subramanian 2012). Gui (2020) sheds light on this question.

Specifically, Gui (2020) focuses on the recycling infrastructure and studies the joint investment decisions from two producers under the take-back regulation. To capture the complicated collaborative dynamics between the producers under the collective system, the paper adopts a Nash bargaining model. Intuitively, since the collective system has the merit of spreading out the fixed investment costs among producers to ease the economic burden, it should be able to encourage higher investment in recycling infrastructures. Nevertheless, the paper reveals that the collective system may instead reduce the investment and compromise the development outcome compared to the individual system, due to a significant challenge in the collaboration: The joint investment under the collective system has to reconcile the different recycling needs from all participating producers to be mutually agreed on. Therefore its effectiveness may be undermined by the

heterogeneity across products. In contrast, each producer enjoys the flexibility to make the optimal decision on its own under the individual system. Consequently, the collective system can be disadvantageous to only result in recycling infrastructures with lower capacity and efficiency.

Meanwhile in the recycling operations, the specific process technologies often work closely with the product designs in a complementary way to realize the potential environmental benefits of the latter. For example, electronic producers may improve their product design for recycling by using alloys that self-separate under proper stimuli to increase disassembly. Yet the adoption of advanced technology in recycling (which provides the corresponding stimuli) will also be needed to effectively achieve high material recovery. The complementarity between the improvements in product and process technology is also supported by the EPR implementation for, for example, the Japanese Home Appliance Recycling Law (HARL), where the coordinated efforts from both product designers and recycling facility engineers have been a contributing factor to the significant enhancement in the recycling efficiency and resulting environmental benefits (Atasu et al. 2010). As such, incentives on process technology improvements of processors should also be accounted for, along with those on product designs of producers, to establish a comprehensive assessment of the regulation and the implementation models. Rahmani et al. (2021) provide useful insights.

To capture the interactive relation between the decisions on improving product and recycling process technology, Rahmani et al. (2021) consider a game-theoretical model with producers and processors that share the total revenue (or cost saving) of the recycling with a given split. While the design of each product (captured by recyclability) is chosen by its own producer, the recycling process technology is set by the processors, and they all, respectively, cover the associated costs. Similar to previous studies, under the individual system, each producer independently contracts with a processor for the recycling and receives the producer split of the recycling revenue (or cost), whereas under a collective system, all producers work with the same processor and share the total producer split of the recycling revenue (or cost) proportionally based on each producer's respective recycling volume. By comparing the producer and processor decisions in equilibrium under the two implementation models, Rahmani et al. (2021) reveal a significant superiority of the collective system over the individual system: Although the collective system suffers from free-riding that suppresses the recyclability improvement in product designs by the producers, it is able to increase the recycling process technology level by the processor. The key rationale is rooted in the scale economy in recycling where products from different producers are processed in an aggregated manner. It does not only help reduce the recycling cost as has already been widely recognized but also enhance the processor incentive in improving the recycling process as the return of such investment is magnified by the large recycling volume. Furthermore, the more efficient process may in turn encourage product design improvements from the producers due to their complementarity in determining the recycling revenue. As a result, under the collective system, the compromised product design incentives due to the free-riding problem can be more than compensated. As such, the collective

system can lead to higher overall environmental and economic benefits, especially when the producers' recycling volumes are similar.

These studies echo the importance of accounting for operational details on the ground in the formulation and evaluation of EPR regulation. The realization of better product design incentives, which constitute one of the most significant strengths of EPR, needs to be supported by every step along the reverse supply chain including the recycling infrastructure and the process technology. Each of these considerations may create different challenges (such as the conflicting interests among producers due to their product heterogeneity) or nuanced influences (such as the complementarity between product and process technology improvements on the recycling efficiency) and therefore has to be reckoned with by the reverse supply chain managers accordingly in order to establish a well-functioning regulatory compliance system.

8.10 Local Regulation or Global Goals?

Another major concern in dealing with end-of-life products is their leakage from the regulated systems due to exports: As end-of-life product waste is costly to process, exporting the waste to developing regions such as Africa and Asia may become an attractive option to producers. The processing costs in these areas are typically much lower due to the lack of stringent regulatory processing requirements and the availability of cheap labor. Obviously, this is bad news for the exporting area because when the waste leakage is not properly handled, the effectiveness of regulation in meeting the environmental objectives is seriously undermined. On the other hand, mishandled waste poses an acute hazard to the health and environment in the destination areas of waste export. Note also that such global leakages of environmental externalities are not unique to the take-back regulation but has emerged in the emissions context as well (see Sunar 2016, Drake and Just 2016, and Islegen et al. 2015).

In view of this prominent threat that has already been observed in the practice of electronics recycling (e.g., documented in BAN 2005), the EU has ratified the Basel Convention, which is an export restriction that prohibits export of end-of-life products. Nevertheless, although dumping waste is unacceptable, the export of used but still functioning products to developing regions may be allowed. The rationale is that exporting functional used products can be a win-win solution. It helps manage the unwanted items in the developed countries and provides people in the developing countries with affordable access to used products. This forms the foundation for what we call partial export restrictions, under which used products in working condition are permitted for export while end-of-life products are not. A case in point is the introduction of the Responsible Electronics Recycling Act in the USA in 2011 (US Congress 2011).

Some NGOs, such as the Basel Action Network, however, argue that it may be beneficial to ban exports of even used products, i.e., a full export restriction. As

we compare the partial and full export restrictions, we have to keep in mind that producers may utilize the used products pulled off the secondary market along with the end-of-life products to fulfill the recovery obligation as discussed before. Then allowing used products with remaining lifespans to be exported may accentuate this type of secondary interference that has been shown to be environmentally unfavorable. Therefore, the problem becomes of concern that how do different stringencies in export regulation (none, partial, or full restriction) differ in terms of environmental performance. This problem is addressed in Alev (2015).

In Alev (2015), the model setup and assumptions are the same as those in Alev et al. (2020) explained earlier (in Sect. 8.4). When products are exported from the developed country to a developing country, the associated unit recycling cost is reduced to reflect the lower processing cost in the developing country due to the less stringent requirements or cheaper labor. The study then compares the partial and full export restrictions, where the former forbids the export of only end-of-life products while the latter forbids the export of both end-of-life and used products.

The first finding in Alev (2015) is that when compared to the case with no export restriction, enacting a full export restriction increases the reuse level and reduces the new production volume in the developed country. This result substantiates the need for closing the door for waste product export from the developed country's perspective, because preventing the export of end-of-life and used items reduces a producer's incentive for secondary market interference. The result of a partial export restriction, on the other hand, is surprising. Intuitively, we expect that the partial restriction should partially attain the benefits of a full restriction. However, a partial restriction may lead to even worse outcomes. Specifically, with a partial export restriction, the reuse level may decrease and the total production may increase in the developed country. Moreover, the developing country that receives exports may suffer from the environmental consequences of a higher export volume.

These results once again show that the lack of understanding of the operational mechanisms that underlie producer incentives can compromise the regulation effectiveness, not only in the developed economies where take-back regulation is enacted but also in developing economies importing used products. The detail that critically matters in this case is that reverse supply chains for compliance can in fact have two destinations for recycling, namely, domestic and international. Overlooking the international export as an alternative way to deal with the end-of-life obligation can create loopholes in export restriction mechanisms.

8.11 Conclusion: The Devil in the Details

In this chapter, we reviewed a series of operations management papers dealing with take-back regulation implementations to discuss a broad spectrum of possible issues that may arise in formulating and complying with environmental regulation. This review suggests that the supply chain challenges associated with environmental regulation may be driven particularly by policy implementation decisions often

because of certain assumptions based on abstraction of environmental problems and industrial dynamics. Yet even basic assumptions that are intuitive, sound and applicable in general circumstances may in fact not hold due to various reasons. These may include technology or commodity market dynamics, differences across the regulated industries, nature of the products of interest, or less obvious but valid alternative responses for producers under regulation.

In turn, these observations suggest that the successful formulation and compliance with environmental regulation can only be possible with careful implementation choices that account for the devil in the details. This insight emphasizes the merits and importance of taking a supply chain perspective in the context of environmental regulation. We posit that for environmental regulation to achieve its goals, it must account for its influences on the operational decisions of all stakeholders involved, especially those who are directly targeted by the regulation, e.g., producers and their supply chains. In other words, an operational lens appears to be a key instrument in defining appropriate implementation structures for the regulation and strategic responses for compliance. It is critical that supply chain managers understand the details and nuances of take-back policies as they are often not what they appear to be on the surface.

As environmental regulation gains momentum, it will inevitably have to deal with more problems emerging from a wider scope of fields. To triumph in these new challenges calls for more supply chain research in different directions. Looking again at the take-back regulation, although extensive work already exists as discussed, its continuous development may still give rise to new questions to be answered. One instance is that, as recycling activities flourish, encouraged by the more stringent recovery targets, an increasing number of entities are entering the recycling industries to share the pie and make profit. To ensure regulatory effectiveness in promoting environmentally superior treatment of end-of-life products, current regulation on recycling standards may need to be modified and extended to also cover the quality of recycling by recyclers. As demonstrated by research studies discussed in this chapter, crafting more efficient regulation will require a thorough understanding of the operational details involved. The same is also true for other forms of environmental regulation, such as the RoHS and the REACH (Registration, Evaluation, Authorization, and Restriction of Chemicals) Directives of the European Union that regulate the use of certain substances and chemicals, and the variety of emissions regulations around the globe. In particular, the emissions issues have a wide range of influences because pollutants can be transported to other areas easily. As a result, dealing with emissions problems may involve extensive collaboration and coordination between various supply chain stakeholders and lead to different operational problems.

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Chapter 9

Food Loss, Food Waste, and Sustainability in Food Supply Chains



Renzo Akkerman and Frans Cruijssen

9.1 Introduction

Food supply chains are a major part of many discussions about sustainability. Considering that these supply chains account for more than a quarter of global greenhouse emissions (Poore and Nemecek 2018), this attention certainly seems warranted. From a broader sustainability perspective, food supply chains are arguably even more essential, considering that about half of the world's habitable land is used for agriculture, 70% of global freshwater withdrawals are used for agriculture (see Chap. 4 by Hoekstra (2024) for more details), 78% of waterway pollution with nutrient-rich pollutants (i.e., freshwater eutrophication) is the result of agriculture, and 94% of mammal biomass (excluding humans) is livestock (Ritchie and Roser 2020; FAO 2011; Poore and Nemecek 2018; Bar-On et al. 2018). The management of food supply chains therefore has major impacts on climate change, land use, water management, pollution, and biodiversity (Ritchie and Roser 2020).

With regards to climate change, different parts of the food system contribute to greenhouse emissions, as shown in Fig. 9.1. Clearly, raising animals and growing crops account for most emissions (31% and 27%, respectively), especially if we also consider the land use involved (an additional 24%). The part of the supply chain after primary production (processing, transport, packaging, and retail) contributes the remaining 18% of the food system impacts.

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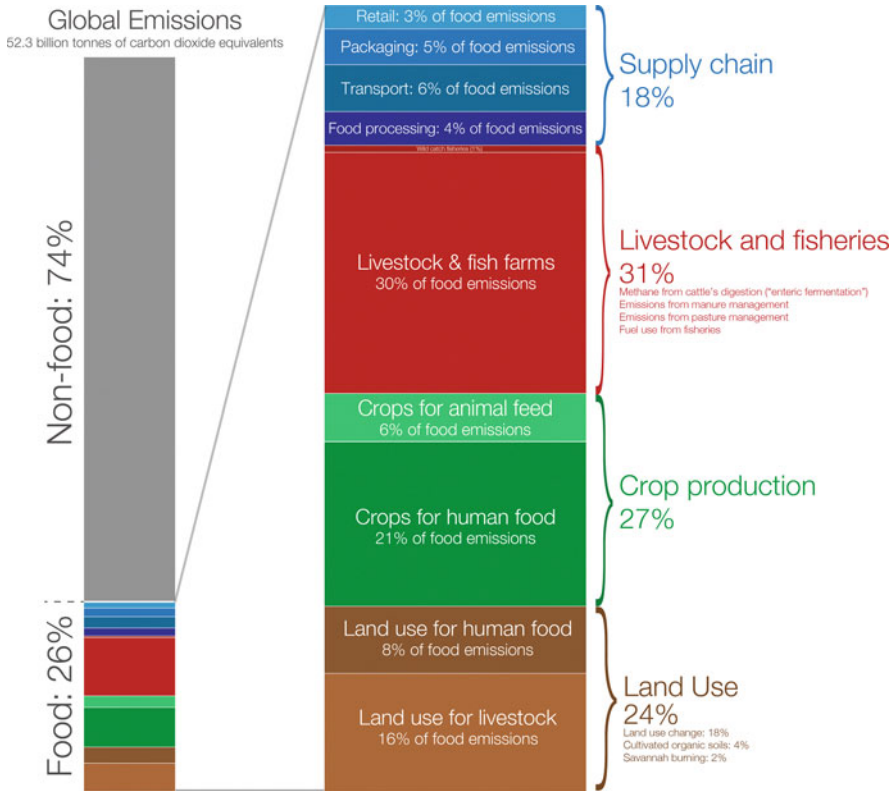


Fig. 9.1 Global greenhouse gas emissions from food production (Ritchie and Roser 2020, based on Poore and Nemecek 2018). Cropped from original figure licensed under CC BY 4.0

The fact that a significant share of emissions is related to animal production systems (including the related animal feed production and land use) does not only mean that emissions are mostly caused early in the supply chain. It also means that compared to other food products, meat consumption has relatively high impacts. Reduction of meat consumption is therefore also one of the main discussion topics in relation to the sustainability of the human diet (refer to Chap. 3 by Boukherroub et al. (2024) for a detailed discussion about carbon footprinting).

In addition to the environmental concerns, food supply chains also have important social and economic impacts. Many people around the world depend on agriculture and other food supply chain activities for their income (refer to Chap. 21 by Sodhi and Tang (2024) for a discussion about social responsibility in supply chains). Even in high-income countries, in which the number of people involved in agriculture is significantly lower than in low-income countries, many people are still involved in downstream food supply chain activities (such as food processing, retail, or foodservice). And even though many people earn their livelihood in food and agriculture, a key socio-economic concern is food insecurity: many people do

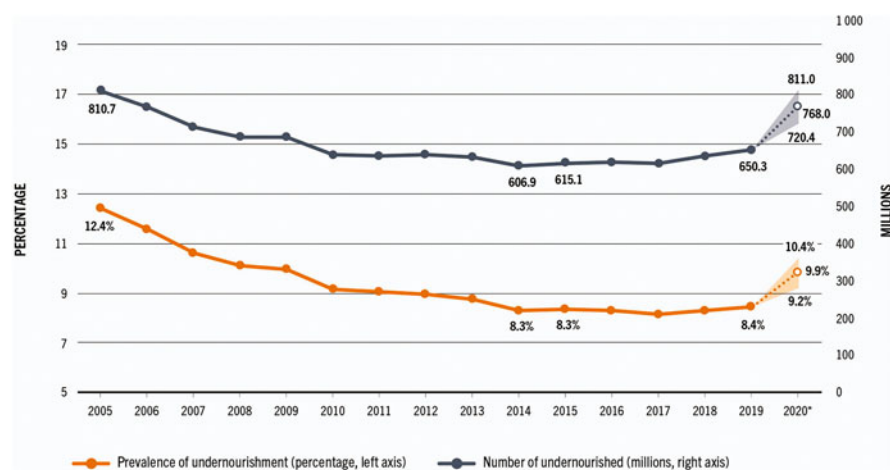


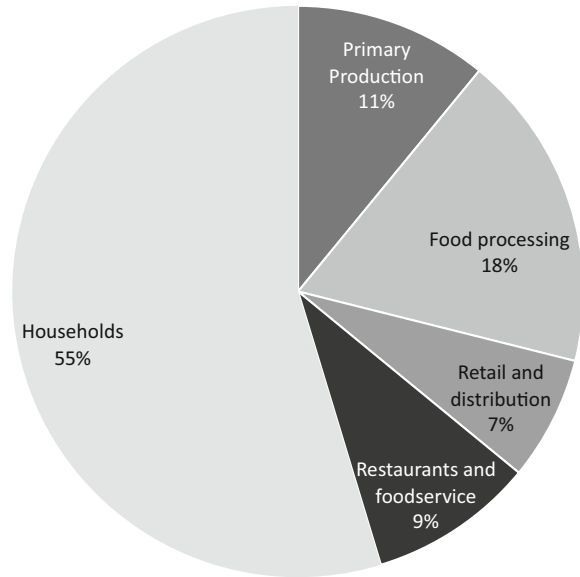
Fig. 9.2 Development of the number of undernourished people in the world (projected values for 2020 in the figure are illustrated by dotted lines; shaded areas show lower and upper bounds of the estimated range) (FAO 2021). Adapted from original figure licensed under CC BY-NC-SA 3.0

not have sufficient access to food. As illustrated in Fig. 9.2, almost 10% of the world's population are considered undernourished (FAO 2021). It should be noted that this is not just referring to developing countries. For instance, recent 2020 statistics show that 10.5% of US households were considered food insecure for at least part of the year (Coleman-Jensen et al. 2021). In their 2030 Agenda for Sustainable Development, the United Nations (UN) therefore also has Zero Hunger as the second of their 17 sustainable development goals (SDGs) (United Nations 2015).

Considering the environmental concerns regarding climate change and the social concerns regarding food insecurity, it is worrying that a lot of the food that is produced does not make it to consumption. Throughout the food supply chain, a significant share of food products is lost or wasted. In a seminal study for the UN's Food and Agriculture Organization (FAO), Gustavsson et al. (2011) report that a third of the globally produced food is lost or wasted at some point during the food supply chain. This means that related environmental impacts were made for nothing, and that many people remain food insecure. Reducing the food that is lost or wasted is a key part of the twelfth UN SDG: Responsible Consumption and Production.

From a supply chain perspective, this also changes the view on the environmental impacts of food production presented in Fig. 9.1. Even though most of the environmental impacts are made early in the food supply chain, if the resulting food products get lost or wasted in later supply chain stages, the actors involved in these stages have an impact on a larger share of environmental impacts than the statistics in Fig. 9.1 would suggest. For instance, in Europe, more than half of the food losses and waste occur at the household level (Stenmarck et al. 2016; Eurostat 2022), as can be seen in Fig. 9.3. Another 34% of food loss and waste occurs somewhere

Fig. 9.3 Division of food loss and waste by sector for the European Union (Eurostat 2022)



along the supply chain between primary production and consumers (in processing, retail and distribution, or in restaurants and foodservice).

In general, studies show that the total amount of food loss and waste is comparable between countries and regions (e.g., Gustavsson et al. 2011). However, why and where this loss and waste occurs varies significantly. Typically, in developing countries, more food products are lost in the beginning of the supply chain, due to a lack of proper postharvest handling and storage technologies. Refrigeration of perishable products is for instance more likely to be insufficient. In developed countries, these early stages are much more professionalized, and more products are lost and wasted later in the supply chain, as seen in Fig. 9.3. Here, consumer behavior plays a much more important role.

Preventing or reducing the amount of food lost or wasted from farm to fork is therefore essential. This is also reflected on the agenda of many policy makers. For instance, in 2023, the European Commission aims to introduce legally binding targets to reduce food waste (European Commission 2022). Also, in 2015, the US Department of Agriculture (USDA) and the Environmental Protection Agency (EPA) set a goal to cut food loss and waste in the US in half by 2030 (EPA 2022a).

In the remainder of this chapter, we discuss food loss and food waste in the context of sustainable supply chain management. In Sect. 9.2, we further introduce the concepts of food loss and food waste. Then, in Sect. 9.3, we provide an overview of the drivers behind food loss and food waste. Section 9.4 subsequently goes into potential actions to prevent or reduce loss and waste in food supply chains. Finally, Sect. 9.5 concludes this chapter with a brief outlook towards the future of sustainable food supply chains.

9.2 Distinguishing Food Loss and Food Waste

The terms food loss and food waste are both used to describe food that does not end up making it to the intended human consumption. The two terms are, however, sometimes used interchangeably, and sometimes they are used to point to different things. To clarify, we briefly discuss two of the leading definitions that have been used: one by FAO and another by the European Commission.

A key difference between the FAO definition and the EC definition is that FAO distinguishes between food loss and food waste, whereas the EU only refers to food waste. FAO refers to food loss as “the decrease in the quantity or quality of food resulting from decisions and actions by food suppliers in the chain, excluding retailers, foodservice providers and consumers” (FAO 2022a) and to food waste as “the decrease in the quantity or quality of food resulting from decisions and actions by food suppliers in the chain, including retailers, foodservice providers and consumers” (FAO 2022b). Basically, loss is used in the initial stages of the supply chain, and waste is used in the later stages.

In the overview of the food supply chain shown in Fig. 9.4, this distinction is also illustrated. As can be seen, both the FAO and EU perspective start at primary production, after products are harvested, up to the point of consumption.

The EU instead uses food waste to refer to all waste streams throughout the supply chain (as also shown in Fig. 9.4). Formally, the EU, in its waste directive, simply defines food waste as all food that has become waste (EU 2018). Here, it refers to the Food Law to define food as “any substance or product, whether processed, partially processed or unprocessed, intended to be, or reasonably expected to be ingested by humans” (EU 2002).

In the remainder of this chapter, we will follow the FAO terminology as much as possible, distinguishing between food loss and food waste where useful, and otherwise refer to food loss and waste or the acronym FLW.

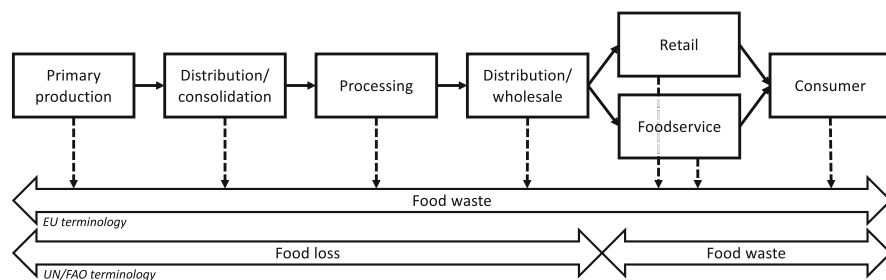


Fig. 9.4 General structure of food supply chains with an illustration of the distinction between food loss and food waste based on UN/FAO and EU terminology (partly based on Akkerman et al. 2010; Cruijsen et al. 2022)

9.3 Drivers of Food Loss and Food Waste

In recent years, many studies have been performed to identify the sources and drivers behind FLW. Here, we do not aim to repeat such a literature study, but instead focus on summarizing the main findings from recent studies. Previous overviews sometimes had a broad scope covering most of the supply chain, but some studies also zoomed in on specific supply chain stages. The content of this section is mainly based on the reviews done by Bhattacharya et al. (2021), de Moraes et al. (2020), Dora et al. (2021), Juvan et al. (2021), Magalhães et al. (2021), Raak et al. (2017), Shafiee-Jood and Cai (2016), Spang et al. (2019), and Surucu-Balci and Tuna (2021).

9.3.1 General Classification of FLW Drivers

In general, the literature on drivers of FLW discusses their findings across different product categories, not necessarily identifying different drivers behind different products. In this section, we structure our discussion around the supply chain stages identified in the previous section. The focus will be on identifying what specific drivers play a role in each supply chain stage. However, we first introduce several broader, overarching driver categories that lead to FLW across the different supply chain stages. These more general driver categories do often have specific dynamics in different supply chain stages, which we then reflect on in the individual sections. Also, some of the drivers might not apply to every stage (e.g., drivers related to consumer behavior are mostly relevant in the retail, foodservice, and consumption stages). Table 9.1 presents an overview of driver categories, together with the main underlying drivers mentioned in the literature.

9.3.2 Primary Production

Primary production is the stage in which drivers related to environmental conditions are arguably more relevant than in any other stage. This is related to, for example, weather conditions, the possibility of disease and contamination, as well as damages by insects or birds.

As in any of the supply chain stages, standards and regulations play a key role. Already this early in the supply chain, products that do not satisfy often rigorous quality demands will be discarded. Here, quality demands can relate to weight, size, shape, and appearance. This either means that products are left on the field or that they are sorted out soon after they are harvested. Standards and regulations related to food safety might also play a role here, as food can also be wasted due to safety concerns.

Table 9.1 General food loss and waste drivers

Driver category	Key drivers
Environmental conditions	Conditions during, e.g., drying or storage; product disease or contamination; product damage by insects or rodents
Standards and regulation	Failure to meet quality standards (weight, shape, size, visual defects); food safety concerns; social norms
Lack of infrastructure or technology	Lack of adequate or efficient storage, refrigeration, transport, or handling technology; inadequate packaging; inadequate waste management services
Lack of skills or expertise	Improper handling of products; poor organization culture regarding FLW; labor shortages; lack of product knowledge; lack of planning skills
Economic factors and incentives	Lack of managerial incentives to reduce FLW; market conditions, social norms and attitudes, high transportation costs
Supply chain management issues	Inefficient procurement channels; poor forecasting practices (uncertain demand); inefficient inventory management; logistical inefficiencies, lack of communication and collaboration; mismanagement of cold chains
Consumer behavior	Lack of FLW awareness; overbuying discounted products, lack of inventory management and planning skills; attitudes towards leftovers

The lack of appropriate infrastructure or technology is also prevalent. In primary processing, this mainly relates to technology related to quality preservation (e.g., refrigeration) and harvesting. In addition, the use of lower quality input materials is mentioned, as this can affect the quantity and quality of the products being produced. It is interesting to note that this would imply a broader perspective on FLW, as such unrealized yield would typically not be measured. In general, the importance of these drivers related to infrastructure also depends on the context, as these drivers are often much stronger in developing countries.

In addition to technology, the human factor also plays a key role. This for instance concerns the potentially improper handling of delicate products. More specifically, for primary processing, the literature points out the lacking knowledge of cultivation techniques and harvesting timing as important drivers. Like the technological drivers, these expertise-based drivers are often stronger in developing countries.

Economic factors also play a significant role in the primary processing stage. The main issue here is that crops are sometimes left on the field due to unfavorable market conditions: low prices can mean that harvesting efforts are not worth it.

Finally, the literature reports different drivers related to the complexities of supply chain management. Uncertainties related to demand forecasts or harvest yields can lead to overproduction, which in turn leads to products being wasted. This also includes aspects related to how different supply chain actors collaborate and communicate. For instance, to meet supply agreements with retail, overproduction might be unavoidable.

9.3.3 *Distribution/Consolidation*

After harvesting, during the initial distribution and consolidation of products, many of the food waste drivers present during primary production, especially since in many cases there is still a lack of adequate infrastructure and technology, and similar environmental conditions influence product degradation. Also, the sorting and possible rejection of products due to quality standards continue during this stage, potentially even more so.

During distribution and consolidation, environmental conditions also refer to storage conditions, especially related to products that require drying during this stage. During storage, damages by insects and rodents are also a relevant factor. Especially in developing countries, these environmental conditions play a key role in these early parts of the supply chain.

In terms of skills and expertise, improper handling of delicate products is still a key driver, but during distribution and consolidation, there is also more emphasis on the lack of investment in knowledge regarding postharvest technologies. Furthermore, the improper stacking and overfilling of bins during the distribution process is mentioned as an additional driver.

From a supply chain management perspective, overstocking still plays a significant role, but in this stage, it is not only linked to poor forecasting and uncertainty in demand, but also specifically linked to the existence of take-back agreements. Finally, when temperature-controlled supply chains are used, they are not always managed appropriately, also contributing to FLW.

9.3.4 *Processing*

The literature on FLW does typically not distinguish between primary and secondary processing, even though some FLW drivers mentioned in the literature seem more applicable to the primary processing stage in which food products are transformed to commodities (e.g., cutting and trimming losses or processing by-products).

High quality standards for products remain a challenge in this stage of the supply chain, but now also extends to processed food products, which often also have strict guidelines on visual characteristics.

Lack of adequate infrastructure and technology concerns processing and packaging aspects here, which includes lacking technology for quality preservation, defective equipment, and inadequate packaging technology. Skills and expertise of operators in processing environments also contribute to FLW: process inefficiencies and defective products can be caused by a lack of skilled staff (or general human error).

In addition to lacking processing technology, several inherent characteristics of processing activities lead to food waste drivers. During processing, parts of the food products used as ingredients might be trimmed, leading to losses. This includes

both edible food losses and inedible by-products that might still have other uses. In many processing environments, part of the production volume is furthermore lost due to product sampling for destructive quality-control tests and due to losses around product changeovers and cleaning operations.

Finally, in addition to the overstocking due to uncertainty and take-back agreements, the lack of information sharing as well as inadequate inventory management is mentioned as FLW drivers. Also, the lack of attention to by-product flows in supply chain management activities is relevant here as processing activities are a major source of by-products.

9.3.5 Distribution/Wholesale

After processing, in the distribution and wholesale stage, drivers related to quality standards and improper product handling skills remain to be somewhat relevant. Product handling issues in this stage specifically relate to moving products in and out of storage and to product damage by transportation equipment. Also, some economic factors and incentives are mentioned: high transportation costs and a lack of incentives to reduce food waste are part of these discussions.

In this stage, two categories of food waste drivers are however most prominent in the literature: drivers related to lacking infrastructure and technology and drivers related to supply chain management.

The lack of infrastructure and technology is mostly related to inadequate cold chain technology. Proper cold chain logistics equipment for storage and transportation is clearly important and lacking or inadequate transportation equipment or cold storage facilities is a key driver mentioned by many authors. In addition, poor product packaging material is also mentioned within this category.

Since this supply chain stage purely deals with logistics activities, it might not be surprising that the literature mentions many FLW drivers related to supply chain management issues. Like other supply chain stages, lacking communication and supply chain collaboration are often mentioned. More specific logistics-related drivers are found in relation to poor forecasting practices and inefficient inventory management, but the literature also mentions aspects like the proliferation of product portfolios, the management of promotional activities, and handling seasonality in supply or demand.

Many of the drivers mentioned in this stage are more relevant in developed countries, as the distribution and wholesale stages are often more prominent in the more professionalized, longer supply chains found in those countries.

9.3.6 Retail

In general, in the retail stage, we continue to see quality standards play a similar role as in other supply chain stages, but we see some retail-specific aspects mentioned in relation to other driver categories. The FLW literature does not specifically exclude certain types of retail but seems to mostly be relying on data from traditional brick-and-mortar retail environments.

In relation to infrastructure and technology, lacking or inadequate cold chain equipment, as well as poor packaging, are still prominent FLW drivers mentioned by many authors. The lack of infrastructure, however, also extends to managing waste streams, and some of the literature specifically mentions the lack of access to recycling and collection for composting, as well as the lack of storing and transporting infrastructure for the donation of surplus product to organizations such as food banks.

The lack of a donation infrastructure is also mentioned in relation to economic factors and incentives. More specifically, the literature points to the fear of lawsuits in relation to donations: if donated food would lead to food safety issues after donation, retailers could be held responsible or incur reputational damage. In addition to a general lack of incentives to reduce waste, this even adds an incentive against redistribution. A similar incentive exists due to a fear of finding donated products back on grey markets, which retailers would also like to prevent.

FLW drivers in the skills and expertise category relate to improper handling of delicate fresh products. Some of the literature also emphasizes the lack of knowledge on recycling and waste minimization, as well as the presence of a poor organizational culture with regards to waste reduction.

Many supply chain management issues are mentioned as food waste drivers in the retail stage. Inefficiencies in inventory management and resulting expiry date problems are a key driver here. Also connected to poor forecasts and lacking communication and collaboration with other supply chain actors, managing inventories can be difficult.

Finally, the literature also identifies some drivers related to specific retail practices. Especially in this stage of the supply chain, a wide range of products and brands is often present, complicating the supply chain management issues mentioned above. More specifically, factors such as promotions lead to significant fluctuations in sales and providing a visually attractive store sometimes also leads to overstocking.

9.3.7 Foodservice

For the foodservice channel, the literature also identifies a variety of specific FLW drivers. In terms of skills and expertise, FLW drivers also relate to inadequate skills related to planning and preparing meal production. This, for instance, includes

lacking skills to manage leftovers, poor cutting and trimming practices, and inadequate judgment of portion sizes. Like in the retail channel, the literature here also mentions poor organization culture regarding waste reduction and a lack of knowledge on reuse and recycling options.

Several characteristics of foodservice practices are also linked to food waste drivers. More specifically, choices in relation to the menu, the amount of food on display, as well as serving sizes are all mentioned in the literature as key FLW drivers. Some other drivers found in the retail channel are also relevant here: overstocking of food for visual purposes as well as pricing strategies and promotional activities leading to fluctuations in supply and demand for certain products.

In the context of foodservice environments, FLW drivers related to consumer behavior also play an important role. A general lack of food waste awareness among consumers can lead to significant plate waste, especially in environments in which consumers get to serve themselves (e.g., in a buffet setting). Here, overestimating portion sizes plays a role as well as the customer's attitude regarding whether or not to take leftovers home from the foodservice setting.

9.3.8 Consumer

As discussed earlier in this chapter, the consumer stage is known to be a significant source of food waste, especially in developed countries. As in previous supply chain stages, some lack of infrastructure or technology is still relevant here. More specifically, inadequate storage and refrigeration can be a driver, as well as a lack of access to recycling or composting options.

In the category related to lacking skills and expertise, the literature identifies several specific drivers. In addition to more general product handling skills, there is specific mention of lacking skills in relation to food safety assessment, understanding product (date) labels, lack of food preparation knowledge, and understanding of how to use leftovers. Also, the lack of awareness regarding FLW is mentioned as a driver in this category.

The purchasing behavior of customers also relates to a series of FLW drivers. Difficulties with inventory management can be a key driver, partly related to the management of uncertainties in demand and current inventory (e.g., not using a shopping list), but also due to the sizing of packages, and overbuying behavior caused by discounted products (especially related to volume discounts). This FLW driver connects to retailer practices, and misalignment of incentives, as additional sales are clearly beneficial for retailers, and waste prevention at the consumer level might not be.

Many socio-cultural aspects further add FLW drivers. A lower perceived value of food is, for instance, mentioned in the literature, as well as having experience with specific food products (loose), attitudes towards food waste, social norms around eating, and individual eating habits.

Especially in the consumer stage, the complex combination of many behavioral and socio-cultural factors leads to significant FLW, and the overview of drivers included here only provides a brief overview of the main categories and drivers. For a more detailed discussion of individual drivers, we therefore also refer to the references mentioned earlier.

9.4 Reduction and Reuse of Food Loss and Food Waste

In this section, we first discuss some common frameworks related to reduce and reuse strategies, followed by reduce and reuse strategies, and finally discuss the not so straightforward connection between reducing FLW and addressing food insecurity.

9.4.1 Reduce and Reuse Hierarchies

Throughout the supply chain, many FLW streams are reused or repurposed in one way or another, for instance, in energy production by anaerobic digestion, as an ingredient in the production of animal feed, or as donations to food banks that might still be able to repurpose towards human consumption. Figure 9.5 illustrates the hierarchy of interventions that is also recognized in EU policy. A similar hierarchy has been introduced in the US by the EPA, based on: (1) source reduction, (2)

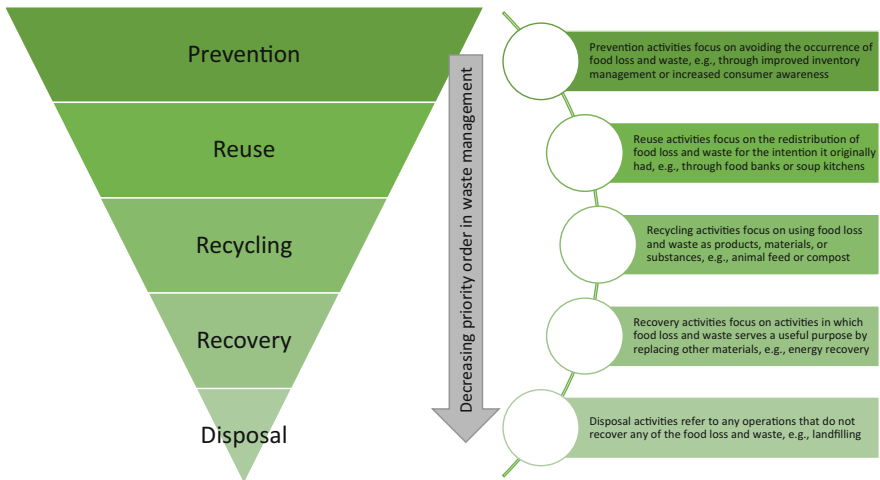


Fig. 9.5 Food waste hierarchy including selected potential interventions (based on EU 2008)

feed hungry people, (3) feed animals, (4) industrial uses, and (5) composting (EPA 2022b).

Using such hierarchies and prioritizations makes clear that there are different things we can do with FLW and that it matters which one we use, as higher levels of valorization are preferable. In most cases, this is also a good proxy for the best use in terms of the underlying environmental impacts. However, in terms of measuring the impact of efforts to address FLW, the variety of reuse options is often not appropriately captured: in most cases, if food is not used for human consumption, it is still considered FLW. This means that interventions to reuse FLW might have large positive environmental impacts, but that it is might not be appropriately incentivized by policies that aims to reduce FLW. This prevents the efficient reuse of FLW streams, see also Teigiserova et al. (2020).

The discussion of reduce and reuse strategies does however seem to gain more and more attention, also related to developments regarding the circular economy, a concept that describes an economy in which the natural environment is protected by minimizing resource consumption and increasing reuse (Beames et al. 2021). The abovementioned hierarchies and prioritizations follow the logic behind the circular economy concept.

Figure 9.6 illustrates several key ideas behind a circular economy approach in relation to FLW (Wang et al. 2021). Interested readers can refer to Chap. 16 by Saman et al. (2024) for more details about circular supply chain implementation. Most of the interventions included in the hierarchies and prioritizations are present in this approach. It should however be noted that this circular system does not need to be a closed system within the food sector, as some products derived from FLW streams might have uses outside of the food system (e.g., the production of bioplastics from FLW).

9.4.2 Food Loss and Food Waste Reduction and Reuse Strategies

Reducing or reusing FLW can be done in different ways. Many initiatives to prevent or reduce FLW aim to remove or decrease one or more of the drivers mentioned in the previous Sect. 9.3. However, considering FLW might be difficult to reduce, there are a variety of initiatives aiming to reuse FLW. Table 9.2 provides an overview of strategies, categorized by Caldeira et al. (2019).

The first category of strategies concerns redistribution, which aims for FLW to be redirected in such a way that it still ends up being consumed by humans. The example of gleaning addresses FLW in primary production, by collecting products that are left on the field after harvesting. Often, these collected products are then donated to organizations such as food banks (see, e.g., Lee et al. 2017). Similarly, surplus food in other supply chain stages can also be redistributed to such organizations aiming to address food insecurity. Finally, there are also many

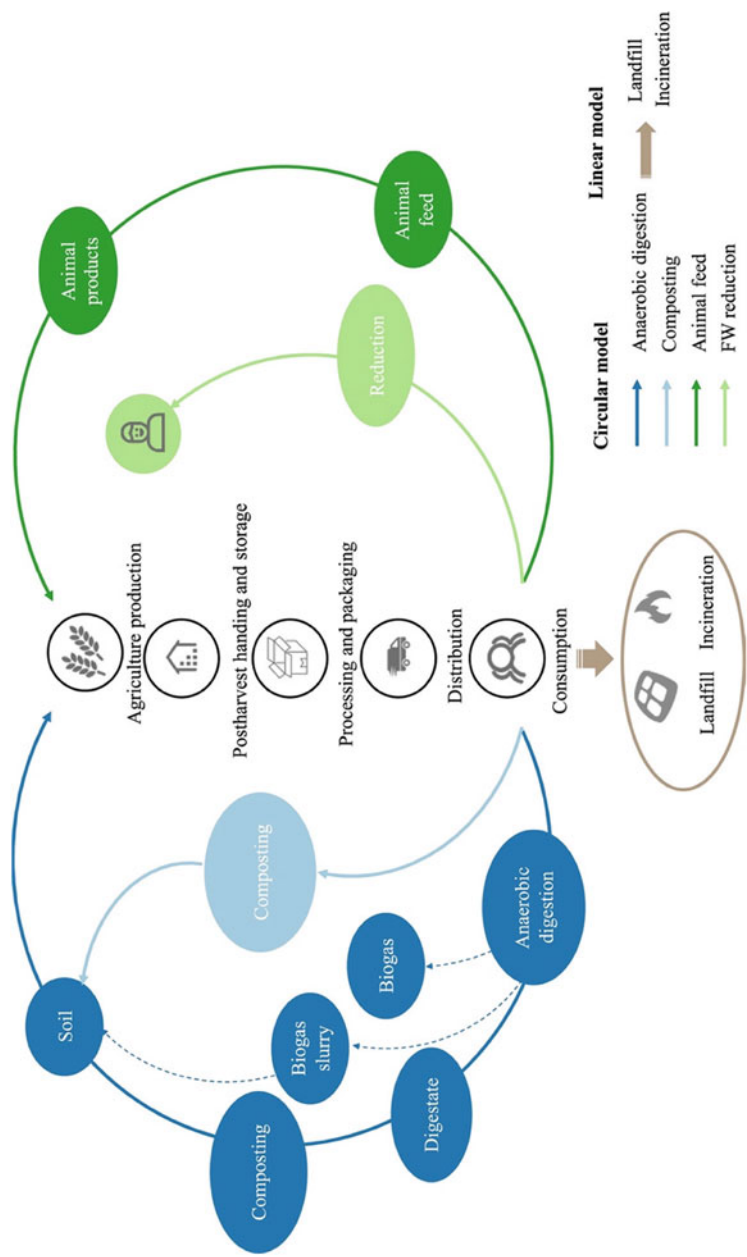


Fig. 9.6 A circular economy approach to food loss and waste (Wang et al. 2021). Original figure licensed under CC BY-NC-ND 4.0

Table 9.2 Food loss and food waste reduction and reuse strategies (based on Caldeira et al. 2019)

Category	Key strategies
Redistribution	Gleaning
	Surplus food redistribution
	Digital tools for redistribution
Food valorization	Value-added processing
	Animal feed
Consumer behavior change	Awareness/educational campaign
	School programs
	Awards
	Digital tools for behavior change
Supply chain efficiency	Process innovation
	Innovation of products—packaging
	Innovation of products—date marking
	Training & guidelines
	Price discounting
	Imperfect product sale
	Certification
	Public procurement
	Digital tools for supply chain efficiency
Food waste prevention governance	Voluntary agreement
	Regulatory framework/policy
	National food waste prevention program
	Fiscal incentives

technological innovations related to redistribution, for instance developing apps in which surplus food is offered at discounts. Well-known examples are Too Good To Go, Food For All, and Olio. Whereas some of these apps focus mainly on offering surplus food in retail and foodservice environments to consumers at a discount, some of the apps also have the option for people to offer their surplus food.

Valorization of FLW streams is another important reuse strategy. Partly, this might still include strategies that end up with products for human consumption, for instance by processing fruit waste streams into fruit juice. After redistribution for consumption by humans, valorization as animal feed is often considered the most valuable use of FLW. It avoids dedicated food production for animal feed and as such prevents significant environmental impacts.

Following up on the important role that consumers play in causing FLW, strategies aiming to influence behavioral change are also highly relevant. This includes educational efforts and campaigns that hope to make consumers more aware of the environmental and social impacts of FLW. Awareness is also the main goal behind school programs and competitions and awards for FLW reduction innovations. Finally, digital tools might also help change consumer behavior, for instance to help with food planning, food acquisition, and food storage (see, e.g., Vogels et al. 2018).

Reducing waste is a classic operations and supply chain management principle. Even though this often takes the form of wasted capacity or wasted time, the principles still translate to the reduction of FLW. Supply chain efficiency is therefore also a sizeable category of FLW reduction strategies. This entails process innovation, which could be the increase of technical efficiency in terms of producing a more stable product quality, or managerial innovations such as more intelligent processes for production and inventory management. Supply chain efficiency could also be achieved by improved packaging technology, such as for instance shelf-life-extending modified atmosphere packages. Improved date marking on packaging might also help reduce FLW in later stages in the supply chain. Training and guidelines could also help increase efficiency, especially in situations where lack of skills or expertise is a key FLW driver, such as in retail and foodservice environments. Especially for retail environments, discounting practices can significantly reduce FLW, even though this sometimes just shifts waste downstream, as discounting is also considered a key driver of waste at the consumer level. Another strategy that is mostly relevant for retailers is the sale of imperfect products. Also offering the 'ugly' or 'wonky' to consumers is something many retailers have implemented. Certification could also be a strategy, as this would institutionalize the measurement and continuous reduction of waste. Certification organizations such as Bureau Veritas and the British Standards Institution already offer such certifications. An additional strategy especially relevant for the public sector is the inclusion of FLW reduction considerations in public procurement. This could for instance relate to including FLW aspects and other sustainability considerations in tenders regarding foodservice operations in public institutions. Finally, the development of digital tools to support supply chain efficiency is a strategy that uses digitalization to help improve the matching of supply and demand in food supply chains, which would help reduce the FLW caused by supply chain inefficiencies. Also, these tools can help implement the process innovations and training mentioned earlier.

A final category of FLW reduction and reuse strategies is FLW prevention governance. This is a relatively broad category and includes several public or private initiatives. First, voluntary agreements could help companies across the food industry to implement efficiency improvement. Several additional strategies build on regulatory support and/or pressure, which might be required to achieve ambitious reduction goals. This entails regulatory frameworks and policy facilitating all the FLW reduction strategies mentioned in this section, national food waste prevention programs to combine and promote the FLW reduction strategies, as well as fiscal incentives for companies along the food supply chain to become more active in reducing FLW.

Many of the FLW reduction and reuse strategies discussed in this section impact the way we design and operate food supply chains. In addition to the technological challenges, consumer awareness issues, and policy requirements, many of the strategies also build on advances in operations and supply chain management (see also Akkaş and Gaur 2022; Do et al. 2021; Muth et al. 2019). This ranges from designing new supply chains for effective redistribution of surplus food to the

integration of FLW prevention in the planning and control of food manufacturing operations and the consideration of FLW drivers in managing retail operations.

9.4.3 Food Loss and Food Waste in Relation to Food Security

It can be argued that reducing FLW can have a beneficial effect on food security. After all, the natural resources that are needed to produce food can be used to feed people who are acutely food insecure. Reduction of FLW could contribute to an increase in the productivity of food production advocated for in the Zero Hunger SDG. On the other hand, if FLW that is currently redistributed to people that are food insecure would be prevented from occurring at all, such FLW prevention initiatives might have unintended negative side effects for food security.

Clearly, the relation between food security and FLW reduction deserves to be discussed in more detail. Although FLW reductions can affect food security in a positive way, the relation is certainly not straightforward.

First, it is widely acknowledged that, on an aggregated level, there should be enough food for everybody on the planet. Clearly, this significantly changes when a third of food products does not find its way to human consumption. Feeding the world population and avoiding hunger is thus a matter of efficient and effective food supply chains, and a fair allocation of resources and food. When left to the free market however, food products tend to flow to those with the highest willingness to pay, which is the wealthiest part of the population. This results in an unbalanced (and arguably unfair) allocation among the world population. This is easily illustrated by observing that, next to the 700 million undernourished people (as we saw in Fig. 9.2), an estimated number of 1.6 billion people are overweight. It would be a severe simplification to just state that redistributing food between the mentioned groups of people would solve hunger. This relates to the third topic that we will discuss below on measurement of FLW in terms of nutrients instead of volume. In fact, obesity is caused by complex relationships between genetic, socio-economic, and cultural influences, and when the dietary intake is concerned, obesity is more due to the overconsumption of energy-dense foods than to the overall mass of the consumed food (Apovian 2016). However, it is a fact that cheap food is widely available in some parts of the world, while there are severe shortages in other parts of the world. An interesting question is therefore: will we still see hunger if we succeed in eliminating FLW and as a result a significantly larger food supply would stay available for consumption on the global market?

Second, the interaction between food security and FLW reduction is explicitly present in initiatives like food banks and other charitable organizations. These initiatives typically have two goals: to provide essential food items to people facing hunger and to avoid or reduce FLW by redistributing surplus food. Most of the food products donated to food banks originate from food supply chain actors such as retailers and processing companies. These are however also supply chain actors that

are looking into reduction of FLW and/or interventions that reuse FLW in different ways (e.g., utilizing it as animal feed). Such reduction and reuse interventions are often also economically more interesting than donating to a charitable organization like a food bank. Such trade-offs between different FLW prevention, redistribution, and reuse interventions should clearly also be considered in addition to the identified synergies between FLW reduction and food security.

A third friction between food security and FLW reduction lies in their respective measurement. Whereas the most common measurements of FLW simply focus on weight, this in a way is a simplification. It assumes that the burden of discarding a kilo of cucumber is the same as discarding a kilo of beef or a kilo of candy. This is clearly not true. When looking through the lens of food security, people in hunger do not care about weight of food, but about nutrients. This makes a case for FLW measurements to also look at nutrients rather than weight. Also, it means we should be even more careful with high-nutrient foods than with food that is low in nutrients. And even more so with food that contains nutrients that generally people are lacking. Another concern with the focus on weight measurement is that it does not consider the natural resources that are used and emissions that were caused in the supply chain of various types of food. And research has shown that there are significant differences between the environmental impacts of products. In other words, it is important to always translate FLW measurements into environmental impacts, as we might otherwise not create the incentives for food supply chains to become more sustainable.

9.5 Conclusions

To develop more sustainable food supply chains, it is essential to reduce its environmental impacts, while still providing sufficient food for a growing population. A key challenge in this discussion is the presence of significant food loss and waste (FLW). In this chapter, we briefly discussed sustainability in food supply chains and the role of FLW in this context. We then provided an overview of the drivers behind FLW, what strategies so far have been used to reduce or reuse FLW, and how FLW reduction strategies possibly interact with strategies to address food insecurity.

From an operations and supply chain perspective, many challenges remain for practitioners and researchers alike. A lot of the FLW reduction strategies lead to the development of novel and different supply chains, for instance dealing with redistribution of surplus food, valorization of FLW streams, management of novel processing and packaging technologies, developing intelligent pricing strategies, extending product assortments, and changing procurement practices.

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Chapter 10

Demand Management for Sustainable Supply Chain Operations



Niels Agatz and Moritz Fleischmann

10.1 Introduction

Supply chain management (SCM) is about fulfilling demand. Based on given estimates of future demand, SCM invests the appropriate resources and then uses these resources to match supply to demand. The traditional SCM perspective takes demand as exogenous. The goal of SCM is then to serve the forecasted or materialized demand effectively and efficiently. How difficult it is to achieve this goal depends on the characteristics of that demand. For example, serving a stable, predictable demand is relatively cheap, whereas serving an unpredictable, strongly fluctuating demand may imply less efficient operations characterized by high inventory built-up and low capacity utilization.

In the same way, demand characteristics impact not only the financial performance of the supply process but also its environmental impact. For example, satisfying demand for fresh produce during the harvesting season results in lower emissions than serving off-season demand which requires substantial storage and/or long-distance shipments from other growing regions.

Supply chain planning tasks commonly take demand as a given. For example, most of the models presented in this book treat demand as an exogenous parameter. However, demand itself is shaped by long- and short-term decisions on, for example, assortments and prices, but also lead times and service quality. These demand management decisions interact with supply (chain) management decisions in an iterative fashion, as visualized in Fig. 10.1. For example, long-

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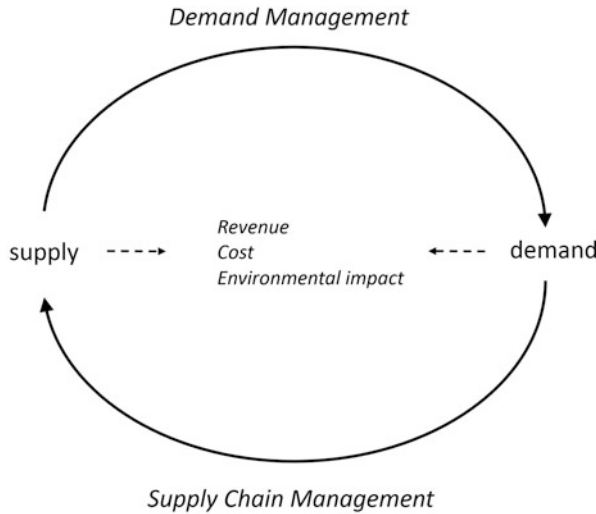


Fig. 10.1 Interaction between supply chain management and demand management

term assortment decisions drive demand expectations. Corresponding forecasts then serve as input not only for long-term production capacity decisions but also for short-term production and inventory planning. For a given available supply, short-term demand management, such as dynamic pricing, may seek to maximize revenues by exploiting differences in valuations between customers and over time. Such short-term demand management for a fixed supply is at the heart of revenue management (Klein et al. 2020), with prominent applications in the airline, car rental, and hospitality industries. For corresponding approaches in a production setting, see Quante et al. (2009).

The most common objective of demand management is revenue. However, as sketched above, the resulting demand patterns may also impact the cost of the corresponding supply operations. Managing these interactions to achieve alignment is at the heart of a firm's sales and operations planning (S&OP) processes. On the operations side, the focus of S&OP is often on production processes. More recently, the interaction with short-term demand management, akin to traditional revenue management, has also been addressed for delivery operations, especially in online retailing (Waßmuth et al. 2022).

In addition to revenues and costs, demand management also has the potential to impact a firm's environmental performance in the broadest sense, i.e., emissions, waste, and material and energy consumption. While the idea as such is not new, real-life applications of sustainability-oriented demand management are only starting to emerge. Several factors drive this development. First, growing consumer concerns, pressure from both NGOs and investors (requesting reporting of ESG criteria), and expanding legislation (such as the recent German Supply Chain Due Diligence Act and the planned EU Corporate Sustainability Due Diligence Directive) put

more weight on corporate social responsibility (CSR) in general, including firms' environmental performance. Second, companies that underestimate the impact of their demand management decisions on their operations are struggling. For example, the competition among online retailers for the ever shorter delivery times leaves many delivery services unprofitable, thereby repeatedly forcing some of them to close their doors. Third, new information and communication technology makes it easier to track and predict customer behavior and to respond by changing prices and service offerings in real time.

One example of using demand management to encourage more sustainable behavior is the use of "green labels" in grocery home delivery to steer customers toward sustainable delivery options (Agatz et al. 2021). Another example is the Amazon Day Delivery service, which allows for the consolidation of packages on the last mile to the customer's home. In addition to the transport and service dimensions, product assortment decisions also drive demand and impact the environmental sustainability of operations. The decision of French grocery retailer Carrefour to stop selling strawberries in the off-season winter period (Carrefour 2021) exemplifies this type of demand management.

Most of the current supply chain literature models demand as an exogenous factor. This also holds true for sustainable supply chain management, as addressed in the preceding chapters of this book. In this chapter, we complement this perspective and discuss opportunities for endogenizing demand. We proceed as follows. In Sect. 10.2, we discuss the impact of relevant demand characteristics on a firm's environmental performance. In Sect. 10.3, we first introduce common demand management levers and then review examples of how they have been applied to manage environmental objectives. Section 10.4 synthesizes our main observations and discusses potential avenues for future research.

10.2 Environmental Impact of Customer Demand

In this section, we highlight different ways in which customer demand impacts a firm's environmental performance. The discussion serves as a basis for investigating opportunities for sustainability-oriented demand *management* in Sect. 10.3.

In principle, the motivation for all of a firm's processes is to meet demand. Therefore, demand ultimately drives a firm's entire environmental performance. Dependent on the firm's position in the supply chain, this demand comes from final users (B2C) or from intermediaries (B2B). For a discussion of coordination issues arising from the interdependence of demand at different supply chain stages, we refer to Mentzer et al. (2007). To structure our assessment, we follow the steps of the classical SCOR model—i.e., source, make, deliver, and return—and discuss them separately. We include demand for both physical products and services, e.g., home delivery, and comment on the differences between both cases where needed.

Note that while, for ease of exposition, we discuss environmental impact by process, the individual effects may interact. For example, switching suppliers may

Table 10.1 Environmental impact of selected demand dimensions by process step

Process	Demand dimension	Potential impact
Generic	How much?	(Potentially nonlinear) scaling of all environmental impacts
Source	Inbound transport, storage, supplier operations	
	What?	Component substitution: Durability, composition, production processes
	From whom?	Alternative suppliers: Supplier performance
	From where?	Local versus global: Transportation
	Mode of transport?	Lead time: Speed vs. sustainability
Make	Process emissions, material consumption, energy consumption	
	What?	Product substitution: Large/small, new/re-manufactured
	How?	Production process choice: Cost vs. sustainability
	When?	Production scheduling: Inventory build-up, cleaning
Deliver	Outbound transport, storage, scrapping	
	What?	Product substitution: Product substitution: Expiration
	How?	Mode of transport, channel choice, business model: Transport emissions, used product recovery
	When?	Delivery timing, lead time: Transport consolidation/bundling
Return	Return transport, recycl./reman. operations, energy/material recovery	
	What?	Core quality: Reprocessing intensity
	When?	Acquisition timing: Product wear, demand potential

reduce shipping distances in sourcing but lead to higher emissions in production, due to inferior product quality. Effects may also interact across geographical regions and across different players in the supply chain. For example, switching from home delivery to click and collect in online retailing reduces the retailer's transport emissions but likely increases those of the consumer. Furthermore, environmental impact by itself is fundamentally multidimensional. Reducing the impact on one dimension can lead to negative effects in other dimensions. To identify the environmentally most favorable solution, demand management must take these inter- and intrafirm dependencies into account, as in a life cycle assessment (LCA) (see Chap. 2, by Guinée and Heijungs (this volume)). In addition, demand management, as with any sustainability instrument, also must consider the interactions between economic, environmental, and social performance. We return to this issue in Sect. 10.3.

In what follows, we discuss the potential impact of different demand attributes on the environmental performance of a firm's source, make, deliver, and return processes. Table 10.1 summarizes our main observations. We start by highlighting two generic effects that apply to all process steps.

The first generic relationship is between demand *volume* and environmental performance. Since any supply chain activity entails a certain environmental impact, there is no truly "green" product: more demand inevitably means a higher impact (Zink and Geyer 2016) even though economies of scale might reduce the impact *per unit* of demand. Environmentally oriented demand management thus faces a fundamental inherent trade-off that makes the question of the "right" demand

volume an intricate one. Note that in traditional supply chain models this issue does not typically appear since those models take demand as exogenous. Demand management, however, endogenizes demand, and thus the question of an appropriate target demand volume arises naturally. Its answer does not reside within the domain of supply chain management, however, but rather touches upon a more fundamental economic and ecological debate related to degrowth (Kallis et al. 2018) and the planetary boundaries (Steffen et al. 2015), which is beyond our scope. We highlight the issue here and refer to the literature on rebound effects as a helpful source of inspiration from a supply chain perspective (Zink and Geyer 2017).

The second generic effect concerns product substitution. Demand management may affect a firm's environmental performance by shifting demand from one product variant to another. The argument as such applies to all supply processes. Relevant differences between products are process specific. We provide examples of the different supply processes in the discussion below.

10.2.1 *Source*

We highlight three sources of environmental impact, associated with a firm's sourcing processes: (i) transportation, (ii) storage of input materials and components, and (iii) suppliers' operations. The main physical processes on the supply side of the firm are inbound shipments and storage. That is, it is the emissions and resource consumption of these processes that matter. Among these, the impact of transportation processes, i.e., emissions and energy consumption for shipping items to the firm, are dominant, in general. This holds true especially in the case of global sourcing, with items being shipped over large distances. Energy consumption for storage matters especially for items requiring special conditions, such as refrigerated items. For a detailed discussion of the environmental impact of general transportation and storage processes, we refer to Chap. 5 by Blanco and Sheffi and Chap. 6 by Marklund and Berling of this book, respectively. Additional environmental impacts arise further upstream in the value chain, from the source, make, and deliver processes of the firm's suppliers (WBCSD 2004). For a discussion of managerial issues arising from the supplier interaction, see Chap. 11 by Tan et al. (this volume).

The aforementioned environmental impacts interact with demand in manifold ways. However, due to the upstream position of the sourcing processes, the demand influence is more indirect in many cases than, for example, for distribution (see Sect. 10.2.3 below). In addition to driving *how much to source* (see the introductory discussion to Sect. 10.2), demand affects the environmental performance of the sourcing processes by influencing:

- *What to source*: Alternative available components may differ in their environmental footprint in manifold ways, including their material composition, durability, and underlying production process. Whether demand drives the firm's

choice among these alternatives depends on whether the customer is aware of the differences.

- *From whom to source*: The argument is very similar to the previous one but concerns differences in environmental performance between alternative suppliers.
- *From where to source*: Global versus local sourcing has a strong impact on transportation-related environmental performance. Again, transparency toward the customer is a prerequisite for steering demand toward the more sustainable alternative. It is worth noting that the available sourcing options may be time dependent, e.g., for fresh produce. Therefore, shifting demand over time may also be a way to manage sustainability in sourcing—offering strawberries in summer versus in winter.
- *Mode of transport*: The promised delivery lead time toward the customer may influence the mode of transport of inbound shipments, invoking a trade-off between speed and sustainability. However, the argument holds true only for order-specific sourcing, as in make-to-order production, since the sourcing lead time is otherwise sunk.

10.2.2 Make

The environmental impact of a firm's make processes, i.e., production processes in a broad sense, includes (i) process emissions, (ii) material consumption, and (iii) energy consumption. They require little explanation. In addition, the "make" stage also includes transportation and storage between production steps. The assessment of these logistics processes is very similar to the corresponding sourcing processes discussed above; thus, we do not address them separately. Naturally, which of the mentioned impacts is most relevant is highly product specific.

For a discussion of how to manage these "make"-related environmental impacts through technology choice, we refer to Chap. 12 by Ovchinnikov and Mackinnon (this volume). Hereunder we discuss again their dependence on selected demand attributes.

Like for the "source" processes, demand directly drives environmental impact through the decision of *how much to make*. Note that in a make-to-stock (MTS) setting, it is not only the expected demand but also the degree of demand uncertainty that is relevant to safety buffers. Other relevant demand derivatives include the following:

- *What to make*: Different product variants may differ in the environmental impact of their production. This includes, e.g., not only new products versus remanufactured products but also large versus small product variants. Demand management may stimulate substitution effects between these product variants. Similarly, product specifications may also be a relevant factor. Again, demand management may aim to push more favorable specifications, e.g., by avoiding production waste. More generally, a broader assortment tends to have a higher

environmental impact, due to missing economies of scale and lack of demand pooling. Demand management may thus improve environmental performance by concentrating demand on fewer product variants.

- *How to make:* There may be multiple options for making a given product, which differ in their environmental impact. For example, different ways of catching tuna fish differ in how they affect other sea life, cleaning of production equipment may use biodegradable or toxic solutions, and energy can be generated from renewable or fossil resources. Demand management may favor more sustainable production process options, provided they can be communicated effectively and credibly to customers.
- *When to make:* The timing of production also influences sustainability. Preproduction for seasonal peaks that exceed capacity results in inventory, i.e., items being stored and potentially heated or cooled. Appropriate sequencing of the items produced on a shared production line may reduce the need for cleaning and thereby water consumption and/or the use of hazardous chemicals. Production timing may also synchronize the production process with the availability of renewable energy (Materi et al. 2020). In all of these cases, demand management may seek to shift demand to the most environmentally favorable time.

10.2.3 Deliver

Akin to sourcing, a firm's main physical processes associated with delivery include (i) transportation and (ii) storage, with particularly intense transportation operations arising for online delivery services. Another relevant factor is (iii) the scrapping of obsolete items. Concerning the environmental impact of transportation and storage, the discussion in Sect. 10.2.1 applies analogously. In the case of scrapping, obsolescence implies wasted resources, including material and energy, as well as wasted production and transportation emissions but also, potentially, additional emissions from waste processing and from accelerating new production.

Like for the preceding processes, demand drives the scale of the delivery-related environmental impacts by affecting *how much to deliver*. More interestingly, demand influences the environmental footprint of a firm's deliver processes through the decisions discussed below. While some of the effects are analogous to those for a firm's "source" and "make" processes, differences arise from the fact that "deliver" is a pure service process, whereas the output of the "source" and "make" processes are often storable.

- *What to deliver:* Demand substitution between products may help reduce obsolescence. For perishable products, shifting demand toward items that are approaching their expiration date may avoid having to scrap them.
- *How to deliver:* This includes the choice not only between transportation modes, e.g., cargo bikes versus vans in home delivery services, but also between distribution channels, such as home delivery versus pickup versus a traditional

store. These choices largely shape the delivery-related transport operations and thus the associated environmental impact. They also impact the required amount of packaging material. More fundamentally, how to deliver also includes the strategic choice of the firm's business model, notably the choice between leasing and selling. From an environmental perspective, this choice fundamentally impacts opportunities for used product recovery and thus the "return" processes discussed below.

- *When to deliver:* Relevant aspects include the timing of delivery within the week and within the day, as well as the delivery lead time. The former may have a strong impact on routing efficiency by allowing or prohibiting joint delivery to geographically close destinations. This is a particularly relevant issue in attended home delivery where narrow time windows are common. Similarly, the delivery lead time also determines the potential for consolidating shipments, e.g., for the same-day delivery services. In addition, the delivery timing also influences the need for storage, as discussed for the "make" processes in Sect. 10.2.2.

10.2.4 *Return*

"Return" refers to the collection, handling, and processing of products from a customer, either before or after their use. Examples are commercial returns of unwanted items, e.g., in online retailing, and the recycling or remanufacturing of used items. For an in-depth discussion of such "product recovery" strategies we refer to Chap. 16 by Saman et al. (this volume).

The potential environmental impacts of return processes include (i) transportation, (ii) emissions from recovery operations, and (iii) material and/or energy recovery. Transportation processes arise in the collection of items from the customer. Analogous to the "last mile" in distribution, it is the "first mile" in the return channel that is the most transportation intense. In addition to the direct transportation-related impact, returns may also involve additional packaging material, see also Chap. 17. Recovery operations are a special case of a "make" process, such that the discussion of Sect. 10.2.2 applies. The impact of material and energy recovery is different from those discussed in the preceding sections in that they represent a potential environmental benefit, rather than a burden, as a replacement for some new production. However, for judging the overall environmental performance, it is crucial to carefully assess the extent of replacement versus market expansion (Zink and Geyer 2017).

Another difference from the preceding sections is with regard to the role of the customer. Due to the "reverse" direction of the corresponding supply chain flows, the customer is situated on the supply side of the return process. Thus, steering customer behavior boils down to "supply management" (denoted as product acquisition management in the literature (Guide et al. 2003)) rather than "demand management." Alternatively, one may interpret supply from the customer as negative demand. We include this setting in our discussion because of the conceptual analogy

and the similarity of the relevant approaches. The aspects through which supply management can influence environmental performance of return processes include the following:

- *How much to return:* This refers to the customer's choice between returning an item—i.e., making it available for recovery—and disposing or storing it. Ensuring a sufficient supply of used items of appropriate quality is a serious bottleneck in many product recovery systems (Guide and Van Wassenhove 2009).
- *What to return:* The required recovery operations, and, thus, the associated environmental impact, may depend on the quality of the returned product. For example, worn-out products require more extensive remanufacturing steps than lightly used products. Supply management may seek to shape the quality mix of the available supply, e.g., by influencing how the customer is using the product.
- *When to return:* Related to the previous point, the timing of returning a product also influences the available recovery options, either due to shifts in demand or due to progressing product deterioration during usage. For example, used consumer electronics may partially substitute for new products in the market sufficiently early in the product life cycle, whereas they have to be recycled at a later stage, due to a lack of demand.

10.3 Sustainability-Oriented Demand Management

10.3.1 Demand Management Levers

Demand management encompasses a wide range of decisions. In line with Waßmuth et al. (2022) and Agatz et al. (2013), we map these along two dimensions: (i) the demand management lever and (ii) the hierarchical planning level. We distinguish three main levers: (i) offering, (ii) pricing, and (iii) communication. The first two of these, offering and pricing, capture the main characteristics of the product or service by specifying the offered product variants and the associated prices. In the context of sustainable behavior, we identify a third dimension that is related to communication. This refers to the information provided to customers on the environmental performance of products or services.

As is common in many areas of supply chain planning, we can also distinguish between different planning levels, notably long-term strategic, medium-term tactical, and short-term operational demand management. Furthermore, the customer order decoupling point influences the way demand management interacts with supply chain management decisions (compare Fig. 10.1). For example, in a make-to-stock setting, emissions from production processes are sunk when customer orders materialize. Thus, short-term, customer-specific demand management only affects the environmental impact of delivery (and potentially return) processes, and production performance can only be managed through forecast-based tactical demand management. In contrast, in a make-to-order setting, short-term demand

Table 10.2 Demand management approaches by lever and planning horizon

	Offering	Pricing	Communication
Strategic/tactical	What (green) products and services to offer?	How to price the offered options?	How to communicate the offering?
Operational	How to dynamically adjust the offering?	How to dynamically update prices?	How to dynamically adjust communication?

management can steer the environmental impact of an individual order at the production stage and potentially even at the sourcing stage. In what follows, we elaborate further on the different demand management levers. Table 10.2 summarizes the resulting distinct settings.

Offering refers to both the long-term design of the product and service assortment and the short-term management of their availability. The assortment decision involves trade-offs between revenues, costs, and sustainability aspects. As in traditional assortment planning, these problems are challenging as they involve modeling customer behavior while taking into account different practical constraints and business rules. This includes, for example, modeling different preferences and customer flexibilities with respect to the speed of delivery. Our context adds an additional layer of complexity related to evaluating the environmental impact of different assortment offerings and substitution behavior between products and firms.

Naturally, companies can also influence demand through their *pricing* decisions. We use “pricing” to denote a wide range of incentives for steering customer choice behavior. Pricing can be used to manage demand to better match supply. As such it is complementary to supply chain efforts focusing on improving the flexibility to handle demand seasonalities and shocks. Often, pricing is used as a tool to exploit differences in product valuations between customers and across time to maximize revenues from a given supply (comp. Gallego et al. 2019). However, pricing can also be used as a tool to improve the environmental performance, e.g., reduce waste or emissions.

Finally, companies can manage demand by means of *communication*. There is a large body of empirical research on encouraging pro-environmental behavior through the use of eco-feedback on resource consumption. Research suggests that feedback on costs and environmental impacts can reduce the energy and water consumption of households (Abrahamse et al. 2005). A popular form of eco-feedback in the context of energy consumption is to report the associated carbon emissions or the number of trees required to offset these emissions (Sanguinetti et al. 2018).

10.3.2 Models for Sustainability-Oriented Demand Management

In this subsection, we highlight models from the Operations Research (OR) and Operations Management literature that support sustainability-oriented demand management. We also point out potential trade-offs between different performance metrics, notably between profit and environmental performance. We structure the discussion again along the steps of the SCOR model, analogous to Sect. 10.2. Contributions to this stream of literature are few and far between. In addition, they are distributed very unevenly over the SCOR elements, with a dominant focus on delivery. Table 10.3 lists the corresponding papers that we have been able to identify.

Source

We are aware of a single demand management model geared toward the environmental performance of a source process. Hovelaque and Bironneau (2015) consider pricing as a means for steering demand so as to manage carbon emissions of a generic batch supply process. Emissions have an EOQ-type structure, consisting of a fixed amount per replenishment and a batchsize-dependent amount. Reflecting environmentally conscious customers, demand decreases in total emissions, in addition to price. Assuming EOQ-type costs, the paper points out a tension between profit and emissions. In fact, minimizing emissions would force demand to zero. This reflects the general demand–volume issue, sketched in Sect. 10.2. In response to this issue, the paper assigns the task of controlling emissions to a policymaker

Table 10.3 Published optimization models for sustainability-oriented demand management

	Offering	Pricing	Communication
Source		Emissions from batch supply (Hovelaque and Bironneau 2015)	
Make		Pricing of green portfolio (Yenipazarli and Vakharia 2015; Hong et al. 2018; Ovchinnikov et al. 2014) Pricing of renewable energy (Dutta and Mitra 2017)	Carbon footprint product labels (Cohen and Vandenberghe 2012)
Deliver	Time slot management for routing emissions (Manerba et al. 2018)	Subscription pricing for routing emissions and food waste (Belavina et al. 2017) Pricing for reduced food waste (den Boer et al. 2022; Sanders 2022; Yang et al. 2022)	Green time slot labels (Agatz et al. 2021) Green labels for intermodal transport (Heinold et al. 2022)
Return		Used product acquisition pricing (Guide et al. 2003)	Advertising to stimulate used product returns (Hong et al. 2015)

rather than to the firm and investigates the impact of different policy instruments on the emissions ensuing from the firm's profit-maximizing price.

We are not currently aware of available optimization models for demand management addressing any of the other environmental impacts discussed in Sect. 10.2.1.

Make

There is a growing interest in the carbon footprint labeling of products. Evidence suggests that credible labels can potentially change customer purchase behavior and steer customers away from certain polluting products. However, the overall potential reduction in carbon emissions due to product labeling is still unclear due to complex feedback and rebound effects in demand and supply (Cohen and Vandenberghe 2012). Customers, for example, may increase their consumption of more environmentally friendly products, thereby partially offsetting the corresponding carbon reductions.

An emerging stream of research focuses on studying the behavioral responses to carbon labels, focusing on food products, restaurant menus, air travel, and detergents. From a modeling perspective, several papers have considered product portfolio decisions for manufacturers that intend to add green products to their product line. Yenipazarli and Vakharia (2015) address pricing decisions for a manufacturer expanding her product line with a green offering. The paper explores the conditions under which a firm can use pricing to limit the impact of the cannibalization of green products on existing products. Hong et al. (2018) consider the pricing strategies of a green product based on consumer environmental awareness and reference behavior. The results suggest that the optimal price of the green product is not always higher than that of the regular product. The green manufacturer should adopt a pricing strategy based on the quality of the green product and on consumers' levels of reference recognition and environmental awareness.

In a similar vein, there is a substantial body of work that addresses the decision for (original equipment) manufacturers to offer remanufactured products alongside new ones, taking into account cannibalization, competition, market expansion, and the number of available used products (Rizova et al. 2020). Several papers in this line of research address pricing and market segmentation (Debo et al. 2005; Abbey et al. 2015b,a), primarily to maximize profits. Ovchinnikov et al. (2014) specifically focus on the tension between profit maximization and environmental performance in remanufacturing and provide a modeling framework to analyze when these two goals align. Their findings suggest that remanufacturing frequently aligns firms' economic and environmental performance by increasing profits and decreasing the total environmental impact.

Renewable energy is an example involving significant supply constraints. Specifically, supply is often unpredictable and volatile. In this setting, pricing can be used to temporally shift demand for electricity to better match supply (Dutta and Mitra 2017). A better balance between demand and supply can help reduce wasting renewable energy in peak supply periods and reducing conventional energy resources in peak demand periods. In the context of industrial make processes, we see the advent of demand response initiatives in which production systems

automatically vary their electricity demand over time based on the availability of supply, e.g., by interrupting noncritical processes or by shifting time-flexible processes (Shoreh et al. 2016).

The work in the “make” stage primarily addresses questions on the strategic/tactical planning horizon. That is, most research focuses on decisions related to adding products to the product portfolio based on expected demand and market forecasts.

Deliver

Most work on sustainability-oriented demand management focuses on the “deliver” processes. A prominent example is the offering of delivery options for attending home delivery. Several papers focus on the impact of time slot length on the expected route distances. The use of longer time windows for delivery is generally associated with fewer vehicle miles, as there is more flexibility in route planning (Lin and Mahmassani 2002). Focusing specifically on the environmental impact, Manerba et al. (2018) show that longer delivery time windows can significantly reduce carbon emissions. A related, very active line of research is focusing on the real-time management and pricing of delivery time slots (Campbell and Savelsbergh 2005, 2006; Yang et al. 2016; Vinsensius et al. 2020). This means dynamically deciding which time slots to offer at what price for a specific customer order, based on, e.g., the customer location, already accepted orders, and forecasts of future orders. The offered time slots may all have the same length or consist of a mix of different lengths (Köhler et al. 2020). Campbell and Savelsbergh (2005) show that dynamically integrating order capture and order delivery may significantly improve the profitability of grocery delivery operations. Campbell and Savelsbergh (2006) study the use of price incentives to influence consumer behavior in real time to reduce delivery costs. They suggest that small incentives may suffice to change customer behavior in this context. Research in this area often focuses on profit maximization, taking into account differences in both revenues and travel distances. As is often the case for transportation operations, routing costs tend to be highly correlated with environmental objectives as fuel consumption and emissions are strongly dependent on driving distances. We are not aware of papers that explicitly study the trade-offs between environmental and economic objectives in this context.

In a similar setting, Yildiz and Savelsbergh (2020) focus on using discounts to persuade customers to allow for flexibility in the previously agreed time windows as this enhances opportunities for consolidating delivery orders that are geographically close. The paper develops a dynamic programming algorithm to compute optimal discount offers to customers. Their numerical experiments show that offering discounts in return for delivery time flexibility can generate cost (distance) savings of more than 30% and that customized discounts lead to much better performance than offering the same discount to all customers.

In addition to price incentives, some researchers have investigated the use of communication as a lever for managing transport emissions of delivery operations. Based on a stated-preference survey, Ignat and Chankov (2020) conclude that communication can indeed be an effective means in this context. Nogueira et al.

(2021) support this conclusion but show that customer's willingness to sacrifice delivery speed for more sustainable delivery, i.e., fewer carbon emissions, differs per product category and customer type.

Agatz et al. (2021) study the use of green labels that specify time slots as environmentally friendly in an e-grocery home delivery setting. They specifically compare the use of green labels that intrinsically motivate customers to choose a specific delivery time slot to price incentives based on extrinsic motivation. Their experimental findings suggest that green labels are effective in steering customer choice behavior in different environments and work in fundamentally different ways than price incentives. In addition, simulation findings suggest that green slots, compared to price incentives or no incentives, offer providers a way to effectively steer consumer time slot choices to yield shorter routes, fewer delivery vehicles used, and more per-customer revenue. One challenge with eco-labels is that the message should be simple and credible, which is difficult to achieve here, given the complex and multidimensional nature of the environmental impact.

Finally, it is worth mentioning subscription models (Bray et al. 2021) and proactive customer contacting (Schwamberger et al. 2022) as recent demand management approaches for home delivery services. Belavina et al. (2017) study the environmental impact of a subscription model in which customers pay a fixed fee to receive free grocery deliveries over a given period. They find that subscription incentivizes smaller and more frequent grocery orders, which increases delivery-related emissions but may reduce food waste on the customer side. Agrawal et al. (2012) address a related issue outside of the field of delivery services by comparing the environmental performance of leasing versus selling. We are not aware of any approach that explicitly considers environmental performance of proactive customer contacting initiatives.

Apart from last-mile operations in e-commerce, carbon or climate labels have also been investigated in other retail settings as a means to provide information on the environmental impact of offered products.

In addition to the labeling of products, we also see the advent of labels for logistics services. Kirschstein et al. (2022) propose an eco-labeling system that communicates emissions rate categories for different types of freight transportation. Heinold et al. (2022) consider the operational planning of an intermodal transport network that lets customers select eco-performance requirements. This gives rise to dynamically arriving shipment orders with different eco-labels. They model the problem as a multiobjective sequential decision process taking into account expected costs and eco-label violations.

Another delivery-related topic in retailing that has received significant attention in the context of environmentally oriented demand management is centered on food waste. For perishable food items, retailers may use discounts to encourage customers to buy products that are close to their expiration date. This can help prevent food waste while also enhancing revenues. Most academic papers on the dynamic pricing of perishable items specifically focus on maximizing profitability; see, e.g., Chatwin (2000); Lu et al. (2018); Azadi et al. (2019). However, the goals

of reducing waste and increasing revenues tend to be well aligned, as selling at a discount generally leads to more revenues than scrapping a product.

A recent study suggests that simple dynamic pricing policies can help to simultaneously reduce waste and increase profit (den Boer et al. 2022). In another recent paper, Sanders (2022) shows that dynamic pricing is likely to be more effective in reducing retail food waste than an increase in disposal costs. Based on the structural econometric analysis of sales, perishability, and marginal cost data from a grocery retailer's artisan bread category, the author estimates that the use of dynamic pricing would reduce the retailer's waste by 21% and increase the chain's gross margins by approximately 3%. In particular, dynamic pricing Pareto-dominates organic waste bans that prohibit retailers from disposing of waste in landfills and instead require an alternative means of disposal, such as composting. In a similar setting, Yang et al. (2022) focus on product quality and investigate the combination of price discounts and information disclosure. To this end, they consider three different pricing policies, namely, quality-independent pricing, quality-dependent pricing, and quality-dependent pricing with disclosure of quality information. Based on a simulation study, they find that combining dynamic pricing with information disclosure allows for a decrease in food waste while simultaneously increasing profits.

In the "deliver" stage, most contributions focus on short-term operational decisions related to adjusting prices, labels, and product offerings based on real-time demand. In this phase, it is specifically helpful to adapt future demand based on already materialized orders and updates on the expected future demand.

Return

Recall that for return operations, customers act as suppliers. Acquisition management denotes the active management of this supply source, as opposed to treating returns of used products as exogenous (Guide and Van Wassenhove 2009). Many researchers have proposed optimized acquisition strategies (Wei et al. 2015). Most approaches use pricing as a lever to influence the quality, quantity, and timing of product returns (Guide et al. 2003). Hong et al. (2015) alternatively investigate the use of advertising as a way to steer product returns. For both levers, particular complexities arise from the need to align used product acquisition and the remarketing of recovered products. To the best of our knowledge, all currently available acquisition models take a profit-maximizing perspective. Whether this objective is aligned with environmental performance is far from obvious, given manifold interactions between forward and reverse channels, new and recovered products, and different environmental impacts. The work in this space focuses on both operational acquisition control and strategic and tactical return strategies.

10.4 Conclusions

While traditional supply chain planning treats demand as exogenous, demand itself can be shaped not only by, for example, assortment and pricing decisions but also by tailored communication. This type of demand management provides new levers for managing a firm's environmental performance since different demand patterns vary in how environmentally costly they are to serve.

Compared to sustainable supply (chain) management, sustainable demand management is still at a nascent stage. Following the categories of the SCOR framework, available studies focus primarily on the "deliver" stage. However, even these contributions are few and far between, with a notable cluster on e-commerce last-mile transportation operations. Sustainable demand management approaches for other SCOR processes, namely source, make, and return, are even more scant. The predominant focus in demand management on delivery processes is intuitive, given that these are the most downstream, customer-facing processes. Shaping demand to facilitate more upstream processes further away from the customer is more difficult, due to diluting effects from the interaction with intermediate processes and supply chain actors. As we argue in this chapter, demand patterns nevertheless have a significant impact on the environmental cost also of upstream supply processes. In conclusion, we see manifold opportunities for relevant future research on environmentally oriented demand management in supply chains.

The trade-offs underlying such types of demand management differ from those on the supply side. In particular, they crucially involve customer behavior, which is not typically the case for supply planning. Moreover, the relationship between environmental and financial performance also changes. Most noteworthy, we observe a fundamental trade-off between both performance dimensions with respect to demand volume. Since any supply chain process has an environmental impact, the impact-minimizing demand management would be to eliminate demand—which obviously is not economically sustainable. How to systematically deal with this fundamental tension is largely an open question.

While one may argue that the same is true for costs—the cost-minimal solution is again zero demand—we have revenues as a counterweight in this case, which is missing in environmental performance. This asymmetry renders setting an appropriate objective for sustainable demand management nontrivial.

For some other demand attributes, economic and environmental performances go hand in hand. This is the case, for example, for temporal shifts in demand that allow for reducing inventories and for consolidating shipments to neighboring destinations, thereby reducing delivery mileage. Both of these effects can be observed, for example, in online retailing. In this case, companies may be facing another challenge, namely, being accused of greenwashing when emphasizing the environmental benefits of certain actions that are also financially beneficial.

To conclude, demand management complements supply (chain) management by providing an additional set of levers for managing the sustainability of a firm's operations. Both perspectives are important and must be aligned to make strides on

the path toward sustainability. To date, sustainability-oriented demand management is much less developed, both in the academic literature and in practice, than its supply-side counterpart. In this chapter, we have attempted to outline the potential of this nascent field. We hope that our discussion will contribute to tapping this potential, to the benefit of a sustainable future.

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Part III
Business Models and Strategy in
Sustainable Supply Chains

Chapter 11

Supply Chain Collaboration for Sustainability



Tarkan Tan, Yann Bouchery, and Joerg Hofstetter

11.1 Introduction

11.1.1 *From If to How*

In recent years, and especially in the last decade, we have been witnessing a rapid increase in sustainability awareness everywhere in the world, making sustainability no longer just a nice attribute that firms may choose to pay attention to, but more and more a necessity, or even a legal requirement as regulations are quickly evolving in this direction. Also on the consumer side, the common belief as to “people will not pay for more sustainable products or services” is rapidly changing (see, e.g., Statista 2021). Therefore, it is imminent that the global pressure will increase and more actions by businesses will be demanded to fight against global warming and other sustainability issues. Accordingly, sustainability will—and should—increasingly drive business decisions. So, at least from the environmental sustainability point of view, the question has been shifting from if to how. The sheer dimension of the challenges from sustainability makes any individual action too small to reach the required momentum, putting special attention on the aspect of collaboration. In this chapter, we focus on supply chain collaboration.

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11.1.2 A Focus on Internal Improvements Is No Longer Adequate

An increasing number of firms disclose and exert efforts to improve their sustainability impacts. Although these efforts have to be recognized, these companies have traditionally taken actions in reference to their “own” (internal) operations. This is questionable as considering only internal operations is myopic and can have unintended consequences. Upstream supply chains that are contributing to a major share in the production of goods or services are often located in developing parts of the world with many social and environmental problems. This detachment from sustainability violations, supported by the legal liability concerning sustainability holding, at most, for internal activities only, has long been the key argument for companies to limit their focus to their own activities and hold direct suppliers responsible for sustainability issues in the supply chain.

However, the pressure from stakeholders such as NGOs, media, customers, communities, and governments has recently been increasing to hold companies responsible for the sustainability impacts of Global Value Chain (“footprint perspective”) that they are a part of Hartmann and Sabine (2014). Regulations are also becoming more and more stringent in this direction.

11.1.3 Collaboration via Industry Initiatives

Over the past three decades, most companies have been responding to external pressures to address sustainability issues with a risk management approach and by trying to avoid legislation. This approach is based on the belief that sustainability issues are of such a large scale that only collective action on industry level generates sufficient power to realize changes. Such industry collaboration also ensures a “level playing field,” so any company in an industry is equally hit by costs from sustainability action. Collaboration has become a corner stone of the sustainability agenda. When the UN developed the Sustainable Development Goals (United Nations 2015) to succeed the Millennium Development Goals, collaboration was considered so important that it became a goal for itself: the SDG 17 “Partnerships for the Goals.” Policymakers were (and remain) convinced that the challenges in the world can only be solved by working together—instead of against one another. Today’s dominant approach in the Global North to sustainability in supply chains is participating in powerful voluntary industry initiatives and adhering to their respective rules. The most important thing for the power of a voluntary industry initiative is the size and quality of its membership. Generally, these voluntary industry initiatives cover four areas of activities:

1. The definition of what is considered sustainable (e.g., targets, minimum requirements, and trajectories of development)

2. The establishment of an efficient supply chain monitoring system (e.g., codes of conduct, reporting formats, auditing systems, and sharing databases for audit results)
3. The provision of capability development offers (e.g., economic development programs, consultant networks, and experience sharing)
4. Communications

Frequently, companies implement their membership in voluntary industry initiatives by including their code of conduct in supplier contracts, by making supplier performance monitoring (e.g., auditing) a mandatory requirement (as long as suppliers are not powerful enough to deny such contractual elements), and by enforcing compliance (often, if possible, threatening suppliers with contract termination in case audit results are below a certain threshold).

11.1.4 The Way to Hell Is Paved with Good Intentions

Many codes of conducts of voluntary industry initiatives turned out as a Global North perspective on how to solve sustainability issues in the Global South. The lack of understanding of the context in which companies operate upstream in the supply chain limited the effectiveness of many past sustainability activities—and thus their impact. Overall, coercive approaches of forcing suppliers to stop business practices that violate the voluntary industry initiative's code of conduct have not delivered the results public policymakers and society were aspiring. In consequence, new laws on supply chain liability emerge across Europe and beyond. For instance, legal liability determinations might assign supply chain emissions to a particular company. The Hague District Court recently ordered Royal Dutch Shell to reduce its total carbon dioxide emissions—which include those of its suppliers and customers (i.e., Shell's Scope 3 emissions)—by 45% by 2030 (Rechtspraak 2022). Besides, the finance industry is putting attention on sustainability in companies' supply chains (Sarkis et al. 2020), pressuring companies to shift their attention from “sustainability actionism” to realizing impact.

11.1.5 Supply Chain Collaboration

Based on the past experiences of industry collaboration for sustainability, a new understanding emerges, considering that solutions with high impact need to be developed and implemented together with (and not for) the actors in the supply chain and include further stakeholders along value chains. Supply chain collaboration implies aligning the decisions made locally by different members of the supply chain to achieve an agreed-upon centralized solution and to share the benefits among the actors. This practice is often proposed by practitioners and academics as a

solution that achieves better results in various sustainability dimensions (including profitability). For instance, a report from CDP (2015) notes that companies that engage with one or more of their suppliers, consumers, or other partners are more than twice as likely to see a financial return from their emission reduction investments, and almost twice as likely to reduce emissions, than those who do not engage with their value chain. Besides, supply chain collaboration is necessary to account for the dynamics of supply chain interactions and their effects on operations both upstream and downstream. For example, altering the manufacturing process of a product to reduce internal carbon emissions, such that the product requires different storage conditions, likely affects emissions downstream, whether by eliminating some activities or requiring additional ones (such as heat treatment or temperature-controlled storage). The result might be a net increase or decrease in the total carbon footprint of the product. Even if a company's internal carbon abatement effort results in a diminished total carbon footprint in the supply chain, exerting abatement efforts upstream or downstream in the supply chain might still be more effective and/or efficient. In this chapter, we discuss specifically how collaboration among members of the same supply chain (vertical collaboration) can materialize and how this could affect supply chain sustainability. We note that while horizontal collaboration—which goes beyond industry collaboration discussed in Sect. 11.1.3—might be of interest as well, it is outside our scope.

In the remainder of this chapter, we introduce three cases in which a focal firm takes the initiative to improve sustainability in its supply chain. The cases complement each other in the sense that each case refers to collaboration with different tier levels (up- and downstream) in the supply chain. In Sect. 11.2 we discuss how Eastman Chemical Company decided on where to exert carbon abatement efforts in its *downstream* supply chain for a particular product. In Sect. 11.3 we discuss how Philips assesses and improves sustainability in its *upstream* supply chain, focusing on direct (first-tier) suppliers. In Sect. 11.4 we show how Barry Callebaut influences sub-supplier (a.k.a. lower-tier supplier) practices to improve sustainability in its *further upstream* supply chain. Finally, Sect. 11.5 is devoted to takeaways from the cases and to further discussions.

11.2 Where to Exert Carbon Abatement Efforts in Eastman's Supply Chain?

Eastman is a global specialty chemicals company that manufactures chemicals, fibers, and plastic materials.¹ The case that we introduce here focuses on hydrocarbon resins sold by Eastman to a set of customers. Eastman makes these hydrocarbon resins in molten states, and customers use them in the same state. However, the products must be kept within 160°–180°C to remain molten. Two options are

¹ The text in this section is an adaptation of Koomen et al. (2023).

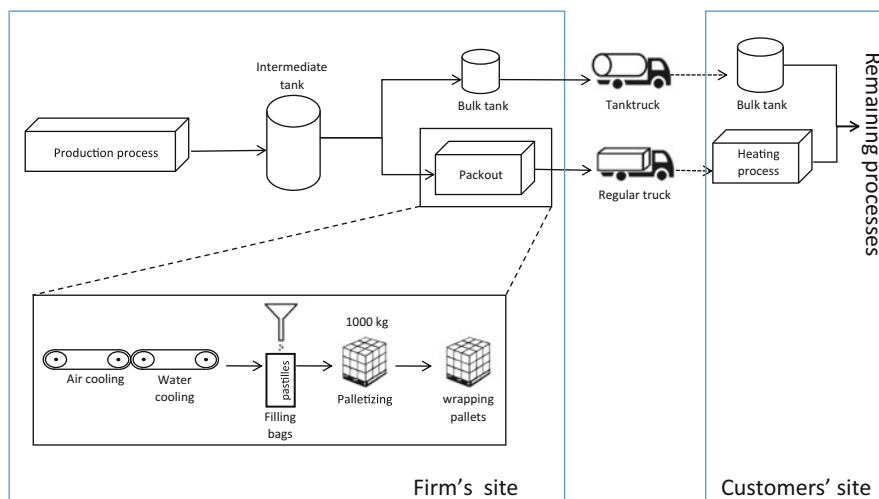


Fig. 11.1 Process flow for molten and solid product

available for shipping the product. The first option is to keep the products molten all the way from production at Eastman's site to use at the customer's site, which requires storing the products in heating tanks, at both Eastman and customer sites. Besides, transportation then needs to be done via dedicated, heated trucks. However, some customers lack capabilities to store the products in their molten state, so the second option for Eastman is to cool the product, make pastilles (solid state) that can be stored at ambient temperatures, and package them for shipment; customers bring the pastilles back to a molten state prior to processing. On the one hand, shipping molten product increases transport and storage emissions due to temperature control requirement, but, on the other hand, it eliminates cooling, pastillation, and packaging processes at Eastman and the unpacking and heating processes at the customers. We refer to Fig. 11.1 for an overview of the process flow for both options.

The products have substantial environmental impacts that Eastman sought to reduce and wanted to identify the most promising path. The project, commissioned by the supply chain department, aimed to determine whether keeping the product molten at all points would be less carbon intensive, and thus whether it would be worthwhile to invest in enabling this situation for some customers that lack capabilities to store the products in molten state. Even though these customers use the resin in molten state, they did not need heated bulk tanks, because they process the product immediately after heating the pastilles. Thus, they were willing to receive molten shipments only if Eastman invested in heated storage at their locations.

Eastman applied a four-step framework during the project. Step 1 consists of selecting the boundaries for carbon emission accounting. Step 2 involves identifying

the abatement actions available. Step 3 focuses on the selection of the optimization strategy to be followed. Finally, step 4 aims at deciding the level of carbon abatement effort to exert on each action. During on-site visits and interviews with technicians and supply chain experts, the project team specified the boundaries for carbon emission accounting (step 1), namely, starting where the processes for solid and molten products are different and ending where they are similar again. This definition thus includes Scope 1, Scope 2, and Scope 3 downstream emissions. The project team gathered primary and secondary data to evaluate these greenhouse gas emissions. The carbon abatement actions under consideration (step 2) are the decisions about the product state (molten or solid) for each customer. Note that shipping a product in molten state to the customer requires investing in heated bulk tanks at the customer's location. Accordingly, the effort by Eastman corresponds to investments in heated bulk tanks installed at selected customers. This downstream action could refer to each customer, but differences in ordering patterns (which affect shipment size and storage) and locations (which affect distances, emissions, and energy consumption) mean the carbon abatement obtained by investing in heated bulk tanks varies for each customer. Although calculations were performed for all customers, the cut-off point appears when three customers receive their shipments in molten state. There was no carbon abatement for other customers, which accounted for smaller orders, and for which carbon emissions would have increased because the trucks could not be fully utilized. Step 3 defines the optimization strategy to follow. Eastman was willing to abate as much emission as possible, subject to its budget. Finally, step 4 consists of identifying the customers to invest in heated bulk tank, so as to ship the product molten. This decision was heavily dependent on the budget available and the project team provided an analysis of the optimal effort and carbon abatement for different budgets to Eastman executives. On the basis of the analysis provided, Eastman chose where to invest in carbon abatement efforts. The framework and analysis provided Eastman with a clearer view of the supply chain dynamics of its carbon abatement efforts. First, they found that the total abatement is increasing with diminishing returns, so finding valuable abatement projects might become harder over time (Blanco et al. 2020). However, this property does not hold if only the carbon emissions from Eastman are accounted for. In this sense, this application highlights the importance of defining the right boundaries. If Eastman had focused only on transportation costs and emissions, shipping molten resin would not have been perceived as valuable, because it requires a heated truck that consumes more fuel than a regular truck, implying an increase in transportation costs and emissions. Eastman also could have focused solely on the process steps after production and before transportation. In this case, it would be better to sell more molten products to customers, because packaging and the pack out result in more emissions and higher costs than keeping the product stored in heated tanks. Second, as this assessment indicates, it is important to analyze the impact for each customer individually. Indeed, the optimal decision on heated storage investment depends on the location and ordering pattern of each individual customer as these features affect transportation and shipment sizes. This illustrates that there is no one-size-fits-all carbon abatement strategy.

Overall, the results showed that investing in abatement options in the downstream supply chain enables reducing emissions both from Eastman's and its customers' operations. Besides, the largest abatement was obtained for Eastman's Scope 1 and 2 emissions. This shows that external abatement efforts are worth investigating in practice.

11.3 Supplier Sustainability Assessment and Improvement at Philips

Traditional auditing has been common practice for multinational companies to monitor their suppliers against sustainability violations and enforce them to comply with their code of conduct.² However, auditing activities are typically outsourced to third parties and financed by suppliers themselves, at least partially. This conflict of interest encourages biased evaluations by the auditors to mediate negative findings and exaggerate positives (Locke et al. 2007). Furthermore, suppliers are subject to repeated audits, possibly by multiple buyers, resulting in "audit fatigue." Leading MNCs try to alleviate this problem by constructing alliances to jointly conduct audits and/or share audit results, see, e.g., Stauffer (2018), Caro et al. (2018), and Fang and Soo-Haeng (2020). Nevertheless, audits are based on a pass/fail mentality to check supplier compliance, instead of addressing the root problems and building up continuous improvement plans. Porteous et al. (2015) empirically show that traditional supplier monitoring via auditing is not effective to eliminate sustainability violations. Jiang (2009) suggests that supplier sustainability compliance can only be achieved by combining supplier assessments with continuous improvement and consultancy support from buyers (see also Vachon and Klassen (2008)).

Consequently, a paradigm shift from policing the suppliers to collaborating with the suppliers is required. Based on our research collaboration with Philips, we advocate supplier self-assessments and propose a conceptual framework that we refer to as CARE, consisting of Collect, Assess, React, and Enhance phases. CARE is a generic framework that can be adopted based on the capabilities, resources, strategic priorities, and preferences of the buyer. Different methods can be used in each phase of the framework and some of the steps can be skipped, resulting in high flexibility in implementation.

The *Collect* phase is meant to gather supplier data that is essential at the consequent phases of CARE, which could be constructed via Self-Assessment Questionnaires (SAQs), on-site visits by the buyer or publicly available data sources. As on-site visits are unlikely to do for all suppliers, pre-classification can be made to determine the most critical suppliers to approach first, the strategic suppliers being ideal candidates to be shortlisted. Philips's SAQ consists of a multitude of questions divided into five categories: general information of the supplier,

² The text in this section is an adaptation of Tan et al. (2023).

environment, health and safety, human capital, and business ethics. For a self-assessment approach to work, the buyer should commit to not penalizing suppliers for self-declared sustainability violations, since the aim is improvement via supply chain collaboration.

The *Assess* phase is meant to create sustainability profiles of the suppliers to evaluate their sustainability levels and categorize them. This involves three intertwined steps: (i) measuring the sustainability level of suppliers, (ii) sustainability level prediction, and (iii) supplier classification. As a measurement metric, we propose a combination of sustainability topic scores which are derived from the data gathered in the *Collect* phase. Each data point is classified according to the sustainability topic it relates to (e.g., environmental, social, and economic) and convex combination constructs sustainability topic scores. When there are suppliers who are not (or cannot be) monitored, machine learning is used to predict sustainability levels and suggest most likely improvement actions. Once a sufficient number of suppliers are assessed, cf. “training data” is collected for machine learning, the sustainability levels of other suppliers, who have not undergone an assessment yet, cf. “test data,” can be predicted. For this purpose, we propose feature selection models to identify the most relevant supplier characteristics that predict supplier sustainability scores. Supplier characteristics indicate “all possible variables that reflect a supplier’s characteristics, which might impact or relate with supplier’s sustainability,” such as the location, activities, worker composition, and a number of employees of the supplier.

The *React* phase is meant to predict the improvement potential and future sustainability levels of suppliers, which is pivotal in determining the actions to take for suppliers’ sustainability improvement. Supplier characteristics that show a promise for sustainability improvement can be extracted at this stage using similar data-driven algorithms as in the *Assess* phase.

Finally, the *Enhance* phase is meant to improve supplier sustainability by making the best use of resources based on the outcomes of the former phases of CARE. Among the set of possible improvement actions, a choice should be made according to the improvement potentials based on supplier characteristics and the resources that those actions require at the buyer and/or the suppliers. The set of improvement actions ranges from introduction of protective gears for manufacturing to training activities at a supplier’s site to enable know-how sharing, as well as from the advancement level of procedures for disposal of hazardous wastes to performing regular occupational illness checks for the employees. When compared to a typical MNC’s financial power, the investment on some of these actions may yet not be affordable for some suppliers on their own, or they might be lacking the know-how to implement the suggested improvement. Hence, the involvement of the buyer can be a game changer not only to improve supplier sustainability but also to decrease costs through increased productivity. In parallel, to make the selection of the improvement actions that yield the highest sustainability improvement at suppliers, we propose a knapsack-type model. The objective is to maximize the total improvement potential of all suppliers, while accounting for the capacity and budget constraints of suppliers and the buyer. Buyer solves this problem to identify

the improvement actions to be prescribed to each supplier and determine the extent of involvement.

Results of the implementation of CARE at Philips generate significant insights and sustainability improvement. We unveil that the most important supplier characteristics that contribute to higher sustainability (improvement) level prediction are quality certifications (ISO 14001, OHSAS 18001, ISO 9001, SA 8000), supplier activities and facilities, workforce structure, and the buyer's assessment frequency. Since we utilize feature selection techniques, these are the most significant features in our models. In the Enhance phase, we compare the results of the mathematical model with Philips's current rule which is an automated selection of improvement actions based on scores. This rule results in 21%, 20%, and 52% improvement potential for small, medium, and big suppliers, respectively. Application of the knapsack-like model shows that a small supplier can achieve 53% improvement when the buyer is not involved, but 88% with buyer involvement. Similarly, the achievable supplier sustainability improvement is 56% versus 83% for a medium supplier, 65% versus 96% for a big supplier, and without versus with buyer involvement. Clearly, the greatest improvement can be achieved when the buyer is involved. Nevertheless, even when the buyer does not invest, suppliers significantly benefit from an efficient use of resources. Philips independently reports that the overall 2020 year-on-year improvement in sustainability performance has been 36% for suppliers that entered the sustainability assessment and improvement program in 2019, which utilizes CARE, indicating the efficacy of the suggested paradigm shift and the proposed framework.

11.4 Improving Sustainability in the Upstream Supply Chain: How Barry Callebaut Influences Sub-supplier Practices

Food supply chains relate to some of the most pressing sustainability challenges on the Earth, including environmental issues like soil fertility or loss of biodiversity, social issues like poverty or hunger, and economic issues like inequality or stability. These challenges are both local and global and originate from both local and global context. Often, these issues have a large distance to consumer brands. Since food products at their final stage address end consumers and are considered key to both nourish people and keep them healthy, they are at the heart of societal and political conversations on sustainability around the globe. Food supply chains can be among the most complex and most intransparent, with sustainability issues occurring in different world regions at many steps in the value chain—from inputs for farming to retailing or food services.

Food companies in the Global North have long come under public pressure to make their supply chains more sustainable—way beyond direct supplier levels. Some of these companies publicly report how they address these challenges, and

many others try to keep out of the public debate. Some state in their corporate strategy that sustainability in their supply chains is a core prerequisite for their future success, while many others rather go with the flow. Some companies develop and realize their own strategy, while many others follow industry mainstream. What can be learned by those pioneers who have become leaders and outspoken advocates of sustainable food supply chains, engaging beyond direct supplier levels?

Barry Callebaut, one of the world's largest traders and producers of cocoa and chocolate intermediate goods, has worked hard over the past years to become recognized as a global leader in sustainable chocolate. Joining the company in 2015 after a career at Unilever, former CEO Antoine de Saint-Affrique decided for a differentiation strategy, developing a unique offer to customers that is reliable and broadly accepted as sustainable. This program, called "Forever Chocolate," intends "creating a movement to make sustainable chocolate the norm." Activities range across all parts of the company and reach far up- and downstream in the supply chain: from hundreds of thousands of distant smallholder farmers upstream in the supply chain over internal operations to global distribution, but also internal support functions. It aspires to end poverty of farmers, eradicate child labor, and become carbon and forest positive. At that time, its main competitors still followed their traditional strategies built on market mechanisms and coercive power. In the remainder of this subsection, we focus on the upstream supply chain.

Chocolate consists of many different raw materials. Its making includes various production, logistics, and traditionally countless trading steps in different world regions. The key raw materials are cocoa, sugar, milk, fats, spices like vanilla, and various nuts. Depending on the production location and a product's unique formulation, ingredients and processes vary. Each of these raw materials and processes and their respective origin have specific sustainability challenges requiring a dedicated approach.

In November 2016, Barry Callebaut "committed to having 100% sustainable ingredients in all of our products by 2025 as part of our Forever Chocolate strategy." To reach this objective in a situation where most issues occur at distant sub-tier levels in the supply chain, different approaches are being applied and combined.

Procurement at Barry Callebaut is divided into an organizational entity for the core raw material cocoa and another one for all other input materials. For cocoa, the company buys raw materials and also operates numerous production sites that process cocoa into intermediate goods like cocoa liquor, cocoa butter, etc. To ensure coordination, cocoa procurement and cocoa processing are part of the same business unit. All other input materials are procured in a processed stage from producers or traders, by a global procurement organization. Both procurement entities follow the same objective: to ensure that by 2025 all ingredients are 100% sustainable—throughout the supply chain.

For cocoa, most sustainability issues occur on the farming and early trading levels—often unregistered smallholders in remote areas—with whom Barry Callebaut traditionally has no contractual relationships. This situation makes it difficult to identify the many parties involved, control and document their environmental and social practices, and, in case of performance gaps, ensure remedy. Intermediate

actors restrict access and define the business conditions for smallholder farmers. Since they sell to many customers, it is often assumed that concerted industry actions are needed to impact upstream supply chains. Barry Callebaut is an active participant in numerous industry initiatives that work with cocoa farmers to develop capacity in order to root out sustainability issues in the sourcing countries in general. The company further uses collective approaches to realize changes in public policy and law enforcement. The company buys certified goods to both benefit from the achievements of these certifying organizations and bring scale to and support them. Collective initiatives helped to control some sustainability issues but were not able to fulfill aspirations. Experiences with collective approaches quickly showed that accompanying individual changes were necessary—including changes to the business practices in procurement and supply chain management.

To influence sub-suppliers, Barry Callebaut engages with and supports intermediaries, established and funds the Cocoa Horizons Foundation to understand and collaborate with cocoa farmers, and is changing its own internal business practices for sourcing cocoa, to align conditions upstream in the supply chain. To respond to the strong requirement for supply chain transparency, the company is shifting from short-term to longer-term business relationships with cooperatives and those traders who themselves have long-term relationships with the farmers they source from. The shift from exchangeability of suppliers to establishing longer-term partnerships allows all parties involved to benefit from investments that are necessary to achieve shared sustainability targets. The trainings and the help provided by the Cocoa Horizons Foundation, aimed at improving in the areas of productivity (“prospering farmers”), community (“zero child labor”), and environment (“no deforestation & carbon positive”), are promoted and monitored by Barry Callebaut’s procurement practices. This way, the direct suppliers experience the company’s interest first hand both in achieving the targets (economic, environmental and social) and in using Cocoa Horizons’ services. When suppliers fall behind performance expectations, Barry Callebaut’s buyers are quick to intervene and work on remedy with the respective supplier. Still, the challenge is to roll out this approach to cover all cocoa the company sources.

For non-cocoa materials, Barry Callebaut has developed and follows customized approaches that fit the specific relationship constellation it has with the farmers and the respective intermediaries (Hofstetter 2018). For beet sugar (predominantly in Europe), solutions are pursued and implemented collectively as part of the Sustainable Agriculture Initiative. For cane sugar (originating from different countries with different industry structures), the company procures certified material (e.g., Bonsucro) to support this certifying organization’s work and its achievements, and works with civil society (e.g., Solidaridad) to address issue this certifying organization did not cover as required. For coconut oil, predominantly farmed by smallholders in Indonesia and the Philippines and traded through complex structures, Barry Callebaut sees little direct access to smallholders and thus decided for a collective approach, joining forces with other multinationals in the Sustainable Coconut & Coconut Oil Roundtable. For milk, the company has developed its VisionDiary program in alignment with the Sustainable Agriculture Initiative to

ensure its suppliers' supplier requirements align with its own targets. For vanilla, Barry Callebaut has teamed up with its main supplier to jointly develop and establish business models for smallholder farmers, broadening their activities beyond the traditional focus on vanilla and progressing their capabilities. A key element of this strategy is to enable smallholders to diversify into other fruits (including cocoa).

Some of Barry Callebaut's customers (chocolate consumer brands and retailer private labels) share commitments in improving sustainability in the chocolate supply chain, while many others do not. As customer sustainability requirements and strategies are little aligned, but scale effects remain essential, Barry Callebaut had to find compromises—with their customers and with their other stakeholders. Some customers engage with Barry Callebaut on progressing in sustainability, putting the company in the role of the intermediary between the chocolate brands and the upstream cocoa supply chain.

The levels of sustainability performance Barry Callebaut has achieved so far are not undisputed. Some products have a much higher sustainability performance compared with others in the product portfolio. Also, progress is high for some sustainability issues, while others lag behind aspirations. Overall, it is remarkable that a company of this size and market dominance has massively changed procurement practices and relationship types with suppliers—from short term to longer term—and has reached out to supply chain partners upstream beyond direct suppliers. Collaboration plays a key role now: with suppliers, sub-suppliers, local organizations, certification organizations, and even competitors (in sector initiatives), in essence, multi-stakeholder collaboration.

11.5 Conclusions

We start this section by highlighting some takeaways from the three cases presented through the chapter. One of the main results is that supply chain collaboration can take different forms to be effective. The cases highlight several of these forms including supplier-induced development (Eastman), co-development with suppliers through self-assessment and improvement (Philips), and influencing suppliers and sub-supplier through a combination of multi- and bilateral collaboration practices (Barry Callebaut)—all indicating some shift of the horizon from “coercive short-term” to “collaborative longer-term,” as illustrated in Fig. 11.2. This relates to the vast body of literature related to supplier involvement in innovation. For instance, Kim et al. (2015) demonstrate that different types of buyer–supplier relationships can lead to distinct innovation activities. In addition, Luzzini et al. (2015) demonstrate that companies that place a greater emphasis on innovation are more likely to be involved in sustained collaborations with suppliers. The example from Barry Callebaut is particularly insightful with this respect as the same firm has decided to undertake various types of collaborative actions to find the best alignment between the supply chain structure and available sustainability improvement options. Furthermore, the Eastman case shows that even when focusing on

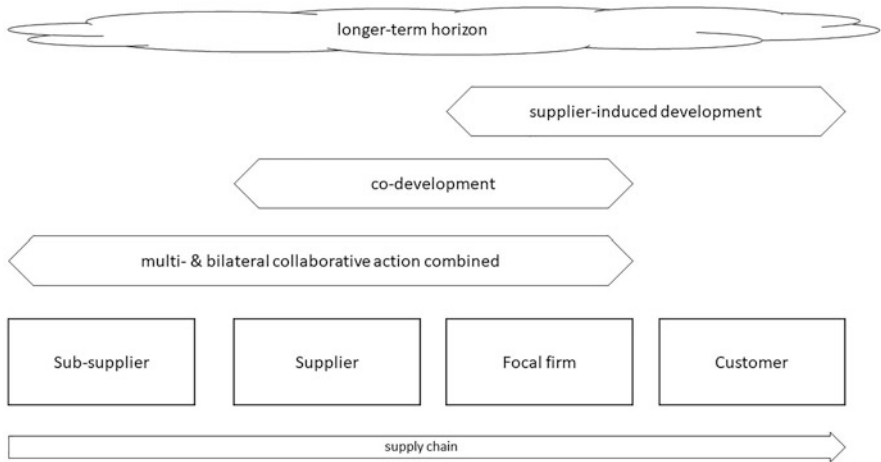


Fig. 11.2 Some examples of supply chain collaboration practices

a single product from a single firm, the strategy could differ from one customer to another depending on organizational and operational features. This demonstrates that there is no one-size-fits-all strategy when it comes to supply chain collaboration for sustainability.

The cases also share key similarities. All three firms experienced that collaboration was more effective than having separate actions taken by different supply chain members. This enables aligning decisions while taking the complex dynamics of supply chain interactions into account. Interestingly, this approach was effective even when the firm’s original motivation was to improve its “own” sustainability performance (see the Eastman case). Further improvements can be achieved by joint efforts of multiple parties in a supply chain. The Philips and Barry Callebaut cases highlight that collaboration is about mutual understanding and shared objectives, and not, as often found in practices and conversations, policing suppliers, and sub-suppliers. All three companies actively engage in and commit to the respective collaboration. They do not “outsource” the collaborative practices to third parties (e.g., voluntary sector initiatives) and continue their conventional (competitive) practices in the relationship with the business partner. The commitment and engagement are key sources for trust that allow the business partners to make specific adjustments.

Various other issues of interest are frequently mentioned by companies implementing a collaborative project to reach sustainability objectives. We discuss two of them below. First, the objectives of supply chain collaboration for sustainability should be stated clearly. We highlight in Sect. 11.1 that supply chain collaboration can achieve better results in various sustainability dimensions. However, this does not guarantee that all the dimensions of sustainability will be optimized simultaneously. There are some inherent trade-offs between sustainability dimensions and supply chain collaboration for sustainability is not immune from this. For instance, a

pure cost-driven collaboration project might lead to some positive sustainability side effects such as a reduction in carbon emissions, for instance. However, Bouchery et al. (2017) prove that in some particular cases, cost-driven coordination for lot sizing decisions leads to an increase in total emissions due to a reduction in the order quantity at the upstream echelon of the supply chain. In such situations, they show how to obtain a solution that decreases both costs and carbon emissions by avoiding the reduction in the order quantities. This proves that pure cost-driven coordination might not always be beneficial for the environment and that agreeing upon clear objectives is pivotal when implementing supply chain collaboration for sustainability.

Another key concern relates to *hold-up risks*. The hold-up problem is characterized by the fear of a firm to contribute to quality improvements at a partner if this latter can use these improvements to increase its bargaining power (Williamson 1975). This hold-up risk is very high when contributing to sustainability improvements for another supply chain member as the returns created will also benefit competitors. This can be viewed as a spill-over effect as competitors can benefit from the actions undertaken without participating to the collaboration project. This might bias fair competition and many companies are afraid of this hold-up risk and make conservative decisions. This issue is worth further investigation and several solutions to mitigate the hold-up risks have been proposed (see, e.g., Miller and Andrea (2016), Haruvy et al. (2019), and Lambertini (2018)).

Vertical collaboration in supply chains, in particular for supply chains with major sustainability concerns, is rather exceptions than the norm. The cost-focused mindset in procurement and outsourcing that created most of the today's supply chains has established relationships in supply chains that are challenging to turn into collaborations. Short contract durations, online auctions, coercive negotiation strategies, etc. are common practices that go counter the idea of trust and interdependence. The perceived complexity, dynamics, and sheer size of today's supply chains make them appear to managers as impossible to establish collaborative relationships with business partners. The frequency of this mindset in senior management may be one of the most important roadblocks toward transforming into collaborative approaches in supply chains.

While the three cases presented above provide an overview of the variety of supply chain collaboration projects for sustainability, we acknowledge that they represent only a very small subset of the many ongoing initiatives in this domain. Besides, they all are improvements in the never ending journey toward more sustainable supply chains and they cannot be considered as final achievements. We hope that these examples will inspire executives and will pave the way for additional research and practical initiatives to reinforce supply chain collaboration for sustainability. This is an existing challenge that can only be successful if many parties in the supply chain align and act together in the same direction.

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Chapter 12

Green Technology Choice for Deep Decarbonization



Anton Ovchinnikov and Jay Mackinnon

12.1 Introduction: Definition, Motivation, and Structure of the Chapter

Without a radical change in trajectory, the earth and its inhabitants will face devastating environmental, social, and economic consequences from an inevitable climate crisis caused by the growing concentration of greenhouse gases (carbon dioxide and equivalents, hereinafter and interchangeably “GHG,” “CO₂,” or “carbon”) in the atmosphere since the early 1900s. Prior generations’ failure to act has created an immense challenge for current and future generations to overcome or face the consequences of. However, in this challenge, we also observe an opportunity to transform large sectors of the economy from fossil-fuel-based ways of producing and delivering products and services into ones with dramatically lower emissions—a transition called “decarbonization.” Note that Chap. 3 by Boukherroub et al. (2024) focuses entirely on carbon footprinting.

In this chapter we address the mechanics of deep decarbonization. “Deep decarbonization” refers to a process of implementing major structural changes to all facets of the global economy with the goal of minimizing carbon emissions. Consider an analogy to a car: deep decarbonization is not about simply improving the car’s fuel efficiency, but also about transitioning all cars to zero-carbon powertrains—in fact,

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transitioning the car's entire lifecycle to a zero-carbon, or even a carbon-negative, process.

Fortunately, in many heavy-polluting industries, green technologies for deep decarbonization already exist, and many of the policy instruments needed to enable deep decarbonization—such as cap and trade, carbon taxation, and emissions standards—are well understood. For example, a McKinsey report on decarbonization in Europe (d'Aprile et al. 2020) suggests that more than 85% of the emissions abatement needed to achieve net-zero emissions by 2050 could come from technologies already demonstrated, including 28% that are now mature and 32% that are in an early phase of adoption.

Despite this, the deployment of green technologies is lagging. Another McKinsey report (Henderson et al. 2020) estimates that the number of solar panels installed globally per week should be approximately eight times higher than the number today; the same ratio is fivefold for wind turbine installations. Combining the conceptual possibility of decarbonizing with the evident gap in progress to date makes green technology choice—the topic of this chapter—critically important.

This chapter starts with a macro-level picture of decarbonization (Sect. 12.2) and then describes two main carbon-emitting sectors of the economy: energy (Sect. 12.3) and industry (Sect. 12.4).

For the macro-level analyses, we start in Sect. 12.2.1 with the landscape of decarbonization (sectors, geographies, etc.), then in Sect. 12.2.2 proceed to the opportunity for decarbonization—the so-called carbon arbitrage—and finish in Sect. 12.2.3 by discussing the macro-level cost of failing to decarbonize: the mitigation, adaptation, and build-back investments, and related technological and (operations and supply chain) research findings and opportunities.

For the energy and industry sectors, we present frameworks to think about decarbonization in these sectors and discuss the main technological and research findings and opportunities. These frameworks naturally differ, and hence the two sections do not have the same structure.

For energy, we start in Sect. 12.3.1 by presenting data on the costs of different technologies, which suggest that zero-carbon renewables, such as solar and wind, are reaching parity with other sources, favoring their broad adoption. Then, in Sect. 12.3.2 we discuss intermittency (including at a system level), which limits adoption. Energy storage connects the two and thus holds the key to decarbonizing energy; thus in Sect. 12.3.3 we discuss the main technologies for energy storage and the recent operations and supply chain management research.

For industry, the framework involves triangulation of demand, supply, and policy mechanisms. For demand, we explore the nontrivial differences between thinking about the end consumers' demand for lower carbon products (Sect. 12.4.1.1) versus thinking about the supply chain partners' demand for lower carbon inputs (Sect. 12.4.1.2). For supply, we discuss technologies in two main carbon-emitting industries: steel (Sect. 12.4.2.1) and cement manufacturing (Sect. 12.4.2.2). We proceed to investigate carbon capture and storage (CCS) technology and hydrogen technologies (Sect. 12.4.2.3), which the experts in heavy carbon-emitting industries nearly unanimously highlight as key elements of their decarbonization. Finally,

regarding policy mechanisms, we discuss tax, rebate, and subsidy instruments as applied to individual firms (Sect. 12.4.3.1) and competitive industries (Sect. 12.4.3.2). As with the other sections, throughout this discussion, we focus on recent operations and supply chain management research.

12.2 Macro-Level Analyses

12.2.1 Landscape of Decarbonization

According to the most recent assessment by the International Panel on Climate Change (IPCC; Dhakal et al. 2022), global GHG emissions reached 59 ± 6.6 Gt in 2019, averaging 56 Gt for the decade of the 2010s. This was the highest decadal average on record, although the average rate of increase dropped from 2.1% in the 2000s to 1.3% in the 2010s. During the COVID-19-affected year 2020, global emissions dropped by about 5.8%, or approximately 2.2 Gt, but they have rebounded since then.

Figure 12.1 presents the breakdown of the 59 Gt total by economic sectors. The main insight is that the biggest emissions come from either the electricity and heat (energy) sector (~33%) or the industry sector (~34%) if one allocates the energy used in this sector to it. Agriculture, forestry, and other land use (AFOLU) (~22%), transportation (~15%), and buildings (~5%) are the other main polluters, yet for parsimony, this chapter will discuss technologies only for energy and industry—specifically, metals and cement production, which represent the largest emitters.

Geographically, Fig. 12.2 shows the 2019 global emissions by country. China has the largest carbon emissions (~25% of the world’s total), followed by the United States (~14%), India, and Russia (~5–7% each). European Union countries

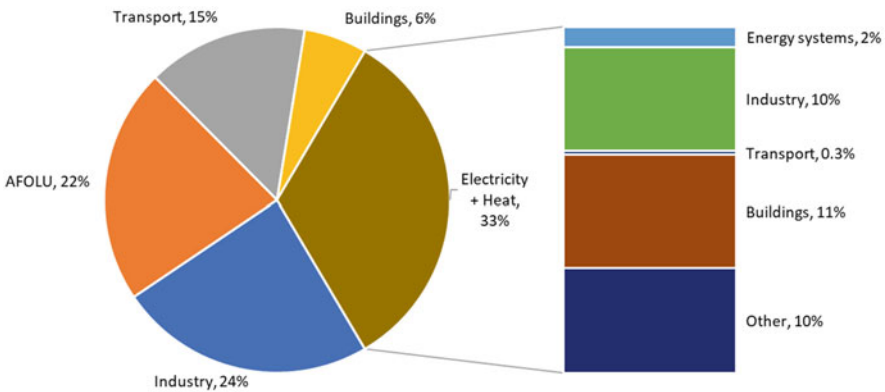


Fig. 12.1 Direct and indirect emissions by sector (Source: Adapted from Dhakal et al. 2022, Fig. TS.6)

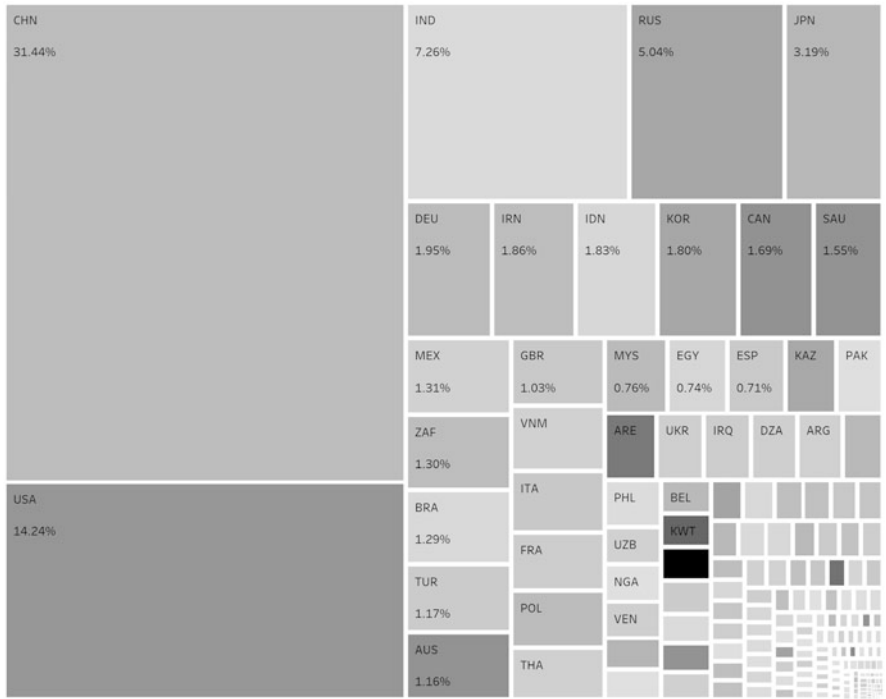


Fig. 12.2 Yearly GHG emissions by country (Source: Data from The World Bank, <https://data.worldbank.org/indicator/EN.ATM.CO2E.PC>)

together correspond to approximately 10%. However, this should be compared to the countries’ vastly different populations: India, for example, has nearly ten times more people than Russia. Darker colors in Fig. 12.2 correspond to larger emissions per capita. Qatar leads on a per-capita basis; of the large countries, Canada, Russia, and the United States are the biggest per-capita emitters.

In contrast to Fig. 12.2, which presents annual emissions, Fig. 12.3 presents cumulative emissions (although from fossil fuels and cement only) since the beginning of the industrial revolution. Darker colors represent the larger per-capita annual emissions (as in Fig. 12.2). The key insight is that China’s contribution to the cumulative total is much smaller, while the roles of the United States and western European countries, like Germany and the United Kingdom, are much larger. For example, the United Kingdom (labeled as GBR in the figures) contributes only about 1% of the current emissions yet is responsible for approximately 5% of the cumulative total. This underlies the concept of “loss and damage,” which was discussed at the United Nations Climate Change Conference COP27¹ at the time of writing this chapter.

¹ <https://unfccc.int/event/cop-27?item=7>

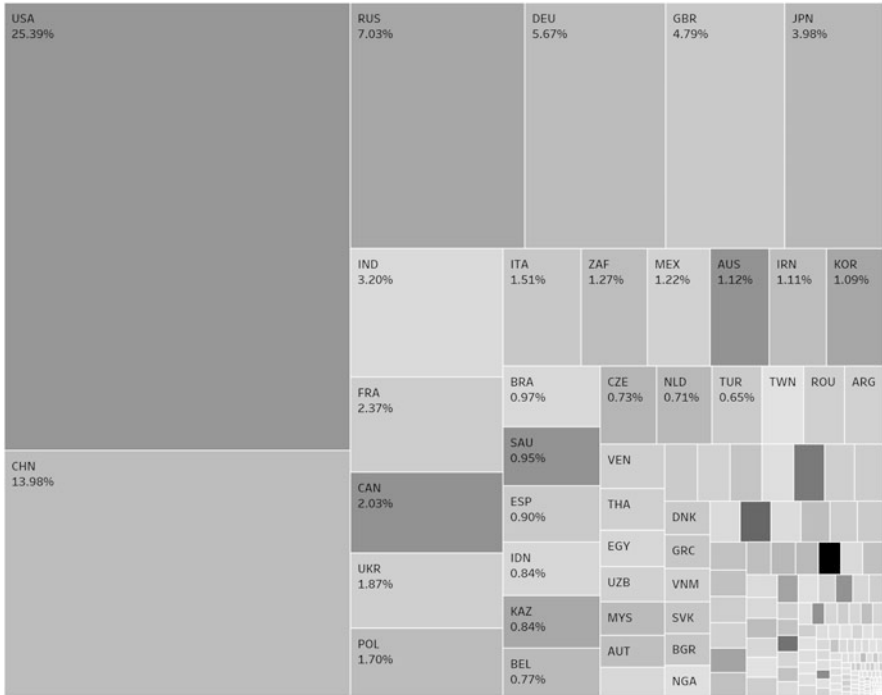


Fig. 12.3 Cumulative GHG emissions (Source: Data from the Global Carbon Project, <https://globalcarbonbudget.org/archive/>)

12.2.2 Opportunity for Decarbonization: Carbon Arbitrage

Having established where, sector-wise and location-wise, most carbon emissions are coming from, we now turn to the business case for decarbonization, still at the global scale.

The International Monetary Fund’s working paper by Adrian et al. (2022) estimates the gain that the world would realize by phasing out coal. Using the proprietary dataset on the vast majority of the world’s coal producers from Asset Resolution, a provider of data for climate-related financial data (<https://asset-resolution.com/>), and utilizing the NGFS (Network for Greening the Financial System) Net Zero 2050 scenario (see Elderson and Breeden 2021), they calculate what they call carbon arbitrage: benefits from phasing out coal minus the costs of doing so. The benefits are calculated as the total (discounted) emissions multiplied by the social cost of carbon (SCC). The costs consist of the forgone profits of the coal companies and the cost of building the renewable infrastructure (they consider only solar and on- and offshore wind; more on that later in this chapter). Using an SCC of \$75 per ton of CO₂, which is in the lower range of currently available estimates, they calculate the net present value of a policy replacing coal

with cleaner energy sources to be nearly US\$78 trillion in net social benefits. To put this staggering number in perspective, it represents over 1% of the global GDP each year until 2100 and corresponds to US\$125 per ton of coal; for comparison, the median profitability of the top ten coal companies is under \$1 per ton.

So, if the global opportunity is that large, why is there relatively little action? Basically, the world leaders failed to agree to include phasing out coal in the COP26 agreement. Three factors are at play. First, the required financing to build the renewable infrastructure is estimated to be nearly \$30 trillion, or nearly 15 times larger than the current investments.² Second, much of the financing is front-loaded. Third, most of this financing is needed for emerging market countries, which have little means to generate it. Requiring wealthy countries to provide this financing would require up to 6% of their GDP to be devoted to this, according to the analyses by Adrian et al. (2022). However, since such countries are responsible for a large fraction of the accumulated carbon, their disproportional contribution to decarbonization should be expected; such an approach is in line with the so-called just transition; e.g., see Newell and Mulvaney (2013) and McCauley and Heffron (2018).

12.2.3 *Cost of Failing to Decarbonize*

A report by Neufeldt et al. (2021) for the United Nations Environment Programme (UNEP) emphasizes the importance of separating investments in reducing emissions to stabilize the levels of heat-trapping greenhouse gases in the atmosphere (mitigation) from investments in adapting to the climate change already in the pipeline (adaptation). A report by Westaway et al. (2022) for Vanguard, an investment management firm, also specifies three kinds of climate-change-related investments: (1) mitigation, (2) adaptation, and (3) build-back. The study estimates these investments (on the global scale) in four global warming scenarios: under 1.5 °C, 1.5–2 °C, 2–3 °C, and over 3 °C. Green technology investments, like those in renewable power generation to replace coal, are considered mitigation. Adaptation includes preemptive measures, such as sea walls or enhanced irrigation, and build-back is about repairing and reconstructing physical capital that may be damaged by climate-related events. The Vanguard analyses contain two key insights. First, the total amounts of investments in these scenarios are surprisingly similar, but relative shares for the kinds of investments are drastically different. In other words, failure to invest in green technologies to contain global warming will not really save much, as disproportionately high costs would be incurred to build back following the adverse climate events.

² <https://www.iea.org/news/record-clean-energy-spending-is-set-to-help-global-energy-investment-grow-by-8-in-2022>

Interestingly, the Vanguard study also finds that the build-back investments will not necessarily be unproductive. Utilizing the models of Skidmore and Toya (2002), building back from climatic disasters may raise GDP above its no-disaster counterfactual over the medium term as economies upgrade their physical capital and integrate new technologies in a way they would not have done otherwise. For example, following a devastating hurricane, Partanna, a startup company, is building 1000 carbon-negative and hurricane-resistant homes in the Bahamas, as noted in McCartney (2022).

The IPCC sixth Assessment Report (AR6) Summary for Policy Makers (Dhakal et al. 2022), Section C, also emphasizes that many adaptation options exist and are used to help manage projected climate change impacts. Still, their implementation depends upon the capacity and effectiveness of governance and decision-making processes. They highlight the need to establish conditions to enable support of climate-resilient development, while also noting the increased evidence of maladaptation across many sectors and regions since the AR5. Maladaptive responses to climate change can create lock-ins of vulnerability, exposure, and risks that are difficult and expensive to change and exacerbate existing inequalities.

For warming beyond 1.5 °C, and consistent with scenarios in the Vanguard study, a paper by Betts (2020) for the World Adaptation Science Program (WASP) draws on a range of sources that consider either the high CO₂ concentrations scenario (940 ppm by 2100) or analysis of impacts specifically at 4 °C global warming—so-called high-end climate change. The report notes that for some impacts within this century, human adaptation to high-end climate change may be possible, for example, by increasing indoor cooling to reduce heat stress risks, by building flood defenses or floodplain management in areas at increased risk of river or coastal flooding, by improving water management and storage practices, and by increasing extraction of deep groundwater, in areas under increased drought risk. However, they also note the disparate spatial, temporal, and sectoral barriers and limits to adaptation: well-educated and wealthy societies have a high capacity to adapt, while areas with high unemployment, poverty, or civil unrest may have low adaptive capacity. Table 12.1 summarizes how mitigation and adaptation strategies compare on spatial,

Table 12.1 Comparison of mitigation and adaptation on spatial, time, and sectoral dimensions

	Mitigation	Adaptation
Spatial scale	Primarily an international issue, as mitigation provides global benefits	Primarily a local issue, as adaptation provides benefits mainly at the local scale (e.g., Eijgenraam et al. 2014, 2017)
Time scale	Long-term effects because of the inertia of the climatic system	Affects both short- and long-term reduction of vulnerability
Sectors	Priority in the energy and industry sectors (discussed in this chapter) as well as transportation and agriculture	Priority in the water and health sectors and in specific geographical areas (like low-lying coastal areas, those at risk of wildfires, etc.)

time, and sectoral dimensions. Likewise, the effectiveness of adaptation depends on whether it can take place on the appropriate time scale, and since much adaptation is local and often reactive or spontaneous rather than planned, it is crucial to collect information and assess the effectiveness of the “here-and-now” autonomous adaptation. Lastly, the WASP study of the adaptation to high-end climate change makes a sobering statement that “[s]ome coastal regions may not be considered economically worthwhile to protect,” necessitating research to determine which regions (and ecosystems, broadly speaking) are worthwhile to protect and which are not. This could be perceived as being at odds with the just transition discussed earlier.

An example of a study combining the “here-and-now” adaptation with rigorous technical and economic analyses is the 2013 INFORMS Franz Edelman Award-winning work by Eijgenraam et al. (2014) and Eijgenraam et al. (2017) that showcases a state-of-the-art solution for adaptation to sea-level rise in the Netherlands. The researchers applied nonlinear integer programming to optimize flood protection standards and then stress-tested the resultant solutions via Monte Carlo simulation. Compared to the earlier recommendations, their solution yields both a highly significant increase in flood protection and a large cost savings. Importantly, the developed methods can also be used in decision-making for other flood-prone areas.

Summarizing this discussion, recall an infamous remark by Stuart Kirk, then head of responsible investment at HSBC Asset Management: “Who cares if Miami is six meters underwater in 100 years? Amsterdam has been six meters underwater for ages, and that’s a really nice place.” Per Vetter (2022), this comment “caused no small amount of embarrassment at HSBC,” which stated its commitment to net zero by 2050 or sooner in its own operations and in its financed emissions. Mr. Kirk subsequently resigned, but this episode emphasizes an important point: alongside mitigation—where much of the green technology choice research in operations and supply chains resides—studies on adaptation are also interesting and relevant.

Having established the macro-level case for decarbonization, we now consider technologies in the main polluting industries: energy and industry.

12.3 Decarbonizing Energy

The production of energy (“Electricity + Heat” in Fig. 12.1) is either the first or second largest source of global emissions, corresponding to 33% of the total, or 23% if indirect emissions from energy used in industrial processes are allocated to the industrial sector, which is discussed in the subsequent section.

Figure 12.4 illustrates the total energy supply (in terajoules, TJ), by source and region. Three observations are in order:

1. Fossil fuels (coal, oil, and natural gas) are by far the leading energy sources; apart from Africa, where biofuels are the leading source, in all other regions, fossil

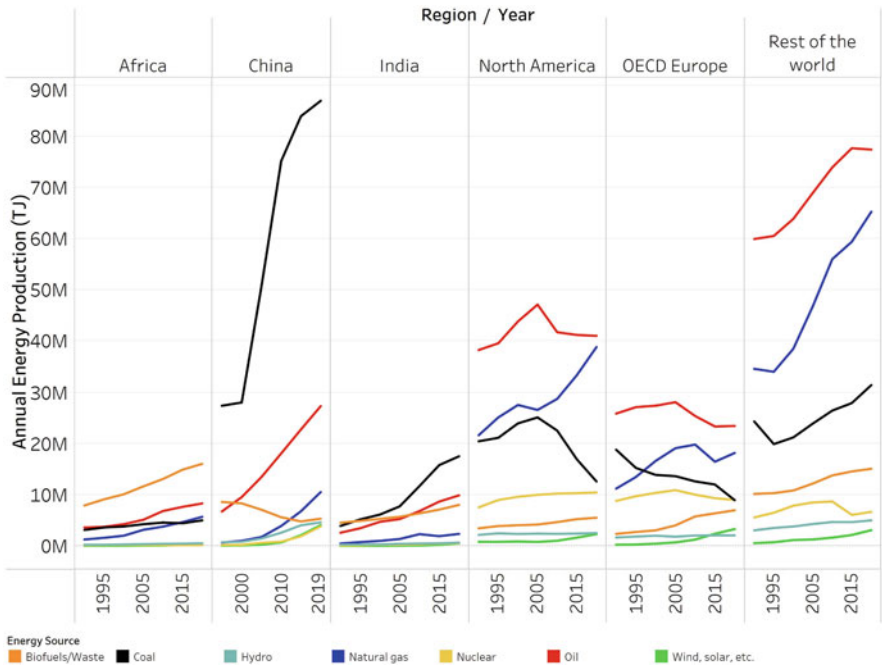


Fig. 12.4 Annual energy production by source/region/year, 1990–2019 (Source: Data from IEA’s Statistics Data Browser, <https://www.iea.org/data-and-statistics/data-product/world-energy-statistics>)

- fuels are the top three energy sources. This is widely known: decarbonization is often used interchangeably with transitioning away from fossil fuels.
- China’s use of coal is the single biggest source of energy. Coupled with the fact that carbon emissions per TJ from coal exceed those from oil by about 50% and those from natural gas by about 100% (US Energy Information Administration [EIA] 2022), China’s coal-related *emissions* look even more striking than the energy production reported in Fig. 12.4.
 - Low/no-carbon emitting sources (hydro, nuclear, wind, solar, etc.) are small but growing. The growth was speeding up prior to 2019 (where the data in Fig. 12.4 end) and has accelerated further since the International Energy Agency’s 2021 report on renewables (IEA 2021).

Next, we zoom in further on the low/no-carbon sources.

12.3.1 *Cost of Low/Zero-Carbon Energy Sources Reached Parity with Fossil Fuels*

“Renewables” commonly denotes wind or solar power. Lazard’s (2021) Levelized Cost of Energy (LCOE) analysis estimates the unsubsidized cost of approximately \$30–\$40 for utility-scale photovoltaic solar installations and about \$25–\$50 for (onshore) wind, which is smaller than that of the new-built conventional power sources (about \$45–\$70 for gas, \$65–\$150 for coal, and \$130–\$200 for nuclear). However, many of these costs are fixed costs, which are amortized over time. Considering that many existing power plants have exhausted their planned useful life (such useful lives are frequently extended), another relevant comparison is between the (total, levelized) cost of a new renewable source and the marginal cost of operating a fully amortized traditional source. The latter is estimated at \$24 for gas, \$29 for nuclear, and \$42 for coal, implying that the new renewables are cost competitive even with such a lowered benchmark. This is because between 2009 and 2021, the renewables LCOE declined by approximately 80% for wind and 90% for solar (although the rate of decline slowed more than twofold since 2016); see the 2011, 2016, and 2021 figures for solar and wind in Fig. 12.5.

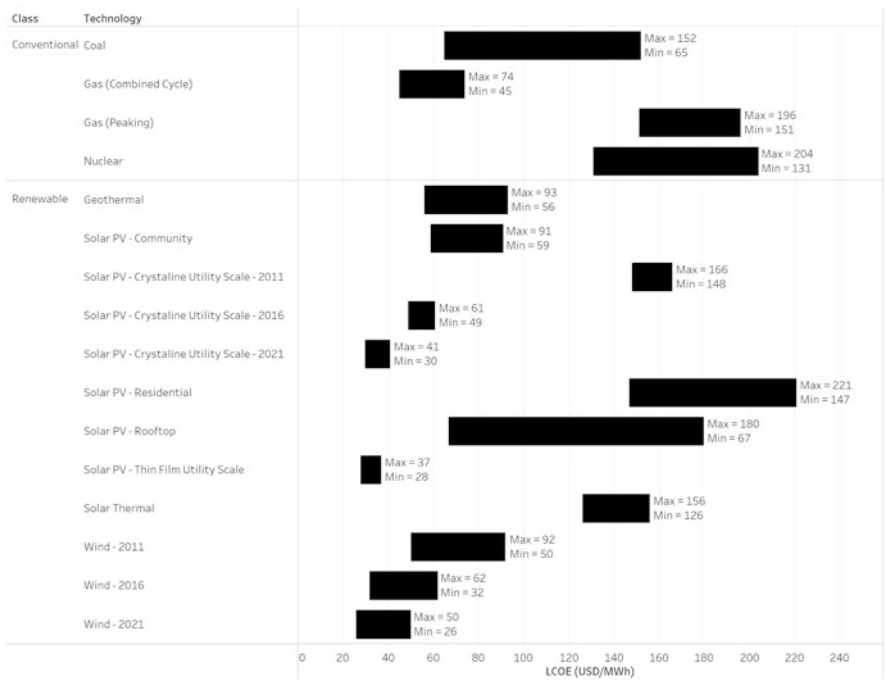


Fig. 12.5 US Levelized Cost of Energy (LCOE) for renewable and conventional sources (Source: Data from Lazard 2021)

Taken together, the International Renewable Energy Agency (IRENA) shows that “almost two-thirds [...] of newly installed renewable power in 2021 had lower costs than the world’s cheapest coal-fired option” (IRENA 2022). Total renewable power installations have outpaced the conventional ones (fossil fuel, nuclear, and hydro), and the International Energy Association (IEA) predicts that by 2026, the global renewable electricity capacity “will rise more than 60% from 2020 levels” and be “equivalent to the current total global power capacity of fossil fuels and nuclear combined” (IEA 2022).

12.3.2 Intermittency Is a Major Challenge for Renewables

Solar and wind power are not without limitations. A main one is intermittency, from both anticipated disruptions (the sun does not shine at night) and unanticipated ones (variations in wind and cloud patterns). A particularly challenging phenomenon known as “dunkelflaute” (as first described in Germany) refers to periods with calm wind and overcast conditions across a large geographical area. Figure 12.6, adapted from Li et al. (2021), provides an illustration of the phenomenon, with light (dark) gray depicting the daytime (nighttime) dunkelflaute. Clearly, such periods can last

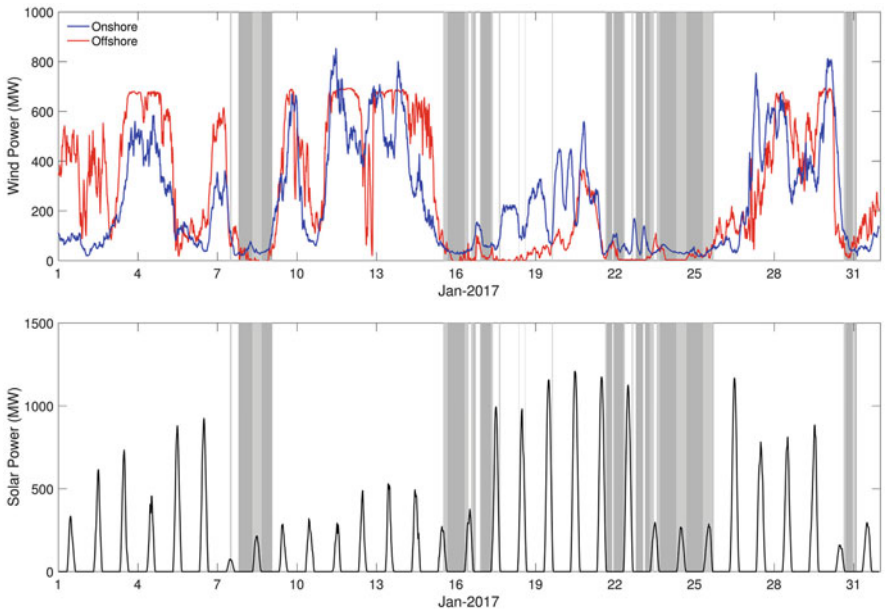


Fig. 12.6 Time series of wind and solar energy production with dunkelflaute periods in gray (Source: Created using <https://opendata.elia.be> following the description in Li et al. 2021, Fig. 12.1)

for several days—meaning that energy should be produced with other means during such periods, and there are two main technological solutions: reactive (backup) generation and storage.

Aflaki and Netessine (2017) consider strategic investments in renewable energy in light of intermittency and service guarantee. In their model, intermittency is represented as a two-point distribution³ (i.e., the energy from the renewable source is either available with full capacity or completely unavailable), and the service guarantee is assumed at 100% (that is, an equal amount of renewable and nonrenewable capacity must be installed to offset intermittence). Their main question is about the impact of carbon pricing (tax, specifically; see subsequent sections) on intermittent renewable energy investments.

Their main finding is that the impact of carbon pricing consists of two counteracting effects. The first effect is a decrease in carbon tax obligations from a cleaner renewable technology, which positively affects the installed renewable capacity. However, a second impact is that as more clean renewable capacity is installed, more “dirty” backup capacity is needed to offset intermittency. Hence if the tax is low, or the clean technology is rarely not producing, the former effect dominates, which motivates higher-capacity investments, but if the tax is high or the technology is often down, the latter effect dominates. This finding is qualitatively similar to the non-monotone response to the environmental tax in Krass et al. (2013) but is driven by the increased tax burden of intermittency as opposed to the trade-off between shrinking demand and fixed costs.

12.3.3 *Energy Storage Technologies for Decarbonizing Energy*

Several papers consider the joint operation of renewable generation and storage, e.g., Zhou et al. (2019), Peng et al. (2021), Kaps et al. (2022), and K  k et al. (2020). The storage technology could be different in these papers, and for the sake of simplicity, we refer to a broad set of technologies as batteries (we discuss other storage technologies later).

Peng et al. (2021) consider the problem of optimally charging or discharging a battery using either renewable (e.g., solar) or conventional but flexible, reactive sources (e.g., natural gas). Figure 12.7 illustrates several of their solutions, and it is clear that this is a rich and nontrivial problem. For example, at the same time of the day (e.g., at 6:00 am), the battery could be either charged or discharged, depending on the cost (and hence the optimal installed capacity) of solar. The source/usage of the resultant power would also be different. In some scenarios, the battery is charged using a nonrenewable source to save on importing external and more expensive power. In other situations, the battery would be discharged (to

³ Kok et al. (2018) also modeled intermittency as a two-point distribution, while Hu et al. (2015) modeled it as a discrete random variable with 20 mass points on a $[0,1]$.

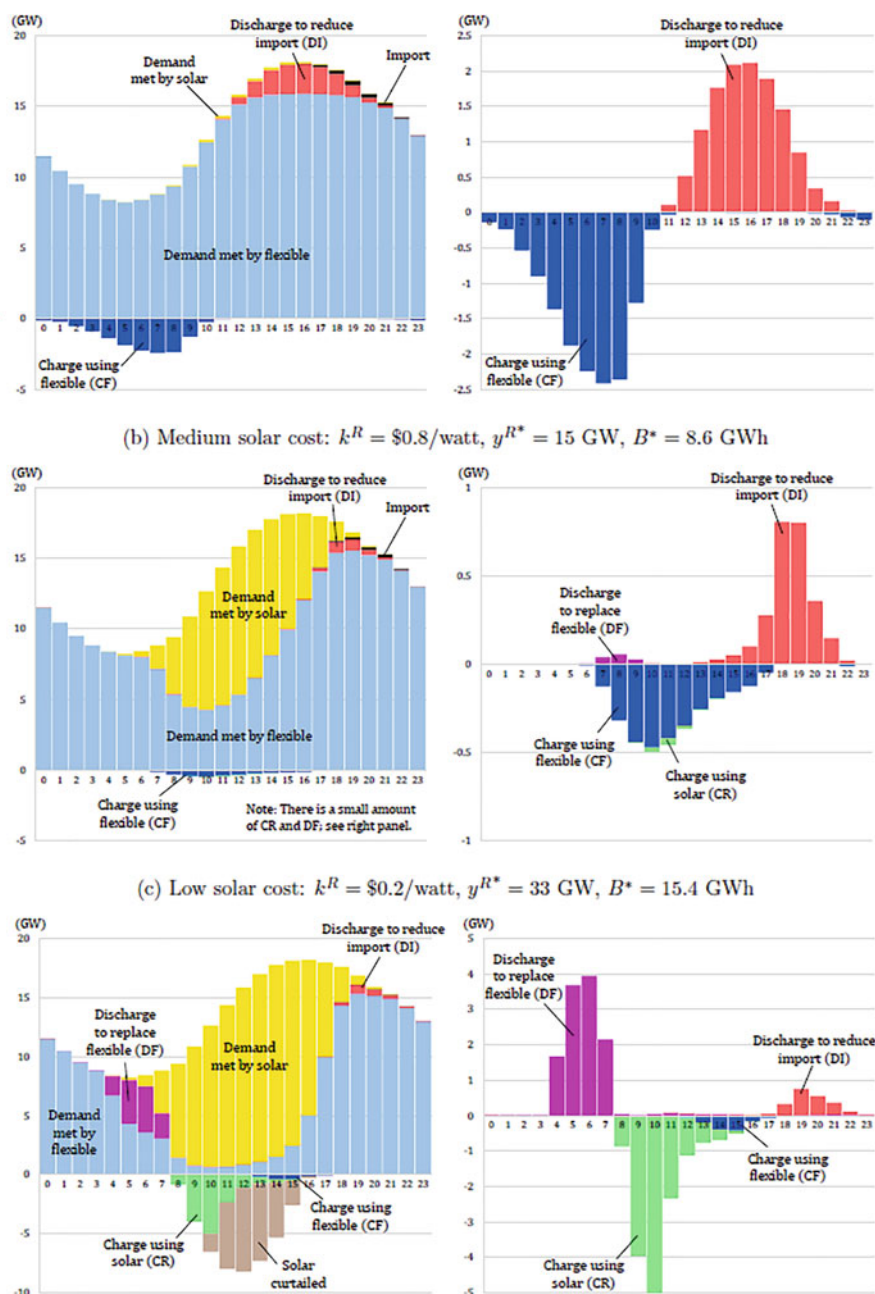


Fig. 12.7 Battery charging illustrations (Source: Peng et al. 2021, Fig. 4, used with permission)

reduce the nonrenewable generation) in anticipation of the excess renewable power later.

Kaps et al. (2022) consider a less granular setting with the “day” and “night” periods, with positive and zero renewable production, respectively, with primary application to small, off-grid systems, such as in isolated communities or small islands. They do not consider conventional sources.

Kök et al. (2020) present a newsvendor-like model for the capacity investment between renewable and conventional sources, but they do not consider storage, which allows the problem to decouple. As Kaps et al. (2022) note, storage significantly complicates the problem, as one needs to track the state of the storage at any point in time (e.g., before sunrise in the case of solar). The model becomes intractable, and they, therefore, present upper and lower bounds based on full discharge and lost power, respectively.

All these papers fit their models to data and provide empirically grounded numerical simulations. Kaps et al. (2022), in particular, contrast two very different storage technologies: battery-based and thermal-based, and show that a less-efficient thermal technology could be preferred in certain conditions.

Generally, there are several ways to categorize storage technologies, e.g., by their function, response time, or suitable storage duration. One most useful for this chapter is perhaps based on the form of energy stored. A highly cited paper by Luo et al. (2015) identifies several kinds of storage technologies:

[1] Mechanical, which includes:

- a. Pumped hydro. The water is pumped into a reservoir when excess renewable energy is available and then used in a hydro plant when renewable power is insufficient. Interestingly, Dimanchev et al. (2021) show that even without pumping, the existing large-scale hydro plants (which are normally considered sources of base-load power) can still be sources of short- and long-term energy storage that allow for more efficient utilization of intermittent renewables, reduce the need for other reactive capacity (e.g., gas), and reduce the need for carbon capture and storage (CCS) technologies.
- b. Compressed air. Surplus electricity drives a chain of compressors to inject air into under- or overground storage tanks, and when energy is needed, compressed air drives turbines to generate electricity. Such tanks can be located underground (e.g., in decommissioned mines, or overground in specially built tanks), and the compressed element also does not have to be a gas; liquid-based compressed energy systems are also being tested. Canada’s Hydrostor (<https://www.hydrostor.ca/>) undertakes several such projects around the world.
- c. Flywheel. The excess energy is used to spin a wheel, which can then be slowed to release energy. Such systems are being tested for short-term storage and load balancing (as, unlike batteries, their performance does not degrade with frequent charge-discharge cycles), but even a modestly long storage (several hours) is still an area of research.

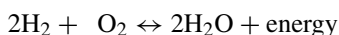
[2] Electrochemical, such as batteries, which include:

- a. Solid-state batteries, such as those based on lithium-ion chemistry, which use solid electrodes and a solid electrolyte to store and release electric charge. They are perhaps the most popular and commonly utilized sources of energy storage, e.g., used in most modern electronics.
- b. Flow batteries, which are devices in which chemical energy is provided by two chemical components dissolved in liquids pumped through the system. They have a longer cycle and useful life than solid-state batteries; the latter lose approximately 50% of their capacity after about 1000 charge-discharge cycles, while the former could last over 10,000 cycles. In turn, the flow batteries have inferior cycle efficiency of about 50–80%, as compared to nearly 100% for leading solid-state technologies. Nevertheless, both have been used in recent applications. In December 2021, the Australian state of Victoria switched on a Tesla Megapack solid-state battery capable of providing temporary power to over 650,000 homes.⁴ In October 2022, the Chinese city of Dalian switched on the flow battery to supply enough backup power for nearly 200,000 residents.⁵

[3] (Purely) electric, such as capacitors, which store static energy. Superconducting magnetic capacitors, as their name suggests, store energy in the form of a magnetic field. They have fast response time (milliseconds), fast full discharge (minutes), high cycle efficiency (>95%), and long useful life (~30 years), but they cannot store energy for long, leading to daily losses exceeding 15%. Therefore, they are used in specialized applications where large amounts of energy must be delivered fast.

[4] Solar fuels, which use solar radiation to produce fuel that can then be used for zero-carbon energy production. One example is artificial photosynthesis: this technology is currently in the development stage. Another example is using solar energy to produce hydrogen, which we will discuss next.

[5] (Purely) chemical, which primarily means hydrogen fuel cells. The chemistry of hydrogen-based energy is enormously appealing for decarbonization:



That is, when excess renewable energy is available, it could be used to generate hydrogen from water with a process known as water electrolysis, and when energy is needed, hydrogen can be burned to produce energy and water. Hydrogen can be stored in high-pressure containers and/or transmitted by pipelines. Both processes are similar, yet more challenging than those for natural gas (i.e., methane), as the

⁴ <https://www.cnn.com/2021/12/08/australia-switches-on-victoria-big-battery-powered-by-tesla-megapacks.html>

⁵ <https://www.pv-magazine.com/2022/09/29/china-connects-worlds-largest-redox-flow-battery-system-to-grid/>

hydrogen molecule is approximately eight times smaller than the methane one, and it is five to ten times more flammable.

Hydrogen can be obtained by means other than electrolysis (for example, from natural gas) and can be used in many other decarbonization applications than energy storage. In fact, nearly all decarbonization scenarios rely significantly on including hydrogen and energy and industrial supply chains.

[6] Thermal, which use excess energy to cool or heat some substance (e.g., liquid air/nitrogen and molten salt, respectively) and then recover the energy in the reverse process. Such systems can store large amounts of energy without major hazards and for a long time, yet their cycle efficiency is relatively low. That said, they are also inexpensive to build, which makes them economically viable and even preferable in some cases (see, for instance, Kaps et al. 2022).

[7] Gravity-based. Excess energy could be used to lift a heavy object so that, when needed, energy can be generated by lowering it back down. Energy Vault, a startup company, has demonstrated the efficacy of this approach by stacking 30-ton concrete blocks (see Energy Vault Holdings, Inc. 2022). This approach reaches a cycle efficiency of approximately 85% and can be built with recycled/reused materials, such as construction waste. Another major advantage of such systems is that their longtime energy loss is negligible—they can store energy for months, which becomes relevant in situations with significant interseasonal variations in energy consumption and production. In Canada, for example, more energy is required for heating in the winter, when less solar energy can be produced.

[8] Hybrid. Last but not least, any practical decarbonization solution will likely involve the combination of the above technologies: some that can store energy for hours to balance daily peaks (e.g., capacitors, flywheels, solid-state batteries), others that store energy for days or weeks to get through the dunkelflaute-like situations (e.g., pumped hydro, compressed air, flow batteries, or hydrogen), and those that can store energy for months (e.g., thermal or gravitational).

To conclude the discussion on technologies for decarbonizing energy, we note that nuclear power could also be considered a low/zero-carbon energy source. Nuclear encompasses three kinds of technologies.

First are the large-scale nuclear fission power plants that have been operating since the 1950s. Several countries (e.g., France) and regions (e.g., the province of Ontario in Canada and parts of eastern Europe) derive most of their electricity from such plants, but they are expensive to build, they produce hard-to-process waste, and highly publicized accidents, like Chernobyl in 1986, have made large-scale fusion controversial. Faced with such challenges, Germany decided to decommission all such power plants despite the zero-carbon electricity they produce.

Second are the so-called small modular reactors (SMR) that can be prefabricated, shipped, and installed on sites that are not suitable for, or where it is not economical to build, large-scale nuclear plants/reactors (Liou 2021). Their designs are generally simpler and rely more on the passive safety features that reduce the likelihoods and consequences of accidents. They also require less frequent refueling and could use the “traditional” reactors’ waste as fuel; some SMRs could operate for up to 30 years without refueling.

Third is nuclear fusion technology. Unlike fission, which derives energy from large atoms breaking apart, fusion energy comes from combining small atoms (or other nuclei). Sun and most other stars “run” fusions; solar energy is therefore that of fusion. Uncontrolled fusion reaction—a hydrogen bomb—has been available since the 1950s, but creating a controlled sun on earth proves to be notoriously difficult. The leading fusion project, called ITER, is a collaborative multinational effort that uses powerful electric magnets to hold plasma (a gas heated to millions of degrees, hotter than the sun’s core) inside a doughnut-shaped reactor. The challenge is to obtain more power from such a reactor than is used to run the magnets and to heat the plasma, and then sustain it. The current “record” is about five minutes. Should these efforts succeed, fusion holds enormous potential as a safe, clean, and essentially inexhaustible power source. Recently, fusion started attracting private investment (The Economist 2019), but as an old joke goes, it “is always 30 years away” (Willmer 2022). For now, the traditional renewables and nuclear fission offer practical decarbonization solutions.

Lastly, decarbonization of energy requires modernization and upgrades to the energy grid to enable distributed production and bidirectional flows. Brown et al. (2022) estimates that the grid upgrade challenge requires “a century of work to do in less than a decade.” Among the key obstacles it lists the lengthy approvals needed, supply chain and labor shortages, and the necessity for a total rethink of the pricing and incentive mechanisms for distributed smart grids. See Dileep (2020) for a recent survey of smart grid technologies and applications, and see Adelman and Uckun (2019), K  k et al. (2018), Singh and Sheller-Wolf (2021), and Gao et al. (2021) as examples of the operations and supply chain papers that contribute to this broader topic.

12.4 Decarbonizing Industry

Per Fig. 12.1, including indirect emissions from electricity and heat, the industrial sector is the biggest polluter, with 34% of GHG emissions. The task of identifying promising technologies and effective policies to decarbonize the industrial sector is therefore critical. With that task in mind, in late 2018, the Aspen Institute organized a week-long workshop, the participants of which jointly produced a review paper by Rissman et al. (2020). As we did with energy technologies, we summarize the technologies for decarbonizing industry below.

The workshop participants also responded to an extensive survey, where they ranked the technologies and policies. Table 12.2 summarizes the top four responses to the main questions (four is a natural cutoff for most questions due to a sharp drop in the number of responses thereafter). Given the limitations of this chapter, we discuss only the top responses in detail.

Based on these responses, the workshop participants identified a three-dimensional approach that could decarbonize the global industry by 2070 in ways consistent with limiting global warming to 2   C. This approach consists

Table 12.2 Summary of responses to the Aspen Institute questionnaire

Question, in reduced form	Response (# of responses)
What are the most important technologies or processes to be pursued in the next five years?	1. Carbon capture and storage (CCS) and CO₂ transport from exhaust streams (17) 2. Hydrogen production, transport, storage (16) 3. New chemistry of cement (11) 4. Electrification (10)
What are the most important policies to be pursued in the next five years?	1. Carbon pricing, border adjustments (19) 2. R&D financing, tax credits, partnerships (13) 3. Energy-efficiency standards (10) 4. Government procurement of low-carbon products (9)
What are the most important research topics to ensure the technologies will be available?	1. Low-carbon cement and steel, or suitable replacements (18) 2. Hydrogen production, transport, storage (15) 3. Policy implementation demonstrations (12) 4. Public reaction, social acceptance, just transition (8)

Source: Rissman et al. (2020)

of (1) supply-side technologies, (2) demand-side technologies, and (3) policy actions and interventions as listed in Table 12.3 (left column) together with the examples of operations and supply chain management research studies related to these technologies and policies (right column).

Table 12.3 and the Aspen Institute's working group report lay out the foundation for decarbonizing industry, but they lack the supply chain view, which connects multiple elements. A first connection is the demand response. This includes both the end consumers, who may or may not be willing to pay more for lower carbon products, and the downstream supply chain players (producers) who, again, may be willing to pay differently for inputs with lower carbon content. The second connection is the supply chain response to government policies and incentives. The third connection is the industry competitive response. We discuss these connections next.

12.4.1 Demand-Side Mechanisms

12.4.1.1 Consumers' Willingness to Pay More for Decarbonized Products

There is some evidence that consumers are willing to pay more for ostensibly "environmentally friendly" products, though much of the research into this topic shows that consumer demand exhibits significant heterogeneity and is influenced by a range of factors, such as social connections and socioeconomic demographics.

Table 12.3 Supply and demand-side technologies and policy interventions to decarbonize industry per the Aspen Institute's working group report

1. Supply-side technologies: a. Energy efficiency (especially at the system level) b. Carbon capture and storage (CCS) c. Electrification d. Zero-carbon hydrogen as a heat source e. Chemical feedstock	Examples: Muthulingam et al. (2013) for energy efficiency; İzlegen and Reichelstein (2011) for CCS; Raz and Ovchinnikov (2015), Cohen et al. (2016) for electrification (although they considered electrification of transportation, not industry); Feng and Menezes (2022) for hydrogen (although in the context of energy storage rather than heat source)
2. Demand-side: a. Material-efficient design b. Reduction in waste c. Substituting low-carbon for high-carbon materials (i.e., storing carbon in finished goods) d. Circular economy (including increasing product longevity, reusability, ease of refurbishment, and recyclability)	Examples: Raz et al. (2013) for material-efficient design; King and Lenox (2001) and Dutt and King (2014) for waste reduction; and Abbey et al. (2014), Atasu et al. (2008), Agrawal et al. (2018), and Van Wassenhove (2019) for circular economy and related topics
3. Policy: a. Carbon pricing with border adjustments or other price signals b. Government support for research, development, and deployment c. Labeling and/or emissions standards d. Government procurement of low-carbon products e. Data collection and disclosure requirements f. Just transition: within sectors, geographies, generations	Examples: Krass et al. (2013), Drake et al. (2016), Huang et al. (2021), Fan et al. (2022) for carbon pricing; Cohen et al. (2016) for government R&D; Agatz et al. (2021) for green labeling; Villena (2019) for the role of procurement in sustainability (although by large multinational corporations, and not government); Kalkanci and Plambeck (2020) and Blanco (2021) for disclosure To the best of our knowledge, there are no operations and supply chain studies addressing just transition; this offers an opportunity for future research

Source: Rissman et al. 2020, with examples of related operations and supply chain management research studies

A McKinsey study by Miremadi et al. (2012) estimated how sensitive consumers' environmental preferences were to price increases. Polling 1000 consumers, the study reports that "upward of 70 percent of consumers surveyed said they would pay an additional five percent for a green product if it met the same performance standards as a nongreen alternative," but that if prices were increased by 25%, "less than ten percent of consumers said they would choose green products." The main implication from these results is that while consumers are generally environmentally conscious, environmental impact is only one of the many dimensions that affect demand.

Achtnicht (2012) surveys German car buyers about their vehicle preference across several dimensions, including carbon emissions. The study finds that the car buyers surveyed prefer lower carbon-emitting vehicles, but that there is significant variance in the population for this preference tied to gender, age, and education

level, which all tie to the consumer's level of climate concern. Young people, in particular, had a significantly larger willingness to pay (WTP): "individuals aged under 45 years had more than double or triple the median WTP, compared to the reference group." This paper reinforces the heterogeneity of demand response to carbon emissions. A similar paper by Hackbarth and Madlener (2016) reinforces the heterogeneity of consumer preferences by conducting a discrete choice experiment for 711 German car customers.

Berger (2019) finds evidence using lab experiments to support the "green signaling hypothesis," which states that consumers will increase their consumption of green products when observed by other people. This evidence suggests that part of the consumer's demand for low carbon goods is influenced by their peers and social networks. In light of such effects, a firm may need to factor in the additional utility a consumer receives from observed consumption when that firm is marketing its products. Berger also includes an estimate of the relative price premium consumers would pay for a "green" version of a product and notes that "the WTP for green products was higher than the WTP for conventional products, with an average [willingness to pay more] ranging from 17% up to 27% of the price of the conventional product."

One issue with most studies that measure consumers' willingness to pay more for lower-carbon products is that their supporting data measures stated, rather than revealed, preferences. This means that researchers ask participants questions about theoretical situations and how they would act, instead of observing how consumers act in a real-world situation. Research shows that consumers tend to overestimate their willingness to pay compared to experiments that measure actual willingness to pay (De Corte et al. 2021). Estimates of "green" price premiums may be overestimated in cases where studies employ a stated preference method.

Diederich and Goeschl (2011) asked people to choose between an entry into a real lottery and a reduction in CO₂. They contrasted the revealed willingness to pay with the participants' stated willingness to pay for a reduction in CO₂. The study found that people are much less likely to pay for real reductions in CO₂ than what their stated preferences would suggest. This result implies that researchers should approach estimates of consumers' willingness to pay for carbon reductions with some skepticism and indicates a need for more revealed studies of willingness to pay that employ a revealed preference methodology.

In general, there is a significant amount of evidence that consumers have some preference for low-carbon products. However, these preferences are often relatively small and potentially overstated. If consumers' environmental options are significantly more expensive than the standard version, they will often maintain their status quo consumption.

12.4.1.2 Producers' Willingness to Pay for Decarbonized Inputs

If consumers are willing to pay more for decarbonized products, and if producers' demand for inputs in a supply chain is tied to consumer demand for the final

good, one might speculate that there would be a price premium for low emissions inputs procured farther up in the supply chain. A profit-optimizing manufacturer might choose to source a more expensive input if the carbon emissions are lower to increase demand among environmentally conscious consumers. Relative to the abundance of studies on how environmental impact affects consumer WTP, there is not as much peer-reviewed academic literature that looks at how demand for lower carbon products on the part of the consumer affects demand upstream in manufacturing supply chains.

Let us consider a specific product, such as a car. As discussed previously, many consumers exhibit a preference for lower carbon-emitting vehicles and would prefer an electric or hybrid electric vehicle to a traditional internal combustion engine vehicle. However, even among vehicles with similar fuel consumption and carbon output characteristics, there may be differences in how both vehicles are manufactured that result in the net impact of one vehicle differing from another. Manufacturing a car takes steel, glass, microchips, plastics, and various other materials with their own supply chains.

Each of these inputs can vary in their own carbon emissions. A consumer that cares about their carbon impact could have higher relative demand for a vehicle that not only releases less carbon when driven, but also releases less carbon in the manufacturing process. If demand for a final product is a function of the total carbon released in the manufacturing process, manufacturers should consider carbon emissions when sourcing inputs from upstream in the supply chain. This presents an opportunity for operations and supply chain management researchers.

The Japanese automotive manufacturer Honda has pledged to carbon neutrality, not just for their own operations but also for all upstream and downstream activities as well. The CEO of Honda, Toshihiro Mibe, stated: “Honda will strive to realize carbon neutrality for all products and corporate activities Honda is involved in by 2050.” If Honda is looking to achieve a carbon-neutral supply chain via emission reductions, Honda will need to procure materials, such as steel, that generate less emissions.

One reason that consumer demand for low-impact products may not affect upstream demand for inputs is the lack of a rigorous and consistent carbon accounting framework. A company that buys more expensive, “cleaner” inputs might reduce the carbon footprint of a product, but if it cannot effectively communicate this change to consumers, demand may not increase sufficiently to offset the higher costs. One critical economic issue that hampers rational consumers’ ability to incorporate environmental concerns efficiently into their purchasing decisions is the lack of knowledge about the carbon emissions of competing products.

Muslemeni et al. (2021), for example, explain: “There is a lack of research, or indeed consensus, on the concept of green steel. The term ‘green steel’ has been very rarely used in the academic literature and, when adopted, its definition has been broad.” They go on to explain that there is significant debate about what constitutes green steel given that there are multiple ways it can be manufactured: “A primary (integrated) route which is highly emissions-intensive, and a secondary route which uses recycled steel and is much less emissions-intensive (global averages of 1.85

tCO₂/t steel vs. 0.4 tCO₂/t steel, respectively).” Even if primary steel manufacturers drop emissions by 50%, they will still generate higher emissions than the recycling firms, which raises questions of whether they can claim to be green at all. As opposed to trying to categorize inputs as “green” in a binary fashion, manufacturers would be better off trying to optimize the total carbon emissions across the supply chain.

At the same time, “greenwashing” (a phenomenon where corporations exaggerate or falsify their overall environmental impact) is widespread (de Freitas Netto et al. 2020). The lack of a clear and ubiquitous carbon accounting methodology means that even if consumer demand for low-impact products is very high, information asymmetry may result in an inefficient market equilibrium. Without adequate regulation and consumer protections, consumer preferences may not effectively translate to changes in upstream procurement (Feinstein 2013). Apart from setting standards, regulation could also play a direct role by imposing a price of carbon, thus providing incentives for adopting green technologies.

12.4.2 Supply-Side Mechanisms and Technologies in Main Polluting Industries

12.4.2.1 Steel Industry

As of 2019, steel contributed nearly 5% of global CO₂ emissions. The process typically used to make steel involves adding coal to iron ore in a furnace. As coal burns, it releases carbon. This carbon bonds to the oxygen in the melted iron, purifying the iron and making steel but also making CO₂ in the process (Nidheesh and Kumar 2019). Of course, the process is also energy intensive, and even if the chemical process were improved, there would still be a large environmental impact from steel production.

Some newer furnaces can eliminate chemical emissions by using hydrogen instead of carbon as a method of extracting oxygen from the iron ore. Hydrogen can also bond with oxygen but produces water instead of carbon dioxide. This process would require significantly more hydrogen than is currently available to most steel manufacturers. Hydrolysis is a chemical process where electricity is run through water to split it into oxygen and hydrogen. If a significant amount of clean, cheap electricity were available to steel manufacturers, this might be one way of reducing carbon emissions. This technology is relatively new, and it remains to be seen if this new technology will be able to compete with traditional manufacturing methods.

Muslemani et al. (2021) conducted a survey of key stakeholders in the steel industry and found that there are significant concerns about the ability of “green steel” to compete with lower-price alternatives. The general consensus was that in the near future, the market for “green steel” would remain niche. However, some respondents felt that the larger stakeholders may expand this niche market, with one respondent claiming: “It seems that automakers are moving away from focusing

solely on so-called tailpipe emissions versus the actual embodied carbon that enters into the vehicles. So, I think in the automotive industry the incentive or the direction toward greener products, including steel, probably needs to come from the OEMs themselves, rather than from the consumer, at least at this point.”

12.4.2.2 Cement Industry

Cement is responsible for a significant portion of global CO₂ emissions as well. Unfortunately, a significant portion of these emissions are from the underlying chemical process of the cement bonding, as opposed to energy inputs. The most commonly used cement is called Portland cement. It is used in 98% of concrete globally today, and 4 billion tons are produced each year. Approximately 50% of the carbon emissions in Portland cement production are part of its underlying chemistry: $\text{CaCO}_3 + \text{Heat} \rightarrow \text{CaO} + \text{CO}_2$.

There are also issues with energy consumption for cement, but what is particularly concerning for decarbonization is that even if 100% clean energy were used in cement production, the underlying chemical process would still be a massive source of carbon emissions. Thus, reducing CO₂ emissions will be achieved by reducing the total amount of cement used, by scientific innovation that allows for alternative materials, or via CCS. There are some possible alternatives to Portland cement that generate less CO₂ emissions, but most are not widely used yet due to economic and chemical factors that make them imperfect substitutes. The Oregon Department of Transportation notes: “While alternative cements offer an opportunity to reduce climate impacts, there can be barriers to adoption, including longer curing times to reach required compressive strengths, concerns about toxicity of the alternative materials, increased risk of salt scaling, and prohibitive transportation costs” (ODOT 2020).

12.4.2.3 Carbon Capture and Storage

Decarbonizing supply chains can occur through changes to how companies create products, or by reducing the total amount of production, but for many products (e.g., cement) it will be difficult to reduce emissions beyond a certain point without significant technological leaps forward. One method of reducing emissions at an aggregate level is to invest in technologies that can capture and store carbon, typically from the source of pollution, or by removing ambient CO₂ from the air. There are several ways of implementing carbon capture and storage (CCS) that manufacturers may consider as they work to limit their carbon footprint across their supply chain.

Rubin et al. (2012) notes that the need for CCS is motivated by the realization that “Large emission reductions cannot be achieved easily or quickly simply by using less energy or by replacing fossil fuels with alternative energy sources that

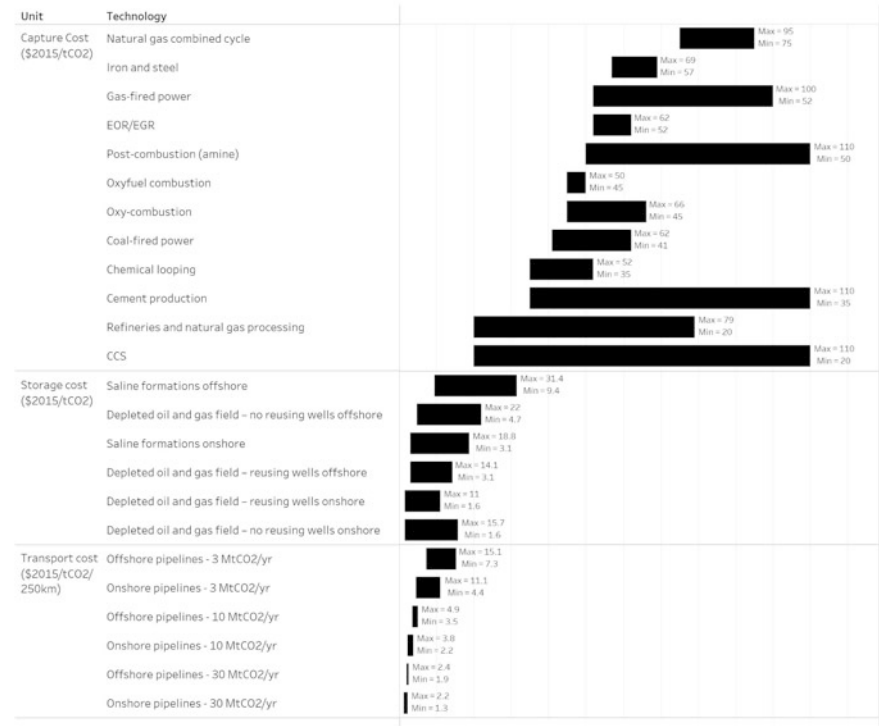


Fig. 12.8 Cost of captured CO₂ for different industrial processes, capture technologies, and transport and storage solutions (Source: Based on data from Budinis et al. 2018)

emit little or no CO₂. The reality is that the world today relies on fossil fuels for over 85% of its energy use.”

One of the key factors influencing the adoption of carbon storage is cost. In 2019, the IEA estimated the per-ton cost of CO₂ abatement across various industries. These estimates suggest that, for many industries, carbon capture can be a cost-effective component of an emissions reduction strategy. The IEA report⁶ also claims that for industries like cement, carbon capture is one of the only cost-effective ways of achieving reductions.

Budinis et al. (2018) note a “mismatch between expectations and reality” with regards to CCS technology. They also note that “no CCS barriers are exclusively technical, with CCS cost being the most significant hurdle in the short to medium term.” The authors go on to provide cost estimates for capture, transportation, and storage depicted in Fig. 12.8.

⁶ <https://www.iea.org/data-and-statistics/charts/levelised-cost-of-co2-capture-by-sector-and-initial-co2-concentration-2019>

Budinis et al. (2018) conclude, “In the long term, CCS is found to be very cost-effective when compared with other mitigation options. Cost estimates exhibit a high range, which depends on process type, separation technology, CO₂ transport technique and storage site.” Interestingly, the sites from which fossil fuels are extracted, such as oil and gas fields, have a comparative cost advantage in terms of CCS storage. This might help decarbonization from a political perspective in the sense of “just transition” (recall Tables 12.2 and 12.3). That is, it is possible that some of the economic pain caused by a reduced production of fossil fuels could be alleviated by new jobs in the CCS industry in the same regions.

CCS storage capacity is one of many constraints affecting deployment of the technology. Fortunately, there is a large capacity for storage in many parts of the world. Celia et al. (2015) note: “While storage capacity estimates are inherently uncertain, a number of estimates have been reported. Recent volumetric-based estimates for North America show storage capacity in deep saline aquifers in the United States to range between 1600 Gt CO₂ and 20,000 Gt CO₂. When compared to annual emissions of about 3 Gt CO₂/year from large stationary sources in the United States, it is clear that sufficient capacity exists to store these emissions over long-time scales.”

12.4.3 Policy Mechanisms

12.4.3.1 Government Incentives and Green Technology Choice Decisions

Government incentives generally come in three different forms. First are the incentives provided to consumers who buy the firm’s product—for example, the US\$7500 tax credit provided to buyers of electric vehicles (EVs) in the United States. We prefer to call them “rebates”; see Krass et al. (2013), Raz and Ovchinnikov (2015), and their case study of the Chevy Volt, a popular EV in North America at the time. In some other papers, these are referred to as subsidies with the implication that those are “consumer subsidies,” e.g., Cohen et al. (2016).

A form of rebate considered in Alizamir et al. (2016) and implicitly in Hu et al. (2015) is a feed-in-tariff (FIT). FIT is linked to the concept of net metering and refers to a price that is paid by the electric utility to the owner of the renewable energy system should the supply of power generated by the system exceed what had been used by the building. It is as if the energy meter rolled backwards, with the consumer receiving money for selling power rather than paying for using it.

The second form are the incentives provided to firms that use green technologies to manufacture their products. These forms of incentives are common in the healthcare industry and somewhat less common in the sustainability arena. We prefer to call such incentives “variable cost subsidies.” A negative subsidy can be viewed as a tax; note that a rescaling may be required if tax is levied on units of emissions rather than production.

Table 12.4 Four different forms of government incentives

	Firm's price	Consumer price	Firm's cost (variable, fixed)	Units action applies to	Government cost/revenue
Rebate, r	P	$p - r$	c, Θ	$\text{Min}[x, D(p - r)]$	$r\text{Min}[x, D(p)]$
Variable cost subsidy, s	P	P	$c - s, \Theta$	x	$s x$
Tax, t	P	P	$c + w(1-r)t, \Theta$	x	$w(1-r)t x$
Fixed cost subsidy, FC	P	P	$c, \Theta - FC$	0/1 if the corresponding technology is chosen	FC

The third form are the incentives provided for adopting the green technology. As an example, consider the 38% profit tax credit in Ovchinnikov and Hvaleva (2012). We prefer to call such credits “fixed cost subsidies.”

These different forms of incentives are summarized in Table 12.4. With these rebate and subsidy incentive tools, researchers considered different mechanisms for incentivizing the adoption of green technologies.

Raz and Ovchinnikov (2015) considered a price-setting newsvendor firm that decides on price and production quantity of a “green product” in response to the policy set by the government regulator. The regulator, in turn, anticipates this response and sets the policy to maximize societal welfare, accounting for the firm’s profit, consumer surplus, externality, and the government cost. Their key analytical finding is that with a combination of rebate and subsidy it is possible to achieve the socially optimal (first-best) welfare, but the optimal subsidy is often negative, i.e., actually a tax. That is, to incentivize the firm to produce the right amount of the green product and price it correctly, the government provides very large rebates to consumers, which generates a windfall of profit to the firm that is then taxed to recover the cost of providing rebates. The intuition for why this approach is optimal is in its impact on the service level: high rebate implies high selling margin, which in turn implies a higher critical fractile. Consumers therefore face less stock-out risk, while the government saves because rebates apply only to units sold (as opposed to subsidies that apply to units made, including those not sold).

Raz and Ovchinnikov (2015) further consider an extended example of the Chevy Volt (discussed in detail in their paper) and show that the current rebate-only incentive of \$7500, although structurally suboptimal, has a very small welfare loss. Overall, their work provides direct support for the use of tax as an incentive mechanism but does not explicitly deal with the technology choice problem.

Krass et al. (2013) consider the impact of tax specifically in the technology choice problem. In their paper the firm is considering whether to invest in one of k green technologies, where, as discussed above, technologies differ in fixed and variable costs as well as in their environmental efficiency. The regulator taxes the firm for pollution, and investing in an efficient technology is a way to reduce tax obligations. Their main result is that the environmental tax has a non-monotone

impact on the green technology adoption: while as expected, small taxes motivate the firm to adopt a green technology, large taxes do not. This is driven by the fixed cost effect: subject to a large tax, the firm must raise prices, which reduces the sales quantity, and so the total additional revenue with green technology may not be sufficient to offset its higher installation cost.

They further consider the regulator's problem, specifically the efficiency of using tax and the need to support it with fixed cost subsidies and rebates. They find that tax is efficient for a medium level of externality parameters (referred to as "environmental concerns" in their paper), but for both small and large externality pollutants, a combined approach with taxes, rebates, and fixed costs subsidies is needed. The only case when the combined approach cannot reach the first-best welfare is when the externality is very small and the socially optimal technology choice is "dirty," while the firm selects "clean."

In the abovementioned papers, the regulation is modeled with the goal of maximizing social welfare. Cohen et al. (2016) study a different approach with the goal of achieving an exogenously given adoption target, considering what we call rebates, although they refer to them as "subsidies." Alizamir et al. (2016) consider both an exogenously given target and a welfare maximization problem, and use the feed-in-tariff, which is a form of rebate as well, as discussed above.

Alizamir et al. (2016) note that when a rebate/subsidy is presented as a schedule (e.g., gradually declining over time as the costs of green technology fall), firms may strategically delay their investments. They study whether a rebate (feed-in-tariff) schedule exists that would result in no strategic delay. They show that for some parameter ranges this is indeed the case, and most such cases involve a variable rebate. When the parameters fall outside of this automatic "no-delay" region, however, and then the government decides to design a rebate policy that would result in no delays, then such a policy would have a constant rebate.

Cohen et al. (2016) study the impact of demand uncertainty on the green technology adoption in more detail. Specifically, they show that ignoring the uncertainty in demand (and optimizing rebates as if demand were deterministic) could lead to significantly missing the adoption target. Additionally, they investigate whether consumer surplus could increase as a result of demand uncertainty—an observation also made by Raz and Ovchinnikov. They show that the situation is somewhat nuanced; there exists a trade-off between lower prices and the possibility to underserve consumers with high valuations (who will receive the highest surplus loss if the product is stocked out).

A third approach to modeling the impact of regulation on the technology choice decision is through capacity allocation among a portfolio of technologies, e.g., Drake et al. (2016). Wang et al. (2013), Hu et al. (2015), and Huang et al. (2021) also implicitly consider portfolios of two technologies, but not from the perspective of incentives and regulation.

Drake et al. (2016) compared the impacts of tax (fixed emissions cost) versus cap-and-trade (variable emissions cost) and show that emissions price uncertainty under cap-and-trade results in greater expected profit than a constant emissions price under an emissions tax, which contradicts common logic. They attribute that to two operational drivers: the firm's option not to operate, which effectively right-censors

the uncertain emissions price; and dispatch flexibility, which effectively allows the firm to arbitrage on the variability of the emissions cost under the cap-and-trade. They also explore policy implications: they show that variable cost subsidies have little impact on the total capacity, but fixed cost subsidies increase capacity. This happens because the firm benefits from a variable cost subsidy only when the corresponding capacity is utilized (as opposed to the fixed cost subsidy that acts even when the capacity is temporarily not utilized, e.g., if the carbon price is too low). Fan et al. (2022) also consider the uncertainty in the policy instruments themselves and show that because of the price certainty they provide, environmental/carbon taxes may be a superior instrument to cap-and-trade mechanisms, which result in inherently uncertain emission prices.

Lastly, the literature has also described practical applications of green technology choice for decarbonization in the industrial context. For example, Bertsimas et al. (2022) outline a pathway to decarbonize OCP—the world's largest phosphate producer, located in Morocco.

12.4.3.2 Impact of Competition on the Efficacy of Government Incentives for Green Technology Choice Decisions

Most of the aforementioned works consider a monopolistic setting. However, government incentives for the adoption of green technologies affect not only firms per se, but also how they compete. A series of recent papers studies that, utilizing a set of common assumptions. Drake (2018) and Anand and Giraud-Carrier (2020) assumed that the industry consists of n players with one technology and m players with another; Huang et al. (2021) and Fan et al. (2022) further simplified $n = m = 1$. All firms with an identical technology are assumed to incur equal variable costs, and those that invested in cleaner technologies with lower per-unit emissions have higher fixed and variable costs than those with dirtier technologies. Lastly, all these studies assumed linear demand.

In many polluting industries, such as metals or cements, discussed earlier, the diversity of geographical locations, energy sources and their costs, and other characteristics lead to a situation where there are many competing firms with many cost and technological (i.e., emission) profiles. ANFRE⁷ and SteelOnTheNet⁸ present cost curves for aluminum and steel, respectively. Figure 12.9 presents the ranges of emissions intensities for several industries, including aluminum and steel. Among the lowest-cost smelters are those in Russia and Canada, which use electric hydropower. At the upper end of the curve are smelters in China, which generate energy from coal. Importantly, (some of) the lowest cost firms are also the cleanest—a setup that cannot be accommodated by the aforementioned papers, which assumed the opposite. Of course, a mixture of the two cases is also possible—

⁷ <http://www.anfre.com/global-aluminium-smelters-production-costs-on-decline/>

⁸ <https://www.steelonthenet.com/kb/world-cost-curve-hrc.html>

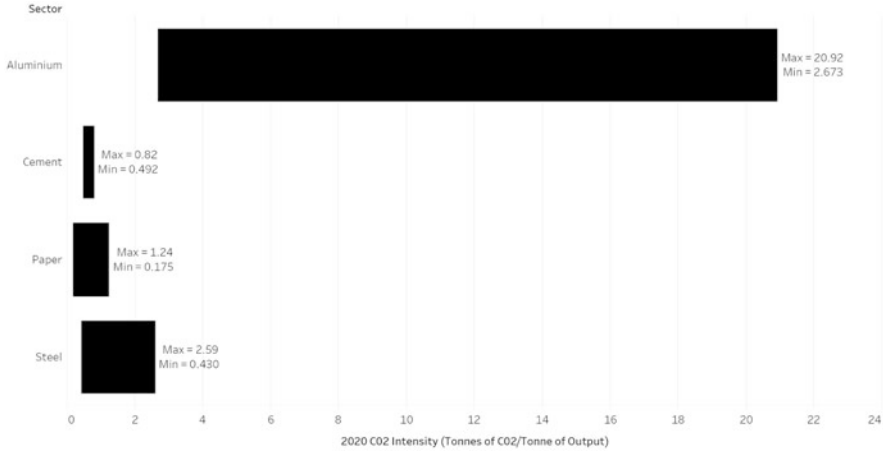


Fig. 12.9 Sample emissions intensity range by industry (Source: [Transition Pathways Initiative, https://www.transitionpathwayinitiative.org/sectors](https://www.transitionpathwayinitiative.org/sectors))

that is, one can be simultaneously high-cost and clean: for example, if one of the coal-based smelters installs a CCS scrubber as discussed above.

A modeling framework that can accommodate this realistic phenomenon is that of Krass et al. (2013). In two sister papers, the authors extend the Krass et al. model to a competitive setting (Krass and Ovchinnikov 2022a, b).

They consider an industry with N firms that have access to K emission-reduction technologies, which is subject to an environmental tax of $\$t$ per unit of emissions, e.g., per ton of carbon. Firm $i = 1, 2, \dots, N$ utilizing technology $j = 1, 2, \dots, K$ has the fixed cost Θ_{ij} and the variable cost c_{ij} , and emits x_i/S_j units of pollutant when producing x_i units. The total production of all firms can be written as $(x_i + x_{-i})$, where x_{-i} is the total production of all firms other than i . The authors consider Cournot competition with demand function $D(p)$ so that the equilibrium market price p satisfies $D(p) = (x_i + x_{-i})$. Assuming a well-behaved monotonic $D(p)$, the inverse equilibrium price function $p(x_i + x_{-i})$ exists and is unique. Then, the firm's profit from producing x_i units with technology j when other firms cumulatively produce x_{-i} equals:

$$\pi_{ij}(x_i + x_{-i}) = \left(p(x_i + x_{-i}) - c_{ij} - t/S_j \right) \times x_i - \Theta_{ij}$$

Observe how this framework captures the realistic scenarios described above:

1. A firm that uses fossil energy without emissions control technology would have high c_{ij} and low S_j as well as some “baseline” (possibly low) Θ_{ij} .

2. A firm that uses fossil energy but invests in emissions control technology such as CCS would have high c_{ij} and S_j , as well as, potentially, high Θ_{ij} . (This depends on whether it outsources CCS or invests in the equipment itself.)
3. A firm that uses renewable energy would have low c_{ij} (as it has no fuel-related costs) and high S_j , as well as, potentially, high Θ_{ij} . The latter again depends on whether it outsources the energy production or invests in the equipment itself; however, per Fig. 12.5 the fixed costs of renewable sources are among the lowest.⁹

With this framework the authors consider two kinds of equilibria: market-only equilibrium (Krass and Ovchinnikov 2022a), a situation in which the technologies are already selected (and their costs are thus sunk) and the firms decide on the production quantities, and technology-and-market equilibrium (Krass and Ovchinnikov 2022b), where the firms choose both the emission-control technologies and production quantities. Further, in both cases, the authors consider two demand functions: iso-elastic and linear (the latter was assumed in all aforementioned sustainable operations papers). The two functions are in some sense at the opposite ends of the spectrum of potential demand relationships: demand elasticity is constant in the former and changes from zero to infinity in the latter, reflecting products that are equally substitutable at any price point (iso-elastic) and those that can become more and more substitutable as price grows, until, eventually, the society can get by completely without the focal product (once the price exceeds the so-called choke price, when the linear function crosses zero).

Krass and Ovchinnikov (2022a) show that the competitive industry's market response to environmental tax is well behaved: at any tax level for either iso-elastic or linear demand function a unique equilibrium exists, and it reflects an intuitive property—that some $N^+ \in N$ firms with the lowest total variable production costs ($c_{ij} - t/S_j$) produce, and the remaining $N \setminus N^+$ firms do not. The latter firms are not unimportant, as they may enter the positive production set at a different tax rate.

Krass and Ovchinnikov (2022a) also study how the structure of the market-only equilibrium and the various quantities of interest (profits, market shares, emissions, etc.) change with tax. They document several curious outcomes, such as that (i) the industry profit can be increasing for a wide range of tax levels; (ii) as tax increases, a clean and low-cost firm may lose market share to higher-cost and dirtier firms; and (iii) the industry-average emissions intensity may be increasing with tax. They also show that the results qualitatively depend on the demand function. For example, with iso-elastic demand, by setting an appropriate tax the regulator can always ensure what the authors call a “k-clean” equilibrium when only the firms with k cleanest technologies produce. This property is practically relevant in the context

⁹ As an example of this situation, consider the solar installation at a steel mill in Pueblo, Colorado, in the United States owned by EVRAZ (a Russian steel conglomerate); <https://www.yahoo.com/news/construction-completed-solar-project-support-213444252.html>. Disclaimer: the first author of this chapter worked with EVRAZ executives on multiple sustainability projects.

of the concept of the “best available technology” advocated in numerous policy documents. Interestingly, the regulator has no control over the value of k (when demand is iso-elastic). In contrast, with linear demand, the k -clean property is not guaranteed to emerge, yet when it emerges, the regulator has direct control over the value of k .

Krass and Ovchinnikov (2022b) consider the technology choice and market equilibrium. With the simultaneous move to a Nash equilibrium (i.e., when all firms select their technologies simultaneously), they show that when demand is iso-elastic, an equilibrium may not exist, and when demand is linear, an equilibrium is guaranteed to exist; in both cases the equilibrium (when it exists) may be non-unique. With the sequential move to a Stackelberg equilibrium, the equilibrium necessarily exists and is unique for both demand functions.

An effective driver of the market and technology response results in Krass and Ovchinnikov (2022a, b) is a realization that the environmental taxes imposed per unit of pollution not only incentivize firms to make investments in clean technologies to reduce tax payments, but also punish firms that have low baseline costs. Indeed, consider the profit equation above and observe that at low t , a firm with low c_{ij} would have a large competitive advantage, and hence may dominate the market. But at high t , much of the cost will be driven by tax payments, hence a firm with high c_{ij} but also high S_j would be nearly equally competitive. A metal smelter that uses solar power is an example of the former; one that uses fossil fuels with CCS is an example of the latter. In other words, Krass and Ovchinnikov (2022a, b) show that while taxes motivate installing CCS, they de-motivate moving away from using fossil fuels to begin with. A high tax eliminates the advantages of operational efficiency.

Two other papers study somewhat similar settings. Anand and Giraud-Carrier (2020) consider the impact of cap-and-trade and emissions tax on the competitive industry where firms have zero production costs and identical emission intensity but can invest in green technologies. The technology is represented by a continuous twice-differentiable function multiplied by a cost constant, which is low for n and high for m firms. In Sect. 6 of their paper, Anand and Giraud-Carrier (2020) state that they analyzed heterogeneous production costs and emission rates, but they do not provide details in either the paper or the supplementary materials. Linear demand is assumed, however, with a cross-elasticity parameter, which reflects the degree of competitiveness in the industry; the classical Cournot model emerges when this parameter equals 1. They prove two main results: first, the cap-and-trade and tax mechanisms are equivalent; second, the industry profit with regulation could exceed that with no regulation.

Fu et al. (2023) consider the production quantity decisions in response to an environmental tax in an industry with $N = 2$ firms with technologies that have symmetric costs but asymmetric emission intensities. In supplementary materials, two separate extensions are also presented: one with asymmetric costs and another with $N > 2$ firms. In the latter case the most transparent results are obtained for the case with two technologies: one, possessed by a single “clean” firm, and another, possessed by $N - 1$ identical “dirty” firms. A fully heterogeneous case as in Krass

and Ovchinnikov (2022a, b) is not considered, and the Cournot competition with linear demand is assumed throughout. Their main result echoes that of Anand and Giraud-Carrier (2020): the industry profit may increase due to taxation, which is driven by the interplay of two effects. The cost increase from taxation has a direct negative effect on the profit of both firms. In parallel, as the dirtier firm is being forced out of the market, the effective competitiveness of the industry decreases, which has an indirect positive effect on the profit of the clean firm as it effectively becomes a monopoly. The latter effect may dominate when the degree of asymmetry is large enough, as Fu et al. (2023) show (the logic in Anand and Giraud-Carrier (2020) is similar: an increase in production costs due to emissions regulation forces firms to reduce overproduction relative to the monopoly level).

With minor deviations, the settings in Anand and Giraud-Carrier (2020) and Fu et al. (2023) are special cases of those in Krass and Ovchinnikov (2022a, b), who naturally also observe that the total industry profits can increase due to taxation, but in the fully heterogeneous industry the profit behavior is more nuanced. The “monopoly” effect is harder to achieve with many firms, and with nonlinear demand it may not be achievable at all. Importantly, at low/zero taxes, the profits are driven by asymmetries in the baseline operating costs, while at high taxes by emission intensities. Thus, as taxes increase, the structure of the equilibrium changes: firms that are not in the equilibrium at low taxes may enter at higher taxes.

Combining these effects, Krass and Ovchinnikov (2022a, b) show that not only can the total profit increase due to taxation, but also the industry-average emissions intensity, which they call per-unit emissions (PUE), can go up. This is certainly a concerning unintended consequence of environmental taxation in competitive settings.

To conclude this discussion, the aforementioned papers studied competitive effects within a polluting industry itself. Interesting extensions exist in considering the supply chain competition. For example, the steel industry is a supplier to the car industry and the construction industry, in both of which firms can adjust how much steel they use vis-à-vis other materials. Then, if better ways to decarbonize the production of steel could exist, then, say for decarbonizing concrete, more steel will be demanded in the construction industry. Despite the price of steel going up (e.g., due to the carbon tax), the demand for it can still grow as it could substitute concrete. In contrast, if aluminum production can be decarbonized further, it would displace steel in the automotive sector, and so on.

12.5 Conclusions

Deep decarbonization is one of the main challenges of our time. The world emits too much carbon into the atmosphere, resulting in unprecedented climate instability that threatens the quality of life for all living things. Decarbonization also presents an enormous opportunity—the so-called carbon arbitrage is estimated to be on the order of a hundred trillion dollars, or about the same as the global annual economic

output. Further, there are no real savings from not decarbonizing: the estimated total cost of mitigation, adaptation, and build-back is essentially the same—less mitigation would imply higher costs of adaptation. Who should carry these costs is contentious and stems big at the current COP27 negotiation: China currently has the highest carbon emissions rate as a result of its large population and large, carbon-intensive economy, yet North America and western Europe are responsible for the biggest cumulative stock of emitted carbon, as they have been carbon-intensive for a longer time.

With this macro-picture in mind, this chapter surveys a technological approach to decarbonization: that is, changing how we do things in ways that emit less carbon, focusing primarily on the two most polluting sectors: energy and industry.

In the past decade, the leveled cost of electricity generated by renewables has rapidly converged with that of fossil fuels, but to ensure that renewables can displace fossil fuels, and overcome intermittency concerns, large-scale investments in energy storage are required. We review the main storage technologies and discuss several recent sustainable operations working papers that analyze the investments in renewable capacity and storage and the operation of the storage.

In industry, we discuss the three-dimensional approach of combining supply-side technologies, demand-side response, and policy instruments, focusing in detail on the main driver of each as identified by experts and the literature. We note that while many studies exist, the work that combines all three is rare, especially in the supply chain context, where nontrivial substitutions could emerge between industries that have different capacities to decarbonize.

Taken together, this chapter offers a comprehensive view of green technology choice for deep decarbonization from the operations and supply chain lens, and we hope it will serve as a valuable reference for scholars and policy makers.

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Chapter 13

Business Implications of Sustainability Practices in Supply Chains



Mark Pagell and Zhaohui Wu

13.1 What Are the Key Questions

This chapter uses a number of examples to detail how a series of path dependent decisions underpins the strategic trajectory of today's leaders and laggards in sustainable supply chain management. A truly sustainable supply chain could, customers willing, operate forever (Pagell and Wu 2009). Such a chain would at a minimum create no harm and might even have positive or regenerative impacts on social and environmental systems while maintaining economic viability (Pagell and Shevchenko 2014). True sustainability is the end goal of SSCM (sustainable supply chain management), a goal which few if any supply chains, especially those with tangible flows, presently meet.

The exemplars presented in this chapter are then leaders when compared to the norms in their industry, but they are not truly sustainable. The first key question even these leaders need to answer is *what has to change in our supply chain to reach true sustainability*. This chapter offers insight, but the leaders of tomorrow will need to build on these examples and those in the rest of the book if they are to survive for multiple generations.

The awareness that today's leaders are far from the end goal of harm free production and distribution is one of two guiding assumptions for this chapter. The second assumption is that the decisions managers make are path dependent. At its simplest path dependency means that previous decisions create constraints for decisions made today (for more information see Pierson 2000). The assumption of

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path dependency indicates that the further a supply chain travels along a strategic path or trajectory the more likely it is to stay on that trajectory because the costs of switching to a different trajectory become ever higher. For example, a supply chain that decides to outsource production has made path dependent decision because it will be difficult if not impossible to bring production back in house in the future. The current geo-political climate finds many firms that outsourced work to China 20 or 30 years ago, looking to re-shore or bring that work back to North American and Europe. However, they are likely to find this task difficult or impossible because in the intervening period, the workers who did those jobs have either taken their skills elsewhere or retired. Even where there is a will to re-shore there is no one to do the work (Cass 2022). Similarly, the initial drivers and values that lead a supply chain to pursue sustainability also create future constraints (Wu and Pagell 2011) and limit future strategic options.

This path dependency also raises questions. For managers in supply chains that have already started to evolve and change to meet the needs of the future the question becomes *how have previous decisions enhanced or limited our options?* For the managers of supply chains just starting on their path to sustainability the questions become *have we waited too long* and if not, *how will the decisions we make today limit our options in the future?* To answer these questions the chapter will cover what we know about how supply chains pursue sustainability, and then detail the four path dependent trajectories that supply chains follow when addressing sustainability; the wait and see trajectory, the balanced trajectory, the focused trajectory and the opportunity first trajectory.

13.2 What Is Known: The How and Why of Sustainability

Regardless of a supply chain's trajectory, managerial efforts to create a more sustainable supply chain typically support three main operational goals; increased efficiency, risk reduction and innovation. Exemplar firms tend to try and achieve all of these goals, and the goals are not mutually exclusive. This section details each of these goals using examples.

One of the main goals of supply chain managers has always been to use resources efficiently to minimize the costs of providing goods and services. This mindset has proven useful in some sustainability related initiatives as well; especially initiatives that attempt to reduce the amount of environmental harm created by the supply chain. Environmental harm tends to come either from using a non-renewable resource or from pollutions that is created in the production, delivery, use or disposal of the chain's goods and services. So, when supply chain managers can reduce the amount of an input needed for production, they reduce the costs of production and the impact on the environment. Similarly, pollution, is in essence wasted resources. For instance, when a firm sends material to the landfill it means it has bought materials that have not been used in production and may even pay a second time for them to be disposed of. Preventing or reducing pollution is often achieved by making

more complete use of inputs or by changing inputs, again reducing waste and environmental impacts. This simultaneous achievement of reduced environmental harm and reduced costs is referred to as *eco-efficiency* (Sharma and Henriques 2005).

One of the more visible and successful eco-efficiency efforts is 3 M's Pollution Prevention Pays programme (3M 2022). This programme is aimed at preventing pollution at the source, rather than paying to clean it up after it is created and was started in 1975. Since then, the company estimates that they have eliminated nearly 2.85 billion short-tons of pollution and saved over 2.34 billion US dollars.

The second way supply chains address sustainability is via *risk reduction* or elimination. Refer to Chap. 19 by Scholten and Stevenson (2024) for more details about supply chain risk and resilience management. Many environmental and social impacts can be viewed as risks, for instance the risk of an industrial accident harming the environment or workers and disrupting the supply chain. Similarly, many supply chains decisions are made to codify, understand or reduce risk. Today, firms face a much broader set of risks as the fundamental assumptions of globalized production networks, consumption patterns and sources of materials and energy are in question (Meichtry and Strasburg 2022).

To address these risks, many firms have codes of conduct for their suppliers. These codes are a form of non-governmental regulation in that they set standards and rules as to what behaviours are unacceptable for suppliers, regardless of local laws. For example, many clothing and footwear supply chains have suffered because of customer boycotts due to the way suppliers treat the environment or their workers (USAS 2022). In response organizations like Nike (2022) and Levi's (2022) have created codes of conduct. In so doing they are trying to mitigate the risks associated with unsustainable actions in the supply chain.

Finally, efforts to become truly sustainable require changing both products and processes. These changes are a form of *innovation* and innovation can also be a source of competitive advantage. For some firms, attempts to make the supply chain sustainable have positive spill over benefits in that the same innovations that reduce environmental and social harm also differentiate the chain from its competitors. For instance, the Broad Group (Broad 2022) has developed innovative products for cooling buildings that do not rely on electricity as well as a complete system for creating more sustainable buildings. The firm differentiates itself by having unique, more sustainable, products and services.

These three interrelated goals; eco-efficiency, risk reduction and innovation are what supply chain managers pursue when making their chains more sustainable. In the next section we detail how the decisions made to improve the chain in one or more of these areas, overtime, create unique path dependent trajectories.

13.3 Findings and Practical Implications: The Four Trajectories

One of the guiding assumptions of this chapter is that it is the series of decisions managers make that determine the supply chain's sustainability trajectory. Every new process, decision to switch to less harmful materials, or redesign of products will not only have operational implications in terms of waste, risk and innovation, but will also affect and be affected by the chain's strategic trajectory.

A supply chain's sustainability strategy can be examined from many perspectives. For instance, the shared value perspective (Porter and Kramer 2011) focuses on how goods and services are tailored to meet the needs of customers while reducing the harm from production and distribution. For example, when Pepsi works to reduce the amount of water used in creating food and beverages, especially in parts of the world where water is scarce (Pepsi 2022), they are creating shared value by allowing customers to continue to enjoy their products while making sure that water is available for more pressing needs.

The strategic perspective taken in this chapter is different, in that we are not focusing on any one decision, product or process. Instead, the focus is on how (if) sustainability was initially operationalized/introduced into the supply chain and how the series of decisions managers make will, overtime, determine how/if true sustainability would develop in that specific supply chain. The decisions managers make, create a path dependent trajectory that has implications for priorities and capabilities, while also setting some limitations on future actions and outcomes.

There are four main trajectories or archetypes of sustainable supply chain management; wait and see, opportunity first, balanced and focused. The trajectories are in essence the way that sustainability is woven into the strategy and culture of the focal firm and its supply chain.

13.3.1 *The Wait and See Trajectory: The Average Firm*

Despite the recent growth in firms making net zero pledges and/or publishing sustainability reports, the reality is that the typical firm or supply chain has not yet really started to address sustainability from a strategic supply chain management perspective. And in many cases, this has been a deliberate strategy of waiting. It is well established that as uncertainty increases, investment decreases (e.g. Baker et al. 2016). Waiting, reduces the risk of choosing the wrong path, requires little to no investment, and does not put current profits or business models at risk. There is a great deal of uncertainty when it comes to sustainability, in terms of future customer expectations, regulatory requirements, and technological advancements. Hence, it is seemingly rational for a currently profitable firm to 'wait and see' before investing in making real changes to make their supply chain more sustainable (e.g. Shevchenko et al. 2016).

At a societal level, waiting is making the need for change greater. Waiting also creates its own path dependencies for firms and industries, and today this trajectory is just as risky, if not riskier than one of the other trajectories. The risks and costs associated with waiting are probably clearest in the auto-industry, which has known since the 1970s that eliminating the reliance on fossil fuels would one day be needed.

Yet it took the entrance of Tesla into the market and the DieselGate scandal to get most auto companies to start to seriously consider electrification; even though the technology has existed for over a century. The end result is that companies such as Volkswagen are investing incredible sums (180 billion for VW alone—Walt 2022) to, in essence, catch up with Tesla. From a strategic standpoint Tesla has a significant first mover advantage.

But Tesla also has a supply chain management advantage that is equally valuable and frequently overlooked. Almost every other company trying to catch Tesla is currently building a new electric vehicle supply chain. But they are also managing their existing traditional supply chain, which they will need to somehow extract profits from, while managing it out of business. Managing Tesla's supply chain is then much simpler. Plus, Tesla does not have the unenviable (if even possible) task of asking suppliers who provide parts for combustion engines to lower prices as volumes are dropping; when suppliers know the end result will be that the customer no longer uses them as suppliers. Waiting to electrify, increased short-term profitability, but likely made long-term survival harder for most car companies.

The same dynamic is playing out in many other industries as well. For instance, in Ireland agriculture is responsible for nearly 40% of the country's emissions (EPA 2022). Irish agriculture is also highly dependent on meat and dairy. Irish cow numbers continue to grow even though it is well known that ruminant meat and to a lesser extent dairy, are sources of both environmental and social harm (e.g. Willett et al. 2019). In other words, the Irish meat and dairy sectors made a series of decision to increase production while knowing their industry would eventually have to radically innovate and or significantly shirk. And they did so in the face of declining global demand for meat, which has led to farmers blockading meat plants because of the low prices on offer for cows (Hutton 2019).

The Irish meat and dairy industry grew instead of shirking and did not significantly invest in their suppliers or innovation. Now they are facing a crisis as they are being asked to contribute their fair share to Ireland's commitments to the Paris Agreement to significantly reduce Ireland's emissions. Yet doing so to the level needed is viewed as an existential crisis by the industry which is now looking for government supports to survive (IFA 2022). The decisions made in the industry not only harmed the key suppliers (the farmers) now the entire industry is at risk. Once more, decisions on how to address sustainability in the past have created constraints today.

There is no denying that the sustainability transition entails a great deal of uncertainty, which explains the reluctance of most firms to seriously address making their supply chain sustainable. Yet waiting creates its own path dependencies and entails its own risks and costs, which are ever increasing.

In supply chains on a *balanced* trajectory sustainable behaviour is the way of going about one's business day in and day out; and it positively and directly benefits the employees, suppliers and local communities. As a result, environmental and social issues are highly integrated and equally important in these supply chains.

Strategically, these supply chains set limits on growth based on access to both environmental and social resources. Typically, these chains would have much more stable patterns of sales than their overall industry. In times of soaring demand they will not increase production beyond what their natural and human resources could maintain in a downturn. This means they miss out on some sales in times of plenty, but they do not deplete natural resources nor do they have to fire people in downturns. Hence, they miss some profit and growth opportunities, at least in the short term that the other chains would likely capitalize on. Supply chains on a balanced trajectory are willing to internalize some of the sustainability costs that are not presently mandated by existing regulations to provide long-term benefits to employees, suppliers, and the communities in which they operate.

Operationally, the balanced approach means that these chains are well placed to be both eco-efficient and innovative. The stability they foster means that workers are a long-term investment. These workers are highly trained and motivated to make both continuous improvements in operations as well as to support innovations when needed. Critically they are willing and able to share their knowledge through the supply chain. Balanced firms can significantly reduce the impact of production processes (theirs and suppliers), while increasing efficiency and offering better working conditions for employees and overseas suppliers. These organizations can attain resource efficiency in many areas of their supply chains.

A leading company on a balanced trajectory is The Collins Companies (Fig. 13.1), a forest products enterprise headquartered in Wilsonville, Oregon. It owns and manages more than 311,000 acres of Forest Stewardship Council (FSC) certified timberland in the United States. While most other wood product companies maximize yield by clear-cutting, Collins lets its forests grow naturally and uses selective harvesting. Since its creation in 1855, it has maintained ethical forestry practices that have allowed the forests to thrive—and provide lumber—for over 150 years. Their overriding operating principle is never log more than the forest grows.

Collins was the first privately owned wood-products company in the U.S. to be certified by the Forest Stewardship Council (FSC), an international organization with membership that includes NGOs, logging and wood-product companies, and environmental groups. Throughout its operations, Collins is committed to maintain the forest ecosystem and support social and economic benefits for its employees and the surrounding communities.

The company's strong stance on environmental and social priorities subjects it to risks and trade-offs. Collins will never use all of their resources—there will always be trees in reserve, but they may not always be able to meet demand. And while FSC certification raises their profile, it is expensive to maintain those standards and it may not repay those costs: lumber and building supplies are considered commodity items, and most builders buy based on price, not environmental ideals.



Fig. 13.1 An example of the balanced trajectory, The Collins Company (<http://www.collinsco.com>). Image from: <https://www.linkedin.com/company/the-collins-companies>

The company's balanced approach to profit, people and planet may limit growth, but the company factors its environmental and social involvement into its business plan. Although the industry has declined in recent years, Collins has maintained a steady work force.

The steady workforce is critical to Collins and the communities they operate in. Forestry is by nature a highly cyclical business with frequent downturns. Many of their facilities are in rural settings where Collins may be the only major employer. If Collins laid off people in every downturn, some small towns would not survive. Stable work then keeps people employed and allows small rural towns to survive industry downturns. Balance allows Collins to protect their workers and the communities they operate in and is a key to their ability to leverage the workforce.

Collins has leveraged the workforce to address sustainability mainly from an eco-efficiency and innovation standpoint, which is exemplified in the journey their Klamath Falls facility (Oregon, U.S.) has made since the mid 1990s. Collins purchased the facility from Weyerhaeuser. Until that point, Collins had concentrated its sustainability efforts on forest stewardship. However, in 1996 they implemented a programme they called Journey to Sustainability (JTS) to bring sustainability to their manufacturing operations. By the next year, employees had begun sustainability training. Morale at Klamath Falls had dropped before the purchase by Collins, and the company used JTS to boost the mood in the plant while improving operations. Supervisors received training in the principles of sustainability and sustainable manufacturing. The company implemented a plan to train all the workers within three months. The company solicited suggestions while assessing improvements to its capital equipment.

One of the first suggestions to arise was to adopt European standards for off-gassing of formaldehyde from finished products (three times as stringent as U.S. requirements). The company believes this has enhanced its position among "enlightened" architects, as well as opening potential new markets for its products.

The company stopped using water from the local river to improve resource efficiency and address regional water shortages. Their initial decision was to treat water and reuse it. They leased a nearby farm and built a wetland to store and treat

the discharge, but it was too small to hold all of the treated water, forcing them to find a way to use some of it immediately. They realized that they could cut electricity use by modifying their heat exchange and cooling systems to use the extra water. The initial decision to stop taking water forced them to re-engineer production. The new process not only uses less water, but also requires less energy to run, freeing up capital. The experience also bolstered morale as workers saw immediate benefits in terms of efficiency and in their community's water supply.

A capital project at the plant significantly reduced power use: they replaced six old electric motors, saving \$118,000 annually. It paid for itself in two years. Other projects were similarly eco-efficient. Condensate from a veneer dryer was used to heat water, saving \$152,000 per year. Maintenance and repair to steam traps saved \$25,000 per year. New equipment allowed sander dust to be incorporated into the finished particleboard. The process saves \$563,000 per year—and using the dust not only improves the appearance of the board, but reduces air emissions.

The Klamath Falls facility not only cut water and energy consumption, it reduced waste, as well. A contest to promote water conservation prompted discovery and repair of a leak, saving over 500,000 gallons per year. The plant no longer discharges warm water into the river. The Journey to Sustainability has now been incorporated into operations throughout the company.

In its first year, Journey to Sustainability projects saved The Collins Companies almost \$1,000,000. Within three years, annual savings were \$1,370,000—totalling over \$three million. Capital purchases, salary increases, seminars, and travel came to about \$50,000—six per cent of the overall savings. The environmental benefits come from reduced use of resources, especially water in a community that faces water shortages. This is also a social benefit in that will allow them to continue production without putting other water uses in the community at risk.

The path dependent decisions Collins has made over the last 150 years allow them to compete by being more sustainable, but these decisions also create constraints. For instance, the stability of their forestry practices combined with the investments in workers and communities means that operationally they are innovative and productive but also not very flexible. Their supply chain is not able to deal with large fluctuations in demand. And given the investments in workers, practices and communities any attempt to alter where work is done or who does it will have to be viewed through the lens of protecting the existing workforce and community. Intuitively this means they never want to close a plant in small town, even if there are no legal constraints on doing so. Instead, the constraint is the intangible. To close a plant in a small town means breaking the bond with the workers and community and undermining their entire philosophy.

A balanced trajectory focuses on conserving social, environmental and economic resources. Decisions to behave in this way often mean higher short-term costs, whether they be investments in FSC or in training the workforce. However, the payoff is the ability to continue to operate over very long time frames. Collins is not going to run out of inputs with their present business model, nor do they have to worry about a dip in demand causing an entire community to collapse.

Many other companies also integrate social and environmental sustainability into a coherent brand positioning and corporate strategy. This integration is easier to communicate to stakeholders and customers when the products are consumer products. For example, all-natural cleaning products convey a firm's concern for the health of the land, biodiversity, and the well-being of consumers, workers and the wider community.

Supply chains on a focused trajectory have capitalized on either environmental or social issues to create a viable business. For them, business success is contingent on the accomplishment of environmental or social objectives. In supply chains on an *environmentally focused* trajectory the founders and managers will have strong environmental values that have been imprinted on the organization. For instance, Patagonia, a privately owned apparel company whose founders are self-proclaimed climbers and surfers recently declared that *Earth is our only shareholder* (Patagonia 2022). These supply chains are a means to carry out a particular environmental agenda. While these companies are engaged in community service and do discuss social issues, tackling environmental issues is an essential part of business operations; dealing with social issues is secondary.

The *socially focused* trajectory is similar but in these supply chains the values are socially oriented. Supply chains that have adopted Fairtrade certification (Fairtrade 2022) as a primary means of certifying themselves or for selecting suppliers would be on this trajectory since the primary objective of Fairtrade is to improve the lives of farmers and farm workers. Socially focused organizations take environmental action when the environmental impact of their business on people becomes a concern. In socially focused organizations environmental initiatives are adopted as a reaction to protect the organization while in environmentally focused organizations social initiatives will be adopted reactively.

The major risk from these trajectories is then the lack of balance, but this can also be a source of strategic strength. The environmental features of the product and service offerings from environmentally focused supply chains communicate certain environmental messages; energy efficiency, reduced carbon footprint via local food, and reductions in pollution associated with chemical fertilizers. Meanwhile, their innovative solutions to environmental challenges benefit the customers directly via reduced energy consumption, a healthy living environment, and high quality products. Similarly, socially focused firms communicate messages such as those associated with fair trade (Fair-trade 2022) or conflict free (CFSI 2022) and are innovating in ways that benefit society in multiple ways such as better working or living conditions. Thus, companies with a focused trajectory can charge a price premium.

Since the price premiums can offset the higher costs associated with social/environmental practices, managers are emboldened to take on more difficult tasks. From the triple-bottom-line perspective of performance, while all show a concern for all elements of the triple bottom line environmentally focused firms are primarily motivated by a concern for the environment while socially focused firms are primarily motivated by a concern for society.



Fig. 13.2 An example of the focused trajectory, Kettle Food (<http://www.kettlebrand.com>). Image from: http://www.kettlebrand.com/about_us/sustainability/

A leading company with an environmentally focused trajectory is Kettle Foods (Fig. 13.2); a multinational producer and distributor of organic and all-natural snack foods. Kettle began in 1978 as the N.S. Khalsa Company, a small wholesaler of natural snacks in Salem, Oregon. The company started producing its Kettle brand potato chips in 1982. By the time the product line was a year old, the company had grown to \$three million in annual sales. Founder Cameron Healy started the business in an effort to earn a living that fit with his spiritual values; those values dictated the “all-natural” approach that still distinguishes the brand and underlies the company’s strong environmental policies. By 1987, the Khalsa Company was growing at a 15% annual rate, with sales doubling every year. The following year, it changed its name to Kettle Foods and established a UK branch in a converted factory in Norwich.

Kettle adopted an overarching operating principle that guides every decision the company makes: the product has to be all natural and must contain what the label says it contains. The company equates this principle with maintaining a strong environmental policy, so it influences how the company designs products, selects suppliers and tracks ingredients in its supply chain. Strategically, they address sustainability from an innovation perspective because their environmental focus allows them to sell innovative/differentiated products at a price premium.

Kettle’s environmental focus also involves trade-offs. Some of their organic ingredients are seasonal, but most grocery retailers do not stock products on a seasonal basis. Therefore, perishable ingredients have to be stored in refrigerated warehouses to maintain a steady supply. The company’s environmental decisions have to be pragmatic, keeping sight of the fact that they must be both economically

and environmentally sustainable. This often means putting profits first, at least in the short term. Nonetheless, Kettle addressed environmental issues in their initial business plan, so their business and environmental agendas reinforce each other.

The environmental focus plays out operationally via efforts at creating eco-efficiency and at reducing risk. Kettle has a supplier certification programme that demands extensive information from suppliers to ensure ingredients meet their standards. This is explicitly a risk reduction activity since ingredients that do not meet these standards would put their competitive advantage at risk. They do not trace money through the supply chain or determine other practices (labour conditions, for example) that are not directly related to a product's all-natural status. They choose to focus on environmental issues and concerns.

The environmental ethos extends to their downstream and upstream supply chains. As part of the commitment Kettle uses sunflower and safflower oils. When the oil is spent, it is converted into biodiesel. By using the fuel in its delivery fleet, the company reduces its CO₂ emissions by 8 tons every year and does not have to purchase fuel, which is a classic example of eco-efficiency. Kettle has taken other steps to reduce emissions. In September 2003, they partnered with the Energy Trust of Oregon and Portland General Electric to install one of the largest grid-tied solar photovoltaic arrays in the Pacific Northwest. Six hundred rooftop solar panels generate more than 120,000 kWh of electricity per year, enough to reduce annual CO₂ emissions by 65 tons. The company offsets the rest of its electricity use in the U.S. by purchasing wind energy credits, preventing more than 13,000 tons of emissions. Citing these achievements, the EPA presented its Green Power Partnership award to Kettle in 2012. The Salem plant has also restored surrounding wetlands, clearing invasive plants and reintroducing native species. As birds and other wildlife returned, the company installed pathways and benches to encourage the public's use and enjoyment of the land.

In 2007, Kettle opened the first LEED® Gold certified food manufacturing plant in the U.S. in Beloit, Wisconsin. The "green building" reduces energy use by 20% compared to conventional construction, saving \$110,000 on natural gas and \$51,000 on electricity. By reclaiming and reusing water, the plant saves over 3.4 million gallons of water a year. The facility also converts 3200 gallons of waste oil to biodiesel each month. Finally, elimination of shipping lines between Oregon and the Midwest further cut the company's CO₂ emissions by more than three million pounds per year.

The decision to protect the firm's environmental values via the all-natural principle has allowed Kettle to grow while selling products at a price premium. But this same path dependent decision also creates serious constraints. At its simplest Kettle cannot use some ingredients even if they are less expensive or have a smaller carbon footprint. It also means that to sell into markets where customers expect a product to be available year round that Kettle ends up storing ingredients, often in temperature controlled settings with large costs and environmental footprints. These actions are antithetical to the firm's environmental ethos, so it is no surprise that they work hard to mitigate all of their environmental impacts. However, like most firms their resources are not unlimited. The efforts they have put into their operations have

created a unique environmental management capability. But every investment in this environmental capability is also a decision not to make a similar sized investment in social sustainability capabilities. The decision to embrace an all-natural principle to protect the environmental values lead to the creation of a proactive environmental management capability which they use to create and maintain their competitive advantage. However, every investment in the all-natural path also makes it harder to embrace/invest in other areas due to resource constraints and knowledge gaps.

Kettle exemplifies the focused trajectory. The organization was founded to meet an environmental objective which translates into their all-natural strategy; a strategy which allows them to charge a price premium which helps to pay for their range of eco-innovations. And over time they have become very adept at recognizing opportunities to make environmental innovations, be it converting a waste stream into fuel or harnessing renewable energy. Similarly, their risk reduction efforts are aimed mainly at protecting their trajectory. This focus has been profitable and allowed them to grow but it is not balanced. They do address social issues, but this is not their strategic advantage. Overtime their focus on the environmental impacts of the supply chain has made the chain an exemplar in reducing its environmental impact, but this focus requires resources, resources that are not applied to social issues in the same manner.

The focused trajectory has a clear mission and creates a strong path dependency. Either social or environmental focus will dictate a set of supply chain and operations norms and practices, production scale, and corresponding characteristics of members along the supply chain including suppliers, distributors, secondary processor and wholesalers. In other words, path dependency also creates the lock-in effect—which can create friction and conflict when companies want to change their business strategy. For example, when Benn and Jerry's ice cream was bought by Unilever, the purchase agreement allowed Benn and Jerry's to maintain independent management, in an attempt to protect the brand's unique focused sustainability attributes. The limits to this agreement were laid bare when Ben and Jerry's sued Unilever (again their parent company) over licensing agreements Unilever had signed for the brand (Stemple 2022).

The *opportunity first* trajectory is largely driven by an economic opportunity and growing stakeholder demand. Unlike the previous two trajectories, where environmental or social values play a leading role in shaping the supply chain's operations, here the causality is reversed. Opportunity first trajectories occur when a business motive pushes an existing chain to adopt additional initiatives that will inculcate environmental or social values into the organization.

For existing supply chains this is the most common trajectory. Organizations such as Coca-Cola, Unilever and GE, that existed for generations with a mainly profit maximizing ethos, have made business decisions to become more sustainable that have put them onto the opportunity first trajectory. The opportunity first trajectory enjoys many of the marketing and branding benefits of the focused trajectory, but these supply chains are not able to charge the same price premiums.

The opportunity first trajectory is different on three main accounts, all of which exemplify the path dependent nature of decisions surrounding sustainability. First,

the sustainability values in this trajectory are relatively recent, which suggests that when faced with decisions where being more sustainable is expensive or risky, that these supply chains are less likely to make the investment. This is buttressed by a second key difference: the opportunity first supply chain was already in the market place as a traditional (not sustainable) competitor. Customers already associate the organization's products and services with specific attributes including prices, and these customer expectations will be slow to change, which can limit the ability to charge a price premium, especially in the short term. Prior decisions create dependencies that influence both how current investments are viewed and interestingly how customers view the supply chain.

Finally, over time customers, employees and other stake-holders of the opportunity first trajectory will come to expect more responsible behaviour. What starts as a business opportunity will have to evolve to something more because stakeholders expect that a supply chain that claims to be more responsible is truly responsible in all of its actions. By announcing their intentions to take advantage of a sustainable business opportunity, supply chains are opening themselves up to much greater scrutiny and higher expectations. Taking the first step towards becoming more sustainable creates a path dependency where the supply chain will have to take many more such steps to maintain support from their stakeholders.

Workers in a company or chain tend to be motivated when the chain becomes more responsible, but they also begin to question why social or environmental issues that make the firm money are addressed while other negative impacts are not. Similarly, customers who respond to a more sustainable message will begin to ask deeper questions about other practices, demanding ever more change. Protecting and growing the brand effectively means that chains on an opportunity first trajectory will likely have to migrate toward the *balanced trajectory*, otherwise they may fail. This is a critical point since the opportunity first trajectory is the one on which most existing firms will address sustainability.

Demands in the marketplace or competitors' strategy can open sustainability opportunities. When Lafarge, one of the largest cement producers in the world, entered the Chinese market in the 1990s, it leveraged its relatively cleaner production process for producing cement. By the 2000s, as Chinese real estate developers began to promote green buildings, Lafarge began to leverage its expertise and started to make energy-saving prefabricated insulated wall, floor and window products. These innovations created a service-oriented business model, a path-dependent choice created by the rapidly developing economy of the time.

Walmart exemplifies the opportunity first trajectory (Fig. 13.3). With approximately 10,500 stores in 24 countries and a global workforce of around 2.3 million, Walmart is the largest retailer in the world. When Walmart undertakes a change that affects its supply chain(s), that change influences industry sectors globally. So it is with the company's sustainability policies. In 2005, CEO Lee Scott announced the company's new business sustainability strategy, designed to meet three sweeping environmental goals: to be powered by 100% renewable energy, to create zero waste and to sell products that "sustain people and the environment". While these were



Fig. 13.3 An example of the opportunity first trajectory, Walmart (<http://www.walmart.com>). Image from: <http://news.walmart.com/news-archive/2010/11/16/walmart-canada-opens-its-first-sustainable-distribution-centre>

“aspirational” rather than concrete goals with definite timelines, their achievement would require major changes in how Walmart managed its global supply chains.

The first two goals (renewable energy and zero waste) are very much about eco-efficiency and innovations in supply chain operations, which align well with Walmart’s existing low price strategy that is based on supply chain efficiencies. Walmart closely tracks numbers across its supply and distribution channels as well as in its retail centres, so it has statistics on its progress toward its measurable goals. At the beginning of 2014, it had over 300 renewable energy projects underway worldwide, providing 2.2 billion kilowatt hours (kWh) a year. Additional purchases of renewable power bring the company to 24.2% of its energy goal. In the U.S., more than 81% of their waste is now diverted from landfills. In Japan and the U.K., the number is over 90%. The company offers electronics recycling at its U.S. stores, along with smartphone and tablet trade-in programmes. While all these projects benefit the environment, they also cut operating costs in terms of energy expenditures and landfill fees. In addition, recycled materials provide new revenue streams—including 56 million pounds of recovered cooking oil used in biodiesel or animal feed.

Energy and waste issues are business opportunities where innovative processes allow them to become more eco-efficient and these initiatives extend to the entire supply chain. In 2008, Scott announced a drive to bring suppliers into compliance with fair labour practices and cut energy consumption by 20% throughout the supply

chain. He also stated that the company would drop suppliers who did not meet these goals.

Walmart offers a range of educational and training programmes to suppliers to help them to meet these targets. Violation Correction Training requires a factory representative to attend classes where approaches to social and environmental issues and compliance methodologies are explained. The suppliers are expected to choose and implement the procedures that fit them. Orange School provides hands-on training to select factories and suppliers. By focusing on the fundamentals of root cause analysis and procedures, Walmart teaches suppliers to identify non-compliance issues proactively and develop systematic approaches to continuous improvement. Walmart has invested heavily in these training programmes, but the onus remains on the suppliers to comply.

Walmart provides a good example of both positives and limits of the opportunity first trajectory. As Walmart becomes more eco-efficient, other issues become more not less pertinent to external stakeholders. The company would of course like to focus on positives like their reduced footprint, decreased reliance on non-renewable energy and everyday low prices. But external stakeholders have raised a litany of serious complaints about the firm; some of which have resulted in legal action.

For instance, Walmart is Mexico's largest private employer, with over 220,000 workers and 2200 locations. Walmart has more than once resorted to bribery to obtain building permits in Mexico—in one case, to open a store just a mile from a culturally significant site: Teotihuacán, an ancient Aztec that contains a temple complex and two pyramids (Bloomberg 2014; NYT 2012, 2014). And in the United States, the company is harshly criticized for its anti-union policies, poor working conditions, inadequate health care and low wages (Edgecliffe-Johnson 2022). Some of these issues arose long before Walmart started on their path to becoming more sustainable, but these issues become more pertinent as the organization tries to capitalize on its sustainability progress.

Walmart has made significant progress and somewhat paradoxically now faces even more pressure to change. They can't have it both ways—if they are truly trying to reduce the impact of their supply chain, they will have to evolve into one of the other trajectories—or lose trust from multiple stakeholders (suppliers, customers, regulators, etc.) as well as sales. In a sense Walmart will soon face a crossroad; they have been very successful at making their existing unsustainable business model more resource efficient, but in so doing they have actually increased their exposure. To be truly sustainable they will have to continue to learn and respond to pressures to make changes that may not create shared value.

The path dependent nature of sustainability decision is most apparent in the opportunity first supply chains. By capitalizing on an opportunity, Walmart, like all supply chains on an opportunity first trajectory has also created new constraints and expectations. The choices Walmart and other opportunity first organizations have made have helped to start to address some of the serious sustainability issues created by supply chains. Yet these activities have done nothing to “quell accusations of ‘greenwashing’, ‘woke-washing’, hypocrisy, and virtue signaling” (Edgecliffe-Johnson 2022). This is the fundamental paradox of the opportunity first supply

chain, the more they do and the more transparent they become the more that will be expected.

Supply chains on an opportunity first trajectory will be indispensable in sustainability for two reasons. First, sustainability has to make sense for companies with such scale to have a significant impact. Second, this is the way the vast majority of existing, presently unsustainable, supply chains will first address sustainability. But, opportunity first organizations also need to recognize that over time they will have to do more than embrace the business opportunity, they will have to fundamentally change.

13.4 Conclusions

This chapter explored the various ways in which supply chains are or could approach sustainability. It starts with two critical pieces of knowledge. First, that no firm has yet created a truly sustainable supply chain and second, that the choices made to reduce a chain's impacts will create path dependencies.

This starting point leads to a pair of questions all managers will need to ask. The first is "*what has to change in our supply chain to reach true sustainability*". Supply chain managers answering this question focus on three interrelated means to reducing impacts; eco-efficiency, risk reduction and innovation. As the chapter has shown many firms are making significant progress in these areas and these examples offer starting points for managers looking for inspiration.

The second question is "*how have previous decisions enhanced or limited our options*". Managers looking to begin the journey toward sustainability or those looking to make further progress in their supply chains need to remain aware of the way past decisions limit future options. The chapter explores the four main paths or trajectories in which sustainability is embedded into a supply chain, and the trade-offs inherent in each. Balanced chains are just that, but by focusing equally on all aspects of the triple bottom line they risk not having the market advantages of focused chains, which are truly leaders at something, even if focused chains are also laggards in other areas. Similarly, chains that use a market opportunity as the initial spur to become more sustainable will, somewhat paradoxically need to be aware that as they make progress more, not less will be demanded of them. Finally, firms who have been waiting, may have made the biggest sustainability trade-off of all. By waiting and seeing they reduced their short-term risks and costs, but potentially at the cost of their own long-term survival.

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Chapter 14

Moving from a Product-Based Economy to a Service-Based Economy for a More Sustainable Future



Ioannis Bellos, Hang Ren, and Mark Ferguson

14.1 Introduction

Traditionally, economic growth and prosperity have been linked with the availability, production, and distribution of tangible goods as well as the ability of consumers to acquire such goods. Early evidence regarding this connection dates back to Adam Smith's *Wealth of Nations* (1776), in which any activity not resulting in the production of a tangible good is characterized as "unproductive of any value." Since then, this coupling of economic value and material production has been prevalent in both developed and developing economies throughout the world.

One unintended consequence of this coupling has been the exponential increase in the amount of solid waste being generated. The reason is that any production and consumption of material goods eventually generate the equivalent amount of (or even more) waste. Exacerbating this problem is the fact that, with today's manufacturing and supply chain management technologies, it has become cheaper to dispose and replace most products rather than to repair and reuse them. This has given rise to what some call a "disposable society."

To put things in perspective: In 2018, households in the U.K. generated approximately 26 million tons of waste, which amounted to 397 kg of waste generated per person (Department for Environment, Food & Rural Affairs 2022). During the same time period, households in the U.S.A. generated 292 million tons of waste, which is equivalent to a person generating approximately 4.9 kg of waste every day

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(U.S. Environmental Protection Agency 2020). Out of these 292 million tons of total waste generated, approximately 20% of the discarded items were categorized as durable goods. The disposal of durable goods is particularly worrisome because they are typically produced using non-renewable resources such as iron, minerals, and petroleum-based raw materials.

It is clear that any business model that relies on a disposable society cannot be a sustainable long-term model. For this reason, recent efforts have been made by organizations (e.g., [Ellen MacArthur Foundation](#)) and policy makers (European Parliament 2022) to enable the creation of a “circular economy,” which minimizes waste by re-using, repairing, refurbishing, and recycling materials and products (see Chap. 17). While doing so can minimize the amount of waste produced, to promote a more efficient and ecofriendly use of overall resources also requires reducing consumption, which results in decreased production and less waste generated in the first place.

One solution to achieve fundamental reductions in resource consumption, proposed by U.S. Environmental Protection Agency (2009), is for traditional product-based firms to move toward more service-based business models that do not focus on selling products but rather on selling the solutions that the products can deliver. A potential benefit of such models is that a firm can always maintain the ownership of the products it manufactures, thus reducing the incentive to increase revenue by simply selling more products. Moreover, switching the base of the transaction, from the product level to the use level, can restructure the economics of consumption and encourage a more sustainable level of product use. In this chapter, we examine the potential of such a solution from both an economic and environmental perspective.

The rest of this chapter is organized as follows. In Sect. 14.2, we first provide examples of firms that have transitioned from a product-based state to a service-based state, and then we introduce the concept of servicizing. In Sect. 14.3, we discuss some reasons why traditional product-based firms may be interested in transitioning to a more service-based delivery strategy and how this transition may impact a firm’s sustainability metrics. In Sect. 14.4, we discuss some of the challenges firms may face when making this transition and some of the key decisions that are required for doing so. We conclude in Sect. 14.5 and provide some possible directions for future research.

14.2 From Products to Services

In this section, we describe how certain companies have made the transition from being product-based to being service-based. We also link this transition to the practice of servicizing.

14.2.1 Leaving the “Product Comfort Zone”

The service sector has been an integral part of economic activity in both developed and developing economies. In 2020, services accounted for approximately 80% of GDP in the U.S.A. (The World Bank 2022b). During the same period, the service sector accounted for approximately 66% of GDP in the European Union (The World Bank 2022a). Although such statistics are commonly used in texts describing the role of services in today’s economy, another observation may help put a more interesting “face” on these numbers. In particular, for many years, organizations that are (or were) typically recognized as manufacturing/product firms have been increasing the service component of their offerings. Examples include BMW’s ConnectedDrive subscription services for accessing software features such as traffic camera and/or on-street parking information (BMW USA 2022), as well as activating hardware functionalities such as front seat heating and automatic high beam control (BMW UK 2022). Caterpillar, the world’s largest construction-equipment manufacturer, now offers subscription-based diagnostic and predictive maintenance services through its Cat Connect Subscriptions (Caterpillar 2022). Dell and Hewlett-Packard created Dell Services and HP Services, respectively, in order to provide IT and business services. These moves came in response to earlier moves by IBM, which has transformed itself from a product-based company into a services company by selling its personal computers and servers business to Lenovo (Lenovo 2005; IBM 2014).

But why do manufacturers choose to leave their “product comfort zone” to engage and invest in the creation of services? In the face of product commoditization and increasing competitive pressure, services can offer a stable and recurring revenue stream stemming from activities such as support, maintenance, and repair, which can extend well beyond the useful life of a product and generate lucrative profit margins. This can also increase the chances of cross-selling or alleviate customers’ hesitation in upgrading to new products and equipment. Additionally, by offering after-sales support services, manufacturers can eliminate the need for third parties, who can erode brand perception, to perform these services. Furthermore, services are more difficult to reverse-engineer (i.e., to be imitated) and can facilitate longer and deeper relationships with customers due to longer contractual agreements. Manufacturers can also obtain a better idea about the customers’ needs and the conditions under which the products operate and possibly customize their offerings accordingly. Finally, by offering services the manufacturers can acquire more accurate feedback about the performance of their products in the field, which can lead to product improvements and redesigns.

14.2.2 *The Strategy of Servicizing*

In the aforementioned examples, regardless of whether the manufacturers offer a support (e.g., maintenance) or an “add-on” (e.g., BMW’s ConnectedDrive) service, the base of the transaction remains at the product level. This is because the service is contingent on the customers purchasing a product. However, in recent years, we have observed a trend where manufacturers from various industries have started switching their focus from selling products to actually selling the solutions that their products offer. In most of the cases, manufacturers maintain the ownership of the products; therefore, the base of the transaction no longer seems to be the product per se, but rather the use of the product. This trend, also known as “servicizing,”¹ indicates the transition from a product-based economy to a functional or solutions economy (Stahel 1994). In such an economy, instead of buying chemicals, for example, customers now have to option to buy chemical management services; instead of buying copiers, they can buy document reproduction services; and instead of buying cars, they can buy mobility services. Such options can be particularly attractive to customers because they free them from the administrative hassles and the operating risks (e.g., maintenance, repair) as well as the financial risks (e.g., depreciation of a product’s market value) associated with product ownership.

The strategy of servicizing involves the transition from an existing business model to one where the focus is on the service or, better put, the solution provided to the end customer. In fact, most business models can be thought of as comprising a combination of a product and a service component (see Fig. 14.1). The academic literature often describes these models under the general term Product–Service Systems (PSS). The definition of PSS varies throughout the literature, but most authors agree that there exist three different PSS types (U.S. Environmental Protection Agency 2009): (i) product-oriented PSS in which a manufacturer, in addition to selling a product, includes extra services (e.g., after-sales service), (ii) use-oriented PSS in which the manufacturer maintains ownership of the product and sells the use or availability of it (e.g., leasing, rentals, car sharing), and (iii) result-oriented PSS in which the manufacturer and the customer agree on a certain result or performance level (e.g., engine up-time).

The above typology is by no means exhaustive, and some business models may fall under more than one type. For instance, the business model of fractional ownership (often found in the private jet industry; NetJets was among the first to offer such a model) where customers own only a fraction of an asset that entitles them to a certain amount of use can be thought of as a combination of product-oriented and use-oriented PSS. In this chapter, we avoid possible categorization conundrums and simply describe models in which the use, rather than the ownership, of the products governs the relationship between manufacturers and customers. To avoid

¹ There is an abundance of similar terms like “servicization” or “servitization.” We have chosen to use the term “servicizing” like many before us, without making any claims about its superiority or grammatical correctness!

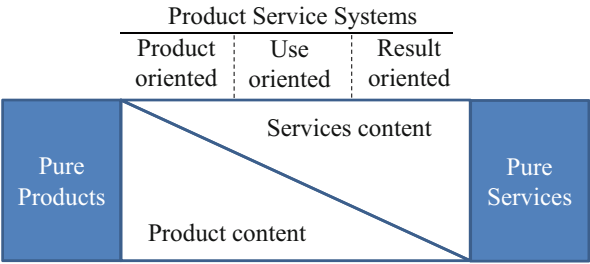


Fig. 14.1 Product–service spectrum and the different types of product–service systems (adapted from Tukker 2004)

an overly cumbersome use of terminology and unnecessary confusion, we refer to such models as *servicizing business models*.

One of the most celebrated examples of a successful *servicizing business model* in the academic literature is Xerox’s document management services (Xerox 2014), where customers are charged on a per-use (per-page-printed) basis (see Fischer et al. 2012). In the automotive sector, *servicizing* has emerged through the business model of car sharing. Under car sharing, customers obtain access to a fleet of vehicles after paying a small annual fee. Upon joining the service, customers can check the availability of the vehicles and make reservations online. After using a vehicle, they pay only for the reserved amount of time. Although the most well-known car sharing provider in the U.S. market is Zipcar, several auto manufacturers such as Volvo, Nissan, and Porsche have entered the car-*servicizing* arena by offering subscription programs, through which customers can gain access to a fleet of vehicles from the manufacturer at a monthly fee that usually also includes the cost of maintenance, insurance, and other services related to vehicle ownership.²

An example of a *servicizing model* in the aviation sector is offered by Rolls-Royce, which pioneered *Power-by-the-Hour* maintenance agreements under which it charges customers based only on the actual flying hours of the engines. Contracts such as *Power-by-the-Hour* are said to align incentives of both the customer and the provider, since both firms benefit from reducing downtime. In a similar spirit, General Electric’s *Predix* system offers remote sensing through IoT devices and analytics capabilities for industrial equipment (e.g., wind turbines) to optimize usage and share the gains from efficiency improvements with customers using outcome-based contracts. Such performance-based contracting has also gained traction in the solar energy sector, where solar panel companies maintain ownership of solar panels and charge customers a per-unit price for each unit of electricity generated (Agrawal et al. 2022). For more examples of *servicizing business models*, we refer the reader to Rothenberg (2007), U.S. Environmental Protection Agency

² Auto manufacturers including BMW, General Motors, and Ford used to offer car sharing services (most notably Car2Go, a subsidiary of Mercedes-Benz Group AG) but have all shutdown operations in the U.S.A.; more on that in Sect. 14.4.1.

(2009), Agrawal and Bellos (2016), Örsdemir et al. (2019), Agrawal and Yücel (2021), and Kim et al. (2022).

Researchers have argued that servicizing business models can lead to higher profitability. In particular, they have proposed that by focusing on the final customer needs and delivering integrated solutions fulfilling these needs, companies can improve their positions in the value chain, enhance the value of their offerings, and improve their innovation potentials (Wise and Baumgartner 1999; Sawhney et al. 2003). Such arguments are in line with our previous mention of the economic benefits of more service-oriented business practices. Besides the potential economic benefits that more service-oriented business models may hold, recently researchers and environmental groups have argued that servicizing business models may also be associated with environmental benefits and, therefore, can be positioned to support the objective of sustainable development (Rothenberg 2007). Throughout the rest of this chapter, we explore some of the rationale for this claim.

In this section, we formally introduced the concept of servicizing and presented some of its most successful implementations in practice. We concluded by mentioning the support that servicizing has received as a potentially win-win (i.e., economically and environmentally superior) business strategy. In the next section, we further explore some of the arguments regarding the economic and environmental potential of servicizing business models.

14.3 The Economic and Environmental Impact of Servicizing Models

In this section, we provide details regarding the economic and environmental appeal of servicizing models and, more specifically, their potential to support the three pillars of sustainability: people, profit, and planet.

14.3.1 *Servicizing: The “Triple-Threat Business Model?”*

The decoupling of customer value from product ownership seems to be at the heart of the arguments in support of the environmental potential of servicizing. The reason is that this decoupling can contribute to the dematerialization of the economy (Heiskanen and Jalas 2000) by requiring less energy or material to generate the same (or even more) customer value. This can sometimes be achieved simply through using novel contractual mechanisms. For instance, in the context of chemical management solutions, shared-savings contracts are sometimes used between suppliers and buyers instead of the typical price-quantity contracts. Under such contracts, a supplier is not rewarded based on the material sold to the buyer but, rather, based on the savings that the buyer enjoys due to a reduction in the consumption of such materials (Corbett and DeCroix 2001; Corbett et al. 2005).

For another context in which dematerialization may also occur, consider the business model of car sharing. By the very nature of this model, car sharing providers such as Zipcar can benefit from a pooling effect and satisfy customers' needs through a smaller number of vehicles because a single vehicle can be used by many customers at different periods of time. This results in better fleet utilization and overall lower production volume, which directly benefits the environment by reducing the production-related emissions and use of raw material. This is true under several servicizing models, the providers of which do not need to dedicate one product to each customer.

The arguments for the environmental superiority of servicizing are also supported by the fact that, in the majority of the business models, customers are charged on a per-use basis (e.g., per-page-printed, per-mile-driven, etc.). Directly linking payment to product usage may incentivize customers to reduce their use of the product, resulting in a lower environment burden. The decrease in usage due to pay-per-use pricing is attributed to what is known as the taxi-meter effect (or flat rate bias; see Lambrecht and Skiera 2006), which postulates that customers enjoy their product usage more at a flat rate and tend to use it less when they are charged for each unit of usage.

Finally, in most of the servicizing business models, the manufacturer maintains the ownership of the products. This may incentivize manufacturers to design and produce products of higher reliability (Guajardo et al. 2012) or with longer useful life, so as to maximize the revenue extracted from each product per unit of time. This can also be achieved through better maintenance, which can typically be accomplished more productively by the manufacturer than the customer. Such practices also affect the end-of-life management of the products since a manufacturer is more likely to choose a more environmentally friendly way to dispose of a product than a customer and may even reduce the number of units that are disposed of in any given period of time by prolonging the useful lifecycles through remanufacturing or harvesting spare parts (see also Chap. 15 by Abbey et al. (this volume)).

Do the above-mentioned arguments imply that servicizing is the “triple-threat business model,” that is, a business model that excels simultaneously in all three pillars of sustainability: people, profit, and planet? Recent research (Bellos et al. 2017; Agrawal and Bellos 2017; Örsdemir et al. 2019) shows that this may not always be the case. But why not? Before we answer this question, we need to understand how the three pillars of sustainability relate to each other and how the performance of a business model with respect to these dimensions is assessed.

14.3.2 Assessing Business Model Performance Against the Triple Bottom Line

With respect to the profit dimension, it is straightforward to argue that, for a given (pay-per-use) price, profit increases as the number of customers who use the firm's

products increases and/or the amount of use that each customer extracts from the product increases (i.e., as customers drive more or print more pages).

With respect to the people dimension, one way to estimate the impact of a new business model is to measure the overall consumer surplus. Assuming that every customer derives a positive utility from using a product, then for a given (pay-per-use) price, higher product usage increases an individual consumer's surplus. Similarly, the overall consumer surplus increases as the number of customers who adopt and use a product increases.

This brings us to the third dimension: that of environmental performance (i.e., the planet). To assess the environmental performance of a business model, a lifecycle approach can be adopted (see Chap. 2 by Guinée and Heijungs (this volume)) and the environmental burden created at the phases of production, use, and disposal can be calculated (Agrawal et al. 2012). The environmental burden during the production and disposal phases depends on the total number of products manufactured and disposed of during each time period, while during the use phase it depends on the amount of the overall use that customers extract from the products (it may already be apparent that what increases profit or consumer surplus may also increase the environmental burden). Each phase may also be characterized by a different per-unit environmental impact. For instance, a certain type of product may cause a higher environmental burden during the use phase as opposed to the production or disposal phase.

14.3.3 Servicizing: Possible Environmental Issues

From an environmental point of view, the implementation of servicizing, as opposed to a traditional sales-only business model, can backfire from a few angles. As an example, consider again the business model of car sharing. It is true that customers may choose to relinquish car ownership and decide to cover their transportation needs by joining a car sharing program, something that would constitute an environmental win. However, at the same time, car sharing may make car usage more attractive to customers who typically use more sustainable modes of transportation (e.g., biking and/or public transportation). While the pooling effect may cause the overall number of vehicles produced to decrease, a larger number of customers adopting car sharing may imply that the overall level of vehicle usage increases (see Bellos et al. 2017). Despite the fact that such an increase in usage may lead to a higher profit and higher consumer surplus, it can also be environmentally problematic because several studies (Sullivan and Cobas-Flores 2001; MacLean and Lave 2003) have shown that the majority of the environmental impact of an automobile occurs during the use phase of its lifecycle. This issue may not be unique to car sharing, as many other types of products are characterized by higher use rather than production impact.

Even the pooling effect, which can be directly linked to the concept of dematerialization, may also be responsible for some unintended environmental drawbacks.

The reason is that a smaller production volume due to pooling results in a smaller overall production cost, which may allow the manufacturer to: (i) lower the prices and/or (ii) invest in improving the product efficiency (the manufacturer may be particularly interested in improving product efficiency because under most servicizing business models the manufacturer is responsible for the operating cost of the product). Since lower prices and improved efficiencies may further increase adoption and overall product usage (and therefore profitability), the well-known Jevons paradox³ may occur and result in higher environmental burden.

In this section, we provided a brief summary of the economic and environmental implications associated with the implementation of a servicizing business model. For an analysis of the environmental performance of servicizing, we refer the reader to Agrawal and Bellos (2017). In what follows, we provide more details on how firms may implement a servicizing strategy and some of the major challenges inherent in doing so.

14.4 Implementing a Servicizing Strategy

Having discussed some of the multiple benefits that a servicizing strategy can provide, we now outline some of the possible implementation challenges that an organization may face during the transformation of its business model from a product-based one to a solutions-based one. In addition, we describe the close relationship of an even newer class of business models, known as collaborative consumption models, to servicizing business models. Finally, we provide information about a design technique that is widely used in practice and which can facilitate the implementation of a servicizing strategy.

14.4.1 *Possible Transition and Implementation Challenges from Adopting Servicizing*

From an economic point of view, the transition to services has not always been successful, as there have been several cases where increasing the service component of a product-based organization decreased the overall firm performance (Gebauer et al. 2005). Interestingly, recent research (Suarez et al. 2013) has identified a non-linear effect between firm profitability and the extent of the service involvement. Specifically, services appear to have a positive effect on profitability after they reach a critical point of contribution to the firm's overall revenue. This may indicate a

³ According to the Jevons paradox (also referred to as the rebound effect; Greening et al. 2000), the overall rate of consumption of a resource increases as the efficiency of the resource improves. That is, as we make the usage of something cheaper, we tend to use more of it.

strategy where firms initially provide their services at a low price in order to attract customers into buying their products. As the number of customers reaches a critical mass, the firms may tend to focus on service delivery and to streamline the relevant processes, resulting in improved profitability.

One may argue that the inability of some product organizations to reap the benefits of providing services relates to their lack of expertise in marketing services to the end customers to fully exploit the profit potential of servicizing. For instance, the profit and environmental benefits from car sharing critically hinge on the aforementioned pooling effect, by which a service provider can satisfy the mobility needs of customers using a smaller number of vehicles than in the traditional ownership model. As such, the servicizing model is more profitable if the provider can attract more customer use. This, however, requires expertise to expand the servicizing business in highly decentralized and heterogeneous markets and can be challenging for traditional auto manufacturers. Such challenges are manifested by the decisions of BMW, General Motors, and Ford to shut down their car sharing operations in 2019-20. To overcome the challenges in marketing services, product organizations may partner with service platforms so the former can focus on designing, producing, and maintaining products that can maximize profitability from servicizing. In the automobile industry, electric vehicles are a promising choice as they have a lower per-mile operating cost, require less maintenance, and generate less emissions during usage than gasoline vehicles. Hence, the ongoing efforts of auto manufacturers in electrifying their fleet can have a synergy with the car sharing business, as evidenced by the recent development of partnership between auto manufacturers (e.g., Tesla and Hyundai Motor) and Uber to electrify its mobility services.

Other challenges for product organizations include the hurdles they face trying to transition from a product-based mentality to a service-based one. These hurdles can be attributed to the inherent differences⁴ between managing products and managing services. For instance, in product-based environments, centralization, efficiency, economies of scale, and standardization share a common underpinning, that of eliminating variability, which is almost always considered to be a necessary condition for successful operational and financial performance. Now compare this with the unique characteristics (Zeithaml et al. 1985) that services are known to hold: (i) intangibility (i.e., lack of specifications), (ii) heterogeneity in the customer requests and/or the service outcome, (iii) perishability (i.e., inability to inventory service performance/outcome), and (iv) co-production (i.e., the need for both the provider and the customer to be engaged to deliver the service outcome). If anything, these characteristics imply an exposure to higher variability. Therefore, the tension between the two worlds immediately becomes apparent. Moving from one mindset to another may arguably require extensive organizational “rewiring.”

⁴ Some of these differences are probably hard-coded due to differences in the training of key stakeholders. Such differences typically fortify silos within organizations. For instance, think of the eternal clash of the “hard” engineering/operations with the “soft” marketing side of the house.

Certain implementation challenges, however, may be unique to (or at least more pronounced under) a servicizing business model that attempts to decouple product ownership from customer value. Such challenges may pertain to:

- Internal resistance from employees, especially salespeople, who under a product regime are typically compensated based on commissions tied to the quantity of products or materials sold. Under servicizing, the base of the transaction is the solution that the product offers, not the product per se. For this reason, incentive mechanisms will have to be redesigned so they do not depend on the quantity of products sold. New incentives may be focused on customer retention, increases in revenue, expansion of the customer base, etc. Without proactively addressing the compensation schemes, a company may realize a loss of talent during the early stages of the transition to servicizing. Toffel (2008) provides a thorough discussion on the agency issues in servicizing business models.
- The complexity of the new contractual agreements. This can be the case especially when such agreements are based on (uncertain) product performance (e.g., engine up-time). The cost of implementing such contracts may be more difficult to estimate since it depends on performance that is unrealized at the time of the agreement. Because of this, manufacturers may be tempted to overpromise on key performance metrics. However, failure to deliver on these promises may impose steep penalties, either by financially compensating the customer or by making last minute arrangements (e.g., through third-party providers) to ensure that customer needs are met.

Changing the base of the transaction from the product to the service and solution level can be challenging for the customers too. The reason is that such a change also affects the basis of the cost–benefit analysis. Assessing the benefit requires a good estimation of the product usage needs (e.g., how many pages we print per year and what we gain from that), whereas assessing the cost requires the holistic estimation of expenses related to functions such as purchasing, operation, maintenance, and disposal, which for most organizations are decentralized (i.e., different parts of the organization are responsible for each function). For instance, see Toktay et al. (2006) and Ferguson and Plambeck (2008) for a detailed discussion of the reasons that Interface’s Evergreen Services AgreementTM (Oliva and Quinn 2003) failed.

- Behavioral effects such as the endowment effect, which postulates that customers tend to place more value on objects that they own than on those that they do not (Thaler 1980; Kahneman et al. 1991) and, for that reason, may not find servicizing as appealing as direct ownership. Although there is already evidence (The Economist 2012) that customers (especially of younger age) now exhibit a more utilitarian attitude toward consumption that is not influenced as much by behavioral effects, this is a cultural/generational change that may require time to fully occur. Also, the lack of product ownership may induce availability or accessibility anxieties, similar to the range of anxieties observed among drivers of electric vehicles (Avci et al. 2015; Lim et al. 2015).

- Faster deterioration due to heavier use or more frequent repairs of the products due to customers' careless use, or even abuse, of the equipment. For instance, car sharing customers may be less mindful of road hazards or may be less conservative with their driving behavior when they are not financially responsible for the maintenance of the vehicle they use. Such behavior can also diminish possible environmental benefits because it may either decrease the efficiency of the products (i.e., products may require more energy to deliver the same amount of use) and/or may require the provider to replace the products more often (i.e., it may increase the number of products required per unit of time). As another example, theft, damages, and vandalism are among the most challenging problems that Vélib, a French bike-sharing provider, has been facing since it started operating in Paris in 2007 (The New York Times 2009; France24 2013). More recently, similar issues led bike share programs in Salem, Oregon, and several UK cities including Bristol and Cardiff, to shut down their operations (Oregon Public Broadcasting 2022; Zag Daily 2022; BBC 2021). Extra security measures, monitoring mechanisms, product redesigns, and more frequent maintenance, however, can partially address such issues, albeit at an increased cost and environmental impact.
- Customer-induced negative externalities. Specifically, the fact that under-servicizing customers do not maintain product ownership implies a loss of control over the product and that the value each customer derives, to a certain extent, depends on decisions made and actions taken (even inadvertently) by other customers. For instance, in a car sharing model, a late return of a vehicle can interfere with the reservations of other customers. Similarly, failure of one customer to refuel or maintain a vehicle's cleanliness can negatively affect the experience of the other users. To counter such negative experiences, the provider may be required to install monitoring mechanisms along with a penalty structure to dissuade such behaviors. However, such "instrumental controls" (Frei 2005) may be perceived as a license to break the rules (e.g., license to be late in the case of car sharing) because it quantifies the implications of doing so, and, therefore, a customer can engage in a cost-benefit analysis, ignoring how their actions will affect other customers. According to Levitt and Dubner (2005), this was the case when some daycare centers in Israel imposed penalties on parents who were late in picking-up their kids. After announcing the fees, the centers observed an increase in the parents' tardiness (the original experiment can be found in Gneezy and Rustichini 2000). On the contrary, "normative controls" rely on subjective measures such as shame, guilt, embarrassment, and a sense of community or sense of duty. A provider's task is thus to devise mechanisms that can deter unacceptable behavior by eliciting such reactions and feelings. For instance, Zipcar always tries to create a sense of community by referring to its members as "Zipsters" or by featuring stories of its members and employees on its online magazine Ziptopia. For an excellent discussion of instrumental versus normative controls, see Frei (2005).

14.4.2 From an Economy Based on Products to Services and Now, to Collaborative Consumption?

Over the recent years, a new class of business models has gained traction by building on what is known as collaborative consumption. The concept of collaborative consumption refers to the type of consumption that takes place through (peer-to-peer) sharing, swapping, lending, or other similar activities (Botsman and Rogers 2011; Benjaafar and Hu 2020; Hu 2021). Such activities can span a wide range of products and services, including bikes (Spinlister), cars (Turo, Getaround), car rides (Uber, Lyft), parking spaces (JustPark), and short-term rentals (Airbnb, Vrbo). In the arguably most well-known business models of Airbnb and Uber, participating members (i.e., qualified hosts and drivers) contribute their properties by listing their apartments/houses and driving services, respectively, to the greater pool in return for monetary compensation, which is typically based on the amount of time the property was used by other customers. Similar to servicizing, the importance of product ownership is diminished in the sense that no ownership rights are transferred to each end user and payment is linked to product usage.

Business models that rely on collaborative consumption may alleviate some of the economic issues identified earlier for the servicizing models by achieving a middle ground. For instance, Uber drivers can still maintain the pride of ownership of their vehicles and, at the same time, improve the utilization for a fair compensation. Essentially, such models achieve a more efficient matching of supply with demand by: (i) increasing both the availability and accessibility of capacity, (ii) encouraging efficient allocation of the supply through centralized and dynamic pricing, and (iii) removing information asymmetries regarding the quality of the product/service through online review systems.

From an environmental point of view, however, it is not clear whether these business models have a positive impact. For instance, one may argue that Uber may be actually contributing to a higher environmental burden because, to a certain extent, it makes vehicle ownership more affordable by creating another source of revenue through which drivers can cover car ownership-related expenses (i.e., gas, maintenance, insurance, etc.) or by worsening traffic conditions (e.g., congestion due to driving and/or vehicle idling between rides). On the other hand, it can also be argued that it contributes to creating a more developed and interconnected network of transportation. As an example, it may make it easier for a traveler to hire an Uber car to head to a Metro station and then use public transportation to travel to the final destination (e.g., the airport). Therefore, there may be segments of the market that find it more economical to cover their transportation needs through Uber (and public transportation) and, for that reason, choose to relinquish car ownership, thus contributing to the dematerialization of the economy.

14.4.3 Design Tools That Can Facilitate Servicizing

Implementing a servicizing strategy to transform from a product organization to a solutions organization will probably come along with an array of critical decisions. For instance, the firm may need to determine: the pricing structure (e.g., how much to charge and whether to include a fixed fee in addition to a pay-per-use fee), the capacity (e.g., in the context of car sharing, the size of the fleet; see Bellos et al. 2017), the efficiency/reliability/durability of its products (see Guajardo et al. 2012 and Agrawal and Bellos 2017), or to what extent (if any at all) the “business-as-usual” model of selling products should be abandoned (e.g., the firm may decide to focus entirely on selling products, offering solutions or offering both solutions and selling products; see Agrawal and Bellos 2017).

However, before a firm starts tackling such issues, certain aspects of which it may have addressed in its existing business model (e.g., pricing issues are not unique to servicizing business models), it is important to identify the key differentiator between the practice of selling products and the practice of selling solutions. To do so it could attempt to answer the question of whether moving from the product business to the solutions business requires a materially different perspective. Namely, does the firm need to use different lenses to look at its offerings, or does it suffice to focus on the aforementioned operational and marketing issues (e.g., determining the capacity and the pricing, respectively)?

To answer these questions requires going through the task of rephrasing and reframing everything in terms of solutions. This can be a revealing exercise. For instance, thinking in terms of selling mobility solutions (as opposed to selling cars) may uncover that the order winners no longer have to do with certain product specifications (e.g., time to increase speed from 0 to 60 mph) and features (e.g., whether a specific car model comes with dual climate control and electronically adjustable seats). This first step can eliminate the product-related tunnel vision and enable the firm to start thinking along the lines of how to best assist their customers in meeting their basic needs.

This implies a holistic perspective that extends before and after the transaction point (i.e., the point where payment occurs) between a customer and a firm. For instance, the satisfaction that customers derive from meeting their mobility needs through a car sharing program is also determined by the ease with which they navigate through the website to find available vehicles and make reservations, make changes in their reservations, find the designated parking lot and the reserved car, address unforeseen situations (e.g., having a flat tire), and, finally, make payments and manage their accounts. The challenge in this case for the firm is to identify the different elements that shape the overall customer experience.

Toward this end, design thinking can be of great value. Design thinking (Brown 2008) refers to a human-centered design approach that relies on direct observation and rapid prototyping (indeed, services can be prototyped; see Thomke 2003) to create solutions that holistically address customer needs. It goes beyond traditional marketing techniques (e.g., focus groups) because it allows customers to be in their

natural environment where they may use a product or receive a service. This is typically achieved through empathic techniques (Leonard and Rayport 1997) such as shadowing, which allow researchers to have a customer-centric view of the firm's offering. One of the most insightful and practical outcomes of these techniques is the creation of the customer journey map.⁵ The customer journey map delineates the series of different steps/stages that customers go through every time they satisfy a need (e.g., mobility need). These different stages are known as touch points (or moments of truth; see Bitner et al. 2008) because they indicate points/moments at which customers possibly interact with some aspect of the firm's offering (e.g., interact with the online reservation system) and derive an experience.

The value of the customer journey rests on the fact that it enables the firm to identify all the different elements that positively contribute to the customer experience along with the points of failure in which the firm needs to improve on. Essentially, it delineates the process through which customers interact with the firms. This process-based view is indicative of the role that the operations management discipline can play in the development of such methods. In addition to the process, the people (i.e., user) dimension is in support of the expansion of the celebrated "4P's" of the marketing mix to "6P's" that define the service mix (Teboul 2006). The customer journey represents a method that is conducive to collaborations among disciplines that can break the silos we mentioned before. IDEO, the innovation and design consulting firm, has pioneered its use as a design method in the domain of both products and services (Bhavnani and Sosa 2006) through the use of diverse and interdisciplinary teams. For an analytical treatment and discussion on the use of the customer journey as a service design framework, we refer the reader to Bellos and Kavadias (2021).

Admittedly, we have presented two sides of the same coin. On one side, the transition to servicizing can be a challenging one as there are many difficulties that an organization may have to tackle. On the other (brighter) side, there are multiple benefits such as a growing demand for innovative business models that depart from conventional product/ownership-based models and the existence of established design methods that can be used to make the implementation of servicizing models easier. In the next section, we conclude this chapter by offering some thoughts on what remains to be done.

⁵ The concept of the customer journey relates to the service blueprints first mentioned in the seminal work of Shostack (1987). For a more recent discussion on the application of service blueprint, see Bitner et al. (2008). Stickdorn and Schneider (2010) describe additional design thinking methods applied on services.

14.5 Conclusion

The increasing number of manufacturing companies that transition to more service- and solution-oriented business models indicates that Theodore Levitt's quote of Leo McGinneva that "People don't want to buy a quarter-inch drill. They want a quarter-inch hole!" (Levitt 1986) is more true and relevant than ever. Done correctly, this transition will be associated with environmental as well as economic benefits. In this chapter, we explored some potential advantages and disadvantages, from both the economic and environmental perspectives, associated with this transition. Identifying these advantages and disadvantages is an important step closer to a service-based economy. However, there is more work that needs to be done from both researchers and practitioners to help overcome the hurdles that firms may face when making this transition. For instance:

- From an academic/research point of view: Existing assessment methodologies such as Lifecycle Analysis (LCA) may have to be further developed and applied in servicizing settings to account for usage patterns, which may differ significantly compared with when customers own the product. Along these lines, it is not clear whether offering certain types of products through servicizing may result in more frequent replacement (due to increased wear and tear) and overall larger production quantity than other types of products. Another environmental consideration pertains to the potential of servicizing in facilitating green technology adoption by providing customers a low-risk opportunity to try the product prior to purchase. In the car sharing industry, major providers including Zipcar and Hertz have included electric vehicles in their fleets, and start-ups such as EnvoyThere and Earth Rides are offering fully electric fleets. These offerings allow customers to try an electric vehicle and potentially alleviate their uncertainties about the driving experience, which differs fundamentally from driving a gasoline vehicle due to the use of electric motors (Agrawal et al. 2023; Ren et al. 2023). Furthermore, more light needs to be shed on the major drivers that determine customers' decisions to forego product ownership. When is the lack of ownership more acceptable to customers, and what kind of technologies need to be developed to alleviate "anxieties" about product unavailability (i.e., the concern about not having access to a product when needed)? Can offering an increased variety of products (e.g., under a car sharing program, this would imply offering an increased variety of brands and trims) make servicizing models more appealing as customers get to experience a broader range of products?
- From a managerial point of view: perhaps the greatest challenge is a cultural one. Manufacturers need to realize that they are in the solutions business and act on this realization. Ford started its transition to a "mobility company" in 2015 (Fortune 2015), and Toyota recently followed suit by announcing its own shift toward a "mobility company" (Toyota 2022). Hopefully more manufacturers will be encouraged to follow these examples. Of course, such a transition cannot and should not happen overnight. As highlighted by several successful cases (Porsche

2020; Volvo 2021), constant experimentation through small-scale pilot programs is likely the safest approach that manufacturers can use to gain momentum and make the transition.

Overall, we hope that our discussion in this chapter will motivate further interest in the exciting topic of servicizing as an alternative business model for product-centric firms.

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Chapter 15

Closed-Loop Supply Chains: A Strategic Overview



James D. Abbey, V. Daniel R. Guide, and Xichen Sun

Managers that consider a closed-loop supply chain just another environmental initiative need to update their thinking. Modern firms that use closed-loop supply chains as a competitive strategy receive many benefits—particularly higher profitability and control over a product’s entire lifecycle. In fact, the market for multiple lifecycle products continues to grow, with estimates holding that remanufactured product sales exceed \$100 billion per year.¹ As a result of analyzing the ever-growing remanufacturing sector through years of working with managers in numerous industries, various levers and themes surrounding effective closed-loop supply chain

¹ The global remanufacturing market has been growing. Current estimates of the remanufactured products market in the U.S. come from the United States International Trade Commission (USITC) report on remanufactured goods. See “USITC. 2012. Remanufactured goods: an overview of the U.S. and global industries, markets, and trade. Public report, U.S. International Trade Commission accessible at <http://www.usitc.gov/publications/332/pub4356.pdf>.” The European Remanufacturing Network estimates that the remanufacturing market in Europe will grow to €90 billion by 2030, tripling the €30 billion market in 2015 (<https://www.remanufacturing.eu/>)

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strategies became apparent. This chapter presents these findings and shows how firms in multiple industries experienced both successes and failures of their closed-loop supply chain strategies, recently also synonymously called circular economy strategies.

15.1 Understanding a Closed-Loop Supply Chain

Over the past few decades, environmental initiatives moved from tertiary goals to integral parts of corporate strategy. The environmental transformation manifested for multiple reasons, ranging from corporate social responsibility initiatives (see also Chap. 20 by Gillai et al. (2024)) to increasing levels of legislative mandates (see also Chap. 8 by Huang and Atasu (2024)).² Whatever the reason for the shift toward environmental initiatives, the outcome is clear: managers need innovative ways to reduce environmental impact while simultaneously improving profitability. Though common wisdom holds that environmental constraints will inevitably reduce profits, numerous counterpoints exist as manifest in firms that successfully implemented closed-loop supply chain (CLSC) strategies.³ This chapter provides a current look

² Corporate social responsibility continues to generate much debate about the role of a corporation as a good citizen. For an excellent discussion of such issues, see “Carroll, A. B. 1999. Corporate social responsibility evolution of a definitional construct. *Business & Society* 38(3): 268–295.” Additionally, for firms less interested in internally promoting environmental initiatives, legislative mandates regarding environmental performance and impact are particularly prevalent in the European Union with the waste electrical and electronic equipment (WEEE: http://ec.europa.eu/environment/waste/weee/legis_en.htm) and reduction of hazardous substances (RoHS: http://ec.europa.eu/environment/waste/rohs_eee/index_en.htm). Both laws seek to control environmental impact. Though the E.U. was among the first to enact such laws, Japan (<http://www.env.go.jp/en/policy/>) and Australia (<http://www.environment.gov.au/about-us/legislation>) have followed with similar legislation aimed at curbing environmental impact. Though much of the strict recycling mandates legislation has not occurred within the U.S., multinational firms already maintain product portfolios that meet the ambitious requirements. Should the legislative programs continue to expand throughout the world, firms holding such product portfolios may hold a competitive advantage over competitors who currently do not meet the requirements of such legislation. Various recent works address the issue of environmental legislation and product take-back. For several such works, see “Atasu, A. and L.N. Van Wassenhove. 2010. “Environmental Legislation Regarding Product Take-Back and Recovery” in ‘Closed-Loop Supply Chains,’ Eds. M. Ferguson, G. Souza. Taylor and Francis.” and works tailored to the extended producer responsibility for electronics waste and durable goods “Atasu, A., R. Subramanian. 2012. Extended Producer Responsibility for E-Waste: Individual or Collective Responsibility? *Production and Operations Management* 21(6): 1042–1059,” “Alev, I., Agrawal, V. V., & Atasu, A. 2020. Extended Producer Responsibility for Durable Products. *Manufacturing & Service Operations Management*, 22(2): 364–382.”

³ If the concept of a closed-loop supply chain is unfamiliar, multiple resources can provide guidance and insights. For example, see “Guide, V.D.R. Jr., and Van Wassenhove, L.N. 2003. “Business Aspects of Closed-Loop Supply Chains” in Guide, V.D.R. Jr., and Van Wassenhove, L.N. (eds.), *Business Aspects of Closed-Loop Supply Chains Exploring the Issues*. Pittsburgh: Carnegie Mellon University Press.” Also, see “Ferguson, M.E. and Souza, G.C. 2010. *Closed-Loop Supply Chains New Developments to Improve the Sustainability of Business Practices*. Boca Raton: CRC Press.” For some readers, perhaps the circular economy is a more familiar concept as

at the ever-evolving body of CLSC knowledge. Through looking at the body of knowledge, the chapter provides strategies to help managers understand the challenges and opportunities of CLSCs in various types of industries under differing strategic forces.

15.1.1 The Core of a Closed-Loop Supply Chain

At the core, a closed-loop supply chain represents a series of processes and flows aimed at some form of reuse or reclamation of products and materials. Specifically, a closed-loop supply chain incorporates design, control, and operation of a system to maximize value creation over the entire lifecycle of a product with dynamic recovery of value from different types and volumes of returns over time.⁴ As the definition states, a CLSC must be dynamic and evolve over time to handle changing market conditions. Such evolution makes managing a closed-loop supply chain an ever-evolving set of processes, which can take on many different forms depending on the product and industry in which a firm operates. Though a CLSC requires decisions at every decision level—operational, tactical, and strategic—this chapter focuses heavily on the broad strategic issues facing a manager considering a CLSC. As will become clear, the decision to operate a CLSC should be based on seeking increased profit—not just reducing costs—and improving corporate social responsibility through improved environmental performance.

Of course, before a CLSC can start a second lifecycle for a product, a first lifecycle must occur followed by the accompanying return of the product into a reuse market. On one extreme, the return of the product into a reuse market can be in the form of an end-of-use product that received considerable use by a prior owner, which is typical in the business-to-business markets for large industrial equipment (e.g., Caterpillar earth moving equipment and Xerox high-speed imaging equipment). On the other extreme, unlike the end-of-use returns that have often experienced extensive use, consumer product returns are often convenience returns—returns due to the customer simply deciding that the product does not suit their needs.⁵ These convenience returns, also called false failures, usually require minimal processing before returning to the market. Staggeringly, these often “convenience” consumer

an alternative to closed-loop supply chains. See “Atasu, A., Dumas, C. and Van Wassenhove, L.N., 2021. The Circular Business Model. *Harvard Business Review*, 99(4): 72–81” for more information on circular economy strategies.

⁴ For a discussion of the continuing challenges in closed-loop supply chains, see “Guide, V. D. R. and L.N. Van Wassenhove. 2009. The evolution of closed-loop supply chain research. *Operations Research* 57(1): 10–18.”

⁵ The rate of false failure returns can vary widely by the nature of the product and industry. As a prime example in the consumer electronics industry, Hewlett-Packard experienced false failure returns rates as high as 80% of all inkjet printer returns. On the other hand, Bosch false failure return rates at a vastly lower 2% of all power drill returns. For more information, see “Ferguson, M., V.D.R. Guide, Jr., G.C. Souza. 2006. Supply chain coordination for false failure returns. *Manufacturing & Service Operations Management* 8(4): 376–393.”

product returns in the United States have grown from \$360 billion in 2019 to over \$760 billion in 2021.⁶ This challenge has become a major area of study with many works addressing issues of return policy design,⁷ return policy behaviors,⁸ return policy reactions by consumers,⁹ and efforts to reduce returns¹⁰ that have not worked as intended in recent years. Other recent literature investigates how returns will change over time¹¹ and how to utilize returns for a profit, not a cost-center loss mentality.¹²

In either case of a heavily used commercial product or a convenience return by a consumer, a CLSC using remanufacturing provides a direct form of reuse that converts returned products into like new condition for resale. Remanufacturing entails disassembling the returned product, replacing any worn or broken components, repairing any remaining defects, and repackaging the product for sale as a remanufactured item.¹³

15.1.2 Overview of Closed-Loop Supply Chain Activities

Before moving into a detailed discussion of the process flows of a CLSC, a high-level view of the three major CLSC activities provides context for readers who are not yet familiar with the nature of a closed-loop supply chain. Figure 15.1 provides a high-level view of the front-end, engine, and back-end of a closed-loop supply chain.

The front-end activities entail collecting returned products through product acquisition management. As noted previously, some products will be heavily

⁶ The National Retail Federation offers free, detailed reports on returns. For recent reports, such as the 2021 report noted, see <https://cdn.nrf.com/sites/default/files/2022-01/Consumer%20Returns%20in%20the%20Retail%20Industry%202021.pdf>

⁷ Abdulla, H., Ketzenberg, M. and Abbey, J.D., 2019. Taking stock of consumer returns: A review and classification of the literature. *Journal of Operations Management*, 65(6), pp. 560–605.

⁸ Ketzenberg, M.E., Abbey, J.D., Heim, G.R. and Kumar, S., 2020. Assessing customer return behaviors through data analytics. *Journal of Operations Management*, 66(6), pp. 622–645.

⁹ Abdulla, H., Abbey, J.D. and Ketzenberg, M., 2022. How consumers value retailer's return policy leniency levers: An empirical investigation. *Production and Operations Management*, 31(4), pp. 1719–1733.

¹⁰ Shang, G., Ferguson, M.E. and Galbreth, M.R., 2019. Where should I focus my return reduction efforts? Empirical guidance for retailers. *Decision Sciences*, 50(4), pp. 877–909.

¹¹ Shang, G., McKie, E.C., Ferguson, M.E. and Galbreth, M.R., 2020. Using transactions data to improve consumer returns forecasting. *Journal of Operations Management*, 66(3), pp. 326–348.

¹² For an examination of seeing returns as a means to profit, see: “Abbey, J., Ketzenberg, M. and Metters, R., 2018. A more profitable approach to product returns. *MIT Sloan Management Review*, 60(1), pp. 1–6.”

¹³ The Remanufacturing Industries Council's (<https://remancouncil.org/>) gives an in-depth look at remanufacturing from various industry perspectives. In particular, the website provides details about remanufacturing processes and the resulting environmental benefits.

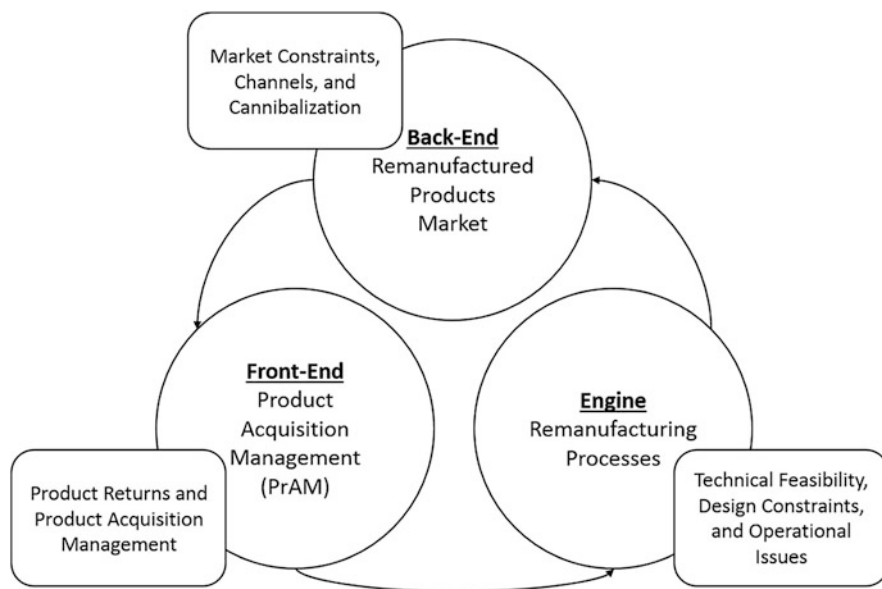


Fig. 15.1 Closed-loop supply chain activities

used while others may have seen minimal use. Such diversity in the nature of returns makes product acquisition management a critical activity for any manager considering a CLSC strategy. If acquiring returned product cores is relatively simple, engaging in CLSC operations can be a fairly straightforward endeavor. Conversely, if acquiring returned product cores is difficult or widely dispersed, even the first step of collecting returned products can prove to be a challenge, which leads many firms to use creative practices, such as “seeding” new products under the guise of being previously used.¹⁴ After returns are collected, the engine activities allow remanufacturing of the returned product to a like new condition.¹⁵ Though this chapter does not delve into the operational details, the topic of the engineering feasibility and technical constraints in a CLSC has received much attention over the past decades.³ After the product has been brought back to a marketable condition,

¹⁴ For an in-depth look at the practice of “seeding,” which means selling a new product under the guise of being previously used and remanufacturing, see: “Abbey, J.D., Geismar, H.N. and Souza, G.C., 2019. Improving remanufacturing core recovery and profitability through seeding. *Production and Operations Management*, 28(3), pp. 610–627.”

¹⁵ For an examination of issues in production planning and control in the engine portion of remanufacturing, see “Guide, V.D.R., Jr. 2000. Production planning and control for remanufacturing: industry practice and research needs. *Journal of Operations Management* 18(4): 467–483.” For detailed information about the issues related to grading the quality of cores acquired through the front-end processes of product acquisition management, see “Ferguson, M., V.D.R. Guide, Jr., E. Koca, and G.S. Souza. 2009. The value of quality grading in remanufacturing. *Production and Operations Management* 18(3): 300–314.”

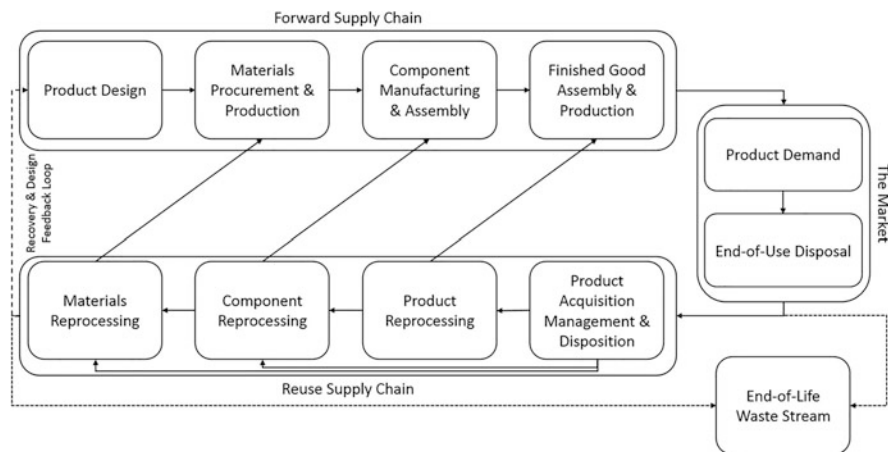


Fig. 15.2 Closed-loop supply chain flows and functions

the back-end activities entail putting the product back into the market for another lifecycle. Until recently, the nature of the market for reused products has received little attention. Fortunately, recent research has started to define the similarities and contrasts with the typical new product market.¹⁶ As will be discussed at length later in this chapter, the three major CLSC activities all work in unison and can be done in-house or as outsourced operations.

15.1.2.1 Examining Closed-Loop Supply Chain Flows

The primary function of a CLSC is to employ some form of reuse at the product, component, or materials level. Each of these choices comes with differing degrees of constraints, both in terms of recovery strategy and remarketing of the multiple lifecycle product. Understanding how these constraints vary by the nature of the industry is critical for managers investigating means to implement or improve their firm's CLSC strategy. Figure 15.2 displays various flows at the product, component, and materials levels.¹⁷ Chapter 18 by Blass et al. (2024) provides more detail on analyzing and managing material flows. Each of these flows offers firms different opportunities for recovery and reuse strategies, which are directly

¹⁶ For an overview of the current state of market-based literature for remanufactured products and pricing structures, and market segments, see, "Abbey, J.D., Meloy, M.G., Blackburn, J. and Guide Jr., V.D.R., 2015. Consumer markets for remanufactured and refurbished products. *California Management Review*, 57(4), pp. 26–42."

¹⁷ This figure is adapted from, "Abbey, J.D., V.D.R. Guide Jr. 2012. "Closed-Loop Supply Chains" in T. Bansal, A. Hoffman, (eds.), *Oxford Handbook on Business and the Natural Environment*. New York: Oxford University Press, 290–309."

influenced by the original product design. In particular, materials reprocessing would typically fall in the realm of recycling. For example, recycled plastics may be used as either like-to-like (e.g., water bottles) or down-cycling (e.g., tires to rubber mulch) materials. Component and product reprocessing, typically called refurbishing or remanufacturing, generally requires far less energy intensity than materials recycling. As such, product and component reuse is generally preferable over the more energy-intensive recycling, which is the least preferred of the options from an environmental standpoint.¹⁸

As shown in Fig. 15.2, some form of a waste stream is inevitable. Simply put, some products, components, and materials have a limited number of lifecycles or only a single use. The goal of a manager trying to implement environmental initiatives through a CLSC is to convert what was previously a stream of waste into profitable reuse. Though this sounds like an ideal win-win scenario—environmental benefits and higher profits—many pitfalls prevent firms from committing to a CLSC strategy. In the consumer products arena, one of the highest hurdles is the challenge of cannibalization of new product sales with the sale of lower-priced remanufactured offerings. As discussed just a few years ago, managers should handle this challenge from a total profitability, portfolio perspective.¹⁹ For most firms considering a CLSC, the problem of reacquiring previously sold products, also known as product acquisition management (PrAM), can be daunting as the reverse supply chain flows may not be a current competence.²⁰ In particular, for firms in the technology sector that face a high marginal value of time—a rapid decline in a product's value during a lifecycle—acquiring, testing, inspecting, and returning the product to market quickly is imperative. On the flip side, for firms facing lesser marginal value of time pressures—minimal decline in the product's value during a lifecycle—the acquisition and resulting remanufacturing activities are not as time sensitive or asset intensive.²¹

Other related research has shown that there is no one size fits all strategy for a closed-loop supply chain. For some firms, such as the Xerox Corporation, product acquisition is quite simple as customers often lease large printing equipment, but

¹⁸ For further information on the reduce, reuse, and recycle (3R) hierarchy, see the Environmental Protection Agency's website at <http://www.epa.gov/recycle>

¹⁹ See "Atasu, A., V.D.R. Guide, Jr., L.N. Van Wassenhove. 2009. "So what if remanufacturing cannibalizes new product sales?" *California Management Review* 52: 56–76."

²⁰ For additional insights into how to operate in a competitive core acquisition market, see "Guide, V. D. R., R.H. Teunter, and L.N. Van Wassenhove. 2003. Matching demand and supply to maximize profits from remanufacturing. *Manufacturing & Service Operations Management* 5: 303–316."

²¹ For a detailed look at the nature of such challenges related to marginal value of time in fast moving industries such as consumer electronics, see "Guide, V. D. R., G.C. Souza, L.N. Van Wassenhove, and J.D. Blackburn. 2006. Time value of commercial product returns. *Management Science* 52: 1200–1214." For potential solutions to maximize profits based on the nature of the industry's marginal value of time, see "Blackburn, J.D., V.D.R. Guide, G.C. Souza, and L.N. Van Wassenhove. 2004. Reverse Supply Chains for Commercial Returns. *California Management Review* 46: 6–22."

lifecycle management and design issues represent major challenges.²² Conversely, many consumer product firms have had great difficulty managing the core acquisition process and reverse supply chain as consumer products typically have a short, single lifecycle.²³ To shed light on the differences in strategy by the nature of the industry, this chapter delves into two major dimensions that help firms match their CLSC strategy with the nature of their market—product design and core competencies.

15.2 A Framework for Matching Product Design and Core Competencies

Product design is a highly complex topic in its own right. Thus, adding yet another layer of complexity in the form of designing for more than one lifecycle may seem daunting. Fortunately, designing for multiple lifecycles tends to have many positive effects, such as reduced return rates, easier reparability, and faster turnaround for warranty claims and returns.²⁴ Of course, when market pressures are high to get an innovative product to market, speed and efficiency of production often take predominant roles as is often the case in the consumer technology industry. Due to such market-based pressures, many firms have naturally evolved core competencies based on the nature of their industries' competitive layout. However, as a product line matures and technological innovation slows, the move toward CLSC operations

²² The Xerox Corporation has been a world leader in remanufacturing systems at their Webster, New York manufacturing/remanufacturing hybrid facilities. In both popular press and academic research, Xerox stands out as an exemplar of environmentally friendly closed-loop supply chain systems. Xerox works diligently from the design phase forward to recycle and remanufacture their equipment with great success in energy, materials, and waste reduction. For instance, in 2020, Xerox's remanufacturing operations diverted over 6030 metric tons of waste from landfills. See a recent corporate sustainability report at <https://www.xerox.com/en-us/about/corporate-social-responsibility>

²³ Reverse supply chain design for consumer products remains a persistent issue for many firms. For a detailed discussion of viable solutions in consumer product industries, see the California Management Review article, "Blackburn, J.D., V.D.R. Guide, G.C. Souza, and L.N. Van Wassenhove. 2004. Reverse Supply Chains for Commercial Returns. California Management Review 46: 6–22."

²⁴ Much recent research focuses on the need for better understanding of design for remanufacturing and reuse. Many resources exist for those interested in the engineering side of design such as the Rochester Institute of Technology's Center for Remanufacturing and Resource Recovery (<http://www.rit.edu/gis/remanufacturing/>). For an excellent summary of research on the topic of product design in a CLSC, see "Bras, B. 2010. "Product Design Issues" in Ferguson, M.E. and Souza, G.C. (eds.), Closed-Loop Supply Chains New Developments to Improve the Sustainability of Business Practices. Boca Raton: CRC Press." Finally, for general resources on remanufacturing, see the Remanufacturing Industries Council website at <http://remancouncil.org/>

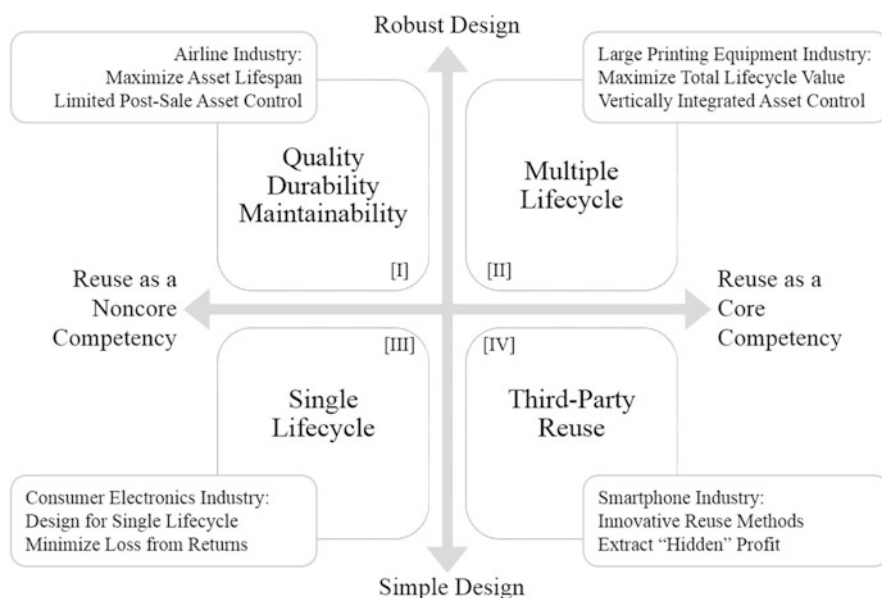


Fig. 15.3 Design and core competence for product reuse in a CLSC

as a core competency may offer increased profit opportunities. Figure 15.3 shows how design and core competencies interact with four resulting strategies.²⁵

Breaking down each quadrant in the manager's matrix of Fig. 15.3 provides insights into the influence that product design has on product acquisition management, reverse logistics, reuse processes, and the nature of the reused product market. First, as shown in Fig. 15.3, product design plays a key role in dictating recoverable value, particularly at the product and component levels. Even at the materials level, product design plays a significant role. For example, recovering heavy metals from smartphones and many other small electronic devices is no easy task as the devices are integral by design.²⁶ Second, reverse logistics entails product acquisition management—the means to reacquire end-of-use returned products—and requires either ownership or outsourcing the transportation network for moving the end-of-use products. Third, the core operational reuse processes, such as the choice of the recovery level (i.e., product, component, or materials), require decisions regarding

²⁵ The major impetus for this chapter and framework for understanding strategies for reuse through remanufacturing comes from ongoing research found in the manuscript, "Abbey, J.D. and Guide Jr., V.D.R., 2018. A typology of remanufacturing in closed-loop supply chains. *International Journal of Production Research*, 56(1–2), pp. 374–384."

²⁶ Reclaiming rare earth and other precious metals from electronics products has been an issue for decades. For a review of issues related to such reclamation, see "Cui, J., and E. Forssberg. 2003. Mechanical recycling of waste electric and electronic equipment: a review. *Journal of Hazardous Materials* 99: 243–263."



Fig. 15.4 Strategic levers in a closed-loop supply chain

investment in remanufacturing capabilities in both capital equipment and people. Finally, the fourth lever describes the means of remarketing products going through more than one lifecycle. If a market for such products does exist, then deciding whether to lease or sell the products has major implications on the CLSC strategy. As will become clear throughout this chapter, a decision in any one of the levers can have significant impacts on all other decisions. Figure 15.4 summarizes the four key levers that either derive from or drive a closed-loop supply chain strategy.¹⁸

15.2.1 *Quadrant I: Quality, Durability, and Maintainability*

In this quadrant of Fig. 15.3, the original manufacturer of the assets tends to focus on selling with limited post-sale support. As a prime example, consider the airline industry. In many cases, airlines prefer to buy the multi-billion dollar fleets of airplanes from the aircraft manufacturers. As such, the aircraft manufacturers accommodate by going through intensive new product design processes focused on quality, reliability, and maintainability in the field. These traits actually mesh with the multiple lifecycle design traits of other industries that do maintain asset control (e.g., Xerox) and lend themselves well to product reuse in a CLSC. However, the loss of asset control post-sale has significant downsides. The most obvious downside is that reacquiring a used product may be difficult if not infeasible. The airplane manufacturers simply have no interest in reacquiring the assets after a sale occurs. Such lack of asset control opened up an entire third-party industry centered on overhauling and remanufacturing products ranging from full airliners to engines

to small components of airplanes.²⁷ In other words, the airplane manufacturers have ceded profits to third-party remanufacturers due to their robust designs combined with a lack of access to the assets after the initial sale.

Another downside is that investing in a reverse logistics network for components/parts is often prohibitive when third-party entrants, such as Delta and Lufthansa, already have a strong presence in the market. Additionally, the airplane manufacturers (e.g., Boeing and Airbus) generally lack both tacit and explicit knowledge of the testing, inspecting, and disposition processes as well as the appropriate asset base for intensive remanufacturing of the assets. Finally, without a strong market presence as a remanufacturing original equipment manufacturer, the airplane manufacturers lack the market cache from which to capitalize. Overall, in this quadrant, firms maintain minimal control over product reacquisition, reverse logistics, and little to no market presence in the multiple lifecycle product market. However, the firms do not need to invest heavily in capital intensive remanufacturing systems.

15.2.2 Quadrant II: Multiple Lifecycle

In this quadrant, the original equipment manufacturer focuses on designing the product for multiple lifecycles from the inception of a product's design. The intention for such products is to extract as many lifecycles as possible to maximize total lifecycle profits. Such designs require strong vertical integration of both the initial sales and end-of-use product reacquisition channels. If such integration of the initial sale and reacquisition is not present, then third-party remanufacturers will gladly enter the market to extract the additional lifecycle value missed by the original equipment manufacturer as seen in the airline industry. Firms playing in this vertically integrated multiple lifecycle design quadrant also need to invest heavily in the reverse logistics systems to maintain appropriate transportation to the asset intensive remanufacturing facilities. At the remanufacturing facilities, the product design naturally leads to rapid testing, inspecting, and disposition of the products for reentry into the market. As exemplified by the Xerox Corporation, markets for such multiple lifecycle products are highly segmented based on the dimensions of product performance and price. The Xerox sales force actively markets the right package of "newly manufactured" (i.e., blended new and remanufactured products) and completely new products to suit any customer type. In fact, at the time of end-of-

²⁷ Delta Airlines and Lufthansa continue to generate significant cost savings and profits through repair and refurbishment of both their own equipment and equipment for other airlines. For details on Delta Airlines' continued strategy of extending lifecycles of aircraft, see "Carry, Susan. 2012. "Delta Flies New Route to Profits: Older Jets," The Wall Street Journal November 16, 2012." Lufthansa Technik AG actively markets their commercial airline refurbishment services for all levels of maintenance, overhaul, engines, components, aircraft systems, and more. For details, see Lufthansa website at <https://www.lufthansa-technik.com/base-maintenance>

use for a product at a customer site, Xerox uses their sales expertise to provide higher performing, remanufactured machines as a means to maintain a strong competitive foothold against their non-remanufacturing competitors who cannot compete at the same price-performance thresholds. In sum, this quadrant represents the pinnacle in maximizing both environmental benefits and profitability through tight control over asset reacquisition, reverse logistics networks, heavy investment in remanufacturing capabilities, and a deep understanding of market segments for the multiple lifecycle products.

15.2.3 *Quadrant III: Single Lifecycle*

The astute reader might have noticed that the prior two quadrants largely focus on high value, business-to-business products. Though some firms still design such high value equipment for a single lifecycle, the most common examples of such single lifecycle products come from the consumer products industries. Consumer tastes and preferences often change rapidly, particularly in fast moving product segments such as consumer electronics that have ambiguity and risk perceptions prevalent among consumers with regard to such previously used products.²⁸ Such fast moving industries can make extracting value from returned products extraordinarily difficult.²¹ In response, many firms have abandoned the idea of designing products for anything beyond a single lifecycle. Instead, such firms focus on extracting maximum profit through efficient design and manufacturing systems with an accompanying focus on preventing returns. Such a strategy has been successful for many companies with the side effect of a massive resulting waste stream. Such a copious waste stream represents a continuing force for increasing legislative mandates to prevent electronics waste even in the United States.²⁹ In effect, though companies found the strategy sustainable for making profits and meeting consumer needs, legislative forces view the strategy as unsustainable from a societal and environmental standpoint. Though little has changed on the design front, the single lifecycle product design may diminish in the coming years.

In sum, firms operating with the single lifecycle design strategy may soon find themselves seeking ways to improve reacquisition of the products through reverse

²⁸ For a discussion of such issues of risk perceptions and ambiguity tolerance among consumers, see: “Abbey, J.D., Kleber, R., Souza, G.C. and Voigt, G., 2019. Remanufacturing and consumers’ risky choices: Behavioral modeling and the role of ambiguity aversion. *Journal of Operations Management*, 65(1), pp. 4–21,” and “Abbey, J.D., Kleber, R., Souza, G.C. and Voigt, G., 2017. The role of perceived quality risk in pricing remanufactured products. *Production and Operations Management*, 26(1), pp. 100–115.”

²⁹ As noted before, the E.U., Japan, and Australia have all moved toward increasing legislative mandates for recycling and waste stream reduction. This increasing legislative pressure has also come to the U.S. in the form of California’s recycling laws aimed to curbing electronics and other waste (<http://www.calrecycle.ca.gov/recycle/>)

logistics networks at least at the materials level. Without a move away from the single lifecycle philosophy, investment in testing, inspection, and disposition technologies and related remanufacturing systems will be a moot point. Interestingly, there is recent evidence that select product types should be designed for a single lifecycle or easy materials reclamation, as a sizeable portion of the consumer market holds no interest in purchasing a multiple lifecycle product.³⁰

15.2.4 *Quadrant IV: Third-Party Reuse*

The final quadrant represents a simple fact: lost profit opportunities on the part of the original equipment manufacturer. As a case in point, ReCellular started their business by refurbishing Motorola cellular phones when Motorola proclaimed that no one could make money off such a venture. Of course, if the third-party reuse occurs at the materials reclamation (i.e., recycling of plastics) level, the original equipment manufacturer may simply cede such markets for a lack of interest in entering a non-core competence market. However, when the third-party reuse occurs at the component or particularly product level, the original equipment manufacturer has inadvertently created a competitor in their own market space—a competitor using the original equipment manufacturer's own product. The common excuse for allowing such third-party entry comes straight from Fig. 15.3: the original equipment manufacturer views the product or component reuse as a non-core competence region. As such, the original equipment manufacturers argue that reacquisition is simply too expensive or difficult, that reverse logistics systems are too hard to manage, and that investment in remanufacturing equipment as well as the related testing, inspection, and disposition processes is simply too risky. Finally, such original equipment manufacturers may also view internal remanufacturing as a source of new sales cannibalization—a fear shown to be questionable at best.³¹ Yet, all these arguments seem to be mythical from the third-party remanufacturer's

³⁰ Two studies shed light on the nature of consumer perceptions of multiple lifecycle products. The first study delves into the various factors that influence how consumers perceive multiple lifecycle (e.g., remanufactured) products (see, “Abbey, J. D., et al. 2015. Remanufactured Products in Closed-Loop Supply Chains for Consumer Goods. *Production and Operations Management* 24(3): 488–503.”). An even more recent work examines how various consumer segments emerge with a sizeable portion—upwards of 35%—of consumers refusing to consider a remanufactured product of any kind (see, “Abbey, J.D., Blackburn, J.D. and Guide Jr., V.D.R., 2015. Optimal pricing for new and remanufactured products. *Journal of Operations Management*, 36, pp. 130–146.”).

³¹ Fear of new product sales cannibalization when offering a remanufactured product has been a long-standing source of opposition for remanufacturing at many firms. For a nice discussion of such opposition and strategies to handle common issues with potential cannibalization, see “Atasu, A., V.D.R. Guide, Jr., L.N. Van Wassenhove. 2009. So what if remanufacturing cannibalizes new product sales? *California Management Review* 52: 56–76.”

perspective as thousands of third-party remanufacturers make billions in profits every year.

15.2.5 *Contrasting the Quadrants*

Table 15.1 provides an accessible summary of the criticality of each major CLSC function available to a manager deciding which quadrant best fits with their design-competence status.

Table 15.1 shows that each quadrant has varying functional dimensions for the closed-loop supply chain. In particular, the quality, durability, and maintainability quadrant has little to no CLSC functions as a result of the sales of products and little if any direct interaction with the product post-sale. Conversely, the multiple lifecycle quadrant has relatively simple reverse logistics and marketability by the very nature of the typically vertically integrated system. Nonetheless, the multiple lifecycle quadrant also requires intensive resource investment to remanufacture and extract value from the returned end-of-use products. The single lifecycle (i.e., consumer products) quadrant usually focuses on minimizing costs related to returned or end-of-use products. As such, third-parties provide most logistics needs to minimize cost, though such a strategy can be a severe mistake for firms that face a high marginal value of time market.²¹ Additionally, the single lifecycle products typically have highly varied reusability due to both the initial design intentions and lack of investment in remanufacturing technologies by the original manufacturer. Further, the marketability of remanufactured single lifecycle products can be questionable due to varying forces from rapid depreciation, quality concerns, and more.

In general, the third-party’s core business is finding innovative ways to extract profit that the original equipment manufacturers either ignored or missed completely. The third-parties typically lack easy access to the returned or end-of-use products, which makes reacquiring the products an intensive endeavor. The remanufacturing and reuse processes must be retrofitted to the decision made by the external entity—the original manufacturer. Finally, as the third-party rarely has the same market power or existing forward supply chain channels as the original

Table 15.1 Summary of CLSC process intensity by quadrant

Quadrant	Product design	Reverse logistics	Reuse processes	Marketability
Quality & durability (I)	Maximal lifespan	Minimal	Minimal	Minimal
Multiple lifecycle (II)	Intentional reuse	Simple	Intensive	Simple
Single lifecycle (III)	Maximal efficiency	Minimal (outsourced)	Simple to intensive	Challenging
Third-party (IV)	Profit extraction	Intensive	Intensive	Challenging

manufacturer, remarketing the products can also be an intensive challenge. Yet, in spite of all these challenges, third-party remanufacturers are plentiful, and more importantly, profitable.

15.3 Design and Core Competence: One Size Does Not Fit All

The above discussion demonstrates that implementing a true, fully closed-loop supply chain requires the vertically integrated, multiple lifecycle design strategy. However, many firms may find that complete vertical integration combined with a multiple lifecycle design is both technically infeasible due to the nature of the product and financially infeasible due to requisite investment in reverse logistics and remanufacturing facilities. Moreover, a sizeable reuse market may not even exist for products that have a short market lifecycle. As such, to maximize profitability and environmental benefits under the various constraints, the original equipment manufacturer should take an active role in establishing the appropriate level of third-party involvement. Recent research describes just such tradeoffs with three major strategies that sit on a continuum from pure outsourcing to pure insourcing. Between these two extreme points sit various levels of hybrid strategies, which are often the best choice for a remanufacturing original equipment manufacturer. Even Xerox, which represents one of the best remanufacturing firms in the world, only maintains a limited reverse logistics fleet with a preference for using third-party logistics carriers to haul end-of-use products back to the centralized remanufacturing facilities. Figure 15.5, adapted from work by Pinar Martin, provides the basic process flow decisions for the vertically integrated remanufacturing, hybrid remanufacturing, and outsourced remanufacturing strategies.³²

The primary difference among the strategies is the intensity of remanufacturing operations. Even in the vertically integrated CLSC, both initial manufacturing and remanufacturing still source some components from an external entity (e.g., microprocessors or memory cells). Such components are not part of the core-competence of the manufacturer and are often either replaced with new components or tested for viability in the remanufactured product. In the hybrid strategy, firms handle some products internally while third-parties handle less complex or less profitable products. Finally, in the outsourced CLSC, the original manufacturer exerts only minimal control in the form of performance specifications or other inputs. Note that in all cases, the original manufacturer plays some role to prevent an unauthorized third-party from entering the market. Obviously, each of the various

³² Pinar Martin has released a book on the various levels of remanufacturing strategy as well as a scholarly paper on the topic. For those interested in the more managerially oriented materials, see “Martin, P. 2010. *Remanufacturing as a Supply Chain Strategy: Business Models and Case Studies*. Dusseldorf, Germany: VDM Verlag.” For those interested in the technical details, see “Martin, P., V.D.R. Guide, and C.W. Craighead. 2010. Supply chain sourcing in remanufacturing operations: an empirical investigation of remake versus buy. *Decision Sciences* 41: 301–324.”

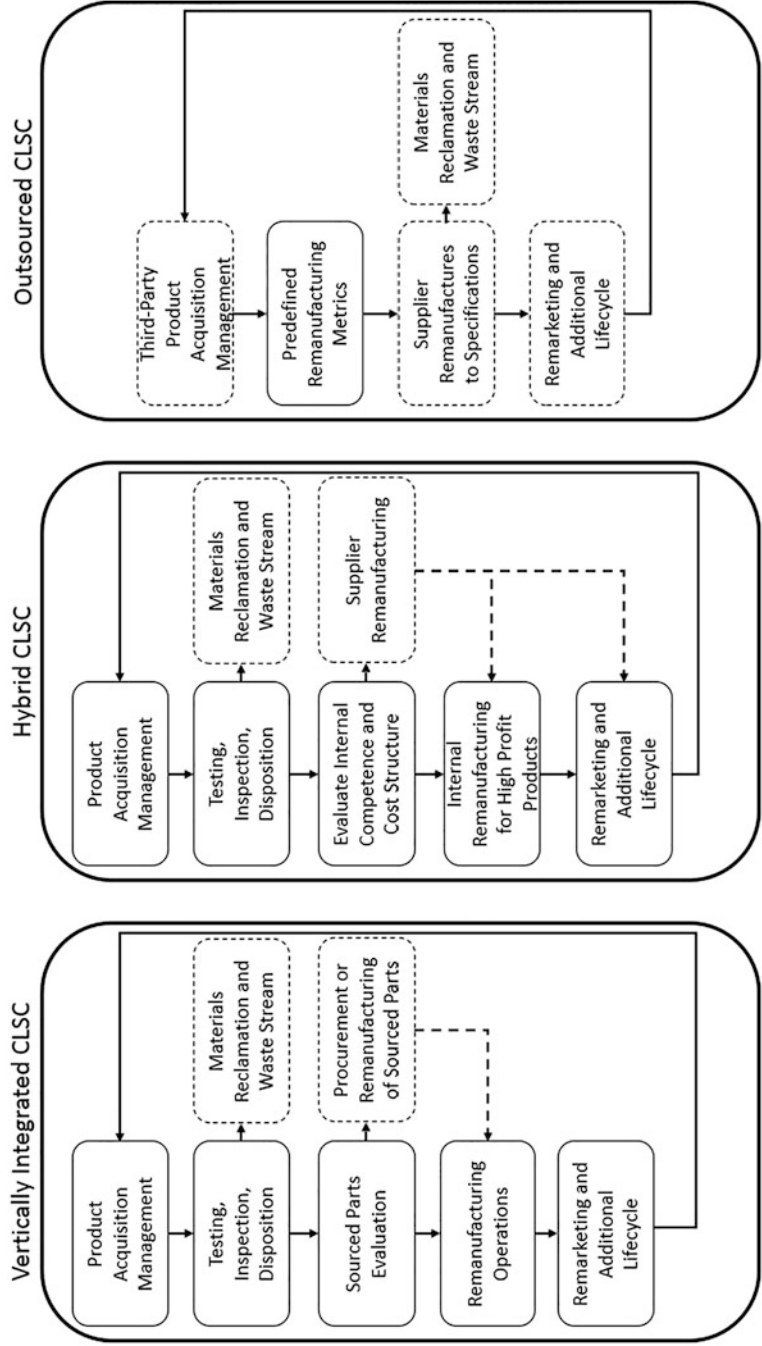


Fig. 15.5 Vertical, hybrid, and outsourced CLSC strategies

choices holds significant tradeoffs. Thus, a quick breakdown of the implications of each choice follows, starting with the endpoints of vertical integration and outsourcing.

15.3.1 Vertically Integrated CLSC Strategy

Control represents a major impetus for the vertically integrated strategy. Under vertical integration, a firm retains control over the brand name, customer service, intellectual property, product acquisition, and even the forward and reverse supply chains. However, control creates challenges. For instance, forward supply chain procurement can represent a major hurdle for the remanufacturing operations. If the procurement division has to acquire additional new parts to support the production of new machines but remanufacturing decreases order quantities for new parts, the bulk rate economies of scale in both purchasing and transportation may disappear. Other issues stemming from metrics decisions can also play a significant role. For example, labor and equipment utilization may actually look better with remanufacturing, but variability in production times may increase. Additionally, overhead allocation to both new and remanufactured product production can be a tricky balancing act. Too much or too little allocation to either new or remanufactured products can create major problems for the manufacturing division's profitability.

15.3.2 Outsourcing CLSC Strategy

If the forward supply chain is largely outsourced, then the reverse supply chain and reuse operations will be outsourced as well. Simply put, if a firm does not manufacture, the firm cannot easily remanufacture. This issue became manifest for Dell. As Dell outsourced more of their forward supply chain production, reuse operations became increasingly more difficult. In the end, Dell eventually outsourced the previously profitable remanufacturing operations.³³ Many other problems emerge when trying to manage a closed-loop supply chain in an outsourced system. Visibility of design flaws and improvements all but vanishes as returned products are not highly accessible for designers to garner feedback. Contractual obligations become vastly more complex as contracting parties must consider both the forward and reverse supply chain contingencies. Moving products that lose value rapidly (i.e., high marginal value of time products) represents a major challenge as the contracted reverse logistics and reuse partners typically do not

³³ For a detailed look at Dell's outsourced remanufacturing/refurbishing strategy, see "Vitasek, K., Ledyard, M., & Manrodt, K. 2013. Vested outsourcing: five rules that will transform outsourcing. Palgrave Macmillan. pp. 185–198."

possess great competence in moving products back to market. In sum, with only a few exceptions, a fully outsourced CLSC rarely achieves the same profitability and environmental benefits as a vertically integrated or hybrid strategy.

15.3.3 *Hybrid CLSC Strategy*

In a hybrid strategy, a firm ideally chooses remanufacturing functions that best mesh with core competencies. However, such hedging can be a double-edged sword as firms may dismiss profit opportunities due to lack of a *current* core competence—one that could easily be developed to generate additional profit. Further, if too many functions are outsourced, the firm loses control over design specifications, which all but assures difficulty for remanufacturing. Other issues include intellectual property, pricing contracts, lack of visibility across the supply chain, and loss of control over both the forward and reverse supply chains. Overall, the hybrid strategy is the most common strategy but always represents a balancing act—in many cases, the hybrid strategy can create cross-divisional conflict as different stakeholders fight for shares of profit.

In the end, there is no single solution for deciding the right level of insourcing or outsourcing. The next section provides some basic guidance and a decision tool when considering the various choices involved in implementing a closed-loop supply chain.

15.4 Choosing a Closed-Loop Supply Chain Strategy

The one clear message so far is that choosing a CLSC strategy is no easy task. As with most decisions, multiple tradeoffs exist when choosing how to compete for more than just maximum profit. In some industries, such as the earth moving and mining equipment, the decision to remanufacture previously produced machinery is a simple matter of materials availability and obvious profitability.³⁴ Unfortunately, the lines blur as the products become less expensive and more widely distributed. The problems compound when the product has a relatively short lifecycle in the market.

Figure 15.6 puts the pieces, originally outlined in Table 15.1, together with a decision flow process for managers considering particular levels of reuse. In some cases, the decision flow outlined in Fig. 15.6 might have exceptions. However, based

³⁴ Caterpillar is proud of their industry-leading remanufacturing and reuse systems. To say that remanufacturing and reuse is a core competence of Caterpillar would be an understatement. In any given year, Caterpillar reuses more than 120 million pounds of iron alone. For a detailed look at Caterpillar's remanufacturing operations, see their website at <http://www.caterpillar.com/en/company/sustainability/remanufacturing.html>

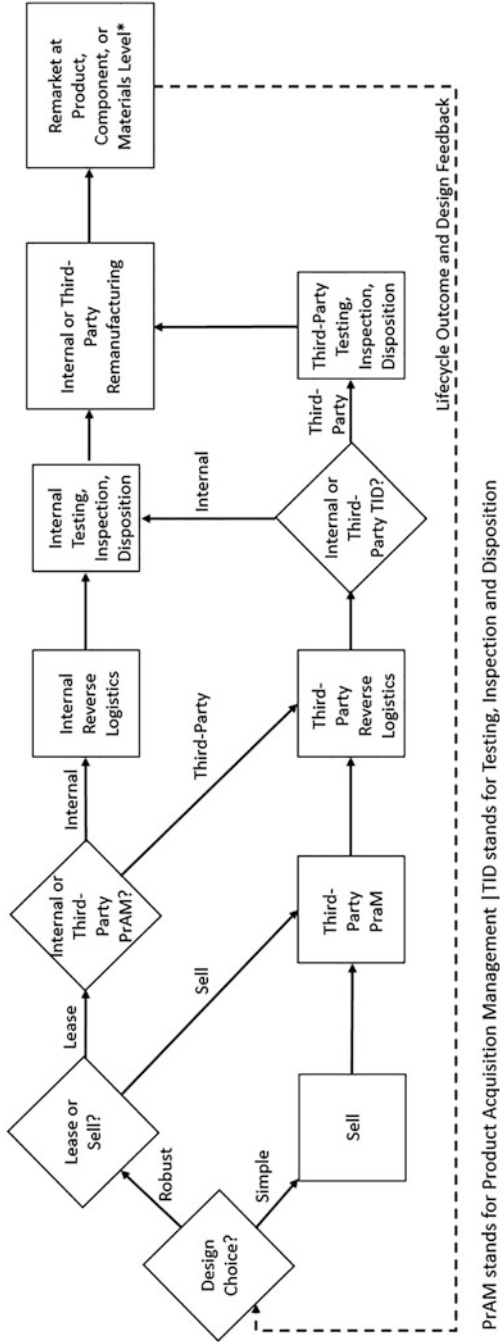


Fig. 15.6 Decision process for major closed-loop supply chain activities

on observations from dozens of companies over dozens of years, the decision flow is largely in line with successful CLSC strategies.

Figure 15.6 summarizes the major levers that all managers must consider when contemplating a CLSC strategy: the product design; product (re)acquisition management; reverse logistics; the testing, inspection, and disposition process as well as the remanufacturing processes; and finally the remarketing of the reused product, components, or materials. As the figure displays, third-parties typically play a role—often significant—in managing at least some elements of a CLSC. The question managers face is choosing the right level of third-party involvement.

15.4.1 The Role of Third-Parties Revisited: When and How to Maintain Control

As noted in the previous section, the choice between internal and third-party activities largely sits on a continuum (see Fig. 15.5). In only a very few cases does a firm do all CLSC activities internally. Xerox and Caterpillar represent uncommon examples of firms that maintained a significant, vertical control over both the forward and reverse supply chains. For both firms, such control has been invaluable in providing improved profitability, design feedback, and environmental benefits. However, such cases of vertically integrated CLSCs are largely the exception.

As outsourcing the forward supply chain gained popularity, insourced remanufacturing became increasingly difficult. Facilities that used to serve for dual new and remanufacturing production were economically unsustainable for only remanufacturing operations. As a case in point, Dell moved nearly all remanufacturing activities to a third-party on a contractual basis as the forward supply chain evolved toward outsourced contract manufacturing.³³ However, Dell still maintains contractual control over the third-party, which is significantly superior to the all too common strategy of simply ignoring the activity of third-parties that will enter the market with or without the original manufacturer's consent. In other words, a complete omission of a remanufacturing strategy opens the door for third-parties to extract profit that could have been controlled by the original manufacturer, which can have significant implications on the original brand's reputation.³⁵

The automotive industry serves as a case in point for ceding control over remanufacturing. The remanufactured automotive parts industry has a long-standing tradition of heavy third-party involvement. Though Ford and other major automotive manufacturers have tried to make inroads into reuse of parts, third-party players

³⁵ For a fascinating look at the adverse and even beneficial consequences of third-party remanufacturing, see “Agrawal, V., A. Atasu, K. van Ittersum. 2015. Remanufacturing, third party competition, and the perceived value of new products. *Management Science* 61(1): 60–72.,” as well as the challenge of selling assortments to third-party remanufacturers in “Mutha, A., Bansal, S. and Guide, V.D.R., 2019. Selling Assortments of Used Products to Third-Party Remanufacturers. *Production and Operations Management*, 28(7), pp. 1792–1817.”

(e.g., Cardone Industries) have maintained a strong control on the automotive parts product acquisition market.³⁶ During a meeting of original equipment automotive manufacturers and their third-party remanufacturing competitors, an interesting argument emerged: the original equipment manufacturer accused the third-party of “stealing parts” to which the third-party responded “stop me”. Though the original manufacturer may felt entitled to “their parts”, the original manufacturer had taken no steps to maintain any vertical control after the time of the initial sale. The automotive industry provides a cautionary message: once an industry cedes control over remanufacturing to third-parties, regaining control over the remanufacturing market can be a difficult and costly proposition.

15.4.2 Leasing as a Strategy to Facilitate CLSC

One strategy to maintain product control over multiple lifecycles while deterring third-party competition is leasing. While leasing is traditionally offered as a financing option for customers who cannot afford an outright purchase or those who choose leasing for tax reasons, manufacturers have found leasing increasingly attractive due to the additional value leasing provides in a CLSC. Many firms started to lease products for multiple lifecycles and remanufacture them during the multiple use cycles.³⁷ By retaining product ownership, manufacturers could reduce the uncertainties of timing and quantity in returned products, which are often a primary challenge in product acquisition management that prevent (re)manufacturers from implementing a profitable CLSC strategy. Additionally, manufacturers with remote monitoring systems could keep track of product usage and key performance metrics. The information collected supports their maintenance activities and improves new product design. The product wear-and-tear information also helps determine the returned product quality and simplifies the remanufacturing process. If the product is sold to customers, manufacturers may not be able to access the product usage information during customer use cycle due to security concerns.

To increase the attractiveness of leasing contracts, manufacturers often bundle services with products to offer customized solutions with product availability guarantees. In many cases, this practice is known as servicizing and creates many dynamics that require coordination of products-and-service bundles within and

³⁶ For a look at the broad range of products—some of which are only available in remanufactured form—that Cardone Industries remanufacturers, see their website at <https://www.cardone.com/>

³⁷ For example, as the leading remanufacturer in the industry, Xerox has adapted its business model to lease photocopiers to customers. Through leasing, the company guarantees that 100% of the equipment is taken back at the end of their leasing terms for remanufacturing or parts recycling (<https://www.xerox.com/corporate-social-responsibility/2018/environment/circular-economy.html>)

across firms.³⁸ For instance, Rolls-Royce's TotalCare solution allows it to lease jet engines to airline companies on a pay-per-flying-hour basis that shifts the engine maintenance responsibility from customers to Rolls-Royce.³⁹ Leveraging monitoring technologies, data analysis, and manufacturing expertise, manufacturers can maintain products better than customers and increase the residual value of used products through planned preventative maintenance. Customers can also benefit from improved operations efficiency and product uptime: Caterpillar claims that their Job Site Solutions help customers improve fuel efficiency by up to 32% and product availability by up to 24%.⁴⁰

This product-service bundle type of contract is prevalent in industries where products are complex machinery, and the stake of product failure is high. Customers are thus willing to pay a premium for solution services to free themselves from burdensome maintenance activities. However, the same contract does not work well when products are of simple design and customers can perform maintenance themselves at a relatively low cost. One such problem occurred at Interface, the commercial flooring solution provider. Interface developed the Evergreen Service Agreement in 1995 to close the loop of floorcoverings. The operating lease-based agreement offers carpet and its installation, maintenance, replacement, and removal as a service package. To maintain profitability, Interface had to charge a service fee higher than the cost of customers performing maintenance services themselves. As a result, although many customers showed interest in the Evergreen Agreement, few purchased it. Interface ended up with a leasing contract that they struggled to sell.⁴¹

Leasing is typically less prevalent in consumer products (1) with a short lifecycle and little to no residual values after use (e.g., electronics), and (2) to which consumers attach additional values to product ownership (e.g., convenience, esteem, etc.). The costs associated with maintenance, reverse logistics, inspection, and remanufacturing could easily surpass the price customers are willing to pay.⁴² Interestingly, leasing can oddly reverse this effect when the product lifecycle is extremely short—for fashion products constantly evolving, for example, low-cost

³⁸ For an excellent discussion of coordination issues incumbent on a firm choosing to servitize, see: "Mutha, A., Bansal, S. and Guide, V.D.R., 2022. What servitizing demands of a company: The need for inter-functional coordination. *Journal of Operations Management* 68:408–420."

³⁹ See Rolls Royce informational online brochure on the program at: <https://www.rolls-royce.com/media/our-stories/discover/2017/totalcare.aspx>

⁴⁰ For more information about Caterpillar's Job Site Solutions, see https://www.cat.com/en_US/support/operations/job-site-solutions.html

⁴¹ For a full documentation of the Evergreen Services Agreement at Interface, see "Oliva R, Quinn J. 2003 Interface's Evergreen Services Agreement. HBS Case 603–112, Harvard Business School, Boston."

⁴² As an example, the white goods manufacturer Gorenje struggled to implement the leasing-and-remanufacturing model, as the total cost of ownership for customers to lease a washing machine could be 40%–100% higher than that of purchasing one owing to substantial repair costs. For more information on Gorenje's journey toward closing the loop, see "van Loon, P., Van Wassenhove, L. N., & Mihelic, A. 2022. Designing a Circular Business Strategy: 7 Years of Evolution at a Large Washing Machine Manufacturer. *Business Strategy and the Environment*, 31(3): 1030–1041."

rental models can be attractive to consumers and manufacturers/dealers. Instead of extending the product lifecycle, manufacturers or dealers explore the economic potential of maximizing product utilization.⁴³

As shown above, there is no simple answer to the lease-or-sell decision. Manufacturers need to weigh the advantages and disadvantages of various levers to determine whether to lease and, perhaps more importantly, how and when to lease. This question is no longer simple as consumers and commercial entities all play a role in the leasing market. This potentially extends into the sharing economy realm due to societal shifts—a topic for the third edition of this chapter.

15.5 Closing the Loop

This chapter provides high-level guidance on the structure of closed-loop supply chain strategies with an overriding theme: third-parties will play a role in nearly all closed-loop supply chain systems. As such, manufacturers in the forward supply chain need to make a conscious decision on their closed-loop supply chain strategy: that decision needs to ask when and how third-parties should play a role. If the forward supply chain original manufacturer does not make such conscious decisions, an *unauthorized* third-party will make the decision for them by entering the market as a competitor. Even if the prospect of investing in resources—capital equipment, skilled labor, and reverse logistics networks—is beyond the core competencies of an original equipment manufacturer, forethought in developing and contracting with third-parties can prove valuable. The value of such contracts comes from improved control over all aspects of the reverse supply chain including product design feedback, environmental reuse opportunities, hedging against future legislative pressures, increased opportunity to reach additional markets, and greater control over brand image. Yet, with all these issues, firms still contend with themselves by creating accounting and inter-functional challenges that continue to fuel the third-party entry into the market.⁴⁴ In the end, if the forward supply chain manufacturer does not have a closed-loop supply chain strategy, a third-party remanufacturer will.

⁴³ Rent The Runaway, an e-commerce platform that provides fast fashion rentals, is one of the retailers pursuing clothes renting strategies. Unfortunately, a recent study has revealed that the rental model aiming at environmental sustainability can result in even higher environmental impact than the basic use-and-dispose model. See “Levänen, J., Uusitalo, V., Härri, A., Kareinen, E., & Linnanen, L. 2021. Innovative recycling or extended use? Comparing the global warming potential of different ownership and end-of-life scenarios for textiles. *Environmental Research Letters*, 16(5): 054069.”

⁴⁴ For insight into the issues of accounting, financials, and inter-functional issues, see: “Mutha, A., Bansal, S. and Guide, V.D.R., 2021. Managing the inter-functional tension between accounting- and financial-profits in remanufacturing multiple-usecycle products. *Production and Operations Management*, 30(9), pp. 2993–3014.”

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Chapter 16

Toward a Circular Economy: A Guiding Framework for Circular Supply Chain Implementation



Saman Amir, Niloufar Salehi, Malvina Roci, Susanne Sweet, and Amir Rashid

16.1 Introduction

Circular economy (CE) has been referred to as one of the main strategies to achieve sustainable development by policymakers (European Commission Directorate-General for Communication 2020; Ministry of Infrastructure and the Environment 2019; Su et al. 2013), practitioners such as the Ellen MacArthur Foundation, and researchers (Korhonen et al. 2018; Murray et al. 2017; Urbinati et al. 2020). CE aims to decouple economic growth from the consumption of limited resources by designing a restorative and regenerative industrial economy that designs out the waste and pollution, keeps the products and materials in use, and regenerates natural resources (Domenech and Bahn-Walkowiak 2019; Ellen MacArthur Foundation 2013, 2020). In manufacturing systems, implementing CE is becoming essential to address the challenges of resource scarcity and increased waste and emission generation (Lieder and Rashid 2016; Murray et al. 2017). Circular manufacturing systems need to be designed intentionally to close the loop of components or

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products through multiple life cycle products, preferably in their original form (Rashid et al. 2020; Roci et al. 2022; Govindan 2022). This approach requires holistic value management through three main functions: value proposition, product design, and supply chain (Asif 2017).

In the CE literature, the supply chain is considered as one of the key enablers of CE implementation (Lahane et al. 2020; Rashid et al. 2020) with a consensus that a CE-oriented supply chain differs from a linear supply chain (Geissdoerfer et al. 2018; Vegter et al. 2020). The difference lies in the fact that circular supply chain (CSC) activities are involved in value creation, delivery, and retention (Geissdoerfer et al. 2018; Vegter et al. 2020; Roci et al. 2022), and their design and performance are significantly influenced by the value proposition and product design (Rashid et al. 2020).

There are several literature reviews on CSC (Masi et al. 2017; Batista et al. 2018b; Batista et al. 2018a; Farooque et al. 2019; González-Sánchez et al. 2020; Lahane et al. 2020) where the integration of forward and reverse flows is pointed out as crucial for the CE concept (Zarbakhshnia and Goh 2022). In these studies, other concepts such as sustainable supply chain (Hussain and Malik 2020) and green supply chain (Nasir et al. 2017) were addressed to contextualize CSC where closing the loop is not the principal focus (Geissdoerfer et al. 2018). In the transition from a linear to a circular supply chain, the concept of closing the loop has been built upon the classical linear supply chain approach where the main focus is on the forward flow management. In this approach, closing the loop is not part of the business strategy development of the companies as such, and the loop is not essentially closed. As a result, value recovery happens by chance because neither the products nor the system is designed for loop closure from the beginning and with the systemic approach of value management, potentially resulting in higher costs for recovery processes and lower recovery efficiency (Guide et al. 2003; Rashid et al. 2020). To successfully close the loop, the forward and reverse flow integration needs to be conducted by design with the consideration of value proposition and product design configurations' effect on the supply chain (Frei et al. 2020). The reverse flow is an important factor of CE adoption (Lopes de Sousa Jabbour et al. 2019) and deals with the uncertainties of product return flow in terms of timing, quality, and quantity (Asif et al. 2012; Rashid et al. 2013). Moreover, ignoring the systemic implementation perspective of CE may result in inefficient value capture and creation (Howard et al. 2019). CSC aims to minimize these uncertainties by designing the products and systems for circularity as well as establishing effective interorganizational collaborations (Kayikci et al. 2022) to ensure maximum value recovery and minimum waste generation. Some of the challenges in the CSC domain are listed below:

- CSC is an emerging area with a nomenclature still in its infancy (Geissdoerfer et al. 2018). There is a discrepancy in the basic terminology of CSC as it has not been explicitly defined in the context of closing the loop for CE.
- Recently published literature focuses on integrating CE strategies within a linear context, often neglecting fundamental CE principles such as holistic value

management and achieving system-wide circularity through seamless forward and reverse supply chain integration. This gap in the literature suggests a need for further exploration and research.

- Decision-makers must understand CSC from a systemic perspective of CE implementation due to the inherent complexity of supply chain management (Liu et al. 2013). This complexity is characterized by diverse supply chain activities, components, and stakeholders (du Toit and Vlok 2014), as well as the strategic importance of supply chain management (Stavrulaki and Davis 2010). Adopting the “circularity” feature in supply chain management adds further complexity, dynamism, and uncertainties to the system.
- Considering the maturity of the field of the supply chain, several frameworks within the supply chain literature deal with a narrow scope such as planning, design, decision-making, or optimization of supply chains. Despite the increasing interest of the companies to implement CE, no CSC frameworks exist for them in making strategic, tactical, and operational decisions.

Before we delve deeper into the challenges of CSC implementation, to scope our work, we define CSC as “closing the loop by intention and design with a zero-waste vision where the forward and reverse flows are integrated and optimized for multiple life cycle products to enable restoration and regeneration of technical cycles.” The primary aim of such a definition is to reduce uncertainties regarding the quality, quantity, and timing of return flows (Thierry et al. 1995; Guide et al. 2003; Flapper et al. 2005; Kleindorfer et al. 2005; Linton et al. 2007; Guide and Van Wassenhove 2009; Asif et al. 2012; Souza 2013), which are inherent in closing the loop of a linear system. The purpose of this chapter is to address the challenges previously mentioned by using our definition of CSC. To achieve this, we aim to connect the latest supply chain concepts and processes with closed-loop thinking and explore various SC frameworks. Finally, we propose a comprehensive decision-making approach for CSC and CE implementation through a new framework. Interested readers can refer to Chap. 15 by Abbey et al. (2024) for a strategic overview on closed-loop supply chains and to Chap. 17 by Transchel et al. (2024) for an application of CSC strategies to product packaging.

This chapter is structured as follows: First, state-of-the-art research based on the literature review of the SC frameworks from the linear, closed loop, and circular perspectives is presented. This is followed by design and development of a CSC framework. In the next step, the applicability and usefulness of the proposed CSC framework is illustrated in two case studies, followed by a final discussion and conclusion.

16.2 Background

16.2.1 *Closing the Loop: Moving Toward Loop Closure by Design or Intention with CSC*

In the linear approach, products at the end of their use cycle are considered as waste, and by closing the loop, resources are recovered by diverting the used products from the waste stream and capturing their remaining value primarily through remanufacturing and recycling (Chari 2015; Wojanowski et al. 2007). For end-of-life (EoL) products, value recovery and remarketing are often handled by third-party remanufacturers that are not involved in the original manufacturing supply chain.

Closed-loop supply chains (CLSCs) have evolved from passive value recovery processes due to the product returns by chance toward an active management of returns. Guide Jr and Van Wassenhove (2009) presented the evolution of CLSC in a five-phase framework from individual activities to a holistic business approach discussing the increase in profits by value recovery. These five phases consist of the golden age of remanufacturing, valuing the reverse logistics process, coordinating the reverse supply chain, closing the loop, and prices and markets. Govindan and Soleimani (2017) defined two main goals of CLSC as managing customers' demand and EoL collection and recovery. The collection of products is not anticipated in the business strategy or value proposition; it is a reactive measure to deal with product returns. Therefore, the types of customer returns dictate the collection strategies and potential value recovery (ibid.).

Companies take products back from end customers for a number of reasons, including commercial, end-of-use, end-of-life, repair, and warranty issues (Flapper et al. 2005; Guide and Van Wassenhove 2009; Tosarkani et al. 2020). In the linear approach of closing the loop, companies passively accept these returns (Guide et al. 2003) and link them to the most suitable recovery process from the business point of view. For example, for end-of-use return, the appropriate recovery strategy is remanufacturing, and end-of-life returns are recovered through recycling (Guide and Van Wassenhove 2009). When returns are managed by chance, neither the system nor the products are designed to intentionally close the loop. As a result, companies may not expect the right information at the right time and cannot gain sufficient control over the quality, quantity, and timing of the returned products. This leads to insufficiently planned value recovery strategies, which are indispensable for efficient loop closure that ensures a high level of certainty in the quality, quantity, and timing of the return flows (Chari 2015).

Figure 16.1 summarizes the CLSC approach by chance, meaning that the return flows in the post-consumption phase are largely dependent on users' willingness to return the products. User decisions, however, can be influenced by the policies and regulations or any potential tangible or intangible benefits they may gain, e.g., money, credits, altruistic, or social values. The dashed lines in Fig. 16.1 represent the various recovery paths and highlight the potential uncertainties that can arise within the system due to the absence of a clear value proposition, as well as products

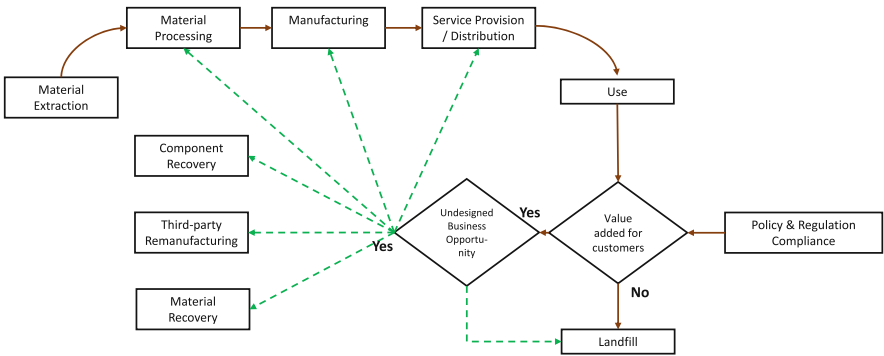


Fig. 16.1 Closed-loop supply chain structure depicting the loop closure by chance

or system designs that can support multiple life cycles. Therefore, the original equipment manufacturers (OEMs) or the third-party actors perform the recovery activities only when they anticipate business opportunities in them.

To address the challenges of unpredictable returns and insufficient value recovery strategy closing the loop by design or intention is an evolution from the linear loop closure thinking. To reliably close the loop, the integrated forward and reverse flow design and the capability of bringing products back at the end of their life cycle are required.

Closing the loop prolongs the lifetime of material, components, and products in use by empowering the inner loops of maintenance for extending the durability, reusing for the same purpose with few or no changes, refurbishing by replacement of components, and remanufacturing by recovery of the components. These inner loops (De Angelis et al. 2018), i.e., maintenance, reusing, refurbishing, remanufacturing, and recycling, not only strengthen the value recovery strategies but also create more economic and environmental benefit by retaining the value (material, energy, and labor) in the loop (Lopes de Sousa Jabbour et al. 2019). Therefore, closing the loop by design or intention promotes designing the products for multiple life cycles from the beginning (Lieder and Rashid 2016). Multiple life cycle products are designed for multiple lives of predetermined periods and are taken back from the customers (Asif et al. 2012) and put into the next cycle through appropriate value recovery strategies based on their quality. These products may have several use cycles with predefined recovery strategies and be required to be taken back after each cycle (Rashid et al. 2013; Shoaib-ul-Hasan et al. 2021).

A robust circular system is necessary to manage products through multiple life cycles. This system should include three main functions—value proposition models, product design, and supply chain management—that can collaborate and interact dynamically with the support of information and communication technologies (ICT).

Following the approach of closing the loop by design to keep the components and products preferably in their original form in the loop, the context of circular

supply chains emerged as an important concept. Masi et al. (2017) examine the relationship between supply chains and the circular economy. They argue that there is a significant lack of supply chain configurations that align with the restorative and regenerative principles of the circular economy. The study also identifies several drivers, enablers, and inhibitors that can either promote or hinder the transition toward a more circular supply chain. Batista et al. (2018a, b) propose a conceptualization of a CSC archetypal form by integrating streams from existing sustainable supply chain narratives such as reverse logistics, green, sustainable, and closed-loop supply chain management to create an understanding of a CSC. The authors refer to CSC as a term for the coordination of both forward and reverse supply chains with an emphasis on restorative processes in the downstream supply chain, taking product design into consideration (ibid.). They underscore the need to understand the linkages between product design, supply chains, and business models as well as stakeholders' collaboration within the regeneration and restoration landscape. To operationalize the CE from a supply chain perspective, Farooque et al. (2019) provide a research direction to substantially enhance the sustainable and green supply chain management field. The authors define the CSC management by integrating terms such as value chains, product service system, logistics activities, and EoL management within the CE literature. Vegter et al. (2020) argue that return flow is not a unique characteristic for CSC. They propose that CSC should focus on achieving resource efficiency while taking into account its economic, environmental, and social impacts. The authors also suggest that end customers should be considered key stakeholders in the implementation of a CSC.

The critical differences of linear, closed-loop, and circular supply chains lie in their origin of material resources. A linear supply chain extracts the material from the primary resources and produces large amounts of waste. On the one hand, CLSC in the linear context partially improves performance by bringing the products back to the producers for recovery. On the other hand, CLSC does not consider the cascading applications that may result in reduced waste and increased resource recovery. Therefore, CLSCs are inefficient in practice from a circularity perspective as they do not maximize resource recovery or support collaboration beyond a single organization.

In contrast, we propose that CSC implementation adopts a systemic and life cycle approach that is closely connected to the circular value proposition model and product design. A systemic view will introduce proactive management of the products' return flow, thereby promoting the restorative, regenerative, and zero-waste capabilities with an added focus on stakeholder collaboration for circularity.

16.2.2 Supply Chain Frameworks and Circularity

In this section, we examine the linear, closed-loop, and circular supply chain frameworks and their main elements to understand their construct, motivation, and focus.

16.2.2.1 Frameworks in the Linear Supply Chain Context

Focusing on supply chain design factors, Melnyk et al. (2014) presented a framework with three key levels: *influencers*, *design decisions*, and *building blocks* that need to be understood in harmony with managing the inherent complexity of the SC. Influencers refer to broader considerations such as life cycle and business model. Design decisions are related to structure and design of the supply chain such as network design. Finally, building blocks are the required investment to fulfill the design decisions such as inventory.

The twenty-first century logistics framework developed at Michigan State University in 1999 helps managers to identify the competencies and capabilities of their companies' supply chain. This framework proposes that a firm has three groups of competencies: *operational*, *planning*, and *behavioral*. The operational processes consist of the customer, supplier, and internal integration. The planning processes include technology, planning, and measurement integration. Finally, the behavioral processes cover the relationship integration between customers and suppliers (Closs and Mollenkopf 2004).

Regarding the operational competencies, Stavroulaki and Davis (2010) highlighted the importance of alignment between the strategic capability of supply chains (leanness, agility, or hybrid) and the operational aspects of the processes (efficiency and flexibility) by developing a conceptual framework. This framework supports manufacturers having a better understanding of supply chain choices while considering the alignment between product, its supply chain processes, and strategies in an integrated perspective. The proposed framework is built upon four manufacturing strategies and their resulting supply chains that are distinguished based on the types of products they make, their production, and logistics processes.

The Supply Chain Operations Reference (SCOR) model evolution clearly shows how the supply chain management systems started from the linear model and are gradually transitioning toward closing the loop (Vegter et al. 2020). The Supply Chain Operations Reference (SCOR) model has been developed to assess the supply chain effectiveness and efficiency in sales and operational planning (Lambert et al. 2005) using five performance attributes: reliability, responsiveness, agility, costs, and asset management efficiency (Bråtenndal and Gustafsson 2020). This framework is a process reference model that links business processes to best practices, metrics, and technology and is designed to describe, measure, and evaluate supply chain configurations. It enables effective communication among supply chain partners (Ming 2011).

The Global Supply Chain Forum (GSCF) is a relationship-oriented framework that focuses on identifying, developing, and maintaining key relationships with customers and suppliers (Lambert et al. 2005) and includes three closely interrelated elements: *business processes*, *management components*, and the *structure of the chain*. It also considers the necessity of information flow throughout the supply chain to improve the overall performance (Ellram et al. 2004). The return management activities in the GSCF framework aim at return avoidance and reducing the number of returns by improving the quality of products (Lambert et al. 2005).

Simatupang and Sridharan (2005) proposed a framework for supply chain collaboration consisting of five connecting features of collaboration that address their interaction. These features are *collaborative performance system*, *information sharing*, *decision synchronization*, *incentive alignment*, and *integrated supply chain processes*.

Parmigiani et al. (2011) developed a conceptual model that focuses on stakeholders' role in the social, environmental, and economic performance of the supply chain. This model integrates the supply chain configuration, stakeholder management, and capability development.

16.2.2.2 Frameworks in the Closed-Loop Supply Chain (CLSC) Context

Carter and Ellram (1998) proposed a guiding framework focusing on defining internal and external forces affecting reverse logistics activities. The reverse logistic in this case was positioned as an enabler for a company to become more environmentally efficient and comprises reusing, recycling, disposal, and landfill processes to reduce resource consumption. The primary forces that drive or constrain the reverse logistics activities in this framework are *input*, *output*, *competitive*, and *regulatory* forces. In this framework, regulations, customers, and policy are considered as the main drivers of reverse logistics activities, while the quality and certainty of inputs and levels of coordination between firms do not directly drive these activities.

Nuss et al. (2015) proposed a conceptual framework for reverse logistics operations and identified relevant planning problems in reverse supply chains and their interrelations. The framework identifies the relevant variables in the reverse supply chain and illuminates the consequences of each decision concerning other parameters. The planning problems were mapped in a matrix of reverse supply chain activities at three planning levels based on five layers of complexity. These layers of complexity cover the *unique problems of planning*; *their interdependencies*; *uncertainty in quantity, quality, and timing*; *cooperation within recovery networks*; and *integrating forward and reverse supply chains*.

Lakhmi et al. (2019) proposed a conceptual framework to structure CLSC operational issues with a focus on returnable transport items (RTI). This framework classifies these operational issues into three categories of long-term, mid-term, and short-term activities that design, manage, and control the returned flows to optimize the performance of the system. In this framework, the long-term issues refer to the design of the system and logistics network, mid-term issues focus on operations planning, and short-term issues focus on scheduling, executing, and controlling the operations.

The Ellen MacArthur Foundation (2016) presented a Reverse Logistics Maturity Model to support companies to assess their reverse logistics and identify focus areas for improvement. The proposed model consists of three components of reverse logistics (*front end*, *engine*, and *back end*) and presents the maturity level of reverse logistics in the three decision-making levels (strategic, tactical, and performance). In their work, reverse logistics was introduced as one of the key building blocks

of CE and a key step to facilitate value-added activities. Moreover, they highlighted that the requirements of reverse logistics differ based on the attributes of product and business model. Therefore, this model considers three demand-driven archetypes on the basis of product type, and each of these archetypes presents product attributes, reverse logistics requirements and implications, solutions, and key success factors.

16.2.2.3 Frameworks in the Circular Supply Chain (CSC) Context

It has been acknowledged by many researchers that supply chains in circular systems differ from those in linear ones, require more stakeholders and new relationships, and are closely connected to circular business models (González-Sánchez et al. 2020; Vegter et al. 2020). However, CSC research still needs special attention in developing innovative frameworks (Lahane et al. 2020), and comprehensive frameworks are needed as the CSC field is developing over time.

Integrating circular business models (CBMs) and CSCs was studied by Geissdoerfer et al. (2018), who proposed a framework that highlights that CSCs are driven by CBMs toward closing, slowing, narrowing, intensifying, and dematerializing loops. Their differences lie in their complexity. The authors claim that the main conditions toward sustainability are based on a long-term perspective including economic, environmental, and social goals, and proactively managing the stakeholders.

Concerning the sustainable supply chain, De Angelis et al. (2018) provided five propositions as a framework categorized into five themes: *strategy* (relationships change as a result of the shift from product ownership to service-based strategies), *structure* (structural flexibility concerning geographical barriers), *flow* (considering technical and biological cycles with closed- and open-loop materials), *focus* (stakeholders' collaboration boundaries), and *scale and scope* (procurement policies).

The conceptual framework proposed by González-Sánchez et al. (2020) addresses CSC performance through four dimensions: *relational*, *logistics and organizational*, *technological*, and *functioning environment*. This framework highlighted the important role of framing new business models to enable closer relationships among actors within an adapted organizational and logistical structure.

Building on the SCOR model, Vegter et al. (2020) focused on describing supply chain processes and performance objectives in transitioning toward circular business models. They added *use* and *recover* as two new processes in addition to the existing six processes of the SCOR model and stressed that certain processes of the linear supply chain need to shift their focus so that they become relevant for CSC.

It can be concluded that several of the existing SC frameworks have been developed with a high-level overview to identify the critical factors of supply chain management (Closs and Mollenkopf 2004; Lambert et al. 2005; Stephens 2001). However, the post-consumer returns at strategic, tactical, and operational levels are considered primarily due to environmental legislations for managing end-of-life product management (Carter and Ellram 1998; Ellen MacArthur Foundation 2016). The SC frameworks lack an overarching consideration of loop closure from the circularity perspective of maximizing value recovery. Yet, a few frameworks

underscore the collaboration between the stakeholders in the SCs (Cooper et al. 1997; Lambert and Cooper 2000; Lambert et al. 2005; Parmigiani et al. 2011; Stavrulaki and Davis 2010) but do not regard the integration of forward and reverse flows as a priority for maximum value recovery (Nuss et al. 2015). Some frameworks also mention the adaptation of an integrative approach and a shift from a process management to a business model perspective, however, not taking into account the interconnection between the value proposition and the design of products.

16.3 Proposed Framework

This section presents the proposed guiding framework for the manufacturing companies producing multiple life cycle products for implementing CE principles. The proposed framework is geared toward the complexities of CSC and addresses them by offering a structured analysis and decision-making approach. It integrates the knowledge from various disciplines such as CE, SC design, CLSC, reverse logistics, and circular value proposition models into a single descriptive framework to enable the analysis, design, and decision-making aspects of the SCs for CE.

The learnings from previous SC literature (Sect. 16.2.1) and frameworks (Sect. 16.2.2) are incorporated in the CSC framework (Fig. 16.3), and where the SC frameworks are lacking, the authors have proposed additions to reflect the changes anticipated for CSC and CE implementation. Table 16.1 categorizes the recurring concepts in the SC literature into main building blocks and their elements, which are further complemented with guiding questions to implement CSC. A salient feature of CSC is the notion of loop closure by design or intention using a systemic approach in CE implementation. When designing a CSC, it is essential that the recovery strategies supported by a circular product design are aligned with the chosen value proposition model. Moreover, the building blocks such as the levels of decision-making and mechanisms of the CSC framework facilitate in achieving the loop closure by design or intention.

The CSC framework can act as a guiding tool in closing the loop by design for an OEM. However, the starting point for any company may differ depending on the capabilities of the company. Any company exploring to close the loop may use this framework and guiding questions (Table 16.1) to transform their SC into a CSC. The guiding questions will facilitate in understanding the appropriate configuration of value proposition, product design, supply chain, navigating the main decisions required for CSC implementation, identifying the drivers that a company may use to close the loop, and the mechanisms that a company would require for the effective implementation of CSC.

Figure 16.2 presents our conceptualization of the overall structure of CSC explained in Sect. 16.3. This figure illustrates the main processes of CSC from sourcing the raw material to implementation of recovery strategies. This structure highlights the necessity of forward and reverse logistics integration to enable

Table 16.1 Building blocks of the CSC framework

Building blocks	Building blocks elements
Systemic approach <i>(A simultaneous configuration of and coordinated developments in business model, product design, and supply chain)</i> Guiding question: “What is the appropriate approach for companies to implement CSC effectively?”	• Integrated approach: Interorganizational integration (Cooper et al. 1997); intra- and intercompany integration and management (Lambert and Cooper 2000); focus shift from minimizing costs to creating value (Kleindorfer et al. 2005); business perspective, coordinated management of product return management, remanufacturing operational issues, and product market development (Guide Jr and Van Wassenhove 2009); integrated perspective of aligning operational activities with strategic decisions (Stavoulaki and Davis 2010); considering the interdependencies of the main tasks (Nuss et al. 2015). • Holistic approach: Life cycle approach (Guide et al. 2003); system-level approach of integrating product design, supply chain, marketing, and manufacturing and remanufacturing (Ellen MacArthur Foundation 2016; Genovese et al. 2017; Asif et al. 2012); system paradigm (Geissdoerfer et al. 2018); simultaneous configuration for CE holistic approach (Hopkinson et al. 2018); value management approach, including value creation, delivery, use, recovery, and reuse with systemic perspective (Asif 2017; Rashid et al. 2020; Roci et al. 2022).
Main drivers <i>(The factors necessary for an organization to achieve its objectives for, e.g., circularity)</i> Guiding question: “What are the main drivers for companies to close the loop and implement CSC?”	• Business and profit: A revenue opportunity for manufacturers (Govindan et al. 2015); business profit (Guide et al. 2003); business drivers (profit, people, planet) (Flapper et al. 2005); customer expectation (Salomon, 1995; Carter and Ellram 1998; Kleindorfer et al. 2005). • Environment: Environmental demands (Thierry et al. 1995); sustainability (Guide et al. 2003); environmental aspects (Flapper et al. 2005); resource conservation, waste reduction, environmental protection, and maximizing the recovery of materials (Rashid et al. 2020). • Circularity: Circular value proposition (Rashid et al. 2020; Roci et al. 2022). • Policy: Environmental legislation, increase in profitability (Mota et al. 2015); extended producer responsibility (Geyer et al. 2007; Mota et al. 2015); regulations (Salomon, 1995; Carter and Ellram 1998); legal requirements (Kleindorfer et al. 2005).
Levels of decision making <i>(The decision processes for implementing CSC)</i> Guiding question: “What are the main decisions that need to be taken to design an effective CSC?”	• Strategic: Product strategic planning processes, physical distribution structures, plant location and production systems, cooperation among remain supply chain stakeholders, financial factors, remanufacturing design, intellectual property, product recovery value, customer product specs, product life cycle costs, core management, brand erosion, regional remanufacturing operation, organization alignment, government regulations (Subramoniam et al., 2009, 2010); long-term planning tasks regarding the design of reverse SC, product and source selection, acquisition strategy, collection strategy, reverse network design, recovery option selection, recovery technology selection, eco-design, reverse network design, market selection and segmentation, remarketing strategy, reverse network design (Nuss et al. 2015).

(continued)

Table 16.1 (continued)

Building blocks	Building blocks elements
	• Tactical: Product acquisition and return, return disposition (Souza, 2013); market, location allocation, SC network flow (Govindan et al. 2015); reverse logistics network structure, returned products inventory control, remarketing planning for secondary markets (Ellen MacArthur Foundation 2016); determining the characteristics of the revenue-sharing contract and return incentive variables, determining the quality of e-waste, assigning the flows between entities (Govindan and Soleimani 2017). • Operational: Disassembly planning, scheduling (Souza 2013); inventory allocations, order fulfilment (Govindan et al. 2015); short-term, regular, and recurring planning tasks including routine and daily business, transportation planning, return handling and inspection, sorting and grading, reprocessing scheduling, redistribution planning, demand fulfilment (Nuss et al. 2015); location decisions, production quantities, transportation quantities, compromised prices for each product, determining the strategical flexibility and operational flexibility, choosing the optimal strategy set in a product life cycle management scenario and total bottles produced (Govindan and Soleimani 2017); production scheduling, disassembly and reassembly control decisions, inventory routing, new items delivery scheduling, recovery operations scheduling, end-of-life disposal scheduling (Lakhmi et al. 2019).
Mechanisms (The steps necessary for the implementation of CSC) Guiding question: “What are the main mechanisms to implement CSC?”	• Recovery strategy: Used products acquisition, reverse logistics, sorting, testing and grading, remanufacturing or repair and remarketing (Kleindorfer et al. 2005; Guide Jr and Van Wassenhove 2009); value creation through the whole life cycle of a product (Guide and Van Wassenhove 2009); recovery, inspection and testing, remarketing in secondary markets (Ellen MacArthur Foundation 2016); remanufacturing and subsequent life cycle (Diallo et al. 2017); cascading (Lopes de Sousa Jabbour et al. 2019); product type to be returned, entities, recovery operations (Lakhmi et al. 2019); reverse logistics processes, recovery, remarketing (Ellen MacArthur Foundation 2020). • Distribution and collection: Product acquisition, reverse logistics, testing, sorting and disposition, refurbishing, remarketing (Guide et al. 2003); collection of used products, EoL products management, remanufacturing, repairing, or disposing (Govindan et al. 2015); value-added processes and collection of EoL products (Govindan and Soleimani 2017). • Network management: Network design (location and size of collection and remanufacturing facilities), collection strategy, OEM as remanufacturer or not, leasing or selling, buy-back programs, SC coordination in terms of incentives and contracts, complying with take-back legislation (See Chap. 8 by Huang and Atasu 2024), recovery activities and their impact on new product design (Souza, 2013); design of facilities configurations and structures (Govindan et al. 2015); reverse distribution planning, coordination between actors, capacity planning, warehouse storage capacity planning, inspection/sorting and handling capacity planning, return/distribution planning, inventory management (Lakhmi et al. 2019); integration of forward and reverse supply chain (D. D. R. Guide et al. 2003; Kleindorfer et al. 2005; Guide and Van Wassenhove 2009; Asif et al. 2012; Govindan et al. 2015; Govindan and Soleimani 2017; Lopes de Sousa Jabbour et al. 2019; Rashid et al. 2020).

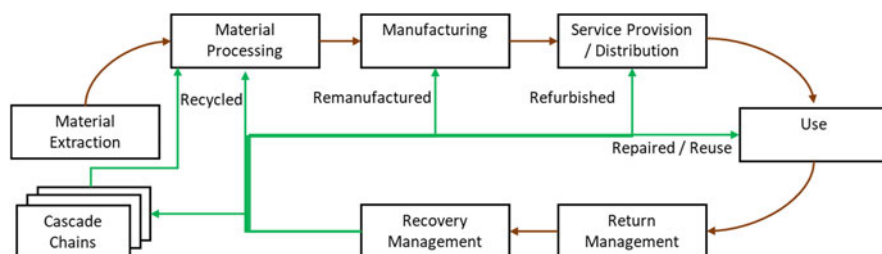


Fig. 16.2 Circular supply chain structure (adapted from Nuss et al. 2015)

closing the loop by design, with uncertainty management in return flows thus enabling multiple use and recovery processes. Moreover, the cascade chains here emphasize the zero-waste vision of CSC management where it collaborates with other organizations to make sure the materials, components, and products that cannot be recovered by the OEM will be used as input flow in other companies. These cascade chains can even take care of so-called production residuals, which may be a valuable resource for other partners in the SC.

To the authors' knowledge, this cross-functional perspective of CSC has not been presented in the literature in the current form, and this framework can lead to additional insights into the sphere of CE implementation. Our proposed framework is presented in Fig. 16.3.

16.4 Case Studies

To demonstrate the application of this framework, we will present two cases of original equipment manufacturers (OEMs) active in the white goods and automotive aftermarket sectors, respectively. They both represent energy- and resource-intensive industries that are relying on virgin materials and generate substantial amounts of waste. In the EU-28, manufacturing generated 10.3 percent of total waste in 2016, being the third most significant contributor after construction and mining (European Commission Eurostat 2014). The OEMs were part of the European Union's Horizon 2020 research and innovation program, and implemented two large-scale demonstrator product service and buy-back systems, respectively. These cases provide an in-depth understanding of CSC implementation and the applicability of the CSC framework that bridges the findings from the literature with a real-world example.

In the following section we give an overview of the case companies and highlight the characteristics of the white goods (Fig. 16.4) and automotive (Fig. 16.5) companies and propose guiding questions for transforming their CSCs in transitioning toward circularity.

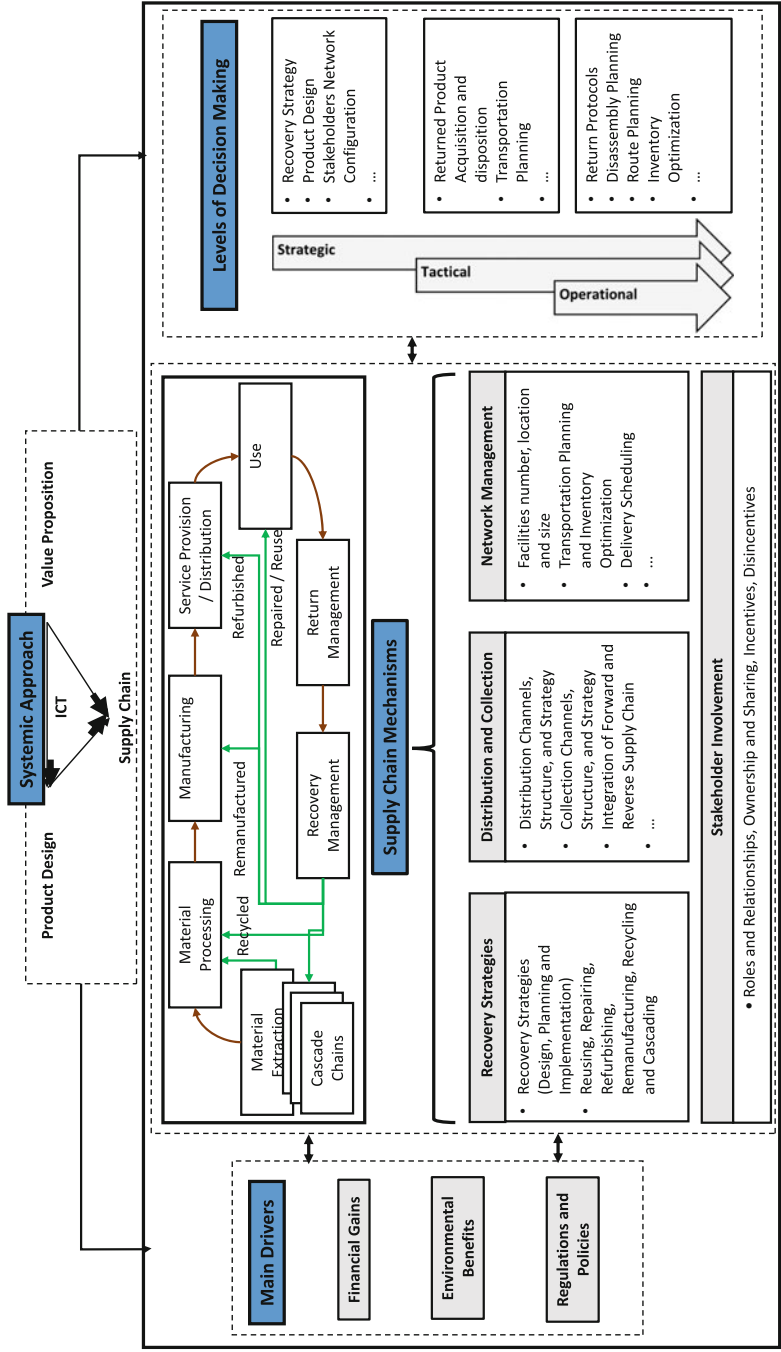


Fig. 16.3 Circular supply chain (CSC) guiding framework

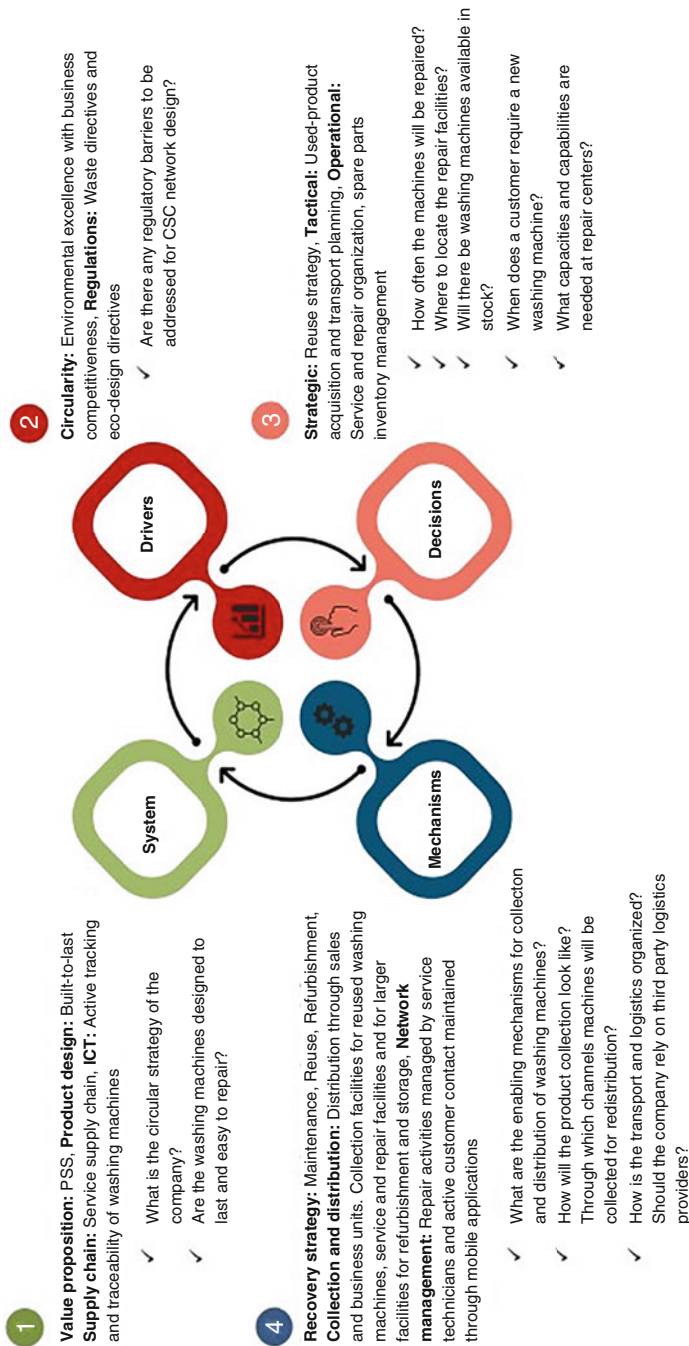


Fig. 16.4 White goods case characteristics and guiding questions

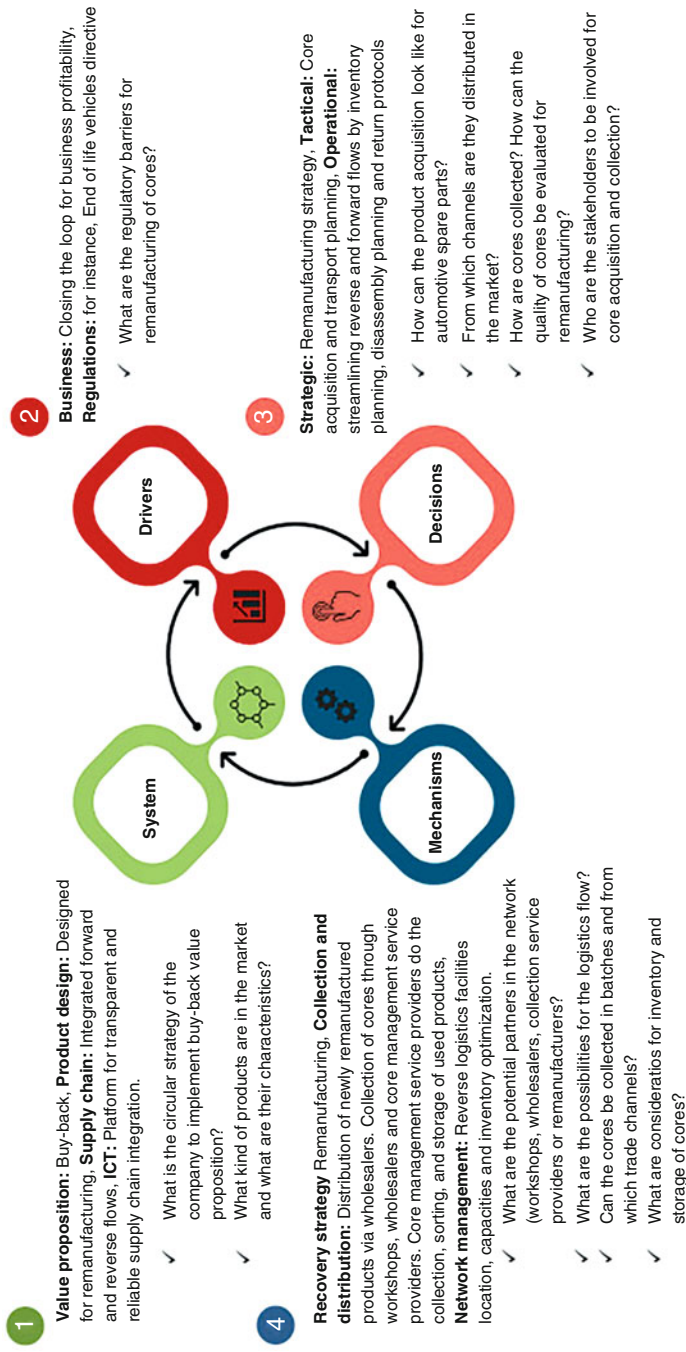


Fig. 16.5 Automotive case characteristics and guiding questions

16.4.1 White Goods Case Company Overview

The first case company in this study is one of the leading European home appliance manufacturers producing ovens, hobs, washing machines, dryers, dishwashers, and so on with a daily output of 19,000 units a day across factories in Slovenia, Czech Republic, and Serbia. This company is currently operating in the linear business logic, which relies on the sales of household products and after-sales services. The company is rethinking the way its products are designed, manufactured, used, and recovered, thereby taking responsibility for its products throughout the entire life cycle. The company has therefore become part of a large-scale demonstrator implementation project, piloting an innovative pay-per-wash scheme for washing machines. The white goods company is planning to develop and implement a pay-per-wash offering for 300 washing machines in four markets—namely, Denmark, Austria, Slovenia, and the Netherlands. These markets were chosen based on sales volume and the existing supply chain infrastructure. The washing machine model relevant for the study is ASKO85, a domestic professional built-to-last model produced in Slovenia. Each washing machine will be refurbished twice and serve over three life cycles of five years. The aim is to reduce costs and environmental impact by combining new sustainable business models with a regenerative approach (e.g., reuse, remanufacture, and recycle). While bringing this innovative service to the market, the company is keeping its ownership of the products, gaining full control of all stages of the value chains (i.e., design, manufacturing, forward supply chains, customer use phase, reverse supply chains, recovery activities, and redistribution). The products designed for this approach are long-lasting washing machines with three life cycles that are capable of reuse, remanufacturing, and recycling. After each use cycle, the washing machine will be refurbished and redistributed to serve another customer. Interested readers can refer to Chap. 14 by Bellos et al. (2024) for an in-depth discussion about product-service systems.

Moreover, these products are also designed for connectivity to enable tracing and tracking throughout the multiple life cycles as well as monitoring the condition of the products to enable predictive and preventive service and repair. After each life cycle, the company needs to take back the washing machines, bring them to as-new condition via value-recovery activities, and then redistribute them to the market.

To support a new business model for washing machines based on multiple use cycles and product refurbishing, the white goods company requires innovation in the reverse and service logistics. The forward supply chain in the sales model begins with the production of washing machines in the factories, and then the machines move to the main warehouse. The products are then sent to distribution centers (DCs) and then further to wholesalers and retailers. The main warehouse is in Velenje (Slovenia) with distribution centers in Celje, Vienna, Duiven, and Brabrand for Slovenia, Austria, the Netherlands, and Denmark, respectively. The wholesalers and retailers have various locations in the chosen four markets. Typically, in the sales model, the washing machine is manufactured in Velenje, then sold in all markets through retailers, whereas in the pay-per-wash offering, the white goods company

will reconsider aspects of location and allocation to recollect and redistribute the products to new customers after refurbishment.

The current supply chain setup at the white goods company will not be sufficient to deal with the product returns, recovery activities, and redistribution. Therefore, the company sees innovation in the reverse logistics operations as an opportunity to be the forerunner in the product-service system domain. Nevertheless, these opportunities require the white goods manufacturer to look closely into reverse supply chains operations for product life extension strategies such as maintenance, repair, reuse, refurbishment, and remanufacturing as well as new customer streams for remarketing the products. This will require the company to set up effective circular supply chains by exploiting the existing infrastructure and by collaborating with other stakeholders such as collection centers, recovery centers, and recycling centers. Therefore, a collaborative stakeholder network capable of integrating the forward and reverse flows are decisive for the company. It is crucial for the company to establish a collaborative network that aims at integrating all the stakeholders involved in the forward and reverse flows. The company intends to expand the existing infrastructure of the service organization that will be responsible for the deployment and management of pay-per-wash offerings in the market. The roles of the service organization can include contract management, machine deployment, maintenance and service, and refurbishment and storage of machines. Moreover, the company is developing an IoT-based platform that enables connection between the washing machines and the value chain stakeholders. The platform will monitor operational and usage data for predictive maintenance as well as process and streamline the data for appropriate stakeholders to use in value management, recovery, and redistribution.

16.4.2 Automotive Case Company Overview

The second case company is an automotive spare parts manufacturer that is active in the automotive aftermarket by remanufacturing a wide range of spare parts. The known benefits of remanufacturing, such as less raw material extraction and waste generation, have motivated the company to design its products for remanufacturing and establish spare parts remanufacturing activities for more than 50 years. Besides product design, successful remanufacturing requires efficient reverse logistics to enable the remanufacturing process. However, with the current supply chain setup of the company, the forward and reverse flows are not integrated, which creates uncertainties in quality, quantity, and timing in reverse flows.

The general structure of the automotive aftermarket is a complex value chain consisting of several stakeholders. These stakeholders include OEM/remanufacturers, wholesalers, and workshops, establishing different trade levels in the value chains. The forward supply chains refer to selling and distributing the new and remanufactured parts to the market through different trade levels. In the forward path, trade levels are the OEM that (re)manufactures the parts, wholesalers are buyers

and distributors to the market, and workshops are the customers of the wholesalers deploying new parts for the end customers (i.e., car owners).

In the reverse supply chains, in addition to the core (used-parts) collection from the end customers (i.e., car owners) delivered to the company, the case company also offers a buy-back value proposition to incentivize the core return flow. The company applies a surcharge to the remanufactured parts; thus, a customer who buys a remanufactured part is credited back the surcharge in case the returned core meets a set of technical and commercial criteria defined by the remanufacturer. These selection criteria consist of core type, condition, and return deadline. Currently the reverse flow of cores (end user to OEM) is not integrated with the forward flow (OEM to the end user); rather, they function as parallel streams, which means that there is a lack of transparency in terms of material and information flows between the OEM and end user.

During the reverse flow of cores, each trade level needs to perform the core identification, evaluation, and sorting to ensure a steady financial flow. It shows that the financial flow is strongly coupled with the physical flow throughout the value chains. This results in inefficiencies in reverse logistics and increases lead time and costs due to the multiple but similar core management and transport activities of different stakeholders for the same cores. To address these inefficiencies and improve the transparency in their value chains, the case company has the ambition to decouple the physical and financial flows by using an ICT platform. The ICT platform aims to offer a reliable and straightforward core return process as well as to manage the financial and information operations. This increased transparency improves the cross-organizational cooperation and collaboration among the involved trade levels as the efficient recovery of used products calls for stakeholder engagement to collect the spare parts and deliver them to the remanufacturer on time and in accordance with the selection criteria. Furthermore, in terms of physical flows, enabling reliable and profitable return process, the company is revisiting its collection structure through consolidating some of the trade levels by optimizing the number of core management facilities and assessing their locations, and capacities.

16.5 Discussion and Conclusions

In this chapter we have explored the role of supply chains in CE implementation and proposed a guiding framework for CSC implementation in the manufacturing industry, followed by two illustrative cases. Circular value proposition models act as the primary driver to move toward circularity, and design for circularity builds the capability in multiple life cycle products to be circulated in the system. A circular supply chain, nevertheless, is a key enabler for CE implementation, in which a systemic approach integrating value proposition, product design, supply chains, and ICT enablers is a prerequisite to close the loop by design or intention.

Based on analysis of the prevailing research in the context of closed-loop and circular-loop closure, it can be concluded that the conventional (linear) approach

of closing the loop addresses product recovery management from the business profitability perspective through waste management (i.e., reactive approach) rather than holistic value management (i.e., proactive approach) and lacks a systemic approach. In the conventional approach, product recovery management refers to the “management of all used and discarded products, components and materials” (Thierry et al. 1995) and proposes reusing, remanufacturing, and material recycling as end-of-life recovery strategies. However, the closed loop in the linear context partially contributes to loop closure without addressing the maximized resource recovery, thereby leaving the loop closure to chance.

Literature review on the SC frameworks revealed that the supply chains configuration aligned with the CE perspective of restoration and regeneration is largely missing. SC management is depicted with several frameworks and conceptual models that differ in their scope, focus, and intention. Many of the proposed frameworks detail the supply chain management activities and consider the return flow management that does not particularly close the loop by intention and through integration of forward and reverse flows. Several researchers acknowledge that CSCs are different than linear SCs and thus require more stakeholders, new relationships, and innovative frameworks to understand and manage the emerging complexities of CSC.

Moreover, analysis of the supply chain frameworks highlighted that the systemic approach is missing in developing these frameworks where value recovery strategies and supply chain structure have not been considered along with the value proposition models and product design. Successful implementation of a circular supply chain strongly advocates for the development of innovative circular value proposition models that enable economically and environmentally feasible ways to close the loop of products and materials. A value proposition model may range from a buy-back, leasing, or a product service system approach that ensures the best value over the life of the product. While the value proposition is considered as a primary driver for CSC implementation, product design also plays a crucial role as these elements of the system operate and evolve dynamically through mutual interactions with the support of ICT infrastructure. Designing products with high durability and reuse capability will facilitate multiple product life cycles. Furthermore, ICT infrastructure is essential for handling the complexity of information management throughout the value chains, especially for tracing and tracking the products over their multiple life cycles. Nonetheless, the value delivery and recovery processes can only be capitalized through an effective CSC design.

In this study, the authors want to emphasize that through closing the loop by intention and working within the inner loops of CE strategies such as repair, reuse, refurbishment, and remanufacturing, we can avoid the SC uncertainties (quality, quantity, and timing of returns). Efficient collection of used products or components is a prerequisite for CSC implementation as they act as input to value recovery operations (e.g., reuse, remanufacture, and recycle). By focusing on the inner loops (reuse, repair, maintenance) of CSC, the recovery strategies can be designed and adapted to a planned value proposition strategy. This in turn minimizes the uncertainties and ensures that the products of right quality are available in the right

quantity at the right timing. Furthermore, for CSC, the vision of zero waste can be realized by further developing cascading application chains instead of disposal. These cascading chains may lead to new CSC collaborations and interorganizational relationships.

The proposed framework emphasizes a systemic approach to implementation of CE, and it further categorizes the circular supply chain elements into main drivers, levels of decision-making, and mechanisms to manage the full loop closure and complexities in the system. To illustrate the applicability and usefulness of the proposed framework, two industrial case studies comprising white goods and automotive spare parts are discussed in light of the CSC framework. Both cases adopt a systemic approach to CE implementation with different value proposition models. The white goods manufacturer is implementing an innovative pay-per-wash service business model in which the user will pay only for the usage of the washing machine. By keeping the ownership of the washing machine, the company gains full control of the product through all stages of the value chain (e.g., design, manufacturing, forward supply chains, customer use phase, reverse supply chains, recovery activities, and redistribution). This will allow the company to predict return flows (quantity, quality, and timing) more accurately for recovering the value and redistribution to the markets. On the other hand, the automotive spare parts manufacturer is operating in a linear business logic built upon the model of product sales and after-sales services. With the intention to develop a closed-loop system and ensure that a large majority of its products are returned for remanufacturing activities, the company has introduced product take-back incentives in the form of deposit schemes. This case relates to a more complex value chain in which the automotive parts manufacturer does not have full control of the product throughout all stages, leading to higher uncertainties in terms of quantity, quality, and timing of reverse flows.

The framework was developed from the literature; thus, the relevance of the framework can be claimed only if the case company can use the framework as guidance to design a CSC. Therefore, we have described the case characteristics and guiding questions to reinforce the building blocks of the framework.

Depending on the company's transition stage toward CE, the starting point for any given company can be different for implementing a CSC. This is further driven by the value proposition approach a company adopts as it allows the OEM to predict reverse flows more accurately. The quantity, quality, and timing of reverse flows can be regulated by contracts—for example, for rented or leased products, or products offered in a pay-per-use scheme—which specify that the provider takes back the product upon contract expiry.

Reflecting on the applicability of the CSC framework from the OEM's perspective—for instance, in terms of decision levels—a company may have a reverse supply chain strategy in place but might be missing tactical-level planning such as location of facilities for efficient reverse flows or reuse procedures on an operational level. However, for closing the loop by intention for effective recovery, both forward and reverse supply chains shall be considered for the product distribution and its management in reverse flows. For effective product collection,

an OEM shall plan its recovery strategy and its channels such as collection done by OEMs themselves, through retailers or third-party partners.

Referring to the white goods case's network management here, the collection strategy of the company may include development of necessary infrastructure such as collection facilities, consolidation centers, and recovery services. The right collection strategy will, for example, enable timely collection of washing machines from customers after their use cycles for necessary inspection, cleaning, and cosmetic changes in the washing machines to prepare them for the next customer.

From a strategic decision level, the automotive case company decides to remanufacture the used parts. This means that strategically the company has a value recovery strategy in place and, on a tactical level, the vision to acquire the cores. However, the operation-level planning might still be missing that is essential for streamlined reverse and forward flows or other operational decisions such as inventory planning or return protocols, which can minimize the SC uncertainties.

The main aim of the chapter was to contribute to CSC research by proposing a guiding framework that clearly and succinctly relates to the proactive approach of closing the loop by design or intention. In contrast to previous SC frameworks, we specifically focused on adopting the systemic approach of CE for CSC implementation through four building blocks: system, drivers, decisions, and mechanisms of CSC. We used our case companies (OEMs) to illustrate the application of the CSC framework. The main contributions of this study can be summarized as:

- Advancing the scholarly discussion on the CSC definition, and its relevance to the existing concepts in the SC literature such as CLSC and CE.
- Consolidating existing SC frameworks to develop a concrete CSC framework that can facilitate in achieving circularity from a systemic implementation perspective and favors the loop closure by design or intention for maximum value recovery.
- Proposing a structured yet flexible framework using the main building blocks, largely in line with earlier SC research (drivers, levels of decision-making, mechanisms), which can guide the decision makers through the whole process of identifying and comparing potential strategies to design CSC.

The framework also has the potential for wider industry implementation, and it can be used to analyze a given CSC design and evaluate its implementation potential. Furthermore, this framework enlists guiding questions that can help OEMs in taking a comprehensive view of CSC design and implementation.

While the CSC framework proposed and illustrated in this work is an important step forward in CE implementation efforts, this work has its limitations. It should be noted that only two cases have been used to illustrate the applicability of the proposed framework. In the future, more industry cases in diverse industrial sectors and contexts should be considered to comprehensively demonstrate the utility of the framework.

This framework addresses mainly the bottom-up approach as it aims at supporting the manufacturing industry in implementing circular supply chains. However, the top-down approach based on social innovation and awareness as well as the involvement of policy makers need to be investigated further since a concurrent

bottom-up and top-down approach is essential to enable an accelerated transition to a circular economy. Essential future developments for CSC implementation will require more extensive work on the development of assessment tools for enhanced decision-making, validating the framework with other companies, and expanding the framework with the pace of CE evolution.

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Chapter 17

Reusable Packaging for B2C Supply Chains



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17.1 Introduction

The world has a tremendous waste problem. Especially packaging waste is growing fast. Most of the packaging waste of the 27 European Union (EU) Member States is made of paper, cardboard, and plastic, contributing 47.6 million tons (around 60%) to the total packaging waste generated in 2019 (Eurostat 2022). Specifically, plastic waste increased tremendously over the last decade. Between 2010 and 2020, the amount of plastic produced globally almost tripled compared to the total amount produced between 1950 and 2010 (Statista 2021). The market for plastic packaging and packaging, in general, is even expected to grow at a compound annual growth rate of 4.2% to 2028 (Grand View Research 2021). However, the value chain of packaging, particularly plastic, is still structured mainly linearly, far from circular. An OECD study revealed that only 9% of the globally produced plastics are effectively reused or recycled. At the same time, 69% of the waste ends up in sanitary landfills or is incinerated, and 22% of the waste leaks into the environment (OECD 2022).

The environmental consequences of insufficient waste treatment are visible around the globe. For example, the Mediterranean Sea is a hot spot for plastic pollution (Burgelman 2021). Plastic particles were also found in very remote regions, e.g., at the bottom of the Arctic Sea (Peeken et al. 2018). Recently,

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microplastics, specifically PET,¹ have been detected in the human blood of almost 80% of tested people (Leslie et al. 2022). Besides personal care products, such as toothpaste, these microplastics are mainly used in packaging materials, packaged food, and plastic beverage bottles (Anderson et al. 2016), and it is still not fully understood to what extent microplastics harm the human body.

Nevertheless, packaging materials belong to our daily lives despite their environmental and human health implications. Paper, cardboard, and plastic are highly versatile materials with many benefits. Firstly, they are light. Compared to other materials, such as glass or metal, the much lower weight can reduce fuel consumption in transportation (McKinnon 2018). A life cycle assessment (LCA, see Chap. 2 by Guinée and Heijungs (2017) for more details) for pasteurized milk finds that recyclable PET bottles are more environmentally friendly than reusable glass bottles in six out of seven indicators (global warming potential, stratospheric ozone depletion, terrestrial acidification, fossil resource scarcity, water consumption, and human carcinogenic toxicity). The main explanations were the high energy demand in production compared to the low number of eight usage cycles and higher emissions from transportation. The only indicator where reusable glass bottles outperformed recyclable PET bottles was marine litter and polluting potential (Stefanini et al. 2021). Secondly, paper, cardboard, and plastic are flexible materials and can be folded and molded into all types of shapes. Such characteristics are desirable for brand customization, specifically relevant to FMCGs. In the food industry, plastic packaging protects food products and can even enhance their shelf lives, e.g., cucumbers wrapped in plastic. While such a plastic wrap creates additional waste, at the same time, it can increase the shelf-life of cucumber for up to 14 days and thus reduce food waste (Dhall et al. 2012). More environmentally friendly wrapping solutions have been developed, such as eatable plant-based layers (Apeel²).

During the past decade, we have observed an increasing demand and legal obligations for more sustainable packaging solutions. One of the first global initiatives was the “anti-plastic bag” movement with the goal to remove plastic shopping bags. Various governments started initiatives to reduce such bags. However, the way in which these initiatives are implemented varies substantially (Clapp and Swanston 2009). At the end of 2020, the European Commission published the “Single-Use Plastic Directive” to reduce the impact of certain plastic products on the environment and thus contribute to Sustainable Development Goal #12 by the United Nations (UN). Since then, the EU member states have widely adopted their national legislations. The packaging law of the EU Packaging Waste Directive,³ implemented by various European countries (e.g., France and Germany) states, for example, that any company selling packaged products to end consumers must

¹ Polyethylene terephthalate (PET).

² Apeel developed a plant-based coating consisting of materials found in peels, seeds, and pulp of fruits and vegetables; <https://www.apeel.com> (July 7, 2022).

³ <https://www.ecosistant.eu/en/eu-packaging-waste-directive-implications-for-e-commerce/>

register their expected annual packaging waste with a public register.⁴ The overall goal is to track all generated waste from packaging materials and thus foster an extended producer responsibility.

While investing in recycling capabilities is one option for reducing packaging waste, business models centered around reusable packaging solutions could ensure that waste is not even produced in the first place (Ellen MacArthur Foundation 2019). However, although the use of reusable packaging appears to be an obvious step toward reducing packaging waste, for which the positive effects on the environment have been scientifically highlighted (Coelho et al. 2020a), the implementation of reusable systems, especially in the business-to-consumer (B2C) sector, is still in its infancy. Moreover, as reusable alternatives are usually sturdier and, thus, more material-intensive than their single-use counterparts, they need to be reused often to yield an environmental and economic benefit. Designing such packaging products and introducing and operating such a system is challenging.

This chapter reviews the development of reusable packaging systems, with a specific focus on the B2C sector. We concentrate on three waste hot spots: (1) the fast-moving consumer goods (FMCG) industry (Sect. 17.3), (2) e-commerce (Sect. 17.4), and (3) takeaway food services (Sect. 17.5). For each application area, we highlight different challenges to be mastered for large-scale adoption and, thus, effective reduction of single-use packaging. Since reusable packaging systems have, to some extent, a common structure across different industries, we first present the theoretical foundation of reusable packaging systems in Sect. 17.2. Finally, in Sect. 17.6, we summarize some future directions for research in reusable packaging systems.

17.2 Theoretical Foundation

The “European Waste Framework Directive” defines “reuse” as “*any operation by which products or components that are not waste are used again for the same purpose for which they were conceived*” (EU Directive 2008). Reusing products multiple times seeks to extend the utilization period and thus “slows” down the flow of virgin resources (Bocken et al. 2016). Reusing packaging is particularly challenging because the actual time of a single usage is typically very short, while the turnover of packaging products is high. Moreover, the packaging is only a vehicle to move the original product through the supply chain and is of little value, so investing in high-value reusable alternatives is often not profitable.

⁴ The packaging register in Germany is “LUCID” <https://www.verpackungsregister.org/en/information-orientation/knowledge-base/using-the-lucid-packaging-register>

17.2.1 Fundamentals of Reusable Packaging

Packaging has various purposes, e.g., protection, preservation, storage, transportation, and information transfer (ten Klooster 2008), and thus affects almost the entire supply chain. To optimally manage packaging waste initiatives, it is crucial to understand the specific function of the packaging and to investigate reusable alternatives.

Every package can, in general, be categorized into one of three packaging types: *primary*, *secondary*, or *tertiary* packaging (Hellström and Saghir 2007). The main purpose of the package implies to which category it belongs. *Primary packaging* has direct contact with the product itself and serves as a container, e.g., the plastic wrap of a vegetable, a container for ice cream, or a plastic bag for pasta. *Secondary packaging* is used as additional packaging, mainly for grouping multiple products in their primary packs, and, besides additional protection, it also provides extra space for branding the product. Finally, the role of *tertiary packaging* is to transport and distribute the products more efficiently through the supply chain (Hellström and Saghir 2007), and it offers additional protection from damage during transport and handling. The most well-known reusable tertiary packaging is probably the standard container (in 20 or 40 ft), with an average lifespan of 25 years.⁵

To understand the relationship between the influencing factors in reusable packaging systems, different conceptual frameworks (reuse frameworks) have been developed. One of the first, the Ellen MacArthur Foundation (EMAF), developed a B2C reuse framework that differentiates reusable packaging systems regarding *package ownership* and the *requirement for users to leave home to refill or return* the packaging (Ellen MacArthur Foundation 2019). They characterized four categories within their framework:

- (i) *Refill at home*: Consumers first purchase the product, including the packaging. Then, for refills, they can buy smaller and more environmentally friendly (recyclable or compostable) refill packages, either online or in the store/restaurant. An innovative refill-at-home example of FMCGs is **Everdrop**, a company that sells cleaning tablets as refillable options for cleaning detergents.⁶ These refill tablets are sold both online as well as in retail stores.
- (ii) *Refill on the go*: Refilling occurs either in a physical store, restaurant, or at some public refilling point. Typical examples are simple water fountains to refill reusable water bottles, the classic coffee-to-go in a reusable coffee mug,

⁵ The lifespan can increase to up to 40 years when the container is properly maintained; <https://www.lotus-containers.com/en/blog/news/lifespan-of-a-shipping-container/>

⁶ The tablet is dissolved in 500 ml of water and glass cleaner, bathroom cleaner, and all-purpose cleaner; <https://www.everdrop.de/en>

- and **Ecover**, which offers refill stations for laundry liquid in retail stores in several European countries.⁷
- (iii) *Return from home*: The empty reusable package is picked up from home, especially relevant to the e-commerce sector. An example is the global reuse platform **Loop**, which allows consumers to purchase grocery products in reusable packaging and return them efficiently with the next delivery.⁸
 - (iv) *Return on the go*: This model is widely used in the beverage industry worldwide. Consumers can return empty packages, typically in retail outlets at pre-defined reverse vending machines (RVMs).

Coelho et al. (2020a) developed a similar framework and classified reusable packing into four types distinguishing the packaging ownership. Their first model, *Refillable by Bulk Dispenser*, is comparable to the *Refill on the go* model of the EMAF framework, where consumers use their own reusable packaging to be refilled in stores or restaurants. The *Refillable Parent Packaging* type is similar to the *Refill at home* of EMAF, where consumers purchase refill packages, which are more environmentally friendly than the parent packaging that will be refilled. The two other packaging types of their framework are *Returnable Packaging*, assuming consumers return reusable packages to the company that offers the packaging or that is responsible for initiating the cleaning and refilling process,⁹ and *Transit Packaging*, typically used to efficiently move products, in B2B markets, for example, pallets, crates, IBCs, and containers.

The framework by Muranko et al. (2021) is more comprehensive as it considers whether and how a reusable product is offered together with the logistical infrastructure. Their framework distinguishes an “*exclusive reuse*,” i.e., a single customer exclusively uses a reusable product, and a “*sequential reuse*,” where different customers use the reusable package sequentially. The possibility of offering an exclusive or sequential reusable product system with or without reusable-enabling infrastructure or the reusable-enabling infrastructure as a stand-alone system results in five different models.

17.2.2 Organization of Reusable Packaging Systems

Most reusable packaging systems have a similar organizational structure independent of the specific application. This generic form is illustrated in Fig. 17.1.

⁷ On their website Ecover provides information in which store consumers can find refill stations. Here is an example for the UK; <https://www.ecover.com/where-can-i-refill>

⁸ Loop is a “circular packaging solution” launched in 2019 by recycling experts Terracycle; <https://buydurable.com/>

⁹ Companies can be retailers (e.g., bottles and cans) but also restaurants or food producers.

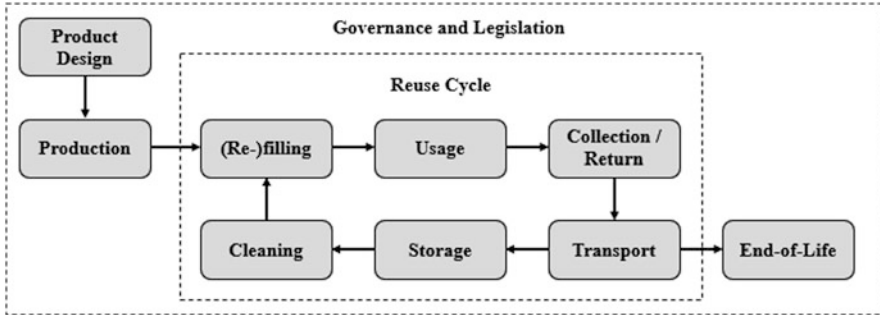


Fig. 17.1 Supply chain (generic) of a closed-loop cycle for reusable packages (adjusted based on Hesseling 2022)

Product/Packaging Design The used material (recyclable) and the right packaging design (affecting the degree of filling and material intensity) are crucial when moving from a linear supply with disposable packaging to an economically viable circular reusable packaging system (Pålsson et al. 2013). Ensuring reusability requires the package to be designed to be resistant to exposure, handling, and transport and can withstand multiple use cycles. At the same time, it requires environmentally friendly materials since reusable products are outdated sooner or later.

Production The production process, mainly influenced by the material used, is one of the main contributors to CO₂ emission. Reusable packaging is typically of higher quality and requires more emission-intense production processes than single-use (disposable) packaging. Therefore, reusable packaging needs multiple use cycles to become more sustainable than disposable packaging (Coelho et al. 2020b).

(Re-)filling The first step within the reuse cycle is the (re-)filling operations. The product design directly influences the efficiency of the (re-)filling process. The more standardized and simpler packaging is designed, the more (cost) efficient the (re-)filling.

Usage The usage (retention) time, i.e., the time a packaging spends with the customer, affects the economic viability of reuse systems. Like the lead-time effect in inventory control systems, the usage time affects how many reusable packages firms need to keep in stock to provide the desired level of availability (Mollenkopf et al. 2005).

Collection/Return Organizing the collection and return process is one of the most challenging tasks in B2C markets. While in B2B, logistics for reusable packaging is typically organized by a single firm—more or less—in a closed system, the B2C market is more fragmented. Moreover, end consumers (responsible for the return) are less predictable if and when they return a package and require sophisticated incentives for returns. The lower the return rate, the fewer reusable packages stay in

the reuse cycle, implying fewer economies of scale of the logistics and operations processes and, at the same time, more frequent new packaging injections. The higher the return rate, the more environmentally viable is entire reusable packaging system (Coelho et al. 2020a). To encourage customers to return reusable packages, the most common incentive systems are *deposit return schemes (DRS)*, where customers pay a small fee added to the product price, and once the customer returns the empty package, the deposit is reimbursed (Calabrese et al. 2021). The challenge is finding the right deposit level. A too low deposit yields that customers may not return the empty package; however, if the deposit is too high, customers may not purchase the product. Another possible incentive scheme is, for example, to offer a price discount for the actual or future purchase. Such incentives are, for example, provided by various coffee store chains.¹⁰

Transport Various studies show that transport distance is a critical factor for the sustainability of reusable packaging systems (Coelho et al. 2020b; Stefanini et al. 2021). If the travel distance is too long, the negative emission impact of transport outweighs the benefits of reusing packaging multiple times. Therefore, more challenging to design a reusable packaging system on a global level. To avoid the adverse effects of long-distance transport, there is common sense that reusable packaging systems should be implemented rather regionally or locally (Ferrara and de Feo 2020).

Storage and Cleaning A sufficient capacity to store and clean the packages is crucial before packaging can be reused. Depending on the reusable package system (e.g., for FMCG, e-commerce, or takeaway food) that determines the size and structure of the network, storage, and cleaning can be more or less challenging. For example, in the brewery and beverage industry, where empty bottles are first stored at retailers and wholesalers, cleaning, and refilling typically occur together at the producer (Simon et al. 2016). In small restaurants or food service providers, however, there is often only little possibility of cleaning and storing a large number of reusable packaging on site.

End-of-Life Similar to single-used packages, also reusable packages reach their end-of-life status sooner or later. Therefore, before deciding about the type of reusable packaging (design and production), it is important to understand the environmental impact at the end-of-life. For a sophisticated life cycle analysis (LCA), these effects (and costs) must be considered (Ferrara and de Feo 2020).

Governance and Legislation So far, there are only a few general legislations and standardizations for reusable packaging systems. One reason is that there are no clear definitions of assessing a package's reusability. The "Single-Use Plastic Directive" released in 2020, for example, strongly impacted the restaurant and food

¹⁰ In June 2022, Starbucks reintroduced personal reusable cups across company operated stores in the USA and is offering customers a \$0.10 discount; <https://stories.starbucks.com/stories/2021/starbucks-brings-back-personal-reusable-cups-to-starbucks-cafes-in-the-us/>

service industry as it banned various products from the market, for example, single-used cutlery (forks, knives, spoons, etc.), plates, straws, or food containers made from polystyrene.¹¹

17.2.3 Drivers and Challenges of Reusable Packaging Solutions

Reusable packaging systems have a clear objective: to conserve resources and reduce waste production. Besides the ecological drivers, there are other advantages to implementing reusable packaging solutions. First, reusable packaging requires only a one-time capital investment and pays for itself over several months or years if return rates are sufficiently high and the return logistic is organized efficiently. This implies that costs for procurement and disposal of single-use packaging can be eliminated. In addition, reusable packaging can boost sales. Consumers are more environmentally concerned about their lifestyle and buying behavior, and sustainability is a key consideration for a variety of consumer segments when making their purchase decisions.¹² Retailers already experienced an increase in demand for sustainable products and shopping formats, and an increase in the willingness to pay for such products. This trend will likely continue in the future due to the changing buying behavior of Gen Z.¹³ Reusable systems have also the potential to promote regional businesses and to establish market advantages for domestic producers since the environmental impact of transporting empty packages is often cited as an argument against reusable packaging (Coelho et al. 2020a). Finally, governmental regulations can drive the implementation of reusable packaging systems, e.g., the obligation to offer reusable packaging for drinks and meals to take away in Germany since 2023.

However, establishing reusable packaging systems in B2C at a larger scale faces various challenges. A successful implementation requires not only general acceptance by all parties involved in the system, i.e., manufacturers, retailers, restaurants/food services, and consumers, but also a well-functioning logistics infrastructure (e.g., return and cleaning locations). The acceptance of consumers is the make-or-break of successful reuse systems in B2C markets. Without consumers' willingness to buy products in reusable packaging and return the empty packaging, reuse systems are neither economically nor environmentally viable. However, to

¹¹ Directive (EU) 2019/904 of the European Parliament and of the Council of 5 June 2019 on the reduction of the impact of certain plastic products on the environment; <https://eur-lex.europa.eu/eli/dir/2019/904/oj>

¹² A study by Deloitte revealed that especially the over 45s are most likely to reduce their consumption of single-use plastic; <https://www2.deloitte.com/ch/en/pages/consumer-business/articles/shifting-sands-sustainable-consumer.html>

¹³ <https://www.forbes.com/sites/gregpetro/2022/03/11/consumers-demand-sustainable-products-and-shopping-formats/>

Table 17.1 Challenges and drivers of reusable packaging solutions for different stakeholders

Factors	Challenges	Drivers
Customer's acceptance and participation	– Hygiene concerns – Quality of packaging – Sensitivity to price and deposit fees – Inconvenience of return	– Environmental benefit – Awareness of sustainability – Enhancing customer's experience (i.e., shifting from low-quality to high-quality packaging) – Efficient collection systems (i.e., deposit system)
Participation of manufacturers, retailers, and restaurants	– Investment costs – Operations cost – Risks concerning customer behavior – Underdeveloped infrastructure (cleaning, return, and logistics networks)	– Green image and reputation – Demand shifting toward more sustainable products – Governmental regulations – Market share concerns due to competition – New business opportunities

reach a high level of consumer acceptance is a difficult task. Many consumers are used to the convenience of single-use (disposable) packaging. However, to achieve a regime change, consumers need to see reusable packaging not simply as waste but as a safe asset that can be conveniently returned (Young 2022).

Table 17.1 represents various challenges and drivers of reusable packaging solutions for both consumers and companies (manufacturers, retailers, or restaurants). Hygiene standards are one of the major consumer concerns (Jiang et al. 2020). During the pandemic, consumers feared that the coronavirus could be transmitted via surfaces and thus tried to avoid reusable packaging, specifically for food. Many countries, in response, restricted the usage of reusable containers. Packaging quality and esthetics are another concern of reusable packaging in B2C. While in B2B markets, the functionality of reusable packaging is of main relevance, in B2C markets also, aesthetical components (e.g., the appearance of the packaging) play an essential role. Companies are concerned that the quality (e.g., due to scratches) could degrade after a few use cycles, and thus affecting customer acceptance. Not to be neglected the effort and cost for the customer to purchase and return the reusable packaging. The price of using a reusable packaging alternative (deposit) and the effort to return it may significantly affect the acceptance level and thus the success of large-scale implementation (Calabrese et al. 2021; Kulshreshtha and Sarangi 2001).

Besides consumer acceptance, also companies in B2C markets must be convinced to invest in reusable packaging systems. Although providing sales opportunities due to a demand shift toward more sustainable products, reusable packaging solutions require significant investments and incur higher costs for operations (inventory management, logistics, and cleaning). The first obstacle is designing the right product with the optimal balance of esthetic (to fulfil marketing purposes), durability (to withstand multiple wash cycles), and standardization (for efficient logistics processes). In addition, companies must invest in a suitable infrastructure

(IT system for booking and/or deposit systems, return, and cleaning networks), which is at this stage not sufficiently developed. Moreover, the investment and operations cost only pay off when customers accept reusable packaging and return them in a reasonable time. Companies are still concerned that high-level investments will not materialize in the future because customers will not use them sufficiently and/or will not be willing to pay a higher price. However, customer acceptance and willingness to pay are still not fully understood and provide a potential for various research in the future.

17.3 Reusable (Primary) Packaging in the FMCG Industry

While FMCG companies try to reduce packaging waste, recycling is still the most common practice in their waste management strategies for primary packaging. Most companies primarily focus on replacing virgin plastics with recycled contents (Kunamaneni et al. 2019). For example, Henkel follows a plan to reduce packaging waste through its “100-50-Zero” strategy. The main aim is to achieve 100% recyclable and reusable packaging, reduce the usage of virgin plastics by 50%, and reach a target of zero plastic waste by 2050 (Gibson 2020). Although recycling is still the most common practice, the low value of recycling and the urgent need to reduce plastic leakage call for alternative ways following the triple-r model (reduce, reuse, and recycle) of the circular economy. However, working examples of “reuse” models for FMCGs are rare.

17.3.1 Development of Reusable Packaging Systems in FMCG Markets

Reusable packaging systems for FMCGs are not new. The origin of reusable bottles dates back to 1903, when beer traders in Frankfurt, Germany, agreed to introduce a joint deposit system to encourage consumers to return empty beer bottles. Later, in 1969, the “Cooperative of German Mineral Water Companies” introduced a uniform reusable glass bottle for mineral water, called the “pearl bottle,” one of the most successful reusable packaging systems.¹⁴ In the 1970s, in Germany, a reusable glass container for yogurt and milk products was introduced, called the “MMP”¹⁵ glass. Besides the beer and beverage bottles, this standardized MMP glass is widely accepted by retailers within Germany and is compatible with most reverse vending machines. Recently, the MMP glass is also used for other (non-dairy) products, for

¹⁴ <https://www.industrialmeeting.club/reusable-pet-pearl-bottle-for-mineral-water-classic-for-german-packaging>

¹⁵ MMP stands for “Milchmehrwegpool,” translated as reusable pool for milk products.

example, tea, rice, or beans. Besides these early examples for beverages, large-scale reusable packaging systems in FMCG are still rare.

In recent years, we can observe a growing number of trials and projects around reusable packaging systems in FMCG markets. Several FMCG manufacturers started corporations with retailers to implement reusable packaging solutions convenient to consumers. In 2019, the USA-based recycling company **TerraCycle**¹⁶ founded **Loop**,¹⁷ a shopping platform for FMCGs in reusable packaging. Loop started as an online platform in Paris (France), cooperating with the French retailer Carrefour and Kroger in the northeast of the USA (Grace 2019). Besides offering a reusable tote used for transporting the products, Loop also collaborated with FMCG manufacturers to offer products like ice cream, shampoo, and laundry detergents in reusable containers. In 2020, Loop started expanding to physical stores, including Carrefour (France), Tesco (UK), AEON (Japan), Kroger (USA), and Woolworths (Australia). Loop estimates that after 3–4 use cycles, the average cost of a reusable container would break even, while the maximum number of use cycles is significantly more (Grace 2019). In 2019, the Rotterdam-based grocery start-up **Pieter Pot**¹⁸ started a package-free online grocery store in the Netherlands, and in August 2022, the German retail chain REWE Group announced cooperating with reusable packaging start-up **Sykell**,¹⁹ which plans to develop a return system compatible with the existing return network in Germany.

All use cases mentioned above belong to the class of *sequential reuse* systems, introduced in Sect. 17.2, where customers buy products in reusable packaging from a retailer and return them either directly to the retailer, through the next online purchase, or to any retailer or return station that belongs to the network. Other business models that are more frequently observed belong to the class *Refill-on-the-go*. For example, in 2018, the Austrian company **Planet Pure**²⁰ collaborated with the retail chain **dm** and installed in-store refilling stations for laundry and dishwashing detergents. Later they expanded their business to Germany, Slovakia, and Croatia. Worldwide we observe an increase in package-free stores where customers purchase products in bulk (typically dry products or oils) and can bring their own containers. Zero Waste Europe (2020) expects sales of package-free stores to triple by the end of 2030. However, further research is needed to understand better the logistical challenges of a large-scale adoption (Zero Waste Europe 2020).

¹⁶ <https://www.terracycle.com>

¹⁷ Loop expanded to Canada, the UK, and Japan; <https://exploreloop.com>

¹⁸ <https://www.forbes.com/sites/jeroenkraaijenbrink/2021/01/20/how-this-dutch-startup-is-going-to-disrupt-the-supermarket-landscape/?sh=7cec9d31469d>

¹⁹ <https://www.sykell.com>

²⁰ <https://www.planetpure.com>

17.3.2 Challenges for Large-Scale Adoption

FMCG supply chains are complex, so investing in large-scale reusable packaging systems is capital intense. Compared to reusable packaging in e-commerce (primarily dealing with tertiary packaging) and takeaway food service (implementable on a local level), managing reusable packaging for FMCGs is far more challenging. First, multinational FMCG manufacturers operate on global markets, characterized by high-volume manufacturing, high turnover rates, and low margins due to intense competition (Meyr and Stadler 2008). Introducing reusable primary packaging for FMCGs requires large-scale investments along the entire chain, e.g., in the reusable package itself, which is typically more expensive than single-use packaging, new packaging (filling) machines, collection/reverse logistics (e.g., reverse vending machines), and cleaning processes. Such investments must be planned well ahead of time, typically several years (van Velzen and Brouwer 2022).

Many FMCGs are sold globally; however, the feasibility of the reusable packaging solution differs from region to region (Bocken et al. 2022). The technical and economic feasibility of reusable packaging seems more challenging in regions suffering inefficient collection processes or customers not concerned about packaging sustainability. On the other hand, reusable packaging solutions seem more affordable in European countries as consumers are more interested in sustainable solutions. Of the 15,620 consumers across the USA, Europe, and South Africa, 74% of consumers are willing to pay more for products with reusable packaging. Twenty-five percent of consumers are willing to pay an additional 10% or more. Seventy-four percent of European participants are eagerly looking forward to sustainable packaging solutions. Furthermore, the demand for sustainable packaging is growing and robust (Trivium Packaging 2020). Another study conducted in Portugal shows that 70% of consumers are willing to pay additional between 1 and 5%, and 22% are not willing to pay more (Martinho et al. 2015). These studies reveal that although consumers are willing to pay more, the extent of additional payment is still low and the price of the product is still the first and foremost factor in consumers' purchasing behavior. As highlighted in Sect. 17.2, transport distance is a fundamental concern with reusable packaging systems, and the FMCG industry is characterized by long-distance transport. An LCA study for glass bottles (beverage industry) indicated that a reuse system only outperforms single-use packaging when the total travel distance of the reusable packages does not exceed 150 km (Nessi et al. 2012). While this number is certainly not precise because it depends on various factors, such as transport mode, load factors, and network structure., it still provides a reasonable indicator. In the FMCG industry, truck transport is still the dominant transport mode and thus has a significant impact on CO₂ emissions (Palmer et al. 2018). Since the FMCG industry is transportation-intensive, introducing (economically and environmentally) feasible reuse ecosystems is challenging.

Compared to tertiary reusable packaging in e-commerce, the primary packaging of FMCGs is an important marketing (brand) instrument. On the one hand, the packaging needs to provide all functionalities (as explained in Sect. 17.2), but

on the other hand, FMCG companies are concerned that their products grab the most consumer attention, which is significantly influenced by the packaging design (Chen et al. 2020). Therefore, introducing products in reusable packaging may benefit FMCG companies by growing consumer loyalty, enhancing the green image, and increasing product variety; however, uncertainty about consumer acceptance, willingness to pay for the packaging, and additional investment and operations costs can quickly outweigh the benefits.

Reusable packaging systems in FMCG businesses are usually managed via deposit systems. Governments increasingly use deposit-based return systems worldwide to tackle the waste problem (Reloop 2020). The deposit system has already successfully implemented to manage beverage packaging in several countries (i.e., Germany, Iceland, and Switzerland). However, the deposit value depends on the packaging size and varies from country to country (Calabrese et al. 2021). While the collection rate of beverages is relatively high (in Germany more than 95%, see Oltermann 2018), Loop's deposit initiative in Paris reaches a collection rate of around 70%.²¹ In general, the value of the deposit fee plays a key role in incentivizing return rates (Mahmoudi and Parviziomran 2020). While a high deposit fee ensures a high collection rate but may adversely affect customers' willingness to buy products in reusable packaging, a low deposit fee encourages more customers toward the reusable packaging but may cause a lower collection rate because of missing incentive to return them (Zhou et al. 2020).

Not-returned packages are a fundamental concern, especially for manufacturers that benefit from the economies of scale in production and operations. Moreover, it requires that new packages are produced, which is twice negative from an economic and environmental perspective. However, retailers, on the other side, may not have the same incentive to push for packaging returns as manufacturers. On the one hand, not-returned packages may drive revenue for retailers from forfeited deposits, and on the other hand, when retailers are responsible for collecting the packages, they face additional costs. From a supply chain management perspective, this may lead to misaligned incentives. Reusable packaging ecosystems thrive only when all stakeholders sufficiently cooperate and share the cost and benefits of managing such systems.

17.4 Reusable Packaging in E-Commerce

Around the globe, e-commerce is on the rise. After 2 years of unpredictably high growth during the COVID-19 pandemic in 2020/2021, the growth of global e-commerce spending has slowed down a bit. However, it is likely to exceed \$5 trillion in 2022 and is expected to surpass \$7 trillion by 2025. Worldwide, e-commerce

²¹ <https://www.olivierdauvers.fr/2021/06/13/exclu-le-premier-bilan-de-la-consigne-loop-chez-carrefour>

accounts for over 20% of retail sales (Cramer-Flood 2022). As a result, also the global parcel volume saw high growth rates. In 2020 alone, it grew by 31% in China, 46% in Brazil, and 33% in the UK. Keeping up with the continuing e-commerce growth, the global parcel volume is forecasted to double in the next 5 years (Dies 2021).

Every e-commerce transaction triggers a delivery process that causes a range of environmental impacts: storing, commissioning, packing, and shipping the product to the consumers, for example, produces emissions, consumes resources, and leads to packaging waste (Escursell et al. 2021). The shipping packaging (tertiary packaging) can account for 10–30% of the CO_{2eq} emissions, depending on the product category (Zimmermann and Bliklen 2020). In Germany alone, the mountain of cardboard used for shipping parcels in 2018 weighed 830,000 tons (Reitz 2020). Similar to packaging in FMCG and takeaway food, almost all shipping packages used today are single-use items. Typical types of packaging are flexible bags used for textiles (made from different types of plastics, approximately 20–30 g) or rigid boxes used for most other product categories (made from cardboard, about 50–430 g) (Zimmermann and Bliklen 2020).

Before the background of this continuously growing waste problem, several attempts to develop and launch reusable packaging for e-commerce have been made over the past years (Coelho et al. 2020a). They all aim to reframe packaging from a single-use waste source to a multi-use asset. **RePack** (Finland),²² for example, leases shipping bags of different sizes that, after use, can be folded and sent back via mail to central reverse logistics hubs. They are made from recycled polypropylene and weigh between 80 and 180 g. The company claims that a single bag can be reused up to 20 times. **Limeloop** (USA)²³ operates a similar model but equips its shipping bags with GPS sensors to track shipment status and increase return rates. **LivingPackets** (France)²⁴ provides a foldable shipping box equipped with temperature and motion sensors and an e-ink display. They claim that the box can last for 1000 journeys. These are just three examples from the broader start-up landscape around reusable e-commerce packaging. They all provide very different packaging solutions for specific use cases, some with added services.

17.4.1 Factors Influencing the Environmental Benefits

Like the FMCG example before, whether reusable packaging is actually better for the environment than a single-use item depends on many factors, for example, the number of use cycles, the material the packaging is produced from, and the distances it travels between distribution centers, customer, and cleaning facility (Escursell

²² <https://www.repack.com/>

²³ <https://thelimeloop.com/>

²⁴ <https://livingpackets.com/>

et al. 2021). So far, only a few LCA studies have been published that shed light on the environmental merits of reusable e-commerce packaging under different assumptions (Coelho et al. 2020a).

The key question of these studies is how many cycles a usually more material-intensive reusable packaging has to run through (including return logistics and cleaning) until it creates an actual environmental benefit (Mahmoudi and Parviziomran 2020). For example, Zimmermann and Bliklen (2020) analyze the carbon footprints associated with returnable boxes from hard plastics and plastic bags and compare them with their single-use counterparts. For the hard plastic boxes (polypropylene, 14 l volume, approximately 1.7 kg), they find that it needs 82 cycles until the reusable packaging produces fewer emissions per voyage than a single-use carton (recycled cardboard, same volume, 180 g). If the single-use carton is not made from recycled cardboard but from primary material, it takes 61 cycles to break even. If, however, the hard plastic box is made from recycled polypropylene, the number of required cycles lowers to 32. These results exemplify the immense impact of material choice on life cycle emissions. For the plastic bags (polypropylene, 21 l volume, 118 g), they define two alternatives: a single-use shipping bag (low-density polypropylene, same volume, 30 g) and a single-use carton (cardboard, same volume, 180 g). Compared with the carton, the reusable bag performs better from the first use. It takes eight cycles until it also has a better per-voyage carbon footprint than the single-use bag. Overall, Zimmermann and Bliklen (2020) show that material choices matter significantly. The heavier a packaging is and the less it contains recycled materials, the more often it needs to be reused to break even with single-use alternatives. In principle, these results are in line with the findings for FMCG packaging introduced before. However, the cleaning process has a lower impact on the feasibility of reusable e-commerce packaging than it has in the FMCG sector.

A second example is the LCA by Hugill et al. (2021). The authors focus on fashion e-commerce and compare a single-use low-density polypropylene shipping bag (30% recycled content) with a reusable alternative from 70% recycled polypropylene. Their base case assumes that the average distance to the next cleaning node is 1500 km from the consumer and that the non-return rate is as low as 5%. With these assumptions, the authors find that the environmental break-even is reached after four cycles. They additionally conduct sensitivity analyses to shed light on the influence on transport distance and return rate. Assuming 30 cycles, they find that for large distances of 2000 km each between the distribution center and consumer as well as between consumer and cleaning node, the emissions footprint of reusable and single-use packaging is almost equally high. If consumers, distribution centers, and cleaning nodes are located closer to each other (up to 500 km), the reusable shipping bag saves 75% of emissions per voyage compared to the single-use counterpart. They also analyze the impact of the non-return rate and find that a non-return rate of 20% already leads to higher per-voyage emissions for reusable bags. If compared with a single-use bag with a higher recycling content (80%), the reusable bag only performs better for a non-return rate of below 5% and rather short transport distances. Overall, the findings of Hugill et al. (2021)

underline the large impact that material choices during product design have on the environmental evaluation of single use vs. reusable shipping bags. Further, they clearly show the crucial impact of a very high return rate. Of note, both studies focus on the climate effect of packaging and do not investigate other environmental impacts.

17.4.2 Challenges for Large-Scale Adoption

Both studies introduced above clearly showcase the environmental benefits of available solutions (especially in the case of reusable plastic bags). Still, reusable packaging plays a minuscule role in e-commerce (Escursell et al. 2021). Some large e-commerce retailers like Walmart in the USA and Tchibo in Germany have been trialing solutions. However, such solutions (if pursued at all) are far from breaching the mainstream (Coelho et al. 2020a). Two main challenges are seen to inhibit a larger uptake of reusable packaging in e-commerce: a) designing cost-efficient return logistics systems and b) achieving a high return rate by consumers (Escursell et al. 2021; Mahmoudi and Parviziomran 2020; Zimmermann and Bliklen 2020).

- (a) *Return logistics system*: To achieve a cost-efficient return logistics system, horizontal and vertical collaboration among value chain partners and competitors is seen as a crucial prerequisite (Mahmoudi and Parviziomran 2020). Furthermore, similar to other types of reusable packaging, pooling of resources, standardized packaging types and sizes, and decentral cleaning and maintenance in the distribution centers of retailers are considered to increase the overall efficiency and scalability of the return systems (Hugill et al. 2021; Zimmermann and Bliklen 2020). Also, the questions of how individual packages can be tracked and how a high number of return location points can be achieved need to be addressed. This challenge was specifically addressed in a 2022 pilot study²⁵ in Austria. The Austrian Post collaborated with a few large e-commerce retailers to pilot different reusable packaging alternatives from reusable cardboard boxes to PET bags. The consumers simply dropped off the packaging through the Austrian Post mailboxes or shops found at thousands of locations throughout the country—the consumer-facing end of the return logistics systems was already in place. The pilot was a success (also from an LCA perspective) and will become a regular service in 2023.²⁶
- (b) *Return rate*: As highlighted by the LCA studies (also on FMCG packaging), a high return rate is a prerequisite for creating environmental benefits and operating a financially feasible system. However, even very high return rates of 95% might only enable modest numbers of use cycles (Zimmermann and

²⁵ <https://unsere.post.at/umwelt/post-und-fh-oe-forschen-an-gruener-verpackung,116.html>

²⁶ <https://logistik-heute.de/news/nachhaltigkeit-oesterreichische-post-fuehrt-gruene-verpackung-als-regulaeren-service-ein-38357.html>

Bliklen 2020). While high deposits seem to be a compelling incentive to boost return rates and limit retention time, this might not be feasible in the current highly competitive e-commerce market. Reframing packaging from a waste source to an asset requires fundamental behavioral changes on the consumer side. Therefore, education and assistance are seen to be core prerequisites for an extensive rollout of reusable packaging (Hugill et al. 2021). The pilot project in Austria referenced above also struggled with too low return rates. For the regular service from 2023 on, e-commerce partners are encouraged to implement a deposit system.

While legislation around take-back systems (see Chap. 8 by Huang and Atasu (2017) for more details) has been proven to effectively enforce change in industry practice and consumer behavior (take the European directive on electronics waste of the German deposit system for single-use plastics bottles as an example), legislation to enforce or incentivize the reusability of e-commerce packaging are not yet in place. Also, the public discussions in most countries on this issue are in an early stage (Coelho et al. 2020a; Escursell et al. 2021). However, as the mountains of packaging waste from e-commerce are forecasted to grow fast, legislation might be called for soon.

17.5 Reusable Packaging in Takeaway Food Systems

Takeaway food and drinks are still primarily served in single-use containers. The worldwide consumption of such disposable containers in the food and restaurant industry has increased significantly in recent years. Between 1994 and 2017, Germany saw an increase of 491% in single-use mugs and cups for hot drinks (including vending machine cups and lids); 413% in single-use menu and snack boxes, e.g., lunch and pizza boxes; and 1098% in single-use containers for food, e.g., soup, fruit, and cereal cups (Kleinhückelkotten et al. 2021). The strict lockdown restrictions and the high hygiene and safety regulations during the pandemic in 2020 have even raised this consumption (Song et al. 2021). We can expect that the number of food delivery services and take-home food sales will remain high as many customers have become accustomed to the convenience of such food services (Durbin 2021). However, this surge has dramatic implications for the environment.

A promising alternative to single-use food packages is a reusable alternative, namely the reusable cup and container (RC). However, like FMCG and e-commerce, the RC is only a valuable alternative if production, usage, cleaning, and disposal are sustainable from both an environmental and an economic perspective (Gallego-Schmid et al. 2019). Moreover, many single-use packages in the food sector cannot be recycled as they typically contain certain non-recyclable plastic coating (e.g., in single-use plastic cups, which is needed to strengthen the cup, prevent leakages, and provide additional insulation). In addition, food packaging can contaminate

recyclable materials due to food residues, making them unusable for recycling (Geueke et al. 2018).

17.5.1 Regulations and Governance

In May 2021, the German Federal Parliament voted in favor of an amendment to the EU Packaging Act, requiring from January 2023, restaurants and retailers that sell food and beverage for immediate consumption to offer the products in reusable packages. The only exception was made for small restaurants with less than eighty square meters of selling space and less than five employees.²⁷ The new law, however, does not ban single-use packages altogether; instead, it only requires offering reusable packages as an alternative to single-use packages, meaning that restaurants are also allowed to offer both reusable and single-use (disposable) packages in parallel. While the law aims in the right direction for restaurant and food service providers, many questions arise regarding its implementation and operation, for example: How should they implement this law correctly in their own operations? How many reusable packages should be held in stock? How should the restaurant (chain) organize collection and cleaning operations? Should it be managed internally, or should it be outsourced? How should the restaurant price them correctly? While such a law sets the overall requirements, it leaves various challenges for restaurants and food service providers to set up and organize the operation.

Another form of market regulation is the introduction of levies or taxes. The Netherlands, for example, announced a new rule by July 2023 that requires customers to pay a fee for disposable plastic cups and food packages when they take away their food and drinks or when they get delivered (Netherlands Chamber of Commerce 2022). In addition, the government guidelines include price recommendations, for example, 0.25 € for cups and 0.50 € for a meal. While the rule also points out that municipalities can develop stricter rules for specific locations or events (Netherlands Chamber of Commerce 2022).

Although regulated on a national level, the waste problem typically shows its consequences on a local level. Municipalities play a central role in implementing circular economy (reuse) systems to avoid single-use packaging locally. In 2022, Tuebingen, a city in Germany, for example, introduced a packaging tax on disposable tableware at the municipality level. By charging a 50-cent levy on disposable beverage containers and tableware, and 20 cents on disposable cutlery, their goal was to tackle the flood of disposable waste in cities. However, a successful lawsuit

²⁷ Other rules included mandatory deposit schemes for certain single-use beverage containers and a registration obligation for companies that place packaging filled with goods on the German market; <https://www.verpackungsgesetz.com/themen/das-neue-verpackungsgesetz-verpackg2-2021/>

filed by the owner of a McDonald's branch²⁸ showed that implementing a local tax is complicated and requires substantial coordination with federal tax laws. The city of Berkeley in the USA was more successful in breaking people's immense throwaway habit. Since January 2020, Berkeley has initiated a new rule that requires all restaurants and cafes to charge 25 cents for each disposable cup (McCormick 2020).

Instead of using malus systems such as levies or taxes, some municipalities use subsidies to incentivize restaurants to invest in central reusable systems. For example, the county of Ravensburg and Springe in Germany offers restaurants a starting aid to promote introducing a reusable system.²⁹ The administrative county office of Ravensburg decided to collaborate with the external pooling system provider Vytal and restaurants, cafes, and other food service providers are encouraged to participate in this system.

17.5.2 Classification of Reusable Packaging Systems in Takeaway Food Services

Changing the takeaway food system from a linear to a circular system by implementing reusable containers requires significant changes in the supply chain. While in linear systems, the food is served in disposal containers, which are discarded right after food consumption, in circular systems, the reusable containers must be cleaned and transferred back to the restaurant for refilling. While reverse logistics has been extensively studied in the academic literature (Dekker et al. 2014; Govindan et al. 2015), the literature on reverse logistics of reusable containers in the takeaway food business is very limited. One reason is that reusable packaging as a business model is a new field and is driven by new laws, regulations, and business potentials. A market scan revealed various new business models for reusable containers in the takeaway food market. A first classification can be made regarding who is mainly responsible for managing the reusable container within the reuse system. Here, “*Customer-Managed Systems*” (CMS) and “*Firm-Managed Systems*” (FMS) can be distinguished. CMS and FMS have a similar meaning to the “*exclusive reuse*” and “*sequential reuse*” models by Muranko et al. (2021), described in Sect. 17.2.

In *Customer-Managed Systems*, the customer is responsible for purchasing the RCs and bringing them to the restaurant or food service provider. The customer is the owner of the RC and is responsible for cleaning and “transport” and bears all effort and costs with it. While the service to bring their own reusable food containers

²⁸ One of the arguments of the McDonalds owner was that the local packaging tax is in conflict with federal waste; https://www.plasteurope.com/news/PACKAGING_TAX_GERMANY_t250108/

²⁹ <https://www.rv.de/ihr+anliegen/abfall/abfallvermeidung/foerderung+mehrweggeschirr> and <https://www.springe.de/portal/seiten/foerderung-von-mehrweggeschirr-pfandsystem-900000292-24600.html>

was shut down to zero in the first year of the pandemic, restaurants, cafes, and food service providers started again offering this service and allowing customers to bring their own containers. Some businesses even offer a price discount when a customer brings her own container (or coffee mug). However, implementing such a system on a large scale would require significant marketing efforts or regulatory changes that discourage using disposable containers (Baumann et al. 2018).

In *Firm-Managed-Systems*, the firm (i.e., the restaurant or food service provider) purchases and offers the RCs to the customers, i.e., the firm remains the owner of the RCs. However, to operate a cost-effective and sustainable RC system, the firm is responsible for providing proper incentives to customers returning the RCs in a reasonable time. Firm-managed systems are particularly challenging from a supply chain and operations management perspective since several stakeholders have to collaborate to implement a well-organized return system. But at the same time, it provides the most potential for a sustainable implementation of reusable container systems on a large scale. Within the class of firm-managed systems, we could identify three subcategories of organizational structures: (1) *Single-Restaurant Systems (SRS)*, (2) *Multi-Restaurant Cooperation (MRC)*, and (3) *External Pooling System (EPS)*.

- *Single-Restaurant System (SRS)*: This system assumes that a single restaurant or a centrally managed restaurant chain purchases and manages the operations of the RCs. The RCs typically contain the name or the logo of the restaurant or the chain and thus cannot be returned at other locations. An example of such a system is the restaurant chain **Beets&Roots** operating a series of small restaurants in big German cities, such as Berlin, Hamburg, and Frankfurt, offering reusable containers that can only be returned to other Beets&Roots restaurants.³⁰
- *Multi-Restaurant Cooperation (MRC)*: This system assumes that multiple restaurants or food service providers jointly operate a local reusable container system. A joint operation implies that restaurants together purchase the RCs and together manage all collection, cleaning, and logistics operations. Compared to the SRS, customers can return their RC to all restaurants in this cooperation or to central collection points if set up. Potential applications are restaurants and food service providers located in food courts or closed systems such as campuses of universities.³¹
- *External Pooling System (EPS)*: The system with the highest market dynamic in recent years are systems where restaurants and food services collaborate with one or multiple EPS providers, which supply the reusable packaging (cups, bowls, etc.) and some of the infrastructure, e.g., logistics and deposit system. Very

³⁰ To note is that Beets&Roots sell the reusable containers to customers meaning that once the RC is bought, the customer can only return the RC by purchasing another meal; <https://www.beetsandroots.de/en/about-us>

³¹ Various German universities collaborate with external providers of reusable container for takeaway food, e.g., <https://stwno.de/en/gastro-en/reusable-packaging/how-does-mehrweg-to-go-work>

often, the RPs are only loaned to the restaurants but stay in the ownership of the EPS provider. Customers buy the food but only lend the RP. After the customer returns the RP, the restaurant or food service provider professionally cleans the RPs so they can be made available for the next customer. The collaborating restaurants/food service providers must pay a fee for using the RC and have access to logistics services. The larger the number of collaborating restaurants and food service providers is, the more convenient it is for consumers to return the container. For example, the two largest fast-food chains, **McDonald's** and **Burger King**, started pilot projects with the platform **Loop** in the UK in March 2022.³² However, although partnering with an external pooling system, the packages—due to the branding—can only be returned to restaurants within their own network. Over the past years, various pooling system providers entered the market, in Germany, for example, **Recup/Rebowl**, **Vytal**, **Recircle**, **Faircup**, and various others.

The market for reusable containers in the takeaway food market is dynamic, and it can be expected to become even more competitive in the future. EPSs are probably the most promising solution for the future, mainly because EPSs can connect with various restaurants and food services and thus can build larger reuse networks. At the same time, increasing return networks increases customers' convenience and hence their acceptance. However, not only does the network density matter but the costs for the customers and food services also drive acceptance.

17.5.3 Cost Structures for Customers and Restaurants

While in FMCG and e-commerce markets, mainly deposit-based incentive systems are in place, i.e., customers pay a deposit upon taking the container and get it reimbursed at the return, in the takeaway food market, two other cost/incentives mechanisms can be observed, *App-based* and *purchase-based* systems. Examples of *deposit-based* solutions are Recup/Rebowl [Germany], reCIRCLE [Switzerland], ECOBOX [Luxembourg], or REkrabička [Czech Republic]. Other providers implemented an App-based system where customers have to download and register on an App of this provider (e.g., Vytal [Germany], Relevo [Germany], Kamupak [Finland], and Bûmerang [Spain], Deliverzero [Netherlands]). When the customer returns the container within a specific time (e.g., 14 days) the container usage is free of charge. Only when the container is not returned within a pre-defined time, the container is assumed to be bought, and costs occur.³³ A downside of this system is

³² <https://packagingeurope.com/news/burger-king-partners-with-loop-to-trial-reusable-packaging-for-burgers-sides-and-drinks/8146.article> and <https://packagingeurope.com/mcdonalds-and-loop-join-forces-on-reusable-coffee-cup-pilot/4179.article>

³³ See, for example, Relevo; <https://stwno.de/en/gastro-en/reusable-packaging/how-does-mehrweg-to-go-work>

that customers must enter personal information such as credit cards into the App, which may be a barrier for some customers. The advantage, however, is that there are no additional deposit costs. The third system found is purchase-based, i.e., the customer or the food service provider purchases the container, and the container either must be brought to the partner restaurant (by the customer)³⁴ or stays with the restaurant, specifically in-house catering or workplaces.³⁵

As noted, in EPSs, providers collaborate with multiple restaurants and food services and charge these partners a service fee for participation. In a research study, we identified different cost structures that EPS providers offer.

- *Charge per sold container*: Only observed with the purchase-based cost. The containers were basically sold to restaurants (**RePlated**).
- *Charge per use of container*: Mostly used in App-based systems, which allow tracking the RC (**Vytal**, **Relevo**, **Bûmerang**).
- *Charge a deposit*: Typically, the same amount that the restaurant or food service provider charges the customer. This deposit can be recovered when RC is returned to EPS (**ECOBIX**, **reCircle**).
- *Charge a fixed monthly fee plus either per use or a deposit* (**Rebowl**, **Tiffin Loop**).

The market for reusable cups and containers is highly fragmented. More and more start-ups enter the market with their specific type of packages and payment systems. While this is a good direction at first glance, it may delay the large-scale adoption of reusable packaging systems. If the network stays fragmented without building up a sufficiently large return network, customers may not be willing to use them. At the same time, the network of cleaning stations is underdeveloped. Currently, cleaning primarily occurs at restaurants. However, they often do not have enough capacity for cleaning and storage to be ready for large-scale implementation. Therefore, in the future, cleaning and logistics operations provide the potential for new business models.

17.6 Future Directions of Reusable Packaging in B2C Markets

The market for reusable packaging in B2C is continuously growing. In all areas, FMCG, e-commerce, and takeaway food service, more and more start-ups offer new systems; however, currently, they operate on a relatively small scale. To implement reusable systems at a large scale and thus sustainably reduce packaging waste needs

³⁴ An example is the German restaurant chain Beets&Roots which only sells their own reusable containers that customers can purchase and replace it with the next purchase. However, there is no reimbursement for the return.

³⁵ An example is Australian firm RePlated; <https://www.replated.co/how-it-works>

a widespread acceptance of all stakeholders in the reuse system, i.e., by customers and by the final distributors (i.e., FMCG and e-commerce companies, retailers, and food service providers). A necessity for such acceptance is to make the entire network convenient and affordable to use, but at the same time, it requires a certain level of compromise by all stakeholders.

Dense Reuse-Enabling Network and Infrastructure Incentivizing customers to return empty packages does not only work with financial incentives (e.g., deposit fees). Another similarly important factor is the accessibility of the return network. Customers will likely return empty packaging only if the return network is sufficiently dense and offers enough return locations (“return everywhere concept”). The pilot project of the Austrian Post on reusable e-commerce packaging achieved this by simply using the existing mailbox and post office infrastructure. In other applications, return systems need to be built from scratch. For example, in takeaway food services, large-scale implementation of reusable container systems is only achievable if the network of cleaning facilities increases. Currently, reusable containers are primarily cleaned at restaurants; however, this has two major drawbacks. First, restaurants do not have sufficient capacity for cleaning and storing, and second, there is a severe safety issue. Customers return packaging (at best) pre-cleaned. However, the restaurant risks a returned packaging could be infested with bacteria, which can have severe consequences for the restaurant. Hence, establishing a well-functioning cleaning infrastructure is necessary to build the fundament of a large-scale implementation of reusable packaging systems.

Standardization Although international standards specify some requirements for reusable packaging (e.g., CEN – EN 13429), the reusable packaging systems in B2C markets are still fragmented and characterized by various packaging types and return systems. The B2B market (e.g., with pallets, IBCs, and containers) has shown that standardizing packaging and procedures allows for more automation and can lead to significant cost savings (Coelho et al. 2020a). Independent of the industry (FMCG, e-commerce, or takeaway food service), standardization in packaging design and reuse systems is critical in implementing large-scale reusable packaging systems. Harmonizing packaging standards are required to support various operational processes, e.g., sorting, cleaning, and inventory management. However, marketing has to make sacrifices regarding individual packaging designs to achieve this. Concerning deposit systems, an existing initiative is the DDRS Alliance, a non-profit organization of leading global industry and technology organizations, which aims to develop an international standard for a *Digital Deposit Return Scheme* (DDRS). Such new standardized return schemes will help digitalize products with common standards, thus easing the customers’ return process and expanding the return network reuse systems.

Regulation and Governance Many NGOs and governments around the globe have committed themselves to foster reusable packaging. It can be expected that regulators will increasingly impose binding requirements on reusable quotas and, at the same time, raise barriers for single-use packaging. However, besides stricter

regulations, governmental organizations should seek to increasingly collaborate with the private sector through public–private partnerships (PPPs). PPPs could be initiated to organize pilot projects to test reusable programs. It is still unclear which reuse structure (organizational and incentive) works best, and private companies, especially large FMCG companies, still hesitate to invest in reusable packaging systems. Public organizations can also incentivize stakeholders and foster awareness through test pilots either on a national level or locally with subsidies and reward programs. In addition, governmental organizations are essential in building the necessary infrastructure and striving for standardization. Altogether, both public and private institutions are needed in a collaborative manner to establish a scalable infrastructure for reusable packaging systems independent of the industry.

Operations and Logistics Management Managing reusable packaging systems in large-scale B2C markets face various operational challenges. We identified four key activities:

- (i) Collection and return logistics: A close-knit network of return stations is necessary to ensure that customers use and return reusable packaging on a large scale. Finding the right network size (and density) that optimally balances customer acceptance and cost-efficient operations will be a major challenge. A reusable packaging network must be convenient for consumers on the one hand and cost efficient to manage on the other hand.
- (ii) Inspection, sorting, and cleaning processes: Before the packaging can be returned to the cycle, it must be freed from contamination and adhesive residues, for example, from the shipping label. This reprocessing involves further costs and time. While in B2B systems inspection, sorting, and cleaning are primarily centrally organized, in B2C there exist many small providers of reusable packaging for which these processes are still a huge cost factor (e.g., reusable packaging for takeaway food). To implement large-scale reuse systems in the future, however, cost-efficient cleaning systems are a necessity, otherwise, reusable packaging systems will not be economically sustainable.
- (iii) Inventory management and replenishment: To optimally manage inventories of reusable products there are various challenges compared to inventory control in linear systems. When planning the injects of new packages, the uncertain return process of used products must be considered. Moreover, due to the limited number of use cycles, the inventory level of reusable packages consists of different age categories, and at the end-of-life cycle (known or unknown), outdated packages must be outdated.
- (iv) Redistribution: A challenge in the redistribution process is to allocate new or cleaned packages to different locations in the network and to decide when to redistribute.

As a final note, this chapter provides insights into how reusable packaging systems can be set up and how they compare to single-use alternatives. In doing so, we assume that the package has a crucial function (as evident for, e.g., a milk bottle) and cannot be eliminated. However, as the design of the packaging and the design of the product it holds are interrelated, packaging must always be evaluated in the larger scheme of things. Replacing a single-use shampoo bottle with a reusable

container might, in the best case, yield a tangible environmental benefit. However, it might be an even better idea to replace the shampoo with a piece of shower soap that needs no packaging at all.

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Chapter 18

Sustainable Non-Renewable Materials Management



Vered Blass, Amit Ashkenazy, and Tzruya Calvão Chebach

18.1 Introduction

The ability of firms to deliver value depends on a consistent supply of material inputs. A particular industry could be more or less resource intensive, requiring a certain material more than others do. However, global resource use has more than tripled since 1970 to reach 92 billion tons in 2017 and is projected to double to the level of 190 billion tons by 2060 (International Resource Panel Report 2019). Therefore, policy makers and firms, especially manufacturers that are highly dependent on natural resources, are increasingly interested in sustainable raw materials management.

Although there are many sustainability issues related to renewable resources, this chapter focuses on management aspects related to non-renewable materials such as metal ores and construction materials (see Fig. 18.1, shaded boxes). Sustainable raw materials management refers to the way nations and firms manage the natural resources they own and use in production and consumption systems from economic, social, and environmental perspectives. In general, within the realm of non-renewable natural resources, while non-metallic resources are more widely available (e.g., sand and gravel, aggregates, and fossil fuels), metallic resources are becoming harder to mine and procure, and so their environmental impacts worsen.

From the firm's perspective, the stability of raw material availability and supply is a key element for supply chain management and forecasting (Xiao and Yu 2006). However, several global trends impede it: (a) rapid depletion of the stocks of certain

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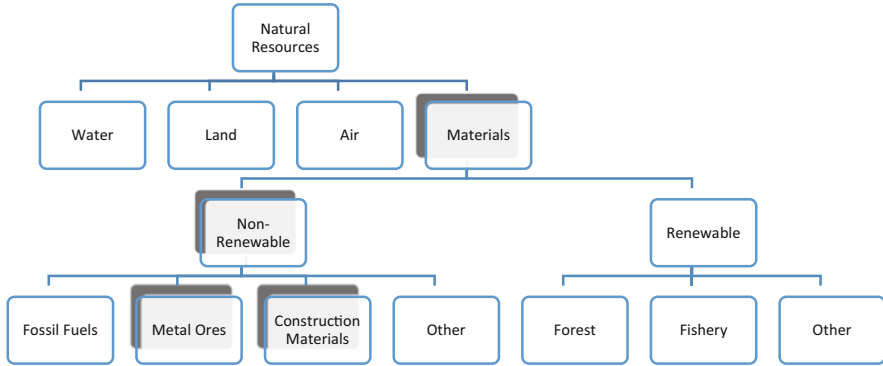


Fig. 18.1 Natural resources chart (adapted from UN SEEA Group)

raw materials, which results in increased effort during extraction, often associated with higher environmental impacts as well as more costly operations; (b) increasing prices of certain global commodities and resources; and (c) a geographic distribution of materials that may signify concentrated control by some countries (such as China) over certain materials, turning them into what is often described as critical materials, vital for various industries, yet originating from only a few source countries. These global trends have the potential to affect the very core of a firm’s production activity, requiring firms to understand both the risks and impacts of their non-renewable resource management strategy and practices, going beyond traditional concepts of material costs or inventory management.

The production and operations management literature has touched on the implications of sustainable management of raw materials from several perspectives: Galbreth et al. (2013) modeled the optimal material strategy for firms, based on potential benefit, required rate of innovation, and actual environmental impact of more sustainable material use strategies, such as integration of product reuse. Pil and Rothenberg (2003) demonstrated that environmental improvements have reciprocal effects on manufacturing quality. Others looked at the role of sustainability in supply chain management and at inspiring new tools and strategies firms can use to improve their decision-making (Kleindorfer et al. 2005). For example, Jacobs and Subramanian (2012) show how sharing responsibility for sustainable materials management along the supply chain can affect environmental performance and social welfare in product recovery schemes.

In this chapter, we first discuss key factors that affect firms’ and countries’ economic resilience based on the stability of sustainable raw material supply and its availability. We then elaborate on the management of material flows using tools such as Material Flow Analysis, providing examples of their use in practice. We continue with policy approaches to ensure sustainable management of raw materials, concluding with guiding questions for firms and policy makers looking to improve their raw material supply chain sustainability and resilience.

18.2 Materiality Risks to Firms and National Economies

Demand for raw materials has soared since the beginning of the twenty-first century. Growing quality of life in emerging economies and industrial expansion there, as well as increased consumption globally, have maintained high levels of pressure on the resource base. At the same time, many emerging economies have deployed strategies to maintain raw materials, especially metallic resources, within the country to ensure their continued ability to expand production and meet local demand. For example, China has introduced export restrictions on yellow phosphorus, bauxite, coke, fluorspar, magnesium, manganese, silicon metal, silicon carbide, and zinc (EU Commission 2010). Furthermore, the prices of metals and other commodities have increased since the 1970s. Raw materials are reported to account for 5–55% of the product's total cost, depending on the industry, region, and product category (PBL 2014; Vogel 2009). Therefore, their prices are critical for product planning and development. Prices are usually affected by demand (which increases every year) as well as by the availability and effort required to produce these metals.

18.2.1 *Growing Demand*

One of the inevitable forces driving resource scarcity is the growing demand for new products and technology in a rapidly urbanizing world that is expected to see three billion new consumers entering the middle class by 2050, most of them in emerging markets (World Economic Forum 2013). During the twentieth century, resource use per capita globally doubled while income increased by a factor of seven, reflecting a relative decoupling of economic growth and resource use (UNEP 2012). The UN International Resource Panel (IRP) projected that resource use per capita would grow from 11.9 tonnes in 2015 to 18.5 tonnes in 2060 (IRP report 2019). However, it warns that current consumption rates are unsustainable and will lead to accelerated resource depletion and further shocks to the economic system. The panel urged governments and companies to act rapidly to increase resource productivity, mentioning in its 2014 report that some forecasts caution that certain rare earth metals will run out in two decades unless recycling rates dramatically improve (IRP report 2014). That said, cities are already moving toward a zero-waste vision, adopting the waste hierarchy, trying to eliminate resources from ending up in landfills, and engaging in urban mining that captures the full potential of material stocks in the city (UNEP 2012). Overall consumption plays an important role in the demand upstream of raw materials, and firms may conclude that their strategy must change and they may need to take an active part in decreasing the rate of product replacements and technology cycles.

18.2.2 Mining Industry Market Structure

The market structure of the mining industry is highly concentrated. According to the Raw Material Group quoted in an EU report, less than 4% of mining companies (149 out of 4173 companies) controlled 83% of the metal mining market (European Commission 2008). Such a centralized market structure is prone to risks emanating from unexpected problems to the stability of these companies and from possible illegal coordination between them to influence market prices. Tackling price volatility, fraud, regulatory uncertainty, mounting government hostility to the mining sector, water/energy availability, worker safety, and a talent gap are all key priority areas presenting major challenges to the mining industry. In order to address these challenges and manage global trends, the mining industry's previous focus on volume has to be replaced with strategic scenario planning as well as innovative new approaches for dealing with supply chain stakeholders. Media and civil society scrutiny of the local environmental impacts of mining companies has expanded in scale and scope and includes monitoring and standard-setting bodies. Mounting hostility to the mining industry has also proliferated, highlighting the unsustainability of the current management of stakeholders in the supply chain. Some countries have reconsidered their resource management resulting in demand for new royalty regimes, concessions, and even expropriation. Examples mainly relate to developing countries, but also to some developed countries, such as the new mining royalty regime introduced in Québec in 2013 (Deloitte 2013).

18.2.3 Price Instability

Commodity prices have not remained stable over the past three decades, with growing demand from emerging economic centers such as China and India pushing prices up, followed by falling prices in the wake of the financial crisis of 2008, and later on with the COVID-19 pandemic and the recent Russian invasion to Ukraine. Changes in oil and coal prices have been particularly volatile, making it harder for companies to predict their long-term costs and revenues (Lin 2009).

Furthermore, the concentration of the metals in the ore decreases over time (IRP report 2019) so mining companies are required to expend more effort in achieving the same yield per every square meter of excavation. As a result, the environmental impacts of mining operations also increase over time (IRP report 2014). Technically, some metals are found alongside other metals in ores and therefore are not extracted separately. If the demand or supply for a certain metal changes sharply, the entire value-added and price structure of the different main and co-materials might change. For example, copper ores commonly include six different elements including copper, silver, gold, arsenic, selenium, and tellurium (Graedel et al. 2015).

Small and medium-sized enterprises (SMEs) are particularly vulnerable to disruptions in the supply of raw materials, as they are often less aware and less equipped to find alternatives when these problems arise. Higher material costs can also negatively affect their competitive positions, unless the government takes action to ensure their survival by, for example, ensuring national material reserves (de Boer and Lammertsma 2012). However, SMEs can also become part of the solution through technological and design innovation, or by taking part in national research programmes (DEFRA 2012).

18.2.4 Geopolitical Constraints

The concentration of metals in the world is also changing geographically due to a shift in mining locations. A review of recent trends in metal mining sites suggests a shift in the areas where mining companies choose to operate. In 2020, while the share of China, Australia, and Russia in gold production increased compared to 2010, the share of the USA decreased. Other countries that now produce gold include Mexico, Peru, Ghana, Canada, and their share over time also changed (Statistica Gold Mining Report 2020). The reasons for such a shift include lower concentration in existing mining sites, increasing environmental regulation of the mining sector in some countries, and overall cost of labor and operation in high-income countries.

China produces 90% of the global supply of rare earth minerals. Its monopoly on these essential resources allows it to control prices and flows by restricting exports. In 2014, the World Trade Organization upheld a ruling that China had violated international trade rules in these restrictions, specifically through tariffs and export quotas. While the WTO decision could facilitate export, illegal mining is still a matter of conflict between local and national governments (Asori et al. 2022). Furthermore, within a growing global demand for rare earth materials, China's own demand exceeds the rest of the world's (de Boer and Lammertsma 2012). When examining import sources for 2017–2020 in the United States, rare earth compounds and metals were almost exclusively imported from China, accounting for 78%; with Estonia, Malaysia, and Japan providing for additional 6%, 5%, and 4%, respectively, and based on mineral concentrates and chemical intermediates which are produced in Australia, China, and elsewhere. According to Patil et al. (2022), only limited quantities, around 2% of these rare earths are recovered, mostly from batteries, permanent magnets, and fluorescent lamps. Estimated uses of rare earth materials listed by the United States Geological Survey included: catalysts, 74%; ceramics and glass, 10%; metallurgical applications and alloys, 6%; polishing, 4%; and other, 6% (Cordier 2022).

Global supply chains are also affected by military tension, as is evident in the 2022 Russian invasion of Ukraine. Because of the war and the economic sanctions on Russian companies, certain manufacturing supply chains that depend on Russia manufactured metal products have been disrupted. For example, Russia accounts for

more than 20% of Palladium and Vanadium exports in the world, the former used in catalytic converters and capacitors, and the latter to improve the stability of metal alloys used in space vehicles, nuclear reactors, and airplanes (OECD 2022).

18.2.5 Effects of COVID-19

The COVID-19 pandemic heavily disrupted global supply chains. On the one hand, manufacturers halted activities as workers had to stay home, in conjunction with a drop in demand for certain products, while struggling to ensure a stable cash flow (Yang et al. 2021). In particular, China's cessation of production activities in the early days of the pandemic had sent ripples throughout the global economy, lowering the demand for metal and with it its price, slowing down economic activity in countries specialized in metal production (Gupta et al. 2020). Furthermore, closing certain manufacturing facilities central to metal processing, such as metal powder plants, created downstream challenges for the automotive industry, for example. The same industry's decreasing demand during the pandemic further drove fluctuation in metal powder prices, fueling instability throughout the supply chain (Kaitwade 2021).

18.2.6 Availability of Alternative Supply Through Recycling

Some of the world's most widely used metals have no easily available substitutes for their current major uses, including, for example, copper, chromium, manganese, and lead (Graedel et al. 2015). Thus, finding design solutions ahead of diminishing supply, or alternatively finding new sources of supply from existing stocks becomes a critical task for companies' long-term planning efforts.

Thus, it is important to look at trends in the recycling rates of different materials. The emergence of concepts such as urban mining, referring to mining elements in old landfills and in discarded products suggests that with the reduction in concentration of the metals in the ore overtime, alternatives are needed. Graedel et al. (2011) reported recycling rates for about 60 metals worldwide and suggested that for many elements, less than 1% is currently recycled. While 18 elements had recycling rates higher than 50%, most were under 25%. It was also evident that for many of the elements, there is insufficient data for reporting and, therefore, a great deal of important information about end-of-life recycling and technologies for many materials is missing.

Furthermore, Reck and Graedel (2012) make the distinction between end-of-life recycling rates for "base metals," reaching 50% and or more, and "specialty elements" which are hardly ever recycled. Base metals are elements that are commonly used in large quantities and relatively pure forms, making them easier to re-smelt, and in products with longer life spans for which suitable recycling

infrastructure is already in place. On the other hand, specialty metals are used in small quantities, mixed with other elements that are hard to separate, and in products with much shorter life spans (such as electronic devices) which require extensive collection and recycling operations to economically justify their extraction at the product's end of life. They suggest that three main improvements have to occur in order for society to reap the environmental benefits embodied in metals recycling: (1) extended collection efforts, (2) better design of products for recycling, and (3) new and better recycling technologies.

18.2.7 Social-Environmental Impact

Raw materials extraction results in different environmental impacts created by the mining equipment, the digging, the chemicals used for extraction, and the transportation activities in and out of the mine area. In the extraction phase, the “Ecological Rucksack” or “Material Intensity” (in kilograms) has become a common unit of measure. This unit of measure (factor), developed by the Wuppertal Institute in Germany, refers to the amount of materials moved to obtain 1 kg of the resource, also known as “Material Input per Service Unit” (MIPS), developed by the Wuppertal institute (last updated on 2015). The MIPS index is published for a variety of materials and provides a quick and good reference to demonstrate the effort and impacts associated with the extraction of different materials.

Raw materials extraction also causes air pollution, soil contamination, water pollution due to energy and water use, as well as waste and emissions emitted in different stages of the process. The mining industry is considered a very energy-intensive sector (Lusty and Gunn 2014) and is therefore particularly prone to emit GHGs. Table 18.1 provides examples of the MIPS factors and a range of primary energy consumption numbers based on a full Life Cycle Assessment (LCA) approach (See Chap. 2 by Guinée and Heijungs (2024) for more details about LCA), from extraction of raw materials to the end of the useful life of a product. This LCA approach is used to demonstrate the differences among several materials and takes into account the product's full impacts. Energy and MIPS factors are available for a variety of mineral fertilizers, resins, construction materials, plastics, chemicals, metals, etc. In order to illustrate this factor, several commonly used materials as well as more rare materials with very high energy and waste intensity are listed in Table 18.1.

Table 18.1 suggests that some rare materials such as precious metals (i.e., gold and platinum) and diamonds are very resource and energy intensive at the extraction and production stages, thousands or even hundreds of thousands times more than more common and abundant materials such as aluminum and lead.

Different LCA studies that take into account the environmental impacts of products across their different life cycle stages reveal that a significant share of the environmental impacts of production actually originate in the raw materials extraction and production phases. Therefore, these stages show great potential

Table 18.1 Materials environmental impact: MIPS and energy consumption

Material (primary)	Rucksack waste factor MIPS	Primary energy range (MJ/Kg)
Aluminum	37	2.6–5.3
Copper	348.5	0.7–1.78
Gold	540,000	2267–4612
Lead	18	0.4–0.7
Platinum	320,301	4827–6658
Silver	7500	5–227
Diamond	5,260,000	168,000–1,445,400
Sand	1.42	0.0008–0.001

Data Sources: Material intensity data sheet by the Wuppertal Institute for Climate, Environment and Energy (http://wupperinst.org/uploads/tx_wupperinst/MIT_2014.pdf, dated February 3, 2014), GaBi 6 LCI database, and Ali (2011). Range is based on different available production processes in different countries, data from GaBi 6 database except for diamonds, which are based on Ali (2011)

for increased eco-efficiency. Due to the high environmental and social impacts of raw materials extraction, more and more countries tighten their environmental regulations in the extraction sector.

18.2.8 Criticality Implications

Based on all of the above parameters, one can start asking questions about the criticality of some materials for everyday economic activities or for specific applications and industries. Graedel et al. (2015) suggested a methodology to assess the criticality of materials at the national level. Figure 18.2 explains their framework for looking at criticality in three dimensions: supply risk, environmental implications, and vulnerability to supply restriction. Several indicators on each axis are aggregated to arrive at a criticality assessment, as indicated on the diagram. Data sources for such criticality assessment could be the following: (a) metal in-use stock determinations, (b) material flow analysis, (c) metal substitution potential, (d) country-level information, and (e) environmental life cycle assessment results. Firms and governments can use this framework in order to assess criticality of specific elements they use in their products.

18.2.9 How Does Raw Material Extraction Affect Different Sectors of the Economy?

Materials scarcity is presumed to affect different sectors in different ways. A survey of senior executives of leading global companies on the impact of minerals and

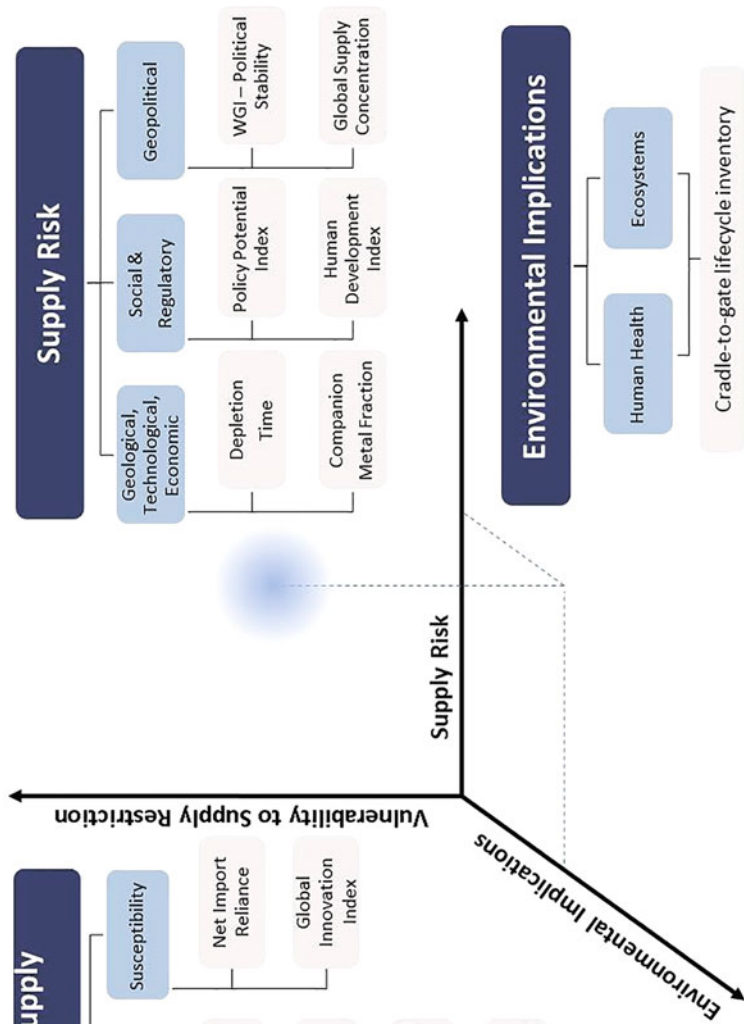


Fig. 18.2 Criticality assessment framework (Source: Graedel et al. 2015)

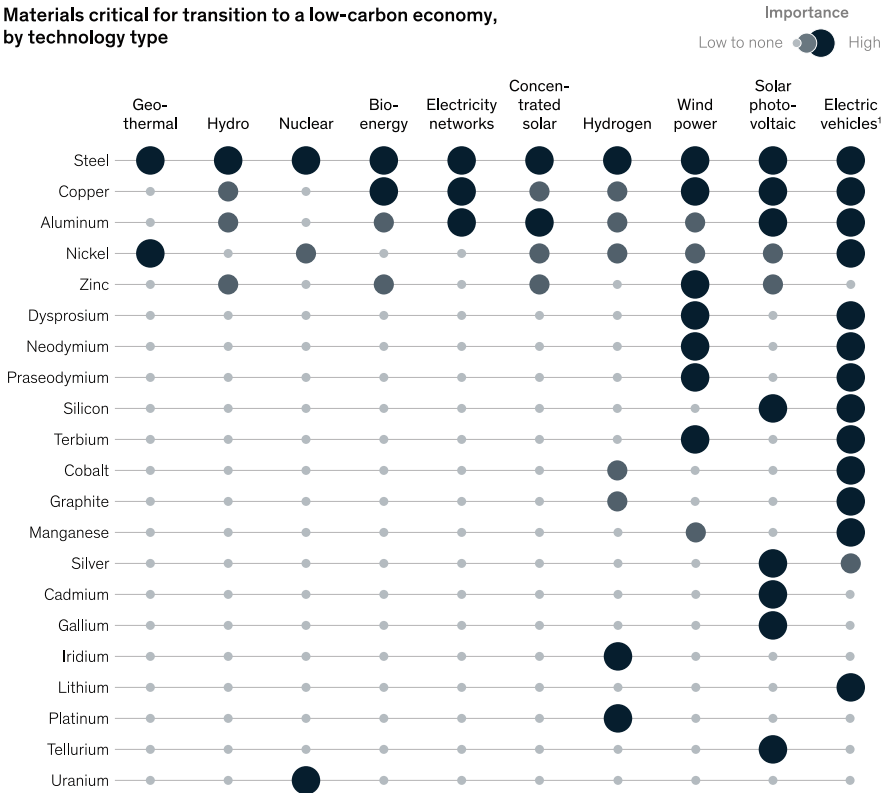
metals scarcity on business, conducted by PricewaterhouseCoopers (PwC 2011), illustrates how some sectors may be perceived to be more susceptible to scarcity. In the survey, 69 senior executives from leading companies were interviewed on different aspects related to scarcity, including awareness, preparedness, impact, causes, risks and opportunities, and responses to mineral and metal scarcity. According to the survey, certain sectors perceived the scarcity of minerals and metals to be a more pressing issue than others did. For example, 82% of respondents in the infrastructure sector considered the issue to be pressing and 78% considered it to be so in the high-tech sector. In the aviation and energy and utilities sectors, only about 50% considered the issue to be pressing. Moreover, it was found that the leading driver of scarcity apart from demand (65%) was geopolitics (54%), while the exhaustion of reserves was rated less highly (30%). In addition, low substitution was also perceived as a major driver of scarcity, particularly for renewable energy (89%), energy and utilities (79%), and chemical (78%) industries. Companies singled out resource efficiency as the key response to the rising scarcity challenge (75%) and also highlighted the importance of collaboration, such as strategic alliances along the supply chain to tackle the challenge. A high level of cooperation with first-tier suppliers was particularly evident in certain sectors. Finally, the level of preparedness at the time the survey was conducted varied according to sectors. For example, 67% of respondents in the renewable energy sector and 64% of respondents in the automotive sector felt prepared, while only 33% in the high-tech and chemical industries felt prepared.

Clean-tech companies are particularly susceptible to scarcity and criticality of raw materials. First, many of the technologies are emerging technologies with a growing number of materials in each product. Second, many of these technologies already rely on critical materials. When it comes to wind turbines, electric vehicles, PV cells, lighting and fuel cells, these technologies rely on critical minerals for their components. Thin film PV panels require tellurium, gallium germanium, indium, selenium, silver, and cadmium. Wind energy generators require neodymium, and dysprosium, and lithium-ion batteries in electric vehicles require lithium and cobalt (Resnick Institute 2011). Platinum and palladium are used for technologies related to emissions prevention. Silver and REEs are used for emissions purification (European Commission 2008). According to some reports, some materials, especially for the EV sector will already be in shortage by 2030–2035. In recent reports, the availability of materials is discussed in the context of transition to low carbon and net-zero energy transition. Figure 18.3 summarizes the projected criticality of different materials in such a transition for each energy technology.

18.3 Managing Risks with Material Flow Analysis

In previous sections, we highlighted the risks involved with production of certain materials, especially metals, due to unexpected changes in demand, technology, or even states' willingness to allow trade that might undermine their material resource

While steel will be crucial as an infrastructure enabler for all technological transition, specific elements will play an important role in each technology.



¹Includes energy storage.
Source: *Critical raw materials for strategic technologies and sectors in the EU*, A foresight study, European Commission, Mar 9, 2020; *The role of critical minerals in clean energy transitions*, IEA, May 2021; McKinsey analysis

Fig. 18.3 Criticality assessment for net-zero transition (Source: McKinsey and Company 2022)

base. These risks require firms as well as governments to evaluate their current and future needs. It also requires assessing how these materials accumulate and move throughout the economy, supply chains, or within their borders as stocks and flows. In this section, we review several tools to conduct these assessments, focusing on the Material Flow Analysis (MFA) methodology. We introduce the concepts and uses of MFA studies as a generic tool that can be used at different levels and continue with the more specific firm-level tool of material flow cost accounting while providing some concrete examples.

18.3.1 Introduction to Material Flow Analysis

Industrial ecology aims to understand the structure and function of industrial or societal metabolism, or how materials move and transform in different systems and processes across space and time (Ayres and Ayers 2002). For example, how does a certain country use fossil fuels? What happens to paper globally in different phases of its life cycle? In order to answer such questions industrial ecologists have created different quantitative methodologies. For example, Life Cycle Assessment (LCA) allows us to measure and compare the different impacts that certain products or services have on the environment in each of the phases of their life cycle—from extraction of raw materials, to production and use, to end of life, such as landfilling or recycling—based on a detailed account of the inputs and outputs invested and created in the system (see also Chap. 2 by Guinée and Heijungs (2024) on LCA and Chap. 4 by Hoekstra (2024) on water footprinting in this book).

A complementary tool that concentrates on materials flows within a given system is material flow analysis (MFA), which studies the flows and stocks of materials across a defined system, with explicit boundaries in a given timeframe. It is the main quantitative tool available today to assess and map the use of raw materials by type, quantity, and location across supply chains, economic sectors, regions, nations, or the economy at large. The analysis can be done in a static way (per system, per given year) or it can be done in a mode dynamic way, looking at trends backward and forecasting to the future (for review, see, for example, Müller et al. 2014). It can point to potential depletion in specific raw materials and gaps in recycling rates and can provide useful predictions that may affect decisions regarding future developments and operations (for example, see Allesch and Brunner 2015). Most MFA studies in recent years were done at the regional and country level, although some aimed at worldwide assessment.

Doing an MFA study can be a lengthy process, mainly due to data availability challenges. Although interest in MFA studies is increasing, it is still not a common tool used by nations and firms. In the practical implications section, we review MFA studies, how they can be used at firm level, and for what purposes.

18.3.2 How Can MFA Be Utilized in Strategic and Operational Planning?

Firms have different opportunities to use resources more efficiently. MFA provides an excellent tool for companies to track and manage their materials management. Tracking the type, quantities, origin, and transformation of materials along the supply chain enables companies to identify opportunities for sustainable materials management and possible hotspots along the flow of materials. Depending on the question or challenge at hand, an MFA can be conducted on different levels. The first step in identifying the opportunities of firms to become more sustainable in their

raw materials management is a review of the actual materials the firm uses, where they come from and how they transform along the supply chain. In other words, this means performing an MFA at different levels, depending on the system in which the firm operates, or the problem the analyst is trying to solve—from a single product to whole supply chains, from factory and neighborhood to city, national and even global analyses. Furthermore, MFA is becoming a customary tool in policy analysis and design. Thus, it may help firms anticipate and prepare for regulatory changes that would mandate a change in operations and strategy. Following are several examples of MFA studies used at different scales along with the risks and opportunities they expose and their possible uses for companies.

18.3.2.1 Using MFA to Optimize Production Processes

Industry has successfully used MFA insights to adjust material flows and waste streams in production processes in order to maximize both economic and environmental gains (Binder 2007). Analyzing the material flows in industrial parks offers an especially salient opportunity to identify opportunities for greater efficiency, as well as possible beneficial use of by-products created by different entities within the park (Sendra et al. 2007). MFA can reflect potential gains in material efficiency at a single factory level as well, disaggregating production into different stages and processes.

Looking at a particular substance flow through the economy can also help identify potential improvements for specific industries and systems. For example, Kowalski and Mazanek (1998) analyzed the flow of sodium chromate in Poland. Their analysis indicated that by technologically modernizing chromium production and using various chromium compounds, industry could improve its chromium waste management solutions so that imported raw materials are reduced, storage fees are reduced and productivity is improved.

18.3.2.2 Using MFA to Identify and Weigh Pollution Sources

Finding the causes of pollution and its detrimental effects can often be an intricate endeavor. Regulators, courts, and communities, as well as firms aiming to diminish their impact on the environment may need to contend with multiple possible sources of pollution in different locations, perhaps even at different times, with multiple materials intertwining downstream. MFA can help identify, classify, and weigh the relative contribution each source and each material may have in creating environmental harm, and thus lead the way in its solution.

For example, Cadmium (Cd) is a dangerous substance known to cause human kidney disease and other physical illnesses through chronic exposure. It is easily absorbed into the body through food, smoking and contaminated air and water, and has a slow rate of excretion. One of the known sources of Cd is agricultural fertilizers, as Cd is present in relatively high concentrations in the phosphate rocks

and acid used in their manufacturing. Kwonpongsagoon et al. (2007) used MFA to examine how Cd is used in the Australian economy and how it dissipates into the environment. By conducting a thorough substance flow analysis, which tracks a single substance, Cd, researchers can find hidden and expected flows of Cd. For example, construction materials were found to be an important Cd sink in Australia in addition to non-ferrous metal production and electricity generation, which are the main sources of Cd emissions. These results suggested that the National Cadmium Minimization Strategy, which focused on crops and soils, should pay greater attention to intentional use of Cd products and to wastes containing Cd. If a change in regulation occurs, firms in these industries would need to take into account the possibility of stricter substance control, which would require a change in their production methods and in their end-of-pipe solutions.

18.3.2.3 Using MFA to Analyze Products and Identify Hidden Trade-Offs

When a particular product or material is found to have harmful effects on the environment or on public health (or both), one of the solutions may be to substitute this material or product for another, more benign alternative. That is also one of the theoretical solutions for dealing with resource depletion or scarcity. MFA can help identify potential problems associated with substitution. For example, in a study of the shift to a lead-free economy, an MFA conducted on the Japanese market, and on solders in particular, found that while the shift to lead-free solders has progressed rapidly with the substitution of silver and copper for lead, it slows down in the products' late life cycle stages, due to the long life spans of electronic products. Thus, not only do emissions still contain lead, but they now also contain silver with its possible impacts on air and water quality. One of the beneficial ways to decrease these emissions could be higher circuit board recycling rates (Fuse and Tsunemi 2012).

Lam et al. (2013) combined environmental impact assessment with dynamic MFA in order to predict the timing of product substitution, the changing quantities of e-waste generation, and the environmental burden the electronic products market may create in coming years. Based on their analysis they proposed a consideration of e-waste pollution prevention measures, targeting specific materials with high toxicity rates taking in mind their model's estimated timing of disposal.

Looking at a specific market, MFA is particularly useful in calculating available stocks and forecasting needs for a specific material in different timeframes. Bergsdal et al. (2007) integrated scenarios about population, persons per dwelling, and floor area per dwelling, with data regarding the Norwegian housing stock to infer wood and concrete demand over the coming century. They found that despite decreasing growth in concrete density, its total use is expected to increase due to continued increase in housing stock. The outgoing flow of concrete is delayed, due to its late introduction as a building material and low renovation rate. Wood, on the other hand, has always been a prominent building material in Norway and undergoes renovation much more frequently, thus creating more output from renovation and demolition.

18.3.2.4 Using MFA to Design a More Efficient and Effective Recycling Scheme

MFA is also useful in looking at waste patterns and designing effective recycling schemes. In Beijing, for example, MFA was used to prepare for upcoming recycling schemes in the electronic appliances sector. Researchers used surveys and existing data to project the amount of electronic units to be discarded in recycling centers versus other end-of-life options. Thus, it allowed a more informed planning process, both by companies expected to take part in the recycling scheme, by recyclers that are dependent on the inflow of end-of-life electronic appliances and by local government, which needs to build proper recycling infrastructure and reshape its waste policies accordingly (Liu et al. 2006).

In Denmark, MFA of NdFeB permanent magnets, which are used in telecommunication products, electronic appliances, and energy technology, revealed that focusing recycling efforts on household consumers could yield minimal results in terms of actual supply of rare earth minerals such as neodymium, a key element in the magnets. The study showed that despite the fact that IT applications comprise 60% of the maximum theoretical recovery potential, overall potential recovery rates of neodymium and dysprosium from NdFeB in Denmark were equivalent to the amount found in five 3 MW direct-drive wind turbines. Thus, it is much more economically sensible and practical to focus efforts on major end users such as wind turbines, rather than hundreds of thousands of decentralized sources (Habib et al. 2014).

One of the most important utilizations of MFA in improving recycling schemes lies in enabling the process of urban mining—the systematic reuse of materials originating in human use in urban areas. Tapping existing stocks of resources, whether in infrastructure, houses, or landfills, requires the preservation of information regarding available materials from production through recovery, and the location of recycling facilities in or close to cities so that they remain both economically and environmentally beneficial (Brunner 2011).

18.3.3 How Could Firms Integrate MFA with Cost Analysis?

The concept of MFA described in the previous section has been further developed into a practical tool for firms, linking the flow of materials to actual costs through material flow cost accounting (MFCA). MFCA, developed in the late 1990s, is a tool for improving material productivity in order to optimize the relative consumption of materials, energy, and water in the production process. It measures the flow and stock of materials in both monetary and physical units such as weight and volume, facilitating the understanding of the interconnectedness of environmental factors such as material use and emissions to economic considerations. Under this approach, the full economic cost of waste is made tangible by examining the cost of

waste in the purchase, production, and disposal stages instead of simply calculating the cost of the disposal stage.

Experience demonstrates that material purchase costs and materials lost as waste and emissions may account for significant, and at times, the most significant cost drivers in different organizations. This may be particularly evident in countries where other cost drivers such as labor costs are low. The MFCA captures all input and output materials in the material flow analysis. Input materials include raw materials, auxiliary materials, merchandise, packaging, operating materials, water, and energy. Outputs are divided into product outputs that go into the final product and non-product outputs (waste, wastewater, and emissions). Successful implementation of MFCA depends on inputs from both accounting and production, two parts of the organization, which do not necessarily share a synchronized information system. Total environmental costs include environmental protection expenditure (end-of-pipe emissions treatment and waste prevention) as well as material flow costs. Together these two account for the total corporate environmental costs (Jasch 2008).

The integrated analysis of costs and the flows and stocks of materials in an organization has proven to yield opportunities for economic and environmental savings through resource efficiency. The ISO 14051 standard published in 2011 was designed as an internal environmental management accounting tool for organizations. It provides a general framework for MFCA on an organizational level through a common terminology, principles, objectives, fundamental elements, and implementation steps. The ISO framework tracks and quantifies in physical units the flows and stocks of materials within an organization and their associated internal costs. External costs are outside the scope of the MFCA but can be included if desired by the organization. It therefore promotes increased transparency of material use practices (Kokubu and Tachikawa 2013). Organizations can then use the information gathered to highlight opportunities to simultaneously generate financial benefits and reduce adverse environmental impacts such as waste production. The standard supports companies by providing a step-by-step guidance to the process through the different stages of an MFCA from guidance on building a materials and energy flow model to guidance on the communication of the MFCA results and the involvement of management. In addition, it includes some real examples to better illustrate the process. Another ISO standard, the ISO 14052 titled “Environmental management—Material flow cost accounting—Guidance for practical implementation in a supply chain” was published in 2017; it was developed and planned to provide guidance on implementation of MFCA along the supply chain.

The framework is relevant to organizations regardless of their products, services, size, structure, location, and existing management and accounting systems. Moreover, it can be extended to other organizations in the supply chain, both upstream and downstream. The supply chain component can then inform the organization and support decisions on an integrated approach to improving material and energy efficiency in the supply chain. For example, waste generation in an organization is

often determined by the nature or quality of materials provided by a supplier, or by the specification of the product requested by a customer.

Dedicated software tools were developed to support the use of the framework (e.g., Umberto[®]). Examples of the use of MFCA encompass the manufacturing and non-manufacturing industries and include opportunities in a multitude of processes, such as textile manufacturing, construction, cleaning, recycling, surface treating and chemical reactions processes, machining, and forming processes as well as logistics and distribution.

18.4 Building Material Resiliency Through Policy

While some countries have set concrete policies to identify critical materials, assess their risk of depletion, and put in place measures to reduce it, they vary in scope of materials and extent of solutions. For example, while the USA has a holistic approach to determine material criticality, its strategy focuses on the energy sector (DEFRA 2012).

In this section, we outline tools that can help to manage and enhance sustainable raw materials management more broadly at national and global levels. We provide insights into more targeted policy aspects and international trade issues.

18.4.1 National Assessments

The EU has recognized the growth in the number of materials used in products and the need to reduce supply risk and strategically secure stable access to these raw materials and resources for manufacturers in Europe. Therefore, as part of its strategy, it initiated the Raw Materials Initiative, which manages raw materials issues at an EU level. A subgroup of the Raw Materials Supply Group, specifically working on defining critical raw materials, published its first criticality analysis for raw materials in 2010. The report focused on non-energy and non-food materials and identified 14 critical raw materials out of a list of 41 materials. The report was updated in 2014, 2017, and again in 2020.

In a review published in 2014, they reviewed a longer list of 54 raw materials in order to enable comparison of criticality over time and implications for economic importance and supply risk criteria and identified 20 critical materials out of the list. In 2017, 27 materials out of 78 were identified as critical, and in 2020 the list was updated to include 30 materials out of the 83 (RMIS website 2022).

The European Union's Critical Material Studies guides decision makers on conflict minerals' policy and industry on scarcity and trade implications. Comparing the reports can be instrumental in understanding what changes took place over 10 years of assessment. Such a comparison clearly indicates that the status of minerals and their criticality cannot be considered as a static status and must be reviewed

periodically. It emphasizes the need for ongoing reviews both on a firm and a national level in order to assure risk and management of minerals are up to date with global fluctuations.

18.4.2 National Materiality Strategies

National level policymaking can help to resolve some of the criticality challenges. Over the years, countries like Japan, Netherlands, China, South Korea, Australia, Canada, and Germany established a variety of strategies in order to stabilize supply of materials, some initiated by the business sector and some by policy makers. For example, some countries promote free trade agreements and MOUs with resource-rich nations; some countries provide financial support for their mining industry and for building up the recycling industry. Countries also perform more criticality assessments and support wider resource efficiency policies. This emphasizes the importance for national analysis of the existing stock for each material, its future use and possible barriers to its production and import, and ways the government and the private sector can overcome these challenges through substitution, recycling and excavation, mine operation and pricing strategies, or through changes in consumption.

18.4.3 National and International Trade Policies

International trade restrictions may present a major challenge for firms as they may create supply risk for manufacturers in one country and may secure preferential use of the materials for manufacturers in another country. In 2012, the USA requested a consultation through the World Trade Organization with the People's Republic of China regarding China's restrictions on rare earths, tungsten, and molybdenum, which are raw materials used in the production of various kinds of electronic goods (WTO 2014). The following countries were third parties to the request: Brazil, Canada, Colombia, European Union, India, Japan, Korea, Norway, Oman, Saudi Arabia, Chinese Taipei, Viet Nam, Argentina, Australia, Indonesia, Turkey, Peru, and the Russian Federation. The listed restrictions included export duties, export quotas, minimum export price requirements, export licensing requirements, and additional requirements and procedures in connection with the administration of the quantitative restrictions. In 2014, the appellate body found that key measures were inconsistent with WTO obligations. In December 2014, China and the USA agreed that China would implement the recommendations and rulings of the Dispute Settlement Body by May 2015 based on the argument that the restrictions were designed to provide Chinese industries that produce downstream goods with protected access to the subject materials. The rare earth dispute came up again in 2018, with continuity of actions, tariff increases, and threats to stop trade between

the USA and China. In 2021, China created a “Megafirm” that merged three REE mining companies and now one company controls 70–80% of the mines in China (Baker Institute 2022). This topic continues to be an issue as the demand for rare earth materials increases. However, the recent findings of different beddings of rare earth in other countries and future discoveries might help the battle.

International trade is also relevant in the case of waste streams and recycling activities. For example, in 2013 China announced that it would tighten the enforcement of its recycling criteria for products and would no longer accept poorly sorted or dirty shipments of recyclable waste from other countries around the world (Green Fence Policy). The customs initiative does not present new legislation but rather better enforcement to assure that the country will not continue to import harmful waste products. China’s initiative immediately provoked a strong reaction within the country and abroad as it reshuffled the global recycling market by putting up a metaphorical fence in order to reject past practices of sending such harmful materials to China. The initiative disrupted existing waste flows and has already required other countries to reconsider their waste export strategy and shift scrap waste to destinations other than China (Bureau of International Recycling 2013).

18.4.4 National Policies on Circularity

In 2020, the EU Commission adopted a new circular economy action plan. The plan proposed a policy framework for creating more sustainable products through a three-pronged approach: First, designing sustainable products, by among different goals, improving product durability, reusability, upgradability and reparability; increasing recycled content in products; enabling remanufacturing and high-quality recycling; and restricting single-use and countering premature obsolescence. Second, the plan called to empower consumers and public buyers, and third, to increase circularity in production processes through strategies such as integration of circular economy practices in Best Available Techniques reference documents; putting in place systems to support and enable industrial symbiosis; and promoting the use of digital technologies for tracking, tracing and mapping of resources (European Commission 2020).

Some EU member states followed suit and created circularity action plans on a national level. For example, the Netherlands set a target of reducing 50% in raw material consumption by 2030, reaching a waste-free economy by 2050. The government adopted a government-wide programme for a circular economy in 2060, and a specific 5-year implementation programme in 2019. It further reached an agreement with 180 industry partners in 2017, providing the basis for a “recycling economy” that integrates plans on how raw materials in each product can be used in its design processes (Government of the Netherlands 2022).

Other member states adopted certain aspects of the circularity action plan into law. For example, in 2019 the government of France passed a law implementing the “right for reparability.” The law mandates that for five categories of electronic

products, manufacturers display clear information about how easy it would be to repair each product, allowing consumers to choose products that would hopefully last longer and thus help reduce the demand for raw materials in the industry (Repair.eu 2021). The basis for this policy is the “right to repair” initiative, a grass-root initiative which is now spread out in different EU countries (<https://repair.eu/>), the USA (www.repair.org), and other countries.

18.5 Summary and Key Lessons

This chapter provided an overview of the main topics related to sustainable management of non-renewable materials such as metallic elements. It reviewed several risk factors to companies’ and countries’ ability to secure a stable supply of raw materials, including growing demand, a concentrated market structure in the mining industry, price instability, geopolitical constraints, the ongoing effects of COVID-19 pandemic, the availability of alternative supply through recycling, and the social-environmental impact of raw material extraction. It further described tools available for firms and policy makers to better understand their material flows (with special attention to MFA) and provided examples of how to use such tools and for what purposes. It concluded with examples of the different ways policy can enhance sustainable management of raw materials at a firm level as well as economy wide.

Based on this chapter review, we outline several questions that can guide firms and other users of raw materials seeking to manage them more sustainably and enhance their resilience (partially adopted from DEFRA 2012):

- ***How can firms internalize material issues:*** Many of the issues discussed in this chapter are still not common knowledge for procurement managers, designers, and supply chain managers. In order to understand the local and global trends and take actions, firms need simple tools that will help them assess their situation and act on it in the most relevant way, both internally within the firm and externally to it.
- ***Practical use of MFA at the firm and supply chain levels:*** As presented, there is already a vast body of research and findings on the use of MFA methodology, and firms can use these studies in various ways. However, thus far, MFAs have mainly been conducted at the national and regional levels. Much less, work has been done at the firm level. Further work that concentrates on the supply chain level and the firm level will give higher value to firms and will increase their understanding of the importance of this tool for decision-making. It will help them to better understand such questions as: “what are the specific implications and concerns with regards to the materials I use in my products.”
- ***Where do our raw materials come from?*** Are we relying on domestic or imported materials? Domestic supply offers greater security, while imported resources may be at risk due to geopolitical issues, or protectionist policies from key supplying countries.

- ***What innovation is needed more specifically?*** Three forms of innovation should be encouraged by nations and by industrial associations:
 - New recycling technologies for materials currently not recycled.
 - Small-scale recycling operations that will be economic and will help countries that are poor in local resources to maintain the materials within the country at the end-of-life stage.
 - New substitutes for existing and new industries. Businesses will need to identify alternative sources domestically and/or internationally, while also bearing in mind issues relating to the time lag of developing a new mine, for example.
- ***Urban mining:*** The concept of urban mining is becoming much more relevant, especially within regions poor in local resources. At one end, new collection and recycling schemes for different product waste streams must be developed on a large scale. At the other end, new technologies for mining landfills must be developed and deployed. Landfill sites can become a major source of materials and employment.
- ***Needed collaboration:*** Since some material issues are global, some regional, and some sector specific, sector level “think-tank” groups could be very effective in reaching a solution (for example, the process of finding substitutes for the ROHS banned materials in the electronics sector was a collaborative action in some cases). Can the government help us secure the availability of some materials? Regulatory and government policy may have resource availability implications as it may affect supply or increase demand for particular resources.

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Chapter 19

Supply Chain Risk and Resilience Management as Enablers for Sustainability



Kirstin Scholten and Mark Stevenson

19.1 Introduction

The COVID-19 pandemic has highlighted to many of us how uncertain our world is and, importantly, how vulnerable our operations and supply chains are—from empty toilet paper and flour shelves in our supermarkets, to shortages of personal protective equipment such as face masks in the medical field, to an exaggerated shortage of microchips, to problems with the supply and recycling of cardboard boxes due to increased e-commerce, and increased food waste in restaurants due to mandatory closures. Beyond COVID-19, newspaper headlines highlight the consequences of risks and the globally connected nature of our world to us on a daily basis—oil price rises due to the war in Ukraine in 2022, staff shortages in Dutch and German airports in the summer of 2022, or earlier the Rana Plaza factory collapse in Bangladesh in 2012, and the volcanic ash eruption in Iceland in 2010 are only some of many examples that have caused disruptions to supply chains globally.

The vast majority of companies (>75%) experience at least one supply chain disruption each year, while in 2020 about 28% reported as many as 10 or more

The first edition of this chapter was written with Professor Brian Fynes (1959–2020). We are hugely grateful to Brian for his contribution to the first edition, and it is with deep sadness that he has not been able to contribute to this second edition. We dedicate the chapter to Brian's memory and hope he would have approved of the new material.

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disruptions, with the causes ranging from workforce strikes to transportation delays, adverse weather conditions, currency exchange rate fluctuations, energy scarcity, service provision failures and IT breakdowns (Business Continuity Institute 2021). Even a relatively minor problem within a supply chain can have significant consequences: a late delivery of raw materials can affect operations, with knock-on effects for a company's reputation, perceptions of its brands, the ability to win orders, quality, prices, profit margins and lead times (Waters 2011). At worst, these contingencies can threaten the continuity and hence sustainability of organizations, as has been shown in the past (van der Vegt et al. 2015). Hendricks and Singhal (2005) found that companies facing supply chain disruptions experience 33–40% lower stock returns relative to their industry benchmarks over a 3-year period (based on 1 year before and 2 years after the disruption announcement date). At the same time, this might affect society as a downturn in financial performance often results in labor redundancies or it can negatively impact morale and/or the well-being of staff. Other disruptions have severely impacted the environment in more specific ways; for example, the Deep Water Horizon Spill in the Gulf of Mexico in 2010 with subsequent long-term consequences for society in terms of water cleanliness and for the oil and gas industry in terms of regulatory policies. As the examples above illustrate disruptions can affect any of the three dimensions of sustainability. Furthermore, what becomes clear is that concepts of risk and sustainability in supply chains are intimately linked. That is, organizations need to manage sustainability related risks while any given risk can also threaten the long-term sustainability of a company.

Vulnerability in the supply chain centers around the disruption of information, product, service, knowledge, control coordination or monetary flows between organizations (Jüttner 2005; Narasimhan and Talluri 2009) and, as a result, exposes organizations to risks (Craighead et al. 2007). Supply chain risk management and supply chain resilience are the managerial counterparts of the concept of supply chain vulnerability (Jüttner 2005). The ultimate objective is to ensure that supply chains continue to work as planned, with smooth and uninterrupted flows of materials, information and money from initial suppliers through to the final customers (Waters 2011). Hence, risk and resilience management contribute to business survival by limiting vulnerabilities; they are therefore recognized by many organizations as an essential contributor to sustainable operations (Carter and Rogers 2008). While risk and uncertainty are closely linked, they are not the same: risk is something measurable in the sense that estimates can be made of the probabilities of outcomes; uncertainty is not quantifiable and hence probabilities of possible outcomes are unknown (Khan and Burnes 2007). As such, risk is the expected outcome of an uncertain event (Manuj and Mentzer 2008a), whereas vulnerability is the exposure to serious disturbance arising from supply chain risks (Jüttner 2005).

This chapter sets out to examine how supply chain risk management practices support organizations and their supply chains in achieving long-term sustainability, whether it be in relation to economic, environmental or social/societal factors. Firstly, we examine the process of *supply chain risk management* including *risk*

identification, risk assessment and analysis as well as *risk management strategies* and how these enable decisions; we then show how *supply chain resilience* extends traditional supply chain risk management and enables organizations and their supply chains to also deal with unforeseeable events. We will discuss *different levels of resilience* and how they are linked to supply chains. We conclude with a summary and reflection.

19.2 Supply Chain Risk Management

While a crisis in a supply chain may be unpredictable, it may not be unexpected. To deal with and reduce the impact of global supply chain risk, existing theory suggests that businesses follow a course of action from risk identification to strategies specifically aimed at managing risk (Manuj and Mentzer 2008a). Effective supply chain risk management tries to do exactly that by identifying and managing risks in the supply chain through a synchronized approach among supply chain members, to lessen supply chain vulnerability as a whole (Faisal et al. 2006; Jüttner 2005). Proactively identifying where and when vulnerability is present creates the potential to avoid risks, prevent them from actually happening and reduce or mitigate their impact (Waters 2011). Stages in the risk management process (labels differ among authors although the steps are similar, see, e.g., Manuj and Mentzer (2008a) or Jüttner et al. (2003)) include risk identification (or estimation), risk assessment (or analysis), evaluation to make a choice between strategies to manage risk, as well as ongoing risk monitoring (see also ISO 31000) (Fan and Stevenson 2018a; Norrman and Jansson 2004). To be able to conduct the risk management process effectively, top management support is required so that a shared, supply chain wide understanding and awareness of risks is created (Chopra and Sodhi 2004; Sheffi and Rice 2005). Furthermore, an organization needs to focus on: (1) developing and integrating intra-functional risk management processes, (2) cross-functional and (3) inter-organizational relationships to develop a risk management culture and enable visibility across the supply chain, (4) adaptability in order to continuously develop and learn, and (5) technological capabilities that aid in conducting the supply chain risk management process (Norrman and Wieland 2020). The stages of the risk management process will be further outlined in the following sections.

19.2.1 Risk Identification

The identification of risk sources is an important step in the risk management process as decision-makers become aware of events that may cause disturbances (Norrman and Jansson 2004). In other words, it is only by identifying risks that the rest of the supply chain risk management process is triggered (Fan and Stevenson 2018a). It encompasses a comprehensive and structured determination

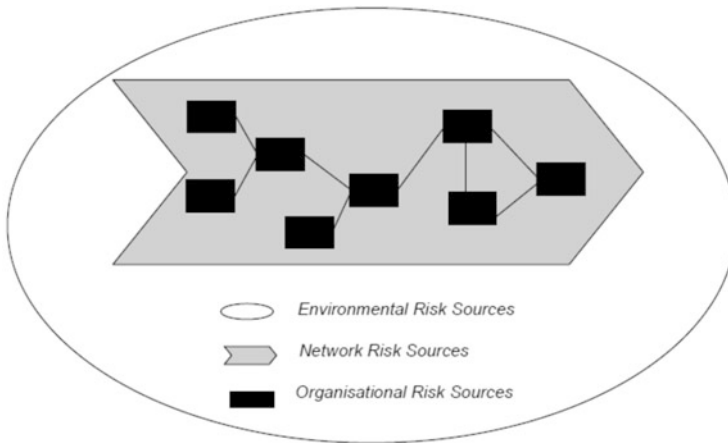


Fig. 19.1 Risk sources in supply chains (Jüttner et al. 2003)

of potential supply chain risks (Tummala and Schoenherr 2011). Risks to the supply chain consist of anything that may interrupt the normal flows of the chain and, as a consequence, expose organizations within the supply chain to economic, environmental and societal consequences (Craighead et al. 2007). Hence, there are a huge number of possible risks that can appear in almost endless variety (Waters 2011): sources of risk can be organizational, supply chain or environmental related factors that affect the supply chain performance (Faisal et al. 2006; Peck et al. 2003; Jüttner 2005)—see Fig. 19.1. Taken together, these risks define the vulnerability of a supply chain (Waters 2011).

- **Organizational risk sources:** Organizational sources of risk affect the operational side of an organization (Manuj and Mentzer 2008a). Hence, they are internal to any organization and relate to processes and controls (Peck et al. 2003, Jüttner 2005). Processes facilitate the management and production of value in an organization in the form of products and services for the end consumer (Christopher 1998), but also payments from suppliers (Fan and Stevenson 2018a). Risk that is inherent in processes can relate to accidents, the reliability of equipment, loss of an information technology system and quality issues (Waters 2011). Variability in processes can be managed via controls, which are assumptions, rules, systems and procedures such as order quantities, batch sizes and safety stocks (Christopher and Peck 2004). Therefore, risks in relation to controls arise more directly from human decision making or behavior via the application or misapplication of these assumptions, rules, systems or procedures due to, e.g., missing skills, a lack of awareness or coordination problems (Peck et al. 2003; Fan and Stevenson 2018b).
- **Network risk sources:** Network risks are internal to the supply network but external to an organization (Jüttner et al. 2003). As such, these sources are prone to affect a number of interdependent parties in a supply chain (Jüttner 2005).

Network risk sources refer to the possibility and impact of a mismatch between supply and demand (Jüttner et al. 2003). Therefore, network risks have been categorized as originating from the demand or the supply side. Demand and supply risks occur from the interactions between members of the supply chain and are often caused by inadequate coordination between members and by a lack of visibility (Waters 2011).

Demand risk relates to vulnerabilities associated with outbound logistics flows and product demand (Jüttner 2005). It can relate to any potential or actual disturbance within the network that affects the supply chain flows downstream (Christopher and Peck 2004), such as variable demand, changes in customer requirements, market price fluctuations, payments, problems with order processing or customized requirements (Waters 2011; Fan and Stevenson 2018a). Demand risks vary with the nature of the product, with functional products generally being less exposed to risks than innovative products (Fisher 1997; Manuj and Mentzer 2008a). At the same time, demand and customer needs are more than ever highly variable and hence difficult to forecast (Sturm et al. 2021). Similarly, on the supply side, risks are associated with possible events that may affect inbound flows (Manuj and Mentzer 2008a). Hence, supply risks are related to supplier actions or supplier relationships (Jüttner 2005). Consequently, the risk manifests itself upstream in potential or actual supply chain disturbances (Peck et al. 2003) caused by variability in the availability of materials, lead times, inbound delivery problems, industrial action, bankruptcy, product redesign, lack of sufficient capacity, price fluctuations or reliability of a supplier (Waters 2011; Fan and Stevenson 2018b).

- **Environmental risk sources:** As environmental risk is associated with disruption that is external to the focal organization and its supply chain network (Jüttner 2005; Peck et al. 2003; Christopher and Peck 2004), it represents a level of risk that is often beyond the direct control of supply chain operators (Peck 2005). Factors influencing the level of environmental risk can be summarized as political, environmental, social, technological, economic and legal aspects (PESTEL).

Other categorizations of sources of risk, rather than organizational, network and environmental are also reported in the literature (see Table 19.1). Whatever categorization one prefers, identifying risks related to such categories is critical so that consequences can be understood and assessed (Tummala and Schoenherr 2011). There are many methods for the identification of specific risks (Norrman and Jansson 2004), including ‘five whys’ analysis, cause and effect diagrams, Pareto analysis, checklists, interviews, process control charts, and supply chain risk mapping. These are only some of the methods that can be used by managers to identify possible risk sources (for further details, see Waters 2011). Care needs to be taken in the risk identification phase as risk sources can never be seen in isolation. They are linked to each other in complex patterns, with one risk source leading to another, or influencing the outcome of other risks (Manuj and Mentzer 2008b; Tukamuhabwa et al. 2017); for example, environmental risks can cause supply or

Table 19.1 Examples of classifications of risk sources in literature

Author (year)	Risk sources
Mason-Jones and Towill (1998)	• Environmental risk • Demand and supply risk • Process risk • Control risk
Jüttner et al. (2003)	• Environmental risk • Organizational risk • Network risk
Chopra and Sodhi (2004)	• Disruption risk • Delay risk • System risk • Forecast risk • Intellectual property risk • Procurement risk • Receivables risk • Inventory risk • Capacity risk
Christopher and Lee (2004)	• Financial risk • Chaos risk • Decision risk • Market risk
Wagner and Bode (2006)	• Demand side risk • Supply side risk • Catastrophic risk
Manuj and Mentzer (2008a)	• Supply risk • Operational risk • Demand risk • Security risk
Wagner and Bode (2008)	• Demand side risk • Supply side risk • Regulatory, legal and bureaucratic risk • Infrastructure risk • Catastrophic risk
Rao and Goldsby (2009)	• Environmental risk • Industry risk • Organizational risk • Problem-specific risk • Decision-maker risk

demand risks (Jüttner 2005; Manuj and Mentzer 2008a). Furthermore, and as a result of complex supply chain designs, there is the risk of additional chaos resulting from over-reactions, unnecessary interventions, second-guessing, mistrust and distorted information throughout the supply chain (Christopher and Lee 2004) which could potentially lead to phenomena such as the bullwhip effect (for further details, see Lee et al. 1997).

Irrespective of the source of the risk, the profile and nature of risk can be highly divergent: the financial default of a supplier (network risk) and a natural disaster

destroying production capacity (environmental risk) are situations with completely different attributes in terms of incubation period, probability, predictability and severity (Wagner and Bode 2006). As such, classifying risks into categories simply indicates the source of the risk and does not indicate its nature, scale or manageability (Ritchie and Brindley 2007a). Hence, the identification and classification of risk sources on its own is meaningless as each risk affects the supply chain differently; at the same time, risk identification is of great importance (Jüttner et al. 2003) by providing the basis for risk quantification, assessment and evaluation that can be used in deriving risk mitigation strategies (Narasimhan and Talluri 2009). This will be further discussed in the following section.

19.2.2 Risk Assessment and Analysis

The main focus of supply chain risk assessment is to recognize and analyze future uncertainties to enable the proactive management of risk-related issues (Norrman and Jansson 2004). This involves determining the consequences of all identified, potential supply chain risks, together with their magnitudes of impact (Tummala and Schoenherr 2011). Impact refers to the significance of the loss to the organization and/or supply chain(s) (Zsidisin et al. 2004). Adverse risk consequence can become manifest in any outcome measure such as loss of or damage to, e.g., assets, income, service levels, injuries or schedules. Supply chain design characteristics such as density, complexity and node critically have been shown to increase the severity of possible risks (Craighead et al. 2007); further example characteristics have been given in the introduction already. Accordingly, supply chain risk assessment should be a formal part of the decision making process at every level—from product design to component availability and lead time (Christopher and Peck 2004). Nonetheless, the consequences of risk do not necessarily need to be negative as the essence of risk-taking is the potential opportunity to produce positive outcomes (Ritchie and Brindley 2007b) although, in the context of this chapter, we define risk outcomes as being inherently negative.

In addition to their severity and impact, organizations need to understand that supply chain disruptions can be frequent or infrequent and short- or long-term (Chopra and Sodhi 2004). Risk is perceived to exist if there is a relatively high likelihood that an event occurs and that the event has a significant associated impact or cost (Zsidisin et al. 2004). Therefore, as some risks will have severe consequences and others will not, it is important for organizations to assess the probability of risk occurrences alongside the possible impact (Cohen and Kunreuther 2007). The likelihood of occurrence, typically expressed as a probability, can be articulated in objective (absolute values) or subjective terms (ranges) (Ritchie and Brindley 2007b). In this context, probability is a measure of the likelihood of the occurrence of a negative event.

Combining the assessment of impact (consequences) and probability allows organizations to classify their supply chain vulnerability, as shown in Fig. 19.2.

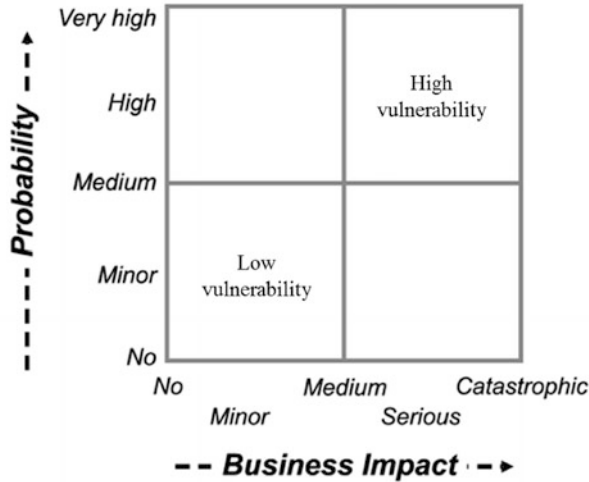


Fig. 19.2 Risk matrix (adapted from Norrman and Jansson 2004)

This is calculated by multiplying the value of the impact of risk by its probability of occurrence (giving the expected value of the impact) or by Monte Carlo simulation, scenario analysis, failure modes and effects analysis, or using network models (Waters 2011). The most important consideration for a risk manager is to ensure they have an advanced plan that indicates a suitable response to any possible event (Spekman and Davis 2004) that has first been identified and then quantified via considerations of impact and probability (vulnerability). Hence, by following these steps in the risk management process, different possible risk management strategies can be considered in order to derive a decision of what to do. Consequently, in the next section, we will describe possible risk management strategies.

19.2.3 Risk Management Strategies

While it is impossible to completely eliminate risks and consequent vulnerabilities from a supply chain, they can be reduced if an organization is proactive and prepared (Faisal et al. 2006). Risk management is the process by which decisions are made to accept a known or assessed risk and by which strategies are developed that reduce the probabilities and impact of negative events and/or their consequences in case they do occur (Cohen and Kunreuther 2007; Manuj and Mentzer 2008b). Depending on the assessment of risks, different strategies can be deployed to manage the risks: risk reduction or mitigation, risk avoidance, risk taking or acceptance and risk sharing or transfer (Fan and Stevenson 2018b; Norrman and Jansson 2004) (see Fig. 19.3). There is no one best strategy for protecting supply chains against risks;

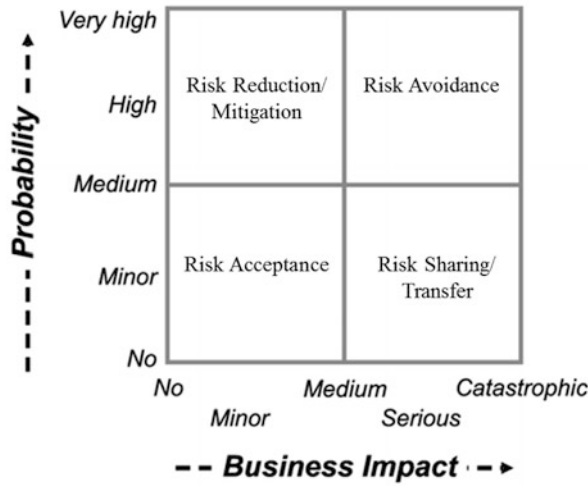


Fig. 19.3 Risk management matrix (adapted from Norrman and Jansson 2004)

instead, managers need to know which strategy works best against any given risk (Chopra and Sodhi 2004).

Notwithstanding the level of vulnerability, managers have two choices, either they can ignore (or accept) the risk and do nothing, or they can respond and do something (Waters 2011). Since it is not feasible nor practical to develop mitigation and sharing/transfer strategies for every risk that can be identified, risk management begins with the examination of the costs required to implement each preventative action to contain and manage the identified supply chain risks (Tummala and Schoenherr 2011). Often, a cost-benefit analysis of low probability low impact risks shows that the cost to mitigate the risk is higher than the cost to bear the risk; for example, in the case of a stationery supply chain for a food processing company, the risk to the stationery supply chain, which is not core to the company’s business or reputation, can be accepted without further action, but it needs to be monitored closely to ensure that the impact remains low in case the risk should manifest itself.

Companies typically focus on managing risks with the greatest impact and the greatest probability. Accordingly, Kleindorfer and Saad (2005) suggest that *risk avoidance*, the strategy used for high probability high impact risks, should precede risk reduction and mitigation. An avoidance strategy is used when the risks associated with operating in a given market (environmental risk) or working with particular suppliers or customers (network risk) is considered unacceptable (Manuj and Mentzer 2008a). To avoid is to eliminate the types of event that could trigger the risk by either driving overall probabilities associated with risk events down to zero or by reducing the frequency and probability of a risk (Manuj and Mentzer 2008b). Being attacked by pirates, for example, is a high risk for cargo ship operators in some parts of the world; to reduce the risk other routes that avoid the most dangerous seas can be used (Waters 2011). Another option for an organization

is to drop specific products, suppliers or geographical markets if supply is seen to be highly vulnerable (Jüttner et al. 2003).

Similarly, *risk reduction and mitigation* are aimed at both reducing the probability and consequences of the possible risk event (Manuj and Mentzer 2008b). Mitigation tactics are those by which an organization takes some action in advance of a disruption (Tomlin 2006) with the aim of achieving robustness, i.e., the ability to resist performance impacts (Brandon-Jones et al. 2014). Table 19.2 gives an overview of risk management strategies that aim to achieve robustness for supply network risks rather than organizational risks only (for further details on each strategy, see Tang 2006). Furthermore, information sharing, aligning incentives, risk sharing and corporate social responsibility are required to successfully implement risk mitigation strategies (Chopra and Sodhi 2004; Faisal et al. 2006; Spekman and Davis 2004).

As illustrated in Table 19.2, supply chain risk mitigation strategies can be beneficial under normal circumstances as well as during a major disruption (Tang 2006). However, at the same time, they can create challenges: every decision that mitigates one risk can end up intensifying another (Chopra and Sodhi 2004). As slack is taken out of supply chains with the aim to reduce costs, the chance that events in one link will affect other links considerably increases. Hence,

Table 19.2 Supply chain risk mitigation strategies (Tang 2006)

Risk mitigation strategy	Benefit(s) under normal circumstances	Benefits after a major disruption
Strategic stock	Improves capability to manage supply	Enables a firm to respond to market demand quickly during a major disruption
Postponement	Improves capability to manage supply	Enables a firm to change the configurations of different products quickly
Flexible supply base	Improves capability to manage supply	Enables a firm to shift production among suppliers promptly
Make and buy	Improves capability to manage supply	Enables a firm to shift production between an in-house production facility and suppliers rapidly
Economic supply incentives	Improves capability to manage supply	Enables a firm to adjust order quantities quickly
Flexible transportation	Improves capability to manage supply	Enables a firm to change the mode of transportation rapidly
Revenue management	Improves capability to manage demand	Enables a firm to influence the customer product selection dynamically
Assortment planning	Improves capability to manage demand	Enables a firm to influence the demands of different products quickly
Silent product rollover	Improves capability to manage supply and demand	Enables a firm to manage the demands of different products swiftly

typical supply chain risk management solutions, such as the maintenance of buffer stocks, built in slack in delivery lead times, and excess profit margins to cover returns, are becoming less viable in a world of outsourcing, just in time (JIT) and materials requirement planning (MRP). As a result, trade-offs between the robustness (benefit) of a supply chain to disruptions and the overall efficiency (cost) of the supply chain under normal operations have to be made (Kleindorfer and Saad 2005). For example, having additional inventory that might buffer a supply failure can simultaneously significantly increase inventory holding costs and, in some cases, the risk of obsolescence.

Risk sharing or transfer shifts some of the risk from one organization in the supply chain to another (typically to an actor more able or willing to handle it) (Waters 2011) via outsourcing, offshoring or contracting (Manuj and Mentzer 2008a). Risk could, for example, be transferred to insurance companies or supply chain partners by moving inventory liability, changing delivery times of suppliers (just-in-time deliveries) and delivery times to customers (make-to-order manufacturing), or by deciding to move a part of the production to an outside party (Manuj and Mentzer 2008b). However, sharing or transferring risk neither eliminates nor reduces the probability or impact of a possible risk event. So, while insurance can cover the costs of disruption, the disruption nonetheless can still occur.

19.2.4 Risk Monitoring

As risks are not static, they need to be continuously monitored, updated, and assessed to determine if any changes to the risk management strategies are needed (Fan and Stevenson 2018b). It has been suggested that specific data management systems and tools can aid in monitoring risks by creating specific routines (Tumala and Schoenherr 2011). Such formalized processes for internal and external monitoring can, on the one hand, provide a platform for reflection and learning to keep on improving the risk management process employed (Norrman and Wieland 2020). On the other hand, they can help to identify changes or trends that might form a new risk in themselves (Manhart et al. 2020).

In conclusion, organizations generally take decisions and plan to protect themselves against recurrent, low-impact risks in their supply chains by making use of one of the strategies above; whereas they generally ignore high-impact, low-likelihood risks (Faisal et al. 2006) simply because of the low probability of occurrence (Waters 2011). Yet the exacerbating frequency, magnitude and impact of disasters presents an increasing threat to the sustainability of communities, supply chains, businesses and their resources. The impact of local or regional events can transcend globally; for example, the war in Ukraine that started in the February of 2022 has not only affected automotive production in Germany, it has had consequences for production as far as Japan. However, such low probability supply chain disruptions are hard to identify, hence they cannot be easily assessed and analyzed, which limits the possibilities of deriving risk management strategies

and decisions. Consequently, they cannot be managed via the risk management process so far; they have to be managed in a different way. Accordingly, Van Hoek (2020) calls for risk management to move beyond risk mitigation to sustainably de-risk supply chain structures. In the following section, we examine how to do this via supply chain resilience.

19.3 Supply Chain Resilience

Today it is no longer about *if*, but *when* a disruption is going to happen. In its extreme form, uncertainty relates to the situation in which there is a total absence of information, knowledge, understanding or awareness of a potential event occurrence (Ritchie and Brindley 2007b). Hence, it is difficult, if not impossible, to follow the risk management process from risk identification through to risk assessment and management. By accepting that not all risks can be foreseen, controlled or eliminated (Jüttner and Maklan 2011), supply chain resilience enhances the traditional risk management process and enables an organization to deal with uncertain factors that can only be identified and predicted to a limited extent (Scholten and Schilder 2015). Whereas risk management tries to decrease vulnerability in terms of the probability of a certain risk, resilience focuses on the ability to absorb or decrease the impact of a disturbance that might stem from any risk by enabling the supply chain to return to stable conditions faster (Peck 2005). The aim is not to analyze risks and find the best strategy to respond, but to consider each part of the supply chain, see what happens when flow (such as a part) is disrupted and set up mechanisms that enable quick recovery of operations (Waters 2011). Resilience has been defined in supply chain terms as “the adaptive capability of the supply chain to prepare for and/or respond to disruptions, to make a timely and cost-effective recovery, and therefore progress to a post-disruption state of operations—ideally, a better state than prior to disruption” (Tukamuhabwa et al. 2015, p. 8). As such, supply chain resilience entails proactivity (i.e., preparedness for an effective and efficient response) and reactivity (i.e., immediate response and recovery actions taken in the long term) (Altay and Green 2006). It is a research field that has had spectacular growth in recent years (Ali and Gölgeci 2019), mainly due to high-profile, global events such as COVID-19 and its global consequences for supply chains.

COVID-19 can be considered a risk with very low predictability that, as of February 21, 2020 had already negatively affected more than 94% of the world’s top 1000 companies (Sherman 2020). As such, it has proven to be a prime example of the need for supply chain resilience. China was the first country affected by COVID-19 and the first to lockdown. With many global supply chains being dependent on China, the rapid shut down of its economy in early 2021 had wide-scale consequences. Those companies that had already mapped their supply chains to establish *visibility* were prepared. They knew which materials were originating in the Wuhan and Hubei areas of China at the center of the pandemic, and hence they were able to respond fast (i.e., *velocity*) as they did not have to search to establish

this information (Linton and Vakil 2020). A little later, a similar situation occurred in Europe, when Italy had rising infection numbers in March 2021 followed by lock and shut downs all over Europe. By this time, over 12,000 factories, warehouses and operations of the world's largest 1000 companies were located in quarantined regions (Linton and Vakil 2020) with little known about what might happen next. Hence, *redundancy* in suppliers was of little help and *flexibility*, to be able to continuously adapt to the evolving situation was required as at that stage *redundancy* in inventory, kept to mitigate demand and supply fluctuations, was in most cases depleted. Simultaneously, on the downstream side many consumer products saw demand surges and consequently empty supermarket shelves. On top of that, there were labor shortages and social distancing measures (Ali et al. 2022). The effects of the pandemic for supply chains worldwide have persisted longer than anyone might have imagined at the start of the virus's spread—well into 2022 (Alexander et al. 2022). While *collaboration* upstream and downstream in the supply chain has always been a key aspect of good supply chain management, it is now more important than ever to be resilient.

Visibility, velocity, flexibility and collaboration, as introduced above, can be considered core resilience strategies that have received the majority of attention in the supply chain resilience literature (Tukamuhabwa et al. 2015). At the same time, there are many other resilience strategies, as the study by Ali et al. (2017) has shown as it identified 13 strategies and 84 managerial practices. On their own, all of these resilience strategies display good supply chain management practices (Jüttner and Maklan 2011). At the same time, investing in resilience strategies has financial implications (Tukamuhabwa et al. 2015) and hence, it is likely that trade-offs have to be made in their implementation (Jüttner and Maklan 2011). Yet, many of these resilience strategies are interrelated where improving one will enforce another one (Dube et al. 2022). For example, investing in collaboration with upstream supply chain members through decision synchronization, resource sharing, collaborative communication or information sharing will automatically also lead to better visibility, flexibility and velocity (Scholten and Schilder 2015). An overview of the linkages between the core resilience strategies is provided in Table 19.3.

The above insights on supply chain resilience are “one-size fits all” where it is assumed that the application of the different resilience strategies will help to secure better supply chain resilience no matter what circumstances, context or company philosophy is encountered (Dittfeld et al. 2022; Scholten et al. 2020). Christopher and Peck (2004), however, highlight that determining the appropriate practices to manage supply chain vulnerabilities and risk appears to be context-specific, dependent amongst other things on the supply chain's response to the need for operational excellence. Hence, while any general framework can provide a road map to guide individual, context-specific supply chain solutions for improving and building resilience and disruption management capabilities (e.g., Pettit et al. 2010, 2013, 2019) these need to be further contextualized. Scholten et al. (2020) differentiate contributions to resilience on four levels: Individuals and teams (Level 1), Organizational (Level 2), Supply chain and network (Level 3), and Sector,

Table 19.3 Resilience strategies, their definitions and relationships between strategies

Resilience strategy	Definition	Relationship between strategies	Practice examples
Flexibility	The ease with which a supply chain can adapt its range number (i.e., the number of possible “options”) and range heterogeneity (i.e., the degree of difference between the “options”) Source: Jüttner and Maklan (2011)	Flexibility supports agility by facilitating quicker reactions to changing customer requirements and hence increase speed Source: Sturm et al. (2021)	Ensure flexible alternative sources
Redundancy	Spare resources such as capacity, suppliers and inventory Sources: Christopher and Peck (2004)	Some redundancy strategies contribute to flexibility and agility, e.g., using backup suppliers and building buffers for essential components, respectively Sources: Lee (2004); Xie et al. (2019)	Inventory buffering; avoid overreliance on single/few factories for supply
Agility	The ability to respond quickly to sudden and unexpected changes in demand and/or supply via visibility and velocity Sources: Christopher and Peck (2004)	1. Improves flexibility by accelerating processes for responding to disruptions Sources: Ali et al. (2017); Christopher and Peck (2004); and 2. Improves situation awareness, warning strategies and recovery times providing a basis for collaboration Sources: Brandon-Jones et al. (2014); Christopher and Peck (2004); Vanpoucke and Ellis (2020)	Use of information technology to improve visibility of demand and transparency of inventory; include near and local sourcing in the supply chain

(continued)

Table 19.3 (continued)

Resilience strategy	Definition	Relationship between strategies	Practice examples
Collaboration	The ability of supply chain members to work together effectively for their mutual benefit Sources: Pettit et al. (2013)	Some collaborative practices improve flexibility and agility, e.g., information-sharing and mutual knowledge creation Sources: Christopher and Peck (2004); Scholten and Schilder (2015); Wieland and Wallenburg (2013)	Negotiate savings with selected suppliers only; focus on ensuring supply with bottleneck suppliers, ensure collaboration with strategic suppliers; active information sharing throughout the supply chain

Adapted from Dube et al. (2022); van Hoek (2020)

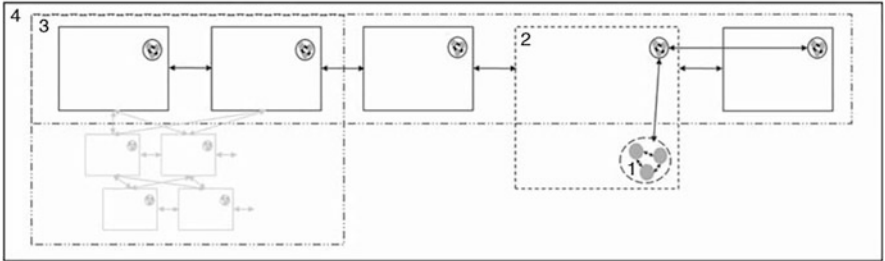


Fig. 19.4 A four-level framework for supply chain resilience (Scholten et al. 2020)

national and supranational (Level 4)—see Fig. 19.4 (Scholten et al. 2020). Each of these levels, and some examples of existing insights, is outlined in the following sections.

Level 1—Individuals and teams: Supply chain management in general, and with it supply chain resilience, is human centric where individuals and teams take decisions and implement strategies that determine if and how a supply chain disruption will impact an organization and its supply chain (de Vries et al. 2022). Recently, a lot of research has taken place at this level. For example, it has been shown that not only interpersonal relationships, commitment, responses and employee tenure play a role in the level of supply chain resilience (Polyviou et al. 2020), but also a manager’s social capital, human capital and cognition (Nikookar and Yanadori 2022). Furthermore, personal characteristics (such as risk propensity) affect the type of risk mitigation strategies individuals might employ en route to building resilience (Vanpoucke and Ellis 2020). Finally, it is known, that regular staff training and employee development has a positive influence on supply chain resilience (Ali et al. 2022).

Level 2—Organization: Most research on supply chain resilience to date has been conducted from an organizational point of view, where the focus is on a firm and its preparation for, response to and recovery from supply chain disruptions via the above resilience strategies (Scholten et al. 2020). Yet, also on this level many insights have been rather generic even though organizations come in different sizes, with different structures (e.g., multi-nationals such as Unilever versus a focused international organization such as Toyota), *raison d'être* (e.g., not-for-profit, social entrepreneurs, commercial) or cultures (e.g., level of hierarchy). Exceptions are the study by Polyviou et al. (2020) that examines medium-sized firms and shows that while such firms do not have the resources that large organizations have, they are also too large to obtain support that may be offered to small businesses. Furthermore, Dittfeld et al. (2022) find that differences in, and trade-offs between, resilience strategies induced by (1) the different configurations of production system characteristics (discrete versus process industries) and (2) the production strategy employed (make-to-stock or make-to-order).

Level 3—Supply chain and network: The resilience of a supply chain as a whole depends on the capability of all of its members—as a supply chain is only as strong as its weakest link (Scholten and Schilder 2015). Hence, organizational resilience is not a proxy for supply chain resilience as the following example also highlights (Novak et al. 2021): at the beginning of the COVID-19 pandemic, it was well known that there were shortages in toilet paper, with empty shelves evident in many supermarkets. Yet the reality was that there were stockpiles in businesses where toilet paper was under-used (business-to-business), and shortages in homes as a result of people working from home (business-to-consumer). Hence, while from an individual organizational perspective it would look like there was a shortage in toilet paper and/or too much, looking at the overall network quickly shows that overall demand and supply still matched, yet there was an availability mismatch in terms of location that is revealed by looking at a higher perspective or level of analysis. There are few studies of supply chain resilience that go beyond a dyadic view. Accordingly, Negri et al. (2021) conclude that supply chain resilience research at this level could learn from other areas where multi-stakeholder perspectives are more commonplace, such as in the sustainable supply chain literature where a true supply chain or network view has been recognized much earlier. An exception is the study by de Sá et al. (2020). These authors study the resilience of two supply chains with low interdependence between actors over three tiers and find that resilience is more dependent on the capacity of downstream actors who are responsible for delivery to the end consumer than on upstream nodes.

Level 4—Sector, national and supranational: With the aim to contextualize existing resilience knowledge, scholars have explored resilience in different sectors, for example, disaster relief (e.g., Day 2014; Scholten et al. 2014), the oil and gas industry (Urciuoli et al. 2014) or the agri-food industry (e.g., Hendry et al. 2019; de Sá et al. 2020). While often times the higher levels influence the lower levels, e.g., via rules and regulations, a study by Rubbio et al. (2020) showed that

resilience at the firm level can become an antecedent for building the resilience of a sector: experience, clinical knowledge and organizational dynamics were found to be antecedents of readiness, flexibility and collaboration in health care. As such, the four levels do not stand in isolation, but rather they are intimately linked.

In the January of 2020, Scholten et al. called for research into threats that have consequences for entire industries, countries or more without knowing what was about to happen in terms of the imminent COVID-19 pandemic. While the effects of the pandemic on supply chains were different in different nations or regions (Seuring et al. 2022), COVID-19 allows for new resilience insights that question the strong focus on global sourcing. Ali et al. (2022) show that domestic plus (food) supply chains are more resilient than those having an excessive reliance on offshoring, relying on global business partners only (i.e., a change in redundancy). It has also been found that social contracts have increased importance since COVID-19 (i.e., a change in collaboration; van Hoek and Loseby 2021). Beyond COVID-19, and as the world around us is constantly evolving, so is the need for and way to build supply chain resilience. Therefore, beyond adaptation to sudden onset events, adaptations to continuous change is also required. As such, and in line with Novak et al. (2021), the emphasis seems to have shifted from maintaining functional efficiency by having, for example, redundant suppliers to promoting functional existence via continuous transformation. While artificial intelligence might play an important role in such transformation (Modgil et al. 2022) learning has always been a key element of supply chain resilience (Eryarsoy et al. 2022; Scholten et al. 2019). Learning also links sustainability actions and resilience, where supply chain sustainability learning is affected by how resilience is implemented (Sauer et al. 2022). Beyond these insights, the clear link between the two concepts and the fact that resilience and sustainability are often discussed jointly, there is a lack of knowledge on practices that can simultaneously improve both resilience and sustainability (Negri et al. 2021). Ultimately, the same authors conclude that the nexus of sustainability and resilience is critical since organizations and supply chains are struggling with both. A company that is successfully able to confront both challenges will most likely be able to have a sustained competitive advantage. Yet, only if social, environmental and economic concerns are considered simultaneously can a supply chain can become truly resilient (Novak et al. 2021).

19.4 Conclusions

COVID-19 has most vividly shown how important risk and resilience management have become. This is now evident not only to organizations and their supply chains, but also to consumers. Today more than ever managers need to be aware of risk and uncertainty that might negatively impact organizational sustainability due to suboptimal supply chain performance. Adopting supply chain risk and resilience

management practices is key to business survival and sustainability. While there is a large body of knowledge and research on risk management and on how to prevent supply chain disruptions via the supply chain risk management process (identification, assessment and analysis, strategies and decisions), in today's global and complex world it is not a question of if a disruption is going to happen but when. Therefore, we need to build resilient supply chains on four different levels. Given that the only constant is change, there is and will continue to be a constant need to refine what we know about risks, risk management and resilience in supply chains to ensure future sustainability. Hence, there are many opportunities for future research, including linkages between sustainability and supply chain resilience, e.g., whether these two concepts are complementary (e.g., both benefitting from having visibility of the supply chain upstream, also in relation to worker conditions) or whether they can also be conflicting (e.g., redundancy of products to deal with possible disruptions that then become waste as they are not used).

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Part IV
The Social Dimension of Sustainable
Supply Chains

Chapter 20

Improving Social and Environmental Performance in Global Supply Chains



Barchi Gillai, Hau L. Lee, and Sonali V. Rammohan

20.1 Introduction: Social and Environmental Problems in Global Supply Chains

In his best seller, *The World Is Flat: A Brief History of the Twenty-First Century*, T.L. Friedman (2005) described how emerging economies like China and India rose as major manufacturing and service centers for the global economy. Over the last few decades, as production shifted from the West to emerging economies, global enterprises were able to leverage cost and other advantages which outweighed factors such as loss of control, longer lead time, higher inventory and other risks. This globalization of production has delivered benefits to corporations (called buyers in this chapter) and to consumers in the form of more affordable products and services.

However, in conjunction with these benefits globalization also brought with it significant negative consequences. The risk consultancy Verisk Maplecroft found that brands face increasing exposure to declining labor rights worldwide (Nazalya 2021). The Global Footprint Network (2021) highlights the fact that current levels of global production and consumption are using 73% more natural resources than ecosystems generate. Weaker law enforcement, corruption, cultural and other factors in developing countries have led to myriad social, environmental and ethical problems at factories which directly or indirectly supply goods and deliver services to global corporations. When factory workers exceed working hour limits and incur excessive overtime, their health can be impacted along with product quality. When factory owners withhold wages for one reason or another, it threatens the

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livelihood of their workers. Factory carbon emissions have climate impacts, and ethical breakdowns such as intellectual property theft by suppliers can affect company sales. In addition to the obvious human and environmental costs associated with these problems, such issues can impact both the reputation and profits of brands.

To combat the social, environmental and ethical problems in global supply chains, governments have increased the level of regulation placed on buyers and suppliers (Westervelt 2012). Consumers, activists and investors are also increasingly vocal about improving supplier responsibility. This has caused global firms to pay closer attention to the issue, and, in many cases, use self-regulation such as supplier codes of conduct to compensate for weak law enforcement.

Despite the push for multinational corporations to develop and enforce supplier codes of conduct to address employee safety, labor, and environmental issues at their suppliers, violations of these codes of conduct remain problematic in many industries, and vary in nature. The tragic collapse of the apparel factory building Rana Plaza in Bangladesh in April 2013, killing over 1000 people, is an example of a major lapse in building safety codes. The building housed suppliers for apparel brands including Benetton, Walmart, Matalan, and Primark. Sadly, this collapse was not a single incident; multiple factory disasters have happened since then, claiming the lives of dozens of additional workers (Chua 2021). The problem of excessive overtime in factories is widespread, with 85% of factories assessed by the Fair Labor Association between 2012 and 2019 committing hours-of-work violations (FLA 2021). Wage theft is also a serious issue, one example of which is the hundreds of thousands of garment workers in the Indian state of Karnataka, who for close to 2 years hadn't received their legally mandated minimum-wage increases (Chua 2022). It is estimated that 40 million workers are trapped in modern slavery (Hodal 2019), many of whom are part of global supply chains. On the environmental side, the Institute of Public and Environmental Affairs, a prominent research nonprofit in China which collects pollution data on factory environmental violations in China, cited over 186,000 environmental administrative penalties for manufacturers in 2018 (IPE 2019). IPE is known for making public evidence that suppliers of world-renowned brands such as Apple (Mozur and Dou 2013) and Marks & Spencer (IPE 2012) polluted rivers and the air in China.

Buyers have many motivations for improving supply chain responsibility, including adherence to regulations, avoidance of supply side disruptions (e.g., a factory closure resulting from a health and safety violation), negative media coverage, and pressure from external stakeholders (Cousins et al. 2004; Newman and Breeden 1992). Additionally, consumers are becoming increasingly aware of and concerned about responsible supply chain practices, thus influencing demand for responsibly made products. Li and Kallas (2021) conducted a meta-analysis of 80 worldwide studies dealing with consumers' willingness to pay (WTP)—that is, the maximum amount that a current or potential consumer is willing to pay for a product or good—for sustainable food products. They found an overall 29.5% WTP premium for sustainability, on average. While WTP estimates varied for different sustainability attributes, product categories, consumer characteristics and geographic regions, they

always had a positive value. Beyond food products, in a survey of more than 1000 consumers (First Insight and Baker Retailing Center 2022), over two thirds of respondents (68%) indicated their willingness to pay more for sustainable products while 72% said that sustainability is a very or somewhat important purchase consideration.

Lee et al. (2012) found that, among 1281 supply chain executives surveyed, 49% were somewhat or very concerned about unsatisfactory social and environmental standards at suppliers, and the corresponding percentage on breach of intellectual property rights was at 53%. An even higher percentage, 58%, worried about counterfeit products from the supply network.

O'Rourke (2014) calls for better data, decision-support tools, and ultimately incentives to move from policing supply chains to predicting and preventing unsustainable practices. Encouragingly, corporations, governmental and nongovernmental organizations, consumers and other groups have made progress in recent years to understand the activities in global supply chains and implement strategies aimed at making an impact.

Research such as Rao and Holt (2005), Vachon and Klassen (2006), Lewis et al. (2014) and Gimenez and Tachizawa (2012) suggests that various supplier management strategies aimed at improving social and environmental performance can also improve economic performance. This complements evidence that socially responsible firms yield higher returns (Derwall et al. 2005) and that social and environmental incidents and non-compliances can lead to subsequent financial losses (Klassen and McLaughlin 1996; PwC 2010). Rao and Holt (2005) empirically find significant positive relationships between “green” supply chain management across the entire supply chain and economic performance measures. Wu and Pagell (2011) find their sample of firms maintains business viability while pushing for improved environmental performance. Cousins et al. (2004) look at actions that can be taken by a buying firm to manage the environmental performance of its suppliers, comparing the resources available and perceived losses from environmental non-compliances. They suggest that incentives and supplier monitoring schemes typically require more resources but are adopted by proactive companies who wish to gain competitive advantage through improved environmental supply chain performance.

Still, debate exists on the link between responsible supplier management practices and the resulting economic benefits (Corbett and Klassen 2006). We will examine the relationship between responsible supplier management practices and social, environmental and, when possible, economic performance using a “sense” and “response” framework. But first, we'll discuss how buyers and suppliers can align on expectations for responsible practices.

20.2 Setting Expectations Between Buyers and Suppliers

It is important to have a common understanding of what responsible supply chain management means, to establish expectations between suppliers and buyers and build a responsible supply chain management strategy. Seuring and Muller (2008) define sustainability in supply chain management as the “management of material, information and capital flows as well as cooperation among companies along the supply chain while taking goals from all three dimensions of sustainable management, i.e., economic, environmental and social, into account which are derived from customer and stakeholder requirements.” Similarly, in operations management literature, Bowen et al. (2001) and Handfield et al. (1997) discuss “green supply” and “green value chain practices” respectively, “which are used to characterize environmental aspects of supplier arrangements; all of these implicitly or explicitly focus on improved environmental performance through better supplier management” (Corbett and Klassen 2006).

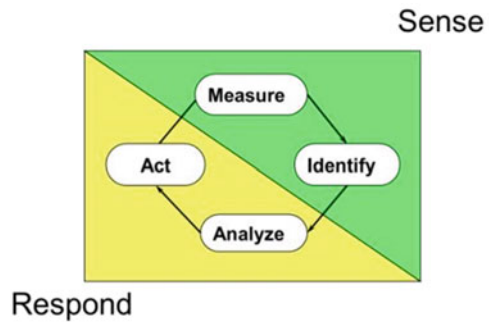
When implementing a responsible supply chain program, the first step buyers should take is to set expectations and clear goals with suppliers (see Chap. 11 by Tan et al. (2024) for examples of buyer-supplier relationship in the context of supply chain sustainability). A common way for buyers to do that is through a supplier code of conduct, which is a written guideline that may cover such areas as labor practices, environmental policy, ethics, and documentation policy. In a survey of 159 procurement and sustainability executives, summarized in Gillai et al. (2021), 74% of respondents said that they use a supplier code of conduct. About half (47%) of the same surveyed executives reported using a specific contract clause relating to sustainability, which is yet another way for communicating expectations with suppliers.

While corporations are the ones to formally set expectations with suppliers, consumers have the power to impact buying organizations’ priorities and actions. A survey of more than 900 supply chain executives from around the world (Gartner® 2021) revealed that social and environmental issues are top of mind with consumers, with more than half of responding supply chain leaders (54%) anticipating their customers to intentionally only purchase or order products or services that are sourced, made, packaged, delivered, returned and destroyed in a socially responsible and environmentally sustainable way, even when it costs more. Such sentiments no doubt impact the requirements supply chain and procurement leaders place on their suppliers.

20.3 A Framework for Continuous Improvement: Sense and Response

A useful guide to address responsible supply chain management is the sense and response framework. Haeckel (1992) described this framework as a system for companies to respond to rapidly changing customer needs. In larger firms, this

Fig. 20.1 Sense and respond framework



can mean that networks of skills, assets, cross-functional processes, information, and knowledge are linked into capabilities, which are in turn linked into processes for creating product and service responses to customer needs. The framework can also be applied to continually evolving social, environmental and ethical issues in global supply chains, such as human trafficking, pollution, and intellectual property protection, as argued by Gillai et al. (2015). Kapoor et al. (2005) discuss the sense and response model as a management tool for managing risk and unpredictability in operations. These dimensions mirror the structure adopted by Gimenez and Tachizawa (2012), who define assessment and collaboration as the two governance structures for management of supplier responsibility, and include a further dimension of managerial and external “enablers” that influence the implementation and success of responsible supply chain practices. A buyer must first gain visibility, or “sense,” into issues in the supply chain. After understanding the present state by measuring and identifying problems, a buyer can then “respond” by analyzing the problem and taking action. This process can be repeated for continuous improvement (see Fig. 20.1). The framework mimics the familiar Six Sigma management cycle for eliminating defects and minimizing variability in manufacturing and business processes¹.

Buyers can gain a “sense” of activities and impacts in the supply chain through practices including:

- Traceability: the ability to trace the points of origin of materials used in a product
- Visibility: knowledge of social, environmental and ethical performance at suppliers
- Monitoring: the action of examining supplier performance

Once buyers have a “sense” of the supplier’s behavior, there are various ways to “respond.”

¹ Originally based on Motorola’s work to eliminate quality defects, the Six Sigma process is a management process used by many companies such as Motorola, GE and others. For an introduction to Six Sigma, see Harry and Schroeder (2000).

The following are typical practices commonly used in industry:

- Reactions to violations once they have occurred (e.g., root cause analysis, and penalties such as fines, supplier warning, reduced business, contract termination)
- Incentives (e.g., preferred supplier status, increased business, price premium)
- Supplier capacity building (e.g., productivity improvement, capability expansion)
- Proactive product and/or process design (e.g., design for the environment)
- Shared value chain strategies (e.g., extended value creation through community development)
- Cascading responsible practices to the supplier network (e.g., training and motivating the suppliers to adopt incentives, capacity building and design principles to improve the sustainability of their own supply network)
- Advocacy with government officials to address issues at the industry or national level

20.4 Sense and Respond Practices

A number of surveys, some of which are mentioned in this chapter, found that companies are investing in multiple “sense” and “response” strategies. How can companies improve sustainability, and in some cases business performance through such strategies? This section offers some examples.

20.4.1 *Sense*

20.4.1.1 Traceability

Buyers cannot effectively control the sustainability of the supply network if they do not even know exactly where the materials in their products come from. Being able to trace the points of origin of materials used in the product is a crucial step in being able to sense the sustainability status of a supply chain. A growing number of supply chain executives recognize the importance of traceability, as is demonstrated in Gartner (2021), where 58% of surveyed executives reported investing in developing and embedding enterprise-wide transparency and traceability sustainability metrics.

To illustrate the importance of traceability and how it can be achieved, consider the case of Levi Strauss & Co. (Levi Strauss 2014). Early in 2008, the company faced pressure from external stakeholders—media, worker-rights and environmental nongovernmental organizations, socially responsible investment firms, and retailers—about the cotton used in their products, as there were reports that forced child labor had been used to harvest cotton in Uzbekistan. This prompted the

company into action. Tracing the exact origin of a commodity like cotton is difficult; the company had to reach out to the textile mills that supply the cotton fabric. Textile suppliers and licensees were informed that, unless there was clear evidence that the use of forced child labor had been eliminated, Uzbek cotton would be forbidden in the production of branded products of Levi Strauss. But, as cotton went through the apparel supply chain, there was little transparency into its country origins. In order to trace the country of origin of the cotton in their products, the company partnered with external organizations with expertise in supply chain traceability to implement a tracking system from the level of the yarn spinner to the product manufacturer. This provided confidence that Uzbek cotton was not being used in the Levi Strauss supply chain. At the same time, the company joined NGOs, the socially responsible investment community, major U.S. apparel and retail trade associations and the U.S. Department of State in engaging the Uzbek government to address the problem.

A traceability challenge similar to cotton applies to “conflict minerals,” such as [cassiterite](#) (for tin), [wolframite](#) (for tungsten), [coltan](#) (for tantalum), and [gold](#) ore. These minerals are mined in various regions of the world, and then passed through many intermediaries before they are used by electronic companies to produce consumer electronics such as mobile phones and laptop computers. Many of these minerals are mined in Eastern Congo, a region dominated by conflicts and human rights violations. Various international efforts have been made to reduce trading of conflict resources. A prominent effort is the 2010 Dodd–Frank Act that requires manufacturers to audit their supply chains and report conflict minerals usage (SEC [2012](#)).

20.4.1.2 Visibility

As discussed above, traceability refers to knowing which suppliers are providing materials to a firm. While knowing a firm’s suppliers is important, it is also critical to understand what those suppliers are doing. Visibility refers to having knowledge of a supplier’s sustainability performance. While efforts to gain visibility into the supply chain can be associated with improved social and environmental performance (Awaysheh and Klassen [2010](#)), it can be a challenge for buyers to obtain this visibility. In a survey by Gillai et al. ([2021](#)), procurement and sustainability executives reported having somewhat limited visibility into the supply chain with regard to sustainability. Specifically, 34% of respondents reported having visibility of immediate (tier 1) suppliers, while an additional 34% had at least some visibility of their extended supply network. At the same time, close to 32% reported having no visibility into their supply chain beyond their internal operations, as seen in Fig. [20.2](#).

Gaining visibility of violations or improved sustainability performance requires firms to carefully establish mechanisms to detect them. Such mechanisms can include (1) suppliers sharing data with the buyer, which requires a high degree of trust and a collaborative relationship; (2) direct monitoring by buyers; and (3) reporting from interested parties. One example of the latter is the nongovernmental

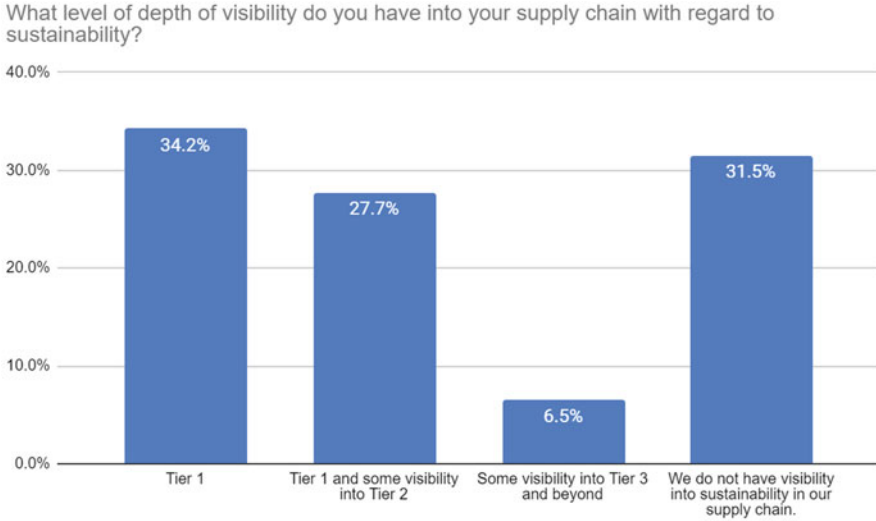


Fig. 20.2 Supply chain visibility with regards to sustainability. Source: Sustainable Procurement Barometer 2021, Gillai et al. (2021), page 24 (see source for full chart)

group IPE (Shao and Yatsko 2010), which uses a vast volunteer network to collect extensive data and report environmental violations throughout China on a public website. Another example of reporting from interested parties is the effort of groups such as Sedex and Elevate (through its Labor Link mobile platform) which enable crowdsourcing of potential labor and safety issues in factories via mobile phone calls or texts (Kraft and Zheng 2021). More in-depth programs such as the International Labor Organization’s Better Work initiative have also shown that elevating worker voice can lead to improvements in a number of compliance areas (Pike 2020).

20.4.1.3 Monitoring

While there are myriad regulations in developing countries addressing labor rights, employee health and safety, environmental standards and other issues, private sector supply chain interventions can act as complements or substitutes, depending on the context and factors such as regional governance (Locke et al. 2013). Before a buyer enters into a contract with a supplier, there is typically a supplier certification process which involves examining the supplier’s product quality, ability to deliver the product, pricing, financial status, communication capabilities and other factors. Increasingly, companies are also assessing supplier social and environmental practices as part of this certification process. The survey summarized in Gillai et al. (2021) demonstrates the popularity of such practices, with 69% of responding buyers stating that they take sustainability performance into account

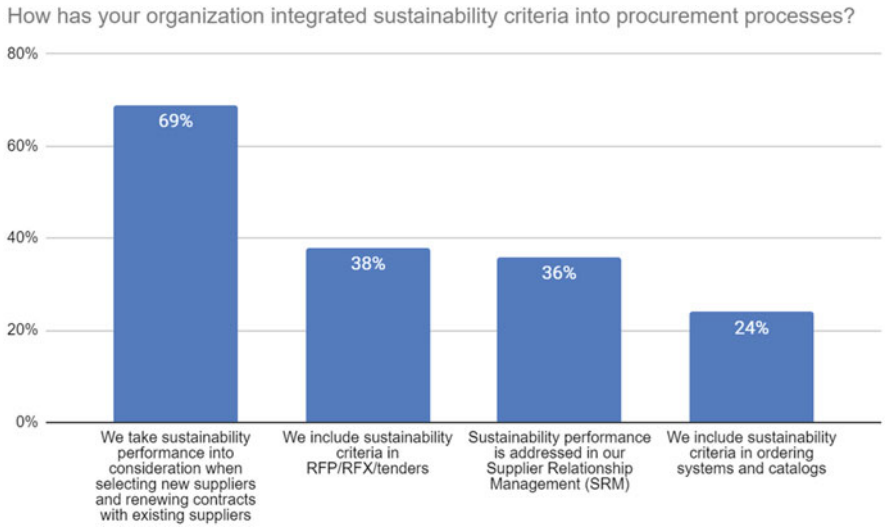


Fig. 20.3 Integrating sustainability criteria into procurement processes. Source: Sustainable Procurement Barometer 2021, Gillai et al. (2021), page 18 (see source for full chart)

when selecting new suppliers and renewing contracts with existing suppliers. Moreover, some companies will refuse to work with suppliers who do not meet certain sustainability criteria: In Gartner (2021) 46% of responding supply chain executives reported exclusively partnering with suppliers and third-party vendors that are committed to socioeconomic sustainability and equity initiatives, while 40% exclusively partner with first- and sub-tier suppliers who demonstrate achievement of aligned sustainability ambitions.

Once a contract is established, the supplier may be asked to provide self-assessments of its social and environmental practices. Buyers may also assess supplier sustainability practices and performance through annual supplier evaluations, by including sustainability criteria in Requests For Proposals or tenders, or by addressing sustainability performance in their supplier relationship management. Figure 20.3 shows how common each of these practices are, based on Gillai et al. (2021).

The buyer may also choose to conduct audits to verify that the supplier is conforming to the buyer’s code of conduct, and/or contract with a third party to obtain this assurance.

While audits are a common industry practice, research suggests the effectiveness of using audits (a “sense” tool) alone is limited. For instance, Sanders et al. (2018) analyzed data from internal factory audit reports of 832 factories audited between 2003 and 2010 and found no significant improvement in factory working conditions over time, while Lindholm et al. (2016) found auditing to be statistically correlated with improvements only at factories that have undergone numerous audits.

There can be multiple reasons for the limited effectiveness of audits. For suppliers that work with multiple buyers, when each buyer uses a different scale to measure and rate the supplier's performance it can create confusion as to the expectations and the actions the suppliers must take to improve performance. The level of resources required for performing each audit may limit their frequency or force buyers to limit audit programs to just a few key suppliers. There can also be flaws in the audit programs. For instance, they can be too brief to uncover all violations or the auditors may lack the expertise to identify a range of violations due to insufficient training. Another issue may be fraud, in the form of factory managers bribing auditors to give them a good rating, or factories keeping alternative records of wages and working hours to show auditors. Research suggests that the extent of unreliable information given to auditors is quite high: data shared by a leading auditing company classified the data collected in more than half of the audits they performed in China, India and Ethiopia between 2011 and 2017 as not trustworthy (Judd and Kuruvilla 2021).

Kraft and Zheng (2021) offer a number of ways for buyers to mitigate some of the challenges associated with audits. One such strategy is to collaborate with independent organizations that work within a supplier's region and can therefore conduct more frequent, unannounced audits. Multiple buyers can also pool their resources to conduct joint audits of common suppliers, or they can share audit information through trustworthy third parties. Technologies such as the internet of things and blockchain can also help increase visibility into the supply chain. Robertson (2020) finds that transparency, whereby factory-level audit information is made publicly available, can improve compliance.

Beyond "sense" mechanisms, buyers are encouraged to take more proactive actions, some of which are discussed in the next paragraph, to make an impact on sustainability.

20.4.2 Response

Given weak enforcement mechanisms in many developing countries and the fact that monitoring alone can have limited effectiveness, many buyers have adopted additional practices to motivate suppliers to improve social and environmental practices. Some companies use penalties and/or incentives tied to social and environmental performance in supplier contracts (Porteous et al. 2015). Some collaborate with suppliers to increase their awareness of issues and provide them with tools needed to address social and environmental challenges. Another growing practice is to collaborate with various actors in the value chain to create shared value. We examine below key response methods and the research regarding their effectiveness.

20.4.2.1 Reactions to Violations

Once a violation has occurred, buyers can react with consequences such as root cause analysis, and penalties such as fines, supplier warning, reduced business, or contract termination. Chen and Lee (2015) modeled supplier behavior under the premise that non-compliance is a result of unexpected and uncertain costs faced by a supplier, which threaten profits. When faced with unexpected and uncertain cost increases, for instance due to fluctuating input material costs or unexpected external disruptions, a supplier may engage in non-compliant activities to save money. The propensity of a supplier to do so is a function of the ethical standard of the supplier, which may or may not act as a deterrent to his/her urge to violate. Accordingly, penalties could discourage violations by “increasing the stakes” to suppliers. One form of monetary penalty is the use of contingencies. A buyer can withhold a portion of payment, which is subject to forfeiture if a supplier violation is found through an audit (the base payment can be paid to the supplier either upfront or at the end of the contract). Alternatively, the withheld payment can also be construed as a bonus, i.e., the supplier would receive an additional payment if no violation is found through an audit.

Lee et al. (2012) finds evidence of company intolerance for sustainability violations. Examples of penalties enacted for violations include monetary fines, reduced business or termination of business relationships (with or without an initial warning). Those surveyed reported that monetary fines were not as common as reducing or terminating business relationships. Gillai et al. (2021) offer a similar observation, with 23% of respondents saying they discontinue cooperation with suppliers that fail to improve their sustainability performance. Many companies have a “zero tolerance” policy for serious issues such as child labor, and will terminate business relationships if such issues are detected. Still, Oka et al. (2020) observed that even the threat of termination of business does not always give suppliers sufficient incentive to address violations. This is especially true with suppliers that have very large production capacity, which makes it hard for individual buyers to wield much influence. One way for buyers to overcome this challenge is through inter-brand collaborations that can result in greater leverage.

20.4.2.2 Incentives

In addition to penalties, incentives can also be used to motivate suppliers to invest in social and environmental improvements. As seen in the case study of Starbucks’ “C.A.F.E. Practices” scheme (Lee 2008), Starbucks has used positive incentives like preferred supplier status as well as price premiums to reward coffee farmers that achieve high sustainability standards. Sustainability performance impacted the degree of support some buyers offered to their suppliers during the COVID-19 pandemic, as observed by Gillai et al. (2021): close to half of responding buyers (49%) indicated that sustainability criteria were “important” or “very important” to them when deciding how to support their suppliers during the COVID-19 crisis,

What are the top three strategies that your clients could implement that would encourage your participation in their sustainability initiatives?

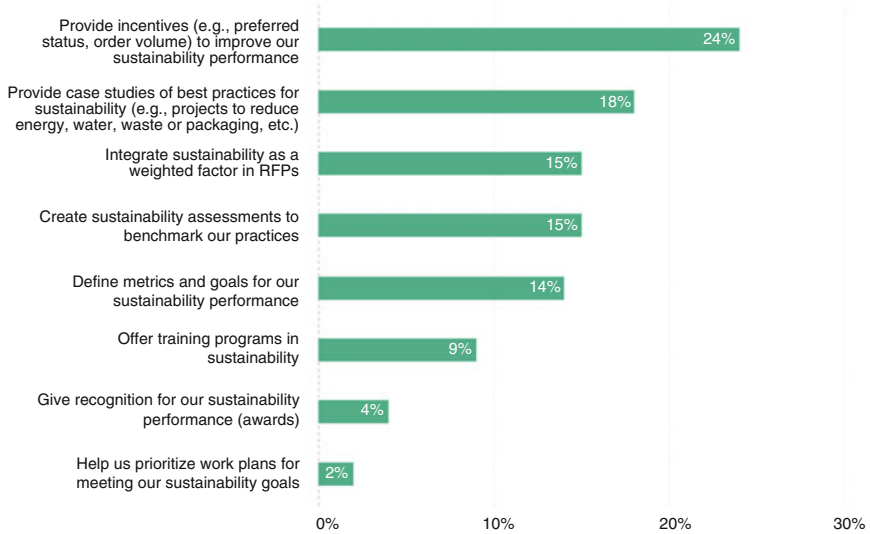


Fig. 20.4 Top strategies preferred by suppliers. Source: Sustainable Procurement Barometer 2021, Gillai et al. (2021), page 43

e.g., by providing more business or offering better payment terms. Motivated by Starbucks’ “C.A.F.E. Practices” scheme, Lewis et al. (2014) show that supply contracts can facilitate the long-term commitment of supply chain partners to collaborate with a firm to achieve social and environmental performance.

Lee et al. (2012) find that companies have used various incentive schemes to induce suppliers to be sustainable. Only 10% of executives surveyed used price premiums as a reward. Instead, giving suppliers special status (58%), increased businesses (48%), recognition (32%) and better terms and conditions (25%) were more commonly used. Of these companies, 42% also invested in training and education of suppliers.

Incentives such as a preferred status or higher order volume are also viewed by suppliers as a top motivator for encouraging them to participate in their clients’ sustainability initiatives. In a survey of more than 200 suppliers, 24% of respondents selected buyer incentives as their top motivator (Gillai et al. 2021). Suppliers also valued access to case studies of best practices as well as sustainability assessments, both of which can help them identify strategies to improve their sustainability practices, as well as the integration of sustainability as a weighted factor in RFPs (see Fig. 20.4).

20.4.2.3 Supplier Collaboration and Capacity Building

Since suppliers may not have the means and know-how to improve sustainability on their own, buyers often find they must implement collaborations with their suppliers (see Chap. 11 by Tan et al. (2024) for more details). Oka et al. (2020) find that buyers increasingly see capacity building as a promising alternative or complement to traditional auditing, as it can help address deeper issues and bring about change. Such collaboration often takes the form of education and training, capability enhancement, and the introduction of better production methods for productivity improvement. Research has shown the effectiveness of certain collaborative mechanisms to encourage improved supplier performance (Locke et al. 2012; Vachon and Klassen 2006; Locke and Romis 2006; Bowen et al. 2001). Gimenez and Tachizawa (2012) find the implementation of both supplier assessment and collaboration with suppliers improves environmental and social performance. Furthermore, Krause et al. (1998) investigate firms' supplier development processes (not specific to SER) comparing reactive approaches to strategic efforts to increase supplier capabilities, and thus a firm's competitive advantage. The development process includes identifying suppliers, issues and opportunities, and also collaborative efforts to increase capabilities, training, rewards and recognition, and continuous improvement programs. Krause et al. (2000) find direct involvement of a firm, including providing supplier incentives, is a key enabler of supplier development.

Chen and Lee (2015) find that a supplier can be vulnerable to non-compliant activities when faced with unexpectedly high costs that threaten its profit margins. One strategy is to invest in improving the productivity of suppliers, so that they can “weather the storm” and maintain margins despite cost hikes. Based on a study of 25 Indian textile plants from 2009 to 2011, Bloom et al. (2013) examined the effect of training and support of suppliers. The plants were separated into experimental and control groups, with experimental groups receiving diagnosis, training and consultation on factory operations, quality control, inventory, human resource, and sales and order management. The experimental groups were found to have significant improvements in quality, inventory performance, and finally, productivity. This study suggests that collaborating with and training factory management can influence a factory's performance. Related to this, Distelhorst et al. (2016) found that, by introducing Lean² production practices to factories supplying to Nike, labor compliance significantly improved. The study, based on labor compliance data from 2009 to 2013 at 300 factories 2 years prior and 2 years after the introduction of Lean practices, suggests that stronger performance can enable a factory to be more compliant. Finally, Lollo and O'Rourke (2018) find that compensation systems aligned with advanced Lean principles can increase productivity, profits and worker wages.

² The Lean production process was championed by Toyota, and has been widely adopted by companies as a way to eliminate waste, improve productivity, and increase the efficiency of production systems.

20.4.2.4 Proactive Product and Process Design

In quality management, it is well known that product and process design, in addition to product inspection and testing, can be an effective way to assure quality. Motorola's Six Sigma process (Pande et al. 2000) was based on the premise that, by improving product or process design and management approaches, the variability of processes could be reduced and quality control assured, thereby reducing the reliance on inspection. Similarly, by improving the design of products or processes, a supplier's manufacturing, farming or other processes can be made less vulnerable to cost shocks and other external disturbances, resulting in a lower risk of sustainability breaches. In Chap. 19, Scholten and Stevenson (2024) provide a framework for identifying and managing risks in supply chains.

In the textile industry, Esquel (Peleg and Lee 2013) invested in R&D on the scientific development of cotton seeds that are more pest-resistant, so that cotton farmers could use fewer pesticides and insecticides. The company also conducted research on ways in which fabrics could be dyed with less chemicals and water, so that the fabric mills could reduce pollution. Similar "design for the environment" techniques are now being used by firms to lower consumption of water and energy and reduce waste at various parts of a product's life cycle. In the agribusiness sector, McDonald's (Rammohan 2013) worked with its major supplier in India to experiment with the right types of potato seeds and growing methods that would achieve an optimal quality and texture, and increase productivity of potato farmers.

20.4.2.5 Shared Value: Extended Value Creation and Community Development

The previous sections described how buyer "response" strategies such as positive incentives and investments in suppliers could improve productivity, reduce non-compliances, improve sustainability performance, and in some cases, increase the incomes of suppliers. Increasingly, firms such as Johnson & Johnson, Wal-Mart, Coca-Cola, Unilever and others are also using shared value strategies to address issues in the supply chain. Shared value is a management strategy focused on companies creating measurable business value by identifying and addressing social problems that intersect with their business (Porter and Kramer 2011). One aspect of this concept involves increasing productivity of the company or its suppliers by addressing the social and environmental constraints in its value chain. Li & Fung (Melvin 2015) is a consumer goods management firm using a shared value approach with the goal of increasing the long-term income of factory workers in addition to factory owners, which in turn, is intended to lead to economic development in regions where suppliers are based. Sometimes, its buyers have also invested in community development such as building infrastructure, engaging in education activities, and creating other income-generating activities in supplier communities. Such shared value creation—delivering value to buyers, to suppliers, to workers, and ultimately, to communities—can be considered the highest level of sustainability,

and a growing goal of leading firms. Sodhi and Tang (2024), in Chap. 21, go deeper into how the stakeholder resource-based view can make concepts such as shared value more tangible.

20.4.2.6 Cascading Responsible Practices to the Extended Supply Network

In 2007, Mattel had a massive recall of its toys due to the tainting of toys by lead paint (Hoyt 2008). The root of the problem turned out to be a pigment supplier using pigment containing lead. The tainted pigments were provided to a paint supplier, whose paint was then supplied to a contract manufacturer who manufactured the toys for Mattel. The violation occurred at the third tier of Mattel's supplier network. As seen by this example, it is often a daunting job for a buyer to effectively check so many layers of a supply chain. This problem points to the need to cascade responsible practices to each supplier tier.

Cascading practices to the extended supply network refers to transforming suppliers to be as vigilant as the buying firm in managing and assuring sustainability. If suppliers recognize the value and importance of maintaining the sustainability of their own supply network, and are motivated to engage in similar types of "response" strategies as the buyer, then it will increase the chances of the whole supply network being able to effectively address sustainability problems. In other words, the accountability of sustainability must be cascaded to the next level, which in turn should cascade to another level, and so on. As highlighted in a Hewlett-Packard case study (Rammohan 2008), the company engaged intensively with its key suppliers, with the intent of guiding them in being equally focused on sustainability improvements with their own supply networks.

By using a cascading approach, Intel has worked to permeate socially and environmentally responsible practices throughout its supply network (Intel and BSR 2013). The company advocates the concept of supplier ownership of issues, so that suppliers will take a proactive approach in determining their own sustainability strategies and objectives. The company has experienced positive results when key suppliers such as Murata and Schneider Electric were able to take on the role of sustainability advocates in managing their own supply network.

20.4.2.7 Advocacy

At times, identified issues are more systemic and are therefore better addressed at the industry or national level, rather than with individual suppliers. Oka et al. (2020) distinguish between ad hoc and more institutionalized advocacy. The former can take the form of boycotts, letter writing and face to face meetings with government officials, and can be quite effective at times. For instance, when in 2017 a group of brands threatened to boycott the Dhaka Apparel Summit in response to a violent

crackdown on striking garment workers, it resulted in an immediate release of detained activists (Oka et al. 2020).

An example of institutionalized advocacy is the ACT initiative,³ which involves a coalition of global brands and retailers seeking to leverage collective bargaining at the industry level in pursuit of living wages for workers in key garment-producing countries. They aim to negotiate a living wage that applies to the whole industry in each of these countries. By leveraging their collective bargaining power, buyers are more likely to obtain meaningful results than if each of them attempts to tackle these issues on their own.

20.5 Business Benefits of Sustainable Supply Chain Programs and Other Insights

20.5.1 Effectiveness of Sense and Response Practices

Many companies incorporate both penalties and incentives into supplier evaluations. How effective are these practices? One study suggests that incentives and penalties (“response”) have a stronger influence on supplier compliance and buyer operating costs compared to visibility and monitoring efforts (“sense”) (Porteous et al. 2015). The study suggests that firms can improve supply chain social and environmental performance by ceasing business with continually violating suppliers, while using specific incentives, such as supplier training, increased business, and public recognition, to motivate and further build capabilities of higher-performing suppliers. This aligns with research that finds monitoring through supplier audits, when not complemented with supplier collaboration, can be ineffective in addressing persistent supplier violations (Locke et al. 2007).

20.5.2 Business Benefits of Sustainable Supply Chain Programs

Naturally, the top priority of sustainable supply chain programs is to improve social and environmental performance throughout the supply chain, and ensure compliance with rules and regulations. At the same time, such programs can also provide buyers with multiple business benefits, including risk mitigation, cost reduction, increased revenue, the ability to innovate on sustainable products and services, and more. Realizing such business benefits is top of mind with supply chain executives: 53% of responding executives in Gartner (2021) reported investing

³ See <https://actonlivingwages.com/>

How has your organization benefited from your sustainable procurement program? Select any areas that you strongly believe your company is benefiting from, even if you are not specifically measuring it.



Fig. 20.5 Business benefits of sustainable supply chain programs. Source: Sustainable Procurement Barometer 2021, Gillai et al. (2021), page 28

in demonstrating profitability and other benefits of achieving sustainability in the supply chain. Figure 20.5 lists some of the business benefits already realized through such programs.

One form of risk that sustainable supply chain programs can mitigate is the one associated with the damage that revelations of social or environmental violations can bring with them. As an example, consider ultra-fast fashion online retailer Boohoo, which lost more than £1.5 billion (\$1.89 billion) in market value after the Sunday Times revealed that during the COVID-19 pandemic workers making clothes for Boohoo were paid far less than the living wage and had to work in poor conditions with high risk of infection (Paton 2020).

Increased resilience may be attributed to the greater visibility to supplier operations and the closer and collaborative relationship with suppliers that sustainability programs foster, which makes it easier for buyers to assess challenging situations more accurately and respond to crises in a timely manner. The closer relationships with suppliers may also help buyers receive preferential treatment during times of disruption. One example that illustrates this benefit is offered in Gillai et al. (2021), where more than 60% of respondents believed that their sustainable procurement program contributed to their ability to endure the COVID-19 pandemic.

As for the potential of sustainable supply chain programs to increase sales revenues, Buell and Kalkanici (2020) find through a number of experiments that consumers take companies' responsibility efforts into account in their decision making. More specifically, they find that transparency into a company's supply chain responsibility practices can increase a consumer's probability of purchase by 6.40% and 45.85% across two field experiments respectively.

Select initiatives can also bring about cost savings. For example, environmental compliance can reduce costs if fewer resources are wasted in production, and higher productivity may be realized from efforts to improve worker skills and empowerment (Bloom et al. 2013).

There seems to be a correlation between the maturity of a company's sustainable supply chain program and the degree of business benefits it provides. In Gillai et al. (2021), buyers whose program had a wider scope, strong management commitment, covered at least some high-risk and strategic suppliers, and provided visibility to at least tier-1 suppliers, were identified as "Leaders." Those leaders not only made more extensive use of various "sense" and "response" tools and processes, but also a higher percentage of them were able to realize each of the business benefits listed in Fig. 20.5.

As for suppliers, they are likely to realize at least some of the above-mentioned benefits reported by buyers. In addition, a strong commitment to sustainability may help them protect their current business by preventing existing customers from leaving them. Such commitment may also help suppliers attract new customers and expand their business. While some may worry that requiring factories in developing countries to improve working conditions may make it harder for them to survive, an analysis of a relevant dataset found little evidence supporting such claims (Robertson et al. 2021). In fact, the results point to a positive correlation between some types of improvements and factories' chances of survival.

20.5.3 Obstacles to Implementation

While sustainable supply chain programs are adopted by a growing number of organizations, effective implementation may pose various challenges. Some professionals may find it challenging to secure sufficient funding or they may be worried about potentially higher operating costs associated with the more sustainable processes. Some may experience a pushback from suppliers or may find it hard to effectively track supplier sustainability performance. Each of these potential obstacles was identified as a top concern for about a third of survey respondents in Gillai et al. (2021)—see Fig. 20.6. Additional challenges may include lack of sufficient knowledge of best practices to adopt, and no access to relevant technological solutions.

Some of the "response" practices discussed earlier, which support suppliers' transition to more sustainable practices and offer them incentives to do so, may reduce suppliers' resistance. As for financial concerns, the understanding of the

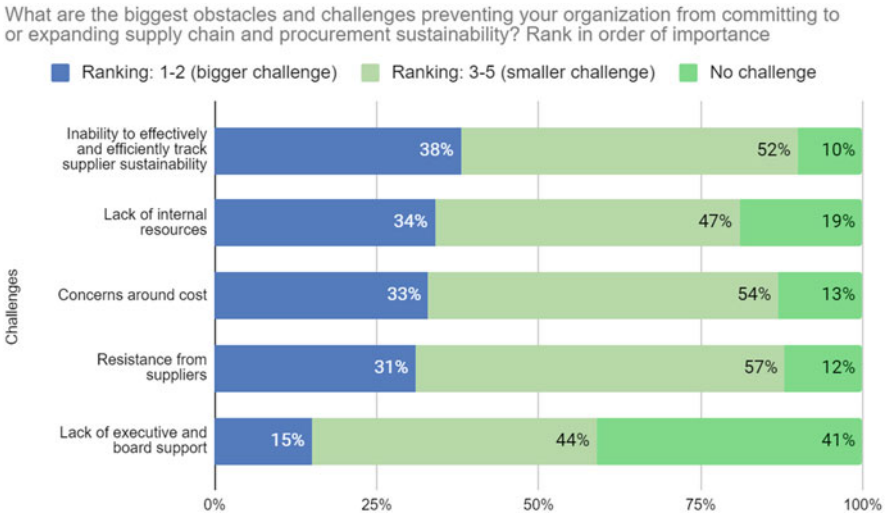


Fig. 20.6 Obstacles to implementation of sustainable supply chain programs. Source: Sustainable Procurement Barometer 2021, Gillai et al. (2021), page 26

business benefits that sustainable supply chain practices may provide may make it easier for supply chain leaders to secure the required funding. Moreover, Gillai et al. (2021) show that mid-size companies, which are likely to have more limited resources they can dedicate to sustainable procurement programs, were still able to successfully implement such programs and realize meaningful benefits. Supply chain leaders should therefore not be deterred by limited resource availability.

20.6 Summary

This chapter offers strategies for improving social and environmental performance in global supply chains using a sense and response framework, and discusses the effectiveness of “sense” and “response” practices based on published research. It suggests that monitoring and other “sense” practices on their own can have limited effectiveness, but that pairing them with various “response” practices such as incentives and penalties can result in meaningful improvement in supplier compliance. The chapter also highlights the fact that sustainable supply chain programs can offer buyers and suppliers a range of business benefits, including financial benefits, risk mitigation, and more. Moreover, those buyers with more mature and expanded sustainable supply chain programs are likely to reap a wider range of benefits.

There is a need for more research on the impact of various sense and response sustainability strategies on social, environmental and economic performance. As

responsible supply chain management continues to evolve and incorporate practices that go beyond monitoring efforts, there is an opportunity to better integrate management into core supply chain operations in order to have a stronger impact on conditions in global supply chains. For firms which engage the entire supply chain in the search for breakthroughs and risk reduction, there are opportunities to gain competitive advantage (Lee 2010). Calls for increased collaboration with suppliers and stronger incentives (Plambeck et al. 2012) will likely continue to grow, as will the call to improve monitoring methods, seek opportunities to build shared value, and ultimately cascade these practices to suppliers throughout a firm's network in order to create lasting change.

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Chapter 21

Social Responsibility in Supply Chains



ManMohan S. Sodhi and Christopher S. Tang

21.1 Introduction

Companies are increasingly confronted with social responsibility questions in the media throughout the supply chain they are seen to control, and they find it particularly challenging. We look at how companies can design and operate supply chains to fulfill their social responsibility and aim to address four key questions that face managers and researchers.

One question is about the choice of the level and scope of supply chain operations to consider. Supply chain operations comprise coupled processes that, in turn, include coupled sub-processes, whether within a department or across many companies. Moreover, for social responsibility, the context is always a large company. Such a company would typically have many global supply chains entailing many other large and small companies, so the context for any analysis or application must be chosen carefully. Finally, the observed operations may be considered “socially responsible” at one level and in one context but not so at another level or in another context.

An example is the 2014 spat between Oxfam and American actress Scarlett Johanssen, then brand ambassador for the charity. Johanssen also became a brand ambassador for SodaStream, an Israeli company manufacturing in the occupied West Bank, claiming the company contributed to peace by giving jobs to Palestinian people under occupation. At another level, Oxfam maintained that “businesses, such

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as SodaStream, that operate in settlements further the ongoing poverty and denial of rights of the Palestinian communities that we work to support” (BBC 2014).

A second question pertains to the many alternative definitions and the diverse objectives of social responsibility. Dahlsrud (2008) has identified 37 definitions of CSR from various researchers and industry bodies (mainly in the period 1998–2003). In the operations literature, Carter and Jennings (2004) take CSR to include business ethics, philanthropy, community, workplace diversity, safety, human rights, and the environment. Lists of objectives tend to be rather long—see, for instance, Carroll (1979, 1999) and Bowen (1953: pp. 8–12), although Friedman (1970) advocates the single-objective view that “the social responsibility of business is to increase its profits.” Dealing with multiple objectives raises the question of Pareto efficiency (thus, having to trade one objective off another) as well as *maximizing* versus *satisficing* (meeting some threshold value of) these objectives (Ackoff 1970). Shareholders may also be content with meeting threshold shareholder value (Monsen and Downs 1965).

A third question is how companies can be socially responsible by working directly with the weaker members of society, the “poor,” who comprise the majority in every society. Social responsibility refers to the responsibility of business to society. In the business-and-society discourse, “business” really means large companies (corporations) because “the powerful are given closer scrutiny” (Carroll and Buchholtz 2012: p. 6; see also Bowen 1953: p. 6), narrowing further on the senior managers who make strategic decisions at these companies (cf. Bowen 1953). Equally, while “society” is a broad concept, political, media, and research attention is focused on those at the other end of the power spectrum, i.e., groups of people without economic or political power: employees, small suppliers, and local communities. Therefore, the question is: can companies with economic power run supply chains to meet the needs of the weakest in society, the poor, not just by selling to them but also by improving their economic level by employing them or buying from them?

A final question for managers and OM researchers is to decide whether to or how to develop overarching frameworks to guide a company’s strategy to incorporate social responsibility. An overarching framework requires consistency with theoretical frameworks already used in the strategy, OM, and social responsibility literature and practices. For instance, strategists may use the resource-based view, OM modeling may entail utility theory, and social responsibility practitioners may draw on stakeholder theory. The alternative would be either to develop an entirely new framework or to separate social responsibility efforts from its operations by, for instance, relying solely on philanthropy.

21.2 Literature Review

21.2.1 *Corporate Social Responsibility and Environmental, Social, and Governance*

The social obligations of the business are generally codified as corporate social responsibility (CSR). Rangan et al. (2015) describe two other “theatres” for a company doing CSR besides philanthropy: operational improvement and new business models. In the OM literature, there is interest in how CSR initiatives impact purchasing and supply chain management (Cruz 2009) and how supply chain managers incorporate or implement CSR (Carter and Jennings 2004; Carter 2005; Maloni and Brown 2006). London et al. (2010) examine value creation with social enterprises, while Sodhi and Tang (2011) take a supply chain perspective on these enterprises.

Furthermore, Sodhi and Tang (2018) provide a thematic analysis of the literature on corporate social sustainability in supply chains. They identify eight themes in the pertinent literature: (1) stakeholder pressure, (2) governance, (3) contingencies, (4) practices, (5) partnerships, (6) barriers and enablers, (7) performance (including data and metrics), and (8) optimization. They also provide a thematic map, connecting these themes (Fig. 21.1).

In general, there remains considerable skepticism about whether a modern corporation can or does fulfill its social obligations (Banerjee 2007; Devinney 2009). Such skepticism may explain why research in “socially responsible operations” tends to focus on social enterprises, small farmers, foundations, etc. Only a handful of papers in the operations or supply chain literature focus on large company initiatives such as

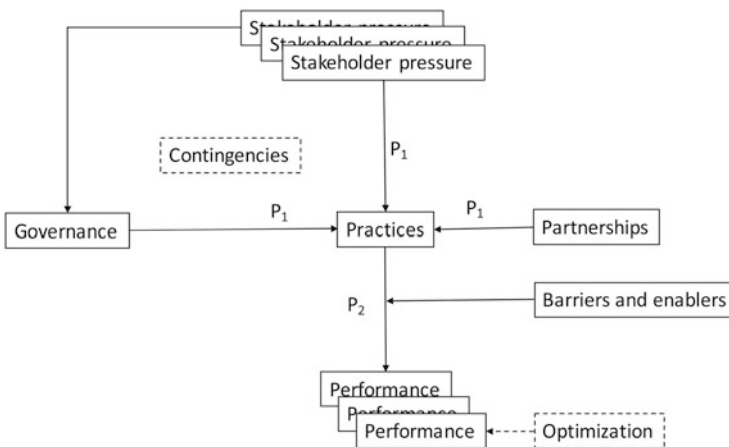


Fig. 21.1 A thematic map of the themes in the literature on social responsibility in the supply chain. P₁ and P₂ correspond to propositions on influence. Source: Sodhi and Tang (2018)

ITC's e-choupal, an electronic platform to provide farmers the company's purchase price 1 day in advance, or Unilever's Shakti Amma, training women in rural areas to sell Unilever's products with financial help from microfinance NGOs; see Chap. 20 by Gillai et al. (2024) for additional examples.

Ever since the release of a United Nations report in 2006, environmental, social, and governance (ESG) issues have garnered significant attention from academics, businesses, and governments (Atkins 2020). Unlike CSR, which pressures a company to be "accountable" for its actions, ESG is intended to make a company's actions and their outcomes "measurable" so that investors can make their investment decisions and the public can form a judgment. The US Securities and Exchange Commission (SEC) established an ESG subcommittee, which proposed in 2022 that all publicly traded companies disclose material ESG information (SEC 2022).

The "Environmental" pillar focuses on carbon emissions and consumption of natural resources such as water, not only in the firm but in the supply chain as well, the "Social" pillar includes customer satisfaction, employee engagement, and supplier labor standards (Dai and Tang 2022). Finally, the "Governance" pillar refers to managers overseeing suppliers through supplier selection, supplier comparison, supplier monitoring, and policies on how the company and its suppliers deal with whistleblowers (Sodhi and Tang 2018).

21.2.2 Sustainability

'Sustainability' has become an instrument for large companies to subsume diverse company initiatives for CSR, environment, and profitability. It allows companies to align their CSR and environmental efforts with profitability. Initially, the concept was conceived as being broader in scope (cf. Elkington 1998): The Brundtland Commission defined it "as development that meets the needs of the present without compromising the ability of future generations to meet their needs" (Carter and Rogers 2008).

Even before the formalization of ESG, Elkington (1998) presented sustainability as having three "pillars"—economic, environmental, and social—with overlapping zones. Such overlapping zones are helpful for a company because it can report initiatives as serving the environmental (or social) cause even when these are economically motivated. Indeed, the most reported initiatives in companies' sustainability reports are about reducing energy consumption. However, the focus on overlaps avoids the awkward question of how to make trade-offs between profitability and social objectives—see Pagell and Shevchenko (2014) in this regard. There are also trade-offs between environmental and economic sustainability: shutting down coalmines (to reduce pollution) adversely affects mining communities' economic and social sustainability.

The OM literature has considered sustainability from an environmental perspective without explicitly incorporating social aspects (Carter and Rogers 2008). Sustainability has entered the OM literature as "sustainable operations manage-

ment” (Kleindorfer et al. 2005) or “sustainable supply chain management” (cf. Linton et al. 2007; Seuring and Müller 2008; Pagell and Wu 2009). Pagell and Shevchenko (2014) argue that sustainability should be entrenched in all aspects of supply chains. In a similar spirit, Dai and Tang (2022) articulate the importance of unifying ESG and supply chain thinking and present research opportunities in ESG and supply chain management.

21.2.3 The Poor, the Bottom of the Pyramid, and Shared Value

The “bottom-of-the-pyramid” (Prahalad 2006) approach entails large companies seeking to increase profits by selling goods and services to the “poor” and possibly using them as suppliers or distributors. Such operations can require redesigning goods and packaging, such as Unilever selling shampoo or skin color-lightening products in small sachets to poor consumers in developing countries (Karamchandani et al. 2011). However, Karnani (2007) provides economic arguments against marketing consumer goods to the bottom-of-the-pyramid poor. A broader approach is the “base-of-the-pyramid” (cf. London and Hart 2011), where “shared value” is highlighted. An example is German-based Bayer, selling agricultural chemicals in small packets to smallholder farmers in developing countries (Karamchandani et al. 2011), creating profits for itself and the farmers. Redesigning production and delivery modes using the poor as suppliers and distributors can help them become economically better off (Sodhi and Tang 2014). More recently, Fransoo et al. (2017) discuss how certain innovative logistics strategies can sustain the operations of 50 million “nanostores” operating in many neighborhoods of large cities across the developing world, enabling the poor to gain access to consumer-packaged goods and supporting and livelihood of those small family operated stores.

Engagement with the “base” of the pyramid requires new business models to engage with the “poor” as customers or suppliers. Karamchandani et al. (2011) discuss challenges for companies wishing to engage profitably with the bottom of the pyramid: (1) uncertainty of cash flows given a large number of low-margin and low-value transactions, (2) gauging demand and working in “informal” markets, (3) sales and distribution, (4) providers being neither aggregated nor capable enough to provide quality or volume, and (5) business ecosystems not being able to support initiatives.

Although sustainable development has been studied extensively in the development economics literature (Ray 1998; Lal 2000; Hayami 2005), *operational* issues in this context have yet to be explored much. Sodhi and Tang (2011) have looked at social enterprises from a supply chain perspective: These are not large companies, but the work of social enterprises has practical implications for what large companies can do. They explore this idea further by looking at the poor as suppliers or distributors in the supply chain rather than as consumers (Sodhi and Tang 2014).

The relationship between the company and those at the bottom of the pyramid can create “shared value” (Porter and Kramer 2006), which recalls the overlaps between the pillars of sustainability “by reconceiving the intersection between society and corporate performance.” However, they avoid discussing how to divide the “shared” value (Coff 1999); Crane et al. (2014) provide some more limitations of the shared value concept.

21.2.4 *Stakeholder Resource-Based View*

Sodhi (2015) outlines the “stakeholder resource-based view” (SRBV), building on the resource-based view, utility theory, and stakeholder theory to better understand how to incorporate social responsibility in supply chain operations.

One model for companies to build lasting competitive advantage is the **resource-based view** (RBV) or its extensions (cf. Hart 1995; Lavie 2006), whereby part of the resources are bundled as firm-specific “capabilities” that the firm develops in a static economic setting (cf. Wernerfelt 1984; Barney 2001). The resources must raise barriers to entry to others if a competitive advantage is to be durable (Rumelt 1984). In a dynamic economic setting with high uncertainty, resources must be changed using “dynamic capabilities” as the firm seeks competitive *survival* in a rapidly changing environment (cf. Teece et al. 1997). Dynamic capabilities “are the organizational and strategic routines by which firms achieve new resource reconfigurations as markets emerge, collide, split, evolve, and die” (Eisenhardt and Martin 2000). However, many other definitions exist (Ambrosini and Bowman 2009). “Dynamic” refers to the external environment rather than the capabilities built around “routines” that are the organization’s processes. Being able to deliver on social responsibility could be a firm’s capability or dynamic capability. However, dynamic capability, say, with the company moving facilities from one low-cost country to another lower-cost one continually, could also be tied to social *irresponsibility*.

Garriga and Melé (2004) classify theories in the social responsibility literature as (1) *instrumental theories* with the corporation solely as an instrument for wealth creation, so any social responsibility activity serves to further that aim; (2) *political theories* about the responsible use power of corporations in society and the politics; (3) *integrative theories* on how business integrates social demands based on the assumption that business depends on society for its existence, continuity and growth; and (4) *ethical theories* based on ethical responsibilities of corporations to society. These categories can overlap, and integrative theories can potentially be reconciled with instrumental theories. If a company can meet some social demands only to make more profits in the long run, it is trying to integrate social aspects into its decisions. If political power, such as that exerted by large companies on governments to increase wealth, an instrumental view can also subsume political views.

One integrative theory is the **stakeholder theory**. The assumption is that managers have fiduciary duties to the corporation, not just to the shareholders. The stakeholders are all the people and groups interested in the corporation. According to Donaldson and Preston (1995), the interests of all stakeholders are of intrinsic value, and “each group of stakeholders merits consideration for its own sake and not merely because of its ability to further the interests of some other group, such as the shareowners.”

In the analytical operations and supply chain literature, **utility theory** is used in the economics of decision-making. Here, the assumption is that we have rational players who seek to maximize their utility—their preference for goods and services—given the possible/actual moves of the other players. Utility theory is consistent with the corporation as an instrument for wealth creation for wealth maximizing shareholders. If the concept of utility can extend to the *means* for acquiring goods and services, it can cover both resources (as in the resource-based view) under known conditions and dynamic capabilities under uncertainty.

Building on RBV, utility theory, and stakeholder theory, Sodhi (2015) proposes the **stakeholder resource-based view** (SRBV), defined as:

SRBV is a model to guide the decision-making of managers towards maximizing their utility by developing their organization’s capabilities—dynamic capabilities, resources, and routines—while recognizing the need to improve the respective utilities of other groups of the organization’s stakeholders, possibly by helping them develop their capabilities as extensions of the company’s capabilities. Under SRBV, stakeholders for the organization are those whose utility is significantly dependent on these managers’ decisions.

SRBV helps managers recognize, whether for a company or particular operations within a company or across companies, that *there are different groups of stakeholders with their respective resources, routines, and dynamic capabilities, seeking to maximize their respective utilities under uncertainty and over time horizons*. Stakeholders for a large corporate include those involved in operations: *suppliers* such as smallholder farmers and contract laborers, *employees*, *mid-level managers*, *senior managers*, and *distributors/wholesalers/franchisees*. *Shareholders*, *government*, *communities* where facilities are located, and *consumers* are also stakeholders. Note that the “company” or “corporation” is not a monolith. Instead, we have senior managers, mid-level managers, and shareholders of companies, although our focus is on senior managers as decision makers. Under SRBV, each stakeholder (individually or as a group) is treated on par with other stakeholders from a research perspective.

SRBV allows the manager as well as the researcher to tackle the four key questions listed in Sect. 21.1:

1. *The choice of the level and scope of supply chain operations to consider:* Under SRBV, there is no explicit restriction. The manager could consider any subset of stakeholders as long as the utility and capabilities of each (type of) stakeholder is fully accounted for. In the Johanssen-Oxfam example, both sides could agree on the utility of Palestinians obtained from gainful employment at Israeli companies operating in the occupied territories. Equally, they could agree that there are other

(non-working) Palestinians whose utility is affected by living conditions under occupation, which is only solidified through the operations of Israeli companies. Under SRBV, the manager (or researcher) has to treat all stakeholders of interest on a par with each other to understand them at an economic level. Different managers may select different subsets of stakeholder groups, but they can agree on any stakeholder's utility derived from the operations in scope. Furthermore, the unit of observation is the "operation," whatever its scope. A particular manager researcher will have to scope out the breadth and level of the operations over which they can make decisions.

2. *The many alternative definitions and the diverse objectives of social responsibility:* Under SRBV, we have a broad conceptual view of utility. Managers must recognize that other stakeholders have their objectives underlying their respective utility. Under SRBV, the different objectives from the broad CSR literature are split across the various stakeholders, and the manager should recognize other stakeholders' objectives as their respective utility functions. Understanding utility by stakeholder allows the manager to focus on and differentiate stakeholder-specific drivers of utility maximizing effort.
3. *How companies could be socially responsible by working directly with the weaker members of society:* The poor may be stakeholders if part of the company's supply chain as suppliers, distributors, consumers, or even as members of communities where the company has supply chain operations. If their utility (or disutility) is affected by the company's operations—indirectly by the managers' decisions—then the poor are stakeholders. Developing their capabilities may be part of or align with the managers' efforts to build their company's capabilities. However, philanthropic efforts unrelated to the company's operations are not included in SRBV. This is because the intended beneficiaries of philanthropy may not be stakeholders whose utility depends significantly on a company's operations. Still, there is a gray area if the philanthropic effort is considered helpful for marketing.
4. *Deciding how to develop overarching frameworks to guide a company's strategy to incorporate social responsibility:* SRBV provides a framework that enables decision-making consistent with utility theory, resource-based view, and stakeholder theory. Concepts like "shared value" are rendered more tangible under SRBV in terms of increasing utility for the managers, their shareholders, and their suppliers or the communities in which these suppliers operate. Using SRBV, a manager can have a long-term view focusing on developing their company's capabilities (dynamic capabilities, resources, and routines) and those of the company's stakeholders.

21.3 Related Findings and Practical Implications

We find many operations and supply chain configurations being tried by different organizations, mainly targeting the economic improvement of the poor. These organizations are primarily social enterprises rather than large companies. As such,

we first discuss our findings for social enterprises in Sect. 21.3.1. Next, we discuss our results related to the poor as suppliers (Sect. 21.3.2), as distributors (Sect. 21.3.3), and as borrowers of working capital lending specifically targeting them (Sect. 21.3.4). This sets the stage for us to discuss implications for large companies in Sect. 21.3.5.

21.3.1 *Social Enterprises*

Sodhi and Tang (2011) view social enterprises as enabling the supply chains of micro-entrepreneurs with the supply chain perspective of material, information, and cash flows. Here are some examples of social enterprises: To lend to the poor, Grameen Bank relies on group lending. SKS was a for-profit microfinance organization modeled on Grameen Bank for group lending, using capital markets to scale up its lending operations quickly. It later re-emerged as Bharat Financial Inclusion (<https://www.bfil.co.in>), a subsidiary of a commercial Indian bank. Kiva (www.kiva.org) is a person-to-person online lending organization where volunteers vet potential borrowers.

Then there are the enterprises that consider the poor as suppliers. For instance, Chicago-based Arzu (<https://changingthepresent.org/collections/arzu-inc>) sells custom-designed rugs by Afghan women weavers. Thamel (<https://thamel.com>) is an online portal that allows the Nepali diaspora to send gifts and money to their loved ones in Nepal. Coconut World sells coconut sugar produced by farmers in the Philippines. Another example of a social enterprise is Ecomaximus (<https://ecomaximus.com>), which produces and sells elephant dung paper co-produced by the villagers and elephants in Sri Lanka. Now defunct Men-on-the-Side-of-the-Road (MSR) served as an intermediary to help day laborers in South Africa find jobs in a safe environment.

VisionSpring (<https://visionspring.org>) uses the poor as distributors. It procures cheap reading eyeglasses and sells them through micro-entrepreneurs, thus providing affordable reading glasses for low-income individuals.

Mountain Hazelnut Venture Limited is a social enterprise who distributes young hazelnut plants at no charge to a large number of subsistence farmers in Bhutan. The farmers would plant the trees in fallow or degraded land, tend them, and sell the resulting nuts to the company at a price negotiated between the Bhutanese government and the company. By early 2011, this social enterprise planted over ten million trees in 5 years and involved 15% of the country's population (Hoyt and Lee 2011).

Other social enterprises seek to help micro-entrepreneurs become more productive. KickStart (<https://kickstart.org>) develops and sells mechanical irrigation pumps in sub-Saharan Africa so farmers can improve their crop yields and the number of crops a year, reducing the dependence on annual rainfall.

21.3.2 *The Poor as Suppliers*

A company may be able to buy goods from the poor directly at reasonable prices and, at the same time, be seen as contributing toward poverty alleviation. In developing countries, social enterprises and companies can help the poor as suppliers by using three basic models:

1. **Reducing intermediate echelons to obtain higher selling prices:** Farmers and other small producers in developing countries typically sell their output through layers of intermediaries and consequently get low prices for their products. This creates an opportunity for social enterprises or companies to help the poor by purchasing their output directly. For example, Coconut World buys coconut sugar made by small farmers in the Philippines directly, then sells directly to consumers through its online store and to other retailers in the USA (Cameranesi et al. 2010). Walmart purchases the crops directly from farmers in China to reduce its costs; the farmers also benefit by getting a higher price. The social enterprise Arzu purchases wool rugs directly from Afghan women and sells these in the USA. Organizations such as Fairtrade certify direct purchases from the farmer by manufacturers and retailers (Chen et al. 2022). Doing so enables these manufacturers and retailers to advertise to primarily western consumers that the company is working to ensure the farmer gets a higher price than he would get from intermediaries.
2. **Reducing search cost:** The poor, as suppliers, do not have an easy way to search for customers for their products or services. In South Africa, Men-on-the-Side-of-the-Road (MSR) developed an online portal as a marketplace for day laborers (micro-entrepreneurs) and homeowners, which helps laborers and potential customers find each other (Sodhi and Tang 2011). Likewise, truck owners in India or South Africa are typically micro-entrepreneurs with a single vehicle; private enterprises have created websites offering to match loads from shippers with trucks to help reduce the problem of trucks heading home empty after delivery, e.g., LoadJunction.com in India or 123LoadBoard in South Africa. Chipchase (2006) reports that customer demand for information on mobile phones has helped taxi drivers increase their earnings in Pakistan and Thailand. The same has been said for fishermen in Kerala seeking markets for fish, a perishable product, because of the fishermen's lack of access to cold chain facilities (Jensen 2007). A broader question is that of the value of information. For instance, Chen and Tang (2015) analytically obtain conditions under which the use of (free) public or (costly) private information can be beneficial to a smallholder farmer or not.
3. **Improving productivity:** The poor, especially small farmers, often need more relevant information to enhance productivity and increase selling opportunities. In India, IFFCO disseminates information about weather forecasts and crop advisory information (what to cultivate, when to harvest, and how to improve yield and quality) to farmers via mobile phones so that they can plan their farming activities accordingly (Ghosal and Parbat 2012). Also, Reuters Market

Light (RML) tracks the prices of 50 commodities over 1000 markets and the weather conditions of 2000 locations. It disseminates crop- and location-specific information to subscribed farmers in India using SMS text messages so that farmers can sell their products at a higher price (Preethi 2009). A new platform WeFarm enables farmers to exchange farming techniques, boosting their productivity (Xiao et al. 2020).

21.3.3 *The Poor as Distributors*

In developing countries, the distribution infrastructure is inadequate. A social enterprise or a company can help reduce distribution costs by using micro-entrepreneurs as distributors. For example, Mozambique-based VidaGas uses micro-entrepreneurs to sell propane gas to food stall owners, fishermen, health clinics, etc. (Watson and Kraiselburd 2009). Vision Spring sells affordable reading glasses to low-income individuals through a network of micro-entrepreneurs in developing countries (Bhattacharya et al. 2010). Social enterprises like Living Goods and Solar Sisters, both operating in Uganda, also use women micro-entrepreneurs to distribute household necessities and solar lamps in the last mile, thus emulating the model of the famed Avon Ladies (Economist 2012).

The primary distribution strategy entailing the poor as distributors is a **hub-and-spoke** strategy. An enterprise can set up a center in a larger village as a “hub” from which micro-entrepreneurs (or employees) can travel to the more remote rural areas as “spokes” to sell goods or provide services. Such a distribution network can further benefit from (a) using existing commercial/non-commercial networks for moving goods to the micro-entrepreneurs or (b) providing additional services at the hub or selling more products or services to create more supply chain surplus.

Providing additional services is a **piggyback** strategy. Gramin Suvidha Kendra, a private–public partnership between MCX and Indian Post Office established in 2006, distributes seeds, fertilizers, water purifiers, micronutrients, and solar lanterns to farmers via the ubiquitous post offices in India (Vachani and Smith 2008). In Africa, Cola Life, an independent UK charity, has used a wedge-shaped container that fits between the Coca-Cola bottles in their crates to reduce distribution costs (see www.colalife.org).

21.3.4 *Working Capital Lending for the Poor as Suppliers or Distributors*

Micro-entrepreneurs need more access to credit from traditional banks not only because of a lack of credit history or collateral but also because of the small amounts of money involved relative to the transaction cost for the bank for screening and

collection. Therefore, social enterprises and companies can find ways to finance the working capital if they engage the poor as suppliers or distributors. For example, as already noted, Vision Spring provides each micro-entrepreneur in its supply chain with \$75 worth of eye charts, brochures, and a stock of reading glasses. In general, there are different types of microfinance models for micro-entrepreneurs to obtain micro-loans:

1. **Self-help groups (SHG)/ Rotating Savings and Credits Associations (ROSCAs).** A community of the poor can form self-help groups, where all members bring savings to weekly (or monthly) meetings, and one of the members can take a loan from these savings (Ardener 1995; Snow 1999). While this approach cannot be used for working capital funding, it might be helpful for a micro-entrepreneur to purchase capital goods like a bicycle.
2. **Community banks.** These banks seek to stimulate economic development (in terms of business and job creation) for their communities. Grameen Bank uses “group lending” to reduce its screening, monitoring, and collection costs: all group members are responsible for providing the repayment when one of the members is behind (Foroohar 2010). The benefits of group lending can be extended for working capital funding if all group members are suppliers or distributors in the same supply chain. Both materials and cash flows can be aggregated at the group level.
3. **Peer-to-Peer Networks.** Kiva is a person-to-person online lending organization that enables people in developed countries to provide micro-loans (Flannery 2007). So this could be used for funding micro-entrepreneurs wishing to be suppliers or distributors but more for investment in capital goods rather than working capital.
4. **Commercial MFIs.** SKS is a for-profit, publicly traded microfinance organization in India that uses capital markets to scale up its operations quickly and uses information technology to reduce operating costs (Akula 2008). Such a system could also be helpful for working capital funding if micro-entrepreneurs keep rotating balances.
5. **Data-driven lending schemes for nanostores.** In China, nanostores and other small enterprises lack access to financial loans and credits due to their lack of collateral and limited profitability, which hinders their growth. By leveraging the nanostore’s payment history via the digital wallets of Alipay or WeChat Pay, Alibaba’s and Tencent’s online banks can use data analytics to make loan decisions to small business lenders. As of 2018, Alibaba has offered 1.19 trillion yuan in loans to 9.78 million small business lenders. In Latin America, Tienda Pago, a startup in Latin America, has been working closely with AB InBev in Mexico and Peru to provide lines of credit to nanostores for replenishing their AB InBev stocks based on their historical sales and payments. Upon deliveries, shopkeepers have 1 week to repay the credit loan (Escamilla et al. 2021).

21.3.5 Practical Implications for Large Companies

Large companies can learn from social enterprises. Indeed, a company can offer all three benefits to the poor as suppliers—reducing the number of intermediate echelons, reducing search costs for selling products/services, and improving their productivity—by exploiting both supply chain structure and information technology.

Consider Indian consumer goods giant ITC's e-Choupal initiative: ITC provides farmers with the historical selling prices of different crops at different locations on its web portal, and ITC pre-announces its price for purchasing the crops directly from the farmers before the market opens the next day. These smallholder farmers are least aware of a floor price when bringing their produce to ITC or a commodity marketplace (Anupindi and Sivakumar 2006, 2007; Goyal 2011).

Companies can also use the poor as distributors. Coca-Cola in East Africa delivers over \$500 million of product to 1800 “manual” distribution centers operated by 7500 micro-entrepreneurs. These micro-entrepreneurs use push carts to distribute the product to nanostores in congested areas, making frequent but small deliveries to these cash-strapped micro-retailers. Another example is Hindustan Unilever, a subsidiary of Unilever in India, that started Project Shakti in 50 villages in 2000 with woman entrepreneurs receiving training and stocks of consumer-packaged goods from Unilever's rural distributor to sell the goods to consumers and micro-retailers in 6–10 villages (Rangan and Rajan 2007).

Companies may be pushed by the government to deliver on their social responsibility in areas with unmet social needs. Sodhi et al. (2022) study the case of India, where the government has required large companies to spend a fraction of their income to deliver their social responsibility toward development since 2014. They analyzed the expenses of leading Indian companies, which they found to be statistically similar across companies, especially in the importance of education and healthcare. The spending patterns are quite similar to the government's motivation and the companies' motivations—with unmet social needs driving both the company spending and government needs. The model suggests that governments may find mandatory social spending by companies helpful in achieving development goals. By implication, if companies do not step up voluntarily, they may face mandatory requirements.

Finally, companies can offer microfinance as working capital for the poor as suppliers or distributors, e.g., by pre-paying for supplies from the poor. Collection costs are also reduced because the collection can piggyback on the transfer of goods. Lending transaction costs are significantly reduced if we tie micro-lending to the actual transaction. Moreover, the aggregation of suppliers or distributors can fit the group-lending model well, as we have already noted. A practical way would be to provide micro-retailers inventory on credit till the end of the day: the micro-retailer would effectively get credit for the day, and the company would limit its risk to the value of 1 day's inventory (Sodhi and Tang 2014). Alternatively, a company like ITC could lend to farmers before the sowing season and then get its money back

by receiving the product when the farmer brings it to ITC directly or receiving cash when he sells it on the mandi.

Banks can partner with a key supply chain partner to extend their loan programs to small businesses so that both banks and these small businesses can grow. For example, HSBC has linked up with Alibaba in March 2020 to gain visibility about the background, historical transactions, and projected future transactions of its merchants in Hong Kong who sell on Alibaba's platform Tmall. With access to accurate and timely data about these merchants, HSBC can afford to offer loans (up to US \$500,000) to these merchants quickly without the need for collateral or financial documents (Tang and Yang 2020). In doing so, this form of innovative financing scheme can align incentives among all parties: small businesses, Alibaba, and HSBC can grow together.

21.4 Future Research

Integrating social responsibility into operations and supply chain management practice using SRBV provides many opportunities for socially responsible operations and future research. This is because, as SRBV makes explicit, the researcher can choose any subset of stakeholders for study and suggest norms for their choices rather than take the viewpoint of only a company's managers. Below are some of these opportunities.

21.4.1 Developing Case Studies

We need more well-researched case studies or even descriptions of different operations settings detailing how other groups of stakeholders became better off (or not) because of the operations. One research question can be about the type of operations and how these operations are being economically sustained: What is the business model, and where is the money? Implicitly, this research question can include research objectives tied to *value creation* and *value delivery* (London et al. 2010) and *value sharing* (say, between micro-entrepreneurs and the corporation as between farmers and ITC in the latter's e-Choupal project). Sodhi and Tang attempt to understand how the supply chains of individual micro-entrepreneurs can be strengthened by social enterprises and examine the economic sustenance of such operations. A phenomenological investigation by way of field study and ethnography would be helpful as a foundation for further research.

One aspect of such studies could lead to a better understanding of the success factors of particular operations tied to local communities, NGOs, and the regional government working (or not working) together. Unanticipated *side effects* of seemingly socially responsible operations would stem from studying a broad set of stakeholders. For instance, donated clothes can negatively impact the local apparel

and retail industry, as seen in Africa. Looking at more stakeholders with the SRBV lens can help anticipate “side effects.”

Research in social irresponsibility beyond excellent journalism is limited. The problem of large companies setting up elaborate operations to avoid taxes is not new (Christensen and Murphy 2004) and may even be considered desirable by managers rather than “socially irresponsible.” However, there are other instances of well-documented irresponsible behavior by corporations. Armstrong (1977), using behavioral experiments, suggests that the problem of *irresponsible* behavior among managers may be widespread and is possibly linked to a “stockholder” perspective such as that advocated by Friedman (1970).

In the context of ESG, there is a need to develop more case studies to describe how companies improve the social aspects of their supply chains. For example, in 2021, Unilever announced its plan to pay a living wage to “all of its supply-chain workers” by 2030 (Bowman 2021). Although appreciated, the implementation of this plan is challenging because Unilever’s global supply chain operations entail over 400 brands of consumer goods in 190 countries. Such a case study can be useful for us to understand how to improve social welfare in global supply chains.

21.4.2 Social Enterprise

Social entrepreneurship offers an appealing proposition—making money by doing good. Several topics merit further studies, such as appropriate supply chain and other performance measures for social enterprises working with micro-entrepreneurs; supply chain coordination and collaboration between social enterprises and other organizations; how mutually created value is shared between the social enterprise and its micro-entrepreneurs; and support of government policy for social enterprises. Lee and Tang (2018) describe some research opportunities for developing ways for social enterprises to reduce their financial risks while supporting micro-entrepreneurs.

21.4.3 Developing a Better Understanding of the Poor

There are plenty of opportunities to research the decision-making of the poor in emerging markets. For instance, as the mobile phone penetration rate exceeds 90% in India, companies such as Reuters Market Light (RML) and Nokia offer information services to farmers (cf. Chen and Tang 2015). Some key issues to investigate include identifying the critical drivers for farmers regarding paying for a subscription, how farmers use the information in practice to make farming decisions, and whether or not such market information helps farmers earn more.

Mobile-based finance has been considered a significant breakthrough to help the poor conduct financial transactions (savings, loans, remittances, loan repayments,

and payments) over mobile phones (Lee and Tang 2012). One area of study could be how mobile financial services with instant access change the spending and savings habits of the poor. Also, Escamilla et al. (2021) argue that traditional research in retailing operations are not applicable to nanostore operations in developing countries. This is because the institutional environment of nanostores is fundamentally different: shopkeepers are subsistence entrepreneurs who operate in a harsher environment (inadequate infrastructure, informality, insecurity, etc.) that hinders their operational efficiency and productivity. As such, there is a need to examine these nanostore operations through a different lens using a different modeling paradigm.

21.4.4 Impact Studies

Measuring the alleviation of the targeted social problem across different time frames and scopes requires field study by way of so-called “impact” studies. Current studies do not have consistent results. For instance, Mittal et al. (2010) find that farmers subscribing to market information via mobile phones enjoyed higher income, while Fafchamps and Minten (2012) find no evidence supporting this claim. There is also room for analytical models here: Chen and Tang (2015) show that more accurate market information can affect prices and farmers’ well-being. Incidentally, studies of stock performance are not uncommon. Frooman (1997) does a meta-analysis of event studies to examine the impact of socially responsible announcements on the stock performance of a firm—similar work could be done with companies’ and other stakeholders’ performance. Also, it is of interest to examine the impact of different innovative technological adoptions such as mobile payments affect the poor in developing countries from broader perspective such as the UN Sustainable Development Goals such as gender equality (Tang 2022).

21.4.5 Monitoring Suppliers

Companies that face consumers directly do not wish to be associated in the media with such problems at their suppliers as child labor or poor work conditions of workers. How should companies monitor and motivate their suppliers? Porteous et al. (2015) analyze the responses from practitioners at 334 companies and report that incentives for suppliers rather than penalties are strongly associated with reducing the company’s violations and operating costs.

Monitoring suppliers can be challenging. For example, Patagonia acknowledged that it is very difficult to develop measures for living wages in different countries and to conduct audits to ensure its contract manufacturers pay their workers according to the living wage in their country. Quite often, worker compensations in many developing countries are opaque and difficult to verify (Chen et al. 2022). But

advanced technology can help. For example, Chinese fintech startup JDH leveraged mobile technology and blockchain to track financial transactions of suppliers several tiers deep into various complicated electronics manufacturing supply chains (Tang and Yang 2020). The latest DNA technology can also potentially help to trace products from their factories, so that unauthorized use of subcontractors could be detected (Tullis 2020).

21.4.6 Understanding the Role of Markets and Government in Improving Social Welfare

As groundwater or other natural resources get depleted, trading on the market is considered the best solution. But, does it work? Murali et al. (2015) show that exporting water through a water market with an exogenous price is detrimental to society and the community's environment if we consider "triple bottom line benefits." Their work generalizes to other commodities as well: consider India's ban on the export of cotton in 2012 and a 30% tax to discourage the export of iron ore in 2011. Also, different parties may not participate in the market. For instance, a significant amount of waste currently going to landfills or incinerators could be repurposed. Dhanorkar et al. (2015) consider why such exchanges have had limited take-up. Their work has implications beyond such exchanges to those of workforce such as Men on the Side of the Road in S. Africa and freight boards for truck transportation in Africa or Asia as there may be similar factors affecting lack of take-up.

How should the government balance different interests? This is an important research topic. Park et al. (2015) consider social welfare stemming from the optimal application of carbon taxes with retailers seeking to maximize profit and consumers seeking to maximize utility and show that the government will find carbon taxes more effective as the competition increases.

21.4.7 Improving the Lot of Smallholder Farmers in Developing Countries

Tang and Zhou (2012), Chen et al. (2013), Devalkar et al. (2011), and McCoy (2012) provide initial steps for further research in this area. Aggregating smallholder farmers via cooperative or other aggregations has attracted the attention of policymakers, those interested in social development, and undoubtedly many OM researchers (Chen et al. 2015). But are these always beneficial for farmers? An et al. (2015) find that smallholder farmers' cooperatives (or other aggregations) are not necessarily a silver bullet relative to farmers who choose not to join the cooperative.

One way to develop resources for smallholder farmers is through online or telephone forums. How should such forums be designed and operated? Chen et al. (2015) use game-theoretic analysis to show that knowledgeable farmers never provide answers that are more informative than the experts in equilibrium in a forum with experts and some knowledgeable farmers, Chen and Tang (2015) show that the value of private information providers such as RML in India decreases as public information services improve.

In this context, it would be helpful to study how to optimize different supply contracts (e.g., wholesale price, revenue sharing, or profit-sharing) to profits for the enterprise and poverty alleviation. These contracts would support capital provision to micro-entrepreneurs, for instance, farmers having to buy equipment, seed, and fertilizer. The role of the wholesale auction markets in India, called *mandis*, also needs to be better understood as to how the government can achieve its objectives optimally.

Boyabatli et al. (2022) examine three elements of agricultural supply chains: Planting and Growing, Processing and Selling, and Government Interventions. By examining how individual farmers respond to market information, incentive pricing mechanisms, and different market structures in the trade of agricultural goods, there are research opportunities for policy makers and social enterprises to develop and design market information provision policy, incentive contracts and market structures for improving farmer and consumer welfare.

21.4.8 Distribution Models Using the Poor

Efficient distribution strategies for enabling micro-entrepreneurs in developing countries to buy, distribute, and sell products have yet to be studied much. Moreover, it needs to be clarified how the value created should be shared between the network owner and the enterprise or micro-entrepreneurs for piggyback distribution. For example, how much should Coca-Cola charge Cola Life for distributing its AidPods? How much should India Post charge Gramin Suvidha Kendra? Inventory issues arising from a hub-and-spoke system with many micro-entrepreneurs as spokes provide exciting research opportunities. For example, a hub-based inventory at a centralized warehouse reduces the inventory due to the “pooling” effect. However, it makes it costly for the micro-entrepreneurs to replenish their stocks, especially if they have to do so frequently owing to limited purchasing power. Involving local entrepreneurs as an informal sales force in developing countries creates new research opportunities to extend the existing marketing and the OM literature in salesforce planning, sales-territory design, and incentive design (Lilien et al. 1992).

Escamilla et al. (2021) discuss how innovative distribution models facilitated by new technologies such as smartphones, mobile payments and solar panels can enable nanostores to increase their capabilities to improve accessibility (in terms of

products/services) and affordability (in terms of price) along with some emerging research opportunities for further consideration.

21.4.9 Lending Working Capital to the Poor

Economists have studied microfinance since the early 1990s (cf. Armendáriz and Morduch 2007), and there are different economic theories on group lending for comprehensive reviews. One research opportunity lies in testing the assumption of risk reduction in group lending. The same could be applied to micro-entrepreneurs as distributors when provided with goods on inventory on a credit basis.

A second research opportunity deals with optimal loan repayment: frequent repayment schedule reduces the amount of defaulted loans, but it increases the lenders' cost of collection.

A third research opportunity is screening micro-entrepreneurs for lending to reduce the cost associated with default loans. Developing an effective way to create new credit scoring methods by analyzing the data captured by the financial transactions (remittances, loan repayments, payments) conducted over mobile phones (Lee and Tang 2012) may be a practical way to carry out such research. Researchers have also used Kiva's online portal to examine how this information on financial transactions would affect lending behavior among online lenders (Hartley 2010). This can be specialized to screening for distributors, mainly when the goods are being provided on credit.

Finally, it is less risky for manufacturers to lend to their suppliers because manufacturers have in-depth industry knowledge and informational advantage about their suppliers in developing countries than banks. Tang et al. (2018) have examined and compared the implications between loans offered by banks versus those by manufacturers. They find that manufacturer lending is even more effective when suppliers are risky borrowers. However, there is a need to examine how regulations can affect the development of such lending.

We started with a few questions facing managers and researchers regarding how social responsibility can or should be incorporated into supply chains. Then, we outlined the views in the literature culminating with the stakeholder resource-based view (Sodhi 2015) and how it helps address these questions. Next, we reported the findings regarding how organizations work with the poor, with implications for companies wishing to fulfill their social responsibility using their supply chains. Finally, we described potential areas for application and further research in this critical domain, hoping the readers will be motivated to find solutions in practice or opportunities for extending the literature.

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Chapter 22

Cross-Sector Partnerships for Sustainable Supply Chains



J. Balaisyte, M. Besiou , and L. N. Van Wassenhove 

22.1 Introduction

Sustainability issues are so immense that no single organization can face them alone. Global business, in order to deal with the pressure coming from governments and society, seeks to better manage their supply chains with regards to social and environmental impacts, and contribute to society. They often work with multiple stakeholders such as NGOs, through cross-sector partnerships. The 17th United Nations Sustainable Development Goal (UNSDG) emphasizes even more the importance of partnerships for achieving successful humanitarian actions.¹ Many examples of cross-sector partnership-based initiatives exist such as the TNT and World Food Program partnership, or the partnership between the humanitarian Logistics Cluster and logistics companies (Stadtler and Van Wassenhove 2016).

This chapter focuses on how supply chains can be used to create sustainable value through cross-sector collaboration. The objective of this chapter is twofold. First, we conduct case study research by using three examples of cross-sector partnerships to identify challenges that arise in the process of value creation. Second, we

¹ In order to address the current global challenges and move toward sustainable development, the UN has set up 17 goals: www.un.org/sustainabledevelopment/sustainable-development-goals/.

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discuss how these challenges could be addressed by using Operations Management (OM)/Supply Chain Management (SCM) research.

To achieve the first objective, we select three examples of cross-sector partnerships between pharmaceutical companies and healthcare focused NGOs, since we had the opportunity to follow their development over more than 1 year and they provide good illustrative examples of typical challenges. The pharmaceutical industry has been growing rapidly for the past decades, saturating developed markets, and exploring growth opportunities in emerging markets. Health supply chains in emerging markets face a number of challenges that require local market knowledge like retail distribution and pricing. Collaboration with NGOs can help pharmaceutical companies improve their corporate social responsibility (CSR) performance and close their local knowledge gap, while at the same time support healthcare systems in developing countries. From a supply chain management perspective, the private partners may engage in cross-sector partnerships in order to assess the potential of developing markets, build their reputation and motivate their employees.

The first case involves the Janssen Pharmaceutica hematology department unit of Johnson & Johnson, a multinational medical devices, pharmaceutical and consumer packaged goods manufacturer. The other two of our three cases involve Tibotec, a pharmaceutical company belonging to the Johnson & Johnson group, with a focus on research and development for treatment of infectious diseases. The first partnership was built between Johnson & Johnson and International HIV/AIDS Alliance in Zambia, the second between Tibotec and International HIV/AIDS Alliance in Ukraine, and the third between Tibotec and International HIV/AIDS Alliance in Uganda. International HIV/AIDS Alliance² is a global partnership of nationally based governmental and non-governmental organizations that support community organizations in addressing HIV/AIDS issues in developing countries.

All three cases were part of an Executive Development Programme (EDP) run by INSEAD, a global business school and PEPAL, a global foundation that fosters partnerships between businesses and non-profit organizations to achieve scalable and sustainable social change in developing and emerging markets.³ Hence the NGOs participated in the EDP, together with business executives, in order to get exposed to supply chain and other management related tools that would help them improve their skills. This program was implemented in collaboration with the International HIV/AIDS Alliance (IHAA) headquarters and its regional partner organizations that took part in the program. The EDP involved two 1-week sessions of executive training at INSEAD, one at the beginning and the other at the end of the program and a 1-year project in-between aiming to address challenges identified by NGO participants.

The chapter is organized as follows. In Sect. 22.2 we present an overview of literature on supply chain, cross-sector partnerships and introduce a theoretical

² <http://www.aidsalliance.org>

³ <http://PEPAL.org>

framework depicting factors that affect partnership success. Section 22.3 describes the three cases. Section 22.4 uses the framework to analyze the case studies, summarizes the main findings and practical implications and discusses possible avenues for value creation using OM/SCM research. Finally, Sect. 22.5 presents our conclusions.

22.2 Literature Review

In order to address our question—what are the challenges affecting the success of cross-sector partnerships in value creation for the partners—we first consider the supply chain partnership literature. According to the Supply Chain Management Institute “supply chain management is the management of relationships in the network of organizations, from end customers through original suppliers, using key cross-functional business processes to create value for customers and other stakeholders” (Lambert 2014).

Supply chain partnerships develop through different levels (Lambert 2014). At the first level the partners articulate their objectives. At the second level the partners need to align their expectations by setting the partnership’s objectives and then at the third level they develop the action plan and assign responsibilities. When the action plan is implemented, the partners review performance (fourth level) against expectations in order to decide how to proceed with the partnership.

In the case of cross-sector partnerships, private companies typically engage in such collaborations in order to improve their reputation, motivate their employees and develop or assess potential markets (Maon et al. 2009), while the NGOs’ drivers are the opportunity to increase their resources and the exposure to SCM and management related tools (Van Wassenhove 2006). Cross-sector partnerships refer to the partnerships that involve government, business, non-profits and philanthropies, communities, and/or the public as a whole (Bryson et al. 2006, Cooper et al. 2006, Austin and Seitanidi 2012a, b) and they are seen as necessary for the success of humanitarian operations (Lewin et al. 2018). Figure 22.1 presents the cross-sector supply chain partnership.

The supply chain partnership model was originally built to describe business-to-business collaborations. While partnerships between private companies face



Fig. 22.1 Cross-sector supply chain partnership

multiple challenges, like incentive misalignment or information asymmetry, these challenges are even more complex in the case of cross-sector supply chain partnerships. Health management in developing countries faces a lot of uncertainty due to constraints in funding and skills of the people employed (Thomas 2005; Gustavsson 2003). Hence, NGOs often do not have the necessary resources to acknowledge the importance of supply chain management, regarding it as an auxiliary function (Arminas 2005). While the partnership model provides a good understanding of the private supply chain partnership, it ignores the complex environment of cross-sector collaboration. For this reason, we consult literature on cross-sector collaboration between business and non-profit organizations.

A clear difference between the typical partnership model, described by Lambert (2014), and a cross-sector partnership is that private partnerships are seen as static while Austin (2000a, b) argues there are three stages of collaboration in cross-sector partnerships: philanthropic (unilateral transfer of resources), transactional (reciprocal exchange of resources) and integrative (based on a very close organizational coordination and co-creation of value). Austin and Seitanidi (2012b) build on the work of Austin (2000a, b) and suggest collaboration may evolve to a fourth transformative stage (aimed to co-create change at the societal level). This implies that cross-sector partnerships are dynamically evolving over time based on the nature, intensity and form of interaction.

No matter at which stage partners commence their collaboration, they go through several phases. Selsky and Parker (2005) distinguish three phases of partnership life-cycle: (1) formation, (2) implementation and (3) outcomes, looking at project-based cross-sector partnerships to address social issues. Austin and Seitanidi (2012b) divide partnership life-cycle into the following phases: (1) partnership selection and formation, and (2) partnership implementation and post-formation management. Similarly, Kale and Singh (2009) suggest three phases of alliance: (1) partner selection and alliance formation, (2) alliance governance and design and (3) post-formation alliance management. The expected benefits should be articulated to the partners and society (Austin et al. 2000) and, eventually, decide if the partnership should be continued or terminated. The processes describe how the actions and the dynamics take place during the phases. Finally, the outcomes reflect the value created as the impact of the partnership. Measurement systems and key performance indicators (KPIs) could be of help in achieving this.

Austin and Seitanidi (2012a, b) argue that value of partnering originates from resource complementarity, resource nature, resource directionality and use, and linked interests. It eventually materializes into one of the following types of values: associational, transferred resource, interaction and synergetic. Associational value is a consequent benefit accruing to a partner from having a collaborative relationship with the other one. Transferred resource value arises by receiving a resource from the other partner. Interaction value depicts the intangibles that derive from the collaboration like communication or leadership skills. Synergistic value reflects the value that arises by combining partners' resources; this value is higher than the one that would be accomplished if they would have acted separately.

We use the literature on cross-sector partnerships and supply chain partnerships to build a framework that will help us identify factors leading to successful cross-sector partnerships. Specifically, we look in the literature for factors that influence partnership formation and selection management, partnership implementation management and outcomes. We also examine the managerial challenges and complex environment in which these partnerships occur.

In Sects. 22.2.1 and 22.2.2 we focus on the two phases of partner selection and formation, and partnership implementation and post-formation management, respectively.

22.2.1 Partner Selection and Partnership Formation

Decisions at the partner selection and formation phase influence partnership's future potential to evolve and create value. This phase allows partners to align expectations and determine if a potential relationship is worth time and investment. It includes a range of activities such as problem-setting processes (Gray 1989, McCann 1983), initial conditions of the partners, for example resources they possess (Bryson et al. 2006), and assessments indicating benefits likely to be produced by the collaboration (Clarke and Fuller 2010; Gourville and Rangan 2004). As identified by Lambert (2014) it is also crucial for the supply chain partnership to set objectives and match expectations.

Austin and Seitanidi (2012b) show that initial articulation of the problem, linked interests and resources, partners' motives (Seitanidi and Crane 2009) and missions, history of past interactions and visibility fit are key measures of partnership formation and fit potential. Visibility fit reflects the desire to gain visibility (Gourville and Rangan 2004) that may enhance reputation (Tully 2004), and public image (Alsop 2004; Heap 1998; Rondinelli and London 2003), which are benefits that can be attributed to associational value. Visibility fit is a very important aspect in Corporate Social Responsibility (CSR) based relationships.

One indicator of the potential for value creation is the identification of linked interests through the initial articulation of a social problem relevant to both partners (Bryson et al. 2006; Gray 1989; Waddock 1986). Addressing the social problem often becomes the key objective for the partnership itself. The process of articulation can be challenging and show the incompatibilities between partners signalling the need for realignment (Austin and Seitanidi 2012b). Moreover, when the social problem is linked to the interests of the partners, the probability that they will benefit from the partnership is higher (Le Ber and Branzei 2010). So matching the interests of the partners shows that they better fit with one another and may increase partnership success.

Partners' motives and missions reveal possible linked interests and expected benefits (Seitanidi 2010). According to Kale and Singh (2009), partner complementarity shows how partners contribute non-overlapping resources to the partnership and it may include both tangible and intangible (knowledge, capabilities, management

practices and skills) resources. It also refers to the fit between partner working styles and cultures (Austin and Seitanidi 2012b). Corporations willing to enter developing regions have to understand the unique conditions of this environment and try to fit culturally (Dahan et al. 2010). Checking for compatibility and complementarity early on, by verifying the past history of interactions and visibility fit (Austin and Seitanidi 2012b), can reduce the probability of misunderstandings, misallocation of costs and benefits, mismatches of power, lack of complementary skills, and mistrust (Berger et al. 2004). Social partnerships are inherently fragile also because of individuals involved in the collaboration for whom partnerships may be of secondary concern when compared to their daily jobs (Waddock 1988).

Once the formation measures are set, the next step concerns the partner selection using predefined partnership criteria and the measures of partnership formation and fit potential discussed above. Specific criteria may involve factors such as the industry of interest, resource availability, and scope of operations and may facilitate the process of assessing potential partners. There are cases where some organizations may not agree with all of the activities involved in the partnership formation and selection process or some activities may overlap.

22.2.2 Partnership Implementation and Post-Formation Management

The implementation of the partnership commences when the needs and problems of the partners are clear and the partnership is formed. At the beginning of this phase, the partnership is designed and then the operations follow. Partnership design processes include setting objectives and structures (Austin 2000b; Bryson et al. 2006; Googins and Rochlin 2000), rules and regulations (Das and Teng 1998; Gray 1989), leadership positions (Austin 2000a; Waddock 1986) and agreements on partnership management (Seitanidi and Crane 2009). Moving from design to operations is often followed by experimentation, adaption (Austin 2000a; Gray 1989), operationalization and institutionalization of processes as partners improve and readjust their coordination mechanisms and structural arrangements (Austin and Seitanidi 2012b). This process is facilitated by frequency in communication, professional leadership, evaluation of progress, and ability to set objectives (Googins and Rochlin 2000; Austin and Seitanidi 2012b).

While setting the objectives is an important factor for partnership success (Googins and Rochlin 2000), rules and regulations can also emerge as partnership evolves (Austin and Seitanidi 2012b). Informal communication is more likely to be effective in dealing with tensions between the partners and harmonizing different organizational cultures (Orlitzky et al. 2003). Austin and Seitanidi (2012b) find by conducting a literature review that harmonizing two different organizational cultures, leadership, forms of communication that enable trust, mutual respect, openness, constructive criticism, and open dialogue play an important role in cross-

sector partnerships. Shah and Swaminathan (2008) emphasize that commitment is crucial for partnership success especially when partners have clear expectations from the partnership but vague processes regarding how to achieve them. In this case, partners should be willing to dedicate more resources to the relationship and pledge to work with each other even when they realize that some adaptation might be required. Lu et al. (2018) and Moshtari (2016) identify trust between different organizations as a critical success factor for their collaboration.

The value created by partnerships should then be measured. For example, a group of companies, including pharmaceutical companies like, Abbott, AbbVie, Astra Zeneca, GlaxoSmithKline, uses a framework to assess the real value and impact of their community investment to both business and society.⁴ Other practitioner studies, such as IFC, Community Investment Guidelines⁵ and TPI, Current Practice Evaluation,⁶ provide guidelines on how to measure and communicate community investment for strategic advantage. In this respect, Googins and Rochlin (2000) raise a question that is common among practitioners: how partners can measure results of their partnerships. On an organizational level, by reviewing best practices of social program, Sept et al. (2011) proposes four methods for evaluation: impact evaluation, performance monitoring, process evaluation and social return on investment evaluation. Koza and Lewin (2000) discuss the importance of monitoring performance outcomes. Specifically, they find that in partnerships with less well defined objectives such as exploration alliances, monitoring progress and performance outcomes, and setting clear partnership goals is greatly complicated and requires the design and execution of process controls. This often happens because the performance goals are generally stated in much less specific, causally ambiguous, open-ended terms such as acquiring new capabilities and learning new technologies. Key performance indicators (KPIs) can also be used to show whether the partnership is moving towards its objectives. At that point, the partners need to decide if they should exit the partnership or develop a continuation strategy.

The key processes of the post-formation management and implementation phase described above are used in our framework, presented in Fig. 22.2. These processes highlight also the challenges that partners may face at every phase, like lack of commitment, trust and an open relationship. The two phases are followed by the decision to assess value created and to exit or continue the partnership (continuation strategy). Hence the framework presented in Fig. 22.2 captures the factors that according to the literature affect partnership success. In Sect. 22.3 we use this framework to analyze our three cases.

⁴ LBG Model: <http://www.lbg-online.net/about-lbg/the-lbg-model.aspx>

⁵ Strategic Community Investment: A Good Practice Handbook for Companies Doing Business in Emerging Markets: <http://www.ifc.org/wps/wcm/connect/60a5be8048855226aab4fa6a6515bb18/12014chapter8-.pdf?MOD=AJPERES&CACHEID=60a5be8048855226aab4fa6a6515bb18>

⁶ The Partnering Initiative: Current practice in the evaluation of cross-sector partnerships for sustainable development http://thepartneringinitiative.org/wp-content/uploads/2014/08/WP1_Evaluation.pdf

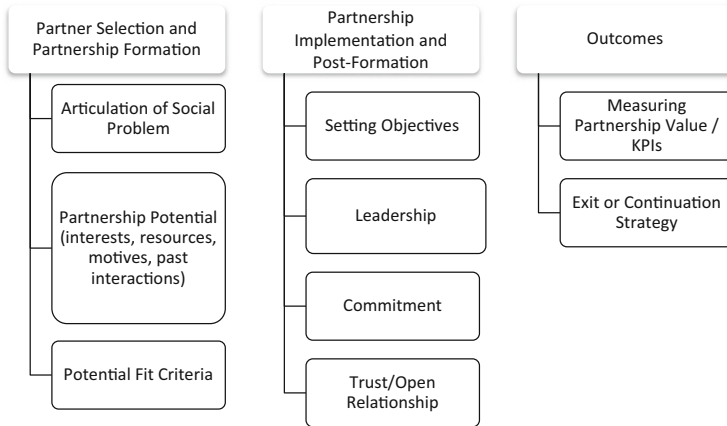


Fig. 22.2 Framework depicting potential factors affecting cross-sector partnership success

22.3 Empirical Study: Case Description

In this section we present three case studies of partnerships between pharmaceutical companies (headquartered in Belgium) and healthcare focused non-profit organizations operating in developing countries. The three partnerships were built and implemented in 2009–2010, during the 12 months Executive Development Program (EDP) commissioned by PEPAL, a social enterprise and carried out by the INSEAD Humanitarian Research Group. Using the framework (Fig. 22.2) developed in the previous section, we identify main challenges that arise in the process of building and implementing the cross-sector partnerships and then we compare the three partnerships.

Partnerships underwent several phases (Austin and Seitanidi 2012b). First, a partnership formation and selection phase took place during which, with assistance of PEPAL, potential candidates were identified and matched. PEPAL following discussions with both NGO and private partners built a list of criteria to match the partners. Thus the three partnerships were formed based on the skills of the private partners, their previous collaboration experience, the mutual interests of both partners and the chemistry developed between the individuals forming the partnership. Table 22.1 presents the partner selection criteria.

Second, partnership implementation and post-formation management followed, throughout which partners carried out partnership management activities as part of the 1-year program. Our three partnerships had no formal governance structure. The collaboration was based on mutual trust and informal relationships. Lastly, the outcomes of the partnerships are also discussed looking at them as the result of the collaboration rather than a separate phase. After 1 year, partners evaluated their progress and outcomes, and made decisions with regard to the future of their collaboration [the full cases can be found in Balaisyte and Van Wassenhove (2011)].

Table 22.1 Partner selection criteria for the three cases

Criterion	Partnership in Zambia	Partnership in Ukraine	Partnership in Uganda
Industry of interest	Previous experience of both partners in healthcare	Previous experience of both partners in healthcare	Previous experience of both partners in healthcare
Previous collaboration experience	No previous cross-sector collaboration experience of either partner	No previous cross-sector collaboration experience of either partner	Previous experience of private partner in healthcare and developing countries
Interests	Mutual interest: healthcare in developing regions	Mutual interests: healthcare in developing regions, cost effective relationship (money/time/investment vs. outcomes)	Mutual interest: healthcare in developing regions
Personal factor	Personal chemistry with some people across the two organizations	Personal chemistry between the core people of the two organizations	

The structure of the mini-cases is as follows. We first describe how the partnership was formed. Then we briefly discuss the evolution of the partnership. Finally we summarize the challenges encountered and outcomes achieved by the partnership.

22.3.1 Case 1: Partnership in Zambia

Around 68% of Zambians live below the recognized national poverty line (Development Indicators Unit, Statistics Division, United Nations 2014). The country is experiencing a generalized HIV/AIDS epidemic, with a national HIV prevalence rate of 17% among adults, with the NGO sector being a crucial player in providing health services to the community. Lack of resources, capacity and training in the public sector for the healthcare services creates demand for NGO services.

22.3.1.1 Formation of the Partnership

International HIV/AIDS Alliance in Zambia, established in 1999, has been a partner organization for the Zambia Integrated Health Program (ZIHPCOMM).⁷ Since

⁷ <http://bixby.berkeley.edu/bixby-internship-zambia-2003/>

it was founded, International HIV/AIDS Alliance in Zambia was challenged by decentralization since it has grown with a need to make the transition from a country office of the International HIV/AIDS Alliance to a fully independent Zambian NGO. Change management and capacity were required to handle this transition. International HIV/AIDS Alliance in Zambia joined this EDP while seeking to benefit from marketing expertise to build the image of the evolving organization, to increase the available resources of its supply chain, and gain benefits through the associational value (by improving its image) and interaction value (acquiring marketing skills from interaction).

At the same time the private partner, backed by its organization, was looking for a 1-year engagement within an NGO operating in the healthcare sector. The private partner had worked in the pharmaceutical industry for the past 10 years, mostly in developed countries, and had extensive experience in the domains required by the NGO. So the private partner could help the NGO improve the management skills of employees and the NGO's operations and hence no longer regard their supply chain as an auxiliary function (Arminas 2005). The private partner was interested to build an understanding of how the healthcare system functions in a less developed context and how the pharmaceutical industry could contribute. Thus the expected benefits of the private partner through this partnership included interaction value (learning how a non-profit organization operates) and satisfy his interest in learning how to operate in a developing country. With the support of PEPAL, the two partners were matched and introduced to each other based on the objectives of the partners and potential fit criteria. For example for the NGO the skill set that the private partner possessed, industry (healthcare) expertise, and interest in developing regions were important criteria.

22.3.1.2 Evolution of the Partnership

The private partner joined Alliance Zambia as an advisor to the senior management team, which allowed him to take a leadership position. Partnership management tools and performance metrics initiated by the private partner through the form of semi-annual surveys were used. Partners established trust, compatibility, complementarity and commitment-based relationships. The relationships were straight, equal, and built on trust. With the private partner being on the ground for the 1-year period, partners had an opportunity to build close relations through daily communication. For example, the private partner supported Alliance Zambia team to prepare for one of the largest events, the National Prevention Convention by supporting and empowering its employees to take on new roles such as public speaking and succeed during the event. This is an example of how with the help of the private partner, the skills of the NGO's employees can be improved and their knowledge gap of business practices can be addressed (Samii and Van Wassenhove 2003).

22.3.1.3 Challenges and Outcomes

Unfortunately, in the midsummer the NGO experienced funding challenges. As a result, there was a change in management and the new Executive Director had different priorities. This is a common phenomenon in NGO supply chains which often face difficulties to attract and retain employees with management experience because of funding issues (Thomas 2005; Gustavsson 2003). The financial difficulties the International HIV/AIDS Alliance in Zambia faced jeopardized project goals. Taking the limited available resources into account, the partners decided not to continue with the collaboration and terminate it at the end of the program at INSEAD.

22.3.2 Case 2: Partnership in Ukraine

Ukraine's major developmental challenges include underdeveloped infrastructure, unstable political environment, corruption and excessive bureaucracy. The country has one of the highest HIV/AIDS prevalence rates in Europe, an area in which the NGO community is very active. The primary issues faced by the supply chains of most NGOs operating in the area, including International HIV/AIDS Alliance in Ukraine, were limited resources and strong competition for funding.

22.3.2.1 Formation of the Partnership

Five years after its foundation, Alliance Ukraine expanded by establishing a subsidiary that hosts the Regional Technical Support Hub (TS Hub) for the Eastern Europe and Central Asia region. The TS Hub provides technical support services to a wide range of NGOs, acting in this way as a new product that Alliance Ukraine can offer to other NGOs, and serves as a mechanism for the International HIV/AIDS Alliance to obtain additional funding for their operations.

Before joining the EDP, the Alliance Ukraine team was struggling to turn the hub into a sustainable business entity. Alliance Ukraine easily identified the TS Hub as the project for this partnership. The potential benefits of engaging in the partnership were of resource nature (to increase their funding) and interaction value (acquiring marketing skills and project development from interaction).

The private partner Tibotec has a strong CSR culture. It was a good match for Alliance Ukraine's needs in terms of skill-set and expertise, as they had extensive experience in marketing and project development. Tibotec was interested in the partnership as a career development opportunity for its senior staff. Moreover, the corporate partners saw this partnership as an opportunity for improving their understanding of the epidemic in Ukraine which may facilitate their future operations in this new market as well as collaboration with the NGO sector. Tibotec partners supported Alliance Ukraine with market growth strategy expertise by sharing market

analysis and its strategic planning knowledge. The potential benefit for the private partner was interaction value (learning how a non-profit organization operates, employees' personal interest and motivation).

22.3.2.2 Evolution of the Partnership

With the support of PEPAL, the two partners were matched and introduced to each other based on their objectives and potential fit criteria. For example for the NGO the skill set that the private partner possessed, their social mission and the interest of the private partner in developing regions were important fit criteria. Table 22.1 presents the partner selection criteria for this partnership.

During the kick-off week at INSEAD, the project team divided roles, leadership and responsibilities, and also agreed on the interaction patterns. The project team had established a routine of bi-weekly telephone conferences that helped partners build a strong relationship and confront all issues openly. After the first 2 months of the partnership the hub manager left the organization and a temporary manager from the International HIV/AIDS Alliance in Ukraine was assigned.

After the new TS Hub manager joined the HIV/AIDS Alliance in Ukraine, the private partners came to Ukraine and spent 1 week working with the new TS Hub manager and her team in order to readjust project goals. The project team developed good relationships. The partners shared the leadership of the project, with the private partner taking the lead on communication and mentoring, and the NGO partner leading the direction of the project.

22.3.2.3 Challenges and Outcomes

The first 2 months of the project were challenging since the International HIV/AIDS Alliance team in Ukraine did not have a clear direction regarding the hub, a typical characteristic of NGO supply chains (Maon et al. 2009). However, partners managed to overcome this challenge by developing their strategic direction. One year later, the partnership proved to be very collaborative with strong commitment and trust coming from both sides. The TS Hub became a fully functional and revenue generating unit and partners maintained good relationships.

22.3.3 Case 3: Partnership in Uganda

Uganda is a landlocked country in East Africa with more than 35% of its population living on less than \$1.25 a day.⁸ Currently, 7.2% of Uganda's population is suffering

⁸ <http://data.worldbank.org/indicator/SI.POV.DDAY>

from HIV/AIDS.⁹ The high poverty of the population combined with the lack of public resources spent on health services, increases the demand for NGO services.

22.3.3.1 Formation of the Partnership

The International HIV/AIDS Alliance in Uganda has been present in the country since 2005 with an objective to improve access to HIV/AIDS prevention means, care treatment and support services to orphans and vulnerable children. Being a young organization, the International HIV/AIDS Alliance in Uganda suffered from their staff lacking project and program implementation skills, a common characteristic of NGO supply chains (Thomas 2005; Gustavsson 2003). Therefore, a partnership with the private sector was considered a good opportunity to improve project management skills and acquire a commercial sector perspective. Tibotec was ready to offer these capabilities and welcomed the project as an opportunity to engage in a meaningful CSR initiative. Through this partnership, Alliance Uganda was seeking to acquire increased resource base (in order to improve the skills of their employees). The benefit for the private partner was associational value (through the CSR initiative).

With the support of PEPAL, the two partners were matched and introduced to each other based on the potential fit criteria and their interests. Past experience of the private partner in working with NGOs and in developing regions was considered an advantage. Table 22.1 presents the partner selection criteria.

22.3.3.2 Evolution of the Partnership

The partnership evolved through several phases. Partnership formation occurred during the EDP at INSEAD where partners agreed to work on the project through frequent communication over the phone and e-mails. At the beginning of the project private partners made a first visit to Uganda, which was an eye opening experience. They realized that defining clear goals was not an easy task. The first 4 months of the project were spent on discussions about how Tibotec could help Alliance Uganda. Then financial challenges arose for the NGO. The International HIV/AIDS Alliance in Uganda office was downsized, including management. The executive director left the NGO, leaving Tibotec without a leadership partner on the ground.

Following 2 months of struggle, a temporary project manager from the headquarters of International HIV/AIDS Alliance stepped in to support the partnership and Alliance in Uganda while the headquarters of the HIV/AIDS Alliance was supporting the hiring of the new executive director and project manager for the partnership. Because of this change, the partners were forced to redefine the goals of the supply chain partnership. Due to the financial challenges, International

⁹ http://www.avert.org/hiv-aids-uganda.htm#footnote3_we9dl36

HIV/AIDS Alliance in Uganda decided to develop a Technical Support Hub (TS Hub), designed to generate revenues as an additional service that Alliance in Uganda could offer to other NGOs. Preparing a business plan and conducting market research became the new partnership objectives. Nine months after the beginning of the project, a new manager arrived at the NGO. The partners had to go through the whole adjustment and relationship building cycle again, which negatively affected the project development dynamics, as described in the next subsection.

22.3.3.3 Challenges and Outcomes

To sum-up, despite a number of adjustments, it was hard for both sides to work on the project with no leadership support coming from the Alliance Uganda side. There was a lack of clarity in responsibilities and understanding by the Alliance Uganda team on how to accomplish the goals of the supply chain partnership, and misalignment of expectations on both sides. When the new executive director was hired, due to financial challenges and other priorities, he was not in a position to engage in the partnership to the full extent. Therefore, without active leadership, commitment and support from the executive director, the project was not able to make progress.

After 1-year of not being able to find a common ground for collaboration the partners re-evaluated their efforts and decided to terminate the partnership. They used the program's final training week at INSEAD to meet face-to-face and take this decision.

22.4 Propositions and Research Implications

Four stages of collaboration in cross-sector partnerships are identified by Austin (2000a, b): philanthropic, transactional, integrative and transformative. Our three cross-sector partnerships are based on transfer of resources to one another. However, our partnerships are not yet so advanced in order to co-create value or to make changes at the societal level, even if these are their future goals. So our three partnerships fall into the transactional stage. The forms of value identified in our partnerships are associational, transferred resource, interaction and synergetic value as presented by Austin and Seitanidi (2012a, b).

In this section the three cases are analyzed using the findings from the literature. Section 22.4.1 presents the propositions that arise, while Sect. 22.4.2 discusses the research implications.

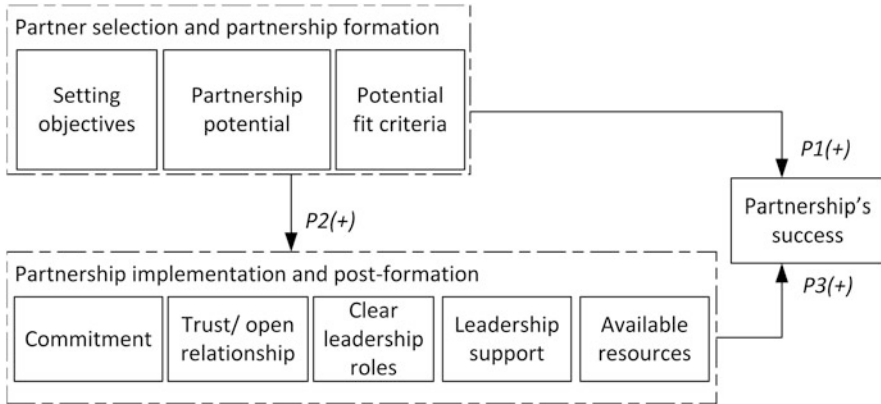


Fig. 22.3 Factors affecting success of cross-sector partnerships. Positive signs (+) indicate that the factor/s at the beginning of the arrow and the variable at the end change in the same direction; for example if the partners put more effort on setting clear objectives for the partnership, then there is higher probability the partnership will be successful

22.4.1 Propositions

In this section we use the framework developed in Sect. 22.2 (Fig. 22.2), which depicts potential factors affecting the success of the cross-sector partnerships, and the case studies conducted to identify the factors that affected the outcomes of the three cross-sector partnerships.

Our analysis of the three cases revealed that during the partnership selection and formation phase setting clear objectives for the supply chain partnership (relevant to all cases), and having compatible and complementary partners regarding skills and available resources (relevant to all cases) improves chances to achieve partnership goals (Fig. 22.3). These findings are supported by the literature on the importance of cultural fit (Dahan et al. 2010), compatibility and complementarity of skills (Austin and Seitanidi 2012a, b; Berger et al. 2004) and resources (Kale and Singh 2009). Taking into account these findings, we form our Proposition 1.

Proposition 1 (P1): During the partner selection and partnership formation phase, improving:

- *Setting objectives*
- *Identifying the partnership potential (clear benefits to all partners)*
- *And identifying fit criteria (between partner working styles and cultures, and clear understanding of the skills and weaknesses on both sides)*

will have a positive impact on the success of cross-sector supply chain partnership.

In all three case studies we found a link between the success of the partner selection and partnership formation, and the partnership implementation and post-

formation management. For example, we find that if the partnership potential is not adequate during the partner selection and partnership formation phase (e.g., regarding available resources of the supply chain partnership) then the partnership during the implementation and post-formation management phase faces challenges (all three cases) and objectives, processes and responsibilities may need to be readjusted. These results are in accordance with the findings of Kale and Singh (2009), Austin and Seitanidi (2012b) and Le Ber and Branzei (2010).

Taking into account these findings, we form our Proposition 2.

Proposition 2 (P2): Improving the planning of the partner selection and partnership formation phase will have a positive impact on the partnership implementation and post-formation phase of the supply chain partnership.

The success of the three cross-sector partnerships depends on factors linked to the partnership implementation and post-formation management phase. In all three cases we find that sustaining the commitment (evident in the Zambia and Ukraine cases) (Shah and Swaminathan 2008), building trust and open relationship (as was the case for all three partnerships) (Austin and Seitanidi 2012b), having a detailed map of leadership roles and responsibilities (evident in the Ukraine case) (Googins and Rochlin 2000; Austin 2000b; Bryson et al. 2006), establishing leadership support (as in the Ukraine case) (Austin 2000a; Waddock 1986), and having available resources for the partnership implementation (which was a barrier for all three cases) (Kale and Singh 2009) has a positive effect on achieving partnership objectives. Based on these observations, we form our Proposition 3.

Proposition 3 (P3): During the partnership implementation and post-formation phase, improving:

- *Commitment*
- *Trust and open relationship*
- *Having a detailed map of leadership roles and responsibilities*
- *Establishing leadership support*
- *And having available resources for the partnership*

will have a positive impact on the success of the cross-sector supply chain partnership.

By comparing the literature findings (Fig. 22.2) with our three case study propositions (Fig. 22.3), we observe that articulation of the social problem did not challenge the partnership's success; perhaps this is due to the set-up of the EDP that allowed to clarify the social problem early in the program. However, there was a difference in the timing of the objectives. The partnership objectives were instrumental for the partner selection and were set before the partnership implementation started. On the contrary, the literature suggests that objectives are typically set later.

Finally, all three partnerships were challenged to measure the success of their supply chain partnerships, both on an organization and partnership level. Corporate partners did not establish any specific measures to track partnership impacts on their organizations. For the NGOs, the positive impacts of the collaboration were attributed to the success of the partnership in general. Observing our partnerships

we found that the main challenge for setting KPIs and monitoring performance was related to the combination of several factors: partners' inability of setting objectives, and constantly changing available resources and engagement of senior management. Partners often were not sure about their ultimate objectives or these objectives were too broad or ill-defined. Initially all partnerships had very ambitious goals such as creating a new marketing strategy or designing a new business plan, that often proved difficult to achieve. These factors affected the progress of the partnership, resulted in a need to readjust or change objectives, and affected the ability to track partnership performance.

22.4.2 Research Implications

The analysis of the three cases also revealed that business and non-profit partnerships are characterized by complexity due to the nature of NGO resource constrained environment, uncertainty, and multiple trade-offs (Besiou and Van Wassenhove 2015). Often NGOs are challenged by the resources that are required in advance for the partnership to function properly. Uncertainty is reflected in fierce competition among NGOs for limited funding, which makes the future of the NGO employees insecure. In our cases the business partners were often left without any support coming from the NGO side. Furthermore, due to the uncertainty of resource availability, the objectives of the partnerships, the support coming from the leadership, the roles of the partners and their commitment change dynamically. There are also time delays in building trust between the partners and improving their skills/capacity. Moreover, the partners need to deal with trade-offs between the short-term losses of investing constrained resources and the long-term benefits of the social cause. In "uneasy" partnerships, like the three cases discussed in this chapter, it may be challenging to capture and monetize the value created by the supply chain partnership. Value is often rather intangible, so setting quantitative KPIs may be hard.

We believe that there is an opportunity for OM/SCM research to dive into the interesting environment of cross-sector partnerships. First, OM/SCM researchers need to go to the field to understand the constraints and the complex environment of these partnerships. After mapping the territory, modeling could be used to better understand the impact of the limited resources (for example lack of funding) on the supply chain partnership.

Then the impact of different decisions and actions could be evaluated to maximize the benefits of working together and building more effective partnerships. Education in OM/SCM could also find research in cross-sector supply chain partnerships beneficial. Examples of cross-sector partnerships could be used contrasting them to examples of commercial partnerships. The students could initially try to explore the impact of constrained resources on the operations of the supply chain partnership at a conceptual level and then capture the right trade-offs with OM/SCM

models. From a pedagogical perspective, students could also try to come up with KPIs that could measure the social impact and the success of such partnerships.

Our examples of cross-sector partnerships are characterized by stakeholders with conflicting goals (private companies, NGOs and donors) (Van Wassenhove and Besiou 2013). The private companies engage in such partnerships because of their CSR strategy, while the NGOs use them as means to improve the skills of their employees and improve their funding. Private sector supply chain partnerships also face multiple challenges like incentive misalignment but what makes these issues more challenging in cross-sector environments are limited resources and higher levels of uncertainty. For example NGO supply chains suffer from high turnover due to limited funding (Thomas 2005; Gustavsson 2003). Moreover, as seen in our cases, many of the partners were engaging for their first time in a cross-sector partnership and they had different expectations. So being part of such a partnership may not be straightforward for all the partners, or even desirable for all employees, an unfamiliar context for private supply chains. OM/SCM can also be used to map the stakeholders' dynamic goals and understand this context. For example Stadler and Van Wassenhove (2016) study the partnership between the humanitarian Logistics Cluster and four logistics companies. Even if the four logistics companies are competitors, when a disaster strikes they activate the supply chain partnership and share resources to optimize the social benefit. Broader issues from a supply chain perspective need to be taken into account, like the strong commitment coming from the partners. Collaborative game theory and system dynamics could be applied in order to study the dynamic changes of the partner roles for different levels of decentralization and competition coming from the funding in an effort to maximize the benefit of the partners.

The complexity and the unfamiliar context of the cross-sector supply chain partnerships can lead to counter-intuitive behavior (Besiou and Van Wassenhove 2015). In our cases even if the partners had to face many challenges, their commitment was really strong due to their belief in a social cause. Pedagogical cases in OM/SCM discipline could also be used here in order to study how the objective functions of the supply chain partnership would change under such conditions.

22.5 Conclusions and Practical Implications

Today increasingly more companies are under pressure to engage in corporate social responsibility initiatives like partnerships with non-governmental organizations. In response to increased practitioner interest on the topic, this chapter studies three examples of cross-sector supply chain partnerships between pharmaceutical companies and healthcare focused NGOs. In this study we look at the factors that affect the success of such partnerships, and identify the avenues for future research on cross-sector partnerships for the OM/SCM discipline.

We hope that our findings will be helpful to master-level students, academia, but foremost to the actual practitioners in healthcare sector both on business and NGO sides, who are already managing or planning to engage in cross-sector partnerships.

As observed from the three partnerships, engaging in cross-sector collaboration requires substantial amount of resources and efforts that may not be straightforward to all partners. Our framework and findings can help practitioners manage the value creation process better through cross-sector partnerships and answer the following questions: How different partnering processes and factors (e.g., setting objectives) affect value generation? How these different factors can be most effectively organized during different phases of partnership (e.g., leadership)? How and in what combination can partners use resources designated for partnering (e.g., available resources)? What actions can help partners improve implementation of the partnership internally, externally and between partners (e.g., communication and commitment)? We hope that this study will support their efforts in setting-up partnerships and will provide insights on how to build more efficient collaboration.

To have a successful partnership, the engagement process needs to be supported by the leadership. The benefits for each partner and for the social cause need to be communicated to all employees; unclear communication and lack of commitment can undermine the partnership's success. If the partners due to external constraints, like lack of funding, lose their direct interest in the partnership, then resetting the objectives of the supply chain partnership or changing the leadership roles may be helpful. For example, in the case of the partnership in Uganda the initial goals concerned how to improve business related skills of employees. However, when the NGO faced funding issues, the partners decided to set sustaining the service of the TS Hub in order to increase their financial resources as the new objective.

In addition, given the specifics and challenges presented by these partnerships, characterized by multiple stakeholders with differing objectives, and the complexity of the system in which they operate, we see ample of space for further research. We believe that the trade-off between short-term goals of spending fewer resources and the long-term goal of capacity building offers interesting research opportunities.

This chapter shows that for research on sustainable supply chains we need to take into account all the other broader issues that should go beyond the traditional supply chain perspective. It is in particular relevant in addressing healthcare management issues in developing countries where actors face uncertainty due to capacity constraints such as funding and skills of their employees (Thomas 2005; Gustavsson 2003).

In addition, the cross-sector supply chain partnerships are very different from the ones developed between private partners. In the NGO context, SCM is of increasing importance but it has been undervalued since NGOs often regard it as an auxiliary function (Arminas 2005). In order to overcome this gap, private companies have an opportunity through cross-sector supply chain partnerships to help NGOs increase their resources and the exposure to SCM and management related tools (Van Wassenhove 2006). In this way, the skills of the NGO employees will improve and their knowledge gap will decrease (Samii and Van Wassenhove 2003). At the same time the pharmaceutical companies engaged in such partnerships will acquire

valuable experience of operations in developing markets. Interdisciplinary research combining OM/SCM discipline with behavioral, management and strategy topics could be beneficial to optimize the desired impact of the partnership with the existing resources.

This research has some limitations. First, it is limited to three cases. This was necessary to be able to focus only on one specific industry in order for the results to be comparable, but the generalization of the findings is limited. Second, more research is needed to test the propositions.

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